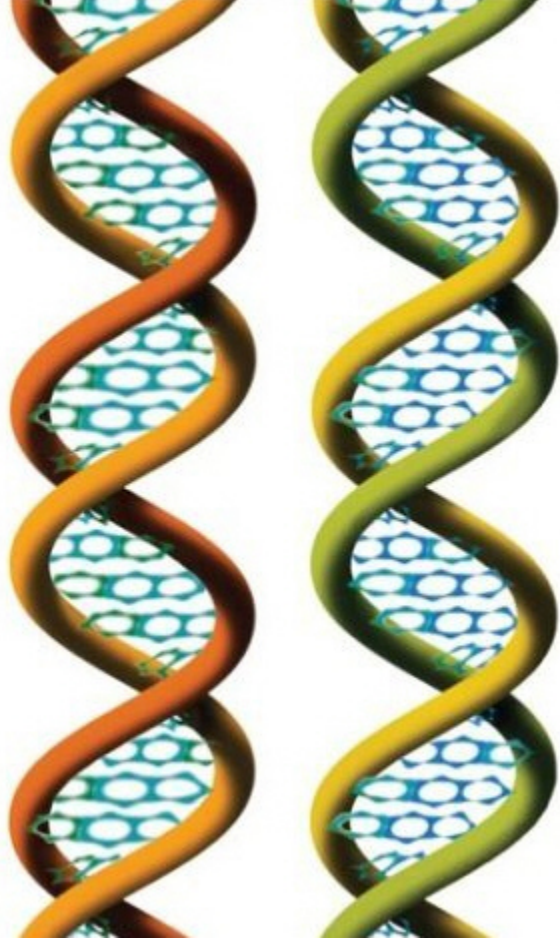


GENETICS

A Conceptual Approach

BENJAMIN A.
PIERCE



1

Introduction to Genetics



Alexis, heir to the Russian throne, and his father Tsar Nicholas Romanoff II.
(Hulton/Archive by Getty Images.)

Royal Hemophilia and Romanov DNA

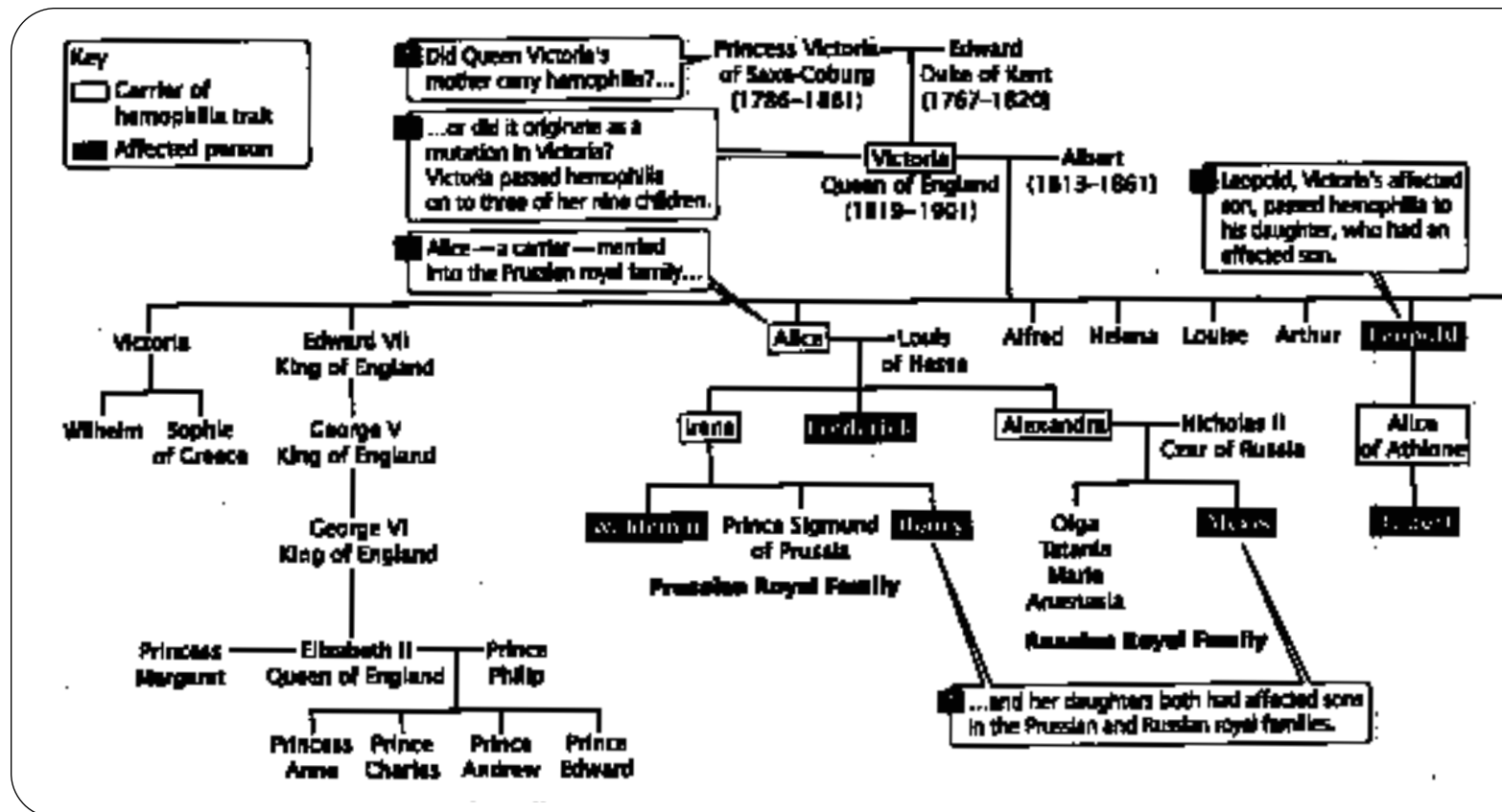
On August 12, 1904, Tsar Nicholas Romanov II of Russia wrote in his diary: “A great never-to-be forgotten day when the mercy of God has visited us so clearly.” That day Alexis, Nicholas’s first son and heir to the Russian throne, had been born.

At birth, Alexis was a large and vigorous baby with yellow curls and blue eyes, but at 6 weeks of age he began spontaneously hemorrhaging from the navel. The bleeding persisted for several days and caused great alarm. As he grew and began to walk, Alexis often stumbled and fell, as

all children do. Even his small scrapes bled profusely, and minor bruises led to significant internal bleeding. It soon became clear that Alexis had hemophilia.

Hemophilia results from a genetic deficiency of blood clotting. When a blood vessel is severed, a complex cascade of reactions swings into action, eventually producing a protein called fibrin. Fibrin molecules stick together to form a clot, which stems the flow of blood. Hemophilia, marked by slow clotting and excessive bleeding, is the result if any one of the factors in the clotting cascade is missing or faulty. In those with hemophilia, life-threatening blood loss can occur with minor injuries, and spontaneous bleeding into joints erodes the bone with crippling consequences.

- Royal Hemophilia and Romanov DNA
- The Importance of Genetics
 - The Role of Genetics in Biology
 - Genetic Variation is the Foundation of Evolution
 - Divisions of Genetics
- A Brief History of Genetics
 - Prehistory
 - Early Written Records
 - The Rise of Modern Genetics
 - Twentieth-Century Genetics
 - The Future of Genetics
- Basic Concepts in Genetics



1.1 Hemophilia was passed down through the royal families of Europe.

Alexis suffered from classic hemophilia, which is caused by a defective copy of a gene on the X chromosome. Females possess two X chromosomes per cell and may be unaffected carriers of the gene for hemophilia. A carrier has one normal version and one defective version of the gene; the normal version produces enough of the clotting factor to prevent hemophilia. A female exhibits hemophilia only if she inherits two defective copies of the gene, which is rare. Because males have a single X chromosome per cell, if they inherit a defective copy of the gene, they develop hemophilia. Consequently, hemophilia is more common in males than in females.

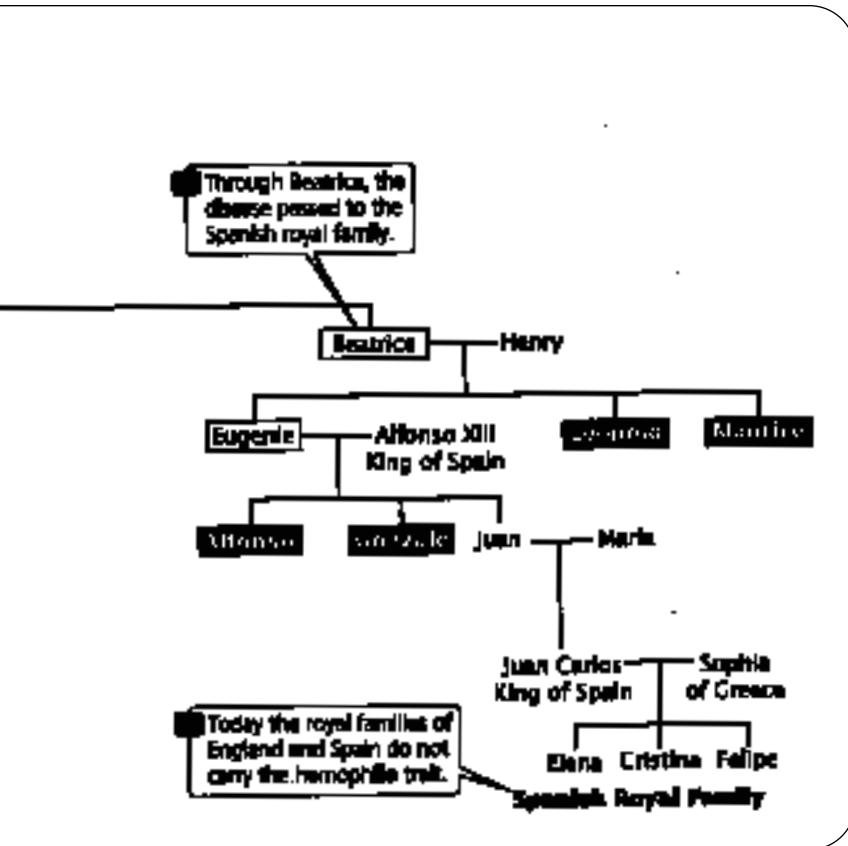
Alexis inherited the hemophilia gene from his mother, Alexandra, who was a carrier. The gene appears to have originated with Queen Victoria of England (1819–1901), (FIGURE 1.1). One of her sons, Leopold, had hemophilia and died at the age of 31 from brain hemorrhage following a minor fall. At least two of Victoria's daughters were carriers; through marriage, they spread the hemophilia gene to the royal families of Prussia, Spain, and Russia. In all, 10 of Queen Victoria's male descendants suffered from hemophilia. Six female descendants, including her granddaughter Alexandra (Alexis's mother), were carriers.

Nicholas and Alexandra constantly worried about Alexis's health. Although they prohibited his participation in sports and other physical activities, cuts and scrapes

were inevitable, and Alexis experienced a number of severe bleeding episodes. The royal physicians were helpless during these crises—they had no treatment that would stop the bleeding. Gregory Rasputin, a monk and self-proclaimed “miracle worker,” prayed over Alexis during one bleeding crisis, after which Alexis made a remarkable recovery. Rasputin then gained considerable influence over the royal family.

At this moment in history, the Russian Revolution broke out. Bolsheviks captured the tsar and his family and held them captive in the city of Ekaterinburg. On the night of July 16, 1918, a firing squad executed the royal family and their attendants, including Alexis and his four sisters. Eight days later, a protsarist army fought its way into Ekaterinburg. Although army investigators searched vigorously for the bodies of Nicholas and his family, they found only a few personal effects and a single finger. The Bolsheviks eventually won the revolution and instituted the world's first communist state.

Historians have debated the role that Alexis's illness may have played in the Russian Revolution. Some have argued that the revolution was successful because the tsar and Alexandra were distracted by their son's illness and under the influence of Rasputin. Others point out that many factors contributed to the overthrow of the tsar. It is probably naive to attribute the revolution entirely to one sick boy, but it is



clear that a genetic defect, passed down through the royal family, contributed to the success of the Russian Revolution.

More than 80 years after the tsar and his family were executed, an article in the *Moscow News* reported the discovery of their skeletons outside Ekaterinburg. The remains had first been located in 1979; however, because of secrecy surrounding the tsar's execution, the location of the graves was not made public until the breakup of the Soviet government in 1989. The skeletons were eventually recovered and examined by a team of forensic anthropologists, who concluded that they were indeed the remains of the tsar and his wife, three of their five children, and the family doctor, cook, maid, and footman. The bodies of Alexis and his sister Anastasia are still missing.

To prove that the skeletons were those of the royal family, mitochondrial DNA (which is inherited only from the mother) was extracted from the bones and amplified with a molecular technique called the polymerase chain reaction (PCR). DNA samples from the skeletons thought to belong to Alexandra and the children were compared with DNA taken from Prince Philip of England, also a direct descendant of Queen Victoria. Analysis showed that mitochondrial DNA from Prince Philip was identical with that from these four skeletons.

DNA from the skeleton presumed to be Tsar Nicholas was compared with that of two living descendants of the

Romanov line. The samples matched at all but one nucleotide position: the living relatives possessed a cytosine (C) residue at this position, whereas some of the skeletal DNA possessed a thymine (T) residue and some possessed a C. This difference could be due to normal variation in the DNA; so experts concluded that the skeleton was almost certainly that of Tsar Nicholas. The finding remained controversial, however, until July 1994, when the body of Nicholas's younger brother Georgij, who died in 1899, was exhumed. Mitochondrial DNA from Georgij also contained both C and T at the controversial position, proving that the skeleton was indeed that of Tsar Nicholas.

This chapter introduces you to genetics and reviews some concepts that you may have encountered briefly in a preceding biology course. We begin by considering the importance of genetics to each of us, to society at large, and to students of biology. We then turn to the history of genetics, how the field as a whole developed. The final part of the chapter reviews some fundamental terms and principles of genetics that are used throughout the book.

There has never been a more exciting time to undertake the study of genetics than now. Genetics is one of the frontiers of science. Pick up almost any major newspaper or news magazine and chances are that you will see something related to genetics: the discovery of cancer-causing genes; the use of gene therapy to treat diseases; or reports of possible hereditary influences on intelligence, personality, and sexual orientation. These findings often have significant economic and ethical implications, making the study of genetics relevant, timely, and interesting.

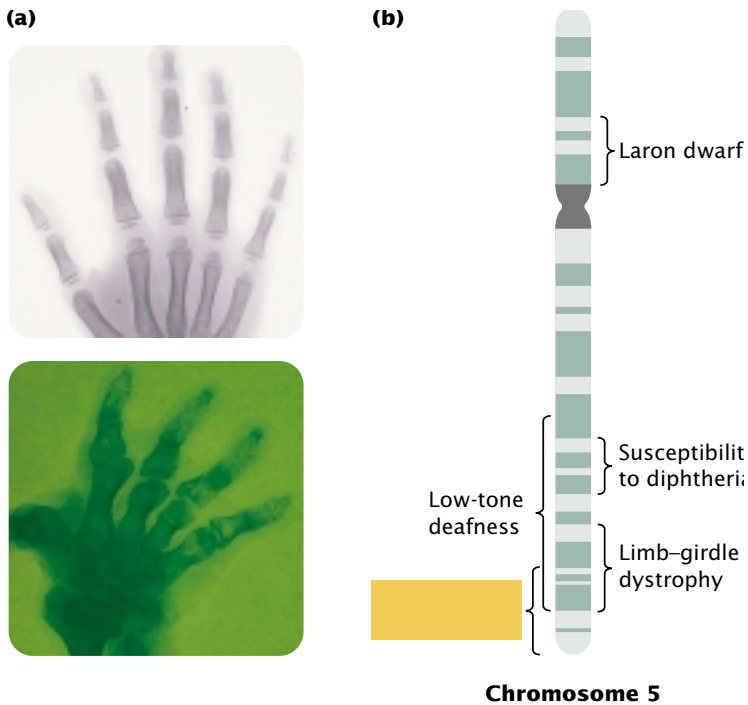
www.whfreeman.com/pierce

More information about the history of Nicholas II and other tsars of Russia and about hemophilia

The Importance of Genetics

Alexis's hemophilia illustrates the important role that genetics plays in the life of an individual. A difference in one gene, of the 35,000 or so genes that each human possesses, changed Alexis's life, affected his family, and perhaps even altered history. We all possess genes that influence our lives. They affect our height and weight, our hair color and skin pigmentation. They influence our susceptibility to many diseases and disorders (FIGURE 1.2) and even contribute to our intelligence and personality. Genes are fundamental to who and what we are.

Although the science of genetics is relatively new, people have understood the hereditary nature of traits and have "practiced" genetics for thousands of years. The rise of agriculture began when humans started to apply genetic principles to the domestication of plants and animals. Today, the major crops and animals used in agriculture have undergone extensive genetic alterations to greatly increase their yields and provide many desirable traits, such as disease and pest



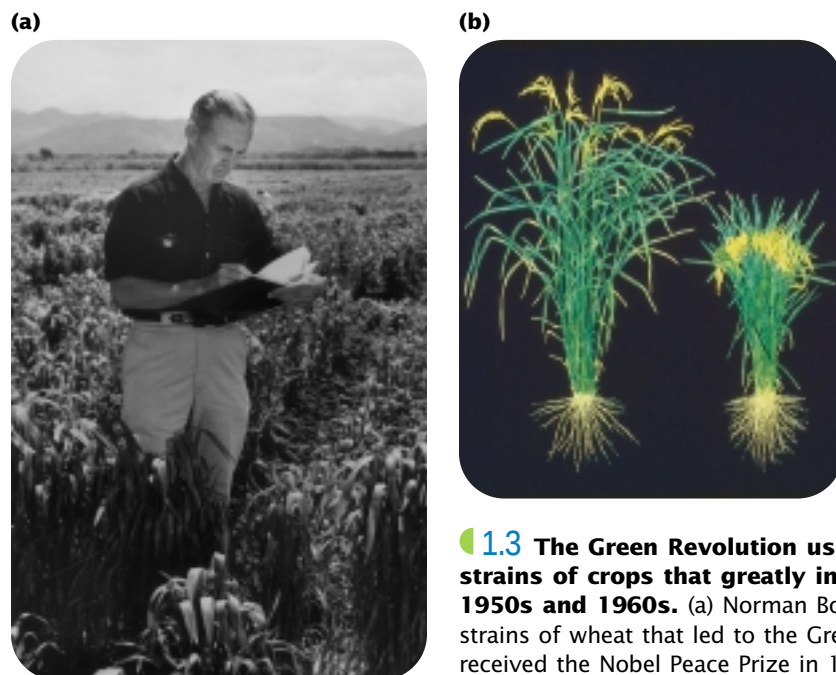
1.2 Genes influence susceptibility to many diseases and disorders. (a) X-ray of the hand of a person suffering from diastrophic dysplasia (bottom), a hereditary growth disorder that results in curved bones, short limbs, and hand deformities, compared with an X-ray of a normal hand (top). (b) This disorder is due to a defect in a gene on chromosome 5. Other genetic disorders encoded by genes on chromosome 5 also are indicated by braces. (Part a: top, Biophoto Associates/Science Source Photo Researchers; bottom, courtesy of Eric Lander, Whitehead Institute, MIT.)

resistance, special nutritional qualities, and characteristics that facilitate harvest. The Green Revolution, which expanded global food production in the 1950s and 1960s, relied heavily on the application of genetics (FIGURE 1.3). Today, genetically engineered corn, soybeans, and other crops constitute a significant proportion of all the food produced worldwide.

The pharmaceutical industry is another area where genetics plays an important role. Numerous drugs and food additives are synthesized by fungi and bacteria that have been genetically manipulated to make them efficient producers of these substances. The biotechnology industry employs molecular genetic techniques to develop and mass-produce substances of commercial value. Growth hormone, insulin, and clotting factor are now produced commercially by genetically engineered bacteria (FIGURE 1.4). Techniques of molecular genetics have also been used to produce bacteria that remove minerals from ore, break down toxic chemicals, and inhibit damaging frost formation on crop plants.

Genetics also plays a critical role in medicine. Physicians recognize that many diseases and disorders have a hereditary component, including well-known genetic disorders such as sickle-cell anemia and Huntington disease as well as many common diseases such as asthma, diabetes, and hypertension. Advances in molecular genetics have allowed important insights into the nature of cancer and permitted the development of many diagnostic tests. Gene therapy—the direct alteration of genes to treat human diseases—has become a reality.

www.whfreeman.com/pierce Information about biotechnology, including its history and applications



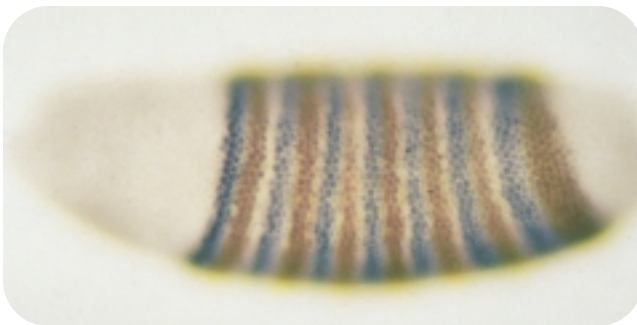
1.3 The Green Revolution used genetic techniques to develop new strains of crops that greatly increased world food production during the 1950s and 1960s. (a) Norman Borlaug, a leader in the development of new strains of wheat that led to the Green Revolution, and a family in Ghana. Borlaug received the Nobel Peace Prize in 1970. (b) Traditional rice plant (top) and modern, high-yielding rice plant (bottom). (Part a, UPI/Corbis-Bettman; part b, IRRI.)



1.4 The biotechnology industry uses molecular genetic methods to produce substances of economic value. In the apparatus shown, growth hormone is produced by genetically engineered bacteria. (James Holmes/Celltech Ltd./Science Photo Library/Photo Researchers.)

The Role of Genetics in Biology

Although an understanding of genetics is important to all people, it is critical to the student of biology. Genetics provides one of biology's unifying principles: all organisms use nucleic acids for their genetic material and all encode their genetic information in the same way. Genetics undergirds the study of many other biological disciplines. Evolution, for example, is genetic change taking place through time; so



1.5 The key to development lies in the regulation of gene expression. This early fruit-fly embryo illustrates the localized production of proteins from two genes, *ftz* (stained gray) and *eve* (stained brown), which determine the development of body segments in the adult fly. (Peter Lawrence, 1992. *The Making of a Fly*, Blackwell Scientific Publications.)

the study of evolution requires an understanding of basic genetics. Developmental biology relies heavily on genetics: tissues and organs form through the regulated expression of genes (FIGURE 1.5). Even such fields as taxonomy, ecology, and animal behavior are making increasing use of genetic methods. The study of almost any field of biology or medicine is incomplete without a thorough understanding of genes and genetic methods.

Genetic Variation Is the Foundation of Evolution

Life on Earth exists in a tremendous array of forms and features that occupy almost every conceivable environment. All life has a common origin (see Chapter 2); so this diversity has developed during Earth's 4-billion-year history. Life is also characterized by adaptation: many organisms are exquisitely suited to the environment in which they are found. The history of life is a chronicle of new forms of life emerging, old forms disappearing, and existing forms changing.

Life's diversity and adaptation are a product of evolution, which is simply genetic change through time. Evolution is a two-step process: first, genetic variants arise randomly and, then, the proportion of particular variants increases or decreases. Genetic variation is therefore the foundation of all evolutionary change and is ultimately the basis of all life as we know it. Genetics, the study of genetic variation, is critical to understanding the past, present, and future of life.

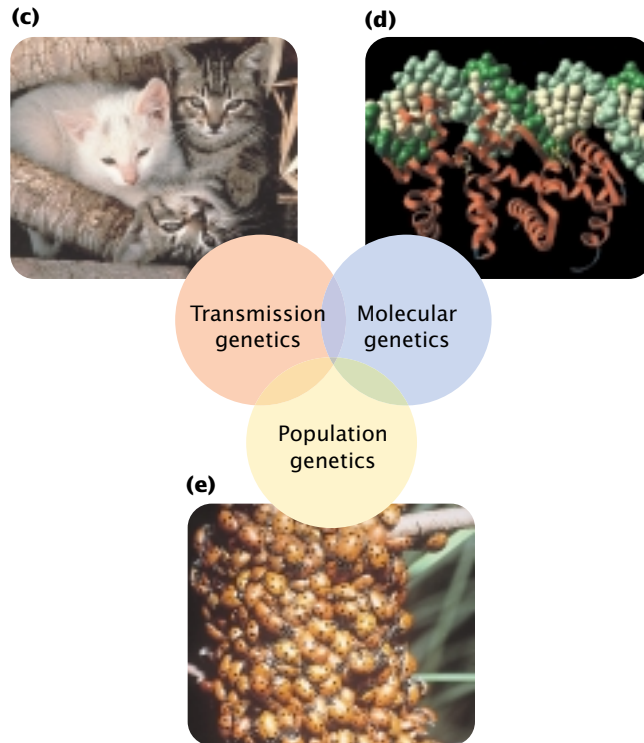
Concepts

Heredity affects many of our physical features as well as our susceptibility to many diseases and disorders. Genetics contributes to advances in agriculture, pharmaceuticals, and medicine and is fundamental to modern biology. Genetic variation is the foundation of the diversity of all life.

Divisions of Genetics

Traditionally, the study of genetics has been divided into three major subdisciplines: transmission genetics, molecular genetics, and population genetics (FIGURE 1.6). Also known as classical genetics, **transmission genetics** encompasses the basic principles of genetics and how traits are passed from one generation to the next. This area addresses the relation between chromosomes and heredity, the arrangement of genes on chromosomes, and gene mapping. Here the focus is on the individual organism—how an individual organism inherits its genetic makeup and how it passes its genes to the next generation.

Molecular genetics concerns the chemical nature of the gene itself: how genetic information is encoded, replicated, and expressed. It includes the cellular processes of replication, transcription, and translation—by which genetic information is transferred from one molecule to another—and gene



1.6 Genetics can be subdivided into three inter-related fields. (Top left, Alan Carey/Photo Researchers; top right, MONA file M0214602.tif; bottom, J. Alcock/Visuals Unlimited.)

regulation—the processes that control the expression of genetic information. The focus in molecular genetics is the gene—its structure, organization, and function.

Population genetics explores the genetic composition of groups of individual members of the same species (populations) and how that composition changes over time and space. Because evolution is genetic change, population genetics is fundamentally the study of evolution. The focus of population genetics is the group of genes found in a population.

It is convenient and traditional to divide the study of genetics into these three groups, but we should recognize that the fields overlap and that each major subdivision can be further divided into a number of more specialized fields, such as chromosomal genetics, biochemical genetics, quantitative genetics, and so forth. Genetics can alternatively be subdivided by organism (fruit fly, corn, or bacterial genetics), and each of these organisms can be studied at the level of transmission, molecular, and population genetics. Modern genetics is an extremely broad field, encompassing many interrelated subdisciplines and specializations.

Concepts

The three major divisions of genetics are transmission genetics, molecular genetics, and population genetics. Transmission genetics

examines the principles of heredity; molecular genetics deals with the gene and the cellular processes by which genetic information is transferred and expressed; population genetics concerns the genetic composition of groups of organisms and how that composition changes over time and space.

www.whfreeman.com/pierce

Information about careers in genetics

A Brief History of Genetics

Although the science of genetics is young—almost entirely a product of the past 100 years—people have been using genetic principles for thousands of years.

Prehistory

The first evidence that humans understood and applied the principles of heredity is found in the domestication of plants and animals, which began between approximately 10,000 and 12,000 years ago. Early nomadic people depended on hunting and gathering for subsistence but, as human populations grew, the availability of wild food resources declined. This decline created pressure to develop new sources of food; so people began to manipulate wild plants and animals, giving rise to early agriculture and the first fixed settlements.

Initially, people simply selected and cultivated wild plants and animals that had desirable traits. Archeological evidence of the speed and direction of the domestication process demonstrates that people quickly learned a simple but crucial rule of heredity: like breeds like. By selecting and breeding individual plants or animals with desirable traits, they could produce these same traits in future generations.

The world's first agriculture is thought to have developed in the Middle East, in what is now Turkey, Iraq, Iran, Syria, Jordan, and Israel, where domesticated plants and animals were major dietary components of many populations by 10,000 years ago. The first domesticated organisms included wheat, peas, lentils, barley, dogs, goats, and sheep. Selective breeding produced woollier and more manageable goats and sheep and seeds of cereal plants that were larger and easier to harvest. By 4000 years ago, sophisticated genetic techniques were already in use in the Middle East. Assyrians and Babylonians developed several hundred varieties of date palms that differed in fruit size, color, taste, and time of ripening. An Assyrian bas-relief from 2880 years ago depicts the use of artificial fertilization to control crosses between date palms (**FIGURE 1.7**). Other crops and domesticated animals were developed by cultures in Asia, Africa, and the Americas in the same period.



1.7 Ancient peoples practiced genetic techniques in agriculture. (Top) Comparison of ancient (left) and modern (right) wheat. (Bottom) Assyrian bas-relief sculpture showing artificial pollination of date palms at the time of King Assurnasirpalli II, who reigned from 883–859 B.C. (Top left and right, IRRI; bottom, Metropolitan Museum of Art, gift of John D. Rockefeller Jr., 1932.

should avoid potential spouses having undesirable traits that might be passed on to one's children. The Talmud, the Jewish book of religious laws based on oral traditions dating back thousands of years, presents an uncannily accurate understanding of the inheritance of hemophilia. It directs that, if a woman bears two sons who die of bleeding after circumcision, any additional sons that she bears should not be circumcised; nor should the sons of her sisters be circumcised, although the sons of her brothers should. This advice accurately depicts the X-linked pattern of inheritance of hemophilia (discussed further in Chapter 6).

The ancient Greeks gave careful consideration to human reproduction and heredity. The Greek physician Alcmaeon (circa 520 B.C.) conducted dissections of animals and proposed that the brain was not only the principle site of perception, but also the origin of semen. This proposal sparked a long philosophical debate about where semen was produced and its role in heredity. The debate culminated in the concept of **pangenesis**, which proposed that specific particles, later called gemmules, carry information from various parts of the body to the reproductive organs, from where they are passed to the embryo at the moment of conception (**FIGURE 1.8a**). Although incorrect, the concept of pangenesis was highly influential and persisted until the late 1800s.

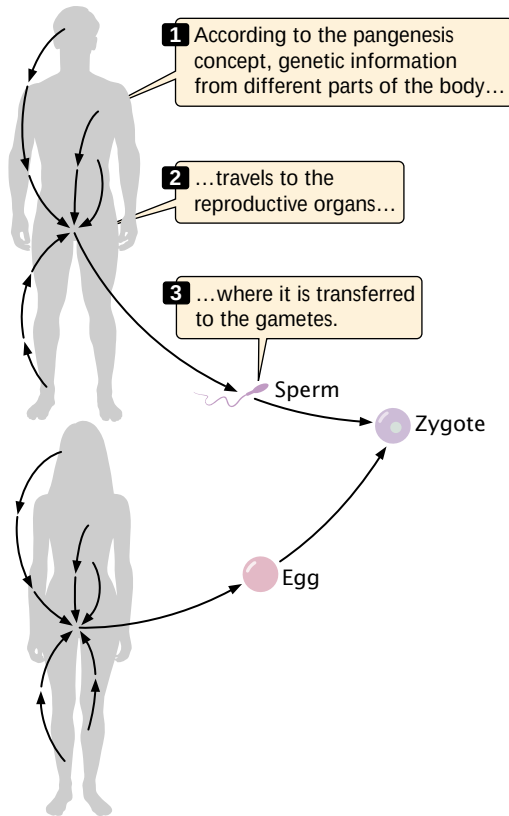
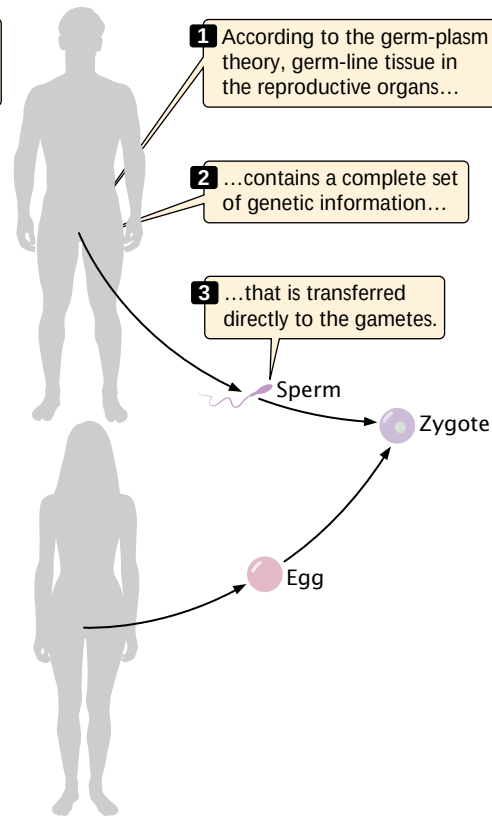
Pangenesis led the ancient Greeks to propose the notion of the **inheritance of acquired characteristics**, in which traits acquired during one's lifetime become incorporated into one's hereditary information and are passed on to

Concepts

Humans first applied genetics to the domestication of plants and animals between approximately 10,000 and 12,000 years ago. This domestication led to the development of agriculture and fixed human settlements.

Early Written Records

Ancient writings demonstrate that early humans were aware of their own heredity. Hindu sacred writings dating to 2000 years ago attribute many traits to the father and suggest that differences between siblings can be accounted for by effects from the mother. These same writings advise that one

(a) Pangenesis concept**(b) Germ-plasm theory**

1.8 Pangenesis, an early concept of inheritance, compared with the modern germ-plasm theory.

offspring; for example, people who developed musical ability through diligent study would produce children who are innately endowed with musical ability. The notion of the inheritance of acquired characteristics also is no longer accepted, but it remained popular through the twentieth century.

The Greek philosopher Aristotle (384–322 B.C.) was keenly interested in heredity. He rejected the concepts of both pangenesis and the inheritance of acquired characteristics, pointing out that people sometimes resemble past ancestors more than their parents and that acquired characteristics such as mutilated body parts are not passed on. Aristotle believed that both males and females made contributions to the offspring and that there was a struggle of sorts between male and female contributions.

Although the ancient Romans contributed little to the understanding of human heredity, they successfully developed a number of techniques for animal and plant breeding; the techniques were based on trial and error rather than any general concept of heredity. Little new was added to the understanding of genetics in the next 1000 years. The ancient ideas of pangenesis and the inheritance of acquired characteristics, along with techniques of plant and

animal breeding, persisted until the rise of modern science in the seventeenth and eighteenth centuries.

The Rise of Modern Genetics

Dutch spectacle makers began to put together simple microscopes in the late 1500s, enabling Robert Hooke (1653–1703) to discover cells in 1665. Microscopes provided naturalists with new and exciting vistas on life, and perhaps it was excessive enthusiasm for this new world of the very small that gave rise to the idea of **preformationism**. According to preformationism, inside the egg or sperm existed a tiny miniature adult, a *homunculus*, which simply enlarged during development. Ovists argued that the homunculus resided in the egg, whereas spermists insisted that it was in the sperm (FIGURE 1.9). Preformationism meant that all traits would be inherited from only one parent—from the father if the homunculus was in the sperm or from the mother if it was in the egg. Although many observations suggested that offspring possess a mixture of traits from both parents, preformationism remained a popular concept throughout much of the seventeenth and eighteenth centuries.

Another early notion of heredity was **blending inheritance**, which proposed that offspring are a blend, or mixture,

Mapping the Human Genome— Where does it lead, and what does it mean?

 by Arthur L. Caplan and Kelly
 A. Carroll

In June 2000, scientists from the Human Genome Project and Celera Genomics stood at a podium with former President Bill Clinton to announce a stunning achievement—they had successfully constructed a sequence of the entire human genome. Soon this process of identifying and sequencing each and every human gene became characterized as "mapping the human genome". As with maps of the physical world, the map of the human genome provides a picture of locations, terrains, and structures. But, like explorers, scientists must continue to decipher what each location on the map can tell us about diseases, human health, and biology. The map accelerates this process, as it allows researchers to identify key structural dimensions of the gene they are exploring, and reminds them where they have been and where they have yet to explore.

What does the map of the human genome depict? When researchers discuss the sequencing of the genome, they are describing the identification of the patterns and order of the 3 billion human DNA base pairs. While this provides valuable information about overall structure and the evolution of humans in relation to other organisms, researchers really wanted the key information encoded in just 2% of this enormous map—the information that makes most of the proteins that compose you and me. Comprised of DNA, genes are the basic units of heredity; they hold all of the information required to make the proteins that regulate most life functions, from digesting food to battling diseases. Proteins stand as the link between genes and pharmaceutical drug development, they show which genes are being expressed at any given moment, and provide information about gene function.

Knowing our genes will lead to greater understanding and radically

improved treatment of many diseases. However, sequencing the entire human genome, in conjunction with sequencing of various nonhuman genomes under the same project, has raised fundamental questions about what it means to be human. After all, fruit flies possess about one-third the number of genes as humans, and an ear of corn has approximately the same number of genes as a human! In addition, the overall DNA sequence of a chimpanzee is about 99% the same as the human genome sequence. As the genomes of other species become available, the similarities to the human genome in both structure and sequence pattern will continue to be identified. At a basic level, the discovery of so many commonalities and links and ancestral trees with other species adds credence to principles of evolution and Darwinism.

Some of the most anticipated developments and potential benefits of the Human Genome Project directly affect human health; researchers, practicing physicians, and the general public eagerly await the development of targeted pharmaceutical agents and more specific diagnostic tests. Pharmacogenomics is at the intersection of genetics and pharmacology; it is the study of how one's genetic makeup will affect his or her response to various drugs. In the future, medicine will potentially be safer, cheaper, and more disease specific, all while causing fewer side effects and acting more effectively, the first time around.

There are however some hard ethical questions that follow in the wake of new genetic knowledge. Patients will have to undergo genetic testing in order to match drugs to their genetic makeup. Who will have access to these results—just the health care practitioner, or the patient's insurance company, employer/school, and/or family members? While the tests were administered for one case,

will the information derived from them be used for other purposes, such as for identification of other conditions/future diseases, or even in research studies?

How should researchers conduct studies in pharmacogenomics? Often they need to group study subjects by some kind of identifiable traits that they believe will assist in separating groups of drugs, and in turn they separate people into populations. The order of almost all of the DNA base pairs (99.9%) is exactly the same in all humans. So, this leaves a small window of difference. There is potential for stigmatization of individuals and groups, of people based on race and ethnicity inherent in genomic research and analysis. As scientists continue drug development, they must be careful to not further such ideas, especially as studies of nuclear DNA indicate that there is often more genetic variation within "races" or cultures, than between "races" or cultures. Stigmatization or discrimination can occur through genetic testing and human subjects research on populations.

These are just a few of the ethical issues arising out of one development of the Human Genome Project. The potential applications of genome research are staggering, and the mapping is just the beginning. Realizing this was simply a starting point, the draft sequences of the human genome released in February 2001 by the publicly funded Human Genome Project and the private company, Celera Genomics, are freely available on the Internet. A long road lies ahead, where scientists will be charged with exploring and understanding the functions of and relationships between genes and proteins. With such exploration comes a responsibility to acknowledge and address the ethical, legal, and social implications of this exciting research.



1.9 Preformationism was a popular idea of inheritance in the seventeenth and eighteenth centuries. Shown here is a drawing of a homunculus inside a sperm. (Science VU/Visuals Unlimited.)

of parental traits. This idea suggested that the genetic material itself blends, much as blue and yellow pigments blend to make green paint. Once blended, genetic differences could not be separated out in future generations, just as green paint cannot be separated out into blue and yellow pigments. Some traits do *appear* to exhibit blending inheritance; however, we realize today that individual genes do not blend.

Nehemiah Grew (1641–1712) reported that plants reproduce sexually by using pollen from the male sex cells. With this information, a number of botanists began to experiment with crossing plants and creating hybrids. Foremost among these early plant breeders was Joseph Gottlieb Kölreuter (1733–1806), who carried out numerous crosses and studied pollen under the microscope. He observed that many hybrids were intermediate between the parental varieties. Because he crossed plants that differed in many traits, Kölreuter was unable to discern any general pattern of inheritance. In spite of this limitation, Kölreuter's

work set the foundation for the modern study of genetics. Subsequent to his work, a number of other botanists began to experiment with hybridization, including Gregor Mendel (1822–1884) (**FIGURE 1.10**), who went on to discover the basic principles of heredity. Mendel's conclusions, which were unappreciated for 45 years, laid the foundation for our modern understanding of heredity, and he is generally recognized today as the father of genetics.

Developments in cytology (the study of cells) in the 1800s had a strong influence on genetics. Robert Brown (1773–1858) described the cell nucleus in 1833. Building on the work of others, Matthias Jacob Schleiden (1804–1881) and Theodor Schwann (1810–1882) proposed the concept of the **cell theory** in 1839. According to this theory, all life is composed of cells, cells arise only from preexisting cells, and the cell is the fundamental unit of structure and function in living organisms. Biologists began to examine cells to see how traits were transmitted in the course of cell division.

Charles Darwin (1809–1882), one of the most influential biologists of the nineteenth century, put forth the theory of evolution through natural selection and published his ideas in *On the Origin of Species* in 1856. Darwin recognized that heredity was fundamental to evolution, and he



1.10 Gregor Mendel was the founder of modern genetics. Mendel first discovered the principles of heredity by crossing different varieties of pea plants and analyzing the pattern of transmission of traits in subsequent generations. (Hulton/Archive by Getty Images.)

conducted extensive genetic crosses with pigeons and other organisms. However, he never understood the nature of inheritance, and this lack of understanding was a major omission in his theory of evolution.

In the last half of the nineteenth century, the invention of the microtome (for cutting thin sections of tissue for microscopic examination) and the development of improved histological stains stimulated a flurry of cytological research. Several cytologists demonstrated that the nucleus had a role in fertilization. Walter Flemming (1843–1905) observed the division of chromosomes in 1879 and published a superb description of mitosis. By 1885, it was generally recognized that the nucleus contained the hereditary information.

Near the close of the nineteenth century, August Weismann (1834–1914) finally laid to rest the notion of the inheritance of acquired characteristics. He cut off the tails of mice for 22 consecutive generations and showed that the tail length in descendants remained stubbornly long. Weismann proposed the **germ-plasm theory**, which holds that the cells in the reproductive organs carry a complete set of genetic information that is passed to the gametes (see Figure 1.8b).

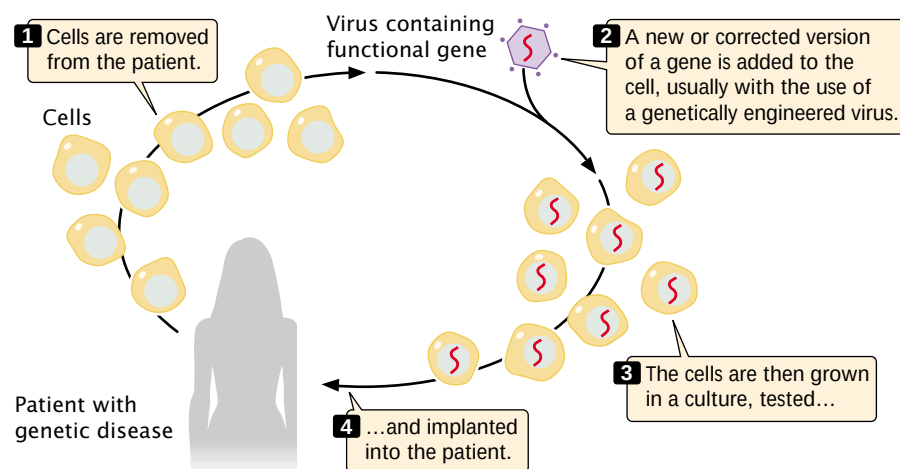
Twentieth-Century Genetics

The year 1900 was a watershed in the history of genetics. Gregor Mendel's pivotal 1866 publication on experiments with pea plants, which revealed the principles of heredity, was “rediscovered,” as discussed in more detail in Chapter 3. The significance of his conclusions was recognized, and other biologists immediately began to conduct similar genetic studies on mice, chickens, and other organisms. The results of these investigations showed that many traits indeed follow Mendel's rules.

Walter Sutton (1877–1916) proposed in 1902 that genes are located on chromosomes. Thomas Hunt Morgan (1866–1945) discovered the first genetic mutant of fruit flies in 1910 and used fruit flies to unravel many details of transmission genetics. Ronald A. Fisher (1890–1962), John B. S. Haldane (1892–1964), and Sewall Wright (1889–1988) laid the foundation for population genetics in the 1930s.

Geneticists began to use bacteria and viruses in the 1940s; the rapid reproduction and simple genetic systems of these organisms allowed detailed study of the organization and structure of genes. At about this same time, evidence accumulated that DNA was the repository of genetic information. James Watson (b. 1928) and Francis Crick (b. 1916) described the three-dimensional structure of DNA in 1953, ushering in the era of molecular genetics.

By 1966, the chemical structure of DNA and the system by which it determines the amino acid sequence of proteins had been worked out. Advances in molecular genetics led to the first recombinant DNA experiments in 1973, which touched off another revolution in genetic research. Walter Gilbert (b. 1932) and Frederick Sanger (b. 1918) developed methods for sequencing DNA in 1977. The polymerase chain reaction, a technique for quickly amplifying tiny amounts of DNA, was developed by Kary Mullis (b. 1944) and others in 1986. In 1990, gene therapy was used for the first time to treat human genetic disease in the United States (**FIGURE 1.11**), and the Human Genome Project was launched. By 1995, the first complete DNA sequence of a free-living organism—the bacterium *Haemophilus influenzae*—was determined, and the first complete sequence of a eukaryotic organism (yeast) was reported a year later. At the beginning of the twenty-first century, the human genome sequence was determined, ushering in a new era in genetics.



1.11 Gene therapy applies genetic engineering to the treatment of human diseases. (J. Coate, MBD/Science VU/Visuals Unlimited.)

The Future of Genetics

The information content of genetics now doubles every few years. The genome sequences of many organisms are added to DNA databases every year, and new details about gene structure and function are continually expanding our knowledge of heredity. All of this information provides us with a better understanding of numerous biological processes and evolutionary relationships. The flood of new genetic information requires the continuous development of sophisticated computer programs to store, retrieve, compare, and analyze genetic data and has given rise to the field of bioinformatics, a merging of molecular biology and computer science.

In the future, the focus of DNA-sequencing efforts will shift from the genomes of different species to individual differences within species. It is reasonable to assume that each person may some day possess a copy of his or her entire genome sequence. New genetic microchips that simultaneously analyze thousands of RNA molecules will provide information about the activity of thousands of genes in a given cell, allowing a detailed picture of how cells respond to external signals, environmental stresses, and disease states. The use of genetics in the agricultural, chemical, and health-care fields will continue to expand; some predict that biotechnology will be to the twenty-first century what the electronics industry was to the twentieth century. This ever-widening scope of genetics will raise significant ethical, social, and economic issues.

This brief overview of the history of genetics is not intended to be comprehensive; rather it is designed to provide a sense of the accelerating pace of advances in genetics. In the chapters to come, we will learn more about the experiments and the scientists who helped shape the discipline of genetics.

www.whfreeman.com/pierce More information about the history of genetics

Concepts

Developments in plant hybridization and cytology in the eighteenth and nineteenth centuries laid the foundation for the field of genetics today. After Mendel's work was rediscovered in 1900, the science of genetics developed rapidly and today is one of the most active areas of science.



Basic Concepts in Genetics

Undoubtedly, you learned some genetic principles in other biology classes. Let's take a few moments to review some of these fundamental genetic concepts.

Cells are of two basic types: eukaryotic and prokaryotic- Structurally, cells consist of two basic types, although, evolutionarily, the story is more complex (see Chapter 2). Prokaryotic cells lack a nuclear membrane and possess no membrane-bounded cell organelles, whereas eukaryotic cells are more complex, possessing a nucleus and membrane-bounded organelles such as chloroplasts and mitochondria.

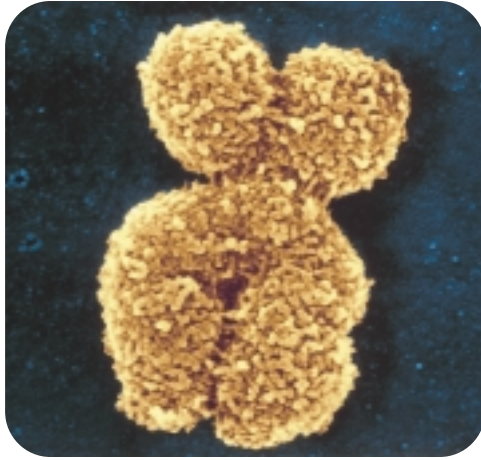
A gene is the fundamental unit of heredity- The precise way in which a gene is defined often varies. At the simplest level, we can think of a gene as a unit of information that encodes a genetic characteristic. We will enlarge this definition as we learn more about what genes are and how they function.

Genes come in multiple forms called alleles- A gene that specifies a characteristic may exist in several forms, called alleles. For example, a gene for coat color in cats may exist in alleles that encode either black or orange fur.

Genes encode phenotypes- One of the most important concepts in genetics is the distinction between traits and genes. Traits are not inherited directly. Rather, genes are inherited and, along with environmental factors, determine the expression of traits. The genetic information that an individual organism possesses is its genotype; the trait is its phenotype. For example, the A blood type is a phenotype; the genetic information that encodes the blood type A antigen is the genotype.

Genetic information is carried in DNA and RNA- Genetic information is encoded in the molecular structure of nucleic acids, which come in two types: deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). Nucleic acids are polymers consisting of repeating units called nucleotides; each nucleotide consists of a sugar, a phosphate, and a nitrogenous base. The nitrogenous bases in DNA are of four types (abbreviated A, C, G, and T), and the sequence of these bases encodes genetic information. Most organisms carry their genetic information in DNA, but a few viruses carry it in RNA. The four nitrogenous bases of RNA are abbreviated A, C, G, and U.

Genes are located on chromosomes- The vehicles of genetic information within the cell are chromosomes (**FIGURE 1.12**), which consist of DNA and associated proteins. The cells of each species have a characteristic number of chromosomes; for example, bacterial cells normally possess a single chromosome; human cells possess 46; pigeon cells possess 80. Each chromosome carries a large number of genes.



1.12 Genes are carried on chromosomes.

(Biophoto Associates/Science Source/Photo Researchers.)

Chromosomes separate through the processes of mitosis and meiosis-

The processes of mitosis and meiosis ensure that each daughter cell receives a complete set of an organism's chromosomes. Mitosis is the separation of replicated chromosomes during the division of somatic (nonsex) cells. Meiosis is the pairing and separation of replicated chromosomes during the division of sex cells to produce gametes (reproductive cells).

Genetic information is transferred from DNA to RNA to protein-

Many genes encode traits by specifying the structure of proteins. Genetic information is first transcribed from DNA into RNA, and then RNA is translated into the amino acid sequence of a protein.

Mutations are permanent, heritable changes in genetic information-

Gene mutations affect only the genetic information of a single gene; chromosome mutations alter the number or the structure of chromosomes and therefore usually affect many genes.

Some traits are affected by multiple factors-

Some traits are influenced by multiple genes that interact in complex ways with environmental factors. Human height, for example, is affected by hundreds of genes as well as environmental factors such as nutrition.

Evolution is genetic change-

Evolution can be viewed as a two-step process: first, genetic variation arises and, second, some genetic variants increase in frequency, whereas other variants decrease in frequency.

www.whfreeman.com/pierce

A glossary of genetics terms

Connecting Concepts Across Chapters



This chapter introduces the study of genetics, outlining its history, relevance, and some fundamental concepts. One of the themes that emerges from our review of the history of genetics is that humans have been interested in, and using, genetics for thousands of years, yet our understanding of the mechanisms of inheritance are relatively new. A number of ideas about how inheritance works have been proposed throughout history, but many of them have turned out to be incorrect. This is to be expected, because science progresses by constantly evaluating and challenging explanations. Genetics, like all science, is a self-correcting process, and thus many ideas that are proposed will be discarded or modified through time.

CONCEPTS SUMMARY

- Genetics is central to the life of every individual: it influences our physical features, susceptibility to numerous diseases, personality, and intelligence.
- Genetics plays important roles in agriculture, the pharmaceutical industry, and medicine. It is central to the study of biology.
- Genetic variation is the foundation of evolution and is critical to understanding all life.
- The study of genetics can be divided into transmission genetics, molecular genetics, and population genetics.
- The use of genetics by humans began with the domestication of plants and animals.
- The ancient Greeks developed the concept of pangenesis and the concept of the inheritance of acquired characteristics.
- Ancient Romans developed practical measures for the breeding of plants and animals.
- In the seventeenth century, biologists proposed the idea of preformationism, which suggested that a miniature adult is present inside the egg or the sperm and that a person inherits all of his or her traits from one parent.
- Another early idea, blending inheritance, proposed that genetic information blends during reproduction and offspring are a mixture of the parental traits.
- By studying the offspring of crosses between varieties of peas, Gregor Mendel discovered the principles of heredity.
- Darwin developed the concept of evolution by natural selection in the 1800s, but he was unaware of Mendel's work and was not able to incorporate genetics into his theory.

- Developments in cytology in the nineteenth century led to the understanding that the cell nucleus is the site of heredity.
- In 1900, Mendel's principles of heredity were rediscovered. Population genetics was established in the early 1930s, followed closely by biochemical genetics and bacterial and viral genetics. Watson and Crick discovered the structure of DNA in 1953, which stimulated the rise of molecular genetics.
- Advances in molecular genetics have led to gene therapy and the Human Genome Project.
- Cells come in two basic types: prokaryotic and eukaryotic.
- Genetics is the study of genes, which are the fundamental units of heredity.
- The genes that determine a trait are termed the genotype; the trait that they produce is the phenotype.
- Genes are located on chromosomes, which are made up of nucleic acids and proteins and are partitioned into daughter cells through the process of mitosis or meiosis.
- Genetic information is expressed through the transfer of information from DNA to RNA to proteins.
- Evolution requires genetic change in populations.

IMPORTANT TERMS

transmission genetics (p. 5)
molecular genetics (p. 5)
population genetics (p. 6)

pangenesis (p. 7)
inheritance of acquired
characteristics (p. 7)

preformationism (p. 8)
blending inheritance (p. 8)
cell theory (p. 10)

germ-plasm theory (p. 11)

COMPREHENSION QUESTIONS

Answers to questions and problems preceded by an asterisk will be found at the end of the book.

1. Outline some of the ways in which genetics is important to each of us.
- * 2. Give at least three examples of the role of genetics in society today.
3. Briefly explain why genetics is crucial to modern biology.
- * 4. List the three traditional subdisciplines of genetics and summarize what each covers.
5. When and where did agriculture first arise? What role did genetics play in the development of the first domesticated plants and animals?
- * 6. Outline the notion of pangenesis and explain how it differs from the germ-plasm theory.
- * 7. What does the concept of the inheritance of acquired characteristics propose and how is it related to the notion of pangenesis?
- * 8. What is preformationism? What did it have to say about how traits are inherited?
9. Define blending inheritance and contrast it with preformationism.
10. How did developments in botany in the seventeenth and eighteenth centuries contribute to the rise of modern genetics?
11. How did developments in cytology in the nineteenth century contribute to the rise of modern genetics?
- * 12. Who first discovered the basic principles that laid the foundation for our modern understanding of heredity?
13. List some advances in genetics that have occurred in the twentieth century.
- * 14. Briefly define the following terms: **(a)** gene; **(b)** allele; **(c)** chromosome; **(d)** DNA; **(e)** RNA; **(f)** genetics; **(g)** genotype; **(h)** phenotype; **(i)** mutation; **(j)** evolution.
15. What are the two basic cell types (from a structural perspective) and how do they differ?
16. Outline the relations between genes, DNA, and chromosomes.

APPLICATION QUESTIONS AND PROBLEMS

- * 17. Genetics is said to be both a very old science and a very young science. Explain what is meant by this statement.
18. Find at least one newspaper article that covers some aspect of genetics. Briefly summarize the article. Does this article focus on transmission, molecular, or population genetics?
19. The following concepts were widely believed at one time but are no longer accepted as valid genetic theories. What experimental evidence suggests that these concepts are incorrect and what theories have taken their place?
(a) pangenesis; **(b)** the inheritance of acquired characteristics; **(c)** preformationism; **(d)** blending inheritance.

CHALLENGE QUESTIONS

20. Describe some of the ways in which your own genetic makeup affects you as a person. Be as specific as you can.
21. Pick one of the following ethical or social issues and give your opinion on this issue. For background information, you might read one of the articles on ethics listed and marked with an asterisk in Suggested Readings at the end of this chapter.
- (a) Should a person's genetic makeup be used in determining his or her eligibility for life insurance?
- (b) Should biotechnology companies be able to patent newly sequenced genes?
- (c) Should gene therapy be used on people?
- (d) Should genetic testing be made available for inherited conditions for which there is no treatment or cure?
- (e) Should governments outlaw the cloning of people?

SUGGESTED READINGS

Articles on ethical issues in genetics are preceded by an asterisk.

- *American Society of Human Genetics Board of Directors and the American College of Medical Genetics Board of Directors. 1995. Points to consider: ethical, legal, psychosocial implications of genetic testing in children. *American Journal of Human Genetics* 57:1233–1241.
- An official statement on some of the ethical, legal, and psychological considerations in conducting genetic tests on children by two groups of professional geneticists.
- Dunn, L. C. 1965. *A Short History of Genetics*. New York: McGraw-Hill.
- An excellent history of major developments in the field of genetics.
- *Friedmann, T. 2000. Principles for human gene therapy studies. *Science* 287:2163–2165.
- An editorial that outlines principles that serve as the foundation for clinical gene therapy.
- Kottak, C. P. 1994. *Anthropology: The Exploration of Human Diversity*, 6th ed. New York: McGraw-Hill.
- Contains a summary of the rise of agriculture and initial domestication of plants and animals.
- Lander, E. S., and R. A. Weinberg. 2000. Genomics: journey to the center of biology. *Science* 287:1777–1782.
- A succinct history of genetics and, more specifically, genomics written by two of the leaders of modern genetics.
- McKusick, V. A. 1965. The royal hemophilia. *Scientific American* 213(2):88–95.
- Contains a history of hemophilia in Queen Victoria's descendants.
- Massie, R. K. 1967. *Nicholas and Alexandra*. New York: Atheneum.
- One of the classic histories of Tsar Nicholas and his family.
- Massie, R. K. 1995. *The Romanovs: The Final Chapter*. New York: Random House.
- Contains information about the finding of the Romanov remains and the DNA testing that verified the identity of the skeletons.
- *Rosenberg, K., B. Fuller, M. Rothstein, T. Duster, et al. 1997. Genetic information and workplace: legislative approaches and policy challenges. *Science* 275:1755–1757.
- Deals with the use of genetic information in employment.
- *Shapiro, H. T. 1997. Ethical and policy issues of human cloning. *Science* 277:195–196.
- Discussion of the ethics of human cloning.
- Stubbe, H. 1972. *History of Genetics: From Prehistoric Times to the Rediscovery of Mendel's Laws*. Translated by T. R. W. Waters. Cambridge, MA: MIT Press.
- A good history of genetics, especially for pre-Mendelian genetics.
- Sturtevant, A. H. 1965. *A History of Genetics*. New York: Harper and Row.
- An excellent history of genetics.
- *Verma, I. M., and N. Somia. 1997. Gene therapy: promises, problems, and prospects. *Nature* 389:239–242.
- An update on the status of gene therapy.

2

Chromosomes and Cellular Reproduction



This is Chapter 2 Opener photo legend. (Art Wolfe/Photo Researchers.)

- The Diversity of Life
- Basic Cell Types: Structures and Evolutionary Relationships
- Cell Reproduction
 - Prokaryotic Cell Reproduction
 - Eukaryotic Cell Reproduction
 - The Cell Cycle and Mitosis
- Sexual Reproduction and Genetic Variation
 - Meiosis
 - Consequences of meiosis
 - Meiosis in the Life Cycle of Plants and Animals

The Diversity of Life

More than by any other feature, life is characterized by diversity: 1.4 million species of plants, animals, and microorganisms have already been described, but this number vastly underestimates the total number of species on Earth. Consider the arthropods—insects, spiders, crustaceans, and related animals with hard exoskeletons. About 875,000 arthropods have been described by scientists worldwide. The results of recent studies, however, suggest that as many as 5 million to 30 million species of arthropods may be living in tropical rain forests *alone*. Furthermore, many species contain numerous genetically distinct populations, and each population contains genetically unique individuals.

Despite their tremendous diversity, living organisms have an important feature in common: all use the same genetic system. A complete set of genetic instructions for any organism is its **genome**, and all genomes are encoded in nucleic acids, either DNA or RNA. The coding system for genomic information also is common to all life—genetic instructions are in the same format and, with rare exceptions, the code words are identical. Likewise, the process

by which genetic information is copied and decoded is remarkably similar for all forms of life. This universal genetic system is a consequence of the common origin of living organisms; all life on Earth evolved from the same primordial ancestor that arose between 3.5 billion and 4 billion years ago. Biologist Richard Dawkins describes life as a river of DNA that runs through time, connecting all organisms past and present.

That all organisms have a common genetic system means that the study of one organism's genes reveals principles that apply to other organisms. Investigations of how bacterial DNA is copied (replicated), for example, provides information that applies to the replication of human DNA. It also means that genes will function in foreign cells, which makes genetic engineering possible. Unfortunately, this common genetic system is also the basis for diseases such as AIDS (acquired immune deficiency syndrome), in which viral genes are able to function—sometimes with alarming efficiency—in human cells.

This chapter explores cell reproduction and how genetic information is transmitted to new cells. In prokaryotic cells, cell division is relatively simple because a prokaryotic

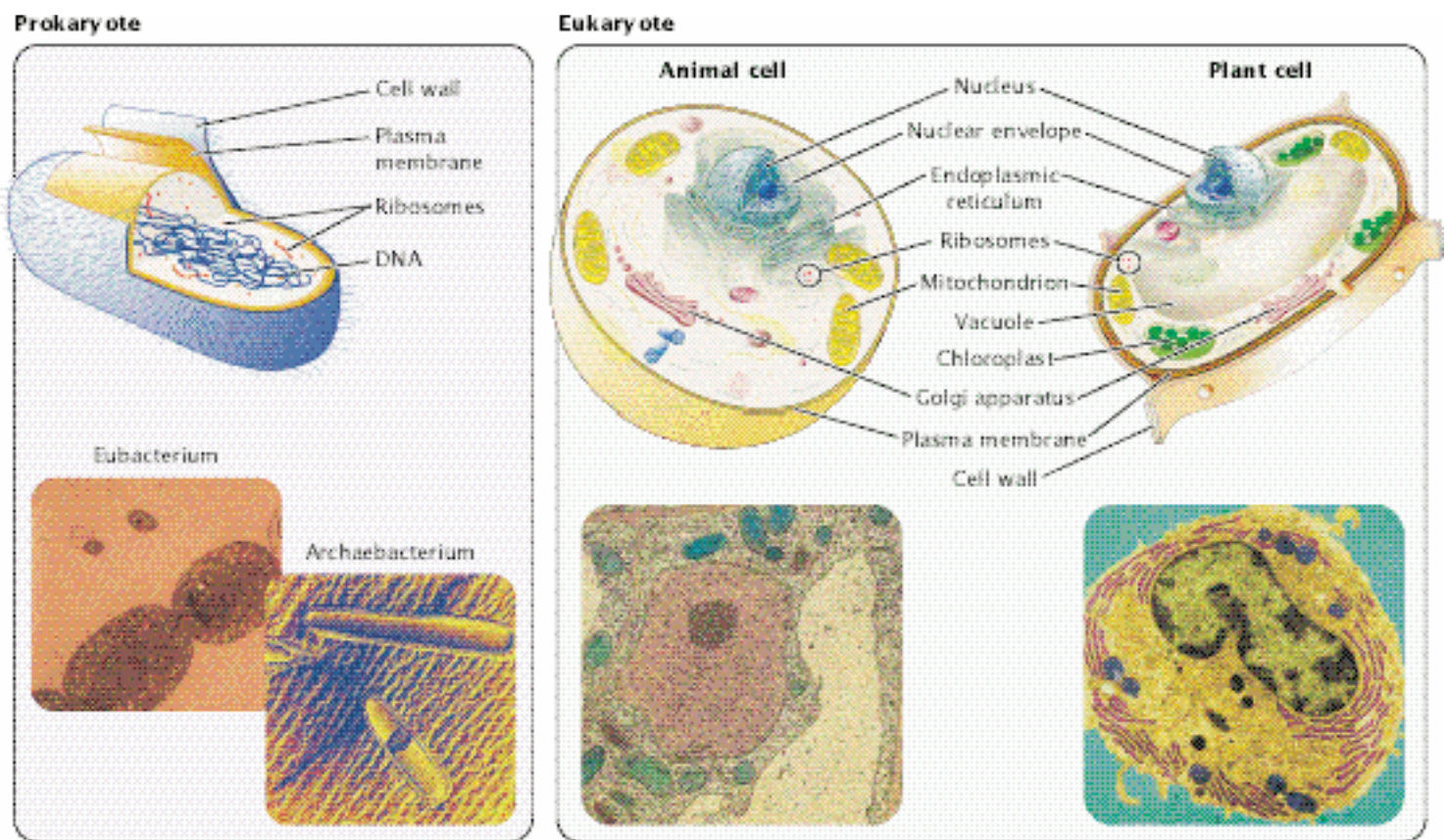
cell usually possesses only a single chromosome. In eukaryotic cells, multiple chromosomes must be copied and distributed to each of the new cells. Cell division in eukaryotes takes place through mitosis and meiosis, processes that serve as the foundation for much of genetics; so it is essential to understand them well.

Grasping mitosis and meiosis requires more than simply memorizing the sequences of events that take place in each stage, although these events are important. The key is to understand how genetic information is apportioned during cell reproduction through a dynamic interplay of DNA synthesis, chromosome movement, and cell division. These processes bring about the transmission of genetic informa-

tion and are the bases of similarities and differences between parents and progeny.

Basic Cell Types: Structure and Evolutionary Relationships

Biologists traditionally classify all living organisms into two major groups, the *prokaryotes* and the *eukaryotes*. A **prokaryote** is a unicellular organism with a relatively simple cell structure (FIGURE 2.1). A **eukaryote** has a compartmentalized cell structure divided by intracellular membranes; eukaryotes may be unicellular or multicellular.



Characteristic	Prokaryotic cells	Eukaryotic cells
Nucleus:	Absent	Present
Cell diameter:	Relatively small, from 1 to 10 μm	Relatively large, from 10 to 100 μm
Genome:	Usually one circular DNA molecule	Multiple linear DNA molecules
DNA:	Not complexed with histones in eubacteria; some histones in archaea	Complexed with histones
Amount of DNA:	Relatively small	Relatively large
Membrane-bounded organelles:	Absent	Present
Cytoskeleton:	Absent	Present

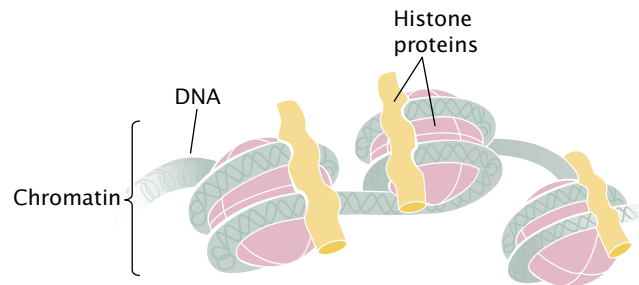
2.1 Prokaryotic and eukaryotic cells differ in structure. (Left to right: T.J. Beveridge/Visuals Unlimited; W. Baumeister/Science Photo/Library/Photo Researchers; Biophoto Associates/Photo Researchers; G. Murti/Phototake.)

Research indicates that dividing life into two major groups, the prokaryotes and eukaryotes, is incorrect. Although similar in cell structure, prokaryotes include at least two fundamentally distinct types of bacteria. These distantly related groups are termed **eubacteria** (the true bacteria) and **archaea** (ancient bacteria). An examination of equivalent DNA sequences reveals that eubacteria and archaea are as distantly related to one another as they are to the eukaryotes. Although eubacteria and archaea are similar in cell structure, some genetic processes in archaea (such as transcription) are more similar to those in eukaryotes, and the archaea may actually be evolutionarily closer to eukaryotes than to eubacteria. Thus, from an evolutionary perspective, there are three major groups of organisms: eubacteria, archaea, and eukaryotes. In this book, the prokaryotic–eukaryotic distinction will be used frequently, but important eubacterial–archaeal differences also will be noted.

From the perspective of genetics, a major difference between prokaryotic and eukaryotic cells is that a eukaryote has a *nuclear envelope*, which surrounds the genetic material to form a **nucleus** and separates the DNA from the other cellular contents. In prokaryotic cells, the genetic material is in close contact with other components of the cell—a property that has important consequences for the way in which genes are controlled.

Another fundamental difference between prokaryotes and eukaryotes lies in the packaging of their DNA. In eukaryotes, DNA is closely associated with a special class of proteins, the **histones**, to form tightly packed chromosomes. This complex of DNA and histone proteins is termed **chromatin**, which is the stuff of eukaryotic chromosomes (FIGURE 2.2). Histone proteins limit the accessibility of enzymes and other proteins that copy and read the DNA but they enable the DNA to fit into the nucleus. Eukaryotic DNA must separate from the histones before the genetic information in the DNA can be accessed. Archaea also have some histone proteins that complex with DNA, but the structure of their chromatin is different from that found in eukaryotes. However, eubacteria do not possess histones, so their DNA does not exist in the highly ordered, tightly packed arrangement found in eukaryotic cells (FIGURE 2.3). The copying and reading of DNA are therefore simpler processes in eubacteria.

Genes of prokaryotic cells are generally on a single, circular molecule of DNA, the chromosome of the prokaryotic cell. In eukaryotic cells, genes are located on multiple, usually linear DNA molecules (multiple chromosomes). Eukaryotic cells therefore require mechanisms that ensure that a copy of each chromosome is faithfully transmitted to each new cell. This generalization—a single, circular chromosome in prokaryotes and multiple, linear chromosomes in eukaryotes—is not always true. A few bacteria have more than one chromosome, and important bacterial genes are frequently found on other DNA molecules called plasmids. Furthermore, in some eukaryotes, a few genes are located on circular DNA molecules found outside the nucleus (see Chapter 20).

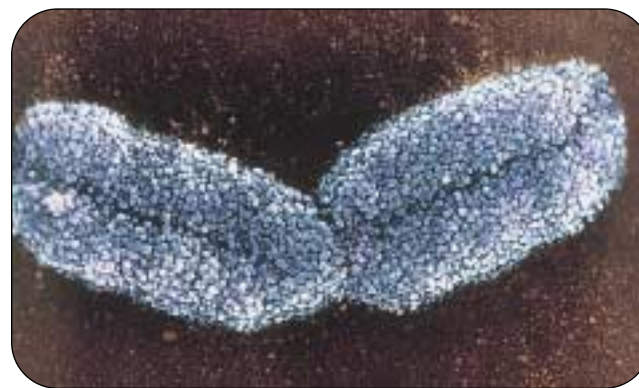


2.2 In eukaryotic cells, DNA is complexed to histone proteins to form chromatin.

(a)

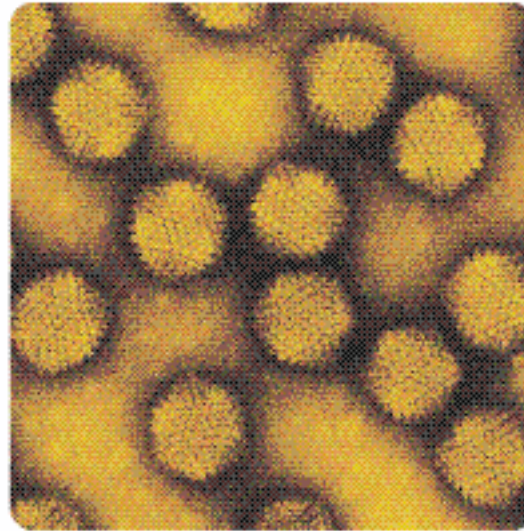
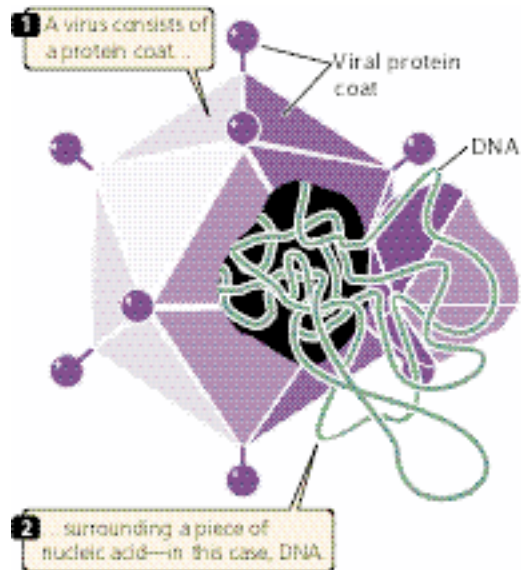


(b)



2.3 Prokaryotic DNA (a) is not surrounded by a nuclear membrane nor is the DNA complexed with histone proteins; eukaryotic DNA (b) is complexed to histone proteins to form chromosomes that are located in the nucleus.

(Part a, Dr. G. Murti/Science Photo Library/Photo Researchers; Part b, Biophoto Associates/Photo Researchers.)



Adenovirus

2.4 A virus consists of DNA or RNA surrounded by a protein coat. (Hans Gelderblam/Visuals Unlimited.)

Concepts

Organisms are classified as prokaryotes or eukaryotes, and prokaryotes comprise archaea and eubacteria. A prokaryote is a unicellular organism that lacks a nucleus, its DNA is not complexed to histone proteins, and its genome is usually a single chromosome. Eukaryotes are either unicellular or multicellular, their cells possess a nucleus, their DNA is complexed to histone proteins, and their genomes consist of multiple chromosomes.

Viruses are relatively simple structures composed of an outer protein coat surrounding nucleic acid (either DNA or RNA; **FIGURE 2.4**). Viruses are neither cells nor primitive forms of life: they can reproduce only within host cells, which means that they must have evolved after, rather than before, cells. In addition, viruses are not an evolutionarily distinct group but are most closely related to their hosts—the genes of a plant virus are more similar to those in a plant cell than to those in animal viruses, which suggests that viruses evolved from their hosts, rather than from other viruses. The close relationship between the genes of virus and host makes viruses useful for studying the genetics of host organisms.

www.whfreeman.com/pierce More information on the diversity of life and the evolutionary relationships among organisms

Cell Reproduction

For any cell to reproduce successfully, three fundamental events must take place: (1) its genetic information must be copied, (2) the copies of genetic information must be separated from one another, and (3) the cell must divide. All cellular reproduction includes these three events, but the processes that lead to these events differ in prokaryotic and eukaryotic cells.

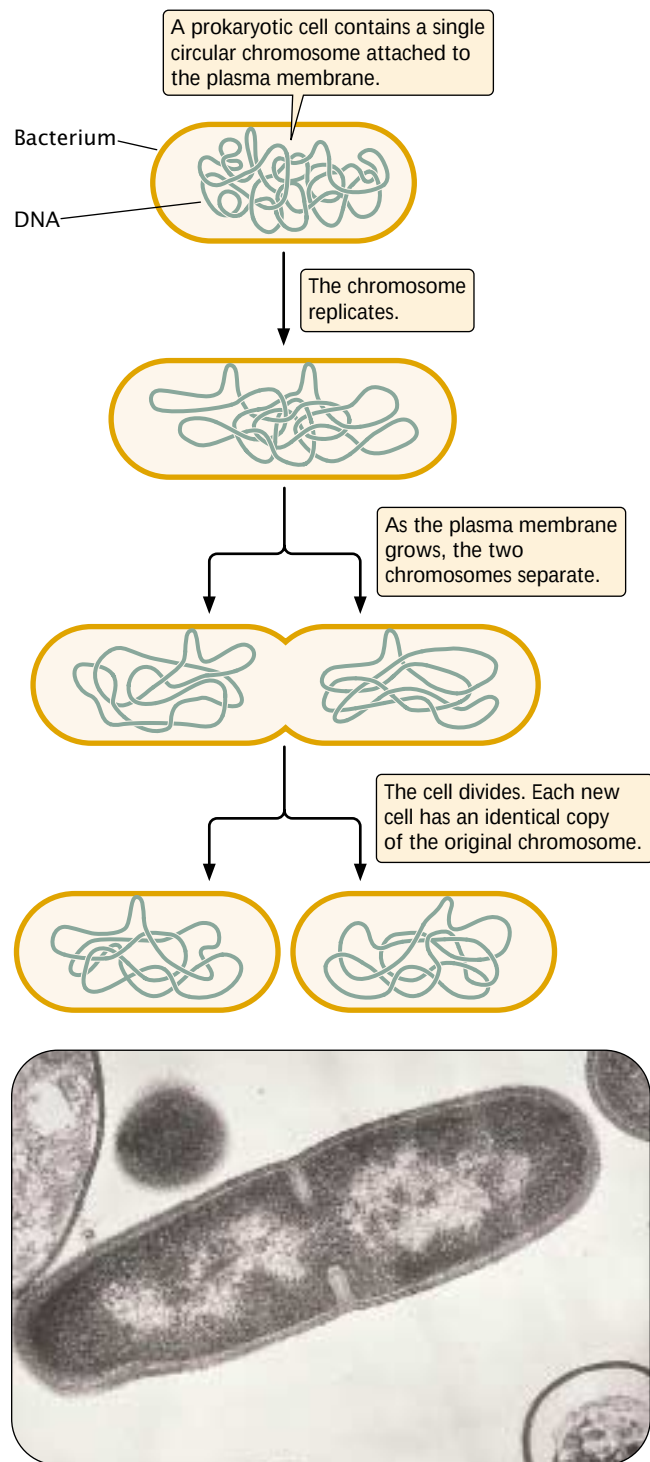
Prokaryotic Cell Reproduction

When prokaryotic cells reproduce, the circular chromosome of the bacterium is replicated (**FIGURE 2.5**). The two resulting identical copies are attached to the plasma membrane, which grows and gradually separates the two chromosomes. Finally, a new cell wall forms between the two chromosomes, producing two cells, each with an identical copy of the chromosome. Under optimal conditions, some bacterial cells divide every 20 minutes. At this rate, a single bacterial cell could produce a billion descendants in a mere 10 hours.

Eukaryotic Cell Reproduction

Like prokaryotic cell reproduction, eukaryotic cell reproduction requires the processes of DNA replication, copy separation, and division of the cytoplasm. However, the presence of multiple DNA molecules requires a more complex mechanism to ensure that one copy of each molecule ends up in each of the new cells.

Eukaryotic chromosomes are separated from the cytoplasm by the nuclear envelope. The nucleus was once thought to be a fluid-filled bag in which the chromosomes



2.5 Prokaryotic cells reproduce by simple division. (Micrograph Lee D. Simon/Photo Researchs.)

floated, but we now know that the nucleus has a highly organized internal scaffolding called the *nuclear matrix*. This matrix consists of a network of protein fibers that maintains precise spatial relations among the nuclear components and takes part in DNA replication, the expression

of genes, and the modification of gene products before they leave the nucleus. We will now take a closer look at the structure of eukaryotic chromosomes.

Eukaryotic chromosomes Each eukaryotic species has a characteristic number of chromosomes per cell: potatoes have 48 chromosomes, fruit flies have 8, and humans have 46. There appears to be no special significance between the complexity of an organism and its number of chromosomes per cell.

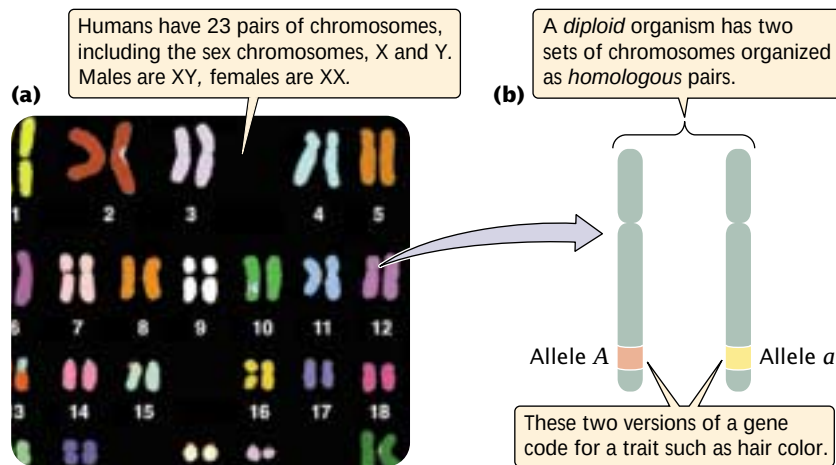
In most eukaryotic cells, there are two sets of chromosomes. The presence of two sets is a consequence of sexual reproduction; one set is inherited from the male parent and the other from the female parent. Each chromosome in one set has a corresponding chromosome in the other set, together constituting a **homologous pair** (FIGURE 2.6). Human cells, for example, have 46 chromosomes, comprising 23 homologous pairs.

The two chromosomes of a homologous pair are usually alike in structure and size, and each carries genetic information for the same set of hereditary characteristics. (An exception is the sex chromosomes, which will be discussed in Chapter 4.) For example, if a gene on a particular chromosome encodes a characteristic such as hair color, another gene (called an *allele*) at the same position on that chromosome's homolog *also* encodes hair color. However, these two alleles need not be identical: one might produce red hair and the other might produce blond hair. Thus, most cells carry two sets of genetic information; these cells are **diploid**. But not all eukaryotic cells are diploid: reproductive cells (such as eggs, sperm, and spores) and even nonreproductive cells in some organisms may contain a single set of chromosomes. Cells with a single set of chromosomes are **haploid**. Haploid cells have only one copy of each gene.

Concepts

Cells reproduce by copying and separating their genetic information and then dividing. Because eukaryotes possess multiple chromosomes, mechanisms exist to ensure that each new cell receives one copy of each chromosome. Most eukaryotic cells are diploid, and their two chromosomes sets can be arranged in homologous pairs. Haploid cells contain a single set of chromosomes.

Chromosome structure The chromosomes of eukaryotic cells are larger and more complex than those found in prokaryotes, but each unreplicated chromosome nevertheless consists of a single molecule of DNA. Although linear, the DNA molecules in eukaryotic chromosomes are highly folded and condensed; if stretched out, some human chromosomes



2.6 Diploid eukaryotic cells have two sets of chromosomes.

(a) A set of chromosomes from a human cell.

(b) The chromosomes are present in homologous pairs, which consist of chromosomes that are alike in size and structure and carry information for the same characteristics. (Courtesy of Dr. Thomas Ried and Dr. Evelin Schrock.)

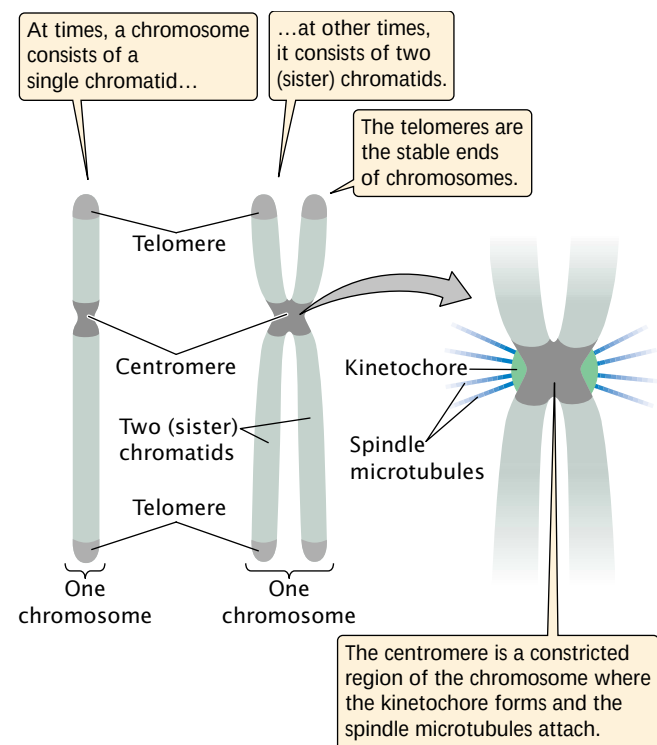
would be several centimeters long—thousands of times longer than the span of a typical nucleus. To package such a tremendous length of DNA into this small volume, each DNA molecule is coiled again and again and tightly packed around histone proteins, forming the rod-shaped chromosomes. Most of the time the chromosomes are thin and difficult to observe but, before cell division, they condense further into thick, readily observed structures; it is at this stage that chromosomes are usually studied (FIGURE 2.7).

A functional chromosome has three essential elements: a centromere, a pair of telomeres, and origins of replication. The *centromere* is the attachment point for *spindle microtubules*, which are the filaments responsible for moving chromosomes during cell division. The centromere appears as a constricted region that often stains less strongly than does the rest of the chromosome. Before cell division, a protein complex called the *kinetochore* assembles on the centromere, to which spindle microtubules later attach. Chromosomes without a centromere cannot be drawn into the newly formed nuclei; these chromosomes are lost, often with catastrophic consequences to the cell. On the basis of the location of the centromere, chromosomes are classified into four types: metacentric, submetacentric, acrocentric, and telocentric (FIGURE 2.8). One of the two arms of a chromosome (the short arm of a submetacentric or acrocentric chromosome) is designated by the letter p and the other arm is designated by q.

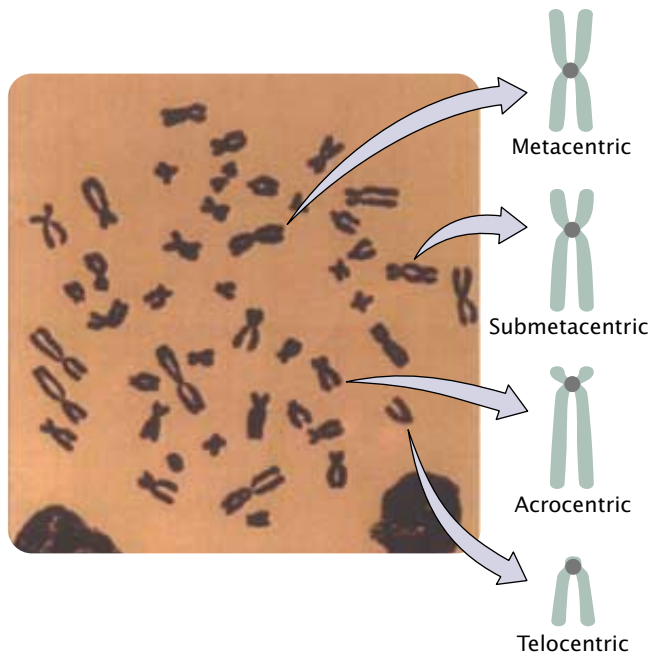
Telomeres are the natural ends, the tips, of a linear chromosome (see Figure 2.7); they serve to stabilize the chromosome ends. If a chromosome breaks, producing new ends, these ends have a tendency to stick together, and the chromosome is degraded at the newly broken ends. Telomeres provide chromosome stability. The results of

research (discussed in Chapter 12) suggest that telomeres also participate in limiting cell division and may play important roles in aging and cancer.

Origins of replication are the sites where DNA synthesis begins; they are not easily observed by microscopy. Their structure and function will be discussed in more detail in Chapters 11 and 12. In preparation for cell division, each



2.7 Structure of a eukaryotic chromosome.



2.8 Eukaryotic chromosomes exist in four major types. (L. Lisco, D. W. Fawcett/Visuals Unlimited.)

chromosome replicates, making a copy of itself. These two initially identical copies, called **sister chromatids**, are held together at the centromere (see Figure 2.7). Each sister chromatid consists of a single molecule of DNA.

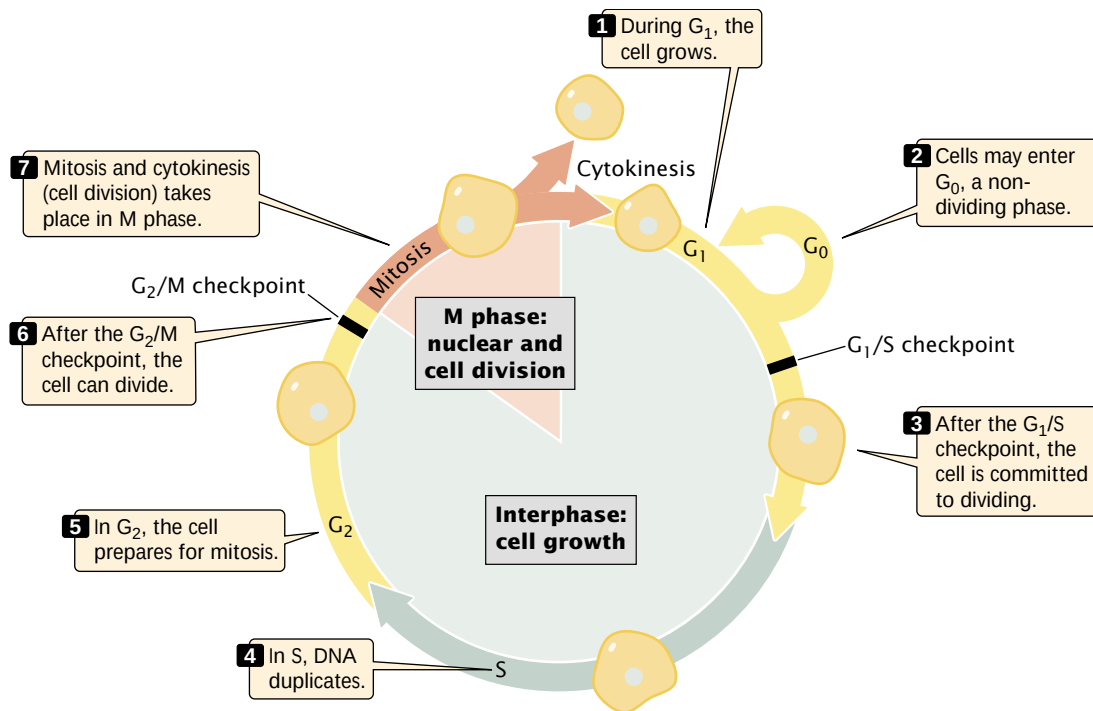
Concepts

Sister chromatids are copies of a chromosome held together at the centromere. Functional chromosomes contain centromeres, telomeres, and origins of replication. The kinetochore is the point of attachment for the spindle microtubules; telomeres are the stabilizing ends of a chromosome; origins of replication are sites where DNA synthesis begins.

The Cell Cycle and Mitosis

The **cell cycle** is the life story of a cell, the stages through which it passes from one division to the next (FIGURE 2.9). This process is critical to genetics because, through the cell cycle, the genetic instructions for all characteristics are passed from parent to daughter cells. A new cycle begins after a cell has divided and produced two new cells. A new cell metabolizes, grows, and develops. At the end of its cycle, the cell divides to produce two cells, which can then undergo additional cell cycles.

The cell cycle consists of two major phases. The first is **interphase**, the period between cell divisions, in which the cell grows, develops, and prepares for cell division. The second is **M phase** (mitotic phase), the period of active cell division. M phase includes **mitosis**, the process of nuclear division, and **cytokinesis**, or cytoplasmic division. Let's take a closer look at the details of interphase and M phase.



2.9 The cell cycle consists of interphase (a period of cell growth) and M phase (the period of nuclear and cell division).

Interphase Interphase is the extended period of growth and development between cell divisions. Although little activity can be observed with a light microscope, the cell is quite busy: DNA is being synthesized, RNA and proteins are being produced, and hundreds of biochemical reactions are taking place.

By convention, interphase is divided into three phases: G_1 , S, and G_2 (see Figure 2.9). Interphase begins with G_1 (for gap 1). In G_1 , the cell grows, and proteins necessary for cell division are synthesized; this phase typically lasts several hours. There is a critical point in the cell cycle, termed the G_1/S *checkpoint*, in G_1 ; after this checkpoint has been passed, the cell is committed to divide.

Before reaching the G_1/S checkpoint, cells may exit from the active cell cycle in response to regulatory signals and pass into a nondividing phase called G_0 (see Figure 2.9), which is a stable state during which cells usually maintain a constant size. They can remain in G_0 for an extended period of time, even indefinitely, or they can reenter G_1 and the active cell cycle. Many cells never enter G_0 ; rather, they cycle continuously.

After G_1 , the cell enters the S phase (for DNA synthesis), in which each chromosome duplicates. Although the cell is committed to divide after the G_1/S checkpoint has been passed, DNA synthesis must take place before the cell can proceed to mitosis. If DNA synthesis is blocked (with drugs or by a mutation), the cell will not be able to undergo mitosis. Before S phase, each chromosome is composed of one chromatid; following S phase, each chromosome is composed of two chromatids.

After the S phase, the cell enters G_2 (gap 2). In this phase, several additional biochemical events necessary for cell division take place. The important G_2/M *checkpoint* is reached in G_2 ; after this checkpoint has been passed, the cell is ready to divide and enters M phase. Although the length of interphase varies from cell type to cell type, a typical dividing mammalian cell spends about 10 hours in G_1 , 9 hours in S, and 4 hours in G_2 (see Figure 2.9).

Throughout interphase, the chromosomes are in a relatively relaxed, but by no means uncoiled, state, and individual chromosomes cannot be seen with the use of a microscope. This condition changes dramatically when interphase draws to a close and the cell enters M phase.

Mphase M phase is the part of the cell cycle in which the copies of the cell's chromosomes (sister chromatids) are separated and the cell undergoes division. A critical process in M phase is the separation of sister chromatids to provide a complete set of genetic information for each of the resulting cells. Biologists usually divide M phase into six stages: the five stages of mitosis (prophase, prometaphase, metaphase, anaphase, and telophase) and cytokinesis (FIGURE 2.10). It's important to keep in mind that M phase is a continuous process, and its separation into these six stages is somewhat artificial.

During interphase, the chromosomes are relaxed and are visible only as diffuse chromatin, but they condense dur-

ing **prophase**, becoming visible under a light microscope. Each chromosome possesses two chromatids because the chromosome was duplicated in the preceding S phase. The *mitotic spindle*, an organized array of microtubules that move the chromosomes in mitosis, forms. In animal cells, the spindle grows out from a pair of *centrosomes* that migrate to opposite sides of the cell. Within each centrosome is a special organelle, the *centriole*, which is also composed of microtubules. (Higher plant cells do not have centrosomes or centrioles, but they do have mitotic spindles).

Disintegration of the nuclear membrane marks the start of **prometaphase**. Spindle microtubules, which until now have been outside the nucleus, enter the nuclear region. The ends of certain microtubules make contact with the chromosome and anchor to the kinetochore of *one* of the sister chromatids; a microtubule from the opposite centrosome then attaches to the *other* sister chromatid, and so each chromosome is anchored to both of the centrosomes. The microtubules lengthen and shorten, pushing and pulling the chromosomes about. Some microtubules extend from each centrosome toward the center of the spindle but do not attach to a chromosome.

During **metaphase**, the chromosomes arrange themselves in a single plane, the *metaphase plate*, between the two centrosomes. The centrosomes, now at opposite ends of the cell with microtubules radiating outward and meeting in the middle of the cell, center at the spindle pole. **Anaphase** begins when the sister chromatids separate and move toward opposite spindle poles. After the chromatids have separated, each is considered a separate chromosome. **Telophase** is marked by the arrival of the chromosomes at the spindle poles. The nuclear membrane re-forms around each set of chromosomes, producing two separate nuclei within the cell. The chromosomes relax and lengthen, once again disappearing from view. In many cells, division of the cytoplasm (cytokinesis) is simultaneous with telophase. The major features of the cell cycle are summarized in Table 2.1.

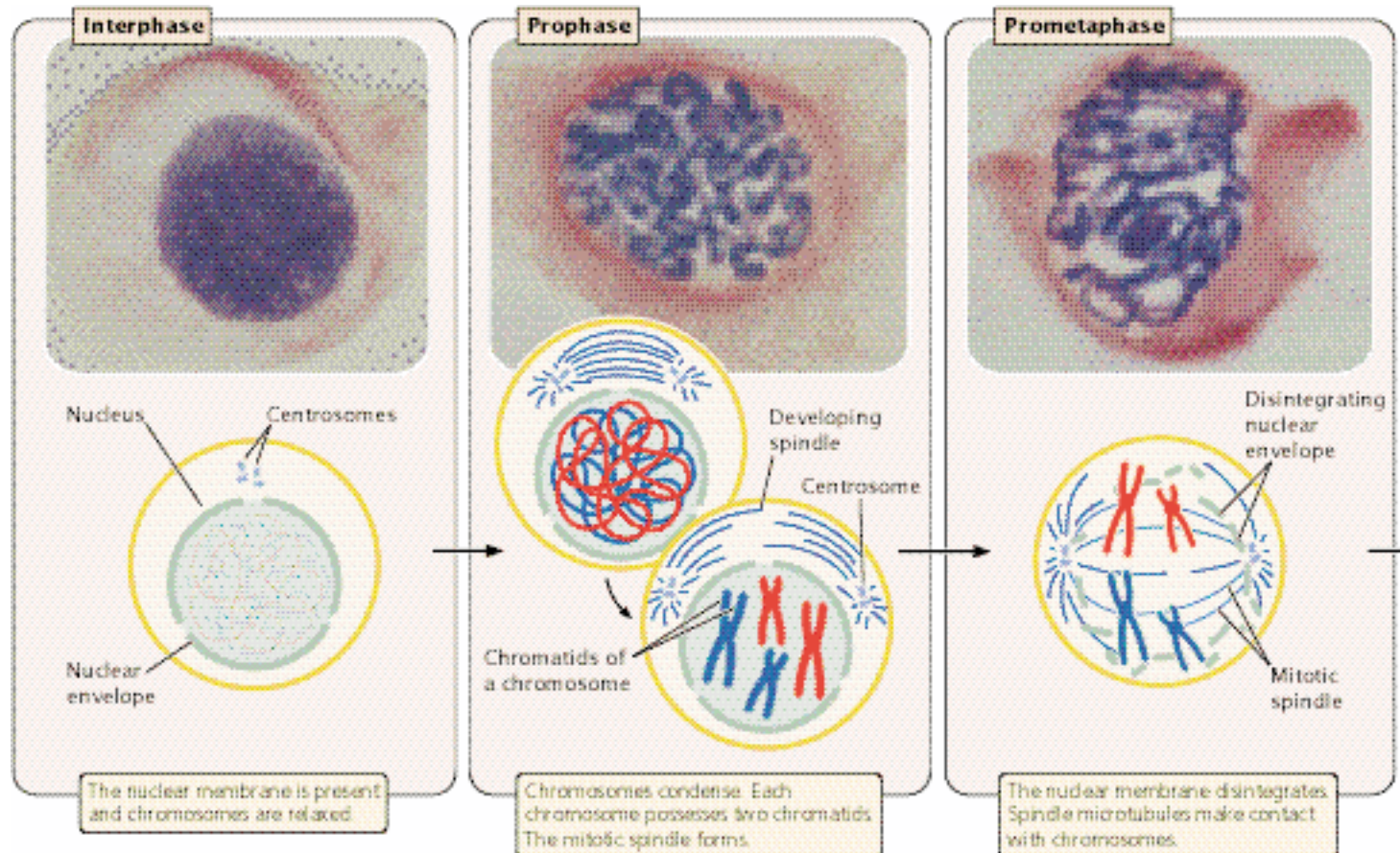
Concepts

The active cell-cycle phases are interphase and M phase. Interphase consists of G_1 , S, and G_2 . In G_1 , the cell grows and prepares for cell division; in the S phase, DNA synthesis takes place; in G_2 , other biochemical events necessary for cell division take place. Some cells enter a quiescent phase called G_0 . M phase includes mitosis and cytokinesis and is divided into prophase, prometaphase, metaphase, anaphase, and telophase.

www.whfreeman.com/pierce Mitosis animations, tutorials, and pictures of dividing cells

Movement of Chromosomes in Mitosis

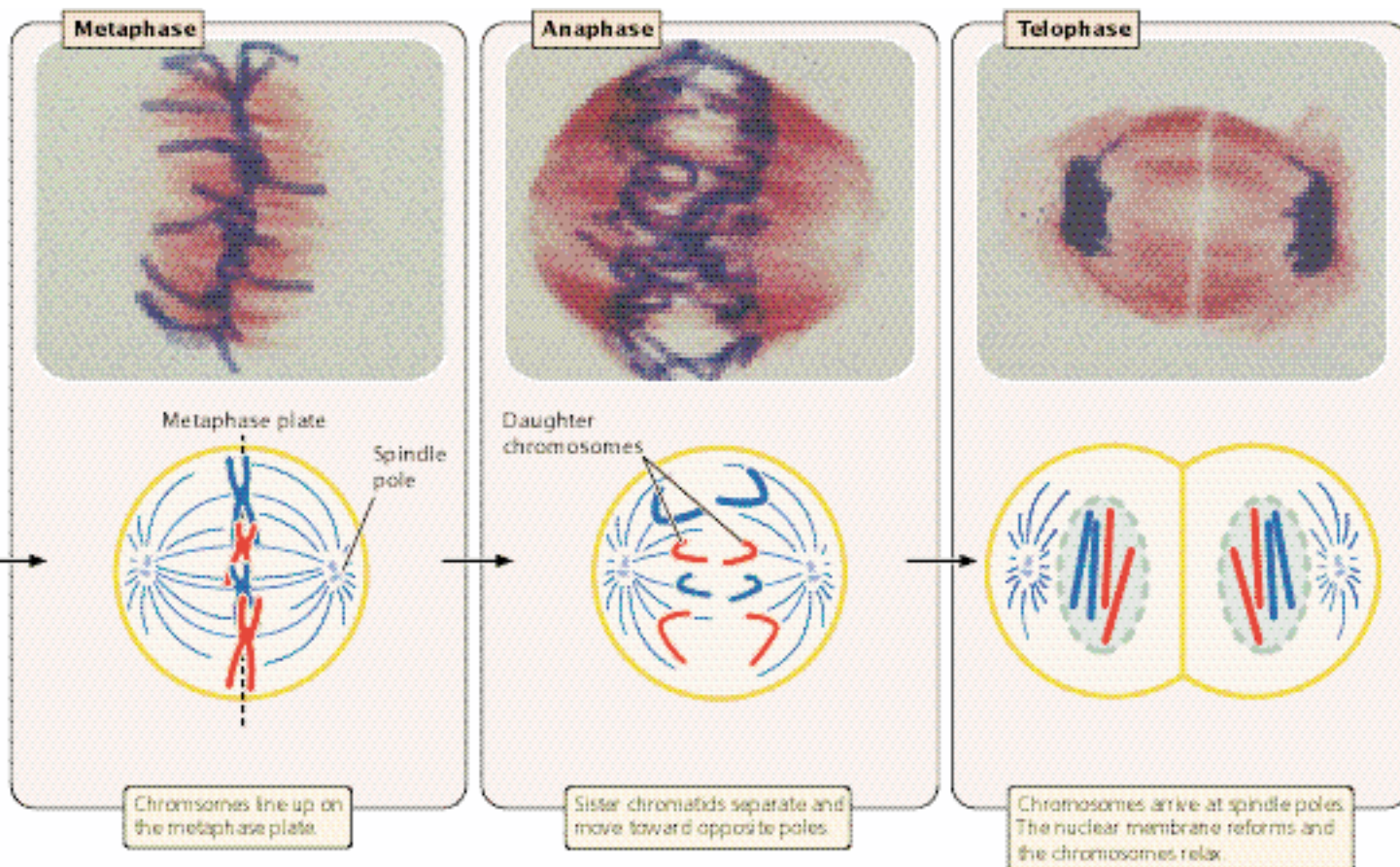
Each microtubule of the spindle is composed of subunits of a protein called tubulin, and each microtubule has direction



2.10 The cell cycle is divided into stages. (Photos © Andrew S. Bajer, University of Oregon.)

Table 2.1 Features of the cell cycle

Stage	Major Features
G ₀ phase	Stable, nondividing period of variable length
Interphase	
G ₁ phase	Growth and development of the cell; G ₁ /S checkpoint
S phase	Synthesis of DNA
G ₂ phase	Preparation for division; G ₂ /S checkpoint
M phase	
Prophase	Chromosomes condense and mitotic spindle forms
Prometaphase	Nuclear envelope disintegrates, spindle microtubules anchor to kinetochores
Metaphase	Chromosomes align on the metaphase plate
Anaphase	Sister chromatids separate, becoming individual chromosomes that migrate toward spindle poles
Telophase	Chromosomes arrive at spindle poles, the nuclear envelope re-forms, and the condensed chromosomes relax
Cytokinesis	Cytoplasm divides; cell wall forms in plant cells



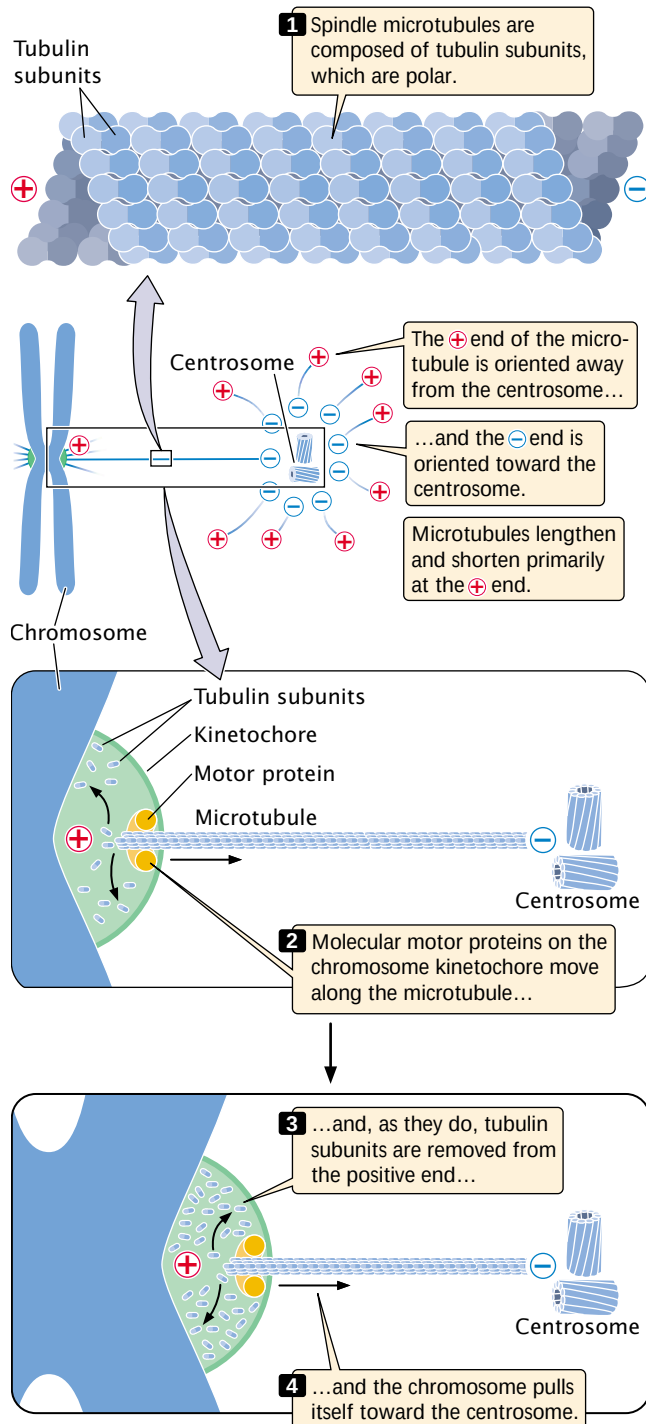
or polarity. Like a flashlight battery, one end is referred to as plus (+) and the other end as minus (−). The “−” end is always oriented toward the centrosome, and the “+” end is always oriented away from the centrosome; microtubules lengthen and shorten by the addition and removal of subunits primarily at the “+” end.

At one time, chromosomes were viewed as passive carriers of genetic information that were pushed about by the active spindle microtubules. Research findings now indicate that chromosomes actively control and generate the forces responsible for their movement in the course of mitosis and meiosis. Chromosome movement is accomplished through complex interactions between the kinetochore of the chromosome and the microtubules of the spindle apparatus.

The forces responsible for the poleward movement of chromosomes during anaphase are generated at the kinetochore itself but are not completely understood. Located within each kinetochore are specialized proteins called *molecular motors*, which may help pull a chromosome toward the spindle pole (FIGURE 2.11). The poleward force is created by the removal of the tubulin primarily at the “+” end of the microtubule.

In mitosis, depolymerization of tubulin and perhaps also molecular motors pull the chromosome toward the pole, but this force is initially counterbalanced by the attachment of the two chromatids. Throughout prophase, prometaphase, and metaphase, the sister chromatids are held together by a gluelike material called cohesion. The cohesion material breaks down at the onset of anaphase, allowing the two chromatids to separate and the resulting newly formed chromosomes to move toward the spindle pole. While the chromosomal microtubules shorten, other microtubules elongate, pushing the two spindle poles farther apart. As the chromosomes near the spindle poles, they contract to form a compact mass. In spite of much study, the precise role of the poles, kinetochores, and microtubules in the formation and function of the spindle apparatus is still incompletely understood.

Genetic consequences of the cell cycle What are the genetically important results of the cell cycle? From a single cell, the cell cycle produces two cells that contain the same genetic instructions. These two cells are identical with each other and with the cell that gave rise to them. They are identical because DNA synthesis in S phase creates an exact copy of each DNA molecule, giving rise to two genetically



2.11 Removal of the tubulin subunits from microtubules at the kinetochore and perhaps molecular motors, are responsible for the poleward movement of chromosomes during anaphase.

identical sister chromatids. Mitosis then ensures that one chromatid from each replicated chromosome passes into each new cell.

Another genetically important result of the cell cycle is that each of the cells produced contains a full complement of chromosomes—there is no net reduction or increase in chromosome number. Each cell also contains approximately half the cytoplasm and organelle content of the original parental cell, but no precise mechanism analogous to mitosis ensures that organelles are evenly divided. Consequently, not all cells resulting from the cell cycle are identical in their cytoplasmic content.

Control of the cell cycle For many years, the biochemical events that controlled the progression of cells through the cell cycle were completely unknown, but research has now revealed many of the details of this process. Progression of the cell cycle is regulated at several checkpoints, which ensure that all cellular components are present and in good working order before the cell proceeds to the next stage. The checkpoints are necessary to prevent cells with damaged or missing chromosomes from proliferating.

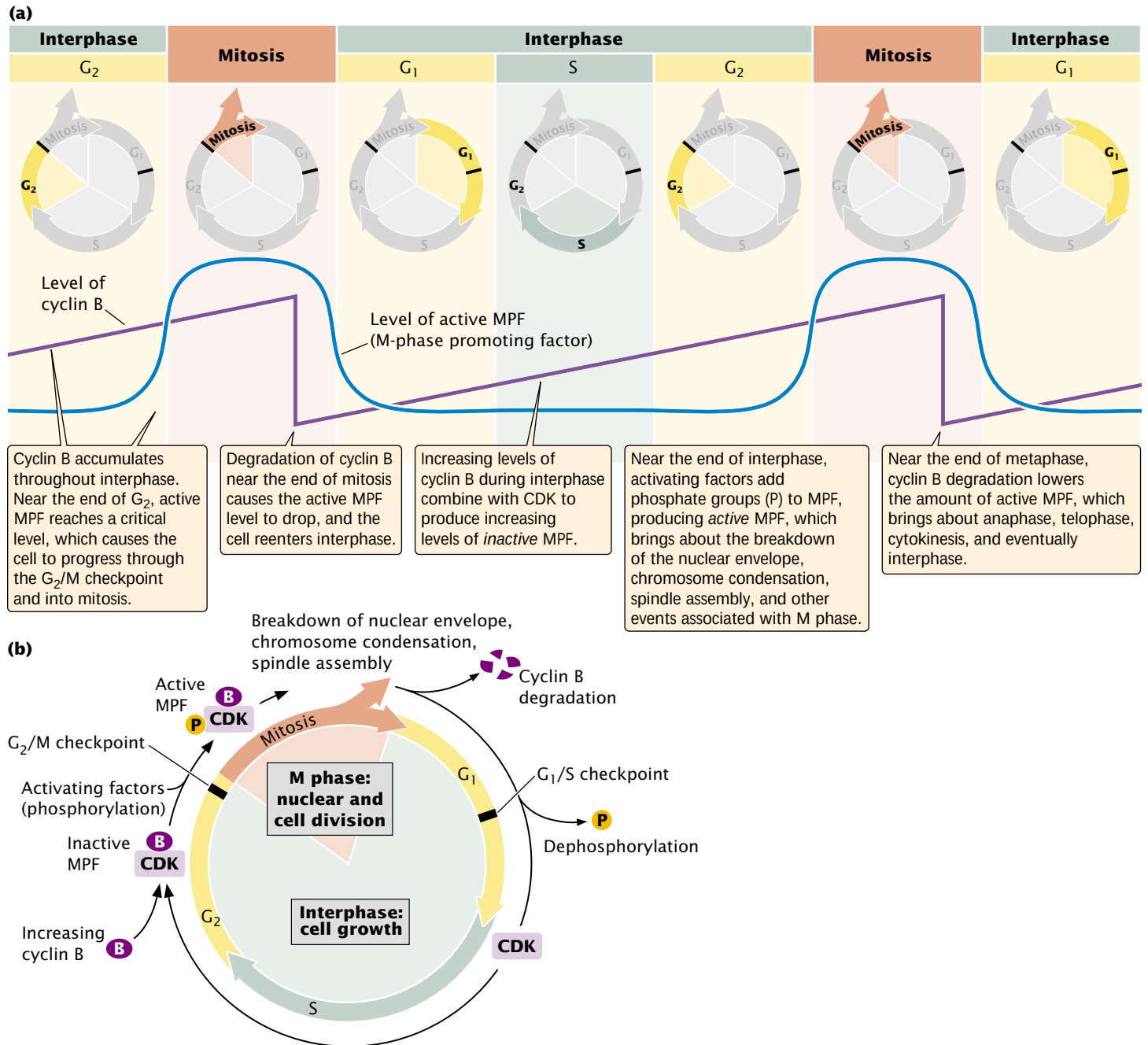
One important checkpoint mentioned earlier, the G_1/S checkpoint, comes just before the cell enters into S phase and replicates its DNA. When this point has been passed, DNA replicates and the cell is committed to divide. A second critical checkpoint, called the G_2/M checkpoint, is at the end of G_2 , before the cell enters mitosis.

Both the G_1/S and the G_2/M checkpoints are regulated by a mechanism in which two proteins interact. The concentration of the first protein, *cyclin*, oscillates during the cell cycle (FIGURE 2.12a). The second protein, *cyclin-dependent kinase* (CDK), cannot function unless it is bound to cyclin. Cyclins and CDKs are called by different names in different organisms, but here we will use the terms applied to these molecules in yeast.

Let's begin by looking at the G_2/M checkpoint. This checkpoint is regulated by cyclin B, which combines with CDK to form *M-phase promoting factor* (MPF). After MPF is formed, it must be activated by the addition of a phosphate group to one of the amino acids of CDK (FIGURE 2.12b).

Whereas the amount of cyclin B changes throughout the cell cycle, the amount of CDK remains constant. During G_1 , cyclin B levels are low; so the amount of MPF also is low (see Figure 2.12a). As more cyclin B is produced, it combines with CDK to form increasing amounts of MPF. Near the end of G_2 , the amount of active MPF reaches a critical level, which commits the cell to divide. The MPF concentration continues to increase, reaching a peak in mitosis (see Figure 2.12a).

The active form of MPF is a protein kinase, an enzyme that adds phosphate groups to certain other proteins. Active MPF brings about many of the events associated with mitosis, such as nuclear-membrane breakdown, spindle formation, and chromosome condensation. At the end of metaphase, cyclin is abruptly degraded, which lowers the amount of MPF and, initiating anaphase, sets in motion a chain of events that ultimately brings mitosis to a close.



2.12 Progression through the cell cycle is regulated by cyclins and CDKs. Shown here is regulation of the G₂/M checkpoint in yeast.

(see Figure 2.12b). Ironically, active MPF brings about its own demise by destroying cyclin. In brief, high levels of active MPF stimulate mitosis, and low levels of MPF bring a return to interphase conditions.

A number of factors stimulate the synthesis of cyclin B and the activation of MPF, whereas other factors inhibit MPF. Together these factors determine whether the cell passes through the G₂/M checkpoint and ensure that mitosis is not initiated until conditions are appropriate for cell division. For

example, DNA damage inhibits the activation of MPF; the cell is arrested in G₂ and does not undergo division.

The G₁/S checkpoint is regulated in a similar manner. In fission yeast (*Shizosaccharomyces pombe*), the same CDK is used, but it combines with G₁ cyclins. Again, the level of CDK remains relatively constant, whereas the level of G₁ cyclins increases throughout G₁. When the activated CDK–G₁–cyclin complex reaches a critical concentration, proteins necessary for replication are activated and the cell enters S phase.

Many cancers are caused by defects in the cell cycle's regulatory machinery. For example, mutation in the gene that encodes cyclin D, which has a role in the human G₁/S checkpoint, contributes to the rise of B-cell lymphoma. The overexpression of this gene is associated with both breast and esophageal cancer. Likewise, the tumor-suppressor gene *p53*, which is mutated in about 75% of all colon cancers, regulates a potent inhibitor of CDK activity.

Concepts

The cell cycle produces two genetically identical cells, with no net change in chromosome number. Progression through the cell cycle is controlled at checkpoints, which are regulated by interactions between cyclins and cyclin-dependent kinases.

Connecting Concepts

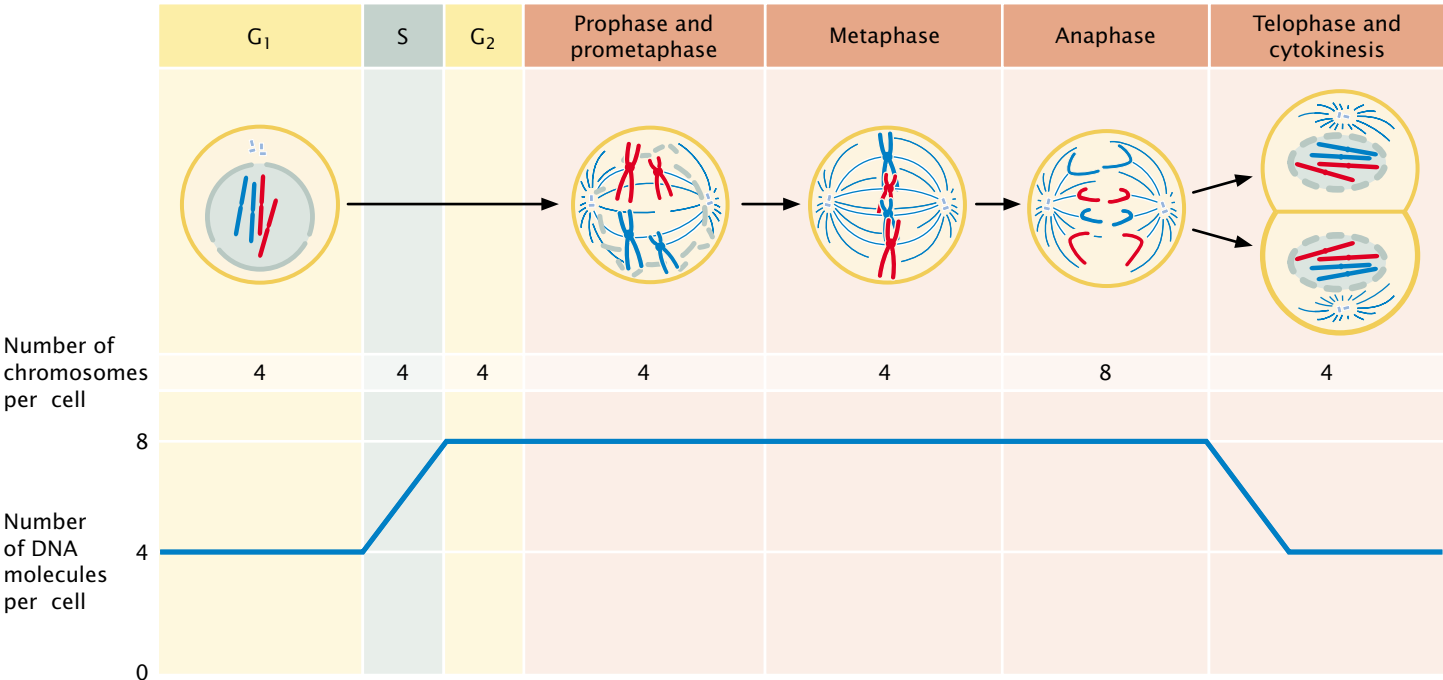
Counting Chromosomes and DNA Molecules

The relations among chromosomes, chromatids, and DNA molecules frequently cause confusion. At certain times, chromosomes are unreplicated; at other times, each possesses two chromatids (see Figure 2.7b). Chromosomes sometimes consist of a single DNA molecule; at other

times, they consist of two DNA molecules. How can we keep track of the number of these structures in the cell cycle?

There are two simple rules for counting chromosomes and DNA molecules: (1) to determine the number of chromosomes, count the number of functional centromeres; (2) to determine the number of DNA molecules, count the number of chromatids. Let's examine a hypothetical cell as it passes through the cell cycle (FIGURE 2.13). At the beginning of G₁, this diploid cell has a complete set of four chromosomes, inherited from its parent cell. Each chromosome consists of a single chromatid—a single DNA molecule—so there are four DNA molecules in the cell during G₁. In S phase, each DNA molecule is copied. The two resulting DNA molecules combine with histones and other proteins to form sister chromatids. Although the amount of DNA doubles during S phase, the number of chromosomes remains the same, because the two sister chromatids share a single functional centromere. At the end of S phase, this cell still contains four chromosomes, each with two chromatids; so there are eight DNA molecules present.

Through prophase, prometaphase, and metaphase, the cell has four chromosomes and eight DNA molecules. At anaphase, however, the sister chromatids separate. Each now has its own functional centromere, and so each is considered a separate chromosome. Until cytokinesis, each cell contains eight chromosomes, each consisting of a single chromatid;



2.13 The number of chromosomes and DNA molecules changes in the course of the cell cycle. The number of chromosomes per cell equals the number of functional centromeres, and the number of DNA molecules per cell equals the number of chromatids.

thus, there are still eight DNA molecules present. After cytokinesis, the eight chromosomes (eight DNA molecules) are distributed equally between two cells; so each new cell contains four chromosomes and four DNA molecules, the number present at the beginning of the cell cycle.

Sexual Reproduction and Genetic Variation

If all reproduction were accomplished through the cell cycle, life would be quite dull, because mitosis produces only genetically identical progeny. With only mitosis, you, your children, your parents, your brothers and sisters, your cousins, and many people you didn't even know would be clones—copies of one another. Only the occasional mutation would introduce any genetic variability. This is how all organisms reproduced for the first 2 billion years of Earth's existence (and the way in which some organisms still reproduce today). Then, some 1.5 billion to 2 billion years ago, something remarkable evolved: cells that produce genetically variable offspring through sexual reproduction.

The evolution of sexual reproduction is one of the most significant events in the history of life. As will be discussed in Chapters 22 and 23, the pace of evolution depends on the amount of genetic variation present. By shuffling the genetic information from two parents, sexual reproduction greatly increases the amount of genetic variation and allows for accelerated evolution. Most of the tremendous diversity of life on Earth is a direct result of sexual reproduction.

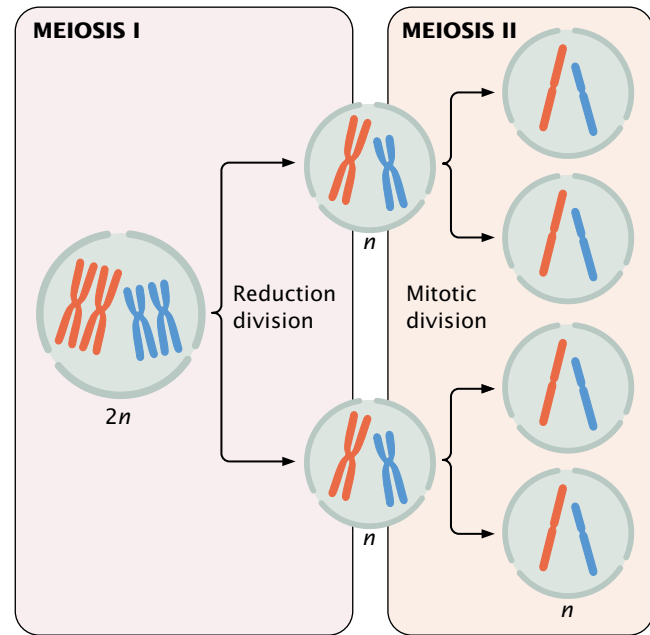
Sexual reproduction consists of two processes. The first is **meiosis**, which leads to gametes in which chromosome number is reduced by half. The second process is **fertilization**, in which two haploid gametes fuse and restore chromosome number to its original diploid value.

Meiosis

The words mitosis and meiosis are sometimes confused. They sound a bit alike, and both include chromosome division and cytokinesis. Don't let this deceive you. The outcomes of mitosis and meiosis are radically different, and several unique events that have important genetic consequences take place only in meiosis.

How is meiosis different from mitosis? Mitosis consists of a single nuclear division and is usually accompanied by a single cell division. Meiosis, on the other hand, consists of two divisions. After mitosis, chromosome number in newly formed cells is the same as that in the original cell, whereas meiosis causes chromosome number in the newly formed cells to be reduced by half. Finally, mitosis produces genetically identical cells, whereas meiosis produces genetically variable cells. Let's see how these differences arise.

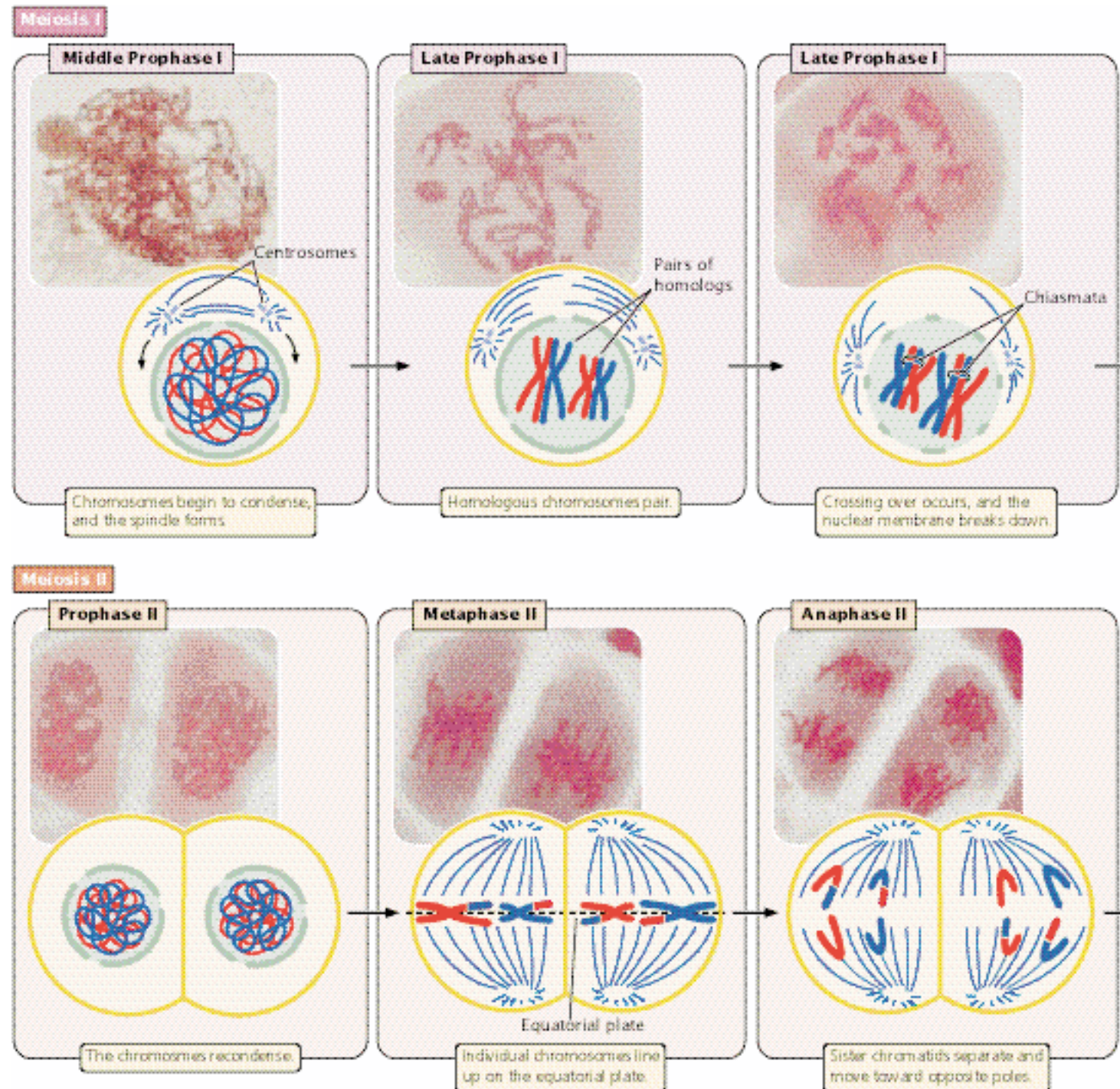
Like mitosis, meiosis is preceded by an interphase stage that includes G_1 , S , and G_2 phases. Meiosis consists of two distinct phases: *meiosis I* and *meiosis II*, each of which



2.14 Meiosis includes two cell divisions. In this figure, the original cell is $2n=4$. After two meiotic divisions each resulting cell $1n=2$.

includes a cell division. The first division is termed the reduction division because the number of chromosomes per cell is reduced by half (FIGURE 2.14). The second division is sometimes termed the equational division because the events in this phase are similar to those of mitosis. However, meiosis II differs from mitosis in that chromosome number has already been halved in meiosis I, and the cell does not begin with the same number of chromosomes as it does in mitosis (see Figure 2.14).

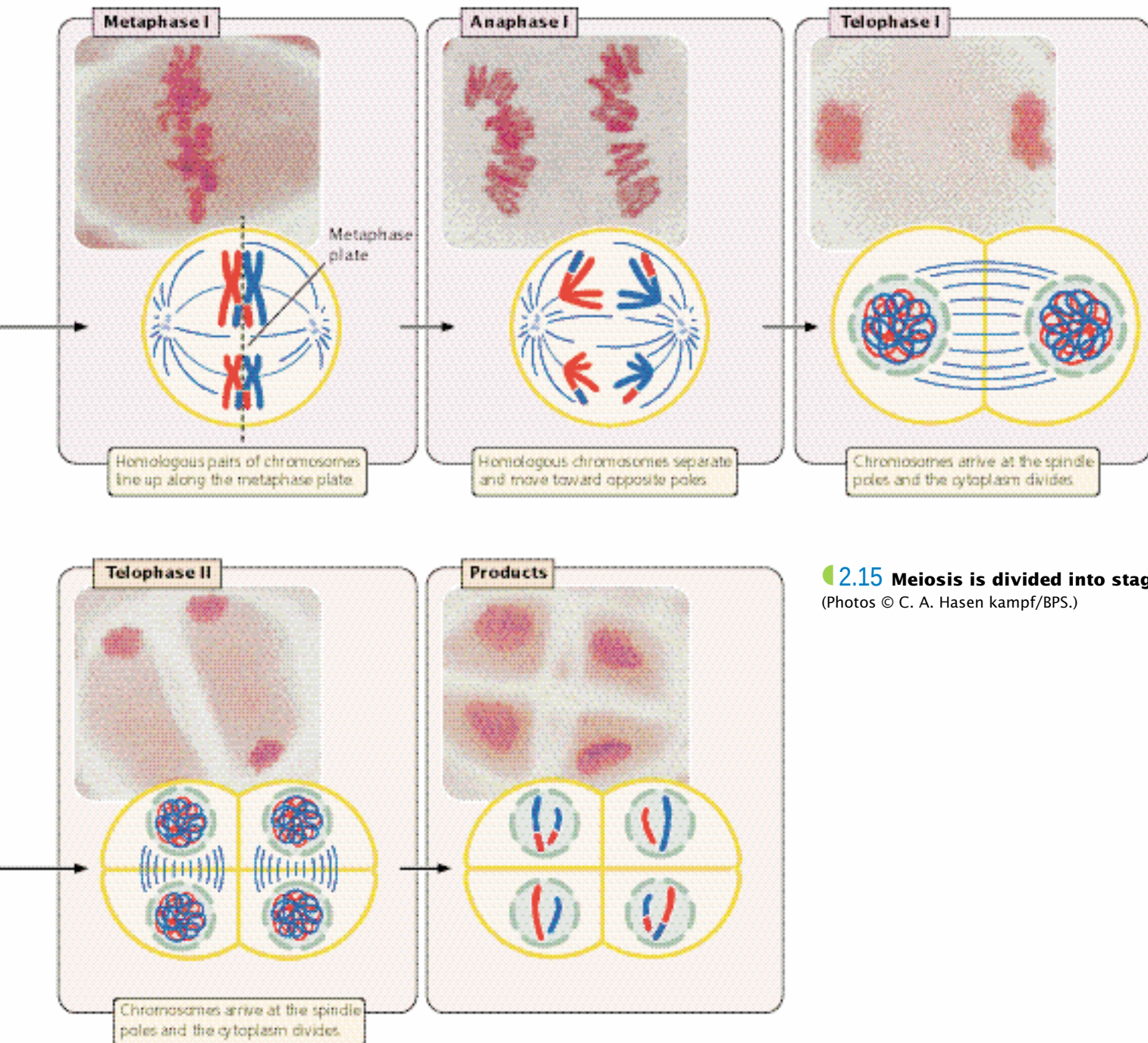
The stages of meiosis are outlined in FIGURE 2.15. During interphase, the chromosomes are relaxed and visible as diffuse chromatin. **Prophase I** is a lengthy stage, divided into five substages (FIGURE 2.16). In *leptotene*, the chromosomes contract and become visible. In *zygotene*, the chromosomes continue to condense; homologous chromosomes begin to pair up and begin **synapsis**, a very close pairing association. Each homologous pair of synapsed chromosomes consists of four chromatids called a **bivalent** or **tetrad**. In *pachytene*, the chromosomes become shorter and thicker, and a three-part *synaptonemal complex* develops between homologous chromosomes. **Crossing over** takes place, in which homologous chromosomes exchange genetic information. The centromeres of the paired chromosomes move apart during *diplotene*; the two homologs remain attached at each *chiasma* (plural, *chiasmata*), which is the result of crossing over. In *diakinesis*, chromosome condensation continues, and the chiasmata move toward the ends of the chromosomes as the strands slip apart; so the homologs remained paired only at the tips. Near the end of prophase I, the nuclear membrane breaks down and the spindle forms.



Metaphase I is initiated when homologous pairs of chromosomes align along the metaphase plate (see Figure 2.15). A microtubule from one pole attaches to one chromosome of a homologous pair, and a microtubule from the other pole attaches to the other member of the pair. **Anaphase I** is marked by the separation of homologous chromosomes. The two chromosomes of a homologous pair are pulled toward opposite poles. Although the homol-

ogous chromosomes separate, the sister chromatids remain attached and travel together. In **telophase I**, the chromosomes arrive at the spindle poles and the cytoplasm divides.

The period between meiosis I and meiosis II is **interkinesis**, in which the nuclear membrane re-forms around the chromosomes clustered at each pole, the spindle breaks down, and the chromosomes relax. These cells then pass through **Prophase II**, in which these events are reversed: the

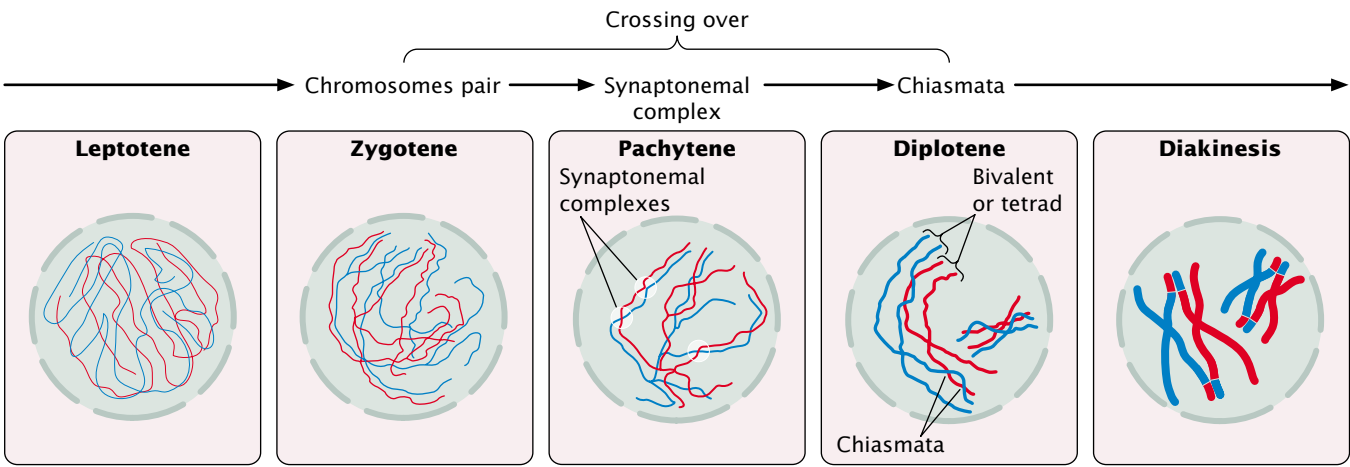


2.15 Meiosis is divided into stages.

(Photos © C. A. Hasenkampf/BPS.)

chromosomes recondense, the spindle re-forms, and the nuclear envelope once again breaks down. In interkinesis in some types of cells, the chromosomes remain condensed, and the spindle does not break down. These cells move directly from cytokinesis into **metaphase II**, which is similar to metaphase of mitosis: the individual chromosomes line up on the metaphase plate, with the sister chromatids facing opposite poles.

In **anaphase II**, the kinetochores of the sister chromatids separate and the chromatids are pulled to opposite poles. Each chromatid is now a distinct chromosome. In **telophase II**, the chromosomes arrive at the spindle poles, a nuclear envelope re-forms around the chromosomes, and the cytoplasm divides. The chromosomes relax and are no longer visible. The major events of meiosis are summarized in Table 2.2.



2.16 Crossing over takes place in prophase I. In yeast, rough pairing of chromosomes begins in leptotene and continues in zygotene. The synaptonemal complex forms in pachytene. Crossing over is initiated in zygotene, before the synaptonemal complex develops, and is not completed until near the end of prophase I.

Table 2.2 Major events in each stage of meiosis

Stage	Major Events
Meiosis I	
Prophase I	Chromosomes condense, homologous pairs of chromosomes synapse, crossing over takes place, nuclear envelope breaks down, and mitotic spindle forms
Metaphase I	Homologous pairs of chromosomes line up on the metaphase plate
Anaphase I	The two chromosomes (each with two chromatids) of each homologous pair separate and move toward opposite poles
Telophase I	Chromosomes arrive at the spindle poles
Cytokinesis	The cytoplasm divides to produce two cells, each having half the original number of chromosomes
Interkinesis	In some cells the spindle breaks down, chromosomes relax, and a nuclear envelope re-forms, but no DNA synthesis takes place
Meiosis II	
Prophase II*	Chromosomes condense, the spindle forms, and the nuclear envelope disintegrates
Metaphase II	Individual chromosomes line up on the metaphase plate
Anaphase II	Sister chromatids separate and migrate as individual chromosomes toward the spindle poles
Telophase II	Chromosomes arrive at the spindle poles; the spindle breaks down and a nuclear envelope re-forms
Cytokinesis	The cytoplasm divides

*Only in cells in which the spindle has broken down, chromosomes have relaxed, and the nuclear envelope has re-formed in telophase I. Other types of cells skip directly to metaphase II after cytokinesis.

Consequences of Meiosis

What are the overall consequences of meiosis? First, meiosis comprises two divisions; so each original cell produces four cells (there are exceptions to this generalization, as, for example, in many female animals; see Figure 2.22b). Second, chromosome number is reduced by half; so cells produced by meiosis are haploid. Third, cells produced by meiosis are genetically different from one another and from the parental cell.

Genetic differences among cells result from two processes that are unique to meiosis. The first is crossing over, which takes place in prophase I. Crossing over refers to the exchange of genes between nonsister chromatids (chromatids from different homologous chromosomes). At one time, this process was thought to take place in pachytene (Figure 2.15b), and the synaptonemal complex was believed to be a requirement for crossing over. However, recent evidence from yeast suggests that the situation is more complex, as shown in Figure 2.16. Crossing over is initiated in zygotene, before the synaptonemal complex develops, and is not completed until near the end of prophase I.

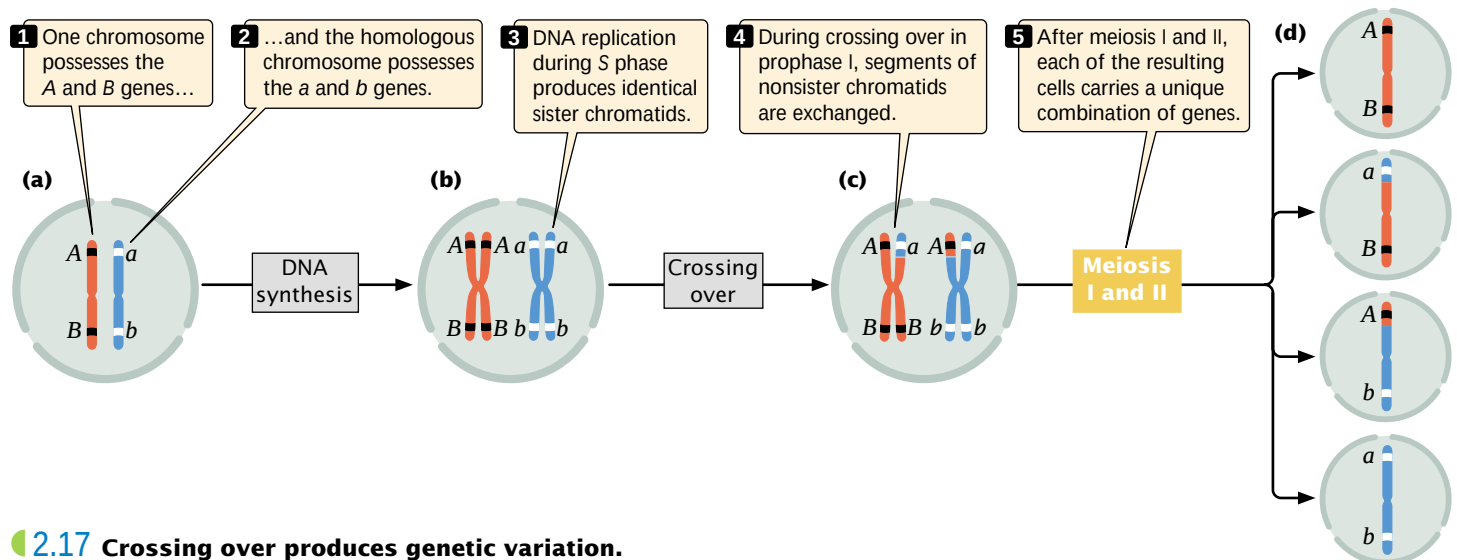
After crossing over has taken place, the sister chromatids may no longer be identical. Crossing over is the basis for intrachromosomal **recombination**, creating new combinations of alleles on a chromatid. To see how crossing over produces genetic variation, consider two pairs of alleles, which we will abbreviate *Aa* and *Bb*. Assume that one chromosome possesses the *A* and *B* alleles and its homolog possesses the *a* and *b* alleles (Figure 2.17a). When DNA is replicated in the S stage, each chromosome duplicates, and so the resulting sister chromatids are identical (Figure 2.17b).

In the process of crossing over, breaks occur in the DNA strands and the breaks are repaired in such a way that segments of nonsister chromatids are exchanged (Figure 2.17c). The molecular basis of this process will be described in more detail in Chapter 12; the important thing here is

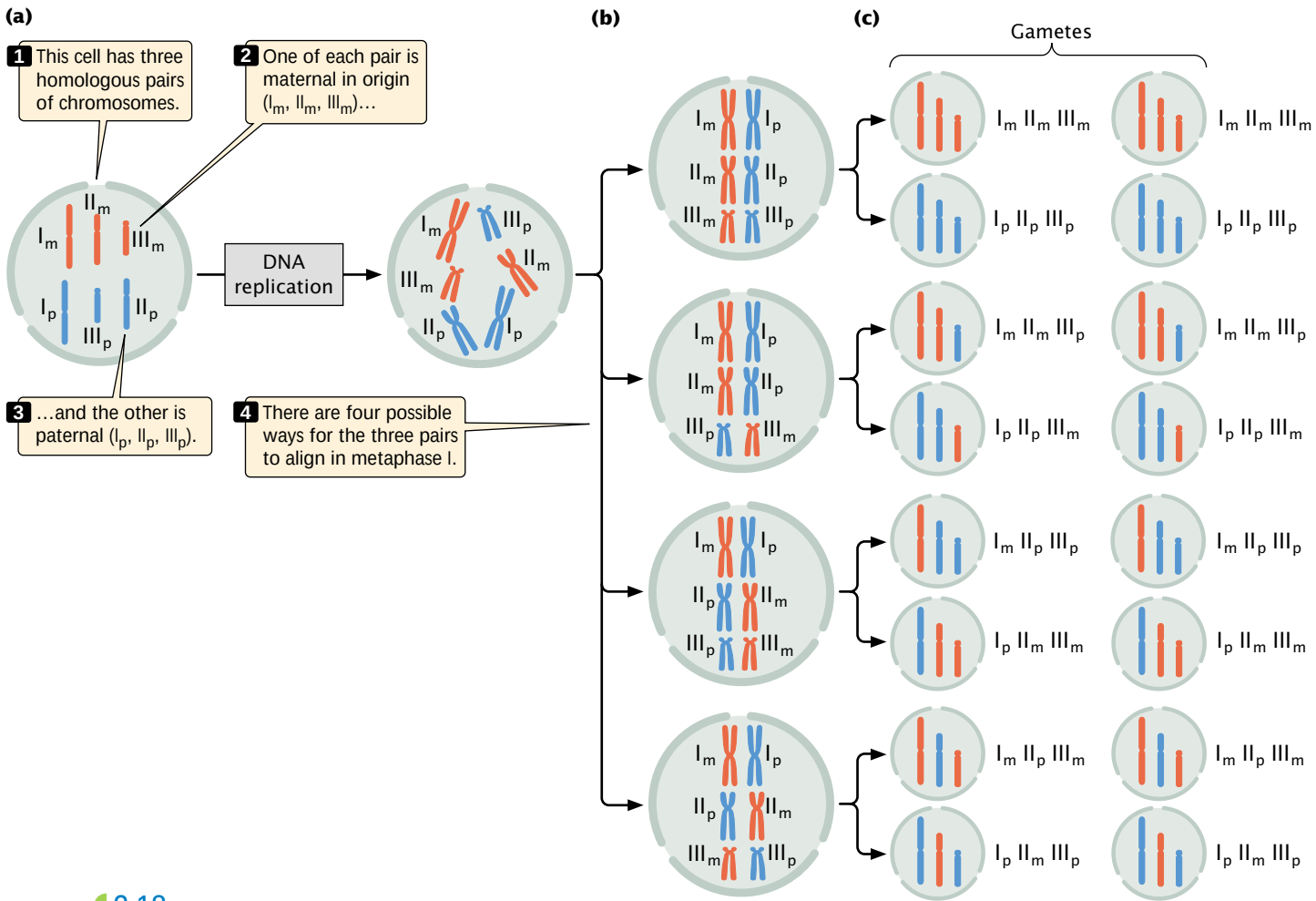
that, after crossing over has taken place, the two sister chromatids are no longer identical—one chromatid has alleles *A* and *B*, whereas its sister chromatid (the chromatid that underwent crossing over) has alleles *a* and *B*. Likewise, one chromatid of the other chromosome has alleles *a* and *b*, and the other has alleles *A* and *b*. Each of the four chromatids now carries a unique combination of alleles: *AB*, *aB*, *Ab*, and *ab*. Eventually, the two homologous chromosomes separate, each going into a different cell. In meiosis II, the two chromatids of each chromosome separate, and thus each of the four cells resulting from meiosis carries a different combination of alleles (Figure 2.17d).

The second process of meiosis that contributes to genetic variation is the random distribution of chromosomes in anaphase I of meiosis following their random alignment during metaphase I. To illustrate this process, consider a cell with three pairs of chromosomes I, II, and III (Figure 2.18a). One chromosome of each pair is maternal in origin (*I_m*, *II_m*, and *III_m*); the other is paternal in origin (*I_p*, *II_p*, and *III_p*). The chromosome pairs line up in the center of the cell in metaphase I and, in anaphase I, the chromosomes of each homologous pair separate.

How each pair of homologs aligns and separates is random and independent of how other pairs of chromosomes align and separate (Figure 2.18b). By chance, all the maternal chromosomes might migrate to one side, with all the paternal chromosomes migrating to the other. After division, one cell would contain chromosomes *I_m*, *II_m*, and *III_m*, and the other, *I_p*, *II_p*, and *III_p*. Alternatively, the *I_m*, *II_m*, and *III_p* chromosomes might move to one side, and the *I_p*, *II_p*, and *III_m* chromosomes to the other. The different migrations would produce different combinations of chromosomes in the resulting cells (Figure 2.18c). There are four ways in which a diploid cell with three pairs of chromosomes can divide, producing a total of eight different



2.17 Crossing over produces genetic variation.



2.18 Genetic variation is produced through the random distribution of chromosomes in meiosis. In this example, the cell shown possesses three homologous pairs of chromosomes.

combinations of chromosomes in the gametes. In general, the number of possible combinations is 2^n , where n equals the number of homologous pairs. As the number of chromosome pairs increases, the number of combinations quickly becomes very large. In humans, who have 23 pairs of chromosomes, there are 8,388,608 different combinations of chromosomes possible from the random separation of homologous chromosomes. Through the random distribution of chromosomes in anaphase I, alleles located on different chromosomes are sorted into different combinations. The genetic consequences of this process, termed independent assortment, will be explored in more detail in Chapter 3.

In summary, crossing over shuffles alleles on the *same* homologous chromosomes into new combinations, whereas the random distribution of maternal and paternal chromosomes shuffles alleles on *different* chromosomes into new combinations. Together, these two processes are capable of producing tremendous amounts of genetic variation among the cells resulting from meiosis.

Concepts

Meiosis consists of two distinct divisions: meiosis I and meiosis II. Meiosis (usually) produces four haploid cells that are genetically variable. The two processes responsible for genetic variation are crossing over and the random distribution of maternal and paternal chromosomes.

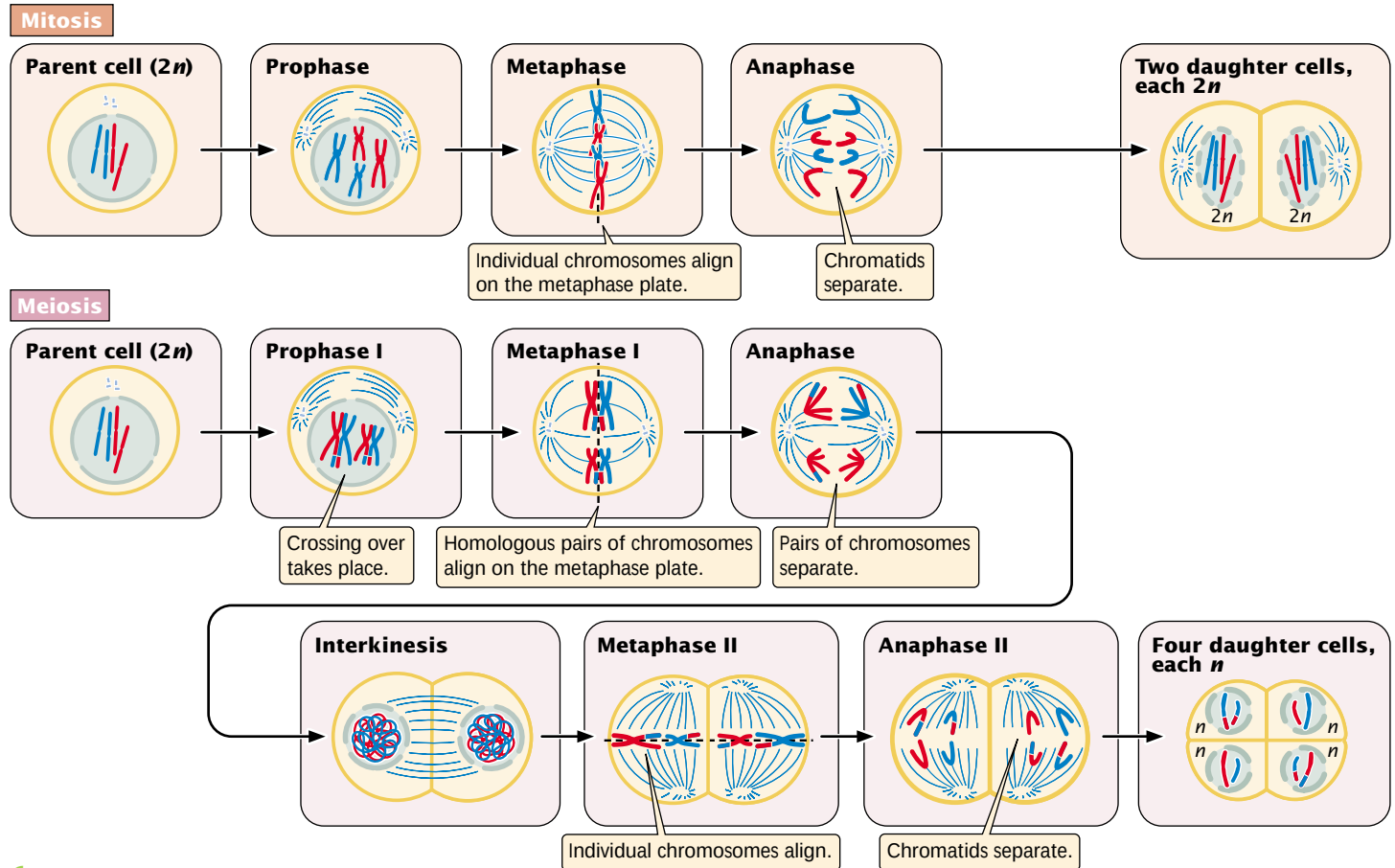
www.whfreeman.com/pierce

A tutorial and animations of meiosis

Connecting Concepts

Comparison of Mitosis and Meiosis

Now that we have examined the details of mitosis and meiosis, let's compare the two processes (FIGURE 2.19). In both mitosis and meiosis, the chromosomes contract and



2.19 Comparison of mitosis and meiosis (female, ♀; male, ♂).

become visible; both processes include the movement of chromosomes toward the spindle poles, and both are accompanied by cell division. Beyond these similarities, the processes are quite different.

Mitosis entails a single cell division and usually produces two daughter cells. Meiosis, in contrast, comprises two cell divisions and usually produces four cells. In diploid cells, homologous chromosomes are present before both meiosis and mitosis, but the pairing of homologs takes place only in meiosis.

Another difference is that, in meiosis, chromosome number is reduced by half in anaphase I, but no chromosome reduction takes place in mitosis. Furthermore, meiosis is characterized by two processes that produce genetic variation: crossing over (in prophase I) and the random distribution of maternal and paternal chromosomes (in anaphase I). There are normally no equivalent processes in mitosis.

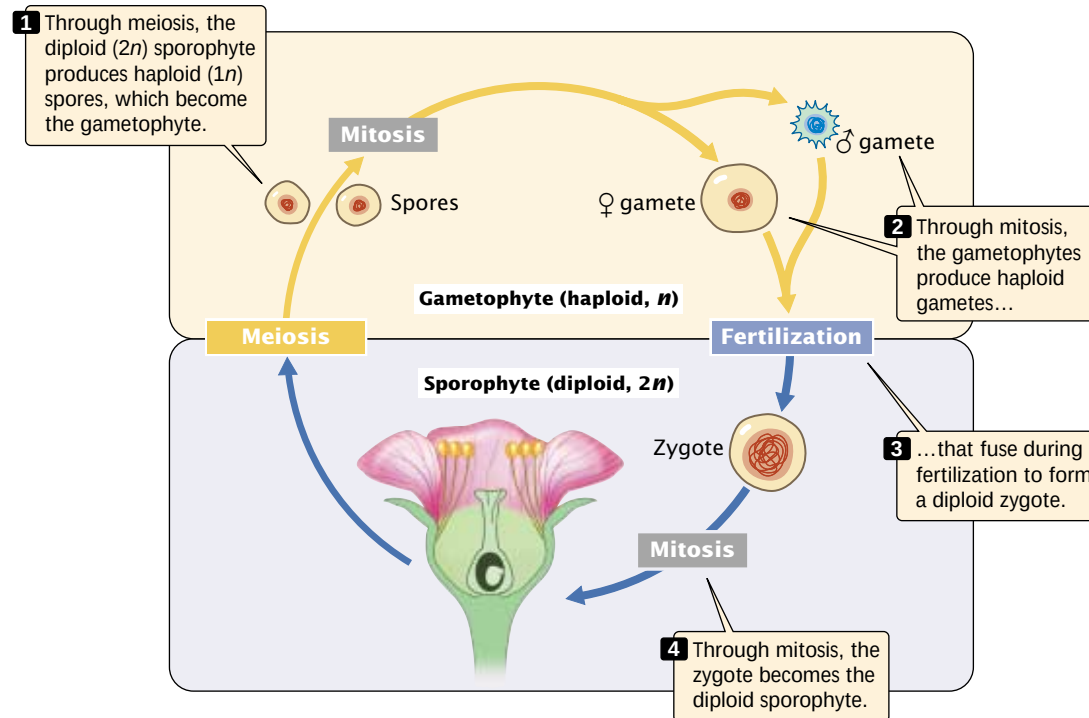
Mitosis and meiosis also differ in the behavior of chromosomes in metaphase and anaphase. In metaphase I of meiosis, *homologous pairs* of chromosomes line up on the metaphase plate, whereas *individual chromosomes* line up on the metaphase plate in metaphase of mitosis (and

metaphase II of meiosis). In anaphase I of meiosis, *paired chromosomes* separate, and each of the chromosomes that migrate toward a pole possesses two chromatids attached at the centromere. In contrast, in anaphase of mitosis (and anaphase II of meiosis), *chromatids* separate, and each chromosome that moves toward a spindle pole consists of a single chromatid.

Meiosis in the Life Cycle of Plants and Animals

The overall result of meiosis is four haploid cells that are genetically variable. Let's now see where meiosis fits into the life cycle of a multicellular plant and a multicellular animal.

Sexual reproduction in plants Most plants have a complex life cycle that includes two distinct generations (stages): the diploid *sporophyte* and the haploid *gametophyte*. These two stages alternate; the sporophyte produces haploid spores through meiosis, and the gametophyte produces haploid gametes through mitosis (FIGURE 2.20). This type of life cycle is sometimes called *alternation of generations*. In this cycle, the immediate products of meiosis



2.20 Plants alternate between diploid and haploid life stages.

are called spores, not gametes; the spores undergo one or more mitotic divisions to produce gametes. Although the terms used for this process are somewhat different from those commonly used in regard to animals (and from some of those employed so far in this chapter), the processes in plants and animals are basically the same: in both, meiosis leads to a reduction in chromosome number, producing haploid cells.

In flowering plants, the sporophyte is the obvious, vegetative part of the plant; the gametophyte consists of only a few haploid cells within the sporophyte. The flower, which is part of the sporophyte, contains the reproductive structures. In some plants, both male and female reproductive structures are found in the same flower; in other plants, they exist in different flowers. In either case, the male part of the flower, the stamen, contains diploid reproductive cells called **microsporocytes**, each of which undergoes meiosis to produce four haploid **microspores** (FIGURE 2.21a). Each microspore divides mitotically, producing an immature pollen grain consisting of two haploid nuclei. One of these nuclei, called the tube nucleus, directs the growth of a pollen tube. The other, termed the generative nucleus, divides mitotically to produce two sperm cells. The pollen grain, with its two haploid nuclei, is the male gametophyte.

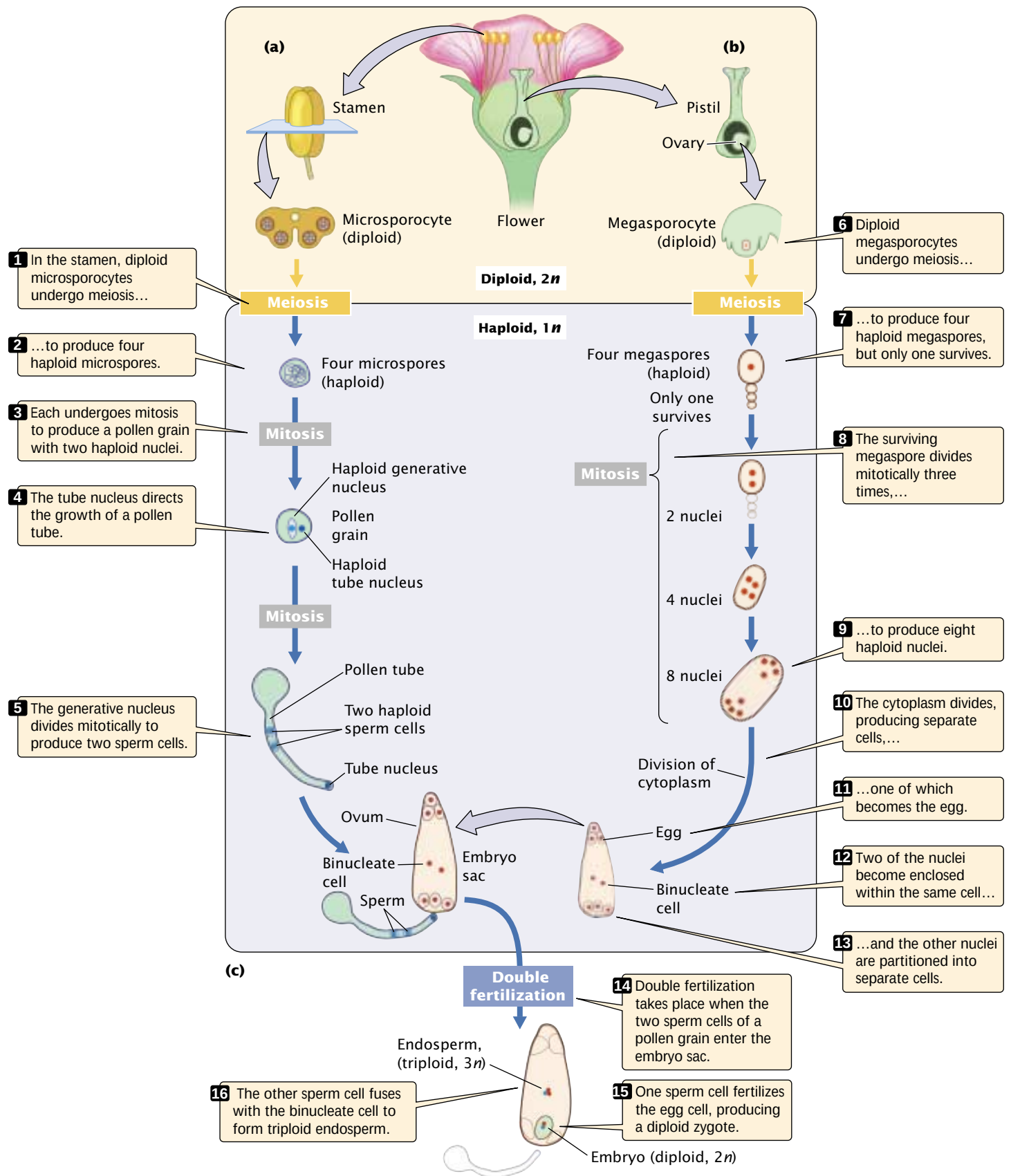
The female part of the flower, the ovary, contains diploid cells called **megaspocytes**, each of which undergoes meiosis to produce four haploid **megaspores** (FIGURE 2.21b), only one of which survives. The nucleus of the surviving

megaspore divides mitotically three times, producing a total of eight haploid nuclei that make up the female gametophyte, the embryo sac. Division of the cytoplasm then produces separate cells, one of which becomes the egg.

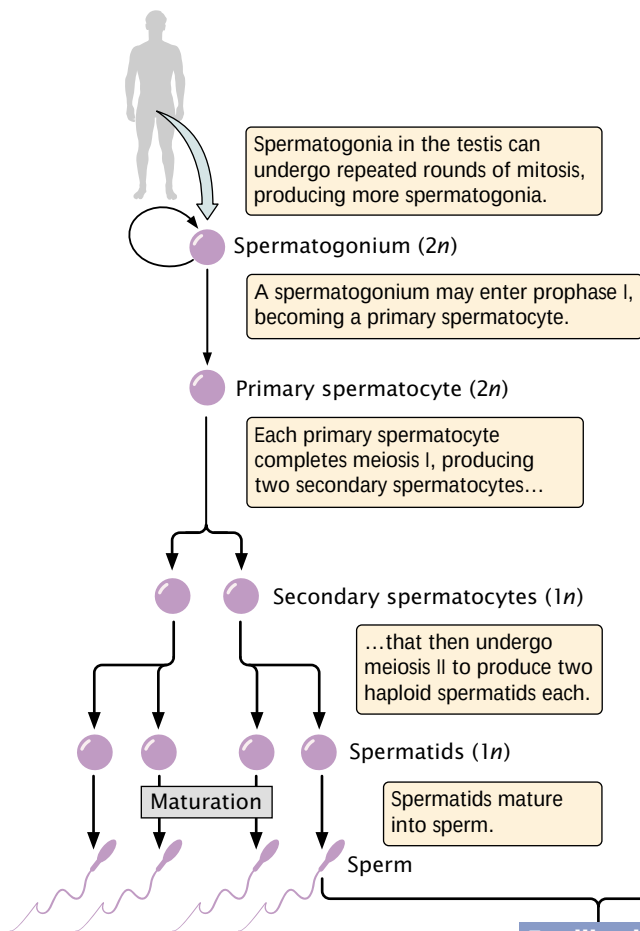
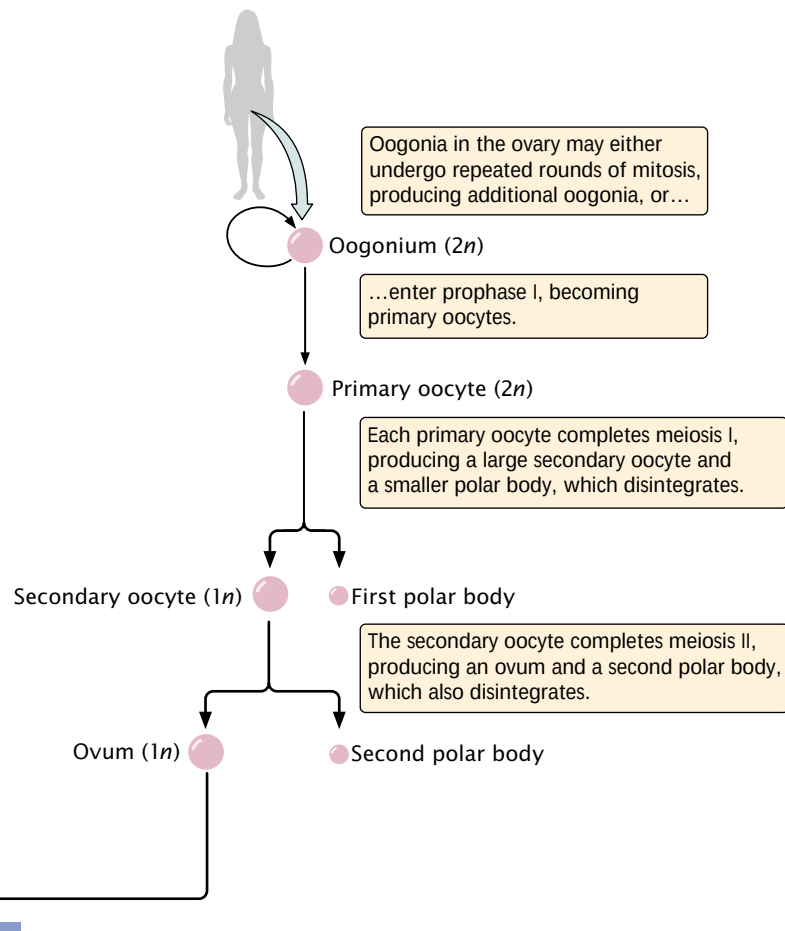
When the plant flowers, the stamens open and release pollen grains. Pollen lands on a flower's stigma—a sticky platform that sits on top of a long stalk called the style. At the base of the style is the ovary. If a pollen grain germinates, it grows a tube down the style into the ovary. The two sperm cells pass down this tube and enter the embryo sac (FIGURE 2.21c). One of the sperm cells fertilizes the egg cell, producing a diploid zygote, which develops into an embryo. The other sperm cell fuses with two nuclei enclosed in a single cell, giving rise to a $3n$ (triploid) endosperm, which stores food that will be used later by the embryonic plant. These two fertilization events are termed *double fertilization*.

Concepts

In the stamen of a flowering plant, meiosis produces haploid microspores that divide mitotically to produce haploid sperm in a pollen grain. Within the ovary, meiosis produces four haploid megaspores, only one of which divides mitotically three times to produce eight haploid nuclei. During pollination, one sperm fertilizes the egg cell, producing a diploid zygote; the other fuses with two nuclei to form the endosperm.



2.21 Sexual reproduction in flowering plants.

(a) Male gametogenesis (spermatogenesis)**(b) Female gametogenesis (oogenesis)****Fertilization**

Zygote (2n)

A sperm and ovum fuse at fertilization to produce a diploid zygote.

2.22 Gamete formation in animals.

Meiosis in animals The production of gametes in a male animal (**spermatogenesis**) takes place in the testes. There, diploid primordial germ cells divide mitotically to produce diploid cells called **spermatogonia** (FIGURE 2.22a). Each spermatogonium can undergo repeated rounds of mitosis, giving rise to numerous additional spermatogonia. Alternatively, a spermatogonium can initiate meiosis and enter into prophase I. Now called a **primary spermatocyte**, the cell is still diploid because the homologous chromosomes have not yet separated. Each primary spermatocyte completes meiosis I, giving rise to two haploid **secondary spermatocytes** that then undergo meiosis II, with each producing two haploid **spermatids**. Thus, each primary spermatocyte produces a total of four haploid spermatids, which mature and develop into sperm.

The production of gametes in the female (**oogenesis**) begins much like spermatogenesis. Diploid primordial germ cells within the ovary divide mitotically to produce **oogonia**

(FIGURE 2.22b). Like spermatogonia, oogonia can undergo repeated rounds of mitosis or they can enter into meiosis. Once in prophase I, these still-diploid cells are called **primary oocytes**. Each primary oocyte completes meiosis I and divides.

Here the process of oogenesis begins to differ from that of spermatogenesis. In oogenesis, cytokinesis is unequal: most of the cytoplasm is allocated to one of the two haploid cells, the **secondary oocyte**. The smaller cell, which contains half of the chromosomes but only a small part of the cytoplasm, is called the **first polar body**; it may or may not divide further. The secondary oocyte completes meiosis II, and again cytokinesis is unequal—most of the cytoplasm passes into one of the cells. The larger cell, which acquires most of the cytoplasm, is the **ovum**, the mature female gamete. The smaller cell is the **second polar body**. Only the ovum is capable of being fertilized, and the polar bodies usually disintegrate. Oogenesis, then, produces a single mature gamete from each primary oocyte.

We have now examined the place of meiosis in the sexual cycle of two organisms, a flowering plant and a typical multicellular animal. These cycles are just two of the many variations found among eukaryotic organisms. Although the cellular events that produce reproductive cells in plants and animals differ in the number of cell divisions, the number of haploid gametes produced, and the relative size of the final products, the overall result is the same: meiosis gives rise to haploid, genetically variable cells that then fuse during fertilization to produce diploid progeny.

Concepts

In the testes, a diploid spermatogonium undergoes meiosis, producing a total of four haploid sperm cells. In the ovary, a diploid oogonium undergoes meiosis to produce a single large ovum and smaller polar bodies that normally disintegrate.

Connecting Concepts Across Chapters

This chapter focused on the processes that bring about cell reproduction, the starting point of all genetics. We have examined four major concepts: (1) the differences that exist in the organization and packaging of genetic material

in prokaryotic and eukaryotic cells; (2) the cell cycle and its genetic results; (3) meiosis, its genetic results, and how it differs from mitosis of the cell cycle; and (4) how meiosis fits into the reproductive cycles of plants and animals.

Several of the concepts presented in this chapter serve as an important foundation for topics in other chapters of this book. The fundamental differences in the organization of genetic material of prokaryotes and eukaryotes are important to keep in mind as we explore the molecular functioning of DNA. The presence of histone proteins in eukaryotes affects the way that DNA is copied (Chapter 12) and read (Chapter 13). The direct contact between DNA and cytoplasmic organelles in prokaryotes and the separation of DNA by the nuclear envelope in eukaryotes have important implications for gene regulation (Chapter 16) and the way that gene products are modified before they are translated into proteins (Chapter 14). The smaller amount of DNA per cell in prokaryotes also affects the organization of genes on chromosomes (Chapter 11).

A critical concept in this chapter is meiosis, which serves as the cellular basis of genetic crosses in most eukaryotic organisms. It is the basis for the rules of inheritance presented in Chapters 3 through 6 and provides a foundation for almost all of the remaining chapters of this book.

CONCEPTS SUMMARY

- A prokaryotic cell possesses a simple structure, with no nuclear envelope and usually a single, circular chromosome. A eukaryotic cell possesses a more complex structure, with a nucleus and multiple linear chromosomes consisting of DNA complexed to histone proteins.
- Cell reproduction requires the copying of the genetic material, separation of the copies, and cell division.
- In a prokaryotic cell, the single chromosome replicates, and each copy attaches to the plasma membrane; growth of the plasma membrane separates the two copies, which is followed by cell division.
- In eukaryotic cells, reproduction is more complex than in prokaryotic cells, requiring mitosis and meiosis to ensure that a complete set of genetic information is transferred to each new cell.
- In eukaryotic cells, chromosomes are typically found in homologous pairs.
- Each functional chromosome consists of a centromere, a telomere, and multiple origins of replication. Centromeres are the points at which kinetochores assemble and to which microtubules attach. Telomeres are the stable ends of chromosomes. After a chromosome is copied, the two copies remain attached at the centromere, forming sister chromatids.
- The cell cycle consists of the stages through which a eukaryotic cell passes between cell divisions. It consists of: (1) interphase, in which the cell grows and prepares for division and (2) M phase, in which nuclear and cell division take place. M phase consists of mitosis, the process of nuclear division, and cytokinesis, the division of the cytoplasm.
- Interphase begins with G_1 , in which the cell grows and synthesizes proteins necessary for cell division, followed by S phase, during which the cell's DNA is replicated. The cell then enters G_2 , in which additional biochemical events necessary for cell division take place. Some cells exit G_1 and enter a nondividing state called G_0 .
- M phase consists of prophase, prometaphase, metaphase, anaphase, telophase, and cytokinesis. In these stages, the chromosomes contract, the nuclear membrane breaks down, and the spindle forms. The chromosomes line up in the center of the cell. Sister chromatids separate and become independent chromosomes, which then migrate to opposite ends of the cell. The nuclear membrane reforms around chromosomes at each end of the cell, and the cytoplasm divides.
- The usual result of mitosis is the production of two genetically identical cells.

- Progression through the cell cycle is controlled by interactions between cyclins and cyclin-dependent kinases.
- Sexual reproduction produces genetically variable progeny and allows for accelerated evolution. It includes meiosis, in which haploid sex cells are produced, and fertilization, the fusion of sex cells. Meiosis includes two cell divisions. In meiosis I, crossing over occurs and homologous chromosomes separate. In meiosis II, chromatids separate.
- The usual result of meiosis is the production of four haploid cells that are genetically variable.
- Genetic variation in meiosis is produced by crossing over and by the random distribution of maternal and paternal chromosomes.
- In plants, diploid microsporocytes in the stamens undergo meiosis, each microsporocyte producing four haploid microspores. Each microspore divides mitotically to produce two haploid sperm cells. In the ovary, diploid megasporocytes

undergo meiosis, each megasporocyte producing four haploid megaspores, only one of which survives. The surviving megaspore divides mitotically three times to produce eight haploid nuclei, one of which forms the egg. During pollination, one sperm fertilizes the egg cell and the other fuses with two haploid nuclei to form a $3n$ endosperm.

- In animals, diploid spermatogonia initiate meiosis and become diploid primary spermatocytes, which then complete meiosis I, producing two haploid secondary spermatocytes. Each secondary spermatocyte undergoes meiosis II, producing a total of four haploid sperm cells from each primary spermatocyte. Diploid oogonia in the ovary enter meiosis and become diploid primary oocytes, each of which then completes meiosis I, producing one large haploid secondary oocyte and one small haploid polar body. The secondary oocyte completes meiosis II to produce a large haploid ovum and a smaller second polar body.

IMPORTANT TERMS

genome (p. 16)
prokaryote (p. 17)
eukaryote (p. 17)
eubacteria (p. 18)
archaea (p. 18)
nucleus (p. 18)
histone (p. 18)
chromatin (p. 18)
homologous pair (p. 20)
diploid (p. 20)
haploid (p. 20)
telomere (p. 21)
origin of replication (p. 21)
sister chromatid (p. 22)

cell cycle (p. 22)
interphase (p. 22)
M phase (p. 22)
mitosis (p. 22)
cytokinesis (p. 22)
prophase (p. 23)
prometaphase (p. 23)
metaphase (p. 23)
anaphase (p. 23)
telophase (p. 23)
meiosis (p. 29)
fertilization (p. 29)
prophase I (p. 29)
synapsis (p. 29)

bivalent (p. 29)
tetrad (p. 29)
crossing over (p. 29)
metaphase I (p. 30)
anaphase I (p. 30)
telophase I (p. 30)
interkinesis (p. 30)
prophase II (p. 30)
metaphase II (p. 31)
anaphase II (p. 31)
telophase II (p. 31)
recombination (p. 33)
microsporocyte (p. 36)
microspore (p. 36)

megasporocyte (p. 36)
megaspore (p. 36)
spermatogenesis (p. 38)
spermatogonium (p. 38)
primary spermatocyte (p. 38)
secondary spermatocyte (p. 38)
spermatid (p. 38)
oogenesis (p. 38)
oogonium (p. 38)
primary oocyte (p. 38)
secondary oocyte (p. 38)
first polar body (p. 38)
ovum (p. 38)
second polar body (p. 38)

Worked Problems

1. A student examines a thin section of an onion root tip and records the number of cells that are in each stage of the cell cycle. She observes 94 cells in interphase, 14 cells in prophase, 3 cells in prometaphase, 3 cells in metaphase, 5 cells in anaphase, and 1 cell in telophase. If the complete cell cycle in an onion root tip requires 22 hours, what is the average duration of each stage in the cycle? Assume that all cells are in active cell cycle (not G_0).

• Solution

This problem is solved in two steps. First, we calculate the proportions of cells in each stage of the cell cycle, which correspond to the amount of time that an average cell spends in each stage. For example, if cells spend 90% of their time in interphase, then, at any given moment, 90% of the cells will be in interphase. The second step is to convert the proportions into

lengths of time, which is done by multiplying the proportions by the total time of the cell cycle (22 hours).

Step 1. Calculate the proportion of cells at each stage. The proportion of cells at each stage is equal to the number of cells found in that stage divided by the total number of cells examined:

Interphase	$\frac{94}{120} = 0.783$
Prophase	$\frac{14}{120} = 0.117$
Prometaphase	$\frac{3}{120} = 0.025$
Metaphase	$\frac{3}{120} = 0.025$
Anaphase	$\frac{5}{120} = 0.042$
Telophase	$\frac{1}{120} = 0.008$

We can check our calculations by making sure that the proportions sum to 1.0, which they do.

Step 2. Determine the average duration of each stage. To determine the average duration of each stage, multiply the proportion of cells in each stage by the time required for the entire cell cycle:

Interphase	$0.783 \times 22 \text{ hours} = 17.23 \text{ hours}$
Prophase	$0.117 \times 22 \text{ hours} = 2.57 \text{ hours}$
Prometaphase	$0.025 \times 22 \text{ hours} = 0.55 \text{ hour}$
Metaphase	$0.025 \times 22 \text{ hours} = 0.55 \text{ hour}$
Anaphase	$0.042 \times 22 \text{ hours} = 0.92 \text{ hour}$
Telophase	$0.008 \times 22 \text{ hours} = 0.18 \text{ hour}$

2. A cell in G_1 of interphase has 8 chromosomes. How many chromosomes and how many DNA molecules will be found per cell as this cell progresses through the following stages: G_2 , metaphase of mitosis, anaphase of mitosis, after cytokinesis in mitosis, metaphase I of meiosis, metaphase II of meiosis, and after cytokinesis of meiosis II?

• **Solution**

Remember the rules that we learned about counting chromosomes and DNA molecules: (1) to determine the number of chromosomes, count the functional centromeres; and (2) to determine the number of DNA molecules, count the chromatids. Think carefully about when and how the numbers of chromosomes and DNA molecules change in the course of mitosis and meiosis.

The number of DNA molecules increases only in S phase, when DNA replicates; the number of DNA molecules decreases only when the cell divides. Chromosome number increases only when sister chromatids separate in anaphase of mitosis and anaphase II of meiosis (homologous chromosomes, not chromatids, separate in anaphase I of meiosis). Chromosome number, like the number of DNA molecules, is reduced only by cell division.

Let us now apply these principles to the problem. A cell in G_1 has 8 chromosomes, each consisting of a single chromatid; so 8 DNA molecules are present in G_1 . DNA replicates in S stage; so, in G_2 , 16 DNA molecules are present per cell. However, the two copies of each DNA molecule remain attached at the centromere; so there are still only 8 chromosomes present. As the cell passes through prophase and metaphase of the cell cycle, the number of chromosomes and DNA molecules remains the same; so, at metaphase, there are 16 DNA molecules and 8 chromosomes. In anaphase, the chromatids separate and each becomes an independent chromosome; at this point, the number

of chromosomes increases from 8 to 16. This increase is temporary, lasting only until the cell divides in telophase or subsequent to it. The number of DNA molecules remains at 16 in anaphase. The number of DNA molecules and chromosomes per cell is reduced by cytokinesis after telophase, because the 16 chromosomes and DNA molecules are now distributed between two cells. Therefore, after cytokinesis, each cell has 8 DNA molecules and 8 chromosomes, the same numbers that were present at the beginning of the cell cycle.

Now, let's trace the numbers of DNA molecules and chromosomes through meiosis. At G_1 , there are 8 chromosomes and 8 DNA molecules. The number of DNA molecules increases to 16 in S stage, but the number of chromosomes remains at 8 (each chromosome has two chromatids). The cell therefore enters metaphase I with 16 DNA molecules and 8 chromosomes. In anaphase I of meiosis, homologous chromosomes separate, but the number of chromosomes remains at 8. After cytokinesis, the original 8 chromosomes are distributed between two cells; so the number of chromosomes per cell falls to 4 (each with two chromatids). The original 16 DNA molecules also are distributed between two cells; so the number of DNA molecules per cell is 8. There is no DNA synthesis during interkinesis, and each cell still maintains 4 chromosomes and 8 DNA molecules through metaphase II. In anaphase II, the two chromatids of each chromosome separate, temporarily raising the number of chromosomes per cell to 8, whereas the number of DNA molecules per cell remains at 8. After cytokinesis, the chromosomes and DNA molecules are again distributed between two cells, providing 4 chromosomes and 4 DNA molecules per cell. These results are summarized in the following table:

Stage	Number of chromosomes per cell	Number of DNA molecules per cell
G_1	8	8
G_2	8	16
Metaphase of mitosis	8	16
Anaphase of mitosis	16	16
After cytokinesis of mitosis	8	8
Metaphase I of meiosis	8	16
Metaphase II of meiosis	4	8
After cytokinesis of meiosis II	4	4

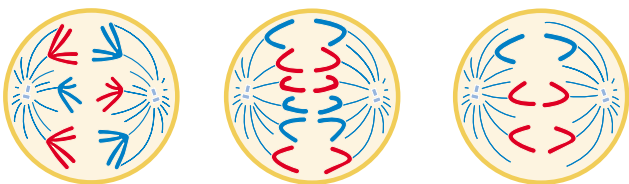
COMPREHENSION QUESTIONS

1. All organisms have the same universal genetic system. What are the implications of this universal genetic system?
2. Why are the viruses that infect mammalian cells useful for studying the genetics of mammals?
- * 3. List three fundamental events that must take place in cell reproduction.
4. Outline the process by which prokaryotic cells reproduce.
5. Name three essential structural elements of a functional eukaryotic chromosome and describe their functions.
- * 6. Sketch and label four different types of chromosomes based on the position of the centromere.
7. List the stages of interphase and the major events that take place in each stage.
- * 8. List the stages of mitosis and the major events that take place in each stage.

- * 9. What are the genetically important results of the cell cycle?
- 10. Why are the two cells produced by the cell cycle genetically identical?
- 11. What are checkpoints? What two general classes of compounds regulate progression through the cell cycles?
- 12. What are the stages of meiosis and what major events take place in each stage?
- *13. What are the major results of meiosis?
- 14. What two processes unique to meiosis are responsible for genetic variation? At what point in meiosis do these processes take place?
- *15. List similarities and differences between mitosis and meiosis. Which differences do you think are most important and why?
- 16. Outline the process by which male gametes are produced in plants. Outline the process of female gamete formation in plants.
- 17. Outline the process of spermatogenesis in animals. Outline the process of oogenesis in animals.

APPLICATION QUESTIONS AND PROBLEMS

- 18. A certain species has three pairs of chromosomes: an acrocentric pair, a metacentric pair, and a submetacentric pair. Draw a cell of this species as it would appear in metaphase of mitosis.
- 19. A biologist examines a series of cells and counts 160 cells in interphase, 20 cells in prophase, 6 cells in prometaphase, 2 cells in metaphase, 7 cells in anaphase, and 5 cells in telophase. If the complete cell cycle requires 24 hours, what is the average duration of M phase in these cells? Of metaphase?
- *20. A cell in G₁ of interphase has 12 chromosomes. How many chromosomes and DNA molecules will be found per cell when this original cell progresses to the following stages?
 - (a) G₂ of interphase
 - (b) Metaphase I of meiosis
 - (c) Prophase of mitosis
 - (d) Anaphase I of meiosis
 - (e) Anaphase II of meiosis
 - (f) Prophase II of meiosis
 - (g) After cytokinesis following mitosis
 - (h) After cytokinesis following meiosis II
- *21. All of the following cells, shown in various stages of mitosis and meiosis, come from the same rare species of plant. What is the diploid number of chromosomes in this plant? Give the names of each stage of mitosis or meiosis shown.



- (a) G₂
- (b) Anaphase of mitosis
- (c) Prophase II of meiosis
- (d) After cytokinesis associated with meiosis II
- 23. A cell in prophase II of meiosis contains 12 chromosomes. How many chromosomes would be present in a cell from the same organism if it were in prophase of mitosis? Prophase I of meiosis?
- *24. The fruit fly *Drosophila melanogaster* has four pairs of chromosomes, whereas the house fly *Musca domestica* has six pairs of chromosomes. Other things being equal, in which species would you expect to see more genetic variation among the progeny of a cross? Explain your answer.
- *25. A cell has two pairs of submetacentric chromosomes, which we will call chromosomes I_a, I_b, II_a, and II_b (chromosomes I_a and I_b are homologs, and chromosomes II_a and II_b are homologs). Allele *M* is located on the long arm of chromosome I_a, and allele *m* is located at the same position on chromosome I_b. Allele *P* is located on the short arm of chromosome I_a, and allele *p* is located at the same position on chromosome I_b. Allele *R* is located on chromosome II_a and allele *r* is located at the same position on chromosome II_b.
 - (a) Draw these chromosomes, labeling genes *M*, *m*, *P*, *p*, *R*, and *r*, as they might appear in metaphase I of meiosis. Assume that there is no crossing over.
 - (b) Considering the random separation of chromosomes in anaphase I, draw the chromosomes (with labeled genes) present in all possible types of gametes that might result from this cell going through meiosis. Assume that there is no crossing over.
- 26. A horse has 64 chromosomes and a donkey has 62 chromosomes. A cross between a female horse and a male donkey produces a mule, which is usually sterile. How many chromosomes does a mule have? Can you think of any reasons for the fact that most mules are sterile?

CHALLENGE QUESTIONS

27. Suppose that life exists elsewhere in the universe. All life must contain some type of genetic information, but alien genomes might not consist of nucleic acids and have the same features as those found in the genomes of life on Earth. What do you think might be the common features of all genomes, no matter where they exist?
28. On average, what proportion of the genome in the following pairs of humans would be exactly the same if no crossing over occurred? (For the purposes of this question only, we will ignore the special case of the X and Y sex chromosomes and assume that all genes are located on nonsex chromosomes.)
- (a) Father and child
 - (b) Mother and child
 - (c) Two full siblings (offspring that have the same two biological parents)
 - (d) Half siblings (offspring that have only one biological parent in common)
 - (e) Uncle and niece
 - (f) Grandparent and grandchild
29. Females bees are diploid and male bees are haploid. The haploid males produce sperm and can successfully mate with diploid females. Fertilized eggs develop into females and unfertilized eggs develop into males. How do you think the process of sperm production in male bees differs from sperm production in other animals?
30. Rec8 is a protein that is found in yeast chromosome arms and centromeres. Rec8 persists throughout meiosis I but breaks down at anaphase II. When the gene that encodes Rec8 is deleted, sister chromatids separate in anaphase I.
- (a) From these observations, propose a mechanism for the role of Rec8 in meiosis that helps to explain why sister chromatids normally separate in anaphase II but not anaphase I.
 - (b) Make a prediction about the presence or absence of Rec8 during the various stages of mitosis.

SUGGESTED READINGS

- Hawley, R. S., and T. Arbel. 1993. Yeast genetics and the fall of the classical view of meiosis. *Cell* 72:301–303.
Contains information about where in meiosis crossing over takes place and the role of the synaptonemal complex in recombination.
- Jarrell, K. F., D. P. Bayley, J. D. Correia, and N. A. Thomas. 1999. Recent excitement about Archaea. *Bioscience* 49:530–541.
An excellent review of differences between eubacteria, archaea, and eukaryotes.
- King, R. W., P. K. Jackson, and M. W. Kirschner. 1994. Mitosis in transition. *Cell* 79:563–571.
A good review of how the cell cycle is controlled.
- Kirschner, M. 1992. The cell cycle then and now. *Trends in Biochemical Sciences* 17:281–285.
A good review of the history of research into control of the cell cycle.
- Koshland, D. 1994. Mitosis: back to basics. *Cell* 77:951–954.
Reviews research on mitosis and chromosome movement.
- McIntosh, J. R., and M. P. Koonce. 1989. Mitosis. *Science* 246:622–628.
A review of the process of mitosis.
- McIntosh, J. R., and K. L. McDonald. 1989. The mitotic spindle. *Scientific American* 261(4):48–56.
Review of the mitotic spindle.
- McIntosh, J. R., and C. M. Pfarr. 1991. Mini-review: mitotic motors. *Journal of Cell Biology* 115:577–583.
Considers some of the experimental evidence concerning the role of molecular motors in the organization of the spindle and in chromosome movement.
- McKim, K. S., and R. S. Hawley. 1995. Chromosomal control of meiotic cell division. *Science* 270:1595–1601.
Reviews evidence that chromosomes actively take part in their own movement and in controlling the cell cycle.
- Morgan, D. O. 1995. Principles of CDK regulation. *Nature* 34:131–134.
An excellent short review of cell-cycle control.
- Nasmyth, K. 1999. Separating sister chromatids. *Trends in Biochemical Sciences* 24:98–103.
Considers the role of cohesion in the separation of sister chromatids.
- Pennisi, E. 1998. Cell division gatekeepers identified. *Science* 279:477–478.
Short review of work on the role of kinetochores in chromosome separation.

Pluta, A. F., A. M. MacKay, A. M. Ainsztein, I. G. Goldberg, and W. C. Earnshaw. 1995. Centromere: the hub of chromosome activities. *Science* 270:1591–1594.

An excellent review of centromere structure and function.

Rothfield, L., S. Justice, and J. Garcia-Lara. 1999. Bacterial cell division. *Annual Review of Genetics* 33:423–428.

Comprehensive review of how bacterial cells divide.

Uhlmann, F., F. Lottespeich, and K. Nasmyth. 1999. Sister-chromatid separation at anaphase onset is

promoted by cleavage of the cohesion subunit Scc1. *Nature* 400:37–42.

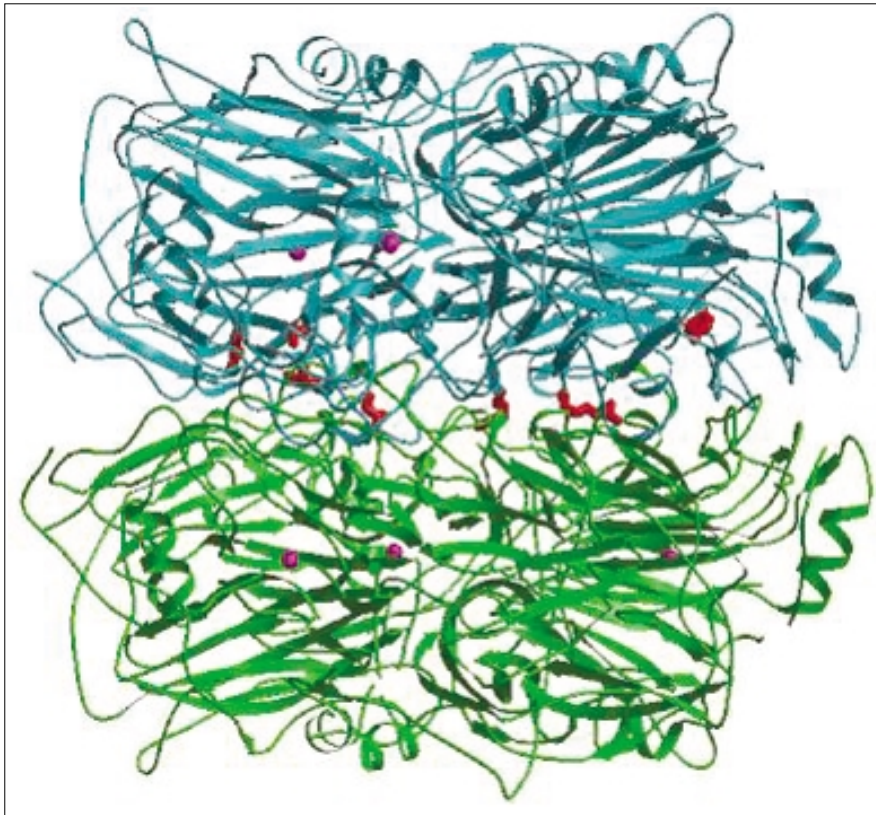
Report that cleavage of cohesion protein has a role in chromatid separation.

Zickler, D., and N. Kleckner. 1999. Meiotic chromosomes: integrating structure and function. *Annual Review of Genetics* 33:603–754.

A review of chromosomes in meiosis, their structure and function.

3

Basic Principles of Heredity



Alkaptonuria results from impaired function of homogentisate dioxygenase (shown here), an enzyme required for catabolism of the amino acids phenylalanine and tyrosine. (Courtesy of David E. Timm, Department of Molecular Biology, Indiana School of Medicine, and Miguel Penalva, Centro de Investigaciones. Biológicas CSIC, Madrid, Spain.)

Black Urine and First Cousins

Voiding black urine is a rare and peculiar trait. In 1902, Archibald Garrod discovered the hereditary basis of black urine and, in the process, contributed to our understanding of the nature of genes.

Garrod was an English physician who was more interested in chemical explanations of disease than in the practice of medicine. He became intrigued by several of his patients who produced black urine, a condition known as alkaptonuria. The urine of alkaptonurics contains homogentisic acid, a compound that, on exposure to air, oxidizes and

turns the urine black. Garrod observed that alkaptonuria appears at birth and remains for life. He noted that often several children in the same family were affected: of the 32 cases that he knew about, 19 appeared in only seven families. Furthermore, the parents of these alkaptonurics were frequently first cousins. With the assistance of geneticist William Bateson, Garrod recognized that this pattern of inheritance is precisely the pattern produced by the transmission of a rare, recessive gene.

Garrod later proposed that several other human disorders, including albinism and cystinuria, are inherited in the same way as alkaptonuria. He concluded that each gene

- Black Urine and First Cousins
- Mendel: The Father of Genetics
 - Mendel's Success
 - Genetic Terminology
- Monohybrid Crosses
 - What Monohybrid Crosses Reveal
 - Predicting The Outcomes of Genetic Crosses
 - The Testcross
 - Incomplete Dominance
 - Genetic Symbols
- Multiple-Loci Crosses
 - Dihybrid Crosses
 - The Principle of Independent Assortment
 - The Relationship of the Principle of Independent Assortment to Meiosis
 - Applying Probability and the Branch Diagram to Dihybrid Crosses
 - The Dihybrid Testcross
 - Trihybrid Crosses
- Observed and Expected Ratios
 - The Goodness of Fit Chi-square Test
 - Penetrance and Expressivity

encodes an enzyme that controls a biochemical reaction. When there is a flaw in a gene, its enzyme is deficient, resulting in a biochemical disorder. He called these flaws “inborn errors of metabolism.” Garrod was the first to apply the basic principles of genetics, which we will learn about in this chapter, to the inheritance of a human disease. His idea—that genes code for enzymes—was revolutionary and correct. Unfortunately, Garrod’s ideas were not recognized as being important at the time and were appreciated only after they had been rediscovered 30 years later.

This chapter is about the principles of heredity: how genes are passed from generation to generation. These principles were first put forth by Gregor Mendel, so we begin by examining his scientific achievements. We then turn to simple genetic crosses, those in which a single characteristic is examined. We learn some techniques for predicting the outcome of genetic crosses and then turn to crosses in which two or more characteristics are examined. We will see how the principles applied to simple genetic crosses and the ratios of offspring that they produce serve as the key for understanding more complicated crosses. We end the chapter by considering statistical tests for analyzing crosses and factors that vary their outcome.

Throughout this chapter, a number of concepts are interwoven: Mendel’s principles of segregation and independent assortment, probability, and the behavior of chromosomes. These might at first appear to be unrelated, but they are actually different views of the same phenomenon, because the genes that undergo segregation and independent assortment are located on chromosomes. The principle aim of this chapter is to examine these different views and to clarify their relations.

www.whfreeman.com/pierce Archibald Garrod’s original paper on the genetics of alkaptonuria

Mendel: The Father of Genetics

In 1902, the basic principles of genetics, which Archibald Garrod successfully applied to the inheritance of alkaptonuria, had just become widely known among biologists. Surprisingly, these principles had been discovered some 35 years earlier by Johann Gregor Mendel (1822–1884).

Mendel was born in what is now part of the Czech Republic. Although his parents were simple farmers with little money, he was able to achieve a sound education and was admitted to the Augustinian monastery in Brno in September 1843. After graduating from seminary, Mendel was ordained a priest and appointed to a teaching position in a local school. He excelled at teaching, and the abbot of the monastery recommended him for further study at the University of Vienna, which he attended from 1851 to 1853. There, Mendel enrolled in the newly opened Physics Institute and took courses in mathematics, chemistry, entomology, paleontology, botany, and plant physiology. It was

probably here that Mendel acquired the scientific method, which he later applied so successfully to his genetics experiments. After 2 years of study in Vienna, Mendel returned to Brno, where he taught school and began his experimental work with pea plants. He conducted breeding experiments from 1856 to 1863 and presented his results publicly at meetings of the Brno Natural Science Society in 1865. Mendel’s paper from these lectures was published in 1866. In spite of widespread interest in heredity, the effect of his research on the scientific community was minimal. At the time, no one seems to have noticed that Mendel had discovered the basic principles of inheritance.

In 1868, Mendel was elected abbot of his monastery, and increasing administrative duties brought an end to his teaching and eventually to his genetics experiments. He died at the age of 61 on January 6, 1884, unrecognized for his contribution to genetics.

The significance of Mendel’s discovery was unappreciated until 1900, when three botanists—Hugo de Vries, Erich von Tschermak, and Carl Correns—began independently conducting similar experiments with plants and arrived at conclusions similar to those of Mendel. Coming across Mendel’s paper, they interpreted their results in terms of his principles and drew attention to his pioneering work.

Concepts

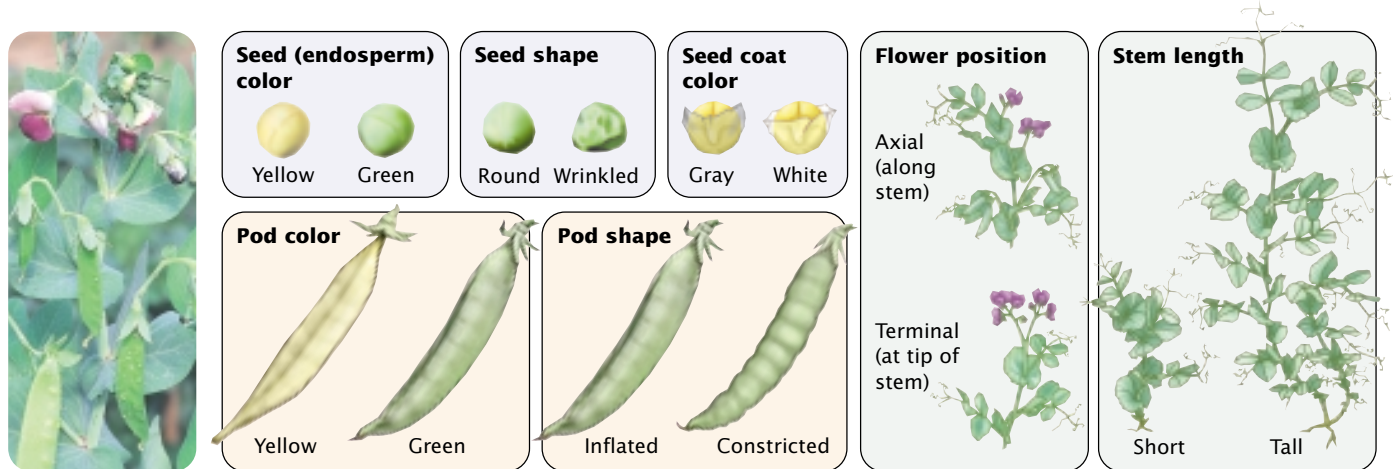
Gregor Mendel put forth the basic principles of inheritance, publishing his findings in 1866. The significance of his work did not become widely appreciated until 1900.



Mendel's Success

Mendel’s approach to the study of heredity was effective for several reasons. Foremost was his choice of experimental subject, the pea plant *Pisum sativum* (FIGURE 3.1), which offered clear advantages for genetic investigation. It is easy to cultivate, and Mendel had the monastery garden and greenhouse at his disposal. Peas grow relatively rapidly, completing an entire generation in a single growing season. By today’s standards, one generation per year seems frightfully slow—fruit flies complete a generation in 2 weeks and bacteria in 20 minutes—but Mendel was under no pressure to publish quickly and was able to follow the inheritance of individual characteristics for several generations. Had he chosen to work on an organism with a longer generation time—horses, for example—he might never have discovered the basis of inheritance. Pea plants also produce many offspring—their seeds—which allowed Mendel to detect meaningful mathematical ratios in the traits that he observed in the progeny.

The large number of varieties of peas that were available to Mendel was also crucial, because these varieties differed in various traits and were genetically pure. Mendel was therefore able to begin with plants of variable, known genetic makeup.



3.1 Mendel used the pea plant *Pisum sativum* in his studies of heredity. He examined seven characteristics that appeared in the seeds and in plants grown from the seeds. (Photo from Wally Eberhart/Visuals Unlimited.)

Much of Mendel's success can be attributed to the seven characteristics that he chose for study (see Figure 3.1). He avoided characteristics that display a range of variation; instead, he focused his attention on those that exist in two easily differentiated forms, such as white versus gray seed coats, round versus wrinkled seeds, and inflated versus constricted pods.

Finally, Mendel was successful because he adopted an experimental approach. Unlike many earlier investigators who just described the *results* of crosses, Mendel formulated *hypotheses* based on his initial observations and then conducted additional crosses to test his hypotheses. He kept careful records of the numbers of progeny possessing each type of trait and computed ratios of the different types. He paid close attention to detail, was adept at seeing patterns in detail, and was patient and thorough, conducting his experiments for 10 years before attempting to write up his results.

www.whfreeman.com/pierce Mendel's original paper (in German, with an English translation), as well as references, essays, and commentaries on Mendel's work

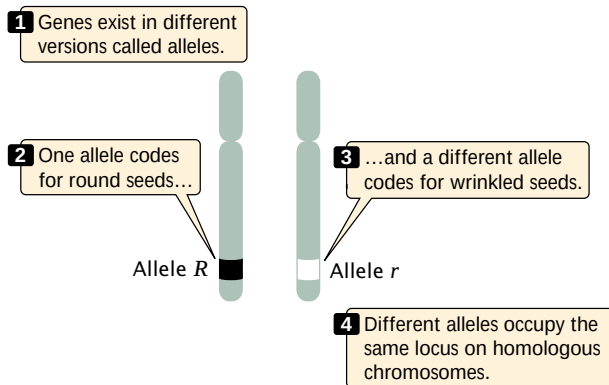
Genetic Terminology

Before we examine Mendel's crosses and the conclusions that he made from them, it will be helpful to review some terms commonly used in genetics (Table 3.1). The term *gene* was a word that Mendel never knew. It was not coined until 1909, when the Danish geneticist Wilhelm Johannsen first used it. The definition of a gene varies with the context of its use, and so its definition will change as we explore different aspects of heredity. For our present use in the context of genetic crosses, we will define a **gene** as an inherited factor that determines a characteristic.

Table 3.1 Summary of important genetic terms

Term	Definition
Gene	A genetic factor (region of DNA) that helps determine a characteristic
Allele	One of two or more alternate forms of a gene
Locus	Specific place on a chromosome occupied by an allele
Genotype	Set of alleles that an individual possesses
Heterozygote	An individual possessing two different alleles at a locus
Homozygote	An individual possessing two of the same alleles at a locus
Phenotype or trait	The appearance or manifestation of a character
Character or characteristic	An attribute or feature

Genes frequently come in different versions called **alleles** (FIGURE 3.2). In Mendel's crosses, seed shape was determined by a gene that exists as two different alleles: one allele codes for round seeds and the other codes for wrinkled seeds. All alleles for any particular gene will be found at a specific place on a chromosome called the **locus** for that gene. (The plural of locus is loci; it's bad form in genetics—and incorrect—to speak of locuses.) Thus, there is a specific place—a locus—on a chromosome in pea plants



3.2 At each locus, a diploid organism possesses two alleles located on different homologous chromosomes.

where the shape of seeds is determined. This locus might be occupied by an allele for round seeds or one for wrinkled seeds. We will use the term *allele* when referring to a specific version of a gene; we will use the term *gene* to refer more generally to any allele at a locus.

The **genotype** is the set of alleles that an individual organism possesses. A diploid organism that possesses two identical alleles is **homozygous** for that locus. One that possesses two different alleles is **heterozygous** for the locus.

Another important term is **phenotype**, which is the manifestation or appearance of a characteristic. A phenotype can refer to any type of characteristic: physical, physiological, biochemical, or behavioral. Thus, the condition of having round seeds is a phenotype, a body weight of 50 kg is a phenotype, and having sickle-cell anemia is a phenotype. In this book, the term *characteristic* or *character* refers to a general feature such as eye color; the term *trait* or *phenotype* refers to specific manifestations of that feature, such as blue or brown eyes.

A given phenotype arises from a genotype that develops within a particular environment. The genotype determines the potential for development; it sets certain limits, or boundaries, on that development. How the phenotype develops within those limits is determined by the effects of other genes and environmental factors, and the balance between these influences varies from character to character. For some characters, the differences between phenotypes are determined largely by differences in genotype; in other words, the genetic limits for that phenotype are narrow. Seed shape in Mendel's peas is a good example of a characteristic for which the genetic limits are narrow and the phenotypic differences are largely genetic. For other characters, environmental differences are more important; in this case, the limits imposed by the genotype are broad. The height that an oak tree reaches at maturity is a phenotype that is strongly influenced by environmental factors, such as the availability of water, sunlight, and nutrients. Nevertheless,

the tree's genotype still imposes some limits on its height: an oak tree will never grow to be 300 m tall no matter how much sunlight, water, and fertilizer are provided. Thus, even the height of an oak tree is determined to some degree by genes. For many characteristics, both genes and environment are important in determining phenotypic differences.

An obvious but important concept is that only the genotype is inherited. Although the phenotype is determined, at least to some extent, by genotype, organisms do not transmit their phenotypes to the next generation. The distinction between genotype and phenotype is one of the most important principles of modern genetics. The next section describes Mendel's careful observation of phenotypes through several generations of breeding experiments. These experiments allowed him to deduce not only the genotypes of the individual plants, but also the rules governing their inheritance.

Concepts

Each phenotype results from a genotype developing within a specific environment. The genotype, not the phenotype, is inherited.



Monohybrid Crosses

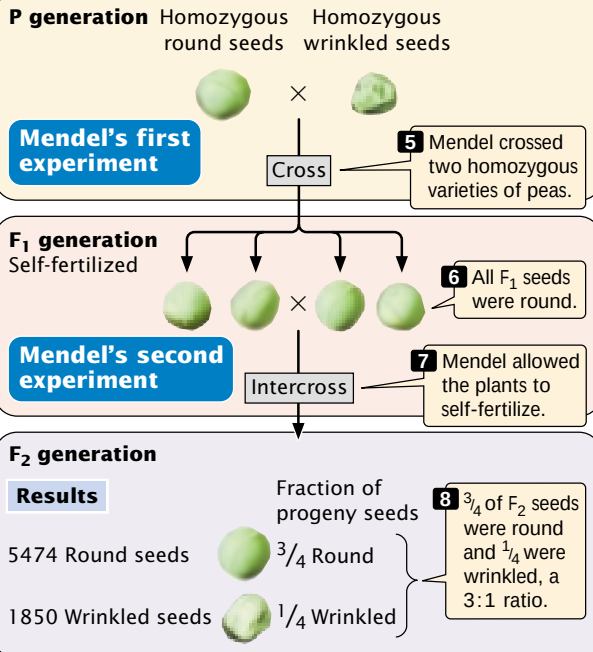
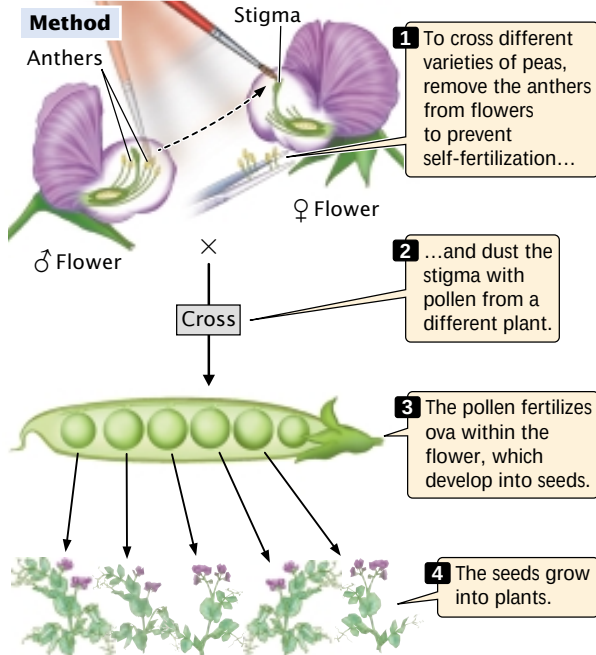
Mendel started with 34 varieties of peas and spent 2 years selecting those varieties that he would use in his experiments. He verified that each variety was genetically pure (homozygous for each of the traits that he chose to study) by growing the plants for two generations and confirming that all offspring were the same as their parents. He then carried out a number of crosses between the different varieties. Although peas are normally self-fertilizing (each plant crosses with itself), Mendel conducted crosses between different plants by opening the buds before the anthers were fully developed, removing the anthers, and then dusting the stigma with pollen from a different plant.

Mendel began by studying **monohybrid crosses**—those between parents that differed in a single characteristic. In one experiment, Mendel crossed a pea plant homozygous for round seeds with one that was homozygous for wrinkled seeds (FIGURE 3.3). This first generation of a cross is the **P (parental) generation**.

After crossing the two varieties in the P generation, Mendel observed the offspring that resulted from the cross. In regard to seed characteristics, such as seed shape, the phenotype develops as soon as the seed matures, because the seed traits are determined by the newly formed embryo within the seed. For characters associated with the plant itself, such as stem length, the phenotype doesn't develop until the plant grows from the seed; for these characters, Mendel had to wait until the following spring, plant the seeds, and then observe the phenotypes on the plants that germinated.

Experiment

Question: When peas with two different traits—round and wrinkled seeds—are crossed, will their progeny exhibit one of those traits, both of those traits, or a “blended” intermediate trait?

Method

Conclusion: The traits of the parent plants do not blend. Although F₁ plants display the phenotype of one parent, both traits are passed to F₂ progeny in a 3:1 ratio.

3.3 Mendel conducted monohybrid crosses.

The offspring from the parents in the P generation are the **F₁** (first filial) **generation**. When Mendel examined the F₁ of this cross, he found that they expressed only one of the phenotypes present in the parental generation: all the F₁ seeds were round. Mendel carried out 60 such crosses and always obtained this result. He also conducted **reciprocal crosses**: in one cross, pollen (the male gamete) was taken from a plant with round seeds and, in its reciprocal cross, pollen was taken from a plant with wrinkled seeds. Reciprocal crosses gave the same result: all the F₁ were round.

Mendel wasn't content with examining only the seeds arising from these monohybrid crosses. The following spring, he planted the F₁ seeds, cultivated the plants that germinated from them, and allowed the plants to self-fertilize, producing a second generation (the **F₂ generation**). Both of the traits from the P generation emerged in the F₂; Mendel counted 5474 round seeds and 1850 wrinkled seeds in the F₂ (see Figure 3.3). He noticed that the number of the round and wrinkled seeds constituted approximately a 3 to 1 ratio; that is, about $\frac{3}{4}$ of the F₂ seeds were round and $\frac{1}{4}$ were wrinkled. Mendel conducted monohybrid crosses for all seven of the characteristics that he studied in pea plants, and in all of the crosses he obtained the same result: all of the F₁ resembled only one of the two parents, but both parental traits emerged in the F₂ in approximately a 3:1 ratio.

What Monohybrid Crosses Reveal

Mendel drew several important conclusions from the results of his monohybrid crosses. First, he reasoned that, although the F₁ plants display the phenotype of only one parent, they must inherit genetic factors from both parents because they transmit both phenotypes to the F₂ generation. The presence of both round and wrinkled seeds in the F₂ could be explained only if the F₁ plants possessed both round and wrinkled genetic factors that they had inherited from the P generation. He concluded that each plant must therefore possess two genetic factors coding for a character.

The genetic factors that Mendel discovered (alleles) are, by convention, designated with letters; the allele for round seeds is usually represented by *R*, and the allele for wrinkled seeds by *r*. The plants in the P generation of Mendel's cross possessed two identical alleles: *RR* in the round-seeded parent and *rr* in the wrinkled-seeded parent (FIGURE 3.4a).

A second conclusion that Mendel drew from his monohybrid crosses was that the two alleles in each plant separate when gametes are formed, and one allele goes into each gamete. When two gametes (one from each parent) fuse to produce a zygote, the allele from the male parent unites with the allele from the female parent to produce the genotype of the offspring. Thus, Mendel's F₁ plants inherited an *R* allele from the round-seeded plant and an *r* allele from the wrinkled-seeded plant (FIGURE 3.4b). However, only the trait encoded by round allele (*R*) was observed in the F₁—all the F₁ progeny had round seeds. Those traits that appeared unchanged in the F₁ heterozygous offspring

The New Genetics

ETHICS • SCIENCE • TECHNOLOGY

Should Genetics Researchers Probe Abraham Lincoln's Genes?

by Art Caplan

Many people agree that no one should be forced to have a genetic test without his or her consent, yet for obvious reasons this ethical principle is difficult to follow when dealing with those who are deceased. There are all sorts of reasons why genetic testing on certain deceased persons might prove important, but one of the primary reasons is for purposes of identification. In anthropology, genetic analysis might help tell us whether we have found the body of a Romanov, Hitler, or Mengele. In cases of war or terrorist attacks, such as those on September 11, 2001, there might be no other way to determine the identity of a deceased person except by matching tissue samples with previously stored biological tissue or with samples from close relatives.

One historically interesting case, which highlights the ethical issues faced when determining genetic facts about the dead, is that which centers on Abraham Lincoln. Medical geneticists and advocates for patients with Marfan syndrome have long wondered whether President Lincoln had this particular genetic disease. After all, Lincoln had the tall gangly build often associated with Marfan's syndrome, which affects the connective tissues and cartilage of the body. Biographers and students of this man, whom many consider to be our greatest president, would like to know whether the depression that Lincoln suffered throughout his life might have been linked to the painful, arthritis-like symptoms of Marfan syndrome.

Lincoln was assassinated on April 14, 1865, and died early the next morning. An autopsy was performed, and samples of his hair, bone, and blood were preserved and stored at the National Museum of Health and Medicine; they are still there. The presence of a recently found genetic marker indicates whether someone has Marfan syndrome. With this

advancement, it would be possible to use some of the stored remains of Abe Lincoln to see if he had this condition. However, would it be ethical to perform this test?

We must be careful about genetic testing, because often too much weight is assigned to the results of such tests. There is a temptation to see DNA as the essence, the blueprint, of a person—that the factor that forms who we are and what we do. Given this tendency, should society be cautious about letting people explore the genes of the deceased? And, if we should not test without permission, then how can we obtain permission in cases where the person in question is dead? In Honest Abe's case, the "patient" is deceased and has no immediate survivors; there is no one to consent. But allowing testing without consent sets a dangerous precedent.



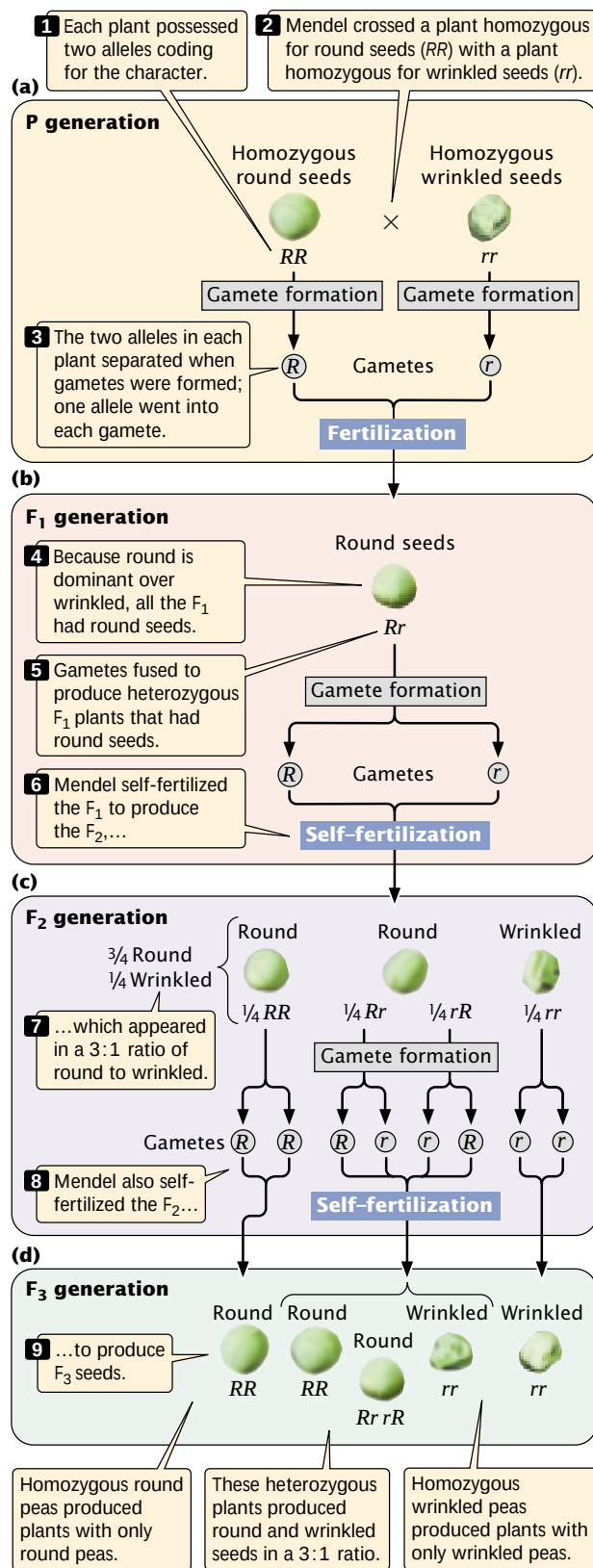
Abraham Lincoln had the tall, gangly build often associated with Marfan syndrome. (Cartoon by Frank Billew, 1864. Bettmann/Corbis.)

It may seem a bit strange to apply the notions of privacy and consent to the deceased. But, considering that most people today agree that consent should be obtained before these tests are administered, do researchers have the right to pry into Lincoln's DNA simply because neither he nor his descendants are around to say that they can't? Are we to say that anyone's body is open to examination whenever a genetic test becomes available that might tell us an interesting fact about that person's biological makeup?

Many prominent people from the past have taken special precautions to restrict access to their diaries, papers and letters; for instance, Sigmund Freud locked away his personal papers for 100 years. Will future Lincolns and Freuds need to embargo their mortal remains for eternity to prevent unwanted genetic snooping by subsequent generations?

And, when it comes right down to it, what is the point of establishing whether Lincoln had Marfan syndrome? After all, we don't need to inspect his genes to determine whether he was presidential timber—Marfan or no Marfan, he obviously was. The real questions to ask are, Do we adequately understand what he did as president and what he believed? How did his actions shape our country, and what can we learn from them that will benefit us today?

In the end, the genetic basis for Lincoln's behavior and leadership might be seen as having no relevance. Some would say that genetic testing might divert our attention from Lincoln's work, writings, thoughts, and deeds and, instead, require that we see him as a jumble of DNA output. Perhaps it makes more sense to encourage efforts to understand and appreciate Lincoln's legacy through his actions rather than through reconstituting and analyzing his DNA.



3.4 Mendel's monohybrid crosses revealed the principle of segregation and the concept of dominance.

Mendel called **dominant**, and those traits that disappeared in the F₁ heterozygous offspring he called **recessive**. When dominant and recessive alleles are present together, the recessive allele is masked, or suppressed. The concept of dominance was a third important conclusion that Mendel derived from his monohybrid crosses.

Mendel's fourth conclusion was that the two alleles of an individual plant separate with equal probability into the gametes. When plants of the F₁ (with genotype Rr) produced gametes, half of the gametes received the R allele for round seeds and half received the r allele for wrinkled seeds. The gametes then paired randomly to produce the following genotypes in equal proportions among the F₂: RR , Rr , rR , rr (FIGURE 3.4c). Because round (R) is dominant over wrinkled (r), there were three round progeny in the F₂ (RR , Rr , rR) for every one wrinkled progeny (rr) in the F₂. This 3:1 ratio of round to wrinkled progeny that Mendel observed in the F₂ could occur only if the two alleles of a genotype separated into the gametes with equal probability.

The conclusions that Mendel developed about inheritance from his monohybrid crosses have been further developed and formalized into the principle of segregation and the concept of dominance. The **principle of segregation** (Mendel's first law) states that each individual diploid organism possesses two alleles for any particular characteristic. These two alleles segregate (separate) when gametes are formed, and one allele goes into each gamete. Furthermore, the two alleles segregate into gametes in equal proportions. The **concept of dominance** states that, when two different alleles are present in a genotype, only the trait of the dominant allele is observed in the phenotype.

Mendel confirmed these principles by allowing his F₂ plants to self-fertilize and produce an F₃ generation. He found that the F₂ plants grown from the wrinkled seeds—those displaying the recessive trait (rr)—produced an F₃ in which all plants produced wrinkled seeds. Because his wrinkled-seeded plants were homozygous for wrinkled alleles (rr) they could pass on only wrinkled alleles to their progeny (FIGURE 3.4d).

The F₂ plants grown from round seeds—the dominant trait—fell into two types (Figure 3.4c). On self-fertilization, about $\frac{2}{3}$ of the F₂ plants produced both round and wrinkled seeds in the F₃ generation. These F₂ plants were heterozygous (Rr); so they produced $\frac{1}{4}$ RR (round), $\frac{1}{2}$ Rr (round), and $\frac{1}{4}$ rr (wrinkled) seeds, giving a 3:1 ratio of round to wrinkled in the F₃. About $\frac{1}{3}$ of the F₂ plants were of the second type; they produced only the dominant round-seeded trait in the F₃. These F₂ plants were homozygous for the round allele (RR) and thus could produce only round offspring in the F₃ generation. Mendel planted the seeds obtained in the F₃ and carried these plants through three more rounds of self-fertilization. In each generation, $\frac{2}{3}$ of the round-seeded plants produced round and wrinkled offspring, whereas $\frac{1}{3}$ produced only round offspring. These results are entirely consistent with the principle of segregation.

Concepts

The principle of segregation states that each individual organism possesses two alleles coding for a characteristic. These alleles segregate when gametes are formed, and one allele goes into each gamete. The concept of dominance states that, when dominant and recessive alleles are present together, only the trait of the dominant allele is observed.



Connecting Concepts

Relating Genetic Crosses to Meiosis

We have now seen how the results of monohybrid crosses are explained by Mendel's principle of segregation. Many students find that they enjoy working genetic crosses but are frustrated by the abstract nature of the symbols. Perhaps you feel the same at this point. You may be asking "What do these symbols really represent? What does the genotype RR mean in regard to the biology of the organism?" The answers to these questions lie in relating the abstract symbols of crosses to the structure and behavior of chromosomes, the repositories of genetic information (Chapter 2).

In 1900, when Mendel's work was rediscovered and biologists began to apply his principles of heredity, the relation between genes and chromosomes was still unclear. The theory that genes are located on chromosomes (the **chromosome theory of heredity**) was developed in the early 1900s by Walter Sutton, then a graduate student at Columbia University. Through the careful study of meiosis in insects, Sutton documented the fact that each homologous pair of chromosomes consists of one maternal chromosome and one paternal chromosome. Showing that these pairs segregate independently into gametes in meiosis, he concluded that this process is the biological basis for Mendel's principles of heredity. The German cytologist and embryologist Theodor Boveri came to similar conclusions at about the same time.

Sutton knew that diploid cells have two sets of chromosomes. Each chromosome has a pairing partner, its homologous chromosome. One chromosome of each homologous pair is inherited from the mother and the other is inherited from the father. Similarly, diploid cells possess two alleles at each locus, and these alleles constitute the genotype for that locus. The principle of segregation indicates that one allele of the genotype is inherited from each parent.

This similarity between the number of chromosomes and the number of alleles is not accidental—the two alleles of a genotype are located on homologous chromosomes. The symbols used in genetic crosses, such as R and r , are just shorthand notations for particular

sequences of DNA in the chromosomes that code for particular phenotypes. The two alleles of a genotype are found on different but homologous chromosomes. During the S stage of meiotic interphase, each chromosome replicates, producing two copies of each allele, one on each chromatid (FIGURE 3.5a). The homologous chromosomes segregate during anaphase I, thereby separating the two different alleles (FIGURE 3.5b and c). This chromosome segregation is the basis of the principle of segregation. During anaphase II of meiosis, the two chromatids of each replicated chromosome separate; so each gamete resulting from meiosis carries only a single allele at each locus, as Mendel's principle of segregation predicts.

If crossing over has taken place during prophase I of meiosis, then the two chromatids of each replicated chromosome are no longer identical, and the segregation of different alleles takes place at anaphase I and anaphase II (see Figure 3.5c). Of course, Mendel didn't know anything about chromosomes; he formulated his principles of heredity entirely on the basis of the results of the crosses that he carried out. Nevertheless, we should not forget that these principles work because they are based on the behavior of actual chromosomes during meiosis.

Concepts

The chromosome theory of inheritance states that genes are located on chromosomes. The two alleles of a genotype segregate during anaphase I of meiosis, when homologous chromosomes separate. The alleles may also segregate during anaphase II of meiosis if crossing over has taken place.

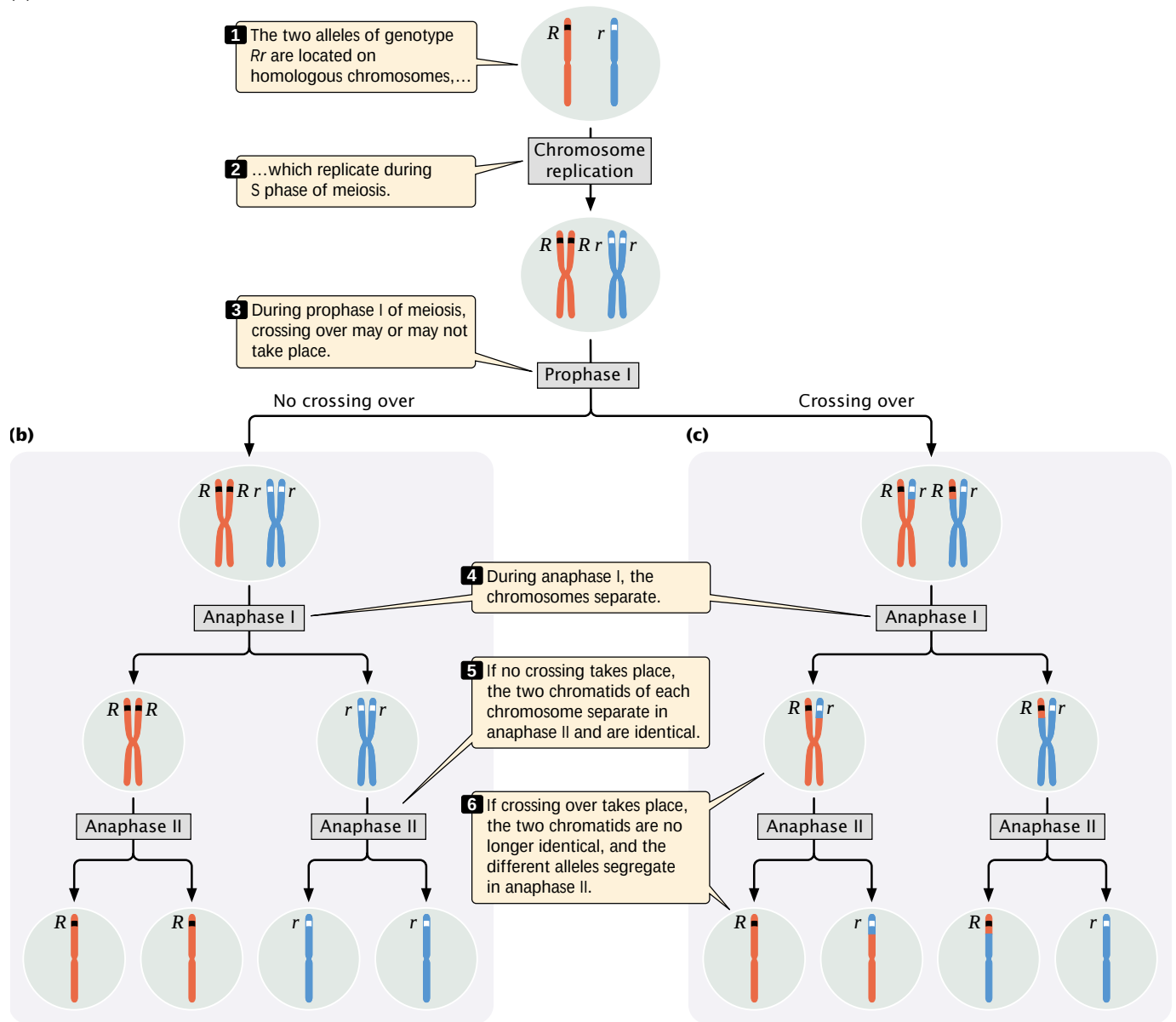


Predicting the Outcomes of Genetic Crosses

One of Mendel's goals in conducting his experiments on pea plants was to develop a way to predict the outcome of crosses between plants with different phenotypes. In this section, we will first learn a simple, shorthand method for predicting outcomes of genetic crosses (the Punnett square), and then we will learn how to use probability to predict the results of crosses.

The Punnett square To illustrate the Punnett square, let's examine another cross that Mendel carried out. By crossing two varieties of peas that differed in height, Mendel established that tall (T) was dominant over short (t). He tested his theory concerning the inheritance of dominant traits by crossing an F_1 tall plant that was heterozygous (Tt) with the short homozygous parental variety (tt). This type of cross, between an F_1 genotype and either of the parental genotypes, is called a **backcross**.

(a)

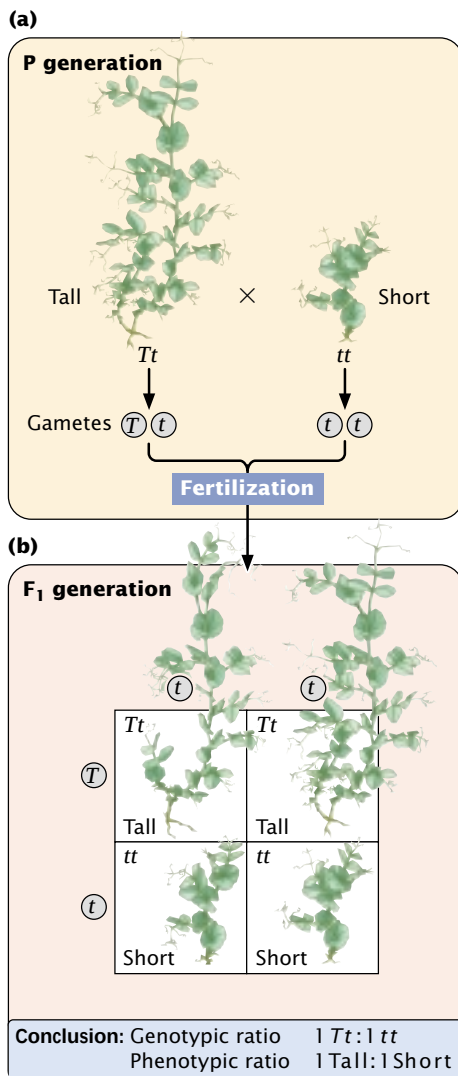


3.5 Segregation happens because homologous chromosomes separate in meiosis.

To predict the types of offspring that result from this cross, we first determine which gametes will be produced by each parent (FIGURE 3.6a). The principle of segregation tells us that the two alleles in each parent separate, and one allele passes to each gamete. All gametes from the homozygous tt short plant will receive a single short (t) allele. The tall plant in this cross is heterozygous (Tt); so 50% of its gametes will receive a tall allele (T) and the other 50% will receive a short allele (t).

A **Punnett square** is constructed by drawing a grid, putting the gametes produced by one parent along the

upper edge and the gametes produced by the other parent down the left side (FIGURE 3.6b). Each cell (a block within the Punnett square) contains an allele from each of the corresponding gametes, generating the genotype of the progeny produced by fusion of those gametes. In the upper left-hand cell of the Punnett square in Figure 3.6b, a gamete containing T from the tall plant unites with a gamete containing t from the short plant, giving the genotype of the progeny (Tt). It is useful to write the phenotype expressed by each genotype; here the progeny will be tall, because the tall allele is dominant over the short allele.



3.6 The Punnett square can be used for determining the results of a genetic cross.

This process is repeated for all the cells in the Punnett square.

By simply counting, we can determine the types of progeny produced and their ratios. In Figure 3.6b, two cells contain tall (Tt) progeny and two cells contain short (tt) progeny; so the genotypic ratio expected for this cross is 2 Tt to 2 tt (a 1:1 ratio). Another way to express this result is to say that we expect $\frac{1}{2}$ of the progeny to have genotype Tt (and phenotype tall) and $\frac{1}{2}$ of the progeny to have genotype tt (and phenotype short). In this cross, the genotypic ratio and the phenotypic ratio are the same, but this outcome need not be the case. Try completing a Punnett square for the cross in which the F_1 round-seeded plants in Figure 3.4 undergo self-fertilization (you should obtain a phenotypic ratio of 3 round to 1 wrinkled and a genotypic ratio of 1 RR to 2 Rr to 1 rr).

Concepts

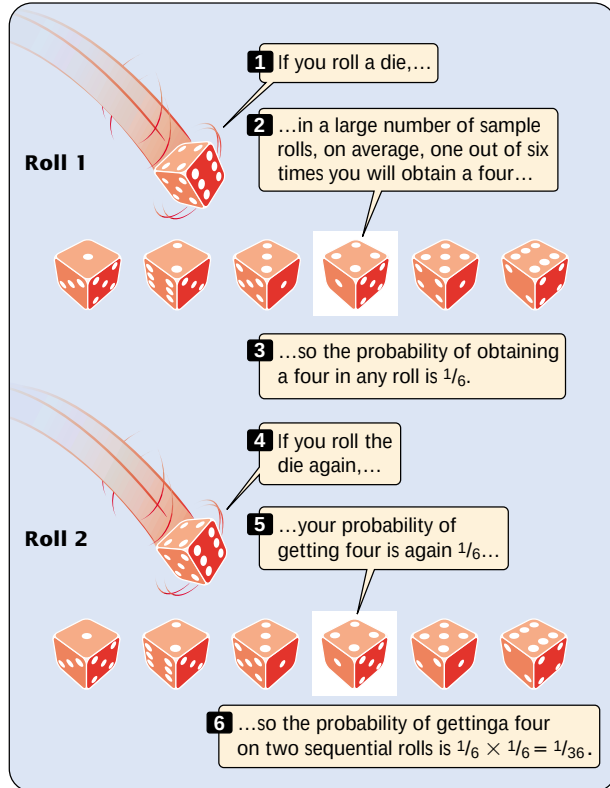
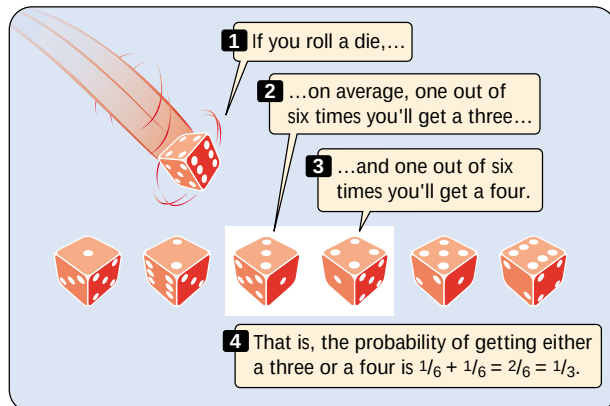
The Punnett square is a short-hand method of predicting the genotypic and phenotypic ratios of progeny from a genetic cross.

Probability as a tool in genetics Another method for determining the outcome of a genetic cross is to use the rules of probability, as Mendel did with his crosses. **Probability** expresses the likelihood of a particular event occurring. It is the number of times that a particular event occurs, divided by the number of all possible outcomes. For example, a deck of 52 cards contains only one king of hearts. The probability of drawing one card from the deck at random and obtaining the king of hearts is $\frac{1}{52}$, because there is only one card that is the king of hearts (one event) and there are 52 cards that can be drawn from the deck (52 possible outcomes). The probability of drawing a card and obtaining an ace is $\frac{4}{52}$, because there are four cards that are aces (four events) and 52 cards (possible outcomes). Probability can be expressed either as a fraction ($\frac{1}{52}$ in this case) or as a decimal number (0.019).

The probability of a particular event may be determined by knowing something about *how* the event occurs or *how often* it occurs. We know, for example, that the probability of rolling a six-sided die (one member of a pair of dice) and getting a four is $\frac{1}{6}$, because the die has six sides and any one side is equally likely to end up on top. So, in this case, understanding the nature of the event—the shape of the thrown die—allows us to determine the probability. In other cases, we determine the probability of an event by making a large number of observations. When a weather forecaster says that there is a 40% chance of rain on a particular day, this probability was obtained by observing a large number of days with similar atmospheric conditions and finding that it rains on 40% of those days. In this case, the probability has been determined empirically (by observation).

The multiplication rule Two rules of probability are useful for predicting the ratios of offspring produced in genetic crosses. The first is the **multiplication rule**, which states that the probability of two or more independent events occurring together is calculated by multiplying their independent probabilities.

To illustrate the use of the multiplication rule, let's again consider the roll of dice. The probability of rolling one die and obtaining a four is $\frac{1}{6}$. To calculate the probability of rolling a die twice and obtaining 2 fours, we can apply the multiplication rule. The probability of obtaining a four on the first roll is $\frac{1}{6}$ and the probability of obtaining a four on the second roll is $\frac{1}{6}$; so the probability of rolling a four on both is $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$ (FIGURE 3.7a). The key indicator for applying the multiplication rule is the word *and*;

(a) The multiplication rule**(b) The addition rule**

3.7 The multiplication and addition rules can be used to determine the probability of combinations of events.

in the example just considered, we wanted to know the probability of obtaining a four on the first roll *and* a four on the second roll.

For the multiplication rule to be valid, the events whose joint probability is being calculated must be independent—the outcome of one event must not influence the outcome of the other. For example, the number that comes up on one roll of the die has no influence on the number that

comes up on the other roll, so these events are independent. However, if we wanted to know the probability of being hit on the head with a hammer and going to the hospital on the same day, we could not simply multiply the probability of being hit on the head with a hammer by the probability of going to the hospital. The multiplication rule cannot be applied here, because the two events are not independent—being hit on the head with a hammer certainly influences the probability of going to the hospital.

The addition rule The second rule of probability frequently used in genetics is the **addition rule**, which states that the probability of any one of two or more mutually exclusive events is calculated by adding the probabilities of these events. Let's look at this rule in concrete terms. To obtain the probability of throwing a die once and rolling *either* a three *or* a four, we would use the addition rule, adding the probability of obtaining a three ($1/6$) to the probability of obtaining a four (again, $1/6$), or $1/6 + 1/6 = 2/6 = 1/3$ (**FIGURE 3.7b**). The key indicator for applying the addition rule are the words *either* and *or*.

For the addition rule to be valid, the events whose probability is being calculated must be mutually exclusive, meaning that one event excludes the possibility of the other occurring. For example, you cannot throw a single die just once and obtain both a three and a four, because only one side of the die can be on top. These events are mutually exclusive.

Concepts

The multiplication rule states that the probability of two or more independent events occurring together is calculated by multiplying their independent probabilities. The addition rule states that the probability that any one of two or more mutually exclusive events occurring is calculated by adding their probabilities.

The application of probability to genetic crosses The multiplication and addition rules of probability can be used in place of the Punnett square to predict the ratios of progeny expected from a genetic cross. Let's first consider a cross between two pea plants heterozygous for the locus that determines height, $Tt \times Tt$. Half of the gametes produced by each plant have a T allele, and the other half have a t allele; so the probability for each type of gamete is $1/2$.

The gametes from the two parents can combine in four different ways to produce offspring. Using the multiplication rule, we can determine the probability of each possible type. To calculate the probability of obtaining TT progeny, for example, we multiply the probability of receiving a T allele from the first parent ($1/2$) times the probability of

receiving a T allele from the second parent ($\frac{1}{2}$). The multiplication rule should be used here because we need the probability of receiving a T allele from the first parent *and* a T allele from the second parent—two independent events. The four types of progeny from this cross and their associated probabilities are:

TT	(T gamete and T gamete)	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	tall
Tt	(T gamete and t gamete)	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	tall
tT	(t gamete and T gamete)	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	tall
tt	(t gamete and t gamete)	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	short

Notice that there are two ways for heterozygous progeny to be produced: a heterozygote can either receive a T allele from the first parent and a t allele from the second or receive a t allele from the first parent and a T allele from the second.

After determining the probabilities of obtaining each type of progeny, we can use the addition rule to determine the overall phenotypic ratios. Because of dominance, a tall plant can have genotype TT , Tt , or tT ; so, using the addition rule, we find the probability of tall progeny to be $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$. Because only one genotype codes for short (tt), the probability of short progeny is simply $\frac{1}{4}$.

Two methods have now been introduced to solve genetic crosses: the Punnett square and the probability method. At this point, you may be saying “Why bother with probability rules and calculations? The Punnett square is easier to understand and just as quick.” For simple monohybrid crosses, the Punnett square is simpler and just as easy to use. However, when tackling more complex crosses concerning genes at two or more loci, the probability method is both clearer and quicker than the Punnett square.

The binomial expansion and probability When probability is used, it is important to recognize that there may be several different ways in which a set of events can occur. Consider two parents who are both heterozygous for albinism, a recessive condition in humans that causes reduced pigmentation in the skin, hair, and eyes (FIGURE 3.8). When two parents heterozygous for albinism mate ($Aa \times Aa$), the probability of their having a child with albinism (aa) is $\frac{1}{4}$ and the probability of having a child with normal pigmentation (AA or Aa) is $\frac{3}{4}$. Suppose we want to know the probability of this couple having three children with albinism. In this case, there is only one way in which they can have three children with albinism—their first child has albinism *and* their second child has albinism *and* their third child has albinism. Here we simply apply the multiplication rule: $\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64}$.

Suppose we now ask, What is the probability of this couple having three children, one with albinism and two with normal pigmentation. This situation is more complicated. The first child might have albinism, whereas the second and third are unaffected; the probability of this sequence of events is $\frac{1}{4} \times \frac{3}{4} \times \frac{3}{4} = \frac{9}{64}$. Alternatively, the



3.8 Albinism in human beings is usually inherited as a recessive trait. (Richard Dranitzke/SS/Photo Researchers.)

first and third children might have normal pigmentation, whereas the second has albinism; the probability of this sequence is $\frac{3}{4} \times \frac{1}{4} \times \frac{3}{4} = \frac{9}{64}$. Finally, the first two children might have normal pigmentation and the third albinism; the probability of this sequence is $\frac{3}{4} \times \frac{3}{4} \times \frac{1}{4} = \frac{9}{64}$. Because *either* the first sequence *or* the second sequence *or* the third sequence produces one child with albinism and two with normal pigmentation, we apply the addition rule and add the probabilities: $\frac{9}{64} + \frac{9}{64} + \frac{9}{64} = \frac{27}{64}$.

If we want to know the probability of this couple having five children, two with albinism and three with normal pigmentation, figuring out the different combinations of children and their probabilities becomes more difficult. This task is made easier if we apply the binomial expansion.

The binomial takes the form $(a + b)^n$, where a equals the probability of one event, b equals the probability of the alternative event, and n equals the number of times the event occurs. For figuring the probability of two out of five children with albinism:

$$a = \text{the probability of a child having albinism} = \frac{1}{4}$$

$$b = \text{the probability of a child having normal pigmentation} = \frac{3}{4}$$

The binomial for this situation is $(a + b)^5$ because there are five children in the family ($n = 5$). The expansion is:

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$$

The first term in the expansion (a^5) equals the probability of having five children all with albinism, because a is the probability of albinism. The second term ($5a^4b$) equals the probability of having four children with albinism and one with normal pigmentation, the third term ($10a^3b^2$) equals the probability of having three children with albinism and two with normal pigmentation, and so forth.

To obtain the probability of any combination of events, we insert the values of a and b ; so the probability of having two out of five children with albinism is:

$$10a^2b^3 = 10(1/4)^2(3/4)^3 = 270/1024 = .26$$

We could easily figure out the probability of any desired combination of albinism and pigmentation among five children by using the other terms in the expansion.

How did we expand the binomial in this example? In general, the expansion of any binomial $(a + b)^n$ consists of a series of $n + 1$ terms. In the preceding example, $n = 5$; so there are $5 + 1 = 6$ terms: a^5 , $5a^4b$, $10a^3b^2$, $10a^2b^3$, $5ab^4$, and b^5 . To write out the terms, first figure out their exponents. The exponent of a in the first term always begins with the power to which the binomial is raised, or n . In our example, n equals 5, so our first term is a^5 . The exponent of a decreases by one in each successive term; so the exponent of a is 4 in the second term (a^4), 3 in the third term (a^3), and so forth. The exponent of b is 0 (no b) in the first term and increases by 1 in each successive term, increasing from 0 to 5 in our example.

Next, determine the coefficient of each term. The coefficient of the first term is always 1; so in our example the first term is $1a^5$, or just a^5 . The coefficient of the second term is always the same as the power to which the binomial is raised; in our example this coefficient is 5 and the term is $5a^4b$. For the coefficient of the third term, look back at the preceding term; multiply the coefficient of the preceding term (5 in our example) by the exponent of a in that term (4) and then divide by the number of that term (second term, or 2). So the coefficient of the third term in our example is $(5 \times 4)/2 = 20/2 = 10$ and the term is $10a^3b^2$. Follow this same procedure for each successive term.

Another way to determine the probability of any particular combination of events is to use the following formula:

$$P = \frac{n!}{s!t!} a^s b^t$$

where P equals the overall probability of event X with probability a occurring s times and event Y with probability b occurring t times. For our albinism example, event X would be the occurrence of a child with albinism and event Y the occurrence of a child with normal pigmentation; s would equal the number of children with albinism (2) and t , the number of children with normal pigmentation (3). The $!$ symbol is termed factorial, and it means the product of all the integers from n to 1. In this example,

$n = 5$; so $n! = 5 \times 4 \times 3 \times 2 \times 1$. Applying this formula to obtain the probability of two out of five children having albinism, we obtain:

$$P = \frac{5!}{2!3!} (1/4)^2 (3/4)^3$$

$$P = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 3 \times 2 \times 1} (1/4)^2 (3/4)^3 = .26$$

This value is the same as that obtained with the binomial expansion.

The Testcross

A useful tool for analyzing genetic crosses is the **testcross**, in which one individual of unknown genotype is crossed with another individual with a homozygous recessive genotype for the trait in question. Figure 3.6 illustrates a testcross (as well as a backcross). A testcross tests, or reveals, the genotype of the first individual.

Suppose you were given a tall pea plant with no information about its parents. Because tallness is a dominant trait in peas, your plant could be either homozygous (TT) or heterozygous (Tt), but you would not know which. You could determine its genotype by performing a testcross. If the plant were homozygous (TT), a testcross would produce all tall progeny ($TT \times tt \rightarrow$ all Tt); if the plant were heterozygous (Tt), the testcross would produce half tall progeny and half short progeny ($Tt \times tt \rightarrow \frac{1}{2} Tt$ and $\frac{1}{2} tt$). When a testcross is performed, any recessive allele in the unknown genotype is expressed in the progeny, because it will be paired with a recessive allele from the homozygous recessive parent.

Concepts

The binomial expansion may be used to determine the probability of a particular set of events. A testcross is a cross between an individual with an unknown genotype and one with a homozygous recessive genotype. The outcome of the testcross can reveal the unknown genotype.

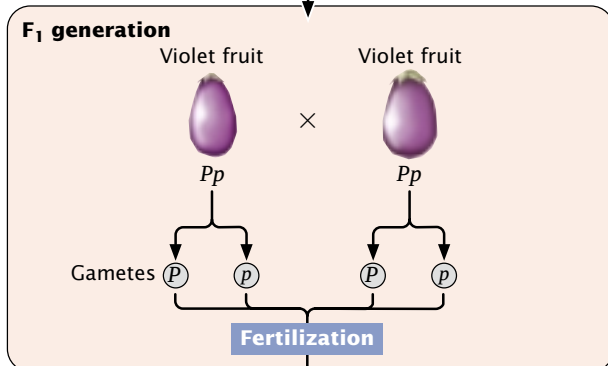
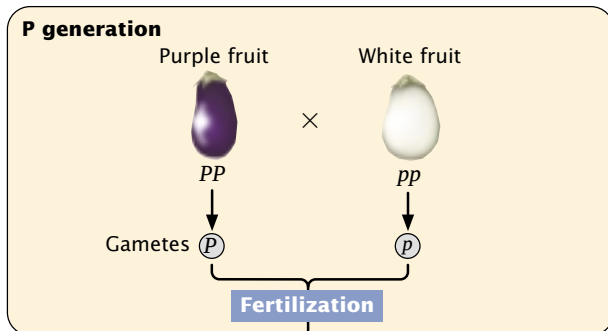
Incomplete Dominance

The seven characters in pea plants that Mendel chose to study extensively all exhibited dominance, but Mendel did realize that not all characters have traits that exhibit dominance. He conducted some crosses concerning the length of time that pea plants take to flower. When he crossed two homozygous varieties that differed in their flowering time by an average of 20 days, the length of time taken by the F_1 plants to flower was intermediate between those of

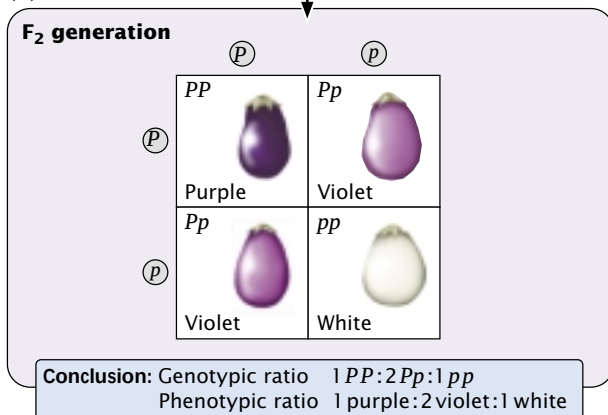
the two parents. When the heterozygote has a phenotype intermediate between the phenotypes of the two homozygotes, the trait is said to display **incomplete dominance**.

Incomplete dominance is also exhibited in the fruit color of eggplants. When a homozygous plant that produces purple fruit (PP) is crossed with a homozygous plant that produces white fruit (pp), all the heterozygous F_1 (Pp) produce violet fruit (FIGURE 3.9a). When the F_1 are crossed with each other, $\frac{1}{4}$ of the F_2 are purple (PP), $\frac{1}{2}$ are violet (Pp), and $\frac{1}{4}$ are white (pp), as shown in FIGURE 3.9b. This 1:2:1 ratio is different from the 3:1 ratio that we would observe if eggplant fruit color exhibited dominance. When a

(a)



(b)



3.9 Fruit color in eggplant is inherited as an incompletely dominant trait.



3.10 Leopard spotting in horses exhibits incomplete dominance. (Frank Oberle/Bruce Coleman.)

trait displays incomplete dominance, the genotypic ratios and phenotypic ratios of the offspring are the *same*, because each genotype has its own phenotype. It is impossible to obtain eggplants that are pure breeding for violet fruit, because all plants with violet fruit are heterozygous.

Another example of incomplete dominance is feather color in chickens. A cross between a homozygous black chicken and a homozygous white chicken produces F_1 chickens that are gray. If these gray F_1 are intercrossed, they produce F_2 birds in a ratio of 1 black: 2 gray: 1 white. Leopard white spotting in horses is incompletely dominant over unspotted horses: LL horses are white with numerous dark spots, heterozygous Ll horses have fewer spots, and ll horses have no spots (FIGURE 3.10). The concept of dominance and some of its variations are discussed further in Chapter 5.

Concepts

Incomplete dominance is exhibited when the heterozygote has a phenotype intermediate between the phenotypes of the two homozygotes. When a trait exhibits incomplete dominance, a cross between two heterozygotes produces a 1:2:1 phenotypic ratio in the progeny.

Genetic Symbols

As we have seen, genetic crosses are usually depicted with the use of symbols to designate the different alleles. Lowercase letters are traditionally used to designate recessive alleles, and uppercase letters are for dominant alleles. Two or three letters may be used for a single allele: the recessive allele for heart-shaped leaves in cucumbers is designated hl , and the recessive allele for abnormal sperm head shape in mice is designated azh .

The normal allele for a character—called the **wild type** because it is the allele most often found in the wild—is of-

Table 3.2 Phenotypic ratios for simple genetic crosses (crosses for a single locus)

Ratio	Genotypes of Parents	Genotypes of Progeny	Type of Dominance
3:1	$Aa \times Aa$	$\frac{3}{4} A_ : \frac{1}{4} aa$	Dominance
1:2:1	$Aa \times Aa$	$\frac{1}{4} AA : \frac{1}{2} Aa : \frac{1}{4} aa$	Incomplete dominance
1:1	$Aa \times aa$	$\frac{1}{2} Aa : \frac{1}{2} aa$	Dominance or incomplete dominance
	$Aa \times AA$	$\frac{1}{2} Aa : \frac{1}{2} AA$	Incomplete dominance
Uniform progeny	$AA \times AA$	All AA	Dominance or incomplete dominance
	$aa \times aa$	All aa	Dominance or incomplete dominance
	$AA \times aa$	All Aa	Dominance or incomplete dominance
	$AA \times Aa$	All $A_$	Dominance

Note: A line in a genotype, such as $A_$, indicates that any allele is possible.

ten symbolized by one or more letters and a plus sign (+). The letter(s) chosen are usually based on the phenotype of the mutant. The first letter is lowercase if the mutant phenotype is recessive, uppercase if the mutant phenotype is dominant. For example, the recessive allele for yellow eyes in the Oriental fruit fly is represented by ye , whereas the allele for wild-type eye color is represented by ye^+ . At times, the letters for the wild-type allele are dropped and the allele is represented simply by a plus sign. Superscripts and subscripts are sometimes added to distinguish between genes: Lfr_1 and Lfr_2 represent dominant alleles at different loci that produce lacerate leaf margins in opium poppies; El^R represents an allele in goats that restricts the length of the ears.

A slash may be used to distinguish alleles present in an individual genotype. The genotype of a goat that is heterozygous for restricted ears might be written El^+/El^R or simply $+/El^R$. If genotypes at more than one locus are presented together, a space may separate them. A goat heterozygous for a pair of alleles that produce restricted ears and heterozygous for another pair of alleles that produce goiter can be designated by $El^+/El^R G/g$.

Connecting Concepts

Ratios in Simple Crosses

Now that we have had some experience with genetic crosses, let's review the ratios that appear in the progeny of simple crosses, in which a single locus is under consideration. Understanding these ratios and the parental genotypes that produce them will allow you to work simple genetic crosses quickly, without resorting to the Punnett square. Later, we will use these ratios to work more complicated crosses entailing several loci.

There are only four phenotypic ratios to understand (Table 3.2). The 3:1 ratio arises in a simple genetic cross when both of the parents are heterozygous for a dominant trait ($Aa \times Aa$). The second phenotypic ratio is the 1:2:1 ratio, which arises in the progeny of crosses between two parents heterozygous for a character that exhibits incom-

plete dominance ($Aa \times Aa$). The third phenotypic ratio is the 1:1 ratio, which results from the mating of a homozygous parent and a heterozygous parent. If the character exhibits dominance, the homozygous parent in this cross must carry two recessive alleles ($Aa \times aa$) to obtain a 1:1 ratio, because a cross between a homozygous dominant parent and a heterozygous parent ($AA \times Aa$) produces only offspring displaying the dominant trait. For a character with incomplete dominance, a 1:1 ratio results from a cross between the heterozygote and either homozygote ($Aa \times aa$ or $Aa \times AA$).

The fourth phenotypic ratio is not really a ratio—all the offspring have the same phenotype. Several combinations of parents can produce this outcome (Table 3.2). A cross between any two homozygous parents—either between two of the same homozygotes ($AA \times AA$ and $aa \times aa$) or between two different homozygotes ($AA \times aa$)—produces progeny all having the same phenotype. Progeny of a single phenotype can also result from a cross between a homozygous dominant parent and a heterozygote ($AA \times Aa$).

If we are interested in the ratios of genotypes instead of phenotypes, there are only three outcomes to remember (Table 3.3): the 1:2:1 ratio, produced by a cross between

Table 3.3 Genotypic ratios for simple genetic crosses (crosses for a single locus)

Ratio	Genotypes of Parents	Genotypes of Progeny
1:2:1	$Aa \times Aa$	$\frac{1}{4} AA : \frac{1}{2} Aa : \frac{1}{4} aa$
1:1	$Aa \times aa$	$\frac{1}{2} Aa : \frac{1}{2} aa$
	$Aa \times AA$	$\frac{1}{2} Aa : \frac{1}{2} AA$
Uniform progeny	$AA \times AA$	All AA
	$aa \times aa$	All aa
	$AA \times aa$	All Aa

two heterozygotes; the 1:1 ratio, produced by a cross between a heterozygote and a homozygote; and the uniform progeny produced by a cross between two homozygotes. These simple phenotypic and genotypic ratios and the parental genotypes that produce them provide the key to understanding crosses for a single locus and, as you will see in the next section, for multiple loci.

Multiple-Loci Crosses

We will now extend Mendel's principle of segregation to more complex crosses for alleles at multiple loci. Understanding the nature of these crosses will require an additional principle, the principle of independent assortment.

Dihybrid Crosses

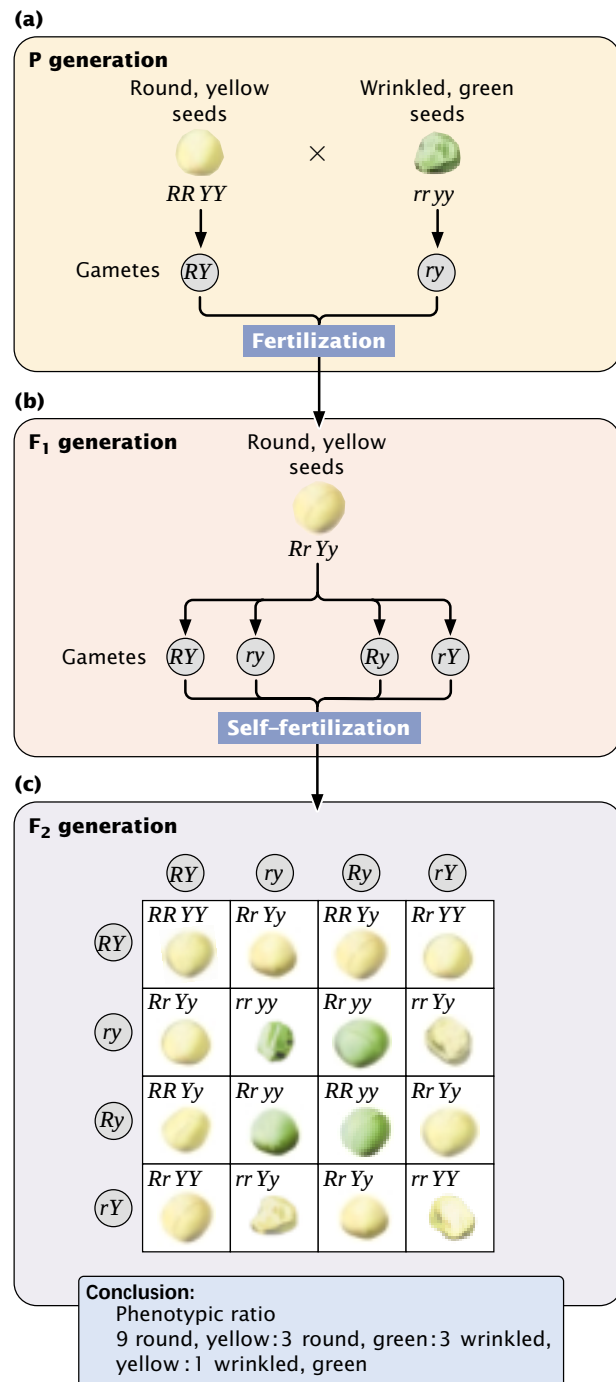
In addition to his work on monohybrid crosses, Mendel also crossed varieties of peas that differed in *two* characteristics (**dihybrid crosses**). For example, he had one homozygous variety of pea that produced round seeds and yellow endosperm; another homozygous variety produced wrinkled seeds and green endosperm. When he crossed the two, all the F_1 progeny had round seeds and yellow endosperm. He then self-fertilized the F_1 and obtained the following progeny in the F_2 : 315 round, yellow seeds; 101 wrinkled, yellow seeds; 108 round, green seeds; and 32 wrinkled, green seeds. Mendel recognized that these traits appeared approximately in a 9:3:3:1 ratio; that is, $\frac{9}{16}$ of the progeny were round and yellow, $\frac{3}{16}$ were wrinkled and yellow, $\frac{3}{16}$ were round and green, and $\frac{1}{16}$ were wrinkled and green.

The Principle of Independent Assortment

Mendel carried out a number of dihybrid crosses for pairs of characteristics and always obtained a 9:3:3:1 ratio in the F_2 . This ratio makes perfect sense in regard to segregation and dominance if we add a third principle, which Mendel recognized in his dihybrid crosses: the **principle of independent assortment** (Mendel's second law). This principle states that alleles at different loci separate independently of one another.

A common mistake is to think that the principle of segregation and the principle of independent assortment refer to two different processes. The principle of independent assortment is really an extension of the principle of segregation. The principle of segregation states that the two alleles of a locus separate when gametes are formed; the principle of independent assortment states that, when these two alleles separate, their separation is independent of the separation of alleles at *other* loci.

Let's see how the principle of independent assortment explains the results that Mendel obtained in his dihybrid cross. Each plant possesses two alleles coding for each characteristic, so the parental plants must have had genotypes $RRYY$ and $rryy$ (FIGURE 3.11a). The principle of segrega-



3.11 Mendel conducted dihybrid crosses.

tion indicates that the alleles for each locus separate, and one allele for each locus passes to each gamete. The gametes produced by the round, yellow parent therefore contain alleles RY , whereas the gametes produced by the wrinkled, green parent contain alleles ry . These two types of gametes unite to produce the F_1 , all with genotype $RrYy$. Because

round is dominant over wrinkled and yellow is dominant over green, the phenotype of the F_1 will be round and yellow.

When Mendel self-fertilized the F_1 plants to produce the F_2 , the alleles for each locus separated, with one allele going into each gamete. This is where the principle of independent assortment becomes important. Each pair of alleles can separate in two ways: (1) R separates with Y and r separates with y to produce gametes RY and ry or (2) R separates with y and r separates with Y to produce gametes Ry and rY . The principle of independent assortment tells us that the alleles at each locus separate independently; thus, both kinds of separation occur equally and all four types of gametes (RY , ry , Ry , and rY) are produced in equal proportions (FIGURE 3.11b). When these four types of gametes are combined to produce the F_2 generation, the progeny consist of $\frac{9}{16}$ round and yellow, $\frac{3}{16}$ wrinkled and yellow, $\frac{3}{16}$ round and green, and $\frac{1}{16}$ wrinkled and green, resulting in a 9:3:3:1 phenotypic ratio (FIGURE 3.11c).

The Relation of the Principle of Independent Assortment to Meiosis

An important qualification of the principle of independent assortment is that it applies to characters encoded by loci located on different chromosomes because, like the principle of segregation, it is based wholly on the behavior of chromosomes during meiosis. Each pair of homologous chromosomes separates independently of all other pairs in anaphase I of meiosis (see Figure 2.18); so genes located on different pairs of homologs will assort independently. Genes that happen to be located on the same chromosome will travel together during anaphase I of meiosis and will arrive at the same destination—within the same gamete (unless crossing over takes place). Genes located on the same chromosome therefore do not assort independently (unless they are located sufficiently far apart that crossing over takes place every meiotic division, as will be discussed fully in Chapter 7).

Concepts

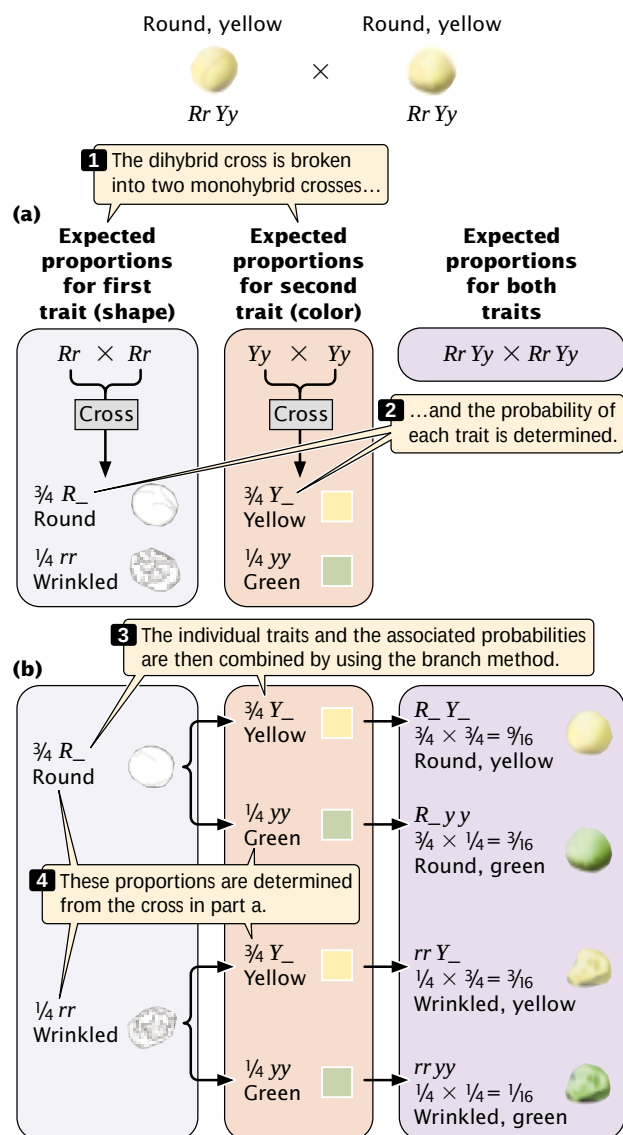
The principle of independent assortment states that genes coding for different characteristics separate independently of one another when gametes are formed, owing to independent separation of homologous pairs of chromosomes during meiosis. Genes located close together on the same chromosome do not, however, assort independently.

Applying Probability and the Branch Diagram to Dihybrid Crosses

When the genes at two loci separate independently, a dihybrid cross can be understood as two monohybrid crosses. Let's examine Mendel's dihybrid cross ($RrYy \times RrYy$) by considering each characteristic separately (FIGURE 3.12a). If we consider only the shape of the seeds, the cross was

$Rr \times Rr$, which yields a 3:1 phenotypic ratio ($\frac{3}{4}$ round and $\frac{1}{4}$ wrinkled progeny, see Table 3.2). Next consider the other characteristic, the color of the endosperm. The cross was $Yy \times Yy$, which produces a 3:1 phenotypic ratio ($\frac{3}{4}$ yellow and $\frac{1}{4}$ green progeny).

We can now combine these monohybrid ratios by using the multiplication rule to obtain the proportion of progeny with different combinations of seed shape and color. The proportion of progeny with round and yellow seeds is $\frac{3}{4}$ (the probability of round) $\times$ $\frac{3}{4}$ (the probability of yellow) = $\frac{9}{16}$. The proportion of progeny with round and green seeds is $\frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$; the proportion of progeny with wrinkled and yellow seeds is $\frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$; and the



3.12 A branch diagram can be used for determining the phenotypes and expected proportions of offspring from a dihybrid cross ($RrYy \times RrYy$).

proportion of progeny with wrinkled and green seeds is $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$.

Branch diagrams are a convenient way of organizing all the combinations of characteristics (FIGURE 3.12b). In the first column, list the proportions of the phenotypes for one character (here, $\frac{3}{4}$ round and $\frac{1}{4}$ wrinkled). In the second column, list the proportions of the phenotypes for the second character ($\frac{3}{4}$ yellow and $\frac{1}{4}$ green) next to each of the phenotypes in the first column: put $\frac{3}{4}$ yellow and $\frac{1}{4}$ green next to the round phenotype and again next to the wrinkled phenotype. Draw lines between the phenotypes in the first column and each of the phenotypes in the second column. Now follow each branch of the diagram, multiplying the probabilities for each trait along that branch. One branch leads from round to yellow, yielding round and yellow progeny. Another branch leads from round to green, yielding round and green progeny, and so on. The probability of progeny with a particular combination of traits is calculated by using the multiplicative rule: the probability of round ($\frac{3}{4}$) and yellow ($\frac{3}{4}$) seeds is $\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$. The advantage of the branch diagram is that it helps keep track of all the potential combinations of traits that may appear in the progeny. It can be used to determine phenotypic or genotypic ratios for any number of characteristics.

Using probability is much faster than using the Punnett square for crosses that include multiple loci. Genotypic and phenotypic ratios can quickly be worked out by combining, with the multiplication rule, the simple ratios in Tables 3.2 and 3.3. The probability method is particularly efficient if we need the probability of only a *particular* phenotype or genotype among the progeny of a cross. Suppose we needed to know the probability of obtaining the genotype $Rryy$ in the F_2 of the dihybrid cross in Figure 3.11. The probability of obtaining the Rr genotype in a cross of $Rr \times Rr$ is $\frac{1}{2}$ and that of obtaining yy progeny in a cross of $Yy \times Yy$ is $\frac{1}{4}$ (see Table 3.3). Using the multiplication rule, we find the probability of $Rryy$ to be $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$.

To illustrate the advantage of the probability method, consider the cross $AaBbccDdEe \times AaBbCcddEe$. Suppose we wanted to know the probability of obtaining offspring with the genotype $aabbccdde$. If we used a Punnett square to determine this probability, we might be working on the solution for months. However, we can quickly figure the probability of obtaining this one genotype by breaking this cross into a series of single-locus crosses:

Cross	Progeny genotype	Probability
$Aa \times Aa$	aa	$\frac{1}{4}$
$Bb \times Bb$	bb	$\frac{1}{4}$
$cc \times Cc$	cc	$\frac{1}{2}$
$Dd \times dd$	dd	$\frac{1}{2}$
$Ee \times Ee$	ee	$\frac{1}{4}$

The probability of an offspring from this cross having genotype $aabbccdde$ is now easily obtained by using the multiplication

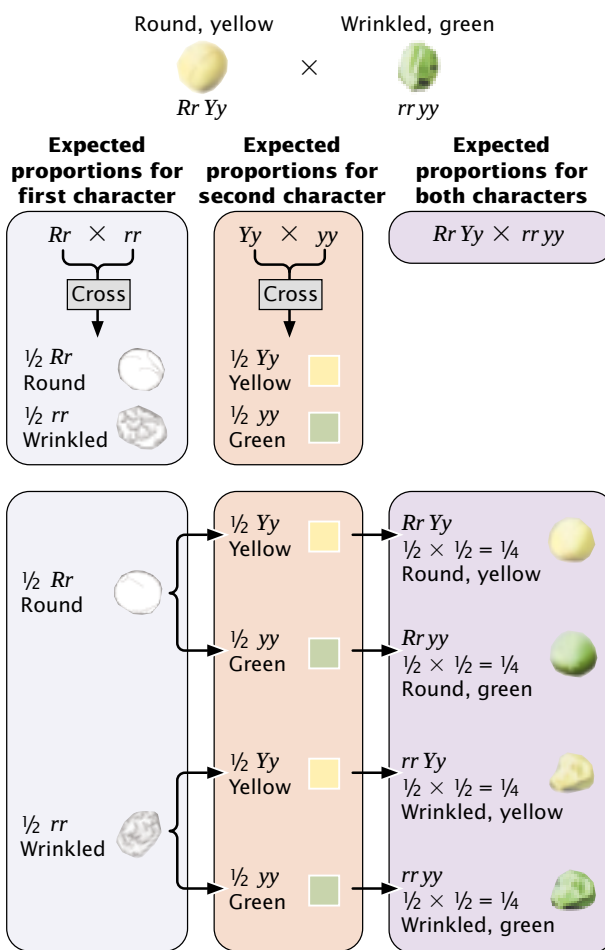
rule: $\frac{1}{4} \times \frac{1}{4} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{4} = \frac{1}{256}$. This calculation assumes that genes at these five loci all assort independently.

Concepts

A cross including several characteristics can be worked by breaking the cross down into single-locus crosses and using the multiplication rule to determine the proportions of combinations of characteristics (provided the genes assort independently).

The Dihybrid Testcross

Let's practice using the branch diagram by determining the types and proportions of phenotypes in a dihybrid testcross between the round and yellow F_1 plants ($Rr Yy$) that Mendel obtained in his dihybrid cross and the wrinkled and green plants ($rryy$) (FIGURE 3.13). Break the cross down into a series of single-locus crosses. The cross $Rr \times rr$ yields $\frac{1}{2}$ round (Rr) progeny and $\frac{1}{2}$ wrinkled (rr) progeny. The cross $Yy \times yy$ yields $\frac{1}{2}$ yellow (Yy) progeny and $\frac{1}{2}$ green (yy)



3.13 A branch diagram can be used for determining the phenotypes and expected proportions of offspring from a dihybrid testcross ($RrYy \times rryy$).

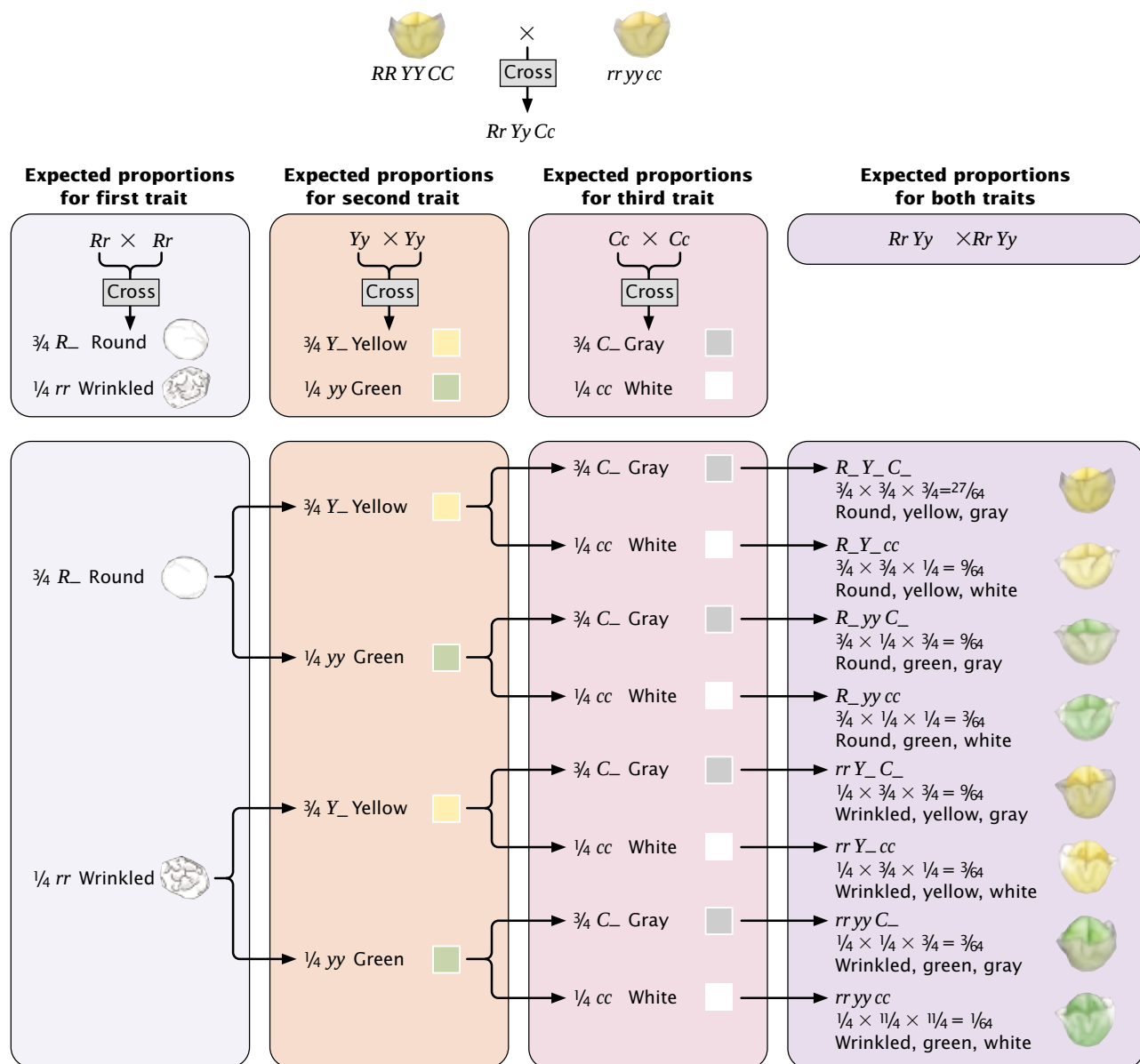
progeny. Using the multiplication rule, we find the proportion of round and yellow progeny to be $\frac{1}{2}$ (the probability of round) $\times \frac{1}{2}$ (the probability of yellow) = $\frac{1}{4}$. Four combinations of traits with the following proportions appear in the offspring: $\frac{1}{4}$ $RrYy$, round yellow; $\frac{1}{4}$ $Rryy$, round green; $\frac{1}{4}$ $rrYy$, wrinkled yellow; and $\frac{1}{4}$ $rryy$, wrinkled green.

Trihybrid Crosses

The branch diagram can also be applied to crosses including three characters (called **tri-hybrid crosses**). In one tri-hybrid cross, Mendel crossed a pure-breeding variety that

possessed round seeds, yellow endosperm, and gray seed coats with another pure-breeding variety that possessed wrinkled seeds, green endosperm, and white seed coats (**FIGURE 3.14**). The branch diagram shows that the expected phenotypic ratio in the F_2 is 27:9:9:9:3:3:3:1, and the numbers that Mendel obtained from this cross closely fit these expected ones.

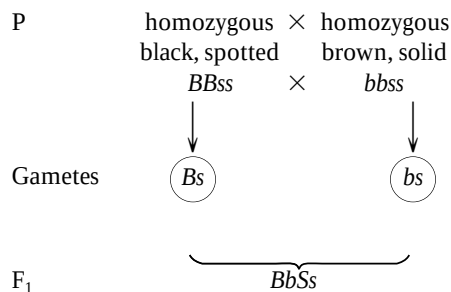
In monohybrid crosses, we have seen that three genotypes (RR , Rr , and rr) are produced in the F_2 . In dihybrid crosses, nine genotypes (3 genotypes for the first locus $\times$ 3 genotypes for the second locus = 9) are produced in the F_2 :



3.14 A branch diagram can be used for determining the phenotypes and expected proportions of offspring from a trihybrid cross ($RrYyCc \times RrYyCc$).

Third, use a branch diagram to determine the proportion of progeny of the testcross with different combinations of the two traits.

Step 5: Work the different parts of problem. Start by determining the genotype of the F_1 progeny. Mendel's first law indicates that the two alleles at a locus separate, one going into each gamete. Thus, the gametes produced by the black, spotted parent contain Bs and the gametes produced by the brown, spotted parent contain bs , which combine to produce F_1 progeny with the genotype $BbSs$:



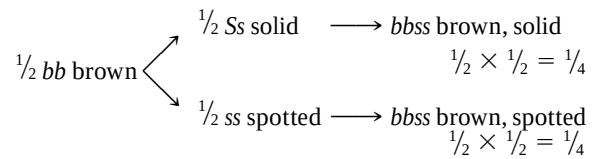
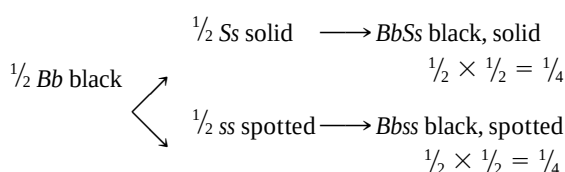
Use the F_1 genotype to work the testcross ($BbSs \times bbss$), breaking it into two single-locus crosses. First, consider the cross for coat color: $Bb \times bb$. Any cross between a heterozygote and a homozygous recessive genotype produces a 1:1 phenotypic ratio of progeny (see Table 3.2):

$$\begin{array}{c}
 BB \times bb \\
 \downarrow \\
 \frac{1}{2} Bb \text{ black} \\
 \frac{1}{2} bb \text{ brown}
 \end{array}$$

Next do the cross for spotting: $Ss \times ss$. This cross also is between a heterozygote and a homozygous recessive genotype and will produce $\frac{1}{2}$ solid (Ss) and $\frac{1}{2}$ spotted (ss) progeny (see Table 3.2).

$$\begin{array}{c}
 Ss \times ss \\
 \downarrow \\
 \frac{1}{2} Ss \text{ solid} \\
 \frac{1}{2} ss \text{ spotted}
 \end{array}$$

Finally, determine the proportions of progeny with combinations of these characters by using the branch diagram.



Step 6: Check all work. As a last step, reread the problem, checking to see if your answers are consistent with the information provided. You have used the genotypes $BBss$ and $bbSS$ in the P generation. Do these genotypes code for the phenotypes given in the problem? Are the F_1 progeny phenotypes consistent with the genotypes that you assigned? The answers are consistent with the information.

Observed and Expected Ratios

When two individuals of known genotype are crossed, we expect certain ratios of genotypes and phenotypes in the progeny; these expected ratios are based on the Mendelian principles of segregation, independent assortment, and dominance. The ratios of genotypes and phenotypes *actually* observed among the progeny, however, may deviate from these expectations.

For example, in German cockroaches, brown body color (Y) is dominant over yellow body color (y). If we cross a brown, heterozygous cockroach (Yy) with a yellow cockroach (yy), we expect a 1:1 ratio of brown (Yy) and yellow (yy) progeny. Among 40 progeny, we would therefore expect to see 20 brown and 20 yellow offspring. However, the observed numbers might deviate from these expected values; we might in fact see 22 brown and 18 yellow progeny.

Chance plays a critical role in genetic crosses, just as it does in flipping a coin. When you flip a coin, you expect a 1:1 ratio— $\frac{1}{2}$ heads and $\frac{1}{2}$ tails. If you flip a coin 1000 times, the proportion of heads and tails obtained would probably be very close to that expected 1:1 ratio. However, if you flip the coin 10 times, the ratio of heads to tails might be quite different from 1:1. You could easily get 6 heads and 4 tails, or 3 and 7 tails, just by chance. It is possible that you might even get 10 heads and 0 tails. The same thing happens in genetic crosses. We may expect 20 brown and 20 yellow cockroaches, but 22 brown and 18 yellow progeny *could* arise as a result of chance.

The Goodness-of-Fit Chi-Square Test

If you expected a 1:1 ratio of brown and yellow cockroaches but the cross produced 22 brown and 18 yellow, you probably wouldn't be too surprised even though it wasn't a perfect 1:1 ratio. In this case, it seems reasonable to assume that chance produced the deviation between the expected and the observed results. But, if you observed 25 brown and 15 yellow, would the ratio still be 1:1? Something other than chance might have caused the deviation. Perhaps the

inheritance of this character is more complicated than was assumed or perhaps some of the yellow progeny died before they were counted. Clearly, we need some means of evaluating how likely it is that chance is responsible for the deviation between the observed and the expected numbers.

To evaluate the role of chance in producing deviations between observed and expected values, a statistical test called the **goodness-of-fit chi-square test** is used. This test provides information about how well observed values fit expected values. Before we learn how to calculate the chi square, it is important to understand what this test does and does not indicate about a genetic cross.

The chi-square test cannot tell us whether a genetic cross has been correctly carried out, whether the results are correct, or whether we have chosen the correct genetic explanation for the results. What it does indicate is the *probability* that the difference between the observed and the expected values is due to chance. In other words, it indicates the likelihood that chance alone could produce the deviation between the expected and the observed values.

If we expected 20 brown and 20 yellow progeny from a genetic cross, the chi-square test gives the probability that we might observe 25 brown and 15 yellow progeny simply owing to chance deviations from the expected 20:20 ratio. When the probability calculated from the chi-square test is high, we assume that chance alone produced the difference. When the probability is low, we assume that some factor other than chance—some significant factor—produced the deviation.

To use the goodness-of-fit chi-square test, we first determine the expected results. The chi-square test must always be applied to numbers of progeny, not to proportions or percentages. Let's consider a locus for coat color in domestic cats, for which black color (*B*) is dominant over gray (*b*). If we crossed two heterozygous black cats (*Bb* × *Bb*), we would expect a 3:1 ratio of black and gray kittens. A series of such crosses yields a total of 50 kittens—30 black and 20 gray. These numbers are our *observed* values. We can obtain the *expected* numbers by multiplying the expected proportions by the total number of observed progeny. In this case, the expected number of black kittens is $\frac{3}{4} \times 50 = 37.5$ and the expected number of gray kittens is $\frac{1}{4} \times 50 = 12.5$. The chi-square (χ^2) value is calculated by using the following formula:

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

where Σ means the sum of all the squared differences between observed and expected divided by the expected values. To calculate the chi-square value for our black and gray kittens, we would first subtract the number of *expected* black kittens from the number of *observed* black kittens ($30 - 37.5 = -7.5$) and square this value: $-7.5^2 = 56.25$. We then divide this result by the expected number of black kittens, $56.25/37.5 = 1.5$. We repeat the calculations on the number of expected gray kittens: $(20 - 12.5)^2/12.5 = 4.5$. To obtain the overall chi-square value, we sum the (observed - expected)²/expected values: $1.5 + 4.5 = 6.0$.

Table 3.4 Critical values of the χ^2 distribution									
	<i>P</i>								
df	.995	.975	.9	.5	.1	.05	.025	.01	.005
1	.000	.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879
2	0.010	0.051	0.211	1.386	4.605	5.991	7.378	9.210	10.597
3	0.072	0.216	0.584	2.366	6.251	7.815	9.348	11.345	12.838
4	0.207	0.484	1.064	3.357	7.779	9.488	11.143	13.277	14.860
5	0.412	0.831	1.610	4.351	9.236	11.070	12.832	15.086	16.750
6	0.676	1.237	2.204	5.348	10.645	12.592	14.449	16.812	18.548
7	0.989	1.690	2.833	6.346	12.017	14.067	16.013	18.475	20.278
8	1.344	2.180	3.490	7.344	13.362	15.507	17.535	20.090	21.955
9	1.735	2.700	4.168	8.343	14.684	16.919	19.023	21.666	23.589
10	2.156	3.247	4.865	9.342	15.987	18.307	20.483	23.209	25.188
11	2.603	3.816	5.578	10.341	17.275	19.675	21.920	24.725	26.757
12	3.074	4.404	6.304	11.340	18.549	21.026	23.337	26.217	28.300
13	3.565	5.009	7.042	12.340	19.812	22.362	24.736	27.688	29.819
14	4.075	5.629	7.790	13.339	21.064	23.685	26.119	29.141	31.319
15	4.601	6.262	8.547	14.339	22.307	24.996	27.488	30.578	32.801

P, probability; df, degrees of freedom.

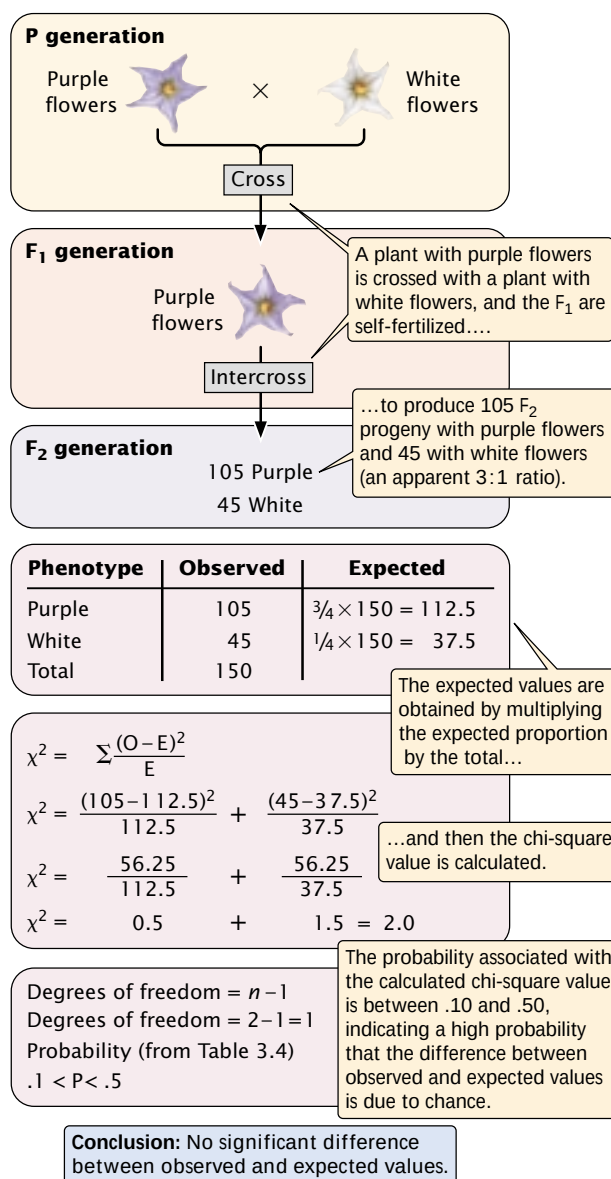
The next step is to determine the probability associated with this calculated chi-square value, which is the probability that the deviation between the observed and the expected results could be due to chance. This step requires us to compare the calculated chi-square value (6.0) with theoretical values that have the same degrees of freedom in a chi-square table. The degrees of freedom represent the number of ways in which the observed classes are free to vary. For a goodness-of-fit chi-square test, the degrees of freedom are equal to $n - 1$, where n is the number of different expected phenotypes. In our example, there are two expected phenotypes (black and gray); so $n = 2$ and the degree of freedom equals $2 - 1 = 1$.

Now that we have our calculated chi-square value and have figured out the associated degrees of freedom, we are ready to obtain the probability from a chi-square table (Table 3.4). The degrees of freedom are given in the left-hand column of the table and the probabilities are given at the top; within the body of the table are chi-square values associated with these probabilities. First, find the row for the appropriate degrees of freedom; for our example with 1 degree of freedom, it is the first row of the table. Find where our calculated chi-square value (6.0) lies among the theoretical values in this row. The theoretical chi-square values increase from left to right and the probabilities decrease from left to right. Our chi-square value of 6.0 falls between the value of 5.024, associated with a probability of .025, and the value of 6.635, associated with a probability of .01.

Thus, the probability associated with our chi-square value is less than .025 and greater than .01. So, there is less than a 2.5% probability that the deviation that we observed between the expected and the observed numbers of black and gray kittens could be due to chance.

Most scientists use the .05 probability level as their cutoff value: if the probability of chance being responsible for the deviation is greater than or equal to .05, they accept that chance may be responsible for the deviation between the observed and the expected values. When the probability is less than .05, scientists assume that chance is not responsible and a significant difference exists. The expression *significant difference* means that some factor other than chance is responsible for the observed values being different from the expected values. In regard to the kittens, perhaps one of the genotypes experienced increased mortality before the progeny were counted or perhaps other genetic factors skewed the observed ratios.

In choosing .05 as the cutoff value, scientists have agreed to assume that chance is responsible for the deviations between observed and expected values unless there is strong evidence to the contrary. It is important to bear in mind that even if we obtain a probability of, say, .01, there is still a 1% probability that the deviation between the observed and expected numbers is due to nothing more than chance. Calculation of the chi-square value is illustrated in (FIGURE 3.15).



3.15 A chi-square test is used to determine the probability that the difference between observed and expected values is due to chance.

Concepts

Differences between observed and expected ratios can arise by chance. The goodness-of-fit chi-square test can be used to evaluate whether deviations between observed and expected numbers are likely to be due to chance or to some other significant factor.

Penetrance and Expressivity

In the genetic crosses considered thus far, we have assumed that every individual with a particular genotype expresses

the expected phenotype. We assumed, for example, that the genotype Rr always produces round seeds and that the genotype rr always produces wrinkled seeds. For some characters, such an assumption is incorrect: the genotype does not always produce the expected phenotype, a phenomenon termed **incomplete penetrance**.

Incomplete penetrance is seen in human polydactyly, the condition of having extra fingers and toes (FIGURE 3.16). There are several different forms of human polydactyly, but the trait is usually caused by a dominant allele. Occasionally, people possess the allele for polydactyly (as evidenced by the fact that their children inherit the polydactyly) but nevertheless have a normal number of fingers and toes. In these cases, the gene for polydactyly is not fully penetrant. **Penetrance** is defined as the percentage of individuals having a particular genotype that express the expected phenotype. For example, if we examined 42 people having an allele for polydactyly and found that only 38 of them were polydactylous, the penetrance would be $38/42 = 0.90$ (90%).

A related concept is that of **expressivity**, the degree to which a character is expressed. In addition to incomplete penetrance, polydactyly exhibits variable expressivity. Some polydactylous persons possess extra fingers and toes that are fully functional, whereas others possess only a small tag of extra skin.

Incomplete penetrance and variable expressivity are due to the effects of other genes and to environmental factors that can alter or completely suppress the effect of a particular gene. A gene might encode an enzyme that produces a particular phenotype only within a limited temperature range. At higher or lower temperatures, the enzyme would not function and the phenotype would not be expressed; the allele encoding such an enzyme is therefore penetrant only within a particular temperature range. Many characters exhibit incomplete penetrance and variable expressivity,



3.16 Human polydactyly (extra digits) exhibits incomplete penetrance and variable expressivity. (Biophoto Associates/Science Source/Photo Researchers.)

emphasizing the fact that the mere presence of a gene does not guarantee its expression.

Concepts

Penetrance is the percentage of individuals having a particular genotype who express the associated phenotype. Expressivity is the degree to which a trait is expressed. Incomplete penetrance and variable expressivity result from the influence of other genes and environmental factors on the phenotype.

Connecting Concepts Across Chapters

This chapter has introduced several important concepts of heredity and presented techniques for making predictions about the types of offspring that parents will produce. Two key principles of inheritance were introduced: the principles of segregation and independent assortment. These principles serve as the foundation for understanding much of heredity. In this chapter, we also learned some essential terminology and techniques for discussing and analyzing genetic crosses. A critical concept is the connection between the behavior of chromosomes during meiosis (Chapter 2) and the seemingly abstract symbols used in genetic crosses.

The principles taught in this chapter provide important links to much of what follows in this book. In Chapters 4 through 7, we will learn about additional factors that affect the outcome of genetic crosses: sex, interactions between genes, linkage between genes, and environment. These factors build on the principles of segregation and independent assortment. In Chapters 10 through 21, where we focus on molecular aspects of heredity, the importance of these basic principles is not so obvious, but most nuclear processes are based on the inheritance of chromosomal genes. In Chapters 22 and 23, we turn to quantitative and population genetics. These chapters build directly on the principles of heredity and can only be understood with a firm grasp of how genes are inherited. The material covered in the present chapter therefore serves as a foundation for almost all of heredity.

Finally, this chapter introduces problem solving, which is at the heart of genetics. Developing hypotheses to explain genetic phenomenon (such as the types and proportions of progeny produced in a genetic cross) and testing these hypotheses by doing genetic crosses and collecting additional data are common to all of genetics. The ability to think analytically and draw logical conclusions from observations are emphasized throughout this book.

CONCEPTS SUMMARY

- Gregor Mendel, an Austrian monk living in what is now the Czech Republic, first discovered the principles of heredity by conducting experiments on pea plants.
- Mendel's success can be attributed to his choice of the pea plant as an experimental organism, the use of characters with a few, easily distinguishable phenotypes, his experimental approach, and careful attention to detail.
- Genes are inherited factors that determine a character. Alternate forms of a gene are called alleles. The alleles are located at a specific place, a locus, on a chromosome, and the set of genes that an individual possesses is its genotype. Phenotype is the manifestation or appearance of a characteristic and may refer to physical, biochemical, or behavioral characteristics.
- Phenotypes are produced by the combined effects of genes and environmental factors. Only the genotype—not the phenotype—is inherited.
- The principle of segregation states that an individual possesses two alleles coding for a trait and that these two alleles separate in equal proportions when gametes are formed.
- The concept of dominance indicates that, when dominant and recessive alleles are present in a heterozygote, only the trait of the dominant allele is observed in the phenotype.
- The two alleles of a genotype are located on homologous chromosomes, which separate during anaphase I of meiosis. The separation of homologous chromosomes brings about the segregation of alleles.
- The types of progeny produced from a genetic cross can be predicted by applying the Punnett square or probability.
- Probability is the likelihood of a particular event occurring. The multiplication rule of probability states that the probability of two or more independent events occurring together is calculated by multiplying the probabilities of the independent events. The addition rule of probability states that the probability of any of two or more mutually exclusive events occurring is calculated by adding the probabilities of the events.
- The binomial expansion may be used to determine the probability of a particular combination of events.
- A testcross reveals the genotype (homozygote or heterozygote) of an individual having a dominant trait and consists of crossing that individual with one having the homozygous recessive genotype.
- Incomplete dominance occurs when a heterozygote has a phenotype that is intermediate between the phenotypes of the two homozygotes.
- The principle of independent assortment states that genes coding for different characters assort independently when gametes are formed.
- Independent assortment is based on the random separation of homologous pairs of chromosomes during anaphase I of meiosis; it occurs when genes coding for two characters are located on different pairs of chromosomes.
- When genes assort independently, the multiplication rule of probability can be used to obtain the probability of inheriting more than one trait: a cross including more than one trait can be broken down into simple crosses, and the probabilities of obtaining any combination of traits can be obtained by multiplying the probabilities for each trait.
- Observed ratios of progeny from a genetic cross may deviate from the expected ratios owing to chance. The goodness-of-fit chi-square test can be used to determine the probability that a difference between observed and expected numbers is due to chance.
- Penetrance is the percentage of individuals with a particular genotype that exhibit the expected phenotype. Expressivity is the degree to which a character is expressed. Incomplete penetrance and variable expressivity result from the influence of other genes and environmental effects on the phenotype.

IMPORTANT TERMS

gene (p. 47)	F ₂ (filial 2) generation (p. 49)	multiplication rule (p. 54)	goodness-of-fit chi-square test (p. 66)
allele (p. 47)	dominant (p. 51)	addition rule (p. 55)	incomplete penetrance (p. 68)
locus (p. 47)	recessive (p. 51)	testcross (p. 57)	penetrance (p. 68)
genotype (p. 48)	principle of segregation (Mendel's first law) (p. 51)	incomplete dominance (p. 58)	expressivity (p. 68)
homozygous (p. 48)	concept of dominance (p. 51)	wild type (p. 58)	
heterozygous (p. 48)	chromosome theory of heredity (p. 52)	dihybrid cross (p. 59)	
phenotype (p. 48)	backcross (p. 52)	principle of independent assortment (Mendel's second law) (p. 60)	
monohybrid cross (p. 48)	Punnett square (p. 53)	trihybrid cross (p. 63)	
P (parental) generation (p. 48)	probability (p. 54)		
F ₁ (filial 1) generation (p. 49)			
reciprocal crosses (p. 49)			

Worked Problems

1. Short hair in rabbits (S) is dominant over long hair (s). The following crosses are carried out, producing the progeny shown. Give all possible genotypes of the parents in each cross.

Parents	Progeny
(a) short $\times$ short	4 short and 2 long
(b) short $\times$ short	8 short
(c) short $\times$ long	12 short
(d) short $\times$ long	3 short and 1 long
(e) long $\times$ long	2 long

• Solution

For this problem, it is useful to first gather as much information about the genotypes of the parents as possible on the basis of their phenotypes. We can then look at the types of progeny produced to provide the missing information. Notice that the problem asks for *all* possible genotypes of the parents.

(a) short $\times$ short 4 short and 2 long

Because short hair is dominant over long hair, a rabbit having short hair could be either SS or Ss . The two long-haired offspring must be homozygous (ss) because long hair is recessive and will appear in the phenotype only when both alleles for long hair are present. Because each parent contributes one of the two alleles found in the progeny, each parent must be carrying the s allele and must therefore be Ss .

(b) short $\times$ short 8 short

The short-haired parents could be SS or Ss . All 8 of the offspring are short (S_{-}), and so at least one of the parents is likely to be homozygous (SS); if both parents were heterozygous, $1/4$ long-haired (ss) progeny would be expected, but we do not observe any long-haired progeny. The other parent could be homozygous (SS) or heterozygous (Ss); as long as one parent is homozygous, all the offspring will be short haired. It is theoretically possible, although unlikely, that both parents are heterozygous ($Ss \times Ss$). If this were the case, we would expect 2 of the 8 progeny to be long haired. Although no long-haired rabbits would be produced among the 8 progeny of the cross.

(c) short $\times$ long 12 short

The short-haired parent could be SS or Ss . The long-haired parent must be ss . If the short-haired parent were heterozygous (Ss), half of the offspring would be expected to be long haired, but we don't see any long-haired progeny. Therefore this parent is most likely homozygous (SS). It is theoretically possible, although unlikely, that the parent is heterozygous and just by chance no long-haired progeny were produced.

(d) short $\times$ long 3 short and 1 long

On the basis of its phenotype, the short-haired parent could be homozygous (SS) or heterozygous (Ss), but the presence of one long-haired offspring tells us that the short-haired parent

must be heterozygous (Ss). The long-haired parent must be homozygous (ss).

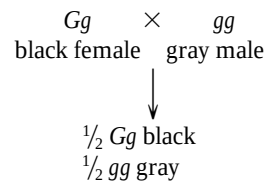
(e) long $\times$ long 2 long

Because long hair is recessive, both parents must be homozygous for a long-hair allele (ss).

2. In cats, black coat color is dominant over gray. A female black cat whose mother is gray mates with a gray male. If this female has a litter of six kittens, what is the probability that three will be black and three will be gray?

• Solution

Because black (G) is dominant over gray (g), a black cat may be homozygous (GG) or heterozygous (Gg). The black female in this problem must be heterozygous (Bb) because her mother is gray (gg) and she must inherit one of her mother's alleles. The gray male is homozygous (gg) because gray is recessive. Thus the cross is:



We can use the binomial expansion to determine the probability of obtaining three black and three gray kittens in a litter of six. Let a equal the probability of a kitten being black and b equal the probability of a kitten being gray. The binomial is $(a + b)^6$, the expansion of which is:

$$(a + b)^6 = a^6 + 6a^5b + 15a^4b^2 + 20a^3b^3 + 15a^2b^4 + 6a^1b^5 + b^6$$

(See text for an explanation of how to expand the binomial.) The probability of obtaining three black and three gray kittens in a litter of six is provided by the term $20a^3b^3$. The probabilities of a and b are both $1/2$, so the overall probability is $20(1/2)^3(1/2)^3 = 20/64 = 5/16$.

3. The following genotypes are crossed: $AaBbCcDd \times AaBbCcDd$. Give the proportion of the progeny of this cross having the following genotypes: (a) $AaBbCcDd$, (b) $aabbccdd$, (c) $AaBbccDd$.

• Solution

This problem is easily worked if the cross is broken down into simple crosses and the multiplication rule is used to find the different combinations of genotypes:

$$\begin{array}{ll}
 \text{Locus 1} & Aa \times Aa = \frac{1}{4} AA, \frac{1}{2} Aa, \frac{1}{4} aa \\
 \text{Locus 2} & Bb \times Bb = \frac{1}{4} BB, \frac{1}{2} Bb, \frac{1}{4} bb
 \end{array}$$

$$\begin{array}{ll}\text{Locus 3} & Cc \times Cc = \frac{1}{4} CC, \frac{1}{2} Cc, \frac{1}{4} cc \\ \text{Locus 4} & Dd \times Dd = \frac{1}{4} DD, \frac{1}{2} Dd, \frac{1}{4} dd\end{array}$$

To find the probability of any combination of genotypes, simply multiply the probabilities of the different genotypes:

$$\begin{array}{ll}\text{(a) } AaBbCcDd & \frac{1}{2} (Aa) \times \frac{1}{2} (Bb) \times \frac{1}{2} (Cc) \times \frac{1}{2} (Dd) = \frac{1}{16} \\ \text{(b) } aabbccdd & \frac{1}{4} (aa) \times \frac{1}{4} (bb) \times \frac{1}{4} (cc) \times \frac{1}{4} (dd) = \frac{1}{256} \\ \text{(c) } AaBbccDd & \frac{1}{2} (Aa) \times \frac{1}{2} (Bb) \times \frac{1}{4} (cc) \times \frac{1}{2} (Dd) = \frac{1}{32}\end{array}$$

4. In corn, purple kernels are dominant over yellow kernels, and full kernels are dominant over shrunken kernels. A corn plant having purple and full kernels is crossed with a plant having yellow and shrunken kernels, and the following progeny are obtained:

purple, full	112
purple, shrunken	103
yellow, full	91
yellow, shrunken	94

What are the most likely genotypes of the parents and progeny? Test your genetic hypothesis with a chi-square test.

• Solution

The best way to begin this problem is by breaking the cross down into simple crosses for a single characteristic (seed color or seed shape):

P	purple × yellow	full × shrunken
F ₁	112 + 103 = 215 purple 91 + 94 = 185 yellow	112 + 91 = 203 full 103 + 94 = 197 shrunken

Purple × yellow produces approximately $\frac{1}{2}$ purple and $\frac{1}{2}$ yellow. A 1:1 ratio is usually caused by a cross between a heterozygote and a homozygote. Because purple is dominant, the purple parent must be heterozygous (Pp) and the yellow parent must be homozygous (pp). The purple progeny produced by this cross will be heterozygous (Pp) and the yellow progeny must be homozygous (pp).

Now let's examine the other character. Full × shrunken produces $\frac{1}{2}$ full and $\frac{1}{2}$ shrunken, or a 1:1 ratio, and so these progeny phenotypes also are produced by a cross between a heterozygote (Ff) and a homozygote (ff); the full-kernel progeny will be heterozygous (Ff) and the shrunken-kernel progeny will be homozygous (ff).

Now combine the two crosses and use the multiplication rule to obtain the overall genotypes and the proportions of each genotype:

P	purple, full × yellow, shrunken
	$PpFf \times ppff$
F ₁	$PpFf = \frac{1}{2} \text{ purple} \times \frac{1}{2} \text{ full} = \frac{1}{4} \text{ purple, full}$
	$Ppff = \frac{1}{2} \text{ purple} \times \frac{1}{2} \text{ shrunken} = \frac{1}{4} \text{ purple, shrunken}$
	$ppFf = \frac{1}{2} \text{ yellow} \times \frac{1}{2} \text{ full} = \frac{1}{4} \text{ yellow, full}$
	$ppff = \frac{1}{2} \text{ yellow} \times \frac{1}{2} \text{ shrunken} = \frac{1}{4} \text{ yellow shrunken}$

Our genetic explanation predicts that, from this cross, we should see $\frac{1}{4}$ purple, full-kernel progeny; $\frac{1}{4}$ purple, shrunken-kernel progeny; $\frac{1}{4}$ yellow, full-kernel progeny; and $\frac{1}{4}$ yellow, shrunken-kernel progeny. A total of 400 progeny were produced; so $\frac{1}{4} \times 400 = 100$ of each phenotype are expected. These observed numbers do not fit the expected numbers exactly. Could the difference between what we observe and what we expect be due to chance? If the probability is high that chance alone is responsible for the difference between observed and expected, we will assume that the progeny have been produced in the 1:1:1:1 ratio predicted by the cross. If the probability that the difference between observed and expected is due to chance is low, the progeny are not really in the predicted ratio and some other, *significant* factor must be responsible for the deviation.

The observed and expected numbers are:

Phenotype	Observed	Expected
purple full	112	$\frac{1}{4} \times 400 = 100$
purple shrunken	103	$\frac{1}{4} \times 400 = 100$
yellow full	91	$\frac{1}{4} \times 400 = 100$
yellow shrunken	94	$\frac{1}{4} \times 400 = 100$

To determine the probability that the difference between observed and expected is due to chance, we calculate a chi-square value with the formula $\chi^2 = \Sigma[(\text{observed} - \text{expected})^2 / \text{expected}]$:

$$\begin{aligned}\chi^2 &= \frac{(112 - 100)^2}{100} + \frac{(103 - 100)^2}{100} + \frac{(91 - 100)^2}{100} \\ &\quad + \frac{(94 - 100)^2}{100} \\ &= \frac{12^2}{100} + \frac{3^2}{100} + \frac{9^2}{100} + \frac{6^2}{100} \\ &= \frac{144}{100} + \frac{9}{100} + \frac{81}{100} + \frac{36}{100} \\ &= 1.44 + 0.09 + 0.81 + 0.36 = 2.70\end{aligned}$$

Now that we have the chi-square value, we must determine the probability of this chi-square value being due to chance. To obtain this probability, we first calculate the degrees of freedom, which for a goodness-of-fit chi-square test are $n - 1$, where n equals the number of expected phenotypic classes. In this case, there are four expected phenotypic classes; so the degrees of freedom equal $4 - 1 = 3$. We must now look up the chi-square value in a chi-square table (see Table 3.4). We select the row corresponding to 3 degrees of freedom and look along this row to find our calculated chi-square value. The calculated chi-square value of 2.7 lies between 2.366 (a probability of .5) and 6.251 (a probability of .1). The probability (P) associated with the calculated chi-square value is therefore $.5 < P < .1$. This is the probability that the difference between what we observed and what we expect is due to chance, which in this case is relatively high, and so chance is likely responsible for the deviation. We can conclude that the progeny *do* appear in the 1:1:1:1 ratio predicted by our genetic explanation.

COMPREHENSION QUESTIONS

- * 1. Why was Mendel's approach to the study of heredity so successful?
2. What is the relation between the terms *allele*, *locus*, *gene*, and *genotype*?
- * 3. What is the principle of segregation? Why is it important?
4. What is the concept of dominance? How does dominance differ from incomplete dominance?
5. Give the phenotypic ratios that may appear among the progeny of simple crosses and the genotypes of the parents that may give rise to each ratio.
6. Give the genotypic ratios that may appear among the progeny of simple crosses and the genotypes of the parents that may give rise to each ratio.
- * 7. What is the chromosome theory of inheritance? Why was it important?
8. What is the principle of independent assortment? How is it related to the principle of segregation?
9. How is the principle of independent assortment related to meiosis?
10. How is the goodness-of-fit chi-square test used to analyze genetic crosses? What does the probability associated with a chi-square value indicate about the results of a cross?
11. What is incomplete penetrance and what causes it?

APPLICATION QUESTIONS AND PROBLEMS

- *12. In cucumbers, orange fruit color (*R*) is dominant over cream fruit color (*r*). A cucumber plant homozygous for orange fruits is crossed with a plant homozygous for cream fruits. The F_1 are intercrossed to produce the F_2 .
 - (a) Give the genotypes and phenotypes of the parents, the F_1 , and the F_2 .
 - (b) Give the genotypes and phenotypes of the offspring of a backcross between the F_1 and the orange parent.
 - (c) Give the genotypes and phenotypes of a backcross between the F_1 and the cream parent.
- *13. In rabbits, coat color is a genetically determined characteristic. Some black females always produce black progeny, whereas other black females produce black progeny and white progeny. Explain how these outcomes occur.
- *14. In cats, blood type A results from an allele (I^A) that is dominant over an allele (i^B) that produces blood type B. There is no O blood type. The blood types of male and female cats that were mated and the blood types of their kittens follow. Give the most likely genotypes for the parents of each litter.

Male parent	Female parent	Kittens
(a) blood type A	blood type B	4 kittens with blood type A, 3 with blood type B
(b) blood type B	blood type B	6 kittens with blood type B
(c) blood type B	blood type A	8 kittens with blood type A
(d) blood type A	blood type A	7 kittens with blood type A, 2 kittens with blood type B
- (e) blood type A blood type A 10 kittens with blood type A
- (f) blood type A blood type B 4 kittens with blood type A, 1 kitten with blood type B
15. In sheep, lustrous fleece (*L*) results from an allele that is dominant over an allele for normal fleece (*l*). A ewe (adult female) with lustrous fleece is mated with a ram (adult male) with normal fleece. The ewe then gives birth to a single lamb with normal fleece. From this single offspring, is it possible to determine the genotypes of the two parents? If so, what are their genotypes? If not, why not?
- *16. In humans, alkaptonuria is a metabolic disorder in which affected persons produce black urine (see the introduction to this chapter). Alkaptonuria results from an allele (*a*) that is recessive to the allele for normal metabolism (*A*). Sally has normal metabolism, but her brother has alkaptonuria. Sally's father has alkaptonuria, and her mother has normal metabolism.
 - (a) Give the genotypes of Sally, her mother, her father, and her brother.
 - (b) If Sally's parents have another child, what is the probability that this child will have alkaptonuria?
 - (c) If Sally marries a man with alkaptonuria, what is the probability that their first child will have alkaptonuria?
17. Suppose that you are raising Mongolian gerbils. You notice that some of your gerbils have white spots, whereas others have solid coats. What type of crosses could you carry out to determine whether white spots are due to a recessive or a dominant allele?

- *18. Hairlessness in American rat terriers is recessive to the presence of hair. Suppose that you have a rat terrier with hair. How can you determine whether this dog is homozygous or heterozygous for the hairy trait?
19. In snapdragons, red flower color (R) is incompletely dominant over white flower color (r); the heterozygotes produce pink flowers. A red snapdragon is crossed with a white snapdragon, and the F_1 are intercrossed to produce the F_2 .
- Give the genotypes and phenotypes of the F_1 and F_2 , along with their expected proportions.
 - If the F_1 are backcrossed to the white parent, what will the genotypes and phenotypes of the offspring be?
 - If the F_1 are backcrossed to the red parent, what are the genotypes and phenotypes of the offspring?
20. What is the probability of rolling one six-sided die and obtaining the following numbers?
- 2
 - 1 or 2
 - An even number
 - Any number but a 6
- *21. What is the probability of rolling two six-sided dice and obtaining the following numbers?
- 2 and 3
 - 6 and 6
 - At least one 6
 - Two of the same number (two 1s, or two 2s, or two 3s, etc.)
 - An even number on both dice
 - An even number on at least one die
- *22. In a family of seven children, what is the probability of obtaining the following numbers of boys and girls?
- All boys
 - All children of the same sex
 - Six girls and one boy
 - Four boys and three girls
 - Four girls and three boys
23. Phenylketonuria (PKU) is a disease that results from a recessive gene. Two normal parents produce a child with PKU.
- What is the probability that a sperm from the father will contain the PKU allele?
 - What is the probability that an egg from the mother will contain the PKU allele?
 - What is the probability that their next child will have PKU?
 - What is the probability that their next child will be heterozygous for the PKU gene?
- *24. In German cockroaches, curved wing (cv) is recessive to normal wing (cv^+). A homozygous cockroach having normal wings is crossed with a homozygous cockroach having curved wings. The F_1 are intercrossed to produce the F_2 . Assume that the pair of chromosomes containing the locus for wing shape is metacentric. Draw this pair of chromosomes as it would appear in the parents, the F_1 , and each class of F_2 progeny at metaphase I of meiosis. Assume that no crossing over takes place. At each stage, label a location for the alleles for wing shape (cv and cv^+) on the chromosomes.
- *25. In guinea pigs, the allele for black fur (B) is dominant over the allele for brown (b) fur. A black guinea pig is crossed with a brown guinea pig, producing five F_1 black guinea pigs and six F_1 brown guinea pigs.
- How many copies of the black allele (B) will be present in *each* cell from an F_1 black guinea pig at the following stages: G_1 , G_2 , metaphase of mitosis, metaphase I of meiosis, metaphase II of meiosis, and after the second cytokinesis following meiosis? Assume that no crossing over takes place.
 - How many copies of the brown allele (b) will be present in each cell from an F_1 brown guinea pig at the same stages? Assume that no crossing over takes place.
26. In watermelons, bitter fruit (B) is dominant over sweet fruit (b), and yellow spots (S) are dominant over no spots (s). The genes for these two characteristics assort independently. A homozygous plant that has bitter fruit and yellow spots is crossed with a homozygous plant that has sweet fruit and no spots. The F_1 are intercrossed to produce the F_2 .
- What will be the phenotypic ratios in the F_2 ?
 - If an F_1 plant is backcrossed with the bitter, yellow spotted parent, what phenotypes and proportions are expected in the offspring?
 - If an F_1 plant is backcrossed with the sweet, nonspotted parent, what phenotypes and proportions are expected in the offspring?
27. In cats, curled ears (Cu) result from an allele that is dominant over an allele for normal ears (cu). Black color results from an independently assorting allele (G) that is dominant over an allele for gray (g). A gray cat homozygous for curled ears is mated with a homozygous black cat with normal ears. All the F_1 cats are black and have curled ears.
- If two of the F_1 cats mate, what phenotypes and proportions are expected in the F_2 ?
 - An F_1 cat mates with a stray cat that is gray and possesses normal ears. What phenotypes and proportions of progeny are expected from this cross?
- *28. The following two genotypes are crossed: $AaBbCcddEe \times AabbCcDdEe$. What will the proportion of the following genotypes be among the progeny of this cross?
- $AaBbCcDdEe$
 - $AabbCcddEe$
 - $aabbccdde$
 - $AABBCCDDEE$

29. In mice, an allele for apricot eyes (a) is recessive to an allele for brown eyes (a^+). At an independently assorting locus, an allele for tan (t) coat color is recessive to an allele for black (t^+) coat color. A mouse that is homozygous for brown eyes and black coat color is crossed with a mouse having apricot eyes and a tan coat. The resulting F_1 are intercrossed to produce the F_2 . In a litter of eight F_2 mice, what is the probability that two will have apricot eyes and tan coats?
30. In cucumbers, dull fruit (D) is dominant over glossy fruit (d), orange fruit (R) is dominant over cream fruit (r), and bitter cotyledons (B) are dominant over nonbitter cotyledons (b). The three characters are encoded by genes located on different pairs of chromosomes. A plant homozygous for dull, orange fruit and bitter cotyledons is crossed with a plant that has glossy, cream fruit and nonbitter cotyledons. The F_1 are intercrossed to produce the F_2 .
- (a) Give the phenotypes and their expected proportions in the F_2 .
- (b) An F_1 plant is crossed with a plant that has glossy, cream fruit and nonbitter cotyledons. Give the phenotypes and expected proportions among the progeny of this cross.
- * 31. A and a are alleles located on a pair of metacentric chromosomes. B and b are alleles located on a pair of acrocentric chromosomes. A cross is made between individuals having the following genotypes: $AaBb \times aabb$.
- (a) Draw the chromosomes as they would appear in each type of gamete produced by the individuals of this cross.
- (b) For each type of progeny resulting from this cross, draw the chromosomes as they would appear in a cell at G_1 , G_2 , and metaphase of mitosis.
32. Ptosis (droopy eyelid) may be inherited as a dominant human trait. Among 40 people who are heterozygous for the ptosis allele, 13 have ptosis and 27 have normal eyelids.
- (a) What is the penetrance for ptosis?
- (b) If ptosis exhibited variable expressivity, what would it mean?
33. In sailfin mollies (fish), gold color is due to an allele (g) that is recessive to the allele for normal color (G). A gold fish is crossed with a normal fish. Among the offspring, 88 are normal and 82 are gold.
- (a) What are the most likely genotypes of the parents in this cross?
- (b) Assess the plausibility of your hypothesis by performing a chi-square test.
34. In guinea pigs, the allele for black coat color (B) is dominant over the allele for white coat color (b). At an independently assorting locus, an allele for rough coat (R) is dominant over an allele for smooth coat (r). A guinea pig that is homozygous for black color and rough coat is crossed with a guinea pig that has a white and smooth coat. In a series of matings, the F_1 are crossed with guinea pigs having white, smooth coats. From these matings, the following phenotypes appear in the offspring: 24 black, rough guinea pigs; 26 black, smooth guinea pigs; 23 white, rough guinea pigs; and 5 white, smooth guinea pigs.
- (a) Using a chi-square test, compare the observed numbers of progeny with those expected from the cross.
- (b) What conclusions can you draw from the results of the chi-square test?
- (c) Suggest an explanation for these results.

CHALLENGE QUESTIONS

35. Dwarfism is a recessive trait in Hereford cattle. A rancher in western Texas discovers that several of the calves in his herd are dwarfs, and he wants to eliminate this undesirable trait from the herd as rapidly as possible. Suppose that the rancher hires you as a genetic consultant to advise him on how to breed the dwarfism trait out of the herd. What crosses would you advise the rancher to conduct to ensure that the allele causing dwarfism is eliminated from the herd?
36. A geneticist discovers an obese mouse in his laboratory colony. He breeds this obese mouse with a normal mouse. All the F_1 mice from this cross are normal in size. When he interbreeds two F_1 mice, eight of the F_2 mice are normal in size and two are obese. The geneticist then intercrosses two of his obese mice, and he finds that all of the progeny from this cross are obese. These results lead the geneticist to conclude that obesity in mice results from a recessive allele.
- A second geneticist at a different university also discovers an obese mouse in her laboratory colony. She carries out the same crosses as the first geneticist did and obtains the same results. She also concludes that obesity in mice results from a recessive allele. One day the two geneticists meet at a genetics conference, learn of each other's experiments, and decide to exchange mice. They both find that, when they cross two obese mice from the different laboratories, all the offspring are normal; however, when they cross two obese mice from the same laboratory, all the offspring are obese. Explain their results.
37. Albinism is a recessive trait in humans. A geneticist studies a series of families in which both parents are normal and at least one child has albinism. The geneticist reasons that both parents in these families must be heterozygotes and that albinism should appear in $1/4$ of the children of these families. To his surprise, the geneticist finds that the

frequency of albinism among the children of these families is considerably greater than $\frac{1}{4}$. There is no evidence that normal pigmentation exhibits incomplete penetrance. Can you think of an explanation for the higher-than-expected frequency of albinism among these families?

38. Two distinct phenotypes are found in the salamander *Plethodon cinereus*: a red form and a black form. Some biologists have speculated that the red phenotype is due to an autosomal allele that is dominant over an allele for black. Unfortunately, these salamanders will not mate in captivity; so the hypothesis that red is dominant over black has never been tested.

One day a genetics student is hiking through the forest and finds 30 female salamanders, some red and some black,

laying eggs. The student places each female and her eggs (about 20–30 eggs per female) in separate plastic bags and takes them back to the lab. There, the student successfully raises the eggs until they hatch. After the eggs have hatched, the student records the phenotypes of the juvenile salamanders, along with the phenotypes of their mothers. Thus, the student has the phenotypes for 30 females and their progeny, but no information is available about the phenotypes of the fathers.

Explain how the student can determine whether red is dominant over black with this information on the phenotypes of the females and their offspring.

SUGGESTED READINGS

Corcos, A., and F. Monaghan. 1985. Some myths about Mendel's experiments. *The American Biology Teacher* 47:233–236.

An excellent discussion of some misconceptions surrounding Mendel's life and discoveries.

Dronamraju, K. 1992. Profiles in genetics: Archibald E. Garrod. *American Journal of Human Genetics* 51:216–219.

A brief biography of Archibald Garrod and his contributions to genetics.

Dunn, L. C. 1965. *A Short History of Genetics*. New York: McGraw-Hill.

An older but very good history of genetics.

Garrod, A. E. 1902. The incidence of alkaptonuria: a study in chemical individuality. *Lancet* 2:1616–1620.

Garrod's original paper on the genetics of alkaptonuria.

Henig, R. M. 2001. *The Monk in the Garden: The Lost and Found Genius of Gregor Mendel, the Father of Genetics*. Boston: Houghton Mifflin.

A creative history of Gregor Mendel, in which the author has used historical research to create a vivid portrait of Mendel's life and work.

Monaghan, F. V., and A. F. Corcos. 1987. Reexamination of the fate of Mendel's paper. *Journal of Heredity* 78:116–118.

A good discussion of why Mendel's paper was unappreciated by his peers.

Orel, V. 1984. *Mendel*. Oxford: Oxford University Press.

An excellent and authoritative biography of Mendel.

Weiling, F. 1991. Historical study: Johann Gregor Mendel 1822–1884. *American Journal of Medical Genetics* 40:1–25.

A fascinating account that contains much recent research on Mendel's life as a scientist.

4

Sex Determination and Sex-Linked Characteristics



This is Chapter 4 Opener photo legend to position here. (Credit for Chapter 4 opening photo allowing 2 additional lines which If we need, if we don't then we can add to depth of photo.) (Historical Picture Archive/Corbis.)

The Toothless, Hairless Men of Sind

In 1875, Charles Darwin, author of *On the Origin of Species*, wrote of a peculiar family of Sind, a province in northwest India,

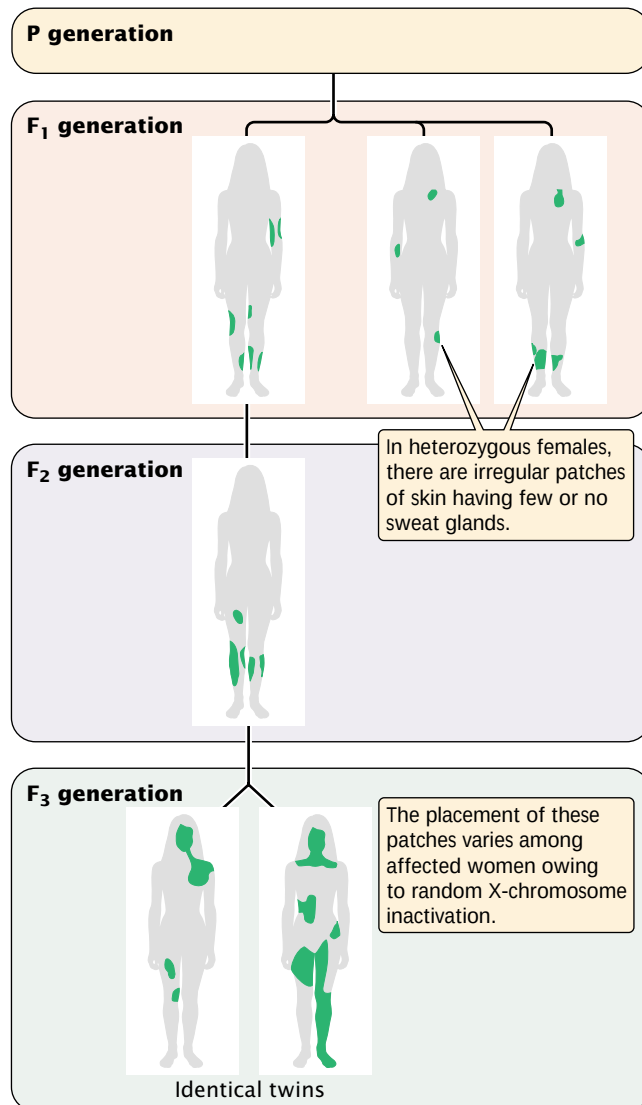
in which ten men, in the course of four generations, were furnished in both jaws taken together, with only four small and weak incisor teeth and with eight posterior molars. The men thus affected have little hair on the body, and become bald early in life. They also suffer much during hot weather from excessive dryness of the skin. It is remarkable that no instance has occurred of a daughter being thus affected. . . . Though daughters in the above family are never affected, they transmit the tendency to their sons; and no case has occurred of a son transmitting it to his sons.

These men possessed a genetic condition now known as anhidrotic ectodermal dysplasia, which (as noted by Darwin) is

characterized by small teeth, no sweat glands, and sparse body hair. Darwin also noted several key features of the inheritance of this disorder: although it occurs primarily in men, fathers never transmit the trait to their sons; unaffected daughters, however, may pass the trait to their sons (the grandsons of affected men). These features of inheritance are the hallmarks of a sex-linked trait, a major focus of this chapter. Although Darwin didn't understand the mechanism of heredity, his attention to detail and remarkable ability to focus on crucial observations allowed him to identify the essential features of this genetic disease 25 years before Mendel's principles of heredity became widely known.

Darwin claimed that the daughters of this Hindu family were never affected, but it's now known that some women do have mild cases of anhidrotic ectodermal dysplasia. In these women, the symptoms of the disorder appear on only some parts of the body. For example, some regions of the jaw are missing teeth, whereas other regions have normal teeth. There are irregular patches of skin having few or no sweat

- The Toothless, Hairless Men of Sind
- Sex Determination
 - Chromosomal Sex-Determining Systems
 - Genic Sex-Determining Systems
 - Environmental Sex Determination
 - Sex Determination in *Drosophila*
 - Sex Determination in Humans
- Sex-Linked Characteristics
 - X-linked White Eyes in *Drosophila*
 - Nondisjunction and the Chromosome Theory of Inheritance
 - X-linked Color Blindness in Humans
 - Symbols for X-linked Genes
 - Dosage Compensation
 - Z-linked Characteristics
 - Y-linked Characteristics



4.1 Three generations of women heterozygous for the X-linked recessive disorder anhidrotic ectodermal dysplasia, which is inherited as an X-linked recessive trait. (After A. P. Mance and J. Mance, *Genetics: Human Aspects*, Sinauer, 1990, p. 133.)

glands; the placement of these patches varies among affected women (FIGURE 4.1). The patchy occurrence of these features is explained by the fact that the gene for anhidrotic ectodermal dysplasia is located on a sex chromosome.

www.whfreeman.com/pierce Additional information about anhidrotic ectodermal dysplasia, including symptoms, history, and genetics

In Chapter 3, we studied Mendel's principles of segregation and independent assortment and saw how these principles explain much about the nature of inheritance. After Mendel's principles were rediscovered in 1900, biologists

began to conduct genetic studies on a wide array of different organisms. As they applied Mendel's principles more widely, exceptions were observed, and it became necessary to devise extensions to his basic principles of heredity.

In this chapter, we explore one of the major extensions to Mendel's principles: the inheritance of characteristics encoded by genes located on the sex chromosomes, which differ in males and females (FIGURE 4.2). These characteristics and the genes that produce them are referred to as sex linked. To understand the inheritance of sex-linked characteristics, we must first know how sex is determined—why some members of a species are male and others are female. Sex determination is the focus of the first part of the chapter. The second part examines how characteristics encoded by genes on the sex chromosomes are inherited. In Chapter 5, we will explore some additional ways in which sex and inheritance interact.

As we consider sex determination and sex-linked characteristics, it will be helpful to think about two important principles. First, there are several different mechanisms of sex determination and, ultimately, the mechanism of sex determination controls the inheritance of sex-linked characteristics. Second, like other pairs of chromosomes, the X and Y sex chromosomes may pair in the course of meiosis and segregate, but throughout most of their length they are not homologous (their gene sequences don't code for the same characteristics): most genes on the X chromosome are different from genes on the Y chromosome. Consequently, males and females do not possess the same number of alleles at sex-linked loci. This difference in the number of sex-linked alleles produces the distinct patterns of inheritance in males and females.



4.2 The sex chromosomes of males (Y) and females (X) are different. (Biophoto Associates/Photo Researchers.)

Sex Determination

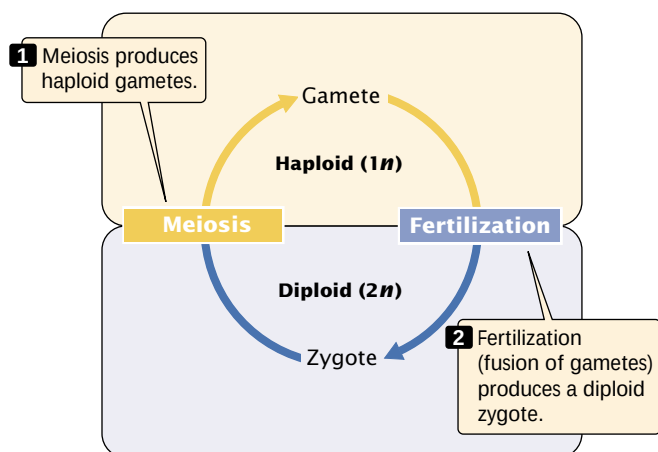
Sexual reproduction is the formation of offspring that are genetically distinct from their parents; most often, two parents contribute genes to their offspring. Among most eukaryotes, sexual reproduction consists of two processes that lead to an alternation of haploid and diploid cells: meiosis produces haploid gametes, and fertilization produces diploid zygotes (● **FIGURE 4.3**).

The term **sex** refers to sexual phenotype. Most organisms have only two sexual phenotypes: male and female. The fundamental difference between males and females is gamete size: males produce small gametes; females produce relatively large gametes (● **FIGURE 4.4**).

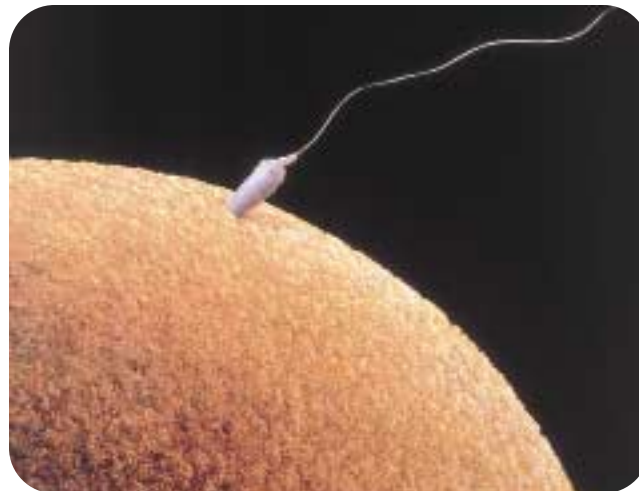
The mechanism by which sex is established is termed **sex determination**. We define the sex of an individual in terms of the individual's phenotype—ultimately, the type of gametes that it produces. Sometimes an individual has chromosomes or genes that are normally associated with one sex but a morphology corresponding to the opposite sex. For instance, the cells of female humans normally have two X chromosomes, and the cells of males have one X chromosome and one Y chromosome. A few rare persons have male anatomy, although their cells each contain two X chromosomes. Even though these people are genetically female, we refer to them as male because their sexual phenotype is male.

Concepts

In sexual reproduction, parents contribute genes to produce an offspring that is genetically distinct from both parents. In eukaryotes, sexual reproduction consists of meiosis, which produces haploid gametes, and fertilization, which produces a diploid zygote.



● **4.3** In most eukaryotic organisms, sexual reproduction consists of an alternation of haploid ($1n$) and diploid ($2n$) cells.



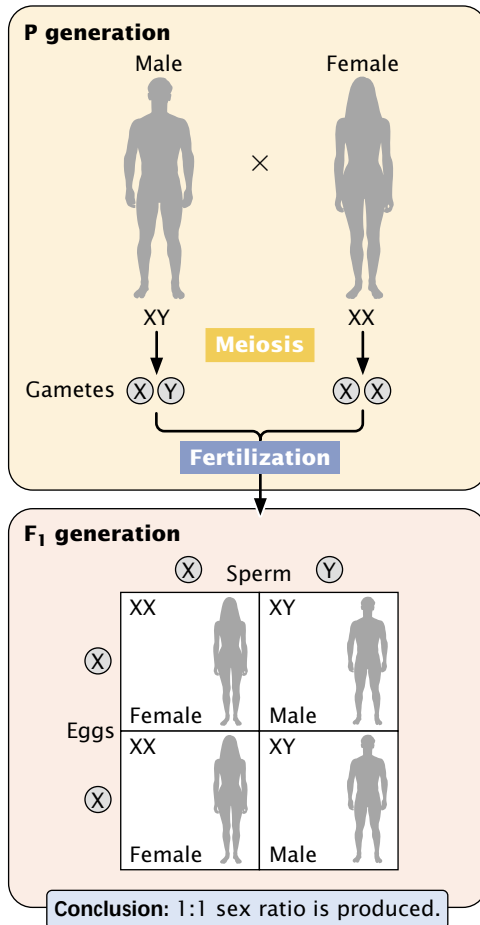
● **4.4** Male and female gametes (sperm and egg, respectively) differ in size. In this photograph, a human sperm (with flagellum) penetrates a human egg cell. (Francis Leroy, Biocosmos/Science Photo Library/Photo Researchers.)

There are many ways in which sex differences arise. In some species, both sexes are present in the same individual, a condition termed **hermaphroditism**; organisms that bear both male and female reproductive structures are said to be **monoecious** (meaning “one house”). Species in which an individual has either male or female reproductive structures are said to be **dioecious** (meaning “two houses”). Humans are dioecious. Among dioecious species, the sex of an individual may be determined chromosomally, genetically, or environmentally.

Chromosomal Sex-Determining Systems

The chromosome theory of inheritance (discussed in Chapter 3) states that genes are located on chromosomes, which serve as the vehicles for gene segregation in meiosis. Definitive proof of this theory was provided by the discovery that the sex of certain insects is determined by the presence or absence of particular chromosomes.

In 1891, Hermann Henking noticed a peculiar structure in the nuclei of cells from male insects. Understanding neither its function nor its relation to sex, he called this structure the X body. Later, Clarence E. McClung studied Henking's X body in grasshoppers and recognized that it was a chromosome. McClung called it the accessory chromosome, but eventually it became known as the X chromosome, from Henking's original designation. McClung observed that the cells of female grasshoppers had one more chromosome than the cells of male grasshoppers, and he concluded that accessory chromosomes played a role in sex determination. In 1905, Nettie Stevens and Edmund Wilson demonstrated that, in grasshoppers and other insects, the cells of females have two X chromosomes, whereas the cells of males have a single X. In some insects, they counted the same number of chromosomes in



4.5 Inheritance of sex in organisms with X and Y chromosomes results in equal numbers of male and female offspring.

cells of males and females but saw that one chromosome pair was different: two X chromosomes were found in female cells, whereas a single X chromosome plus a smaller chromosome, which they called Y, was found in male cells.

Stevens and Wilson also showed that the X and Y chromosomes separate into different cells in sperm formation; half of the sperm receive an X chromosome and half receive a Y. All egg cells produced by the female in meiosis receive one X chromosome. A sperm containing a Y chromosome unites with an X-bearing egg to produce an XY male, whereas a sperm containing an X chromosome unites with an X-bearing egg to produce an XX female (FIGURE 4.5). This accounts for the 50:50 sex ratio observed in most dioecious organisms. Because sex is inherited like other genetically determined characteristics, Stevens and Wilson's discovery that sex was associated with the inheritance of a particular chromosome also demonstrated that genes are on chromosomes.

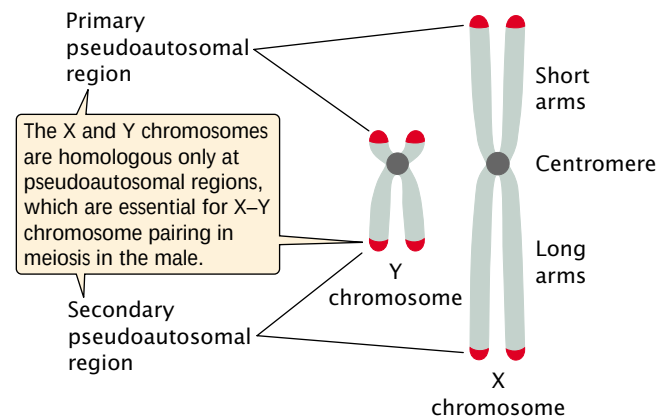
As Stevens and Wilson found for insects, sex is frequently determined by a pair of chromosomes, the **sex chromosomes**, which differ between males and females. The nonsex chromosomes, which are the same for males and females, are called

autosomes. We think of sex in these organisms as being determined by the presence of the sex chromosomes, but in fact the individual genes located on the sex chromosomes are usually responsible for the sexual phenotypes.

XX-XO sex determination The mechanism of sex determination in the grasshoppers studied by McClung is one of the simplest mechanisms of chromosomal sex determination and is called the XX-XO system. In this system, females have two X chromosomes (XX), and males possess a single X chromosome (XO). There is no O chromosome; the letter O signifies the absence of a sex chromosome.

In meiosis in females, the two X chromosomes pair and then separate, with one X chromosome entering each haploid egg. In males, the single X chromosome segregates in meiosis to half the sperm cells—the other half receive no sex chromosome. Because males produce two different types of gametes with respect to the sex chromosomes, they are said to be the **heterogametic sex**. Females, which produce gametes that are all the same with respect to the sex chromosomes, are the **homogametic sex**. In the XX-XO system, the sex of an individual is therefore determined by which type of male gamete fertilizes the egg. X-bearing sperm unite with X-bearing eggs to produce XX zygotes, which eventually develop as females. Sperm lacking an X chromosome unite with X-bearing eggs to produce XO zygotes, which develop into males.

XX-XY sex determination In many species, the cells of males and females have the same number of chromosomes, but the cells of females have two X chromosomes (XX) and the cells of males have a single X chromosome and a smaller sex chromosome called the Y chromosome (XY). In humans and many other organisms, the Y chromosome is acrocentric (FIGURE 4.6), not Y shaped as is commonly assumed. In this type of sex-determining system, the male is the heterogametic sex—half of his gametes have an X chromosome and half have a Y chromosome. The female is the



4.6 The X and Y chromosomes in humans differ in size and genetic content. They are homologous only at the pseudoautosomal regions

homogametic sex—all her egg cells contain a single X chromosome. Many organisms, including some plants, insects, and reptiles, and all mammals (including humans), have the XX-XY sex-determining system.

Although the X and Y chromosomes are not generally homologous, they do pair and segregate into different cells in meiosis. They can pair because these chromosomes are homologous at small regions called the **pseudoautosomal regions** (see Figure 4.6), in which they carry the same genes. Genes found in these regions will display the same pattern of inheritance as that of genes located on autosomal chromosomes. In humans, there are pseudoautosomal regions at both tips of the X and Y chromosomes.

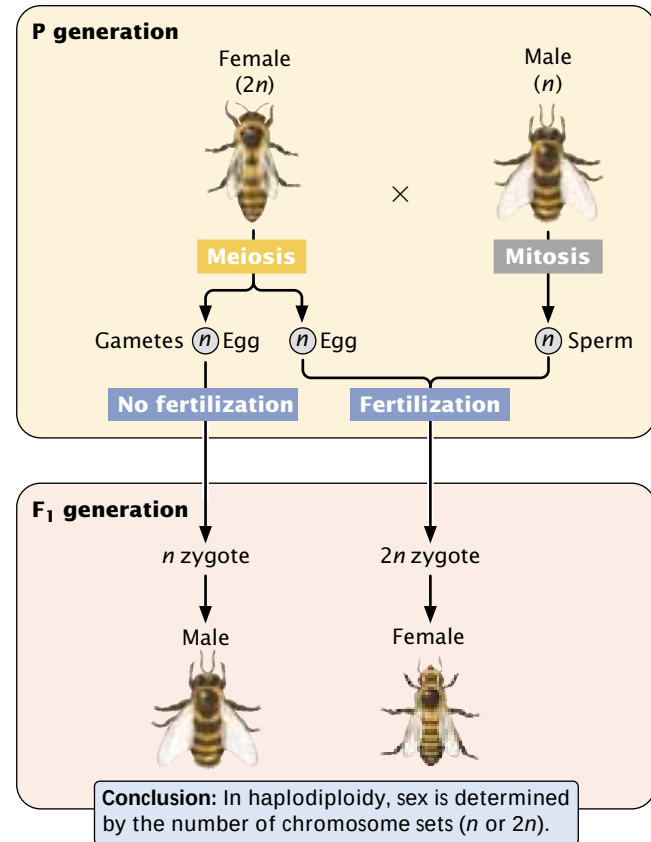
ZZ-ZW sex determination In this system, the female is heterogametic and the male is homogametic. To prevent confusion with the XX-XY system, the sex chromosomes in this system are labeled Z and W, but the chromosomes do not resemble Zs and Ws. Females in this system are ZW; after meiosis, half of the eggs have a Z chromosome and the other half have a W. Males are ZZ; all sperm contain a single Z chromosome. The ZZ-ZW system is found in birds, moths, some amphibians, and some fishes.

Concepts

In XX-XO sex determination, the male is XO and heterogametic, and the female is XX and homogametic. In XX-XY sex determination, the male is XY and the female is XX; in this system the male is heterogametic. In ZZ-ZW sex determination, the female is ZW and the male is ZZ; in this system the female is the heterogametic sex.

Haplodiploidy Some insects in the order Hymenoptera (bees, wasps, and ants) have no sex chromosomes; instead, sex is based on the number of chromosome sets found in the nucleus of each cell. Males develop from unfertilized eggs, and females develop from fertilized eggs. The cells of male hymenopterans possess only a single set of chromosomes (they are haploid) inherited from the mother. In contrast, the cells of females possess two sets of chromosomes (they are diploid), one set inherited from the mother and the other set from the father (FIGURE 4.7).

The haplodiploid method of sex determination produces some odd genetic relationships. When both parents are diploid, siblings on average have half their genes in common because they have a 50% chance of receiving the same allele from each parent. In these insects, males produce sperm by mitosis (they are already haploid); so all offspring receive the same set of paternal genes. The diploid females produce eggs by normal meiosis. Therefore, sisters have a 50% chance of



4.7 In insects with haplodiploidy, males develop from unfertilized eggs and are haploid; females develop from fertilized eggs and are diploid.

receiving the same allele from their mother and a 100% chance of receiving the same allele from their father; the average relatedness between sisters is therefore 75%. Brothers have a 50% chance of receiving the same copy of each of their mother's two alleles at any particular locus; so their average relatedness is only 50%. The greater genetic relatedness among female siblings in insects with haplodiploid sex determination may contribute to the high degree of social cooperation that exists among females (the workers) of these insects.

Concepts

Some insects possess haplodiploid sex determination, in which males develop from unfertilized eggs and are haploid; females develop from fertilized eggs and are diploid.

Genic Sex-Determining Systems

In some plants and protozoans, sex is genetically determined, but there are no obvious differences in the chromosomes of males and females—there are no sex chromosomes. These

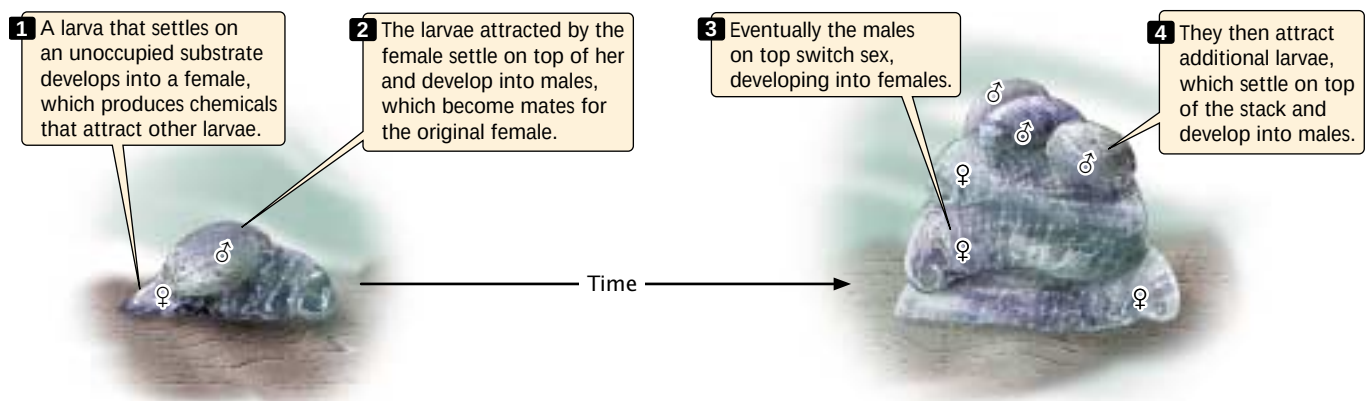
organisms have **genic sex determination**; genotypes at one or more loci determine the sex of an individual.

It is important to understand that, even in chromosomal sex-determining systems, sex is actually determined by individual genes. For example, in mammals, a gene (*SRY*, discussed later in this chapter) located on the Y chromosome determines the male phenotype. In both genic sex determination and chromosomal sex determination, sex is controlled by individual genes; the difference is that, with chromosomal sex determination, the chromosomes that carry those genes *appear* different in males and females.

Environmental Sex Determination

Genes have had a role in all of the examples of sex determination discussed thus far, but sex is determined fully or in part by environmental factors in a number of organisms.

One fascinating example of environmental sex determination is seen in the marine mollusk *Crepidula fornicata*, also known as the common slipper limpet (FIGURE 4.8). Slipper limpets live in stacks, one on top of another. Each limpet begins life as a swimming larva. The first larva to settle on a solid, unoccupied substrate develops into a female limpet. It then produces chemicals that attract other larvae, which settle on top of it. These larvae develop into males, which then serve as mates for the limpet below. After a period of time, the males on top develop into females and, in turn, attract additional larvae that settle on top of the stack, develop into males, and serve as mates for the limpets under them. Limpets can form stacks of a dozen or more animals; the uppermost animals are always male. This type of sexual development is called **sequential hermaphroditism**; each individual animal can be both male and female, although not at the same time. In *Crepidula fornicata*, sex is determined environmentally by the limpet's position in the stack.



4.8 In *Crepidula fornicata*, the common slipper limpet, sex is determined by an environmental factor, the limpet's position in a stack of limpets.

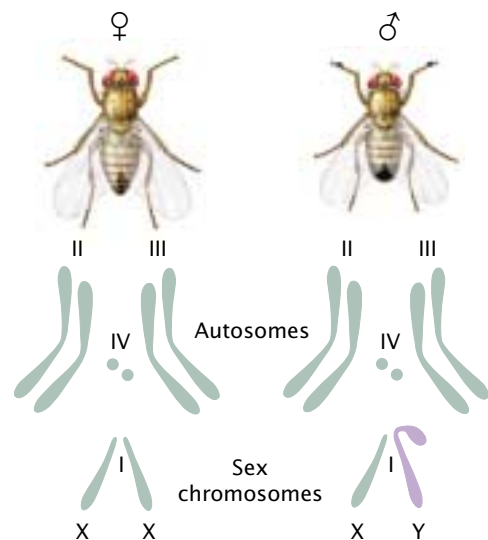
Environmental factors are also important in determining sex in many reptiles. Although most snakes and lizards have sex chromosomes, in many turtles, crocodiles, and alligators, temperature during embryonic development determines sexual phenotype. In turtles, for example, warm temperatures produce females during certain times of the year, whereas cool temperatures produce males. In alligators, the reverse is true.

Concepts

In genic sex determination, sex is determined by genes at one or more loci, but there are no obvious differences in the chromosomes of males and females. In environmental sex determination, sex is determined fully or in part by environmental factors.

Sex Determination in *Drosophila*

The fruit fly *Drosophila melanogaster*, has eight chromosomes: three pairs of autosomes and one pair of sex chromosomes (FIGURE 4.9). Normally, females have two X chromosomes and males have an X chromosome and a Y chromosome. However, the presence of the Y chromosome does not determine maleness in *Drosophila*; instead, each fly's sex is determined by a balance between genes on the autosomes and genes on the X chromosome. This type of sex determination is called the **genic balance system**. In this system, a number of genes seem to influence sexual development. The X chromosome contains genes with female-producing effects, whereas the autosomes contain genes with male-producing effects. Consequently, a fly's sex is determined by the **X:A ratio**, the number of X chromosomes divided by the number of haploid sets of autosomal chromosomes.



4.9 The chromosomes of *Drosophila melanogaster* ($2n = 8$) consist of three pairs of autosomes (labelled I, II, and III) and one pair of sex chromosomes (labelled X and Y).

An X:A ratio of 1.0 produces a female fly; an X:A ratio of 0.5 produces a male. If the X:A ratio is less than 0.5, a male phenotype is produced, but the fly is weak and sterile—such flies are sometimes called metamales. An X:A ratio between 1.0 and 0.50 produces an intersex fly, with a mixture of male and female characteristics. If the X:A ratio is greater than 1.0, a female phenotype is produced, but these flies (called metafemales) have serious developmental problems and many never emerge from the pupal case. Table 4.1 presents some different chromosome complements in *Drosophila* and their associated sexual phenotypes. Flies with two sets of autosomes and XXY sex chromosomes (an X:A ratio of 1.0) develop as fully fertile

females, in spite of the presence of a Y chromosome. Flies with only a single X (an X:A ratio of 0.5), develop as males, although they are sterile. These observations confirm that the Y chromosome does not determine sex in *Drosophila*.

Mutations in genes that affect sexual phenotype in *Drosophila* have been isolated. For example, the *transformer* mutation converts a female with an X:A ratio of 1.0 into a phenotypic male, whereas the *doublesex* mutation transforms normal males and females into flies with intersex phenotypes. Environmental factors, such as the temperature of the rearing conditions, also can affect the development of sexual characteristics.

Concepts

The sexual phenotype of a fruit fly is determined by the ratio of the number of X chromosomes to the number of haploid sets of autosomal chromosomes (the X:A ratio).

www.whfreeman.com/pierce Links to many Internet resources on the genetics of *Drosophila melanogaster*

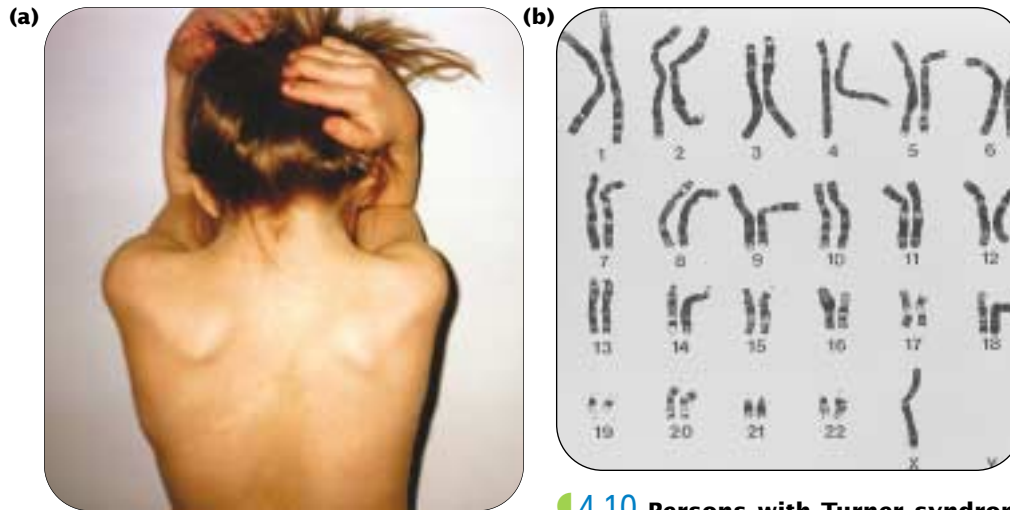
Sex Determination in Humans

Humans, like *Drosophila*, have XX-XY sex determination, but in humans the presence of a gene on the Y chromosome determines maleness. The phenotypes that result from abnormal numbers of sex chromosomes, which arise when the sex chromosomes do not segregate properly in meiosis or mitosis, illustrate the importance of the Y chromosome in human sex determination.

Turner syndrome Persons who have **Turner syndrome** are female; they do not undergo puberty and their female

Table 4.1 Chromosome complements and sexual phenotypes in *Drosophila*

Sex-Chromosome Complement	Haploid Sets of Autosomes	X:A Ratio	Sexual Phenotype
XX	AA	1.0	Female
XY	AA	0.5	Male
XO	AA	0.5	Male
XXY	AA	1.0	Female
XXX	AA	1.5	Metafemale
XXXY	AA	1.5	Metafemale
XX	AAA	0.67	Intersex
XO	AAA	0.33	Metamale
XXXX	AAA	1.3	Metafemale



4.10 Persons with Turner syndrome have a single

X chromosome in their cells. (a) Characteristic physical features.

(b) Chromosomes from a person with Turner syndrome. (Part a, courtesy of Dr. Daniel C. Postellon, Devos Children's Hospital; Part b, Dept. of Clinical Cytogenetics, Addenbrookes Hospital/Science Photo Library/Photo Researchers.)

secondary sex characteristics remain immature: menstruation is usually absent, breast development is slight, and pubic hair is sparse. This syndrome is seen in 1 of 3000 female births. Affected women are frequently short and have a low hairline, a relatively broad chest, and folds of skin on the neck (**FIGURE 4.10**). Their intelligence is usually normal. Most women who have Turner syndrome are sterile. In 1959, C. E. Ford used new techniques to study human chromosomes and discovered that cells from a 14-year-old girl with Turner syndrome had only a single X chromosome; this chromosome complement is usually referred to as XO.

There are no known cases in which a person is missing both X chromosomes, an indication that at least one X chromosome is necessary for human development. Presumably, embryos missing both Xs are spontaneously aborted in the early stages of development.

Klinefelter syndrome Persons who have **Klinefelter syndrome**, which occurs with a frequency of about 1 in 1000 male births, have cells with one or more Y chromosomes and multiple X chromosomes. The cells of most males having this condition are XXY, but cells of a few Klinefelter males are XXXY, XXXXY, or XXYY. Persons with this condition, though male, frequently have small testes, some breast enlargement, and reduced facial and pubic hair (**FIGURE 4.11**). They are often taller than normal and sterile; most have normal intelligence.

Poly-X females In about 1 in 1000 female births, the child's cells possess three X chromosomes, a condition often referred to as **triplo-X syndrome**. These persons have no distinctive features other than a tendency to be tall and thin. Although a few are sterile, many menstruate regularly and are fertile. The incidence of mental retardation among

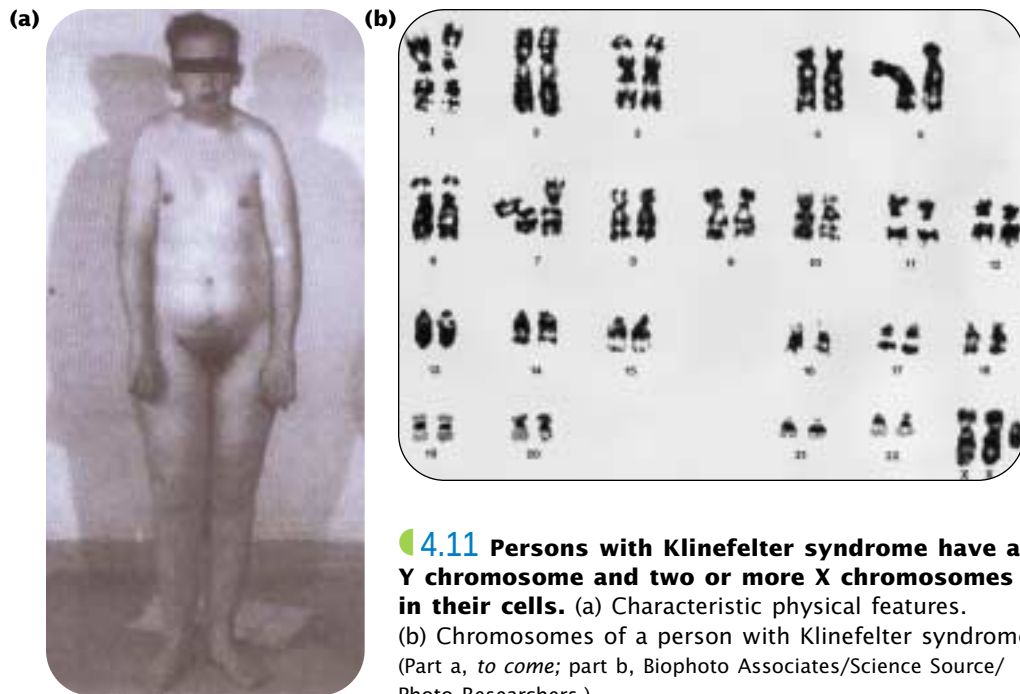
triple-X females is slightly greater than in the general population, but most XXX females have normal intelligence. Much rarer are women whose cells contain four or five X chromosomes. These women usually have normal female anatomy but are mentally retarded and have a number of physical problems. The severity of mental retardation increases as the number of X chromosomes increases beyond three.

www.whfreeman.com/pierce

Further information about sex-chromosomal abnormalities in humans

The role of sex chromosomes The phenotypes associated with sex-chromosome anomalies allow us to make several inferences about the role of sex chromosomes in human sex determination.

1. The X chromosome contains genetic information essential for both sexes; at least one copy of an X chromosome is required for human development.
2. The male-determining gene is located on the Y chromosome. A single copy of this chromosome, even in the presence of several X chromosomes, produces a male phenotype.
3. The absence of the Y chromosome results in a female phenotype.
4. Genes affecting fertility are located on the X and Y chromosomes. A female usually needs at least two copies of the X chromosome to be fertile.
5. Additional copies of the X chromosome may upset normal development in both males and females, producing physical and mental problems that increase as the number of extra X chromosomes increases.



4.11 Persons with Klinefelter syndrome have a Y chromosome and two or more X chromosomes in their cells. (a) Characteristic physical features. (b) Chromosomes of a person with Klinefelter syndrome. (Part a, *to come*; part b, Biophoto Associates/Science Source/Photo Researchers.)

The male-determining gene in humans The Y chromosome in humans and all other mammals is of paramount importance in producing a male phenotype. However, scientists discovered a few rare XX males whose cells apparently lack a Y chromosome. For many years, these males presented a real enigma: How could a male phenotype exist without a Y chromosome? Close examination eventually revealed a small part of the Y chromosome attached to another chromosome. This finding indicates that it is not the entire Y chromosome that determines maleness in humans; rather, it is a gene on the Y chromosome.

Early in development, all humans possess undifferentiated gonads and both male and female reproductive ducts. Then, about 6 weeks after fertilization, a gene on the Y chromosome becomes active. By an unknown mechanism, this gene causes the neutral gonads to develop into testes, which begin to secrete two hormones: testosterone and Mullerian-inhibiting substance. Testosterone induces the development of male characteristics, and Mullerian-inhibiting substance causes the degeneration of the female reproductive ducts. In the absence of this male-determining gene, the neutral gonads become ovaries, and female features develop.

In 1987, David Page and his colleagues at the Massachusetts Institute of Technology located what appeared to be the male-determining gene near the tip of the short arm of the Y chromosome. They had examined the DNA of several XX males and XY females. The cells of one XX male that they studied possessed a very small piece of a Y chromosome attached to one of the Xs. This piece came from a section, called 1A, of the Y chromosome. Because this person had a male phenotype, they reasoned that the male-

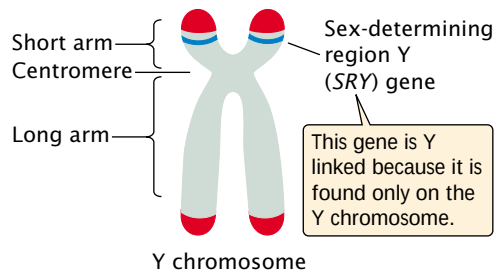
determining gene must reside within the 1A section of the Y chromosome.

Examination of the Y chromosome of a 12 year-old XY girl seemed to verify this conclusion. In spite of the fact that she possessed more than 99.8% of a Y chromosome, this XY person had a female phenotype. Page and his colleagues assumed that the male-determining gene must reside within the 0.2% of the Y chromosome that she was missing. Further examination showed that this Y chromosome was indeed missing part of section 1A. They then sequenced the DNA within section 1A of normal males and found a gene called *ZFY*, which appeared to be the testis-determining factor.

Within a few months, however, results from other laboratories suggested that *ZFY* might not in fact be the male-determining gene. Marsupials (pouched mammals), which also have XX-XY sex determination, were found to possess a *ZFY* gene on an autosomal chromosome, not on the Y chromosome. Furthermore, several human XX males were found who did not possess a copy of the *ZFY* gene.

A new candidate for the male-determining gene, called the **sex-determining region Y (SRY) gene**, was discovered in 1990 (FIGURE 4.12). This gene is found in XX males and is missing from all XY females; it is also found on the Y chromosome of all mammals examined to date. Definitive proof that *SRY* is the male-determining gene came when scientists placed a copy of this gene into XX mice by means of genetic engineering. The XX mice that received this gene, although sterile, developed into anatomical males.

The *SRY* gene encodes a protein that binds to DNA and causes a sharp bend in the molecule. This alteration of DNA structure may affect the expression of other genes that



4.12 The **SRY** gene is on the Y chromosome and causes the development of male characteristics.

encode testis formation. Although *SRY* is the primary determinant of maleness in humans, other genes (some X linked, others Y linked, and still others autosomal) also play a role in fertility and the development of sex differences.

Concepts

The presence of the *SRY* gene on the Y chromosome causes a human embryo to develop as a male. In the absence of this gene, a human embryo develops as a female.

www.whfreeman.com/pierce

Additional information on the *SRY* gene

Androgen-insensitivity syndrome Several genes besides *SRY* influence sexual development in humans, as illustrated by women with androgen-insensitivity syndrome. These persons have female external sexual characteristics and psychological orientation. Indeed, most are unaware of their condition until they reach puberty and fail to menstruate. Examination by a gynecologist reveals that the vagina ends blindly and that the uterus, oviducts, and ovaries are absent. Inside the abdominal cavity lies a pair of testes, which produce levels of testosterone normally seen in males. The cells of a woman with androgen-insensitivity syndrome contain an X and a Y chromosome.

How can a person be female in appearance when her cells contain a Y chromosome and she has testes that produce testosterone? The answer lies in the complex relation between genes and sex in humans. In a human embryo with a Y chromosome, the *SRY* gene causes the gonads to develop into testes, which produce testosterone. Testosterone stimulates embryonic tissues to develop male characteristics. But, for testosterone to have its effects, it must bind to an androgen receptor. This receptor is defective in females with androgen-insensitivity syndrome; consequently, their cells are insensitive to testosterone, and female characteristics develop. The gene for the androgen receptor is located on the X chromosome; so persons with this condition always inherit it from their mothers. (All XY persons inherit the X chromosome from their mothers.)

Androgen-insensitivity syndrome illustrates several important points about the influence of genes on a person's sex. First, this condition demonstrates that human sexual development is a complex process, influenced not only by the *SRY* gene on the Y chromosome, but also by other genes found elsewhere. Second, it shows that most people carry genes for both male and female characteristics, as illustrated by the fact that those with androgen-insensitivity syndrome have the capacity to produce female characteristics, even though they have male chromosomes. Indeed, the genes for most male and female secondary sex characteristics are present not on the sex chromosomes but on autosomes. The key to maleness and femaleness lies not in the genes but in the control of their expression.

www.whfreeman.com/pierce

Additional information on androgen-insensitivity syndrome

Sex-Linked Characteristics

Sex-linked characteristics are determined by genes located on the sex chromosomes. Genes on the X chromosome determine **X-linked characteristics**; those on the Y chromosome determine **Y-linked characteristics**. Because little genetic information exists on the Y chromosome in many organisms, most sex-linked characteristics are X linked. Males and females differ in their sex chromosomes; so the pattern of inheritance for sex-linked characteristics differs from that exhibited by genes located on autosomal chromosomes.

X-Linked White Eyes in *Drosophila*

The first person to explain sex-linked inheritance was the American biologist Thomas Hunt Morgan (**FIGURE 4.13a**). Morgan began his career as an embryologist, but the discovery of Mendel's principles inspired him to begin conducting genetic experiments, initially on mice and rats. In 1909, Morgan switched to *Drosophila melanogaster*; a year later, he discovered among the flies of his laboratory colony a single male that possessed white eyes, in stark contrast with the red eyes of normal fruit flies. This fly had a tremendous effect on the future of genetics and on Morgan's career as a biologist. With his white-eyed male, Morgan unraveled the mechanism of X-linked inheritance, ushering in the "golden age" of *Drosophila* genetics that lasted from 1910 until 1930.

Morgan's laboratory, located on the top floor of Schermerhorn Hall at Columbia University, became known as the Fly Room (**FIGURE 4.13b**). To say that the Fly Room was unimpressive is an understatement. The cramped room, only about 16 × 23 feet, was filled with eight desks, each occupied by a student and his experiments. The primitive laboratory equipment consisted of little more than milk bottles for rearing the flies and hand-held lenses for observing their traits. Later, microscopes replaced the hand-held lenses, and crude incubators were added to maintain the fly

(a)



(b)



4.13 Thomas Hunt Morgan's work with *Drosophila* helped unravel many basic principles in genetics, including X-linked inheritance. (a) Morgan. (b) The Fly Room, where Morgan and his students conducted genetic research. (Part a, World Wide Photos; Part b, American Philosophical Society.)

cultures, but even these additions did little to increase the physical sophistication of the laboratory. Morgan and his students were not tidy: cockroaches were abundant (living off spilled *Drosophila* food), dirty milk bottles filled the sink, ripe bananas—food for the flies—hung from the ceiling, and escaped fruit flies hovered everywhere.

In spite of its physical limitations, the Fly Room was the source of some of the most important research in the history of biology. There was daily excitement among the students, some of whom initially came to the laboratory as undergraduates. The close quarters facilitated informality and the free flow of ideas. Morgan and the Fly Room illustrate the tremendous importance of “atmosphere” in producing good science.

To explain the inheritance of the white-eyed characteristic in fruit flies, Morgan systematically carried out a series of genetic crosses (FIGURE 4.14a). First, he crossed pure-breeding, red-eyed females with his white-eyed male, producing F_1 progeny that all had red eyes. (In fact, Morgan found three white-eyed males among the 1237 progeny, but he assumed that the white eyes were due to new mutations.) Morgan's results from this initial cross were consistent with Mendel's principles: a cross between a homozygous dominant individual and a homozygous recessive individual produces heterozygous offspring exhibiting the dominant trait. His results suggested that white eyes were a simple recessive trait. However, when Morgan crossed the F_1 flies with one another, he found that all the female F_2 flies possessed red eyes but that half the male F_2 flies had red eyes and the other half had white eyes. This finding was clearly not the expected result for a simple recessive

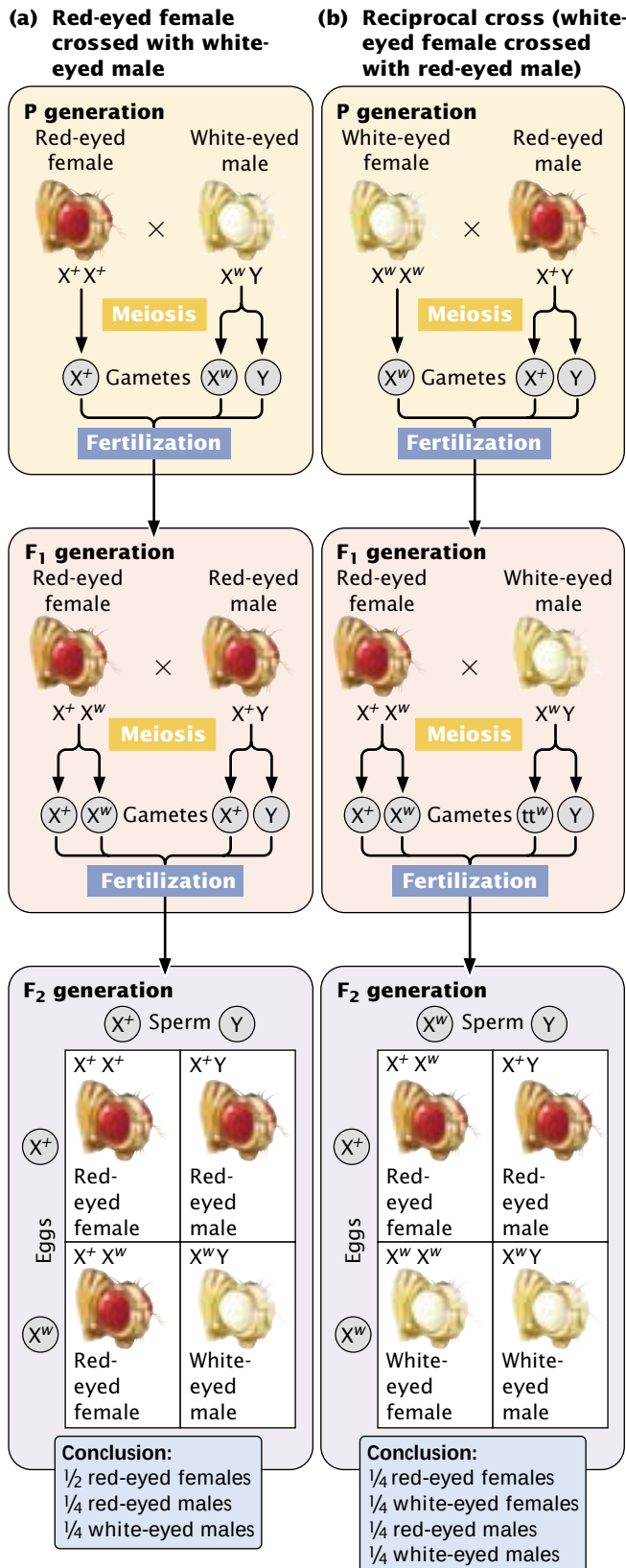
trait, which should appear in $\frac{1}{4}$ of both male and female F_2 offspring.

To explain this unexpected result, Morgan proposed that the locus affecting eye color was on the X chromosome (that eye color was X linked). He recognized that the eye-color alleles were present only on the X chromosome—no homologous allele was present on the Y chromosome. Because the cells of females possess two X chromosomes, females could be homozygous or heterozygous for the eye-color alleles. The cells of males, on the other hand, possess only a single X chromosome and can carry only a single eye-color allele. Males therefore cannot be either homozygous or heterozygous but are said to be **hemizygous** for X-linked loci.

To verify his hypothesis that the white-eye trait is X linked, Morgan conducted additional crosses. He predicted that a cross between a white-eyed female and a red-eyed male would produce all red-eyed females and all white-eyed males (FIGURE 4.14b). When Morgan performed this cross, the results were exactly as predicted. Note that this cross is the reciprocal of the original cross and that the two reciprocal crosses produced different results in the F_1 and F_2 generations. Morgan also crossed the F_1 heterozygous females with their white-eyed father, the red-eyed F_2 females with white-eyed males, and white-eyed females with white-eyed males. In all of these crosses, the results were consistent with Morgan's conclusion that white eyes is an X-linked characteristic.

www.whfreeman.com/pierce

More information on the life of Thomas Hunt Morgan



4.14 Morgan's X-linked crosses for white eyes in fruit flies. (a) Original and F₁ crosses. (b) Reciprocal crosses.

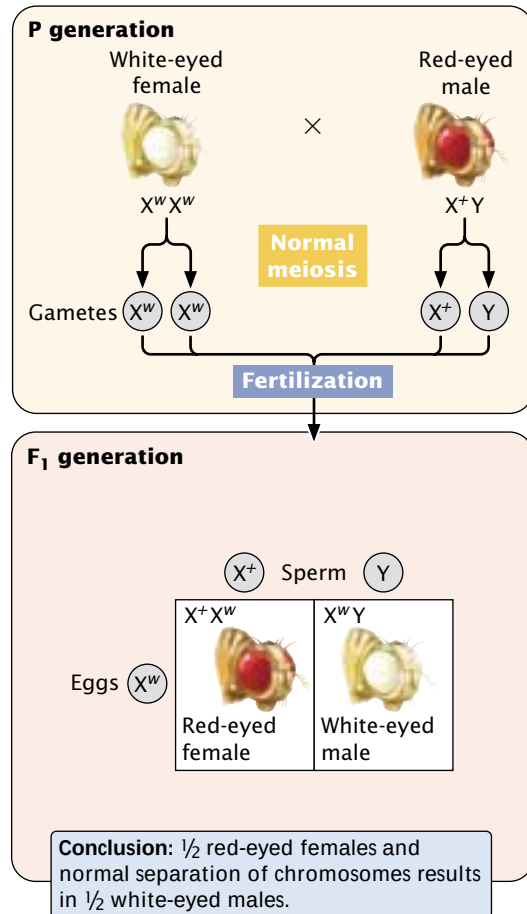
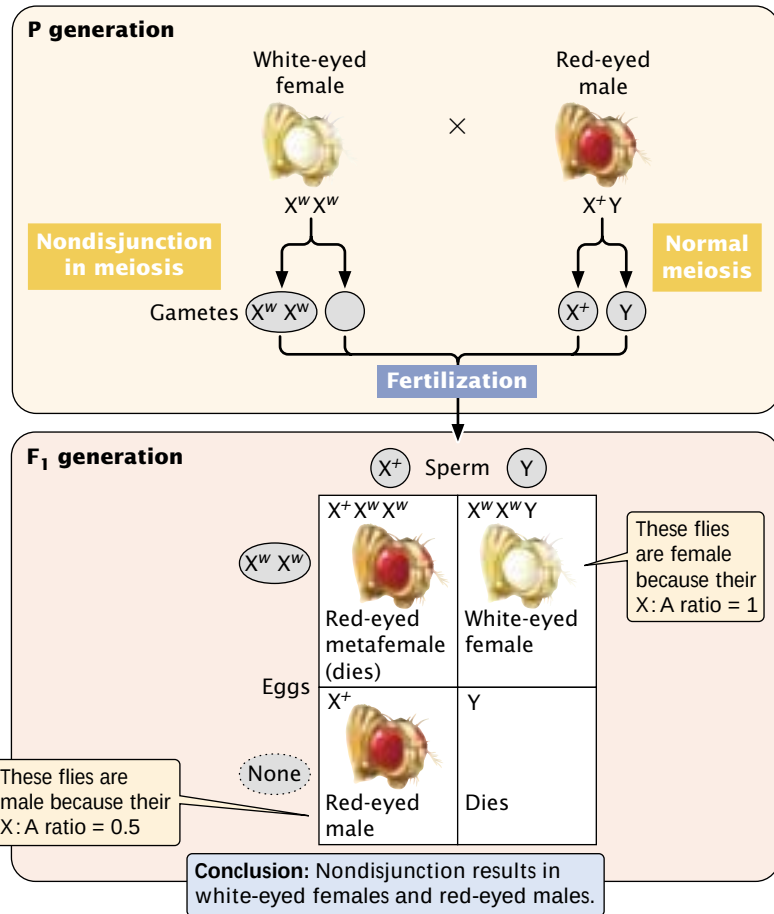
Nondisjunction and the Chromosome Theory of Inheritance

When Morgan crossed his original white-eyed male with homozygous red-eyed females, all 1237 of the progeny had red eyes, except for three white-eyed males. As already mentioned, Morgan attributed these white-eyed F₁ males to the occurrence of further mutations. However, flies with these unexpected phenotypes continued to appear in his crosses. Although uncommon, they appeared far too often to be due to mutation. Calvin Bridges, one of Morgan's students, set out to investigate the genetic basis of these exceptions.

Bridges found that, when he crossed a white-eyed female (X^wX^w) with a red-eyed male (X^+Y), about 2.5% of the male offspring had red eyes and about 2.5% of the female offspring had white eyes (FIGURE 4.15a). In this cross, every male fly should inherit its mother's X chromosome and should be X^wY with white eyes. Every female fly should inherit a dominant red-eye allele on its father's X chromosome, along with a white-eyed allele on its mother's X chromosome; thus, all the female progeny should be X^+X^w and have red eyes. The appearance of red-eyed males and white-eyed females in this cross was therefore unexpected.

To explain this result, Bridges hypothesized that, occasionally, the two X chromosomes in females fail to separate during anaphase I of meiosis. Bridges termed this failure of chromosomes to separate **nondisjunction**. When nondisjunction occurs, some of the eggs receive two copies of the X chromosome and others do not receive an X chromosome (FIGURE 4.15b). If these eggs are fertilized by sperm from a red-eyed male, four combinations of sex chromosomes are produced. When an egg carrying two X chromosomes is fertilized by a Y-bearing sperm, the resulting zygote is X^wX^wY . Sex in *Drosophila* is determined by the X:A ratio (see Table 4.1); in this case the X:A ratio is 1.0, so the X^wX^wY zygote develops into a white-eyed female. An egg with two X chromosomes that is fertilized by an X-bearing sperm produces $X^wX^wX^+$, which usually dies. An egg with no X chromosome that is fertilized by an X-bearing sperm produces X^+O , which develops into a red-eyed male. If the egg with no X chromosome is fertilized by a Y-bearing sperm, the resulting zygote with only a Y chromosome and no X chromosome dies. Rare nondisjunction of the X chromosomes among white-eyed females therefore produces a few red-eyed males and white-eyed females, which is exactly what Bridges found in his crosses.

Bridges's hypothesis predicted that the white-eyed females would possess two X chromosomes and one Y and that red-eyed males would possess a single X chromosome. To verify his hypothesis, Bridges examined the chromosomes of his flies and found precisely what he predicted. The significance of Bridges's study was not that it explained

(a) White-eyed female and red-eyed male**(b) White-eyed female and red-eyed male with nondisjunction****4.15 Bridges conducted experiments that proved that the gene**

for white eyes is located on the X chromosome. (a) A white-eyed female was crossed with a red-eyed male. (b) Rare nondisjunction produced a few eggs with two copies of the X^w chromosome and other eggs with no X chromosome.

the appearance of an occasional odd fly in his culture but that he was able to predict a fly's chromosomal makeup on the basis of its eye-color genotype. This association between genotype and chromosomes gave unequivocal evidence that sex-linked genes were located on the X chromosome and confirmed the chromosome theory of inheritance.

Concepts

By showing that the appearance of rare phenotypes was associated with the inheritance of particular chromosomes, Bridges proved that sex-linked genes are located on the X chromosome and that the chromosome theory of inheritance is correct.

X-Linked Color Blindness in Humans

To further examine X-linked inheritance, let's consider another X-linked characteristic: red-green color blindness in humans. Within the human eye, color is perceived in light-sensing cone cells that line the retina. Each cone cell contains one of three pigments capable of absorbing light of a particular wavelength; one absorbs blue light, a second absorbs red light, and a third absorbs green light. The human eye actually detects only three colors—red, green, and blue—but the brain mixes the signals from different cone cells to create the wide spectrum of colors that we perceive. Each of the three pigments is encoded by a separate locus; the locus for the blue pigment is found on chromosome 7, and those for green and red pigments lie close together on the X chromosome.

The most common types of human color blindness are caused by defects of the red and green pigments; we will refer

to these conditions as red–green color blindness. Mutations that produce defective color vision are generally recessive and, because the genes coding for the red and green pigments are located on the X chromosome, red–green color blindness is inherited as an X-linked recessive characteristic.

We will use the symbol X^c to represent an allele for red–green color blindness and the symbol X^+ to represent an allele for normal color vision. Females possess two X chromosomes; so there are three possible genotypes among females: X^+X^+ and X^+X^c , which produce normal vision, and X^cX^c , which produces color blindness. Males have only a single X chromosome and two possible genotypes: X^+Y , which produces normal vision, and X^cY which produces color blindness.

If a color-blind man mates with a woman homozygous for normal color vision (FIGURE 4.16a), all of the gametes produced by the woman will contain an allele for normal color vision. Half of the man's gametes will receive the X chromosome with the color-blind allele, and the other half will receive the Y chromosome, which carries no alleles affecting color vision. When an X^c -bearing sperm unites with the X^+ -bearing egg, a heterozygous female with normal vision (X^+X^c) is produced. When a Y-bearing sperm unites with the X^+ -bearing egg, a hemizygous male with normal vision (X^+Y) is produced (see Figure 4.16a).

In the reciprocal cross between a color-blind woman and a man with normal color vision (FIGURE 4.16b), the woman produces only X^c -bearing gametes. The man produces some gametes that contain the X^+ chromosome and others that contain the Y chromosome. Males inherit the X chromosome

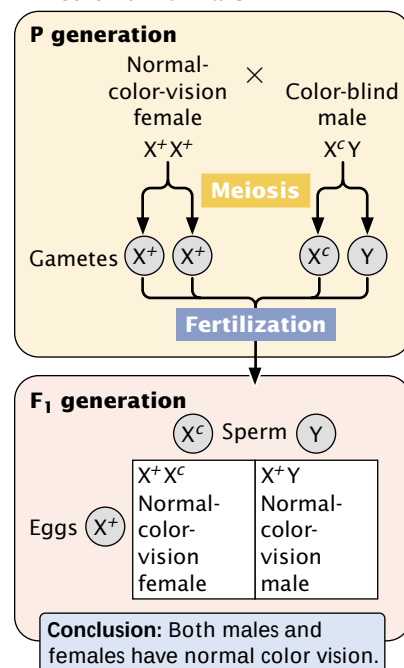
from their mothers; because both of the mother's X chromosomes bear the X^c allele in this case, all the male offspring will be color blind. In contrast, females inherit an X chromosome from both parents; thus the female offspring of this reciprocal cross will all be heterozygous with normal vision. Females are color blind only when color-blind alleles have been inherited from both parents, whereas a color-blind male need inherit a color-blind allele from his mother only; for this reason, color blindness and most other rare X-linked recessive characteristics are more common in males.

In these crosses for color blindness, notice that an affected woman passes the X-linked recessive trait to her sons but not to her daughters, whereas an affected man passes the trait to his grandsons through his daughters but never to his sons. X-linked recessive characteristics seem to alternate between the sexes, appearing in females one generation and in males the next generation; thus, this pattern of inheritance exhibited by X-linked recessive characteristics is sometimes called *crisscross inheritance*.

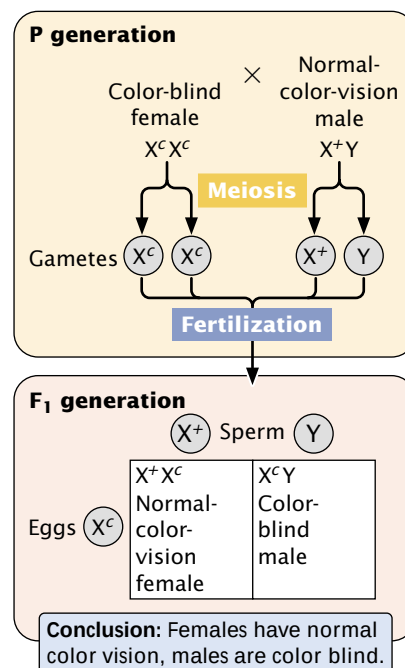
Concepts

Characteristics determined by genes on the sex chromosomes are called **sex-linked characteristics**. Diploid females have two alleles at each X-linked locus, whereas diploid males possess a single allele at each X-linked locus. Females inherit X-linked alleles from both parents, but males inherit a single X-linked allele from their mothers.

(a) Normal female and color-blind male



(b) Reciprocal cross



4.16 Red–green color blindness is inherited as an X-linked recessive trait in humans.

Symbols for X-Linked Genes

There are several different ways to record genotypes for X-linked traits. Sometimes the genotypes are recorded in the same fashion as for autosomal characteristics—the hemizygous males are simply given a single allele: the genotype of a female *Drosophila* with white eyes would be ww , and the genotype of a white-eyed hemizygous male would be w . Another method is to include the Y chromosome, designating it with a diagonal slash (/). With this method, the white-eyed female's genotype would still be ww and the white-eyed male's genotype would be w/Y . Perhaps the most useful method is to write the X and Y chromosomes in the genotype, designating the X-linked alleles with superscripts, as we have done in this chapter. With this method, a white-eyed female would be X^wX^w and a white-eyed male X^wY . Using Xs and Ys in the genotype has the advantage of reminding us that the genes are X linked and that the male must always have a single allele, inherited from the mother.

Dosage Compensation

The presence of different numbers of X chromosomes in males and females presents a special problem in development. Because females have two copies of every X-linked gene and males possess one copy, the amount of gene product (protein) from X-linked genes would normally differ in the two sexes—females would produce twice as much gene product as males. This difference could be highly detrimental because protein concentration plays a critical role in development. Animals overcome this potential problem through **dosage compensation**, which equalizes the amount of protein produced by X-linked genes in the two sexes. In fruit flies, dosage compensation is achieved by a doubling of the activity of the genes on the X chromosome of the male. In the worm *Caenorhabditis elegans*, it is achieved by a halving of the activity of genes on both of the X chromosomes in the female. Pla-

cental mammals use yet another mechanism of dosage compensation; genes on one of the X chromosomes in the female are completely inactivated.

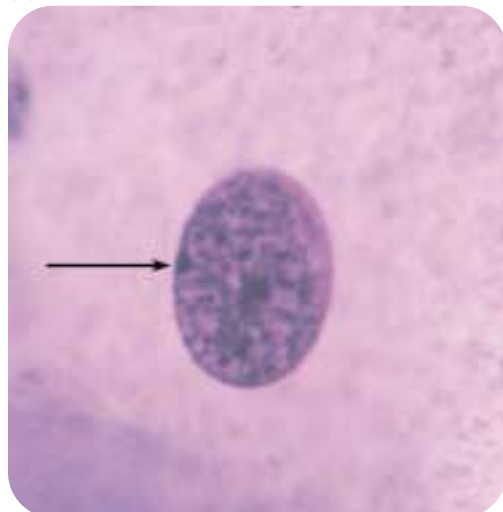
In 1949, Murray Barr observed condensed, darkly staining bodies in the nuclei of cells from female cats (FIGURE 4.17); this darkly staining structure became known as a **Barr body**. Mary Lyon proposed in 1961 that the Barr body was an inactive X chromosome; her hypothesis (now proved) has become known as the **Lyon hypothesis**. She suggested that, within each female cell, one of the two X chromosomes becomes inactive; which X chromosome is inactivated is random. If a cell contains more than two X chromosomes, all but one of them is inactivated. The number of Barr bodies present in human cells with different complements of sex chromosomes is shown in Table 4.2.

As a result of X inactivation, females are functionally hemizygous at the cellular level for X-linked genes. In females that are heterozygous at an X-linked locus, approximately 50% of the cells will express one allele and 50% will express the other allele; thus, in heterozygous females, proteins encoded by both alleles are produced, although not within the same cell. This functional hemizygosity means that cells in females are not identical with respect to the expression of the genes on the X chromosome; females are mosaics for the expression of X-linked genes.

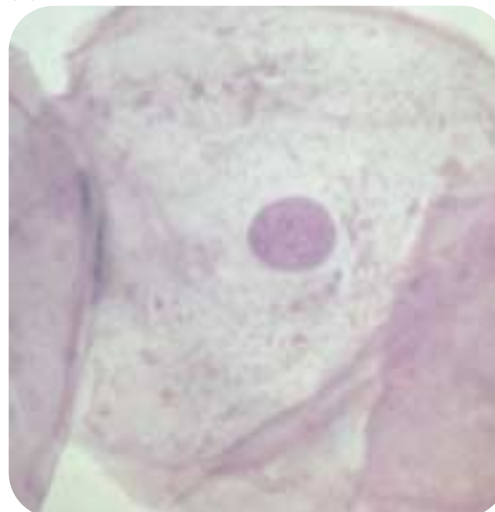
X inactivation takes place relatively early in development—in humans, within the first few weeks of development. Once an X chromosome becomes inactive in a cell, it remains inactivated and is inactive in all somatic cells that descend from the cell. Thus, neighboring cells tend to have the same X chromosome inactivated, producing a patchy pattern (mosaic) for the expression of an X-linked characteristic in heterozygous females.

This patchy distribution can be seen in tortoiseshell cats (FIGURE 4.18). Although many genes contribute to coat color and pattern in domestic cats, a single X-linked locus determines the presence of orange color. There are possible

(a)



(b)



4.17 A Barr body is an inactivated X chromosome. (a) Female cell with a Barr body (indicated by arrow). (b) Male cell without a Barr body. (Part a, George Wilder/Visuals Unlimited; part b, M. Abbey/Photo Researchers.)

Table 4.2 Number of Barr bodies in human cells with different complements of sex chromosomes

Sex Chromosomes	Syndrome	Number of Barr Bodies
XX	None	1
XY	None	0
XO	Turner	0
XXY	Klinefelter	1
XXYY	Klinefelter	1
XXXY	Klinefelter	2
XXXXY	Klinefelter	3
XXX	Triplo-X	2
XXXX	Poly-X female	3
XXXXX	Poly-X female	4

two alleles at this locus: X^+ , which produces nonorange (usually black) fur, and X^o , which produces orange fur. Males are hemizygous and thus may be black (X^+Y) or orange (X^oY) but not black *and* orange. (Rare tortoiseshell males can arise from the presence of two X chromosomes, X^+X^oY .) Females may be black (X^+X^+), orange (X^oX^o), or tortoiseshell (X^+X^o), the tortoiseshell pattern arising from a patchy mixture of black and orange fur. Each orange patch is a clone of cells derived from an original cell with the black allele inactivated, and each black patch is a clone of cells derived from an original cell with the orange allele inactivated. The mosaic pattern of gene expression associated with dosage compensation also produces the patchy distribution of sweat glands in women heterozygous for anhidrotic ectodermal dysplasia (see introduction to this chapter).

Lyon's hypothesis suggests that the presence of variable numbers of X chromosomes should not be detrimental in mammals, because any X chromosomes beyond one should be inactivated. However, persons with Turner syndrome (XO) differ from normal females, and those with Klinefelter syndrome (XXY) differ from normal males. How do these conditions arise in the face of dosage compensation? The reason may lie partly in the fact that there is a short period of time, very early in development, when all X chromosomes are active. If the number of X chromosomes is abnormal, any X-linked genes expressed during this early period will produce abnormal levels of gene product. Furthermore, the phenotypic abnormalities may arise because some X-linked genes escape inactivation, although how they do so isn't known.

Exactly how an X chromosome becomes inactivated is not completely understood either, but it appears to entail the addition of methyl groups ($-\text{CH}_3$) to the DNA. The *XIST*



4.18 The patchy distribution of color on tortoiseshell cats results from the random inactivation of one X chromosome in females.

(David Falconer/Words & Pictures/Picture Quest.)

(for X inactive-specific transcript) gene, located on the X chromosome, is required for inactivation. Only the copy of *XIST* on the inactivated X chromosome is expressed, and it continues to be expressed during inactivation (unlike most other genes on the inactivated X chromosome). Interestingly, *XIST* does not encode a protein; it produces an RNA molecule that binds to the inactivated X chromosome. This binding is thought to prevent the attachment of other proteins that participate in transcription and, in this way, it brings about X inactivation.

Concepts

In mammals, dosage compensation ensures that the same amount of X-linked gene product will be produced in the cells of both males and females. All but one X chromosome is randomly inactivated in each cell; which X chromosome is inactivated is random and varies from cell to cell.

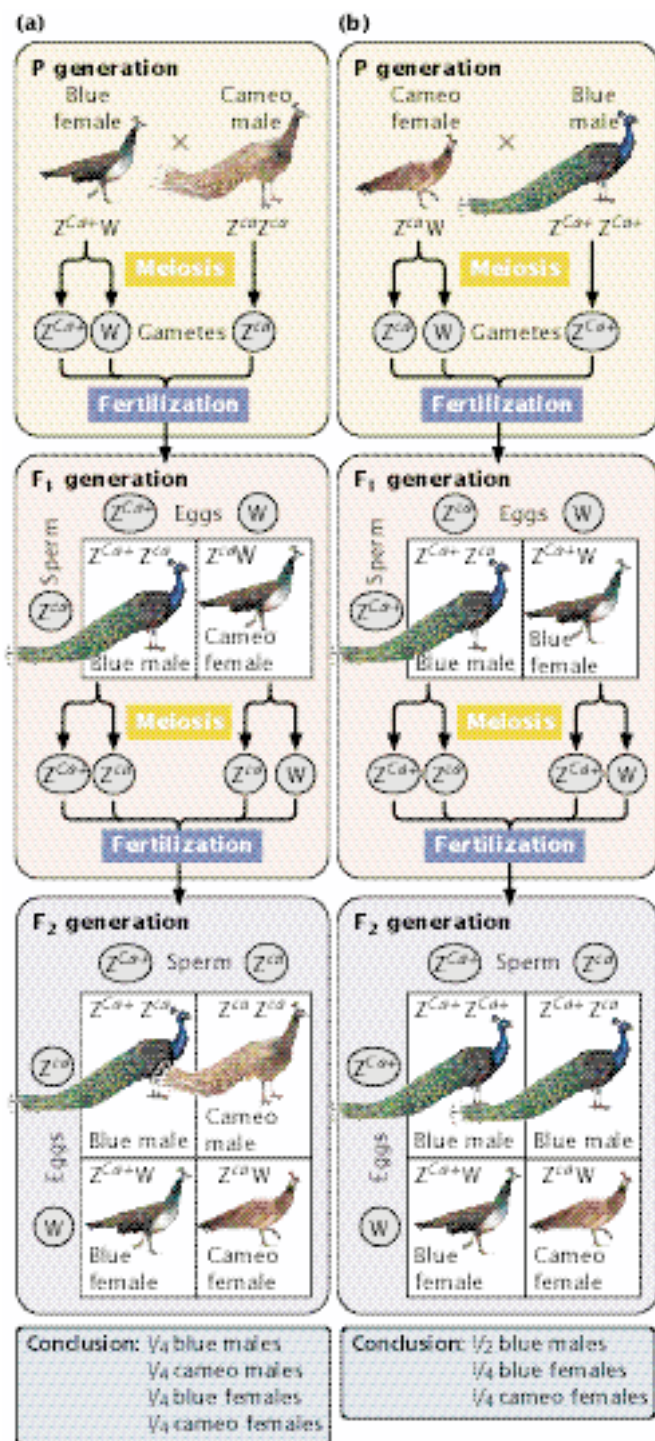
www.whfreeman.com/pierce

Current information on *XIST* and X-chromosome inactivation in humans

Z-Linked Characteristics

In organisms with ZZ-ZW sex determination, the males are the homogametic sex (ZZ) and carry two sex-linked (usually referred to as Z-linked) alleles; thus males may be homozygous or heterozygous. Females are the heterogametic sex (ZW) and possess only a single Z-linked allele. Inheritance of Z-linked characteristics is the same as that of X-linked characteristics, except that the pattern of inheritance in males and females is reversed.

An example of a Z-linked characteristic is the cameo phenotype in Indian blue peafowl (*Pavo cristatus*). In these birds, the wild-type plumage is a glossy, metallic blue. The female peafowl is ZW and the male is ZZ. Cameo plumage,



4.19 Inheritance of the cameo phenotype in Indian blue peafowl is inherited as a Z-linked recessive trait. (a) Blue female crossed with cameo male. (b) Reciprocal cross of cameo female crossed with homozygous blue male.

which produces brown feathers, results from a Z-linked allele (Z^{ca}) that is recessive to the wild-type blue allele (Z^{Ca+}). If a blue-colored female ($Z^{Ca+}W$) is crossed with a cameo male ($Z^{ca}Z^{ca}$), all the F_1 females are cameo ($Z^{ca}W$) and all the F_1 males are blue ($Z^{Ca+}Z^{ca}$) (FIGURE 4.19). When the F_1 are interbred, $\frac{1}{4}$ of the F_2 are blue males ($Z^{Ca+}Z^{ca}$), $\frac{1}{4}$ are blue females ($Z^{Ca+}W$), $\frac{1}{4}$ are cameo males ($Z^{ca}Z^{ca}$), and $\frac{1}{4}$ are cameo females ($Z^{ca}W$). The reciprocal cross of a cameo female with a homozygous blue male produces an F_1 generation in which all offspring are blue and an F_1 consisting of $\frac{1}{2}$ blue males ($Z^{Ca+}Z^{ca}$ and $Z^{Ca+}Z^{Ca+}$), $\frac{1}{4}$ blue females ($Z^{Ca+}W$), and $\frac{1}{4}$ cameo females ($Z^{ca}W$).

In organisms with ZZ-ZW sex determination, the female always inherits her W chromosome from her mother, and she inherits her Z chromosome, along with any Z-linked alleles, from her father. In this system, the male inherits Z chromosomes, along with any Z-linked alleles, from both the mother and the father. This pattern of inheritance is the reverse of X-linked alleles in organisms with XX-XY sex determination.

Y-Linked Characteristics

Y-linked traits exhibit a distinct pattern of inheritance and are present only in males, because only males possess a Y chromosome. All male offspring of a male with a Y-linked trait will display the trait (provided that the penetrance—see Chapter 3—is 100%), because every male inherits the Y chromosome from his father.

In humans and many other organisms, there is relatively little genetic information on the Y chromosome, and few characteristics exhibit Y-linked inheritance. More than 20 genes have been identified outside the pseudoautosomal region on the human Y chromosome, including the *SRY* gene and the *ZFY* gene. A possible Y-linked human trait is hairy ears, a trait that is common among men in some parts of the Middle East and India, affecting as many as 70% of adult men in some regions. This trait displays variable expressivity—some men have only a few hairs on the outer ear, whereas others have ears that are covered with hair. The age at which this trait appears also is quite variable.

Only men have hairy ears and, in many families, the occurrence of the trait is entirely consistent with Y-linked inheritance. In a few families, however, not all sons of an affected man display the trait, which implies that the trait has incomplete penetrance. Some investigators have concluded that the hairy-ears trait is not Y-linked, but instead is an autosomal dominant trait expressed only in men (sex-limited expression, discussed more fully in Chapter 5). Distinguishing between a Y-linked characteristic with incomplete penetrance and an autosomal dominant characteristic expressed only in males is difficult, and the pattern of inheritance of hairy ears is consistent with both modes of inheritance.

The function of most Y-linked genes is poorly understood, but some appear to influence male sexual development

and fertility. Some Y-linked genes have counterparts on the X chromosome that encode similar proteins in females.

DNA sequences in the Y chromosome undergo mutation over time and vary among individuals. Like Y-linked traits, these variants—called genetic markers—are passed from father to son and can be used to study male ancestry. Although the markers themselves do not code for any physical traits, they can be detected with molecular methods. Much of the Y chromosome is nonfunctional; so mutations readily accumulate. Many of these mutations are unique; they arise only once and are passed down through the generations without recombination. Individuals possessing the same set of mutations are therefore related, and the distribution of these genetic markers on Y chromosomes provides clues about genetic relationships of present-day people.

Y-linked markers have been used to study the offspring of Thomas Jefferson, principal author of the Declaration of Independence and third president of the United States. In 1802, Jefferson was accused by a political enemy of fathering a child by his slave Sally Hemings, but the evidence was circumstantial. Hemings, who worked in the Jefferson household and accompanied Jefferson on a trip to Paris, had five children. Jefferson was accused of fathering the first child, Tom, but rumors about the paternity of the other children circulated as well. Hemings's last child, Eston, bore a striking resemblance to Jefferson, and her fourth child, Madison, testified late in life that Jefferson was the father of all Hemings's children. Ancestors of Hemings's children maintained that they were descendants of the Jefferson line, but some Jefferson descendants refused to recognize their claim.

To resolve this long-standing controversy, geneticists examined markers from the Y chromosomes of male-line descendants of Hemings's first son (Thomas Woodson), her last son (Eston Hemings), and a paternal uncle of Thomas Jefferson with whom Jefferson had Y chromosomes in common. (Descendants of Jefferson's uncle were used because Jefferson himself had no verified male descendants.) Geneticists determined that Jefferson possessed a rare and distinctive set of genetic markers on his Y chromosome. The same markers were also found on the Y chromosomes of the male-line descendants of Eston Hemings. The probability of such a match arising by chance is less than 1%. (The markers were not found on the Y chromosomes of the descendants of Thomas Woodson.) Together with the circumstantial historical evidence, these matching markers strongly suggest that Jefferson fathered Eston Hemings but not Thomas Woodson.

Another study utilizing Y-linked genetic markers focused on the origins of the Lemba, an African tribe comprising 50,000 people who reside in South Africa and parts of Zimbabwe. Members of the Lemba tribe are commonly referred to as the black Jews of South Africa. This name derives from cultural practices of the tribe, including circumcision and food taboos, which superficially resemble those of Jewish people. Lemba oral tradition suggests that the tribe came

from “Sena in the north by boat,” Sena being variously identified as Sanaa in Yemen, Judea, Egypt, or Ethiopia. Legend says that the original group was entirely male, that half of their number was lost at sea, and that the survivors made their way to the coast of Africa, where they settled.

Today, most Lemba belong to Christian churches, are Muslims, or claim to be Lemba in religion. Their religious practices have little in common with Judaism and, with the exception of their oral tradition and a few cultural practices, there is little to suggest a Jewish origin.

To reveal the genetic origin of the Lemba, scientists examined genetic markers on their Y chromosomes. Swabs of cheek cells were collected from 399 males in several populations: the Lemba in Africa, Bantu (another South African tribe), two groups from Yemen, and several groups of Jews. DNA was extracted and analyzed for alleles at 12 loci. This analysis of genetic markers revealed that Y chromosomes in the Lemba were of two types: those of Bantu origin and those similar to chromosomes found in Jewish and Yemen populations. Most importantly, members of one Lemba clan carried a large number of Y chromosomes that had a rare combination of alleles also found on the Y chromosomes of members of the Jewish priesthood. This set of alleles is thought to be an important indicator of Judaic origin. These findings are consistent with the Lemba oral tradition and strongly suggest a genetic contribution from Jewish populations.

Concepts

Y-linked characteristics exhibit a distinct pattern of inheritance: they are present only in males, and all male offspring of a male with a Y-linked trait inherit the trait.

www.whfreeman.com/pierce

An overview of the use of Y-linked markers in studies of ancestry

Connecting Concepts

Recognizing Sex-linked Inheritance

What features should we look for to identify a trait as sex linked? A common misconception is that any genetic characteristic in which the phenotypes of males and females differ must be sex linked. In fact, the expression of many *autosomal* characteristics differs between males and females. The genes that code for these characteristics are the same in both sexes, but their expression is influenced by sex hormones. The different sex hormones of males and females cause the same genes to generate different phenotypes in males and females.

Another misconception is that any characteristic that is found more frequently in one sex is sex linked. A number of

autosomal traits are expressed more commonly in one sex than in the other, because the penetrance of the trait differs in the two sexes; these traits are said to be sex influenced. For some autosomal traits, the penetrance in one sex is so low that the trait is expressed in only one sex; these traits are said to be sex limited. Both sex-influenced and sex-limited characteristics will be discussed in more detail in Chapter 5.

Several features of sex-linked characteristics make them easy to recognize. Y-linked traits are found only in males, but this fact does not guarantee that a trait is Y linked, because some autosomal characteristics are expressed only in males. A Y-linked trait is unique, however, in that all the male offspring of an affected male will express the father's phenotype, provided the penetrance of the trait is 100%. This need not be the case for autosomal traits that are sex-limited to males. Even when the penetrance is less than 100%, a Y-linked trait can be inherited only from the father's side of the family. Thus, a Y-linked trait can be inherited only from the paternal grandfather (the father's father), never from the maternal grandfather (the mother's father).

X-linked characteristics also exhibit a distinctive pattern of inheritance. X linkage is a possible explanation when the results of reciprocal crosses differ. If a characteristic is X linked, a cross between an affected male and an unaffected female will not give the same results as a cross between an affected female and an unaffected male. For almost all autosomal characteristics, the results of reciprocal crosses are the same. We should not conclude, however, that, when the reciprocal crosses give different results, the characteristic is X linked. Other sex-associated forms of inheritance, discussed in Chapter 5, also produce different results in reciprocal crosses. The key to recognizing X-linked inheritance is to remember that a male always inherits his X chromosome from his mother, not from his father. Thus, an X-linked characteristic is not passed directly from father to son; if a male clearly inherits

a characteristic from his father—and the mother is not heterozygous—it cannot be X linked.

Connecting Concepts Across Chapters



In this chapter, we have examined sex determination and the inheritance of traits encoded by genes located on the sex chromosomes. An important theme has been that sex is determined in a variety of different ways—not all organisms have the familiar XX-XY system seen in humans. Even among organisms with XX-XY sex determination, the sexual phenotype of an individual can be shaped by very different mechanisms.

The discussion of sex determination lays the foundation for an understanding of sex-linked inheritance, covered in the last part of the chapter. Because males and females differ in sex chromosomes, which are not homologous, they do not possess the same number of alleles at sex-linked loci, and the patterns of inheritance for sex-linked characteristics are different from those for autosomal characteristics. This material augments the principles of inheritance presented in Chapter 3. The chromosome theory of inheritance, which states that genes are located on chromosomes, was first elucidated through the study of sex-linked traits. This theory provided the first clues about the physical basis of heredity, which we will explore in more detail in Chapters 10 and 11.

The ways in which sex and heredity interact are explored further in Chapter 5, where we consider additional exceptions to Mendel's principles, including sex-limited and sex-influenced traits, cytoplasmic inheritance, genetic maternal effect, and genomic imprinting. The inheritance of human sex-linked characteristics will be discussed in Chapter 6, and we will take a more detailed look at chromosome abnormalities, including abnormal sex chromosomes, in Chapter 9.

CONCEPTS SUMMARY

- Sexual reproduction is the production of offspring that are genetically distinct from the parents. Among diploid eukaryotes, sexual reproduction consists of two processes: meiosis, which produces haploid gametes, and fertilization, in which gametes unite to produce diploid zygotes.
- Most organisms have two sexual phenotypes—males and females. Males produce small gametes; females produce large gametes. The sex of an individual normally refers to the individual's sexual phenotype, not its genetic makeup.
- The mechanism by which sex is specified is termed sex determination. Sex may be determined by differences in specific chromosomes, ploidy level, genotypes, or environment.
- Sex chromosomes differ in number and appearance between males and females; other, nonsex chromosomes are termed autosomes. In organisms with chromosomal sex-determining systems, the homogametic sex produces gametes that are all identical with regard to sex chromosomes; the heterogametic sex produces two types of gametes, which differ in their sex-chromosome composition.
- In the XX-XO system, females possess two X chromosomes, and males possess a single X chromosome.
- In the XX-XY system, females possess two X chromosomes, and males possess a single X and a single Y chromosome. The X and Y chromosomes are not homologous, except at

the pseudoautosomal region, which is essential to pairing in meiosis in males.

- In the ZZ-ZW system of sex determination, males possess two Z chromosomes and females possess an L_Z and a L_W chromosome.
- In some organisms, ploidy level determines sex; males develop from unfertilized eggs (and are haploid) and females develop from fertilized eggs (and are diploid). Other organisms have genic sex determination, in which genotypes at one or more loci determine the sex of an individual. Still others have environmental sex determination.
- In *Drosophila melanogaster*, sex is determined by a balance between genes on the X chromosomes and genes on the

autosomes, the X:A ratio. An X:A ratio of 1.0 produces a female; an X:A ratio of 0.5 produces a male; and an X:A ratio between 1.0 and 0.5 produces an intersex.

- In humans, sex is ultimately determined by the presence or absence of the *SRY* gene located on the Y chromosome.
- Sex-linked characteristics are determined by genes on the sex chromosomes; X-linked characteristics are encoded by genes on the X chromosome, and Y-linked characteristics are encoded by genes on the Y chromosome.
- A female inherits X-linked alleles from both parents; a male inherits X-linked alleles from his female parent only.

IMPORTANT TERMS

sex (p. 78)	homogametic sex (p. 79)	X:A ratio (p. 81)	X-linked characteristic (p. 85)
sex determination (p. 78)	pseudoautosomal region (p. 80)	Turner syndrome (p. 82)	Y-linked characteristic (p. 85)
hermaphroditism (p. 78)	genic sex determination (p. 81)	Klinefelter syndrome (p. 83)	hemizygous (p. 86)
monoecious (p. 78)	sequential hermaphroditism (p. 81)	triplo-X syndrome (p. 83)	nondisjunction (p. 86)
dioecious (p. 78)		sex-determining region Y (<i>SRY</i>) gene (p. 84)	dosage compensation (p. 90)
sex chromosomes (p. 79)		sex-linked characteristic (p. 85)	Barr body (p. 90)
autosomes (p. 79)			Lyon hypothesis (p. 90)
heterogametic sex (p. 79)			

Worked Problems

1. A fruit fly has XXXYY sex chromosomes; all the autosomal chromosomes are normal. What sexual phenotype will this fly have?

• Solution

Sex in fruit flies is determined by the X:A ratio—the ratio of the number of X chromosomes to the number of haploid autosomal sets. An X:A ratio of 1.0 produces a female fly; an X:A ratio of 0.5 produces a male. If the X:A ratio is greater than 1.0, the fly is a metafemale; if it is less than 0.5, the fly is a metamale; if the X:A ratio is between 1.0 and 0.5, the fly is an intersex.

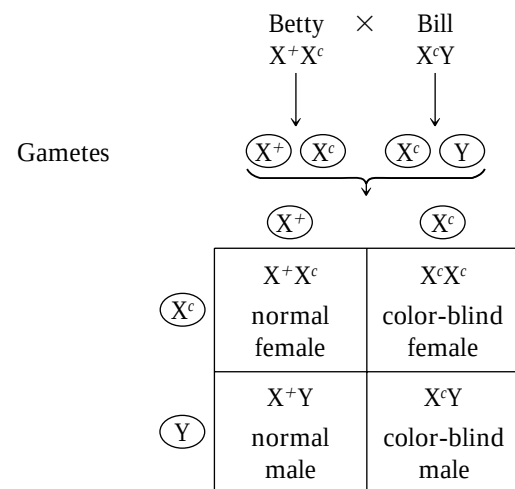
This fly has three X chromosomes and normal autosomes. Normal diploid flies have two autosomal sets of chromosomes; so the X:A ratio in this case is $\frac{3}{2}$ or 1.5. Thus, this fly is a metafemale.

2. Color blindness in humans is most commonly due to an X-linked recessive allele. Betty has normal vision, but her mother is color blind. Bill is color blind. If Bill and Betty marry and have a child together, what is the probability that the child will be color blind?

• Solution

Because color blindness is an X-linked recessive characteristic, Betty's color-blind mother must be homozygous for the color-blind allele (X^cX^c). Females inherit one X chromosome from each of their parents; so Betty must have inherited a color-blind

allele from her mother. Because Betty has normal color vision, she must have inherited an allele for normal vision (X^+) from her father; thus Betty is heterozygous (X^+X^c). Bill is color blind. Because males are hemizygous for X-linked alleles, he must be (X^cY). A mating between Betty and Bill is represented as:

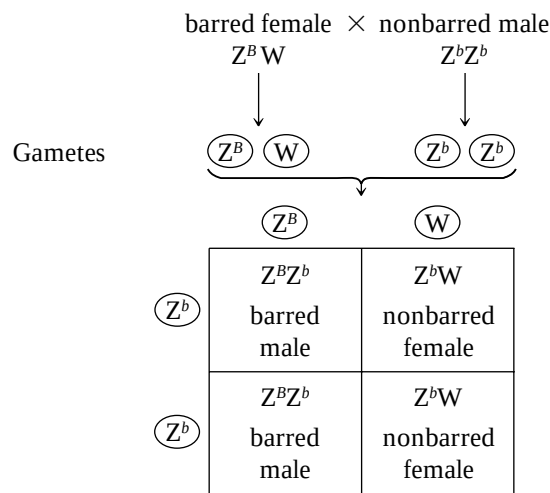


Thus, $\frac{1}{4}$ of the children are expected to be female with normal color vision, $\frac{1}{4}$ female with color blindness, $\frac{1}{4}$ male with normal color vision, and $\frac{1}{4}$ male with color blindness.

3. Chickens, like all birds, have ZZ-ZW sex determination. The bar-feathered phenotype in chickens results from a Z-linked allele that is dominant over the allele for nonbar feathers. A barred female is crossed with a nonbarred male. The F_1 from this cross are intercrossed to produce the F_2 . What will the phenotypes and their proportions be in the F_1 and F_2 progeny?

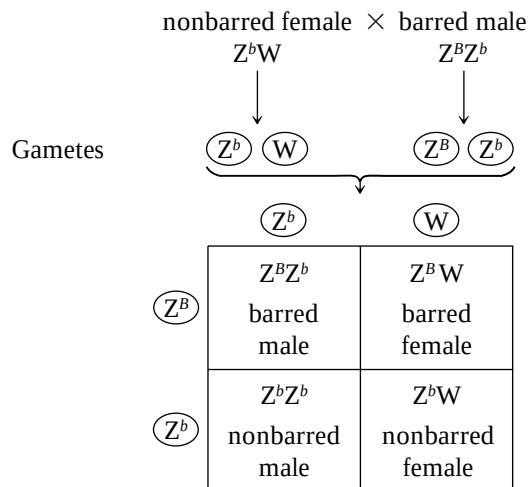
• **Solution**

With the ZZ-ZW system of sex determination, females are the heterogametic sex, possessing a Z chromosome and a W chromosome; males are the homogametic sex, with two Z chromosomes. In this problem, the barred female is hemizygous for the bar phenotype ($Z^B W$). Because bar is dominant over nonbar, the nonbarred male must be homozygous for nonbar ($Z^b Z^b$). Crossing these two chickens, we obtain:



Thus, all the males in the F_1 will be barred ($Z^B Z^b$), and all the females will be nonbarred ($Z^b W$).

The F_1 are now crossed to produce the F_2 :



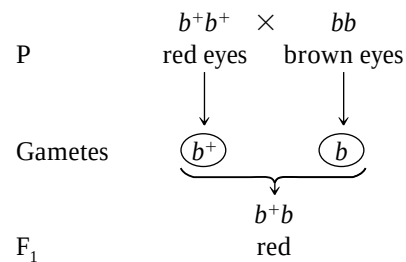
So, $\frac{1}{4}$ of the F_2 are barred males, $\frac{1}{4}$ are nonbarred males, $\frac{1}{4}$ are barred females, and $\frac{1}{4}$ are nonbarred females.

4. In *Drosophila melanogaster*, forked bristles are caused by an allele (X^f) that is X linked and recessive to an allele for normal bristles (X^+). Brown eyes are caused by an allele (b) that is autosomal and recessive to an allele for red eyes (b^+). A female fly that is homozygous for normal bristles and red eyes mates with a male fly that has forked bristles and brown eyes. The F_1 are intercrossed to produce the F_2 . What will the phenotypes and proportions of the F_2 flies be from this cross?

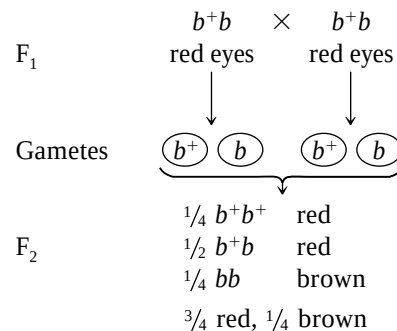
• **Solution**

This problem is best worked by breaking the cross down into two separate crosses, one for the X-linked genes that determine the type of bristles and one for the autosomal genes that determine eye color.

Let's begin with the autosomal characteristics. A female fly that is homozygous for red eyes ($b^+ b^+$) is crossed with a male with brown eyes. Because brown eyes are recessive, the male fly must be homozygous for the brown-eyed allele (bb). All of the offspring of this cross will be heterozygous ($b^+ b$) and will have brown eyes:

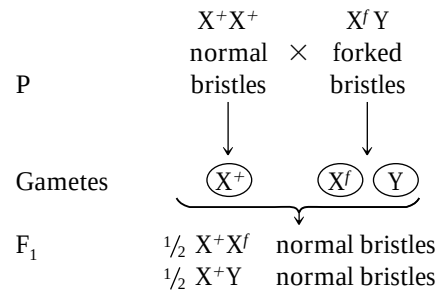


The F_1 are then intercrossed to produce the F_2 . Whenever two individuals heterozygous for an autosomal recessive characteristic are crossed, $\frac{3}{4}$ of the offspring will have the dominant trait and $\frac{1}{4}$ will have the recessive trait; thus, $\frac{3}{4}$ of the F_2 flies will have red eyes and $\frac{1}{4}$ will have brown eyes:

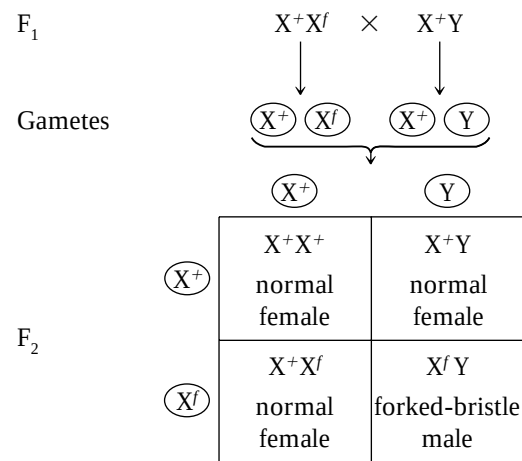


Next, we work out the results for the X-linked characteristic. A female that is homozygous for normal bristles ($X^+ X^+$) is crossed with a male that has forked bristles ($X^f Y$). The female F_1 from this cross are heterozygous ($X^+ X^f$), receiving an X chromosome with a normal-bristle allele from their mother (X^+) and an X chromosome with a forked-bristle allele (X^f) from their father. The male F_1 are hemizygous ($X^+ Y$), receiving an X

chromosome with a normal-bristle allele from their mother (X^+) and a Y chromosome from their father:



When these F₁ are intercrossed, $\frac{1}{2}$ of the F₂ will be normal-bristle females, $\frac{1}{4}$ will be normal-bristle males, and $\frac{1}{4}$ will be forked-bristle males:



$\frac{1}{2}$ normal female, $\frac{1}{4}$ normal male, $\frac{1}{4}$ forked bristle male

To obtain the phenotypic ratio in the F₂, we now combine these two crosses by using the multiplicative rule of probability and the branch diagram:

Eye color	Bristle and sex	F ₂ phenotype	Probability
red ($\frac{3}{4}$)	normal female ($\frac{1}{2}$)	red normal female	$\frac{3}{4} \times \frac{1}{2} = \frac{3}{8} = \frac{6}{16}$
	normal male ($\frac{1}{4}$)	red normal male	$\frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$
	forked-bristle male ($\frac{1}{4}$)	red forked-bristle male	$\frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$
brown ($\frac{1}{4}$)	normal female ($\frac{1}{2}$)	brown normal female	$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8} = \frac{2}{16}$
	normal male ($\frac{1}{4}$)	brown normal male	$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$
	forked-bristle male ($\frac{1}{4}$)	brown forked-bristle male	$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

COMPREHENSION QUESTIONS

- What is the most defining difference between males and females?
- How do monoecious organisms differ from dioecious organisms?
- Describe the XX-XO system of sex determination. In this system, which is the heterogametic sex and which is the homogametic sex?
- How does sex determination in the XX-XY system differ from sex determination in the ZZ-ZW system?
- What is the pseudoautosomal region? How does the inheritance of genes in this region differ from the inheritance of other Y-linked characteristics?
- How is sex determined in insects with haplodiploid sex determination?
- What is meant by genic sex determination?
- How does sex determination in *Drosophila* differ from sex determination in humans?
- Give the typical sex chromosomes found in the cells of people with Turner syndrome, Klinefelter syndrome, and androgen insensitivity syndrome, as well as in poly-X females.
- What characteristics are exhibited by an X-linked trait?
- Explain how Bridges's study of nondisjunction in *Drosophila* helped prove the chromosome theory of inheritance.
- Explain why tortoiseshell cats are almost always female and why they have a patchy distribution of orange and black fur.
- What is a Barr body? How is it related to the Lyon hypothesis?
- What characteristics are exhibited by a Y-linked trait?

APPLICATION QUESTIONS AND PROBLEMS

- * 15. What is the sexual phenotype of fruit flies having the following chromosomes?

	Sex chromosomes	Autosomal chromosomes
(a)	XX	all normal
(b)	XY	all normal
(c)	XO	all normal
(d)	XXY	all normal
(e)	XXX	all normal
(f)	XXYY	all normal
(g)	XXX	all normal
(h)	XX	four haploid sets
(i)	XXX	four haploid sets
(j)	XXX	three haploid sets
(k)	X	three haploid sets
(l)	XY	three haploid sets
(m)	XX	three haploid sets

16. For parts *a* through *g* in problem 15 what would the human sexual phenotype (male or female) be?

- * 17. Joe has classic hemophilia, which is an X-linked recessive disease. Could Joe have inherited the gene for this disease from the following persons?

	Yes	No
(a) His mother's mother	_____	_____
(b) His mother's father	_____	_____
(c) His father's mother	_____	_____
(d) His father's father	_____	_____

- * 18. In *Drosophila*, yellow body is due to an X-linked gene that is recessive to the gene for gray body.

(a) A homozygous gray female is crossed with a yellow male. The F_1 are intercrossed to produce F_2 . Give the genotypes and phenotypes, along with the expected proportions, of the F_1 and F_2 progeny.

(b) A yellow female is crossed with a gray male. The F_1 are intercrossed to produce the F_2 . Give the genotypes and phenotypes, along with the expected proportions, of the F_1 and F_2 progeny.

(c) A yellow female is crossed with a gray male. The F_1 females are backcrossed with gray males. Give the genotypes and phenotypes, along with the expected proportions, of the F_2 progeny.

(d) If the F_2 flies in part b mate randomly, what are the expected phenotypic proportions of flies in the F_3 ?

- * 19. Both John and Cathy have normal color vision. After 10 years of marriage to John, Cathy gave birth to a color-blind daughter. John filed for divorce, claiming he is not the father of the child. Is John justified in his claim of nonpaternity? Explain why. If Cathy had given birth to

a color-blind son, would John be justified in claiming nonpaternity?

20. Red–green color blindness in humans is due to an X-linked recessive gene. A woman whose father is color blind possesses one eye with normal color vision and one eye with color blindness.

(a) Propose an explanation for this woman's vision pattern.

(b) Would it be possible for a man to have one eye with normal color vision and one eye with color blindness?

- * 21. Bob has XXY chromosomes (Klinefelter syndrome) and is color blind. His mother and father have normal color vision, but his maternal grandfather is color blind. Assume that Bob's chromosome abnormality arose from nondisjunction in meiosis. In which parent and in which meiotic division did nondisjunction occur? Explain your answer.

22. In certain salamanders, it is possible to alter the sex of a genetic female, making her into a functional male; these salamanders are called sex-reversed males. When a sex-reversed male is mated with a normal female, approximately $\frac{2}{3}$ of the offspring are female and $\frac{1}{3}$ are male. How is sex determined in these salamanders? Explain the results of this cross.

23. In some mites, males pass genes to their grandsons, but they never pass genes to male offspring. Explain.

24. The Talmud, an ancient book of Jewish civil and religious laws, states that if a woman bears two sons who die of bleeding after circumcision (removal of the foreskin from the penis), any additional sons that she has should not be circumcised. (The bleeding is most likely due to the X-linked disorder hemophilia.) Furthermore, the Talmud states that the sons of her sisters must not be circumcised, whereas the sons of her brothers should. Is this religious law consistent with sound genetic principles? Explain your answer.

- * 25. Miniature wings (X^m) in *Drosophila* result from an X-linked allele that is recessive to the allele for long wings (X^+). Give the genotypes of the parents in the following crosses.

	Male parent	Female parent	Male offspring	Female offspring
(a)	long	long	231 long, 250 miniature	560 long
(b)	miniature	long	610 long	632 long
(c)	miniature	long	410 long, 417 miniature	412 long, 415 miniature
(d)	long	miniature	753 miniature	761 long
(e)	long	long	625 long	630 long

- * 26. In chickens, congenital baldness results from a Z-linked recessive gene. A bald rooster is mated with a normal hen.

The F_1 from this cross are interbred to produce the F_2 . Give the genotypes and phenotypes, along with their expected proportions, among the F_1 and F_2 progeny.

27. In the eastern mosquito fish (*Gambusia affinis holbrooki*), which has XX-XY sex determination, spotting is inherited as a Y-linked trait. The trait exhibits 100% penetrance when the fish are raised at 22°C, but the penetrance drops to 42% when the fish are raised at 26°C. A male with spots is crossed with a female without spots, and the F_1 are intercrossed to produce the F_2 . If all the offspring are raised at 22°C, what proportion of the F_1 and F_2 will have spots? If all the offspring are raised at 26°C, what proportion of the F_1 and F_2 will have spots?
- *28. How many Barr bodies would you expect to see in human cells containing the following chromosomes?
- | | | |
|--------|----------|----------|
| (a) XX | (d) XXY | (g) XYY |
| (b) XY | (e) XXYY | (h) XXX |
| (c) XO | (f) XXXY | (i) XXXX |
29. Red–green color blindness is an X-linked recessive trait in humans. Polydactyly (extra fingers and toes) is an autosomal dominant trait. Martha has normal fingers and toes and normal color vision. Her mother is normal in all respects, but her father is color blind and polydactylous. Bill is color blind and polydactylous. His mother has normal color vision and normal fingers and toes. If Bill and Martha marry, what types and proportions of children can they produce?
- *30. Miniature wings in *Drosophila melanogaster* result from an X-linked gene (X^m) that is recessive to an allele for long wings (X^+). Sepia eyes are produced by an autosomal gene (s) that is recessive to an allele for red eyes (s^+).
- (a) A female fly that has miniature wings and sepia eyes is crossed with a male that has normal wings and is homozygous for red eyes. The F_1 are intercrossed to produce the F_2 . Give the phenotypes and their proportions expected in the F_1 and F_2 flies from this cross.
- (b) A female fly that is homozygous for normal wings and has sepia eyes is crossed with a male that has miniature wings and is homozygous for red eyes. The F_1 are intercrossed to produce the F_2 . Give the phenotypes and proportions expected in the F_1 and F_2 flies from this cross.
31. Suppose that a recessive gene that produces a short tail in mice is located in the pseudoautosomal region. A short-tailed male is mated with a female mouse that is homozygous for a normal tail. The F_1 from this cross are intercrossed to produce the F_2 . What will the phenotypes and proportions of the F_1 and F_2 mice be from this cross?
- *32. A color-blind female and a male with normal vision have three sons and six daughters. All the sons are color blind. Five of the daughters have normal vision, but one of them is color blind. The color-blind daughter is 16 years old, is short for her age, and has never undergone puberty. Propose an explanation for how this girl inherited her color blindness.

CHALLENGE QUESTIONS

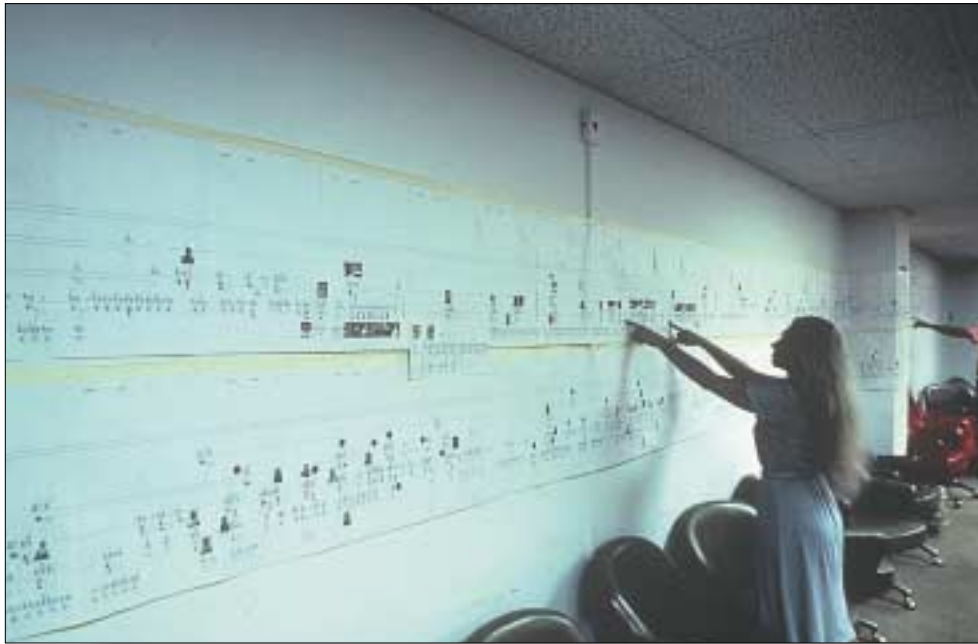
33. On average, what proportion of the X-linked genes in the first individual is the same as that in the second individual?
- (a) A male and his mother
 (b) A female and her mother
 (c) A male and his father
 (d) A female and her father
 (e) A male and his brother
 (f) A female and her sister
 (g) A male and his sister
 (h) A female and her brother
34. A geneticist discovers a male mouse in his laboratory colony with greatly enlarged testes. He suspects that this trait results from a new mutation that is either Y linked or autosomal dominant. How could he determine whether the trait is autosomal dominant or Y linked?
35. Amanda is a genetics student at a small college in Connecticut. While counting her fruit flies in the laboratory one afternoon, she observed a strange species of fly in the room. Amanda captured several of the flies and began to raise them. After having raised the flies for several generations, she discovered a mutation in her colony that produces yellow eyes, in contrast with normal red eyes, and Amanda determined that this trait is definitely X-linked recessive. Because yellow eyes are X linked, she assumed that either this species has the XX-XY system of sex determination with genic balance similar to *Drosophila* or it has the XX-XO system of sex determination.
- How can Amanda determine whether sex determination in this species is XX-XY or XX-XO? The chromosomes of this species are very small and hard for Amanda to see with her student microscope, so she can only conduct crosses with flies having the yellow-eye mutation. Outline the crosses that Amanda should conduct and explain how they will prove XX-XY or XX-XO sex determination in this species.
36. Occasionally, a mouse X chromosome is broken into two pieces and each piece becomes attached to a different autosomal chromosome. In this event, only the genes on one of the two pieces undergo X inactivation. What does this observation indicate about the mechanism of X-chromosome inactivation?

SUGGESTED READINGS

- Allen, G. E. 1978. *Thomas Hunt Morgan: The Man and His Science*. Princeton, NJ: Princeton University Press.
An excellent history of one of the most important biologists of the early twentieth century.
- Bogan, J. S., and D. C. Page. 1994. Ovary? Testis? A mammalian dilemma. *Cell* 76:603–607.
A concise review of the molecular nature of sex determination in mammals.
- Bridges, C. B. 1916. Nondisjunction as proof of the chromosome theory of heredity. *Genetics* 1:1–52.
Bridges's original paper describing his use of nondisjunction of X chromosomes to prove the chromosome theory of heredity.
- Foster, E. A., M. A. Jobling, P. G. Taylor, P. Donnelly, P. de Knijff, R. Mieremet, T. Zerjal, and C. Tyler-Smith. 1998. Jefferson fathered slave's last child. *Nature* 396:27–28.
Report on the use of Y-linked markers to establish the paternity of children of Thomas Jefferson's slave.
- Kohler, R. E. 1994. *Lords of the Fly: Drosophila Genetics and the Experimental Life*. Chicago: University of Chicago Press.
A comprehensive history of *Drosophila* genetics from 1910 to the early 1940s.
- Marx, J. 1995. Tracing how the sexes develop. *Science* 269: 1822–1824.
A short, easy-to-read review of research on sex determination in fruit flies and the worm *Caenorhabditis elegans*.
- McClung, C. E. 1902. The accessory chromosome: sex determinant. *Biological Bulletin* 3:43–84.
McClung's original description of the X chromosome.
- Morgan, T. H. 1910. Sex-limited inheritance in *Drosophila*. *Science* 32:120–122.
First description of an X-linked trait.
- Penny, G. D., G. F. Kay, S. A. Sheardown, S. Rastan, and N. Brockdorff. 1996. Requirement for Xist in X chromosome inactivation. *Nature* 379:131–137.
This article provides evidence that the *XIST* gene has a role in X-chromosome inactivation.
- Ryder, L. C., and A. Swain. 1995. Sex in the 90s. *Cell* 81:483–493.
A review of research findings about sex determination and dosage compensation.
- Thomas, M. G., T. Parfitt, D. A. Weiss, K. Skorecki, J. F. Wilson, M. le Roux, N. Bradman, and D. B. Goldstein. 2000. Y chromosomes traveling south: the Cohen modal haplotype and the origins of the Lemba—the “Black Jews of Southern Africa.” *American Journal of Human Genetics* 66:674–686.
A fascinating report of the use of Y-linked genetic markers to trace the male ancestry of the Lemba tribe of South Africa.
- Williams, N. 1995. How males and females achieve X equality. *Science* 269:1826–1827.
A brief, readable review of recent research on dosage compensation.

5

Extensions and Modifications of Basic Principles



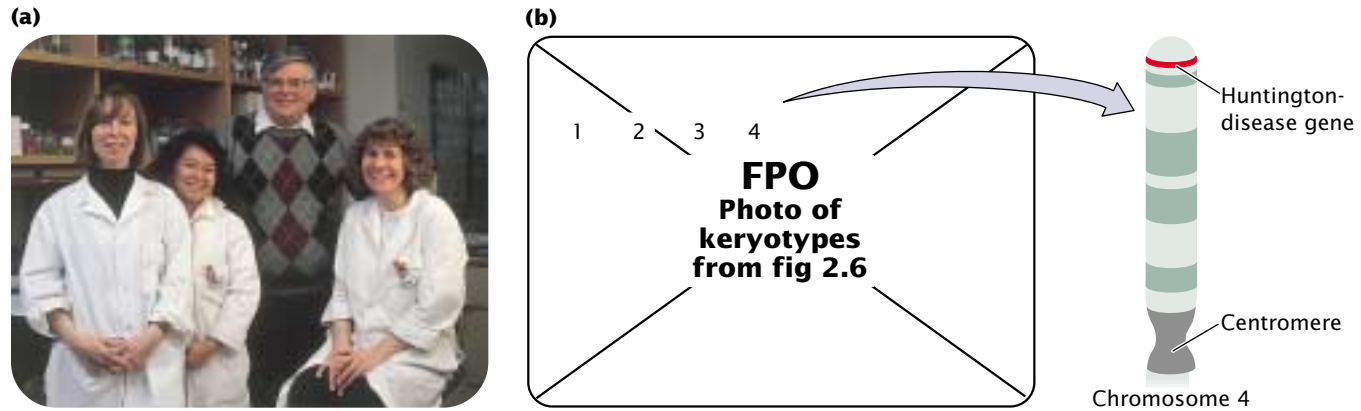
This is Chapter 5 Opener photo legend to position here. (Nancy Wexler, HDF/Neurology, Columbia University.)

- Was Mendel Wrong?
- Dominance Revisited
- Lethal Alleles
- Multiple Alleles
 - Duck-Feather Patterns
 - The ABO Blood Group
- Gene Interaction
 - Gene Interaction That Produces Novel Phenotypes
 - Gene Interaction with Epistasis
 - The Complex Genetics of Coat Color in Dogs
- The Interaction Between Sex and Heredity
 - Sex-Influenced and Sex-Limited Characteristics
 - Cytoplasmic Inheritance
 - Genetic Maternal Effects
 - Genomic Imprinting
- Anticipation
- Interaction Between Genes and Environment
 - Environmental Effects on Gene Expression
 - The Inheritance of Continuous Characteristics

Was Mendel Wrong?

In 1872, a physician from Long Island, New York named George Huntington described a medical condition characterized by jerky, involuntary movements. Now known as Huntington disease, the condition typically appears in middle age. The initial symptoms are subtle, consisting of mild behavioral and neurological changes; but, as the disease progresses, speech is impaired, walking becomes difficult, and psychiatric problems develop that frequently lead to insanity. Most people who have Huntington disease live for 10 to 30 years after the disease begins; there is currently no cure or effective treatment.

Huntington disease appears with equal frequency in males and females, rarely skips generations and, when one parent has the disorder, approximately half of the children will be similarly affected. These are the hallmarks of an autosomal dominant trait—with one exception. The disorder occasionally arises before the age of 15 and, in these cases, progresses much more rapidly than it does when it arises in middle age. Among younger patients, the trait is almost always inherited from the father. According to Mendel's principles of heredity (Chapter 3), males and females transmit autosomal traits with equal frequency, and reciprocal crosses should yield identical results; yet, for juvenile cases of Huntington



5.1 The gene for Huntington disease. (a) James Gusella and colleagues, whose research located the Huntington gene. (b) The gene has been mapped to the tip of chromosome 4. (Part a, Sam Ogden; part b, left courtesy of Dr. Thomas Ried and Dr. Evelin Schrock.)

disease, Mendel's principles do not apply. Was Mendel wrong?

In 1983, a molecular geneticist at Massachusetts General Hospital named James Gusella determined that the gene causing Huntington disease is located near the tip of the short arm of chromosome 4. Gusella determined its location by analyzing DNA from members of the largest known family with Huntington disease, about 7000 people who live near Lake Maracaibo in Venezuela, more than 100 of whom have Huntington disease. Many experts predicted that, with the general location of the Huntington gene pinned down, the actual DNA sequence would be isolated within a few years. Despite intensive efforts, finding the gene took 10 years. When it was finally isolated in the spring of 1993 (**FIGURE 5.1**), the gene turned out to be quite different from any of those that code for the traits studied by Mendel.

The mutation that causes Huntington disease consists of an unstable region of DNA capable of expanding and contracting as it is passed from generation to generation. When the region expands, Huntington disease results. The degree of expansion affects the severity and age of onset of symptoms; the juvenile form of Huntington disease results from rapid expansion of the region, which occurs primarily when the gene is transmitted from father to offspring.

This genetic phenomenon—the earlier appearance of a trait as it is passed from generation to generation—is called anticipation. Like a number of other genetic phenomena, anticipation does not adhere to Mendel's principles of heredity. This lack of adherence doesn't mean that Mendel was wrong; rather, it means that Mendel's principles are not, by themselves, sufficient to explain the inheritance of all genetic characteristics. Our modern understanding of genetics has been greatly enriched by the discovery of a number of modifications and extensions of Mendel's basic principles, which are the focus of this chapter.

An important extension of Mendel's principles of heredity—the inheritance of sex-linked characteristics—

was introduced in Chapter 4. In this chapter, we will examine a number of additional refinements of Mendel's basic tenets. We begin by reviewing the concept of dominance, emphasizing that dominance entails interactions between genes at one locus (allelic genes) and affects the way in which genes are expressed in the phenotype. Next, we consider lethal alleles and their effect on phenotypic ratios, followed by a discussion of multiple alleles. We then turn to interaction among genes at different loci (nonallelic genes). The phenotypic ratios produced by gene interaction are related to the ratios encountered in Chapter 3. In the latter part of the chapter, we will consider ways in which sex interacts with heredity. Our last stop will be a discussion of environmental influences on gene expression.

The modifications and extensions of hereditary principles discussed in this chapter do not invalidate Mendel's important contributions; rather, they enlarge our understanding of heredity by building on the framework provided by his principles of segregation and independent assortment. These modifications rarely alter the way in which the genes are inherited; rather, they affect the ways in which the genes determine the phenotype.

www.whfreeman.com/pierce

Additional information about Huntington disease

Dominance Revisited

One of Mendel's important contributions to the study of heredity is the concept of *dominance*—the idea that an individual possesses two different alleles for a characteristic, but the trait enclosed by only one of the alleles is observed in the phenotype. With dominance, the heterozygote possesses the same phenotype as one of the homozygotes. When biologists began to apply Mendel's principles to organisms other than peas, it quickly became apparent that many characteristics do not exhibit this type of dominance. Indeed, Mendel

himself was aware that dominance is not universal, because he observed that a pea plant heterozygous for long and short flowering times had a flowering time that was intermediate between those of its homozygous parents. This situation, in which the heterozygote is intermediate in phenotype between the two homozygotes, is termed *incomplete dominance*.

Dominance can be understood in regard to how the phenotype of the heterozygote relates to the phenotypes of the homozygotes. In the example presented in **FIGURE 5.2**, flower color potentially ranges from red to white. One homozygous genotype, A^1A^1 , codes for red flowers, and another, A^2A^2 , codes for white flowers. Where the heterozygote falls on the range of phenotypes determines the type of dominance. If the heterozygote (A^1A^2) has flowers that are the same color as those of the A^1A^1 homozygote (red), then the A^1 allele is *completely dominant* over the A^2 allele; that is, red is dominant over white. If, on the other hand, the heterozygote has flowers that are the same color as the A^2A^2 homozygote (white), then the A^2 allele is completely dominant, and white is dominant over red. When the heterozygote falls in between the phenotypes of the two homozygotes, dominance is incomplete. With incomplete dominance, the heterozygote need not be exactly intermediate (pink in our example) between the two homozygotes; it might be a slightly lighter shade of red or a slightly pink shade of white. As long as the heterozygote's phenotype can be differentiated and falls within the range of the two homozygotes, dominance is

Table 5.1 Differences between dominance, incomplete dominance, and codominance

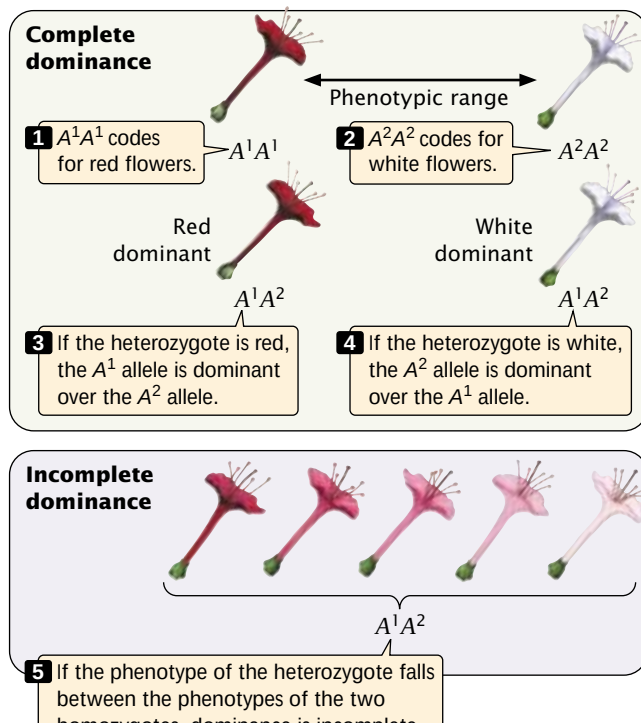
Type of Dominance	Definition
Dominance	Phenotype of the heterozygote is the same as the phenotype of one of the homozygotes
Incomplete dominance	Phenotype of the heterozygote is intermediate (falls within the range) between the phenotypes of the two homozygotes
Codominance	Phenotype of the heterozygote includes the phenotypes of both homozygotes

incomplete. The important thing to remember about dominance is that it affects the phenotype that genes produce, but not the way in which genes are *inherited*.

Another type of interaction between alleles is **codominance**, in which the phenotype of the heterozygote is not intermediate between the phenotypes of the homozygotes; rather, the heterozygote simultaneously expresses the phenotypes of both homozygotes. An example of codominance is seen in the MN blood types.

The MN locus codes for one of the types of antigens on red blood cells. Unlike antigens foreign to the ABO and Rh blood groups (which also code for red-blood-cell antigens), foreign MN antigens do not elicit a strong immunological reaction, and therefore the MN blood types are not routinely considered in blood transfusions. At the MN locus, there are two alleles: the L^M allele, which codes for the M antigen; and the L^N allele, which codes for the N antigen. Homozygotes with genotype L^ML^M express the M antigen on their red blood cells and have the M blood type. Homozygotes with genotype L^NL^N express the N antigen and have the N blood type. Heterozygotes with genotype L^ML^N exhibit codominance and express both the M and the N antigens; they have blood type MN. The differences between dominance, incomplete dominance, and codominance are summarized in Table 5.1.

The type of dominance that a character exhibits frequently depends on the level of the phenotype examined. An example is cystic fibrosis, one of the more common genetic disorders found in Caucasians and usually considered to be a recessive disease. People who have cystic fibrosis produce large quantities of thick, sticky mucus, which plugs up the airways of the lungs and clogs the ducts leading from the pancreas to the intestine, causing frequent respiratory infections and digestive problems. Even with medical treatment, patients with cystic fibrosis suffer chronic, life-threatening medical problems.



5.2 The type of dominance exhibited by a trait depends on how the phenotype of the heterozygote relates to the phenotypes of the homozygotes.

The gene responsible for cystic fibrosis resides on the long arm of chromosome 7. It encodes a protein termed *cystic fibrosis transmembrane conductance regulator*, mercifully abbreviated CFTR, which acts as a gate in the cell membrane and regulates the movement of chloride ions into and out of the cell. Patients with cystic fibrosis have a mutated, dysfunctional form of CFTR that causes the channel to stay closed, and so chloride ions build up in the cell. This buildup causes the formation of thick mucus and produces the symptoms of the disease.

Most people have two copies of the normal allele for CFTR, and produce only functional CFTR protein. Those with cystic fibrosis possess two copies of the mutated CFTR allele, and produce only the defective CFTR protein. Heterozygotes, with one normal and one defective CFTR allele, produce both functional and defective CFTR protein. Thus, at the molecular level, the alleles for normal and defective CFTR are codominant, because both alleles are expressed in the heterozygote. However, because one normal allele produces enough functional CFTR protein to allow normal chloride transport, the heterozygote exhibits no adverse effects, and the mutated CFTR allele appears to be recessive at the physiological level.

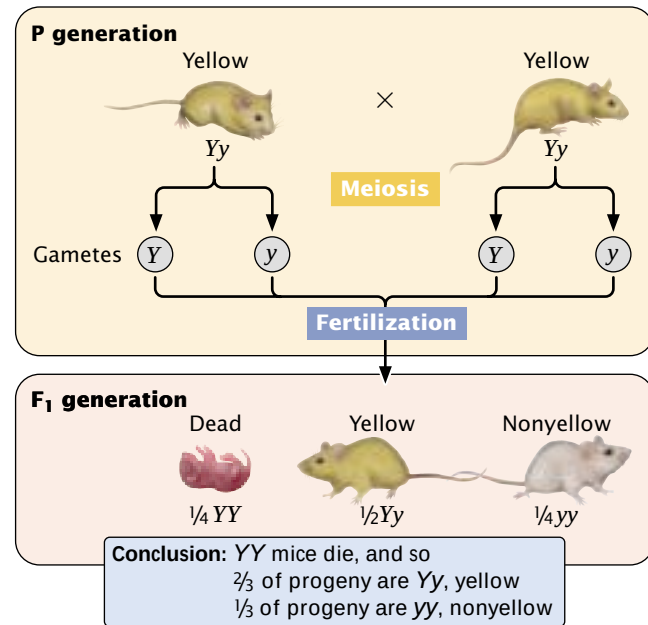
In summary, several important characteristics of dominance should be emphasized. First, dominance is a result of interactions between genes at the same locus; in other words, dominance is *allelic* interaction. Second, dominance does not alter the way in which the genes are inherited; it only influences the way in which they are expressed as a phenotype. The allelic interaction that characterizes dominance is therefore interaction between the *products* of the genes. Finally, dominance is frequently “in the eye of the beholder,” meaning that the classification of dominance depends on the level at which the phenotype is examined. As we saw with cystic fibrosis, an allele may exhibit codominance at one level and be recessive at another level.

Concepts

Dominance entails interactions between genes at the same locus (allelic genes) and is an aspect of the phenotype; dominance does not affect the way in which genes are inherited. The type of dominance exhibited by a characteristic frequently depends on the level of the phenotype examined.

Lethal Alleles

In 1905, Lucien Cuenot reported a peculiar pattern of inheritance in mice. When he mated two yellow mice, approximately $\frac{2}{3}$ of their offspring were yellow and $\frac{1}{3}$ were nonyellow. When he test-crossed the yellow mice, he found that all were heterozygous; he was never able to obtain a yellow mouse that bred true. There was a great deal of



5.3 A 2 : 1 ratio among the progeny of a cross results from the segregation of a lethal allele.

discussion about Cuenot's results among his colleagues, but it was eventually realized that the yellow allele must be lethal when homozygous (FIGURE 5.3). A **lethal allele** is one that causes death at an early stage of development—often before birth—and so some genotypes may not appear among the progeny.

Cuenot originally crossed two mice heterozygous for yellow: $Yy \times Yy$. Normally, this cross would be expected to produce $\frac{1}{4} YY$, $\frac{1}{2} Yy$, and $\frac{1}{4} yy$ (see Figure 5.3). The homozygous YY mice are conceived but never complete development, which leaves a 2:1 ratio of Yy (yellow) to yy (nonyellow) in the observed offspring; all yellow mice are heterozygous (Yy).

Another example of a lethal allele, originally described by Erwin Baur in 1907, is found in snapdragons. The *aurea* strain in these plants has yellow leaves. When two plants with yellow leaves are crossed, $\frac{2}{3}$ of the progeny have yellow leaves and $\frac{1}{3}$ have green leaves. When green is crossed with green, all the progeny have green leaves; however, when yellow is crossed with green, $\frac{1}{2}$ of the progeny are green and $\frac{1}{2}$ are yellow, confirming that all yellow-leaved snapdragons are heterozygous. A 2:1 ratio is almost always produced by a recessive lethal allele; so observing this ratio among the progeny of a cross between individuals with the same phenotype is a strong clue that one of the alleles is lethal.

In both of these examples, the lethal alleles are recessive because they cause death only in homozygotes. Unlike its effect on *survival*, the effect of the allele on *color* is dominant; in both mice and snapdragons, a single copy of the allele in the heterozygote produces a yellow color. Lethal alleles also can be dominant; in this case, homozygotes and

heterozygotes for the allele die. Truly dominant lethal alleles cannot be transmitted unless they are expressed after the onset of reproduction, as in Huntington disease.

Concepts

A lethal allele causes death, frequently at an early developmental stage, and so one or more genotypes are missing from the progeny of a cross. Lethal alleles may modify the ratio of progeny resulting from a cross.

Multiple Alleles

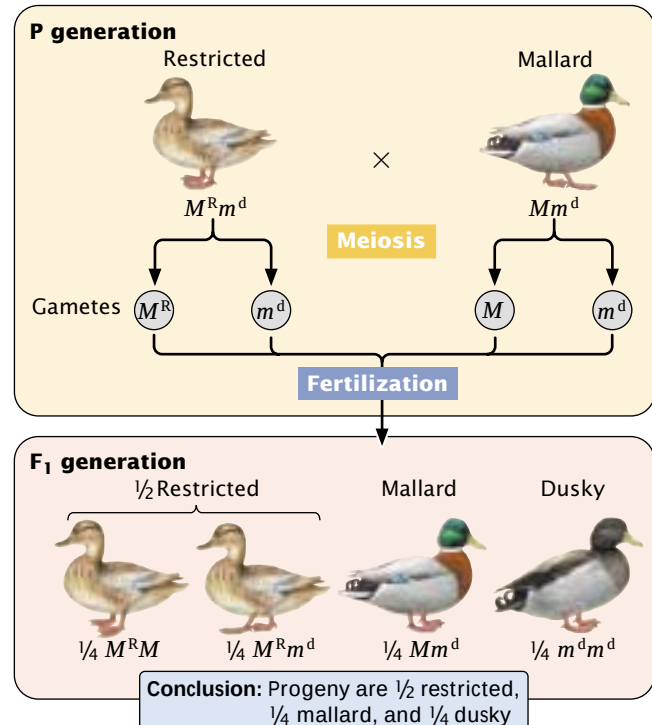
Most of the genetic systems that we have examined so far consist of two alleles. In Mendel's peas, for instance, one allele coded for round seeds and another for wrinkled seeds; in cats, one allele produced a black coat and another produced a gray coat. For some loci, more than two alleles are present within a group of individuals—the locus has **multiple alleles**. (Multiple alleles may also be referred to as an *allelic series*.) Although there may be more than two alleles present within a *group*, the genotype of each diploid *individual* still consists of only two alleles. The inheritance of characteristics encoded by multiple alleles is no different from the inheritance of characteristics encoded by two alleles, except that a greater variety of genotypes and phenotypes are possible.

Duck-Feather Patterns

An example of multiple alleles is seen at a locus that determines the feather pattern of mallard ducks. One allele, M , produces the wild-type *mallard* pattern. A second allele, M^R , produces a different pattern called *restricted*, and a third allele, m^d , produces a pattern termed *dusky*. In this allelic series, restricted is dominant over mallard and dusky, and mallard is dominant over dusky: $M^R > M > m^d$. The six genotypes possible with these three alleles and their resulting phenotypes are:

Genotype	Phenotype
$M^R M^R$	restricted
$M^R M$	restricted
$M^R m^d$	restricted
MM	mallard
Mm^d	mallard
$m^d m^d$	dusky

In general, the number of genotypes possible will be $[n(n+1)]/2$, where n equals the number of different alleles at a locus. Working crosses with multiple alleles is no different from working crosses with two alleles; Mendel's principle of segregation still holds, as shown in the cross between a restricted duck and a mallard duck (● **FIGURE 5.4**).



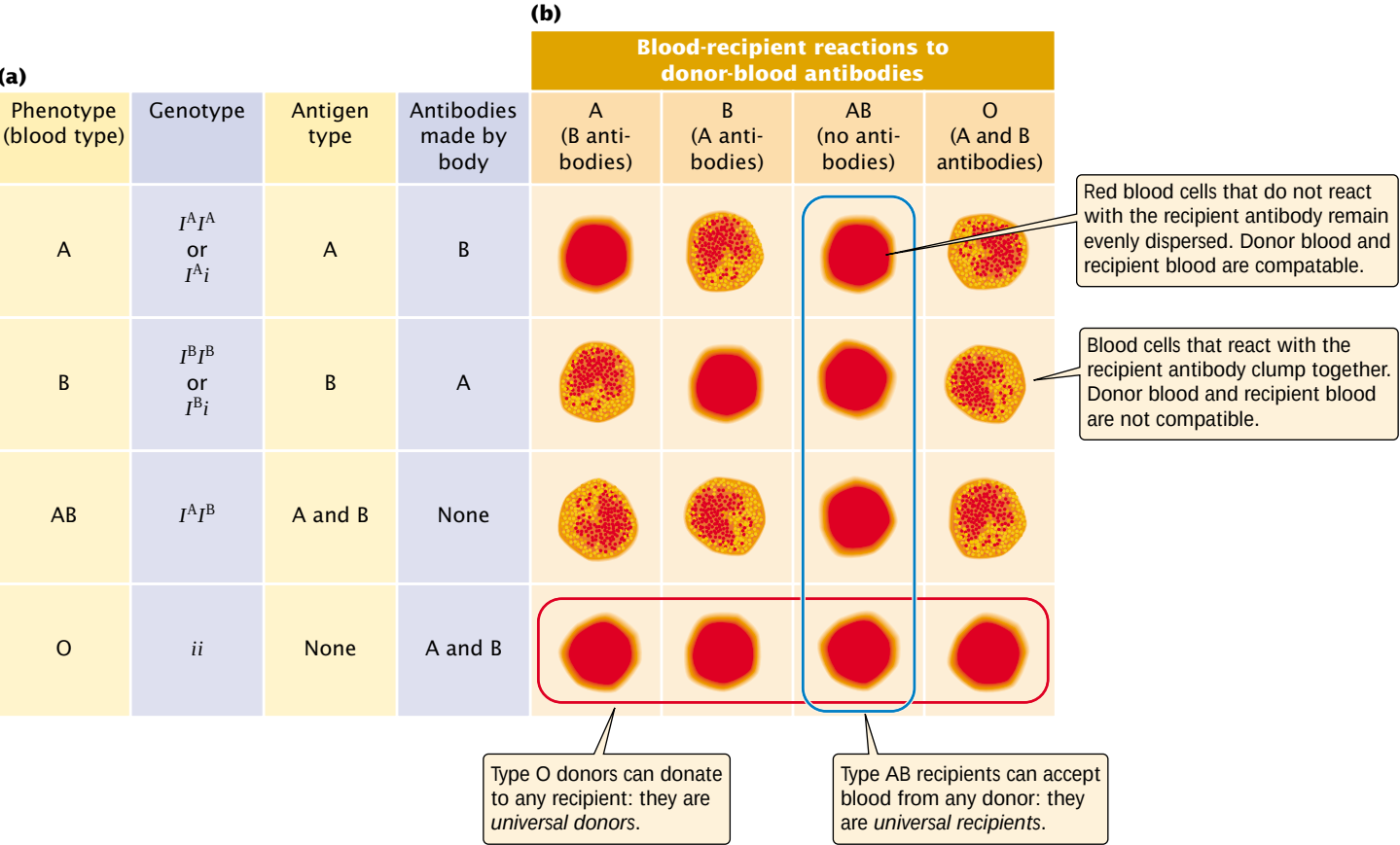
5.4 Mendel's principle of segregation applies to crosses with multiple alleles.

In this example, three alleles determine the type of plumage in mallard ducks: M^R (Restricted) $>$ M (Mallard) $>$ m^d (Dusky).

The ABO Blood Group

Another multiple-allele system is at the locus for the ABO blood group. This locus determines your ABO blood type and, like the MN locus, codes for antigens on red blood cells. The three common alleles for the ABO blood group locus are: I^A , which codes for the A antigen; I^B , which codes for the B antigen; and i , which codes for no antigen (O). We can represent the dominance relations among the ABO alleles as follows: $I^A > i$, $I^B > i$, $I^A = I^B$. The I^A and I^B alleles are both dominant over i and are codominant with each other; the AB phenotype is due to the presence of an I^A allele and an I^B allele, which results in the production of A and B antigens on red blood cells. An individual with genotype ii produces neither antigen and has blood type O. The six common genotypes at this locus and their phenotypes are shown in ● **FIGURE 5.5a**.

Antibodies are produced against any foreign antigens (see Figure 5.5a). For instance, a person having blood type A produces B antibodies, because the B antigen is foreign. A person having blood type B produces A antibodies, and someone having blood type AB produces neither A nor B antibodies, because neither A nor B antigen is foreign. A person having blood type O possesses no A or B antigens; consequently that person produces both A antibodies and B antibodies. The presence of antibodies against foreign ABO antigens means



5.5 ABO blood types and possible blood transfusions.

that successful blood transfusions are possible only between persons with certain compatible blood types (FIGURE 5.5b).

The inheritance of alleles at the ABO locus can be illustrated by a paternity suit involving the famous movie actor Charlie Chaplin. In 1941, Chaplin met a young actress named Joan Barry, with whom he had an affair. The affair ended in February 1942 but, 20 months later, Barry gave birth to a baby girl and claimed that Chaplin was the father. Barry then sued for child support. At this time, blood typing had just come into widespread use, and Chaplin's attorneys had Chaplin, Barry, and the child blood typed. Barry had blood type A, her child had blood type B, and Chaplin had blood type O. Could Chaplin have been the father of Barry's child?

Your answer should be no. Joan Barry had blood type A, which can be produced by either genotype $I^A I^A$ or $I^A i$. Her baby possessed blood type B, which can be produced by either genotype $I^B I^B$ or $I^B i$. The baby could not have inherited the I^B allele from Barry (Barry could not carry an I^B allele if she were blood type A); therefore the baby must have inherited the i allele from her. Barry must have had genotype $I^A i$, and the baby must have had genotype $I^B i$. Because the baby girl inherited her i allele from Barry, she must have inherited the I^B allele from her father. With blood type O, produced only by genotype ii , Chaplin could not have been the father of Barry's child. In the course of

the trial to settle the paternity suit, three pathologists came to the witness stand and declared that it was genetically impossible for Chaplin to have fathered the child. Nevertheless, the jury ruled that Chaplin was the father and ordered him to pay child support and Barry's legal expenses.

Concepts

More than two alleles (multiple alleles) may be present within a group of individuals, although each diploid individual still has only two alleles at that locus.

Gene Interaction

In the dihybrid crosses that we examined in Chapter 3, each locus had an independent effect on the phenotype. When Mendel crossed a homozygous round and yellow plant ($RRYY$) with a homozygous wrinkled and green plant ($rryy$) and then self-fertilized the F_1 , he obtained F_2 progeny in the following proportions:

$\frac{9}{16}$	$R_Y_$	round, yellow
$\frac{3}{16}$	R_yy	round, green
$\frac{3}{16}$	$rrY_$	wrinkled, yellow
$\frac{1}{16}$	$rryy$	wrinkled, green

In this example, the genes showed two kinds of independence. First, the genes at each locus are independent in their *assortment* in meiosis, which is what produces the 9:3:3:1 ratio of phenotypes in the progeny, in accord with Mendel's principle of independent assortment. Second, the genes are independent in their *phenotypic expression*; the *R* and *r* alleles affect only the shape of the seed and have no influence on the color of the endosperm; the *Y* and *y* alleles affect only color and have no influence on the shape of the seed.

Frequently, genes exhibit independent assortment but do not act independently in their phenotypic expression; instead, the effects of genes at one locus depend on the presence of genes at other loci. This type of interaction between the effects of genes at different loci (genes that are not allelic) is termed **gene interaction**. With gene interaction, the products of genes at different loci combine to produce new phenotypes that are not predictable from the single-locus effects alone. In our consideration of gene interaction, we'll focus primarily on interaction between the effects of genes at two loci, although interactions among genes at three, four, or more loci are common.

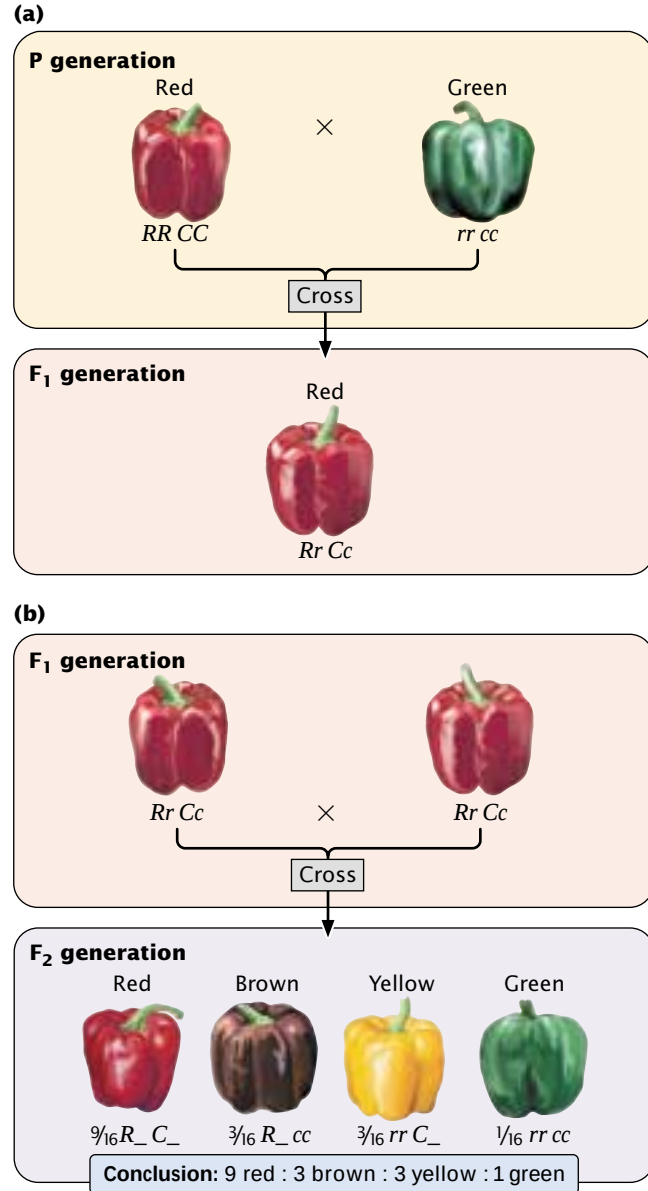
Concepts

In gene interaction, genes at different loci contribute to the determination of a single phenotypic characteristic.

Gene Interaction That Produces Novel Phenotypes

Let's first examine gene interaction in which genes at two loci interact to produce a single characteristic. Fruit color in the pepper *Capsicum annuum* is determined in this way. This plant produces peppers in one of four colors: red, brown, yellow, or green. If a homozygous plant with red peppers is crossed with a homozygous plant with green peppers, all the F_1 plants have red peppers (FIGURE 5.6a). When the F_1 are crossed with one another, the F_2 are in a ratio of 9 red : 3 brown : 3 yellow : 1 green (FIGURE 5.6b). This dihybrid ratio (Chapter 3) is produced by a cross between two plants that are both heterozygous for two loci ($RrCc \times RrCc$). In peppers, a dominant allele *R* at the first locus produces a red pigment; the recessive allele *r* at this locus produces no red pigment. A dominant allele *C* at the second locus causes decomposition of the green pigment chlorophyll; the recessive allele *c* allows chlorophyll to persist. The genes at the two loci then interact to produce the colors seen in F_2 peppers:

Genotype	Phenotype
$R_C_$	red
R_cc	brown
$rrC_$	yellow
$rrcc$	green



5.6 Gene interaction in which two loci determine a single characteristic, fruit color, in the pepper *Capsicum annuum*.

To illustrate how Mendel's rules of heredity can be used to understand the inheritance of characteristics determined by gene interaction, let's consider a testcross between an F_1 plant from the cross in Figure 5.6 ($RrCc$) and a plant with green peppers ($rrcc$). As outlined in Chapter 3 (p. 000) for independent loci, we can work this cross by breaking it down into two simple crosses. At the first locus, the heterozygote Rr is crossed with the homozygote rr ; this cross produces $\frac{1}{2} Rr$ and $\frac{1}{2} rr$ progeny. Similarly, at the second locus, the heterozygous genotype Cc is crossed with the homozygous genotype cc , producing $\frac{1}{2} Cc$ and $\frac{1}{2} cc$ progeny. In accord with Mendel's principle of



5.7 A chicken’s comb is determined by gene interaction between two loci. (a) A walnut comb is produced when there is a dominant allele at each of two loci ($R_P_$). (b) A rose comb occurs when there is a dominant allele only at the first locus (R_pp). (c) A pea comb occurs when there is a dominant allele only at the second locus ($ppR_$). (d) A single comb is produced by the presence of only recessive alleles at both loci ($rrpp$). (Parts a and d, R. OSF Dowling/Animals Animals; part b, Robert Maier/Animals Animals; part c, George Godfrey/Animals Animals.)

independent assortment, these single-locus ratios can be combined by using the multiplication rule: the probability of obtaining the genotype $RrCc$ is the probability of Rr ($\frac{1}{2}$) multiplied by the probability of Cc ($\frac{1}{2}$), or $\frac{1}{4}$. The probability of each progeny genotype resulting from the testcross is:

Progeny genotype	Probability at each locus	Overall probability	Phenotype
$RrCc$	$\frac{1}{2} \times \frac{1}{2} =$	$\frac{1}{4}$	red peppers
$Rrcc$	$\frac{1}{2} \times \frac{1}{2} =$	$\frac{1}{4}$	brown peppers
$rrCc$	$\frac{1}{2} \times \frac{1}{2} =$	$\frac{1}{4}$	yellow peppers
$rrcc$	$\frac{1}{2} \times \frac{1}{2} =$	$\frac{1}{4}$	green peppers

When you work problems with gene interaction, it is especially important to determine the probabilities of single-locus genotypes and to multiply the probabilities of *genotypes*, not *phenotypes*, because the phenotypes cannot be determined without considering the effects of the genotypes at all the contributing loci.

Another example of gene interaction that produces novel phenotypes is seen in the genes that determine comb shape in chickens. The comb is the fleshy structure found on the head of a chicken. Genes at two loci (R, r and P, p) interact to determine the four types of combs shown in **FIGURE 5.7**. A walnut comb is produced when at least one dominant allele R is present at the first locus and at least one dominant allele P is present at the second locus (genotype $R_P_$). A chicken with at least one dominant allele at the first locus and two recessive alleles at the second locus (genotype R_pp) possesses a rose comb. If two

recessive alleles are present at the first locus and at least one dominant allele is present at the second (genotype $rrP_$), the chicken has a pea comb. Finally, if two recessive alleles are present at both loci ($rrpp$), the bird has a single comb.

Gene Interaction with Epistasis

Sometimes the effect of gene interaction is that one gene masks (hides) the effect of another gene at a different locus, a phenomenon known as **epistasis**. This phenomenon is similar to dominance, except that dominance entails the masking of genes at the *same* locus (allelic genes). In epistasis, the gene that does the masking is called the **epistatic gene**; the gene whose effect is masked is a **hypostatic gene**. Epistatic genes may be recessive or dominant in their effects.

Recessive epistasis Recessive epistasis is seen in the genes that determine coat color in Labrador retrievers. These dogs may be black, brown, or yellow; their different coat colors are determined by interactions between genes at two loci (although a number of other loci also help to determine coat color; see p. 000). One locus determines the type of pigment produced by the skin cells: a dominant allele B codes for black pigment, whereas a recessive allele b codes for brown pigment. Alleles at a second locus affect the *deposition* of the pigment in the shaft of the hair; allele E allows dark pigment (black or brown) to be deposited, whereas a recessive allele e prevents the deposition of dark pigment, causing the hair to be yellow. The presence of genotype ee at the second locus therefore masks the expression of the black and brown alleles at the first locus. The

genotypes that determine coat color and their phenotypes are:

Genotype	Phenotype
$B_E_$	black
$bbE_$	brown (frequently called chocolate)
B_ee	yellow
$bbee$	yellow

If we cross a black Labrador homozygous for the dominant alleles with a yellow Labrador homozygous for the recessive alleles and then intercross the F_1 , we obtain progeny in the F_2 in a 9:3:4 ratio:

P	$BBEE \times bbee$
	black yellow
	↓
F_1	$BbEe$
	black
	↓ Intercross
F_2	$\frac{9}{16} B_E_ \text{ black}$ $\frac{3}{16} bbE_ \text{ brown}$ $\frac{3}{16} B_ee \text{ yellow}$ $\frac{1}{16} bbee \text{ yellow}$
	} $\frac{4}{16} \text{ yellow}$

Notice that yellow dogs can carry alleles for either black or brown pigment, but these alleles are not expressed in their coat color.

In this example of gene interaction, allele e is epistatic to B and b , because e masks the expression of the alleles for black and brown pigments, and alleles B and b are hypostatic to e . In this case, e is a recessive epistatic allele, because two copies of e must be present to mask of the black and brown pigments.

Dominant epistasis Dominant epistasis is seen in the interaction of two loci that determine fruit color in summer squash, which is commonly found in one of three colors: yellow, white, or green. When a homozygous plant that produces white squash is crossed with a homozygous plant that produces green squash and the F_1 plants are crossed with each other, the following results are obtained:

P	plants with white squash $\times$ plants with green squash
	↓
F_1	plants with white squash
	↓ Intercross
F_2	$\frac{12}{16}$ plants with white squash $\frac{3}{16}$ plants with yellow squash $\frac{1}{16}$ plants with green squash

How can gene interaction explain these results?

In the F_2 , $\frac{12}{16}$ or $\frac{3}{4}$ of the plants produce white squash and $\frac{3}{16} + \frac{1}{16} = \frac{4}{16} = \frac{1}{4}$ of the plants produce squash having color. This outcome is the familiar 3:1 ratio produced by a cross between two heterozygous individuals, which suggests that a dominant allele at one locus inhibits the production of pigment, resulting in white progeny. If we use the symbol W to represent the dominant allele that inhibits pigment production, then genotype $W_$ inhibits pigment production and produces white squash, whereas ww allows pigment and results in colored squash.

Among those ww F_2 plants with pigmented fruit, we observe $\frac{3}{16}$ yellow and $\frac{1}{16}$ green (a 3:1 ratio). This outcome is because a second locus determines the type of pigment produced in the squash, with yellow ($Y_$) dominant over green (yy). This locus is expressed only in ww plants, which lack the dominant inhibitory allele W . We can assign the genotype $wwY_$ to plants that produce yellow squash and the genotype $wwyy$ to plants that produce green squash. The genotypes and their associated phenotypes are:

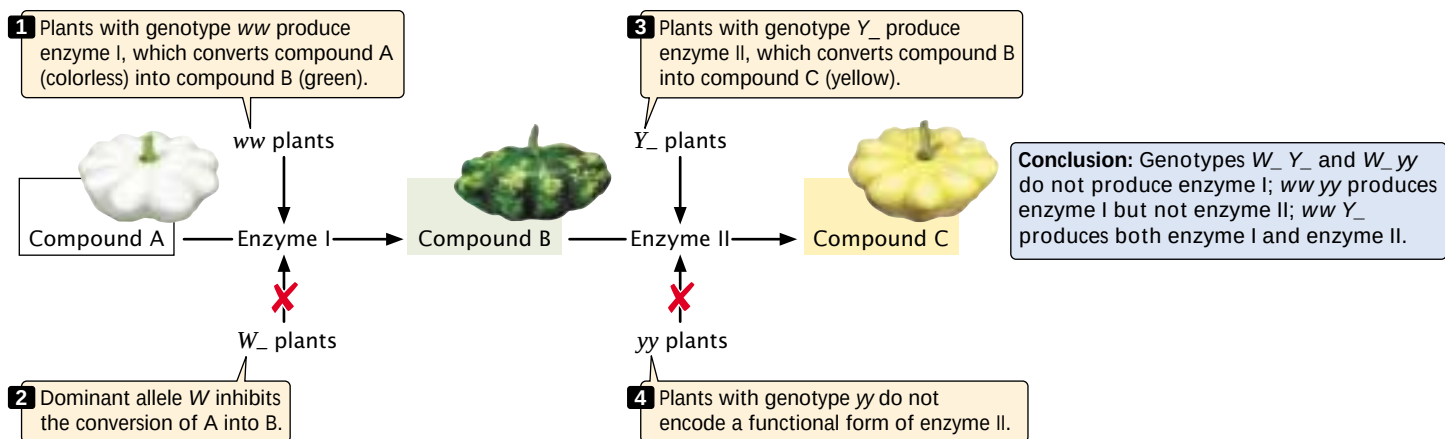
$W_Y_$	white squash
W_yy	white squash
$wwY_$	yellow squash
$wwyy$	green squash

Allele W is epistatic to Y and y —it suppresses the expression of these pigment-producing genes. W is a dominant epistatic allele because, in contrast with e in Labrador retriever coat color, a single copy of the allele is sufficient to inhibit pigment production.

Summer squash provides us with a good opportunity for considering how epistasis often arises when genes affect a series of steps in a biochemical pathway. Yellow pigment in the squash is most likely produced in a two-step biochemical pathway (FIGURE 5.8). A colorless (white) compound (designated A in Figure 5.8) is converted by enzyme I into green compound B, which is then converted into compound C by enzyme II. Compound C is the yellow pigment in the fruit.

Plants with the genotype ww produce enzyme I and may be green or yellow, depending on whether enzyme II is present. When allele Y is present at a second locus, enzyme II is produced and compound B is converted into compound C, producing a yellow fruit. When two copies of y , which does not encode a functional form of enzyme II, are present, squash remain green. The presence of W at the first locus inhibits the conversion of compound A into compound B; plants with genotype $W_$ do not make compound B and their fruit remains white, regardless of which alleles are present at the second locus.

Many cases of epistasis arise in this way. A gene (such as W) that has an effect on an early step in a biochemical pathway will be epistatic to genes (such as Y and y) that affect subsequent steps, because the effect of the enzyme in the later step depends on the product of the earlier reaction.



5.8 Yellow pigment in summer squash is produced in a two-step pathway.

Duplicate recessive epistasis Let's consider one more detailed example of epistasis. Albinism is the absence of pigment and is a common genetic trait in many plants and animals. Pigment is almost always produced through a multistep biochemical pathway; thus, albinism may entail gene interaction. Robert T. Dillon and Amy R. Wethington found that albinism in the common freshwater snail *Physa heterostropha* can result from the presence of either of two recessive alleles at two different loci. Inseminated snails were collected from a natural population and placed in cups of water, where they laid eggs. Some of the eggs hatched into albino snails. When two albino snails were crossed, all of the F_1 were pigmented. On intercrossing the F_1 , the F_2 consisted of $\frac{9}{16}$ pigmented snails and $\frac{7}{16}$ albino snails. How did this 9:7 ratio arise?

The 9:7 ratio seen in the F_2 snails can be understood as a modification of the 9:3:3:1 ratio obtained when two individuals heterozygous for two loci are crossed. The 9:7 ratio arises when dominant alleles at both loci ($A_B_$) produce pigmented snails; any other genotype produces albino snails:

P	$aaBB$ $AAbb$
	albino $\times$ albino
	↓
F_1	$AaBb$
	pigmented
	↓ Intercross
F_2	$\frac{9}{16} A_B_ \text{ pigmented}$ $\left. \begin{array}{l} \frac{3}{16} aaB_ \text{ albino} \\ \frac{3}{16} A_bb \text{ albino} \\ \frac{1}{16} aabb \text{ albino} \end{array} \right\} \frac{7}{16} \text{ albino}$

The 9:7 ratio in these snails is probably produced by a two-step pathway of pigment production (FIGURE 5.9). Pigment (compound C) is produced only after compound A has been converted into compound B by enzyme I and after

compound B has been converted into compound C by enzyme II. At least one dominant allele A at the first locus is required to produce enzyme I; similarly, at least one dominant allele B at the second locus is required to produce enzyme II. Albinism arises from the absence of compound C, which may happen in three ways. First, two recessive alleles at the first locus (genotype $aaB_$) may prevent the production of enzyme I, and so compound B is never produced. Second, two recessive alleles at the second locus (genotype A_bb) may prevent the production of enzyme II. In this case, compound B is never converted into compound C. Third, two recessive alleles may be present at both loci ($aabb$), causing the absence of both enzyme I and enzyme II. In this example of gene interaction, a is epistatic to B , and b is epistatic to A ; both are recessive epistatic alleles because the presence of two copies of either allele a or b is necessary to suppress pigment production. This example differs from the suppression of coat color in Labrador retrievers in that recessive alleles at either of two loci are capable of suppressing pigment production in the snails, whereas recessive alleles at a single locus suppress pigment expression in Labs.

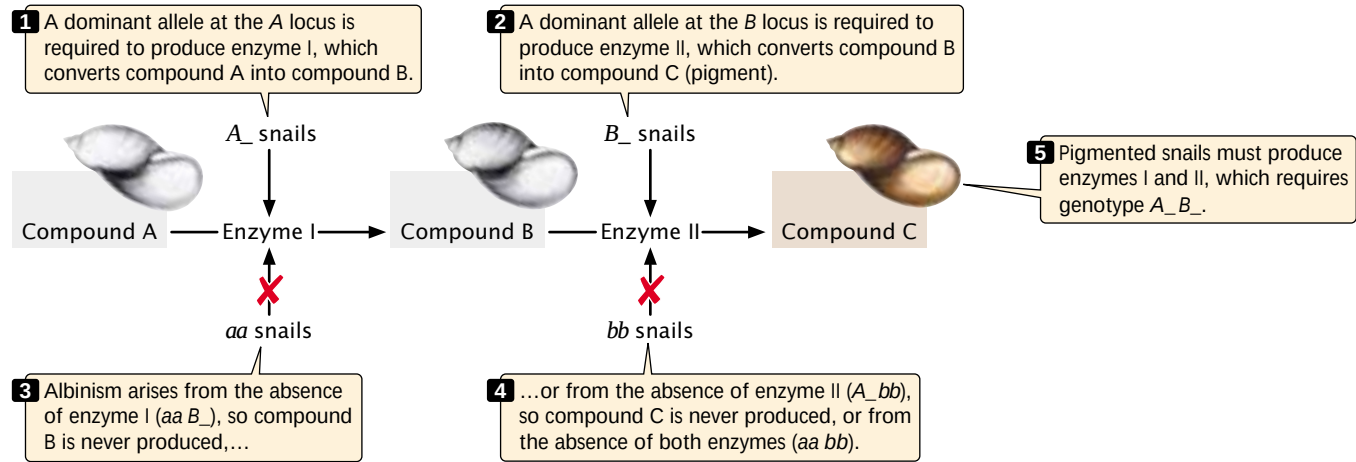
Concepts

Epistasis is the masking of the expression of one gene by another gene at a different locus. The epistatic gene does the masking; the hypostatic gene is masked. Epistatic genes can be dominant or recessive.

Connecting Concepts

Interpreting Ratios Produced by Gene Interaction

A number of modified ratios that result from gene interaction are shown in Table 5.2. Each of these examples represents a modification of the basic 9:3:3:1 dihybrid ratio.



5.9 Pigment is produced in a two-step pathway in snails.

In interpreting the genetic basis of modified ratios, we should keep several points in mind. First, the inheritance of the genes producing these characteristics is no different from the inheritance of genes coding for simple genetic characters. Mendel's principles of segregation and independent assortment still apply; each individual possesses two alleles at each locus, which separate in meiosis, and genes at the different loci assort independently. The only difference is in how the *products* of the genotypes interact to produce the phenotype. Thus, we cannot consider the expression of genes at each locus separately, but must take into consideration how the genes at different loci interact.

A second point is that in the examples that we have considered, the phenotypic proportions were always in sixteenths because, in all the crosses, pairs of alleles segregated at two independently assorting loci. The probability of inheriting one of the two alleles at a locus is $\frac{1}{2}$. Because there are two loci, each with two alleles, the probability of inheriting any particular combination of genes is $(\frac{1}{2})^4 = \frac{1}{16}$. For a trihybrid cross, the progeny proportions should be in sixty-fourths, because $(\frac{1}{2})^6 = \frac{1}{64}$. In general, the progeny proportions should be in fractions of $(\frac{1}{2})^{2n}$, where n equals the number of loci with two alleles segregating in the cross.

Table 5.2 Modified dihybrid—phenotypic ratios due to gene interaction

Ratio	Genotype				Type of Interaction	Example
	<i>A_B_</i>	<i>A_bb</i>	<i>aaB_</i>	<i>aabb</i>		
9:3:3:1	9	3	3	1	None	Seed shape and endosperm color in peas
9:3:4	9	3	4		Recessive epistasis	Coat color in Labrador retrievers
12:3:1	12		3	1	Dominant epistasis	Color in squash
9:7	9	7			Duplicate recessive epistasis	Albinism in snails
9:6:1	9	6		1	Duplicate interaction	—
15:1	15			1	Duplicate dominant epistasis	—
13:3	13		3		Dominant and recessive epistasis	—

*Reading across, each row gives the phenotypic ratios of progeny from a dihybrid cross (*AaBb* × *AaBb*).

Crosses rarely produce exactly 16 progeny; therefore, modifications of a dihybrid ratio are not always obvious. Modified dihybrid ratios are more easily seen if the number of individuals of each phenotype is expressed in sixteenths:

$$\frac{x}{16} = \frac{\text{number of progeny with a phenotype}}{\text{total number of progeny}}$$

where $x/16$ equals the proportion of progeny with a particular phenotype. If we solve for x (the proportion of the particular phenotype in sixteenths), we have:

$$x = \frac{\text{number of progeny with a phenotype} \times 16}{\text{total number of progeny}}$$

For example, suppose we cross two homozygous individuals, interbreed the F_1 and obtain 63 red, 21 brown, and 28 white F_2 individuals. Using the preceding formula, the phenotypic ratio in the F_2 is: red = $(63 \times 16)/112 = 9$; brown = $(21 \times 16)/112 = 3$; and white = $(28 \times 16)/112 = 4$. The phenotypic ratio is 9:3:4

A final point to consider is how to assign genotypes to the phenotypes in modified ratios owing to gene interaction. Don't try to *memorize* the genotypes associated with all the modified ratios in Table 5.2. Instead, practice relating modified ratios to known ratios, such as the 9:3:3:1 dihybrid ratio. Suppose we obtain $15/16$ green progeny and $1/16$ white progeny in a cross between two plants. If we compare this 15:1 ratio with the standard 9:3:3:1 dihybrid ratio, we see that $9/16 + 3/16 + 3/16$ equals $15/16$. All the genotypes associated with these proportions in the dihybrid cross ($A_B_$, A_bb , and $aaB_$) must give the same phenotype, the green progeny. Genotype $aabb$ makes up $1/16$ of the progeny in a dihybrid cross, the white progeny in this cross.

In assigning genotypes to phenotypes in modified ratios, students sometimes become confused about which letters to assign to which phenotype. Suppose we obtain the following phenotypic ratio: $9/16$ black : $3/16$ brown : $4/16$ white. Which genotype do we assign to the brown progeny, A_bb or $aaB_$? Either answer is correct, because the letters are just arbitrary symbols for the genetic information. The important thing to realize about this ratio is that the brown phenotype arises when two recessive alleles are present at one locus.

Concepts

Gene interaction frequently produces modified phenotypic ratios. These modified ratios can be understood by relating them to other known ratios.



The Complex Genetics of Coat Color in Dogs

Coat color in dogs is an excellent example of how complex interactions between genes may take part in the determination of a phenotype. Domestic dogs come in an amazing variety of shapes, sizes, and colors. For thousands of years, humans have

been breeding dogs for particular traits, producing the large number of types that we see today. Each breed of dog carries a selection of genes from the ancestral dog gene pool; these genes define the features of a particular breed.

One of the most obvious differences between dogs is coat color. The genetics of coat color in dogs is quite complex; many genes participate, and there are numerous interactions between genes at different loci. We will consider seven loci (in the list that follows) that are important in producing many of the noticeable differences in color and pattern among breeds of dogs. In interpreting the genetic basis of differences in coat color of dogs, consider how the expression of a particular gene is modified by the effects of other genes. Keep in mind that additional loci not listed here can modify the colors produced by these seven loci and that not all geneticists agree on the genetics of color variation in some breeds.

1. Agouti (A) locus—This locus has five common alleles that determine the depth and distribution of color in a dog's coat:

A^s	Solid black pigment.
a^w	Agouti, or wolflike gray. Hairs encoded by this allele have a “salt and pepper” appearance, produced by a band of yellow pigment on a black hair.
a^y	Yellow. The black pigment is markedly reduced; so the entire hair is yellow.
a^s	Saddle markings (dark color on the back, with extensive tan markings on the head and legs).
a^t	Bicolor (dark color over most of the body, with tan markings on the feet and eyebrows).

A^s and a^y are generally dominant over the other alleles, but the dominance relations are complex and not yet completely understood.

2. Black (B) locus—This locus determines whether black pigment can be formed. The actual color of a dog's fur depends on the effects of genes at other loci (such as the A, C, D, and E loci). Two alleles are common:

B	Allows black pigment to be produced; the dog will be black if it also possesses certain alleles at the A, C, D, and E loci.
b	Black pigment cannot be produced; pigmented dogs can be chocolate, liver, tan, or red.

B is dominant over b .

3. Albino (C) locus—This locus determines whether full color will be expressed. There are five alleles at this locus:

C	Color fully expressed.
c^{ch}	Chinchilla. Less color is expressed, and pigment is completely absent from the base of the long hairs, producing a pale coat.
c^d	All white coat with dark nose and dark eyes.
c^b	All white coat with blue eyes.
c	Fully albino. The dogs have an all-white coat with pink eyes and nose.



5.10 Coat color in dogs is determined by interactions between genes at a number of loci.

(a) Most Labrador retrievers are genotype $A^s A^s CCDDSStt$, varying only at the B and E loci. (b) Most beagles are genotype $a^s a^s BBCCDDs^p s^p tt$. (c) Dalmatians are genotype $A^s A^s CCDD EEs^w s^w TT$, varying at the B locus so that the dogs are black ($B_$) or brown (bb). (Part a, Robert Maier/Animals Animals; part b, Ralph Reinhold/Animals Animals; part c, Robert Percy/Animals Animals.)

The dominance relations among these alleles is presumed to be $C > c^{ch} > c^d > c^b > c$, but the c^{ch} and c alleles are rare, and crosses including all possible genotypes have not been completed.

4. Dilution (D) locus—This locus, with two alleles, determines whether the color will be diluted. For example, diluted black pigment appears bluish, and diluted yellow appears cream. The diluted effect is produced by an uneven distribution of pigment in the hair shaft:

D Intense pigmentation.
 d Dilution of pigment.

D is dominant over d .

5. Extension (E) locus—Four alleles at this locus determine where the genotype at the A locus is expressed. For example, if a dog has the A^s allele (solid black) at the A locus, then black pigment will either be extended throughout the coat or be restricted to some areas, depending on the alleles present at the E locus. Areas where the A locus is not expressed may appear as yellow, red, or tan, depending on the presence of particular genes at other loci. When A^s is present at the A locus, the four alleles at the E locus have the following effects:

E^m Black mask with a tan coat.
 E The A locus expressed throughout (solid black).
 e^{br} Brindle, in which black and yellow are in layers to give a tiger-striped appearance.
 e No black in the coat, but the nose and eyes may be black.

The dominance relations among these alleles are poorly known.

6. Spotting (S) locus—Alleles at this locus determine whether white spots will be present. There are four common alleles:

S No spots.
 s^i Irish spotting; numerous white spots.
 s^p Piebald spotting; various amounts of white.
 s^w Extreme white piebald; almost all white.

S is completely dominant over s^i , s^p , and s^w ; s^i and s^p are dominant over s^w ($S > s^i, s^p > s^w$). The relation between of s^i and s^p is poorly defined; indeed, they may not be separate alleles. Genes at other poorly known loci also modify spotting patterns.

7. Ticking (T) locus—This locus determines the presence of small colored spots on the white areas, which is called ticking:

T Ticking; small colored spots on the areas of white.
 t No ticking.

T is dominant over t . Ticking cannot be expressed if a dog has a solid coat ($S_$).

To illustrate how genes at these loci interact in determining a dog's coat color, let's consider a few examples:

Labrador retriever- Labrador retrievers (FIGURE 5.10a) may be black, brown, or yellow. Most are homozygous $A^s A^s CCDDSStt$; thus, they vary only at the B and E loci. The A^s , C , and D alleles allow dark pigment to be expressed; whether a dog is black depends on which genes are present at the B and E loci. As discussed earlier in the chapter, all black Labradors must carry at least one B allele and one E allele ($B_E_$). Brown dogs are homozygous bb and have at least one E allele ($bbE_$). Yellow dogs are a result of the presence of ee (B_ee or $bbee$). Labrador retrievers are homozygous for the S allele, which produces a solid color; the few white spots that appear in some dogs of this breed are due to other modifying genes. The allele for ticking, T , is presumed not to exist in Labradors; however, Labrador retrievers have solid coats and ticking is expressed only in spotted dogs; so its absence is uncertain.

Beagle- Most beagles are homozygous $a^s a^s BBCCDDs^p s^p tt$, although other alleles at these loci are occasionally present. The a^s allele produces the saddle markings—dark back and sides, with tan head

and legs—that are characteristic of the breed (FIGURE 5.10b). Alleles *B*, *C*, and *D* allow black to be produced, but its distribution is limited by the *a^s* allele. Genotype *ee* does occasionally arise, leading to a few all-tan beagles. White spotting in beagles is due to the *s^p* allele. Ticking can appear, but most beagles are *tt*.

Dalmatian- Dalmatians (FIGURE 5.10c) have an interesting genetic makeup. Most are homozygous *A^sA^sCCDDEEs^ws^wTT*; so they vary only at the *B* locus. Notice that these dogs possess genotype *A^sA^sCCDDEE*, which allows for a solid coat that would be black, if genotype *B₋* is present, or brown (called liver), if genotype *bb* is present. However, the presence of the *s^w* allele produces a white coat, masking the expression of the solid color. The dog's color appears only in the pigmented spots, which are due to the presence of the ticking allele *T*. Table 5.3 gives the common genotypes of other breeds of dogs.

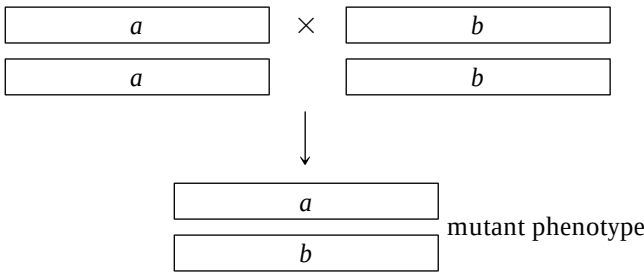
www.whfreeman.com/pierce Information on dog genetics, including the Dog Genome Project

Complementation: Determining Whether Mutations Are at the Same or Different Loci

How do we know whether different mutations that affect a characteristic occur at the same locus (are allelic) or at different loci? In fruit flies, for example, *white* is an X-linked mutation that produces white eyes instead of the red eyes

found in wild-type flies. *Apricot* is an X-linked recessive mutation that produces light orange-colored eyes. Do the white and apricot mutations occur at the same locus or at different loci? We can use the complementation test to answer this question.

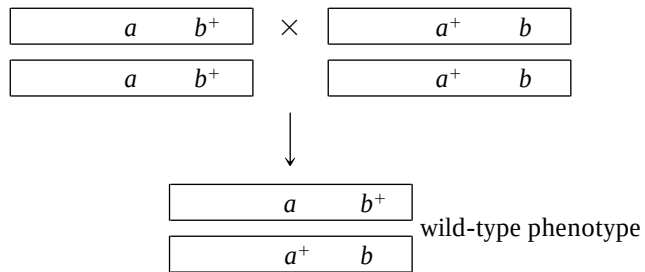
To carry out a **complementation test**, parents that are homozygous for different mutations are crossed, producing offspring that are heterozygous. If the mutations are allelic (occur at the same locus), then the heterozygous offspring have only mutant alleles (*ab*) and exhibit a mutant phenotype:



If, on the other hand, the mutations occur at different loci, each of the homozygous parents possesses wild-type genes at the other locus (*aa b⁺b⁺* and *a⁺a⁺ bb*); so the heterozygous offspring inherit a mutant and a wild-type allele at each locus. In this case, the mutations complement each other and the heterozygous offspring have the wild-type phenotype:

Table 5.3 Common genotypes in different breeds of dogs		
Breed	Usual Homozygous Genes*	Other Genes Present Within the Breed
Basset hound	<i>BBCCDDEEtt</i>	<i>a^y, a^t S, s^p, sⁱ</i>
Beagle	<i>a^sa^sBBCCDDs^ps^ptt</i>	<i>E, e</i>
English bulldog	<i>BBCCDDtt</i>	<i>A^s, a^y, a^t E^m, E, e^{br} S, sⁱ, s^p, s^w</i>
Chihuahua	<i>tt</i>	<i>A^s, a^y, a^s, a^t B, b C, c^{ch} D, d E^m, E, e^{br}, e S, sⁱ s^p, s^w</i>
Collie	<i>BBCCEEtt</i>	<i>a^y, a^t D, d sⁱ, s^w</i>
Dalmatian	<i>A^sA^sCCDDEEs^ws^wTT</i>	<i>B, b</i>
Doberman	<i>a^ta^tCCEESStt</i>	<i>B, b D, d</i>
German shepherd	<i>BBDDSStt</i>	<i>a^y, a^g, a^s, a^t C, c^{ch} E^m, E, e</i>
Golden retriever	<i>A^sA^sBBDDSStt</i>	<i>C, c^{ch} E, e</i>
Greyhound	<i>BBtt</i>	<i>A^s, a^y C, c^{ch} D, d E, e^{br}, e S, s^p, s^w, sⁱ</i>
Irish setter	<i>BBCCDDeeSStt</i>	<i>A, a^t</i>
Labrador retriever	<i>A^sA^sCCDDSStt</i>	<i>B, b E, e</i>
Poodle	<i>SStt</i>	<i>A^s, a^t B, b C, c^{ch} D, d E, e</i>
Rottweiler	<i>a^ta^tBBCCDDEESStt</i>	
St. Bernard	<i>a^ya^yBBCCDDtt</i>	<i>E^m, E sⁱ, s^p, s^w</i>

*Most dogs in the breed are homozygous for these genes; a few individual dogs may possess other alleles at these loci. Source: Data from M. B. Willis, *Genetics of the Dog* (London: Witherby, 1989).



Complementation occurs when an individual possessing two mutant genes has a wild-type phenotype and is an indicator that the mutations are nonallelic genes.

When the complementation test is applied to white and apricot mutations, all of the heterozygous offspring have light-colored eyes, demonstrating that white and apricot are produced by mutations that occur at the same locus and are allelic.

Interaction Between Sex and Heredity

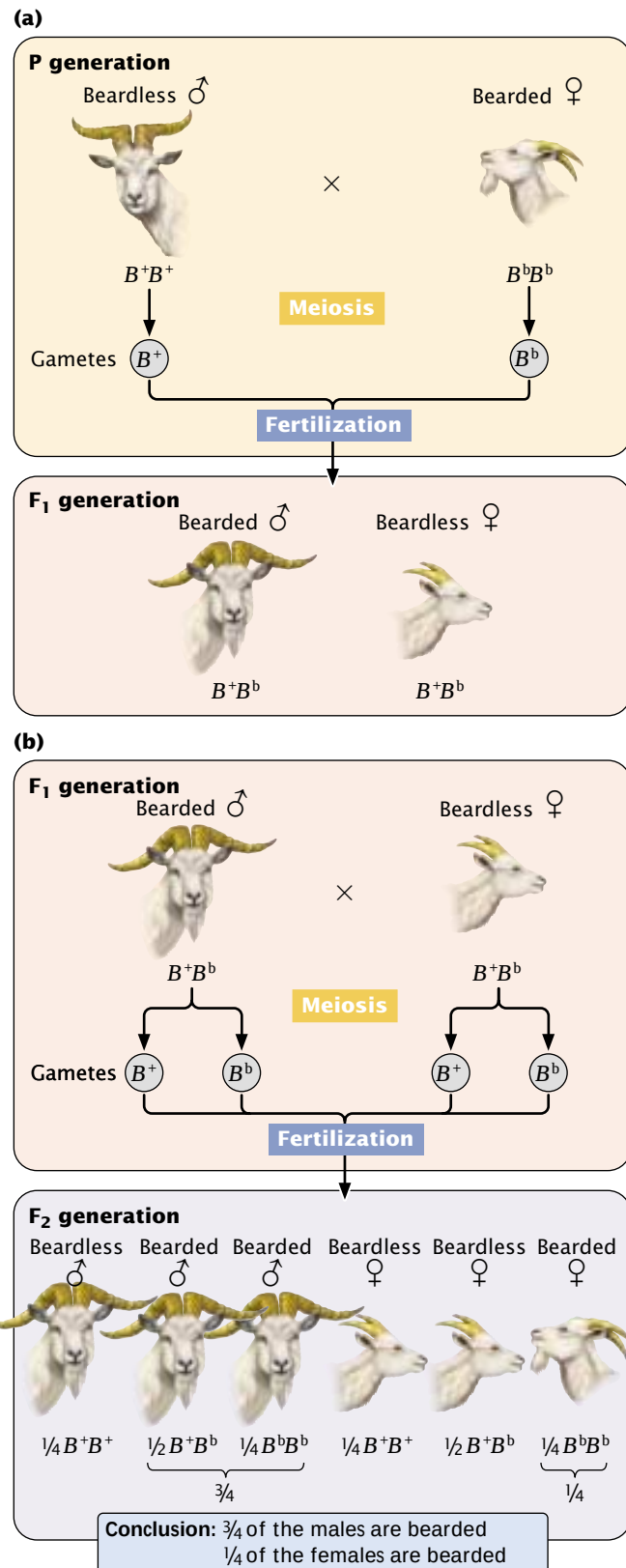
In Chapter 4, we considered characteristics encoded by genes located on the sex chromosomes and how their inheritance differs from the inheritance of traits encoded by autosomal genes. Now we will examine additional influences of sex, including the effect of the sex of an individual on the expression of genes on autosomal chromosomes, characteristics determined by genes located in the cytoplasm, and characteristics for which the genotype of only the maternal parent determines the phenotype of the offspring. Finally, we'll look at situations in which the expression of genes on autosomal chromosomes is affected by the sex of the parent from whom they are inherited.

Sex-Influenced and Sex-Limited Characteristics

Sex influenced characteristics are determined by autosomal genes and are inherited according to Mendel's principles, but they are expressed differently in males and females. In this case, a particular trait is more readily expressed in one sex; in other words, the trait has higher penetrance (see p. 000 in Chapter 3) in one of the sexes.

For example, the presence of a beard on some goats is determined by an autosomal gene (B^b) that is dominant in males and recessive in females. In males, a single allele is required for the expression of this trait: both the homozygote (B^bB^b) and the heterozygote (B^bB^+) have beards, whereas the B^+B^+ male is beardless. In contrast, females require two alleles in order for this trait to be expressed: the homozygote B^bB^b has a beard, whereas the heterozygote (B^bB^+) and the other homozygote (B^+B^+) are beardless. The key to understanding the expression of the bearded gene is to look at the heterozygote. In males (for which the presence of a beard is dominant), the heterozygous genotype produces a beard but, in females (for which the presence of a beard is recessive and its absence is dominant), the heterozygous genotype produces a goat without a beard.

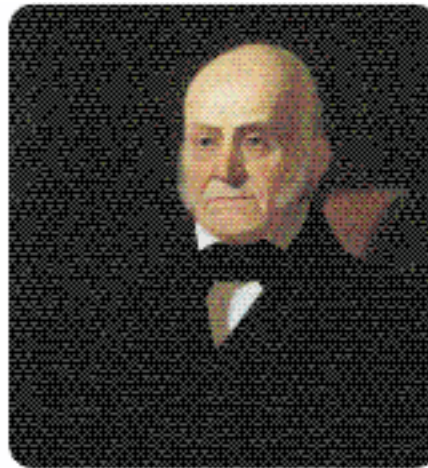
FIGURE 5.11a illustrates a cross between a beardless male (B^+B^+) and a bearded female (B^bB^b). The alleles



5.11 Genes that encode sex-influenced traits are inherited according to Mendel's principles but are expressed differently in males and females.



(a)



(b)



(c)

5.12 Pattern baldness is a sex-influenced trait. This trait is seen in three generations of the Adams family: (a) John Adams (1735–1826), the second president of the United States, was father to (b) John Quincy Adams (1767–1848), who was father to (c) Charles Francis Adams (1807–1886). Pattern baldness results from an autosomal gene that is thought to be dominant in males and recessive in females. (Part (a) National Museum of American Art, Washington, D.C./Art Resource, NY; (b) National Portrait Gallery, Washington, D.C./Art Resource, N.Y.; (c) Bettmann/Corbis.)

separate into gametes according to Mendel's principle of segregation, and all the F_1 are heterozygous (B^+B^b). Because the trait is dominant in males and recessive in females, all the F_1 males will be bearded, and all the F_1 females will be beardless. When the F_1 are crossed with one another, $\frac{1}{4}$ of the F_2 progeny are B^bB^b , $\frac{1}{2}$ are B^bB^+ , and $\frac{1}{4}$ are B^+B^+ (FIGURE 5.11b). Because male heterozygotes are bearded, $\frac{3}{4}$ of the males in the F_2 possess beards; because female heterozygotes are beardless, only $\frac{1}{4}$ of the females in F_2 are bearded.

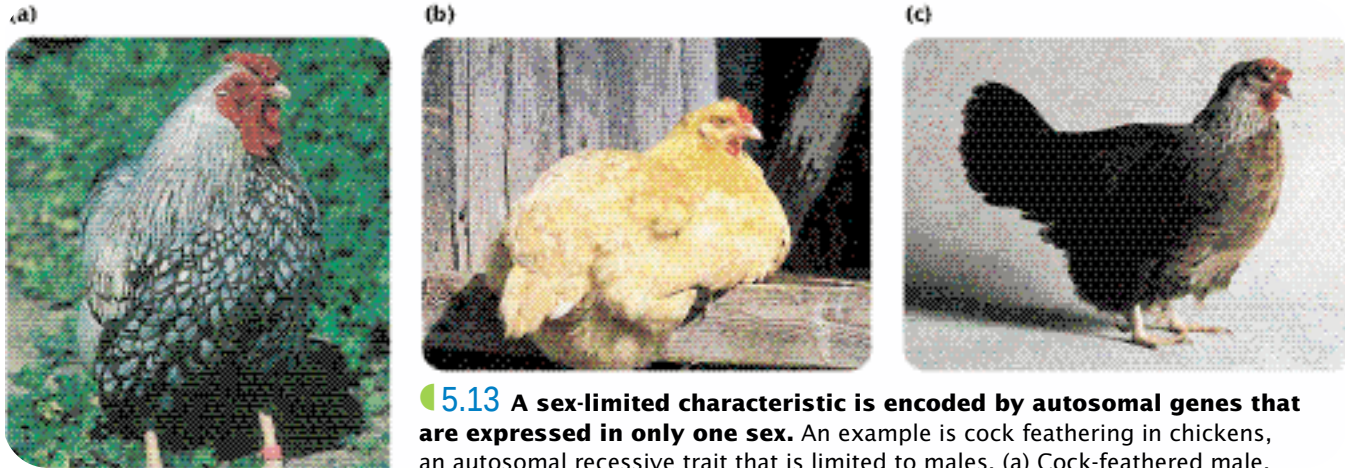
An example of a sex-influenced characteristic in humans is pattern baldness, in which hair is lost prematurely from the front and the top of the head (FIGURE 5.12). Pattern baldness is an autosomal character believed to be dominant in males and recessive in females, just like beards in goats. Contrary to a popular misconception, a man does not inherit pattern baldness from his mother's side of the family (which would be the case if the character were X linked, but it isn't). Pattern baldness is autosomal; men and women can inherit baldness from either their mothers or their fathers. Men require only a single allele for baldness to become bald, whereas women require two alleles for baldness, and so pattern baldness is much more common among men. Furthermore, pattern baldness is expressed weakly in women; those with the trait usually have only a mild thinning of the hair, whereas men frequently lose all the hair on the top of the head. The expression of the allele for pattern baldness is clearly enhanced by the presence of male sex hormones; males who are castrated at an early age rarely become bald (but castration is not a recommended method for preventing baldness).

An extreme form of sex-influenced inheritance, a **sex-limited characteristic** is encoded by autosomal genes that are expressed in only one sex—the trait has zero penetrance in the other sex. In domestic chickens, some males display a plumage pattern called cock feathering (FIGURE 5.13a). Other males and all females display a pattern called hen feathering (FIGURE 5.13b and c). Cock feathering is an autosomal recessive trait that is sex limited to males. Because the trait is autosomal, the genotypes of males and females are the same, but the phenotypes produced by these genotypes differ in males and females:

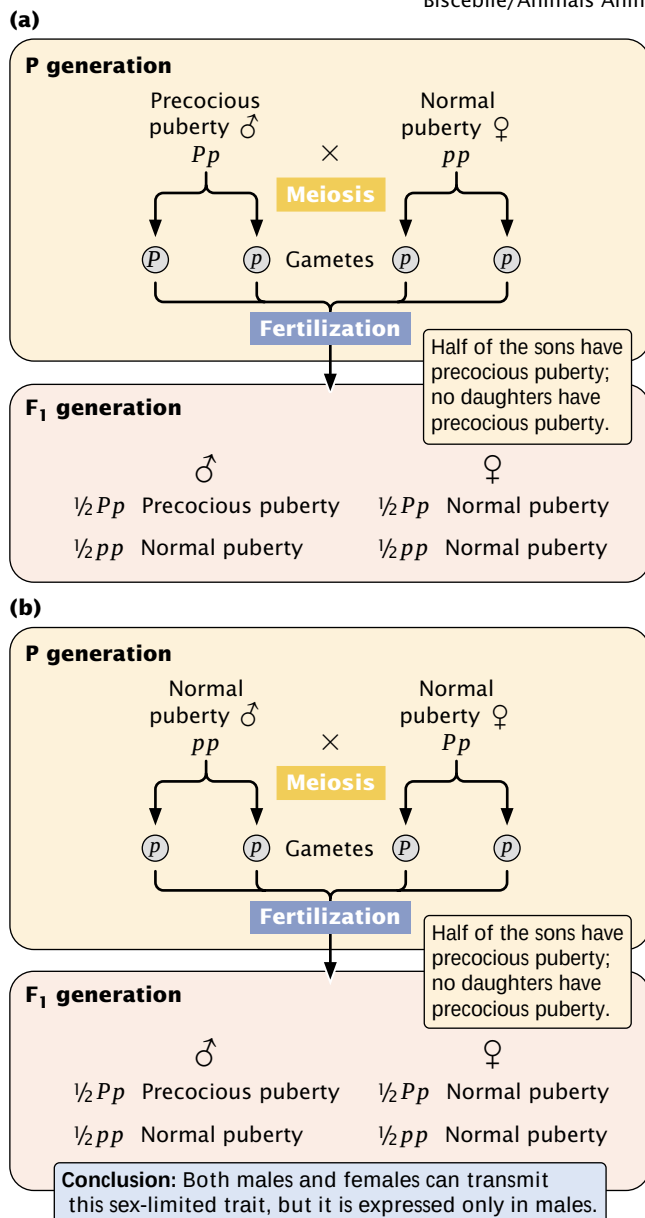
Genotype	Male phenotype	Female phenotype
HH	hen feathering	hen feathering
Hh	hen feathering	hen feathering
hh	cock feathering	hen feathering

An example of a sex-limited characteristic in humans is male-limited precocious puberty. There are several types of precocious puberty in humans, most of which are not genetic. Male-limited precocious puberty, however, results from an autosomal dominant allele (P) that is expressed only in males; females with the gene are normal in phenotype. Males with precocious puberty undergo puberty at an early age, usually before the age of 4. At this time, the penis enlarges, the voice deepens, and pubic hair develops. There is no impairment of sexual function; affected males are fully fertile. Most are short as adults, because the long bones stop growing after puberty.

Because the trait is rare, affected males are usually heterozygous (Pp). A male with precocious puberty who mates



5.13 A sex-limited characteristic is encoded by autosomal genes that are expressed in only one sex. An example is cock feathering in chickens, an autosomal recessive trait that is limited to males. (a) Cock-feathered male. (b) and (c) Hen-feathered females. (Part a, Richard Kolar/Animals Animals; part b, Michael Bisceblie/Animals Animals; part c, R. OSF Dowling/Animals Animals.)



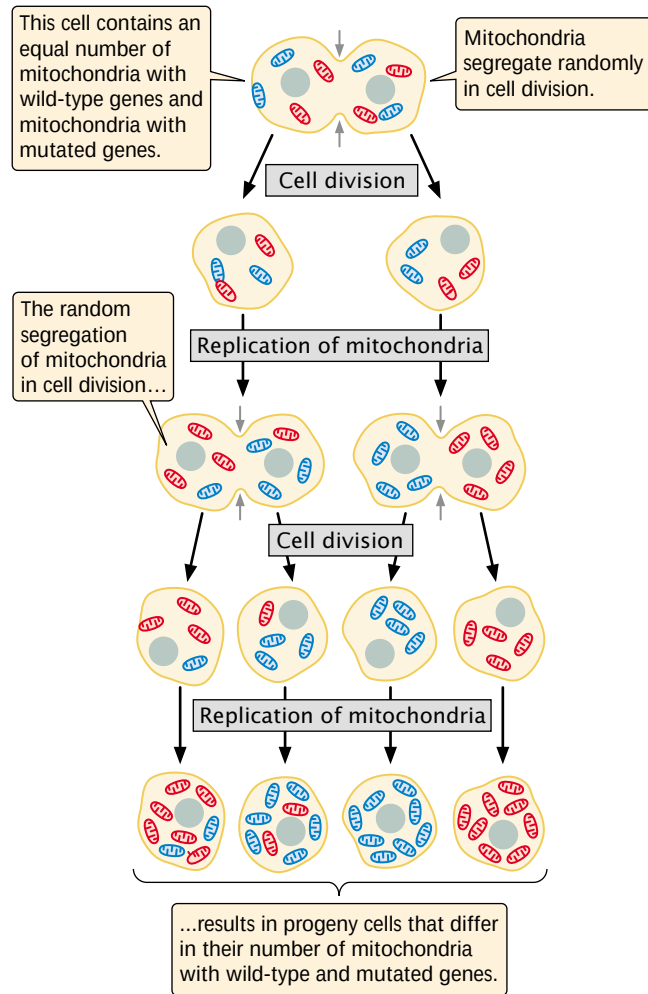
with a woman who has no family history of this condition will transmit the allele for precocious puberty to $\frac{1}{2}$ of the children (FIGURE 5.14a), but it will be expressed only in the sons. If one of the heterozygous daughters (Pp) mates with a male who has normal puberty (pp), $\frac{1}{2}$ of the sons will exhibit precocious puberty (FIGURE 5.14b). Thus a sex-limited characteristic can be inherited from either parent, although the trait appears in only one sex.

The results of molecular studies reveal that the underlying genetic defect in male-limited precocious puberty affects the receptor for luteinizing hormone (LH). This hormone normally attaches to receptors found on certain cells of the testes and stimulates these cells to produce testosterone. During normal puberty in males, high levels of LH stimulate the increased production of testosterone, which, in turn, stimulates the anatomical and physiological changes associated with puberty. The P allele for precocious puberty codes for a defective LH receptor, which stimulates testosterone production even in the absence of LH. Boys with this allele produce high levels of testosterone at an early age, when levels of LH are low. Defective LH receptors are also found in females who carry the precocious-puberty gene, but their presence does not result in precocious puberty, because additional hormones are required along with LH to induce puberty in girls.

Concepts

Sex-influenced characteristics are traits encoded by autosomal genes that are more readily expressed in one sex. Sex-limited characteristics are encoded by autosomal genes whose expression is limited to one sex.

5.14 Sex-limited characteristics are inherited according to Mendel's principles. Precocious puberty is an autosomal dominant trait that is limited to males.



5.15 Cytoplasmically inherited characteristics frequently exhibit extensive phenotypic variation because cells and individual offspring contain various proportions of cytoplasmic genes.

Mitochondria that have wild-type mtDNA are shown in red; those having mutant mtDNA are shown in blue.

Cytoplasmic Inheritance

Mendel's principles of segregation and independent assortment are based on the assumption that genes are located on chromosomes in the nucleus of the cell. For the majority of genetic characteristics, this assumption is valid, and Mendel's principles allow us to predict the types of offspring that will be produced in a genetic cross. However, not all the genetic material of a cell is found in the nucleus; some characteristics are encoded by genes located in the cytoplasm. These characteristics exhibit **cytoplasmic inheritance**.

A few organelles, notably chloroplasts and mitochondria, contain DNA. Each human mitochondrion contains about 15,000 nucleotides of DNA, encoding 37 genes. Compared with that of nuclear DNA, which contains some 3 billion nucleotides encoding perhaps 35,000 genes, the amount

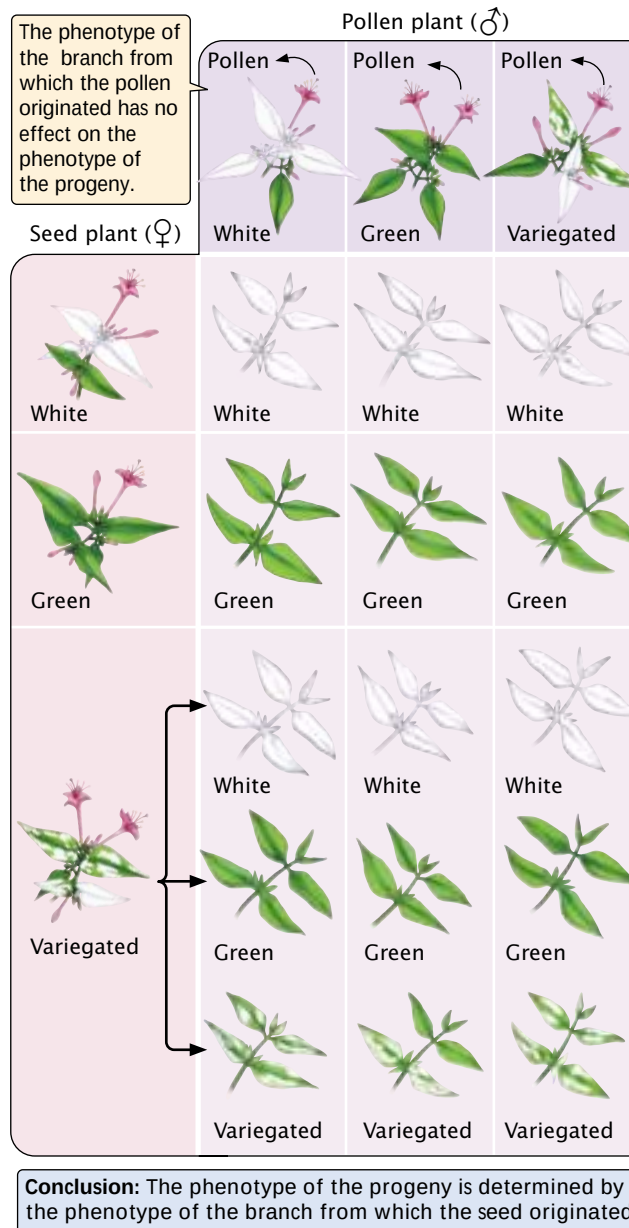
of mitochondrial DNA (mtDNA) is very small; nevertheless, mitochondrial and chloroplast genes encode some important characteristics. The molecular details of this extranuclear DNA are discussed in Chapter 20; here, we will focus on *patterns* of cytoplasmic inheritance.

Cytoplasmic inheritance differs from the inheritance of characteristics encoded by nuclear genes in several important respects. A zygote inherits nuclear genes from both parents, but typically all of its cytoplasmic organelles, and thus all its cytoplasmic genes, come from only one of the gametes, usually the egg. Sperm generally contributes only a set of nuclear genes from the male parent. In a few organisms, cytoplasmic genes are inherited from the male parent, or from both parents; however, for most organisms, all the cytoplasm is inherited from the egg. In this case, cytoplasmically inherited traits are present in both males and females and are passed from mother to offspring, never from father to offspring. Reciprocal crosses, therefore, give different results when cytoplasmic genes encode a trait.

Cytoplasmically inherited characteristics frequently exhibit extensive phenotypic variation, because there is no mechanism analogous to mitosis or meiosis to ensure that cytoplasmic genes are evenly distributed in cell division. Thus, different cells and individuals will contain various proportions of cytoplasmic genes.

Consider mitochondrial genes. There are thousands of mitochondria in each cell, and each mitochondrion contains from 2 to 10 copies of mtDNA. Suppose that half of the mitochondria in a cell contain a normal wild-type copy of mtDNA and the other half contain a mutated copy (FIGURE 5.15). In cell division, the mitochondria segregate into progeny cells at random. Just by chance, one cell may receive mostly mutated mtDNA and another cell may receive mostly wild-type mtDNA (see Figure 5.15). In this way, different progeny from the same mother and even cells within an individual offspring may vary in their phenotype. Traits encoded by chloroplast DNA (cpDNA) are similarly variable.

In 1909, cytoplasmic inheritance was recognized by Carl Correns as one of the first exceptions to Mendel's principles. Correns, one of the biologists who rediscovered Mendel's work, studied the inheritance of leaf variegation in the four-o'clock plant, *Mirabilis jalapa*. Correns found that the leaves and shoots of one variety of four-o'clock were variegated, displaying a mixture of green and white splotches. He also noted that some branches of the variegated strain had all-green leaves; other branches had all-white leaves. Each branch produced flowers; so Correns was able to cross flowers from variegated, green, and white branches in all combinations (FIGURE 5.16). The seeds from green branches always gave rise to green progeny, no matter whether the pollen was from a green, white, or variegated branch. Similarly, flowers on white branches always produced white progeny. Flowers on the variegated branches gave rise to green, white, and variegated progeny, in no particular ratio.



5.16 Crosses for leaf type in four o'clocks illustrate cytoplasmic inheritance.

Corren's crosses demonstrated cytoplasmic inheritance of variegation in the four-o'clocks. The phenotypes of the offspring were determined entirely by the maternal parent, never by the paternal parent (the source of the pollen). Furthermore, the production of all three phenotypes by flowers on variegated branches is consistent with the occurrence of cytoplasmic inheritance. Variegation in these plants is caused by a defective gene in the cpDNA, which results in a failure to produce the green pigment chlorophyll. Cells from green branches contain normal chloroplasts only, cells from white branches contain abnormal chloroplasts only, and cells from variegated branches contain a mixture of normal and abnormal chloroplasts. In the flowers from variegated branches,

the random segregation of chloroplasts in the course of oogenesis produces some egg cells with normal cpDNA, which develop into green progeny; other egg cells with only abnormal cpDNA develop into white progeny; and, finally, still other egg cells with a mixture of normal and abnormal cpDNA develop into variegated progeny.

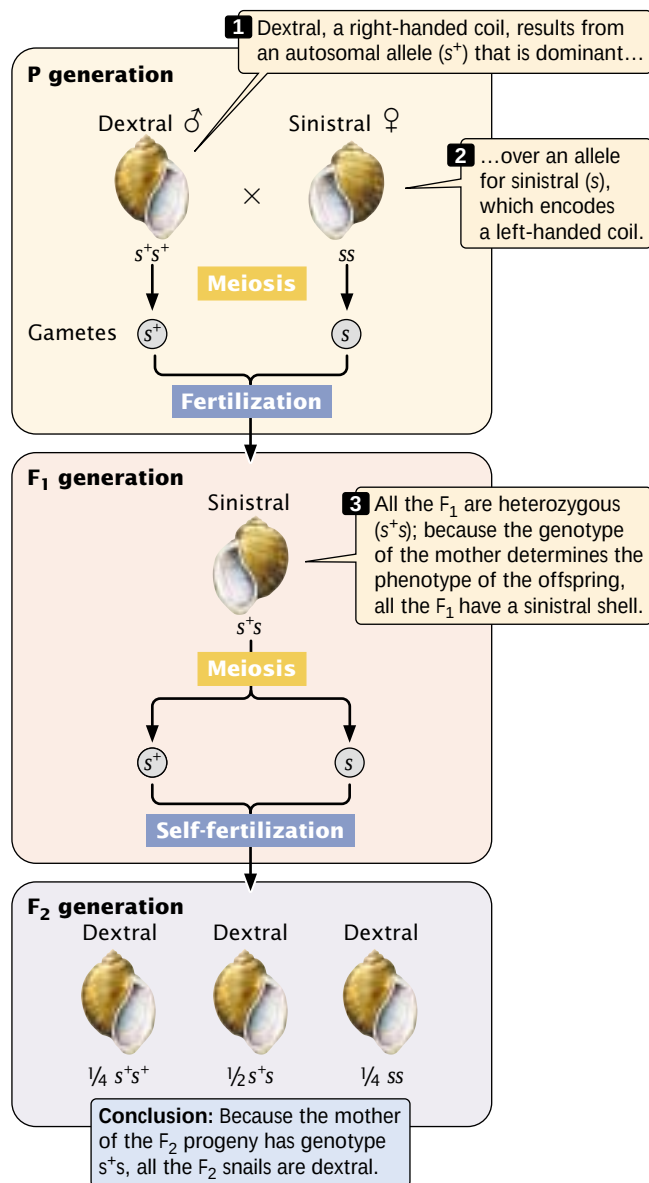
In recent years, a number of human diseases (mostly rare) that exhibit cytoplasmic inheritance have been identified. These disorders arise from mutations in mtDNA, most of which occur in genes coding for components of the electron-transport chain, which generates most of the ATP (adenosine triphosphate) in aerobic cellular respiration. One such disease is Leber hereditary optic neuropathy. Patients who have this disorder experience rapid loss of vision in both eyes, resulting from the death of cells in the optic nerve. Loss of vision typically occurs in early adulthood (usually between the ages of 20 and 24), but it can occur any time after adolescence. There is much clinical variability in the severity of the disease, even within the same family. Leber hereditary optic neuropathy exhibits maternal inheritance: the trait is always passed from mother to child.

Genetic Maternal Effect

A genetic phenomenon that is sometimes confused with cytoplasmic inheritance is **genetic maternal effect**, in which the phenotype of the offspring is determined by the genotype of the mother. In cytoplasmic inheritance, the genes for a characteristic are inherited from only one parent, usually the mother. In genetic maternal effect, the genes are inherited from both parents, but the offspring's phenotype is determined not by its own genotype but by the genotype of its mother.

Genetic maternal effect frequently arises when substances present in the cytoplasm of an egg (encoded by the mother's genes) are pivotal in early development. An excellent example is shell coiling of the snail *Limnaea peregra*. In most snails of this species, the shell coils to the right, which is termed dextral coiling. However, some snails possess a left-coiling shell, exhibiting sinistral coiling. The direction of coiling is determined by a pair of alleles; the allele for dextral (s^+) is dominant over the allele for sinistral (s). However, the direction of coiling is determined not by that snail's own genotype, but by the genotype of its *mother*. The direction of coiling is affected by the way in which the cytoplasm divides soon after fertilization, which in turn is determined by a substance produced by the mother and passed to the offspring in the cytoplasm of the egg.

If a male homozygous for dextral alleles (s^+s^+) is crossed with a female homozygous for sinistral alleles (ss), all of the F_1 are heterozygous (s^+s) and have a sinistral shell, because the genotype of the mother (ss) codes for sinistral (FIGURE 5.17). If these F_1 snails are self-fertilized, the genotypic ratio of the F_2 is $1 s^+s^+ : 2 s^+s : 1 ss$. The phenotype of all F_2 snails will be dextral regardless of their genotypes, because the genotype of their mother (s^+s) encodes a right-coiling shell and determines their phenotype.



5.17 In genetic maternal effect, the genotype of the maternal parent determines the phenotype of the offspring. Shell coiling in snails is a trait that exhibits genetic maternal effect.

Notice that the phenotype of the progeny is not necessarily the same as the phenotype of the mother, because the progeny's phenotype is determined by the mother's *genotype*, not her phenotype. Neither the male parent's nor the offspring's own genotype has any role in the offspring's phenotype. A male does influence the phenotype of the F_2 generation; by contributing to the genotypes of his daughters, he affects the phenotypes of their offspring. Genes that exhibit genetic maternal effect are therefore transmitted through males to future generations. In contrast, the genes that exhibit cytoplasmic inheritance are always transmitted through only one of the sexes (usually the female).

Concepts

Characteristics exhibiting cytoplasmic inheritance are encoded by genes in the cytoplasm and are usually inherited from one parent, most commonly the mother. In genetic maternal effect, the genotype of the mother determines the phenotype of the offspring.

Genomic Imprinting

One of the basic tenets of Mendelian genetics is that the parental origin of a gene does not affect its expression—reciprocal crosses give identical results. We have seen that there are some genetic characteristics—those encoded by X-linked genes and cytoplasmic genes—for which reciprocal crosses do not give the same results. In these cases, males and females do not contribute the same genetic material to the offspring. With regard to autosomal genes, males and females contribute the same number of genes, and paternal and maternal genes have long been assumed to have equal effects. The results of recent studies, however, have identified several mammalian genes whose expression is significantly affected by their parental origin. This phenomenon, the differential expression of genetic material depending on whether it is inherited from the male or female parent, is called **genomic imprinting**.

Genomic imprinting has been observed in mice in which a particular gene has been artificially inserted into a mouse's DNA (to create a transgenic mouse). In these mice, the inserted gene is faithfully passed from generation to generation, but its expression may depend on which parent transmitted the gene. For example, when a transgenic male passes an imprinted gene to his offspring, they express the gene; but, when his daughter transmits the same gene to her offspring, they don't express it. In turn, her son's offspring express it, but her daughter's offspring don't. Both male and female offspring possess the gene for the trait; the key to whether the gene is expressed is the sex of the parent transmitting the gene. In the present example, the gene is expressed only when it is transmitted by a male parent. The reverse situation, expression of a trait when the gene is transmitted by the female parent, also occurs.

Genomic imprinting has been implicated in several human disorders, including Prader-Willi and Angelman syndromes. Children with Prader-Willi syndrome have small hands and feet, short stature, poor sexual development, and mental retardation; they develop voracious appetites and frequently become obese. Many persons with Prader-Willi syndrome are missing a small region of chromosome 15 called q11–13. The deletion of this region is always inherited from the father in persons with Prader-Willi syndrome.

The deletions of q11–13 on chromosome 15 can also be inherited from the *mother*, but this inheritance results in a completely different set of symptoms, producing Angelman

Table 5.4 Sex influences on heredity

Genetic Phenomenon	Phenotype Determined by
Sex-linked characteristic	genes located on the sex chromosome
Sex-influenced characteristic	genes on autosomal chromosomes that are more readily expressed in one sex
Sex-limited characteristic	autosomal genes whose expression is limited to one sex
Genetic maternal effect	nuclear genotype of the maternal parent
Cytoplasmic inheritance	cytoplasmic genes, which are usually inherited entirely from only one parent
Genomic imprinting	genes whose expression is affected by the sex of the transmitting parent

syndrome. Children with Angelman syndrome exhibit frequent laughter, uncontrolled muscle movement, a large mouth, and unusual seizures. The deletion of segment q11–13 from chromosome 15 has severe effects on the human phenotype, but the specific effects depend on which parent contributes the deletion. For normal development to take place, copies of segment q11–13 of chromosome 15 from both male and female parents are apparently required.

Several other human diseases also appear to exhibit genomic imprinting. Although the precise mechanism of this phenomenon is unknown, methylation of DNA—the addition of methyl (CH_3) groups to DNA nucleotides (see Chapters 10 and 16)—is essential to the process of genomic imprinting, as demonstrated by the observation that mice deficient in DNA methylation do not exhibit imprinting. Some of the ways in which sex interacts with heredity are summarized in Table 5.4.

Concepts

In genomic imprinting, the expression of a gene is influenced by the sex of the parent who transmits the gene to the offspring.

www.whfreeman.com/pierce

Additional information about genomic imprinting, Prader-Willi syndrome, and Angelman syndrome

Anticipation

Another genetic phenomenon that is not explained by Mendel's principles is **anticipation**, in which a genetic trait becomes more strongly expressed or is expressed at an earlier age as it is passed from generation to generation. In the early 1900s, several physicians observed that patients with moderate to severe myotonic dystrophy—an autosomal dominant muscle disorder—frequently had ancestors who were only mildly affected by the disease. These observations led to the concept of anticipation. However, the concept quickly fell out of favor with geneticists because there was no obvious mechanism to explain it; traditional genetics held that genes are passed unaltered from parents to offspring. Geneticists tended to attribute anticipation to observational bias.

The results of recent research have reestablished anticipation as a legitimate genetic phenomenon. The mutation causing myotonic dystrophy consists of an unstable region of DNA that can increase or decrease in size as the gene is passed from generation to generation, much like the gene that causes Huntington disease. The age of onset and the severity of the disease are correlated with the size of the unstable region; an increase in the size of the region through generations produces anticipation. The phenomenon has now been implicated in several genetic diseases. We will examine these interesting types of mutations in more detail in Chapter 17.

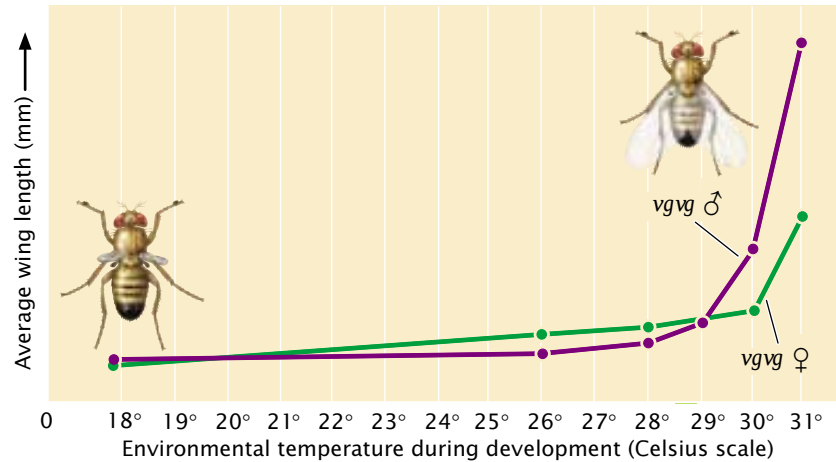
Concepts

Anticipation is the stronger or earlier expression of a genetic trait through succeeding generations. It is caused by an unstable region of DNA that increases or decreases in size.

Interaction Between Genes and Environment

In Chapter 3, we learned that each phenotype is the result of a genotype developing within a specific environment; the genotype sets the potential for development, but how the phenotype actually develops within the limits imposed by the genotype depends on environmental effects. Stated another way, each genotype may produce several different phenotypes, depending on the environmental conditions in which development occurs. For example, genotype *GG* may produce a plant that is 10 cm high when raised at 20°C, but the same genotype may produce a plant that is 18 cm high when raised at 25°C. The range of phenotypes produced by a genotype in different environments (in this case, plant height) is called the **norm of reaction** (FIGURE 5.18).

For most of the characteristics discussed so far, the effect of the environment on the phenotype has been slight.



5.18 Norm of reaction is the range of phenotypes produced by a genotype in different environments. This norm of reaction is for vestigial wings in *Drosophila melanogaster*. (Data from M. H. Harnly, *Journal of Experimental Zoology* 56:363–379, 1936.)

Mendel's peas with genotype *yy*, for example, developed yellow endosperm regardless of the environment in which they were raised. Similarly, persons with genotype $I^A I^A$ have the A antigen on their red blood cells regardless of their diet, socioeconomic status, or family environment. For other phenotypes, however, environmental effects play a more important role.

Environmental Effects on Gene Expression

The expression of some genotypes is critically dependent on the presence of a specific environment. For example, the *himalayan* allele in rabbits produces dark fur at the extremities of the body—on the nose, ears, and feet (FIGURE 5.19). The dark pigment develops, however, only when the rabbit is reared at 25°C or less; if a Himalayan rabbit is reared at 30°C, no dark patches develop. The expression of the *himalayan* allele is thus temperature dependent—an enzyme necessary for the production of dark pigment is inactivated at higher temperatures. The pigment is normally restricted to the nose, feet, and ears of Himalayan rabbits because the animal's core body temperature is normally above 25°C and the enzyme is functional only in the cells of the relatively cool extremities. The *himalayan* allele is an example of a **temperature-sensitive allele**, an allele whose product is functional only at certain temperatures.

5.19 The expression of some genotypes depends on specific environments. The expression of a temperature-sensitive allele, *himalayan*, is shown in rabbits reared at different temperatures.



Reared at 20°C or less



Reared at temperatures above 30°C

Some types of albinism in plants are temperature dependent. In barley, an autosomal recessive allele inhibits chlorophyll production, producing albinism when the plant is grown below 7°C. At temperatures above 18°C, a plant homozygous for the albino allele develops normal chlorophyll and is green. Similarly, among *Drosophila melanogaster* homozygous for the autosomal mutation *vestigial*, greatly reduced wings develop at 25°C, but wings near normal size develop at higher temperatures (see Figure 5.18).

Environmental factors also play an important role in the expression of a number of human genetic diseases. Glucose-6-phosphate dehydrogenase is an enzyme taking part in supplying energy to the cell. In humans, there are a number of genetic variants of glucose-6-phosphate dehydrogenase, some of which destroy red blood cells when the body is stressed by infection or by the ingestion of certain drugs or foods. The symptoms of the genetic disease appear only in the presence of these specific environmental factors.

Another genetic disease, phenylketonuria (PKU), is due to an autosomal recessive allele that causes mental retardation. The disorder arises from a defect in an enzyme that normally metabolizes the amino acid phenylalanine. When this enzyme is defective, phenylalanine is not metabolized, and its buildup causes brain damage in children. A simple

environmental change, putting an affected child on a low-phenylalanine diet, prevents retardation.

These examples illustrate the point that genes and their products do not act in isolation; rather, they frequently interact with environmental factors. Occasionally, environmental factors alone can produce a phenotype that is the same as the phenotype produced by a genotype; this phenotype is called a **phenocopy**. In fruit flies, for example, the autosomal recessive mutation *eyeless* produces greatly reduced eyes. The eyeless phenotype can also be produced by exposing the larvae of normal flies to sodium metaborate.

Concepts

The expression of many genes is modified by the environment. The range of phenotypes produced by a genotype in different environments is called the norm of reaction. A phenocopy is a trait produced by environmental effects that mimics the phenotype produced by a genotype.



The Inheritance of Continuous Characteristics

So far, we've dealt primarily with characteristics that have only a few distinct phenotypes. In Mendel's peas, for example, the seeds were either smooth or wrinkled, yellow or green; the coats of dogs were black, brown, or yellow; blood types were of four distinct types, A, B, AB, or O. Characteristics such as these, which have a few easily distinguished phenotypes, are called **discontinuous characteristics**.

Not all characteristics exhibit discontinuous phenotypes. Human height is an example of such a character; people do not come in just a few distinct heights but, rather, display a continuum of heights. Indeed, there are so many possible phenotypes of human height that we must use a measurement to describe a person's height. Characteristics that exhibit a continuous distribution of phenotypes are termed **continuous characteristics**. Because such characteristics have many possible phenotypes and must be described in quantitative terms, continuous characteristics are also called **quantitative characteristics**.

Continuous characteristics frequently arise because genes at many loci interact to produce the phenotypes. When a single locus with two alleles codes for a characteristic, there are three genotypes possible: *AA*, *Aa*, and *aa*. With two loci, each with two alleles, there are $3^2 = 9$ genotypes possible. The number of genotypes coding for characteristic is 3^n , where n equals the number of loci with two alleles that influence the characteristic. For example, when a characteristic is determined by eight loci, each with two alleles, there are $3^8 = 6561$ different genotypes possible for this character. If each genotype produces a different phenotype, many phenotypes will be possible. The slight differences between the phenotypes will be indistinguishable, and the characteristic will

appear continuous. Characteristics encoded by genes at many loci are called **polygenic characteristics**.

The converse of polygeny is **pleiotropy**, in which one gene affects multiple characteristics. Many genes exhibit pleiotropy. PKU, mentioned earlier, results from a recessive allele; persons homozygous for this allele, if untreated, exhibit mental retardation, blue eyes, and light skin color.

Frequently the phenotypes of continuous characteristics are also influenced by environmental factors. Each genotype is capable of producing a range of phenotypes—it has a relatively broad norm of reaction. In this situation, the particular phenotype that results depends on both the genotype and the environmental conditions in which the genotype develops. For example, there may be only three genotypes coding for a characteristic, but, because each genotype has a broad norm of reaction, the phenotype of the character exhibits a continuous distribution. Many continuous characteristics are both polygenic and influenced by environmental factors; such characteristics are called **multifactorial** because many factors help determine the phenotype.

The inheritance of continuous characteristics may appear to be complex, but the alleles at each locus follow Mendel's principles and are inherited in the same way as alleles coding for simple, discontinuous characteristics. However, because many genes participate, environmental factors influence the phenotype, and the phenotypes do not sort out into a few distinct types, we cannot observe the distinct ratios that have allowed us to interpret the genetic basis of discontinuous characteristics. To analyze continuous characteristics, we must employ special statistical tools, as will be discussed in Chapter 22.

Concepts

Discontinuous characteristics exhibit a few distinct phenotypes; continuous characteristics exhibit a range of phenotypes. A continuous characteristic is frequently produced when genes at many loci and environmental factors combine to determine a phenotype.



Connecting Concepts Across Chapters

This chapter introduced a number of modifications and extensions of the basic concepts of heredity that we learned in Chapter 3. A major theme has been gene expression: how interactions between genes, interactions between genes and sex, and interactions between genes and the environment affect the phenotypic expression of genes. The modifications and extensions discussed in this chapter do not alter the way that genes are inherited, but they do modify the way in which the genes determine the phenotype.



A number of topics introduced in this chapter will be explored further in other chapters of the book. Here we have purposefully ignored many aspects of the nature of gene expression because our focus has been on the “big picture” of how these interactions affect phenotypic ratios in genetic crosses. In subsequent chapters, we will explore the molecular details of gene expression, including transcription (Chapter 13), translation (Chapter 15), and the control of gene expression (Chapter 16). The

molecular nature of anticipation will be examined in more detail in Chapter 17, and DNA methylation, the basis of genomic imprinting, will be discussed in Chapter 10. Complementation testing will be revisited in Chapter 8, and the role of multiple genes and environmental factors in the inheritance of continuous characteristics will be studied more thoroughly in Chapter 22.

CONCEPTS SUMMARY

- Dominance always refers to genes at the same locus (allelic genes) and can be understood in regard to how the phenotype of the heterozygote relates to the phenotypes of the homozygotes.
- Dominance is complete when a heterozygote has the same phenotype as a homozygote. Dominance is incomplete when the heterozygote has a phenotype intermediate between those of two parental homozygotes. Codominance is the result when the heterozygote exhibits traits of both parental homozygotes.
- The type of dominance does not affect the inheritance of an allele; it does affect the phenotypic expression of the allele. The classification of dominance may depend on the level of the phenotype examined.
- Lethal alleles cause the death of an individual possessing them, usually at an early stage of development, and may alter phenotypic ratios.
- Multiple alleles refers to the presence of more than two alleles at a locus within a group. Their presence increases the number of genotypes and phenotypes possible.
- Gene interaction refers to interaction between genes at different loci to produce a single phenotype. An epistatic gene at one locus suppresses or masks the expression of hypostatic genes at different loci. Gene interaction frequently produces phenotypic ratios that are modifications of dihybrid ratios.
- A complementation test, in which individuals homozygous for different mutations are crossed, can be used to determine if the mutations occur at the same locus or at different loci.
- Sex-influenced characteristics are encoded by autosomal genes that are expressed more readily in one sex.
- Sex-limited characteristics are encoded by autosomal genes expressed in only one sex. Both males and females possess sex-limited genes and transmit them to their offspring.
- In cytoplasmic inheritance, the genes for the characteristic are found in the cytoplasm and are usually inherited from a single (usually maternal parent).
- Genetic maternal effect is present when an offspring inherits genes from both parents, but the nuclear genes of the mother determine the offspring's phenotype.
- Genomic imprinting refers to characteristics encoded by autosomal genes whose expression is affected by the sex of the parent transmitting the genes.
- Anticipation refers to a genetic trait that is more strongly expressed or is expressed at an earlier age in succeeding generations.
- Phenotypes are often modified by environmental effects. The range of phenotypes that a genotype is capable of producing in different environments is the norm of reaction. A phenocopy is a phenotype produced by an environmental effect that mimics a phenotype produced by a genotype.
- Discontinuous characteristics are characteristics with a few distinct phenotypes; continuous characteristics are those that exhibit a wide range of phenotypes. Continuous characteristics are frequently produced by the combined effects of many genes and environmental effects.

IMPORTANT TERMS

codominance (p. 103)	complementation (p. 115)	genomic imprinting (p. 120)	continuous characteristic (p. 123)
lethal allele (p. 104)	sex-influenced characteristic (p. 115)	anticipation (p. 121)	quantitative characteristic (p. 123)
multiple alleles (p. 105)	sex-limited characteristic (p. 116)	norm of reaction (p. 121)	polygenic characteristic (p. 123)
gene interaction (p. 107)	cytoplasmic inheritance (p. 118)	temperature-sensitive allele (p. 122)	pleiotropy (p. 123)
epistasis (p. 108)	genetic maternal effect (p. 119)	phenocopy (p. 123)	multifactorial characteristic (p. 123)
epistatic gene (p. 108)		discontinuous characteristic (p. 123)	
hypostatic gene (p. 108)			
complementation test (p. 114)			

Worked Problems

1. The type of plumage found in mallard ducks is determined by three alleles at a single locus: M^R , which codes for restricted plumage; M , which codes for mallard plumage; and m^d , which codes for dusky plumage. The restricted phenotype is dominant over mallard and dusky; mallard is dominant over dusky ($M^R > M > m^d$). Give the expected phenotypes and proportions of offspring produced by the following crosses.

(a) $M^R M \times m^d m^d$

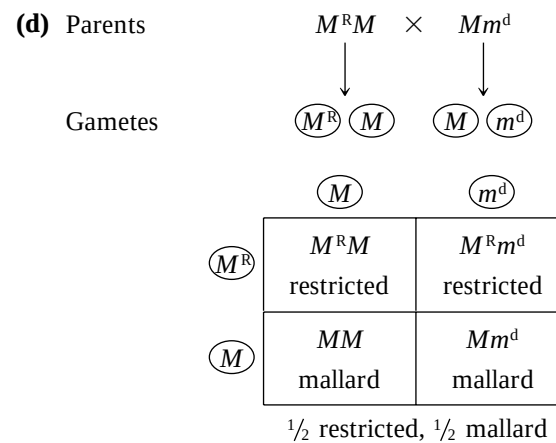
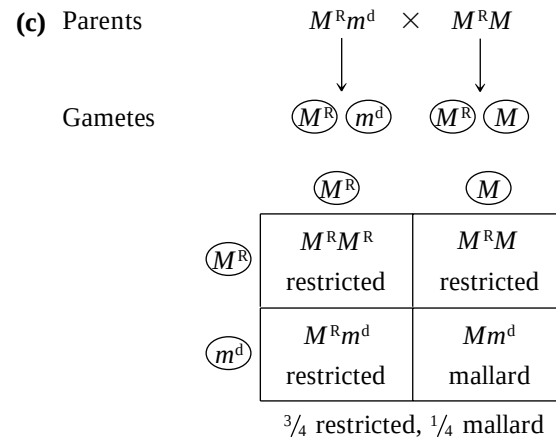
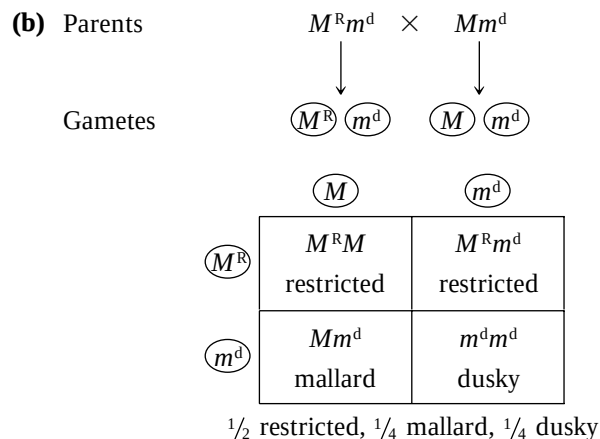
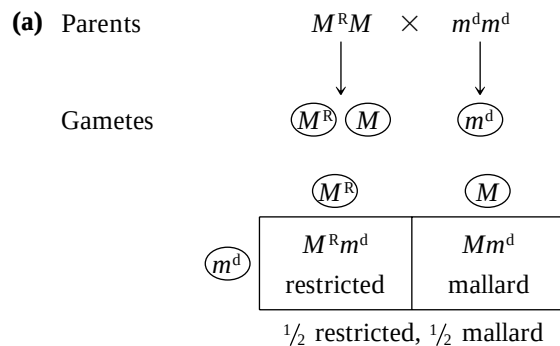
(b) $M^R m^d \times M m^d$

(c) $M^R m^d \times M^R M$

(d) $M^R M \times M m^d$

• Solution

We can determine the phenotypes and proportions of offspring by (1) determining the types of gametes produced by each parent and (2) combining the gametes of the two parents with the use of a Punnett square



2. A homozygous strain of yellow corn is crossed with a homozygous strain of purple corn. The F_1 are intercrossed, producing an ear of corn with 119 purple kernels and 89 yellow kernels (the progeny).

(a) What is the genotype of the yellow kernels?

(b) Give a genetic explanation for the differences in kernel color in this cross.

• Solution

(a) We should first consider whether the cross between yellow and purple strains might be a monohybrid cross for a simple dominant trait, which would produce a 3:1 ratio in the F_2 ($Aa \times Aa \rightarrow 3/4 A_$ and $1/4 aa$). Under this hypothesis, we would expect 156 purple progeny and 52 yellow progeny:

Phenotype	Genotype	Observed number	Expected number
purple	$A_$	119	$3/4 \times 208 = 156$
yellow	aa	89	$1/4 \times 208 = 52$
total		208	

We see that the expected numbers do not closely fit the observed numbers. If we performed a chi-square test (see Chapter 3), we would obtain a calculated chi-square value of 35.08, which has a probability much less than 0.05, indicating that it is extremely unlikely that, when we expect a 3:1 ratio, we would obtain 119 purple progeny and 89 yellow progeny. Therefore we can reject the hypothesis that these results were produced by a monohybrid cross.

Another possible hypothesis is that the observed F₂ progeny are in a 1:1 ratio. However, we learned in Chapter 3 that a 1:1 ratio is produced by a cross between a heterozygote and a homozygote (*Aa* × *aa*) and, from the information given, the cross was not between a heterozygote and a homozygote, because the original parental strains were both homozygous. Furthermore, a chi-square test comparing the observed numbers with an expected 1:1 ratio yields a calculated chi-square value of 4.32, which has a probability of less than .05.

Next, we should look to see if the results can be explained by a dihybrid cross (*AaBb* × *AaBb*). A dihybrid cross results in phenotypic proportions that are in sixteenths. We can apply the formula given earlier in the chapter to determine the number of sixteenths for each phenotype:

$$x = \frac{\text{number of progeny with a phenotype} \times 16}{\text{total number of progeny}}$$

$$x_{(\text{purple})} = \frac{119 \times 16}{208} = 9.15$$

$$x_{(\text{yellow})} = \frac{89 \times 16}{208} = 6.85$$

Thus, purple and yellow appear approximately a 9:7 ratio. We can test this hypothesis with a chi-square test:

Phenotype	Genotype	Observed number	Expected number
purple	?	119	$\frac{9}{16} \times 208 = 117$
yellow	?	89	$\frac{7}{16} \times 208 = 91$
total		208	

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(119 - 117)^2}{117} + \frac{(89 - 91)^2}{91}$$

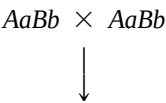
$$= 0.034 + 0.44 = 0.078$$

$$\text{Degree of freedom} = n - 1 = 2 - 1 = 1$$

$$P > .05$$

The probability associated with the chi-square value is greater than .05, indicating that there is a relatively good fit between the observed results and a 9:7 ratio.

We now need to determine how a dihybrid cross can produce a 9:7 ratio and what genotypes correspond to the two phenotypes. A dihybrid cross without epistasis produces a 9:3:3:1 ratio:



<i>A_B_</i>	$\frac{9}{16}$
<i>A_bb</i>	$\frac{3}{16}$
<i>aaB_</i>	$\frac{3}{16}$
<i>aabb</i>	$\frac{1}{16}$

Because $\frac{9}{16}$ of the progeny from the corn cross are purple, purple must be produced by genotypes *A_B_*; in other words, individual kernels that have at least one dominant allele at the first locus and at least one dominant allele at the second locus are purple. The proportions of all the other genotypes (*A_bb*, *aaB_*, and *aabb*) sum to $\frac{7}{16}$, which is the proportion of the progeny in the corn cross that are yellow, so any individual kernel that does not have a dominant allele at both the first and the second locus is yellow.

(b) Kernel color is an example of duplicate recessive epistasis, where the presence of two recessive alleles at either the first locus or the second locus or both suppresses the production of purple pigment.

3. A geneticist crosses two yellow mice with straight hair and obtains the following progeny:

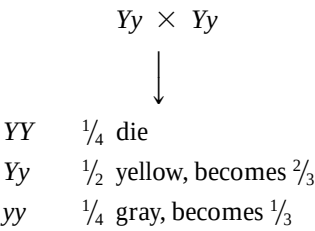
- $\frac{1}{2}$ yellow, straight
- $\frac{1}{6}$ yellow, fuzzy
- $\frac{1}{4}$ gray, straight
- $\frac{1}{12}$ gray, fuzzy

(a) Provide a genetic explanation for the results and assign genotypes to the parents and progeny of this cross.

(b) What additional crosses might be carried out to determine if your explanation is correct?

• Solution

(a) This cross concerns two separate characteristics—color and type of hair; so we should begin by examining the results for each characteristic separately. First, let's look at the inheritance of color. Two yellow mice are crossed producing $\frac{1}{2} + \frac{1}{6} = \frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$ yellow mice and $\frac{1}{4} + \frac{1}{12} = \frac{3}{12} + \frac{1}{12} = \frac{4}{12} = \frac{1}{3}$ gray mice. We learned in this chapter that a 2:1 ratio is often produced when a recessive lethal gene is present:



Now, let's examine the inheritance of the hair type. Two mice with straight hair are crossed, producing $\frac{1}{2} + \frac{1}{4} = \frac{2}{4} + \frac{1}{4} = \frac{3}{4}$ mice with straight hair and $\frac{1}{6} + \frac{1}{12} = \frac{2}{12} + \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$ mice with fuzzy hair. We learned in Chapter 3 that a

3:1 ratio is usually produced by a cross between two individuals heterozygous for a simple dominant allele:

$$\begin{array}{rcl}
 Ss \times Ss & & \\
 \downarrow & & \\
 \begin{array}{l} SS \quad 1/4 \text{ straight} \\ Ss \quad 1/2 \text{ straight} \\ ss \quad 1/4 \text{ fuzzy} \end{array} & \left. \vphantom{\begin{array}{l} SS \\ Ss \\ ss \end{array}} \right\} & 3/4 \text{ straight}
 \end{array}$$

We can now combine both loci and assign genotypes to all the individuals in the cross:

P yellow, straight $\times$ yellow, straight				
$YySs \quad \quad YySs$				
$\downarrow$				
Phenotype	Genotype	Probability at each locus	Combined probability	
yellow, straight	$YyS_$	$2/3 \times 3/4$	$= 6/12 = 1/2$	
yellow, fuzzy	$Yyss$	$2/3 \times 1/4$	$= 2/12 = 1/6$	
gray, straight	$yyS_$	$1/3 \times 3/4$	$= 3/12 = 1/4$	
gray, fuzzy	$yyss$	$1/3 \times 1/4$	$= 1/12$	

(b) We could carry out a number of different crosses to test our hypothesis that yellow is a recessive lethal and straight is dominant over fuzzy. For example, a cross between any two yellow individuals should always produce $2/3$ yellow and $1/3$ gray, and a cross between two gray individuals should produce all gray offspring. A cross between two fuzzy individuals should always produce all fuzzy offspring.

4. In some sheep, the presence of horns is produced by an autosomal allele that is dominant in males and recessive in females. A horned female is crossed with a hornless male. One of the resulting F_1 females is crossed with a hornless male. What proportion of the male and female progeny from this cross will have horns?

• Solution

The presence of horns in these sheep is an example of a sex-influenced characteristic. Because the phenotypes associated with the genotypes differ for the two sexes, let's begin this problem by writing out the genotypes and phenotypes for each sex. We will

let H represent the allele that codes for horns and H^+ represent the allele for hornless. In males, the allele for horns is dominant over the allele for hornless, which means that males homozygous (HH) and heterozygous (H^+H) for this gene are horned. Only males homozygous for the recessive hornless allele (H^+H^+) will be hornless. In females, the allele for horns is recessive, which means that only females homozygous for this allele (HH) will be horned; females heterozygous (H^+H) and homozygous (H^+H^+) for the hornless allele will be hornless. The following table summarizes genotypes and associated phenotypes:

Genotype	Male phenotype	Female phenotype
HH	horned	horned
HH^+	horned	hornless
H^+H^+	hornless	hornless

In the problem, a horned female is crossed with a hornless male. From the preceding table, we see that a horned female must be homozygous for the allele for horns (HH) and a hornless male must be homozygous for the allele for hornless (H^+H^+); so all the F_1 will be heterozygous; the F_1 males will be horned and the F_1 females will be hornless, as shown below:

$$\begin{array}{rcl}
 P & H^+H^+ \times HH & \\
 \downarrow & & \\
 F_1 & H^+H & H^+H \\
 & \text{horned males and hornless females} &
 \end{array}$$

A heterozygous hornless F_1 female (H^+H) is then crossed with a hornless male (H^+H^+):

$$\begin{array}{rcl}
 H^+H & \times & H^+H^+ \\
 \text{horned female} & & \text{hornless male} \\
 \downarrow & & \\
 \begin{array}{l} 1/2 \ H^+H^+ \\ 1/2 \ H^+H \end{array} & & \begin{array}{l} \text{Males} \\ \text{hornless} \\ \text{horned} \end{array} \quad \begin{array}{l} \text{Females} \\ \text{hornless} \\ \text{hornless} \end{array}
 \end{array}$$

Therefore, $1/2$ of the male progeny will be horned but none of the female progeny will be horned.

COMPREHENSION QUESTIONS

- * 1. How do incomplete dominance and codominance differ?
- * 2. Explain how dominance and epistasis differ.
 3. What is a recessive epistatic gene?
 4. What is a complementation test and what is it used for?
- * 5. What is genomic imprinting?
6. What characteristics do you expect to see in a trait that exhibits anticipation?
- * 7. What characteristics are exhibited by a cytoplasmically inherited trait?
8. What is the difference between genetic maternal effect and genomic imprinting?
9. What is the difference between a sex-influenced gene and a gene that exhibits genomic imprinting?
- * 10. What are continuous characteristics and how do they arise?

APPLICATION QUESTIONS AND PROBLEMS

- * 11. Palomino horses have a golden yellow coat, chestnut horses have a brown coat, and cremello horses have a coat that is almost white. A series of crosses between the three different types of horses produced the following offspring:

Cross	Offspring
palomino $\times$ palomino	13 palomino, 6 chestnut, 5 cremello
chestnut $\times$ chestnut	16 chestnut
cremello $\times$ cremello	13 cremello
palomino $\times$ chestnut	8 palomino, 9 chestnut
palomino $\times$ cremello	11 palomino, 11 cremello
chestnut $\times$ cremello	23 palomino

- (a) Explain the inheritance of the palomino, chestnut, and cremello phenotypes in horses.
- (b) Assign symbols for the alleles that determine these phenotypes, and list the genotypes of all parents and offspring given in the preceding table.
- * 12. The L^M and L^N alleles at the MN blood group locus exhibit codominance. Give the expected genotypes and phenotypes and their ratios in progeny resulting from the following crosses.
- (a) $L^M L^M \times L^M L^N$
- (b) $L^N L^N \times L^N L^N$
- (c) $L^M L^N \times L^M L^N$
- (d) $L^M L^N \times L^N L^N$
- (e) $L^M L^M \times L^N L^N$
13. In the pearl millet plant, color is determined by three alleles at a single locus: Rp^1 (red), Rp^2 (purple), and rp (green). Red is dominant over purple and green, and purple is dominant over green ($Rp^1 > Rp^2 > rp$). Give the expected phenotypes and ratios of offspring produced by the following crosses.
- (a) $Rp^1/Rp^2 \times Rp^1/rp$
- (b) $Rp^1/rp \times Rp^2/rp$
- (c) $Rp^1/Rp^2 \times Rp^1/Rp^2$
- (d) $Rp^2/rp \times rp/rp$
- (e) $rp/rp \times Rp^1/Rp^2$
- * 14. Give the expected genotypic and phenotypic ratios for the following crosses for ABO blood types.
- (a) $I^A i \times I^B i$
- (b) $I^A I^B \times I^A i$
- (c) $I^A I^B \times I^A I^B$
- (d) $ii \times I^A i$
- (e) $I^A I^B \times ii$

15. If there are five alleles at a locus, how many genotypes may there be at this locus? How many different kinds of homozygotes will there be? How many genotypes and homozygotes would there be with eight alleles?

16. Turkeys have black, bronze, or black-bronze plumage. Examine the results of the following crosses:

Parents	Offspring
Cross 1: black and bronze	all black
Cross 2: black and black	$\frac{3}{4}$ black, $\frac{1}{4}$ bronze
Cross 3: black-bronze and black-bronze	all black-bronze
Cross 4: black and bronze	$\frac{1}{2}$ black, $\frac{1}{4}$ bronze, $\frac{1}{4}$ black-bronze
Cross 5: bronze and black-bronze	$\frac{1}{2}$ bronze, $\frac{1}{2}$ black-bronze
Cross 6: bronze and bronze	$\frac{3}{4}$ bronze, $\frac{1}{4}$ black-bronze

Do you think these differences in plumage arise from incomplete dominance between two alleles at a single locus? If yes, support your conclusion by assigning symbols to each allele and providing genotypes for all turkeys in the crosses. If your answer is no, provide an alternative explanation and assign genotypes to all turkeys in the crosses.

17. In rabbits, an allelic series helps to determine coat color: C (full color), c^{ch} (chinchilla, gray color), c^h (himalayan, white with black extremities), and c (albino, all white). The C allele is dominant over all others, c^{ch} is dominant over c^h and c , c^h is dominant over c , and c is recessive to all the other alleles. This dominance hierarchy can be summarized as $C > c^{ch} > c^h > c$. The rabbits in the following list are crossed and produce the progeny shown. Give the genotypes of the parents for each cross:

Phenotypes of parents	Phenotypes of offspring
(a) full color $\times$ albino	$\frac{1}{2}$ full color, $\frac{1}{2}$ albino
(b) himalayan $\times$ albino	$\frac{1}{2}$ himalayan, $\frac{1}{2}$ albino
(c) full color $\times$ albino	$\frac{1}{2}$ full color, $\frac{1}{2}$ chinchilla
(d) full color $\times$ himalayan	$\frac{1}{2}$ full color, $\frac{1}{4}$ himalayan, $\frac{1}{4}$ albino
(e) full color $\times$ full color	$\frac{3}{4}$ full color, $\frac{1}{4}$ albino

18. In this chapter we considered Joan Barry's paternity suit against Charlie Chaplin and how, on the basis of blood types, Chaplin could not have been the father of her child.
- (a) What blood types are possible for the father of Barry's child?
- (b) If Chaplin had possessed one of these blood types, would that prove that he fathered Barry's child?

- * 19. A woman has blood type A MM. She has a child with blood type AB MN. Which of the following blood types could *not* be that of the child's father? Explain your reasoning.

George	O	NN
Tom	AB	MN
Bill	B	MN
Claude	A	NN
Henry	AB	MM

20. Allele *A* is epistatic to allele *B*. Indicate whether each of the following statements is true or false. Explain why.

- (a) Alleles *A* and *B* are at the same locus.
- (b) Alleles *A* and *B* are at different loci.
- (c) Alleles *A* and *B* are always located on the same chromosome.
- (d) Alleles *A* and *B* may be located on different, homologous chromosomes.
- (e) Alleles *A* and *B* may be located on different, nonhomologous chromosomes.

- * 21. In chickens, comb shape is determined by alleles at two loci (*R*, *r* and *P*, *p*). A walnut comb is produced when at least one dominant allele *R* is present at one locus and at least one dominant allele *P* is present at a second locus (genotype *R_P_*). A rose comb is produced when at least one dominant allele is present at the first locus and two recessive alleles are present at the second locus (genotype *R_pp*). A pea comb is produced when two recessive alleles are present at the first locus and at least one dominant allele is present at the second (genotype *rrP_*). If two recessive alleles are present at the first and at the second locus (*rrpp*), a single comb is produced. Progeny with what types of combs and in what proportions will result from the following crosses?

- (a) *RRPP* × *rrpp*
- (b) *RrPp* × *rrpp*
- (c) *RrPp* × *RrPp*
- (d) *Rrpp* × *Rrpp*
- (e) *Rrpp* × *rrPp*
- (f) *Rrpp* × *rrpp*

- * 22. Eye color of the Oriental fruit fly (*Bactrocera dorsalis*) is determined by a number of genes. A fly having wild-type eyes is crossed with a fly having yellow eyes. All the *F*₁ flies from this cross have wild-type eyes. When the *F*₁ are interbred, $\frac{9}{16}$ of the *F*₂ progeny have wild-type eyes, $\frac{3}{16}$ have amethyst eyes (a bright, sparkling blue color), and $\frac{4}{16}$ have yellow eyes.

- (a) Give genotypes for all the flies in the *P*, *F*₁, and *F*₂ generations.

- (b) Does epistasis account for eye color in Oriental fruit flies? If so, which gene is epistatic and which gene is hypostatic?

23. A variety of opium poppy (*Papaver somniferum* L.) having lacerate leaves was crossed with a variety that has normal leaves. All the *F*₁ had lacerate leaves. Two *F*₁ plants were interbred to produce the *F*₂. Of the *F*₂, 249 had lacerate leaves and 16 had normal leaves. Give genotypes for all the plants in the *P*, *F*₁, and *F*₂ generations. Explain how lacerate leaves are determined in the opium poppy.

- * 24. A dog breeder liked yellow and brown Labrador retrievers. In an attempt to produce yellow and brown puppies, he bought a yellow Labrador male and a brown Labrador female and mated them. Unfortunately, all the puppies produced in this cross were black. (See p. 000 for a discussion of the genetic basis of coat color in Labrador retrievers.)

- (a) Explain this result.
- (b) How might the breeder go about producing yellow and brown Labradors?

25. When a yellow female Labrador retriever was mated with a brown male, half of the puppies were brown and half were yellow. The same female, when mated with a different brown male, produced all brown males. Explain these results.

- * 26. In summer squash, a plant that produces disc-shaped fruit is crossed with a plant that produces long fruit. All the *F*₁ have disc-shaped fruit. When the *F*₁ are intercrossed, *F*₂ progeny are produced in the following ratio: $\frac{9}{16}$ disc-shaped fruit: $\frac{6}{16}$ spherical fruit: $\frac{1}{16}$ long fruit. Give the genotypes of the *F*₂ progeny.

27. In sweet peas, some plants have purple flowers and other plants have white flowers. A homozygous variety of pea that has purple flowers is crossed with a homozygous variety that has white flowers. All the *F*₁ have purple flowers. When these *F*₁ are self-fertilized, the *F*₂ appear in a ratio of $\frac{9}{16}$ purple to $\frac{7}{16}$ white.

- (a) Give genotypes for the purple and white flowers in these crosses.
- (b) Draw a hypothetical biochemical pathway to explain the production of purple and white flowers in sweet peas.

28. For the following questions, refer to p. 000 for a discussion of how coat color and pattern are determined in dogs.

- (a) Explain why Irish setters are reddish in color.
- (b) Will a cross between a beagle and a Dalmatian produce puppies with ticking? Why or why not?
- (c) Can a poodle crossed with any other breed produce spotted puppies? Why or why not?
- (d) If a St. Bernard is crossed with a Doberman, will the offspring have solid, yellow, saddle, or bicolor coats?
- (e) If a Rottweiler is crossed with a Labrador retriever, will the offspring have solid, yellow, saddle, or bicolor coats?

- *29. When a Chinese hamster with white spots is crossed with another hamster that has no spots, approximately $\frac{1}{2}$ of the offspring have white spots and $\frac{1}{2}$ have no spots. When two hamsters with white spots are crossed, $\frac{2}{3}$ of the offspring possess white spots and $\frac{1}{3}$ have no spots.
- What is the genetic basis of white spotting in Chinese hamsters?
 - How might you go about producing Chinese hamsters that breed true for white spotting?
30. Male-limited precocious puberty results from a rare, sex-limited autosomal allele (P) that is dominant over the allele for normal puberty (p) and is expressed only in males. Bill undergoes precocious puberty, but his brother Jack and his sister Beth underwent puberty at the usual time, between the ages of 10 and 14. Although Bill's mother and father underwent normal puberty, two of his maternal uncles (his mother's brothers) underwent precocious puberty. All of Bill's grandparents underwent normal puberty. Give the most likely genotypes for all the relatives mentioned in this family.
- *31. Pattern baldness in humans is a sex-influenced trait that is autosomal dominant in males and recessive in females. Jack has a full head of hair. JoAnn also has a full head of hair, but her mother is bald. (In women, pattern baldness is usually expressed as a thinning of the hair.) If Jack and JoAnn marry, what proportion of their children are expected to be bald?
32. In goats, a beard is produced by an autosomal allele that is dominant in males and recessive in females. We'll use the symbol B^b for the beard allele and B^+ for the beardless allele. Another independently assorting autosomal allele that produces a black coat (W) is dominant over the allele for white coat (w). Give the phenotypes and their expected proportions for the following crosses.
- $B^+B^b Ww$ male $\times$ $B^+B^b Ww$ female
 - $B^+B^b Ww$ male $\times$ $B^+B^b ww$ female
 - $B^+B^+ Ww$ male $\times$ $B^bB^b Ww$ female
 - $B^+B^b Ww$ male $\times$ $B^bB^b ww$ female
33. In the snail *Limnaea peregra*, shell coiling results from a genetic maternal effect. An autosomal allele for a right-handed shell (s^+), called dextral, is dominant over the allele for a left-handed shell (s), called sinistral. A pet snail called Martha is sinistral and reproduces only as a female (the snails are hermaphroditic). Indicate which of the following statements are true and which are false. Explain your reasoning in each case.
- Martha's genotype *must* be ss .
 - Martha's genotype *cannot* be s^+s^+ .
 - All the offspring produced by Martha *must* be sinistral.
 - At least some of the offspring produced by Martha *must* be sinistral.
 - Martha's mother *must* have been sinistral.
 - All Martha's brothers *must* be sinistral.
34. In unicorns, two autosomal loci interact to determine the type of tail. One locus controls whether a tail is present at all; the allele for a tail (T) is dominant over the allele for tailless (t). If a unicorn has a tail, then alleles at a second locus determine whether the tail is curly or straight. Farmer Baldridge has two unicorns with curly tails. When he crosses these two unicorns, $\frac{1}{2}$ of the progeny have curly tails, $\frac{1}{4}$ have straight tails, and $\frac{1}{4}$ do not have a tail. Give the genotypes of the parents and progeny in Farmer Baldridge's cross. Explain how he obtained the 2:1:1 phenotypic ratio in his cross.
- * 35. Phenylketonuria (PKU) is an autosomal recessive disease that results from a defect in an enzyme that normally metabolizes the amino acid phenylalanine. When this enzyme is defective, high levels of phenylalanine cause brain damage. In the past, most children with PKU became mentally retarded. Fortunately, mental retardation can be prevented in these children today by carefully controlling the amount of phenylalanine in the diet. As a result of this treatment, many people with PKU are now reaching reproductive age with no mental retardation. By the end of the teen years, when brain development is complete, many people with PKU go off the restrictive diet. Children born to women with PKU (who are no longer on a phenylalanine-restricted diet) frequently have low birth weight, developmental abnormalities, and mental retardation, even though they are heterozygous for the recessive PKU allele. However, children of men with PKU do not have these problems. Provide an explanation for these observations.
36. In 1983, a sheep farmer in Oklahoma noticed a ram in his flock that possessed increased muscle mass in his hindquarters. Many of the offspring of this ram possessed the same trait, which became known as the callipyge mutant (*callipyge* is Greek for "beautiful buttocks"). The mutation that caused the callipyge phenotype was eventually mapped to a position on the sheep chromosome 18.
- When the male callipyge offspring of the original mutant ram were crossed with normal females, they produced the following progeny: $\frac{1}{4}$ male callipyge, $\frac{1}{4}$ female callipyge, $\frac{1}{4}$ male normal, and $\frac{1}{4}$ female normal. When female callipyge offspring of the original mutant ram were crossed with normal males, all of the offspring were normal. Analysis of the chromosomes of these offspring of callipyge females showed that half of them received a chromosome 18 with the callipyge gene from their mother. Propose an explanation for the inheritance of the callipyge gene. How might you test your explanation?

CHALLENGE QUESTION

37. Suppose that you are tending a mouse colony at a genetics research institute and one day you discover a mouse with twisted ears. You breed this mouse with twisted ears and find that the trait is inherited. Both male and female mice have twisted ears, but when you cross a twisted-eared male with a normal-eared female, you obtain different results from those obtained when you cross a twisted-eared female

with normal-eared male—the reciprocal crosses give different results. Describe how you would go about determining whether this trait results from a sex-linked gene, a sex-influenced gene, a genetic maternal effect, a cytoplasmically inherited gene, or genomic imprinting. What crosses would you conduct and what results would be expected with these different types of inheritance?

SUGGESTED READINGS

Barlow, D. P. 1995. Gametic imprinting in mammals. *Science* 270:1610–1613.

Discusses the phenomenon of genomic imprinting.

Harper, P. S., H. G. Harley, W. Reardon, and D. J. Shaw. 1992. Anticipation in myotonic dystrophy: new light on an old problem [Review]. *American Journal of Human Genetics* 51:10–16.

A nice review of the history of anticipation.

Li, E., C. Beard, and R. Jaenisch. 1993. Role for DNA methylation in genomic imprinting. *Nature* 366:362–365. Reviews some of the evidence that DNA methylation is implicated in genomic imprinting.

Morell, V. 1993. Huntington's gene finally found. *Science* 260:28–30.

Report on the discovery of the gene that causes Huntington disease.

Ostrander, E. A., F. Galibert, and D. F. Patterson. 2000. Canine genetics comes of age. *Trends in Genetics* 16:117–123. Review of the use of dog genetics for understanding human genetic diseases.

Pagel, M. 1999. Mother and father in surprise agreement. *Nature* 397:19–20.

Discusses some of the possible evolutionary reasons for genomic imprinting.

Sapienza, C. 1990. Parental imprinting of genes. *Scientific American* 263 (October):52–60.

Another review of genomic imprinting.

Shoffner, J. M., and D. C. Wallace. 1992. Mitochondrial genetics: principles and practice [Invited editorial]. *American Journal of Human Genetics* 51:1179–1186.

Discusses the characteristics of cytoplasmically inherited mitochondrial mutations.

Skuse, D. H., R. S. James, D. V. M. Bishop, B. Coppin, P. Dalton, G. Aamodt-Leeper, M. Bacarese-Hamilton, C. Creswell, R. McGurk, and P. A. Jacobs. 1997. Evidence from Turner's syndrome of an imprinted X-linked locus affecting cognitive function. *Nature* 387:705–708.

Report of imprinting in Turner syndrome.

Thomson, G., and M. S. Esposito. 1999. The genetics of complex diseases. *Trends in Genetics* 15:M17–M20.

Discussion of human multifactorial diseases and the effect of the Human Genome Project on the identification of genes influencing these diseases.

Wallace, D. C. 1989. Mitochondrial DNA mutations and neuromuscular disease. *Trends in Genetics* 5:9–13. More discussion of cytoplasmically inherited mitochondrial mutations.

Willis, M. B. 1989. *Genetics of the Dog*. London: Witherby. A comprehensive review of canine genetics.

6 Pedigree Analysis and Applications



This is Chapter 6 Opener photo legend to position here-allowing two lines of caption. (AP/ Wide World Photos.)

Lou Gehrig and Superoxide Free Radicals

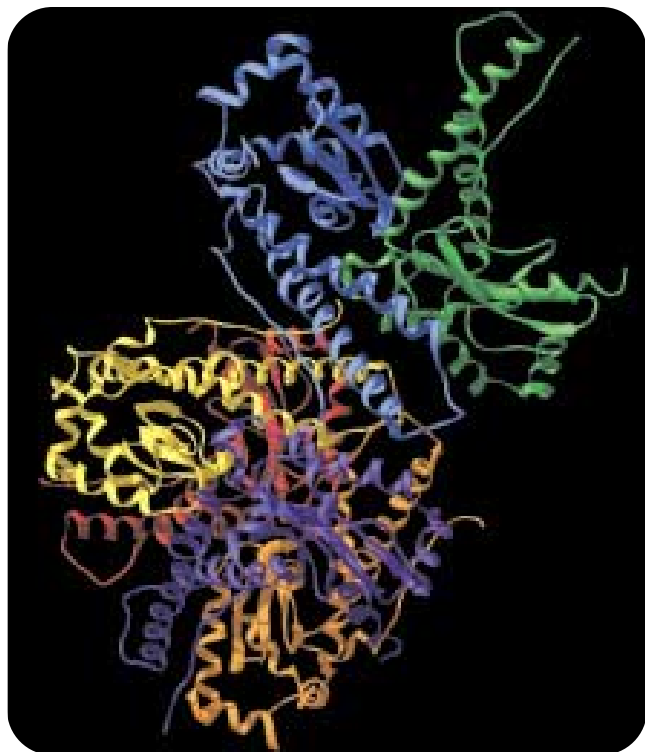
Lou Gehrig was the finest first baseman ever to play major league baseball. A left-handed power hitter who grew up in New York City, Gehrig played for the New York Yankees from 1923 to 1939. Throughout his career, he lived in the shadow of his teammates Babe Ruth and Joe Di Maggio, but Gehrig was a great hitter in his own right: he compiled a lifetime batting average of .340 and drove in more than 100 runs every season for 13 years. During his career, he batted in 1991 runs and hit a total of 23 grand slams (home runs with bases loaded). But Gehrig's greatest baseball record, which stood for more than 50 years and has been broken only once—by Cal Ripkin, Jr., in 1995—is his record of playing 2130 consecutive games.

In the 1938 baseball season, Gehrig fell into a strange slump. For the first time since his rookie year, his batting

average dropped below .300 and, in the World Series that year, he managed only four hits—all singles. Nevertheless, he finished the season convinced that he was undergoing a temporary slump that he would overcome in the next season. He returned to training camp in 1939 with high spirits. When the season began, however, it was clear to everyone that something was terribly wrong. Gehrig had no power in his swing; he was awkward and clumsy at first base. His condition worsened and, on May 2, he voluntarily removed himself from the lineup. The Yankees sent Gehrig to the Mayo Clinic for diagnosis and, on June 20, his medical report was made public: Lou Gehrig was suffering from a rare, progressive disease known as amyotrophic lateral sclerosis (ALS). Within two years, he was dead. Since then, ALS has commonly been known as Lou Gehrig disease.

Gehrig experienced symptoms typical of ALS: progressive weakness and wasting of skeletal muscles due to

- Lou Gehrig and Superoxide Free Radicals
- The Study of Human Genetic Characteristics
- Analyzing Pedigrees
 - Autosomal Recessive Traits
 - Autosomal Dominant Traits
 - X-Linked Recessive Traits
 - X-Linked Dominant Traits
 - Y-Linked Traits
- Twin Studies
 - Concordance
 - Twin Studies and Obesity
- Adoption Studies
 - Adoption Studies and Obesity
 - Adoption Studies and Alcoholism
- Genetic Counseling and Genetic Testing
 - Genetic Counseling
 - Genetic Testing



6.1 Some cases of amyotrophic lateral sclerosis are inherited and result from mutations in the gene that encodes the enzyme superoxide dismutase 1. A molecular model of the enzyme.

degeneration of the motor neurons. Most cases of ALS are sporadic, appearing in people with no family history of the disease. However, about 10% of cases run in families, and in these cases the disease is inherited as an autosomal dominant trait. In 1993, geneticists discovered that some familial cases of ALS are caused by a defect in a gene that encodes an enzyme called superoxide dismutase 1 (SOD1). This enzyme helps the cell to break down superoxide free radicals, which are highly reactive and extremely toxic. In families studied by the researchers, people with ALS had a defective allele for SOD1 (FIGURE 6.1) that produced an altered form of the enzyme. How the altered enzyme causes the symptoms of the disease has not been firmly established.

Amyotrophic lateral sclerosis is just one of a large number of human diseases that are currently the focus of intensive genetic research. This chapter will discuss human genetic characteristics and some of the techniques used to study human inheritance. A number of human characteristics have already been mentioned in discussions of general hereditary principles (Chapters 3 through 5), so by now you know that they follow the same rules of inheritance as those of characteristics in other organisms. So why do we have a separate chapter on human heredity? The answer is that the study of human inheritance requires special techniques—primarily because human biology and culture

impose certain constraints on the geneticist. In this chapter, we'll consider these constraints and examine three important techniques that human geneticists use to overcome them: pedigrees, twin studies, and adoption studies. At the end of the chapter, we will see how the information garnered with these techniques can be used in genetic counseling and prenatal diagnosis.

Keep in mind as you go through this chapter that many important characteristics are influenced by both genes and environment, and separating these factors is always difficult in humans. Studies of twins and adopted persons are designed to distinguish the effects of genes and environment, but such studies are based on assumptions that may be difficult to meet for some human characteristics, particularly behavioral ones. Therefore, it's always prudent to interpret the results of such studies with caution.

www.whfreeman.com/pierce

Information on amyotrophic lateral sclerosis, and more about Lou Gehrig, his outstanding career in baseball, and his fight with amyotrophic lateral sclerosis

The Study of Human Genetic Characteristics

Humans are the best and the worst of all organisms for genetic study. On the one hand, we know more about human anatomy, physiology, and biochemistry than we know about most other organisms; for many families, we have detailed records extending back many generations; and the medical implications of genetic knowledge of humans provide tremendous incentive for genetic studies. On the other hand, the study of human genetic characteristics presents some major obstacles.

First, controlled matings are not possible. With other organisms, geneticists carry out specific crosses to test their hypotheses about inheritance. We have seen, for example, how the testcross provides a convenient way to determine if an individual with a dominant trait is homozygous or heterozygous. Unfortunately (for the geneticist at least), matings between humans are more frequently determined by romance, family expectations, and—occasionally—accident than they are by the requirements of the geneticist.

Another obstacle is that humans have a long generation time. Human reproductive age is not normally reached until 10 to 14 years after birth, and most humans do not reproduce until they are 18 years of age or older; thus, generation time in humans is usually about 20 years. This long generation time means that, even if geneticists could control human crosses, they would have to wait on average 40 years just to observe the F_2 progeny. In contrast, generation time in *Drosophila* is 2 weeks; in bacteria, it's a mere 20 minutes.

Finally, human family size is generally small. Observation of even the simple genetic ratios that we learned in

Chapter 3 would require a substantial number of progeny in each family. When parents produce only 2 children, it's impossible to detect a 3:1 ratio. Even an extremely large family with 10 to 15 children would not permit the recognition of a dihybrid 9:3:3:1 ratio.

Although these special constraints make genetic studies of humans more complex, understanding human heredity is tremendously important. So geneticists have been forced to develop techniques that are uniquely suited to human biology and culture.

Concepts

Although the principles of heredity are the same in humans and other organisms, the study of human inheritance is constrained by the inability to control genetic crosses, the long generation time, and the small number of offspring.

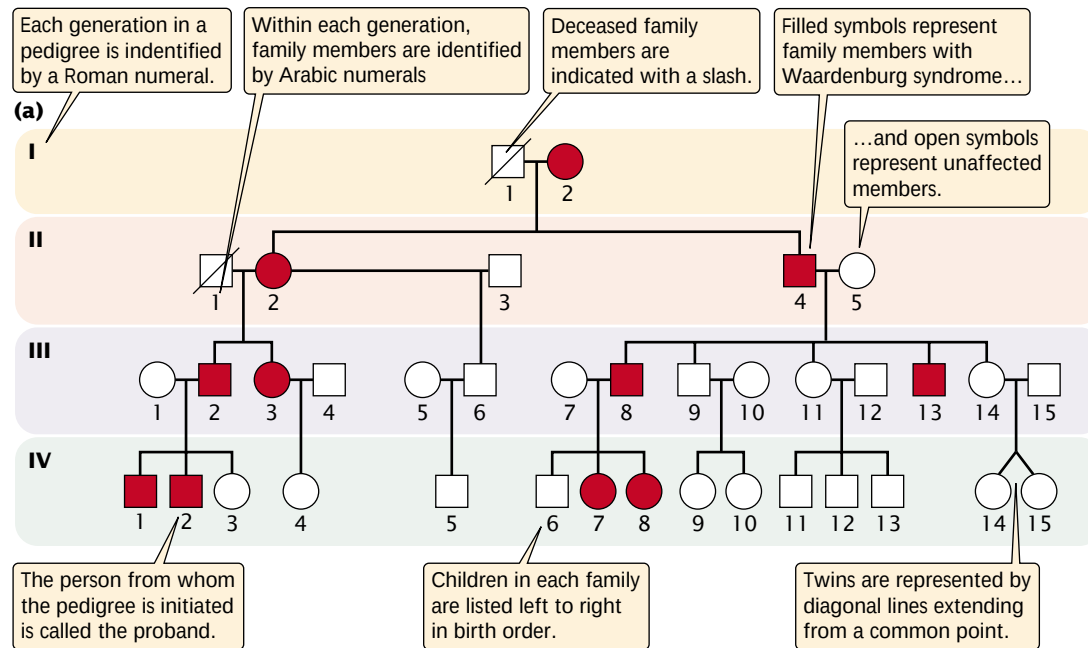
Analyzing Pedigrees

An important technique used by geneticists to study human inheritance is the pedigree. A **pedigree** is a pictorial representation of a family history, essentially a family tree that outlines the inheritance of one or more characteristics. The symbols commonly used in pedigrees are summarized in **FIGURE 6.2**. The pedigree shown in **FIGURE 6.3a** illustrates a family with Waardenburg syndrome, an autosomal dominant type of deafness that may be accompanied by fair skin, a white forelock, and visual problems (**FIGURE 6.3b**). Males in a pedigree are represented by squares, females by circles. A horizontal line drawn between two symbols representing a man and a woman indicates a mating; children are connected to their parents by vertical lines extending below the parents. Persons who exhibit the trait of interest are represented by filled circles and squares; in the pedigree of Figure 6.3a, the filled symbols represent members of the family who have Waardenburg syndrome. Unaffected persons are represented by open circles and squares.

Let's look closely at Figure 6.3 and consider some additional features of a pedigree. Each generation in a pedigree is identified by a Roman numeral; within each generation, family members are assigned Arabic numerals, and children in each family are listed in birth order from left to right. Person II-4, a man with Waardenburg syndrome, mated with II-5, an unaffected woman, and they produced five children. The oldest of their children is III-8, a male with Waardenburg syndrome, and the youngest is III-14, an unaffected female. Deceased family members are indicated by a slash through the circle or square, as shown for I-1 and II-1 in Figure 6.3a. Twins are represented by diagonal lines

	Male	Female	Sex unknown or unspecified
Unaffected individual			
Individual affected with trait			
Obligate carrier (carries the gene but does not have the trait)			
Asymptomatic carrier (unaffected at this time but may later exhibit trait)			
Multiple individuals (5)			
Deceased individual			
Proband (first affected family member coming to attention of geneticist)			
Family history of individual unknown			
Family—parents and three children: one boy and two girls in birth order			
Adoption (brackets enclose adopted individuals. Dashed line denotes adoptive parents; solid line denotes biological parent)			
Twins	Identical 	Nonidentical 	Unknown
Consanguinity (mating between related individuals)			

6.2 Standard symbols are used in pedigrees.



6.3 Waardenburg syndrome is an autosomal dominant disease characterized by deafness, fair skin, visual problems, and a white forelock. (Photograph courtesy of Guy Rowland).

extending from a common point (IV-14 and IV-15; non-identical twins).

When a particular characteristic or disease is observed in a person, a geneticist studies the family of this affected person and draws a pedigree. The person from whom the pedigree is initiated is called the **proband** and is usually designated by an arrow (IV-1 in Figure 6.3a).

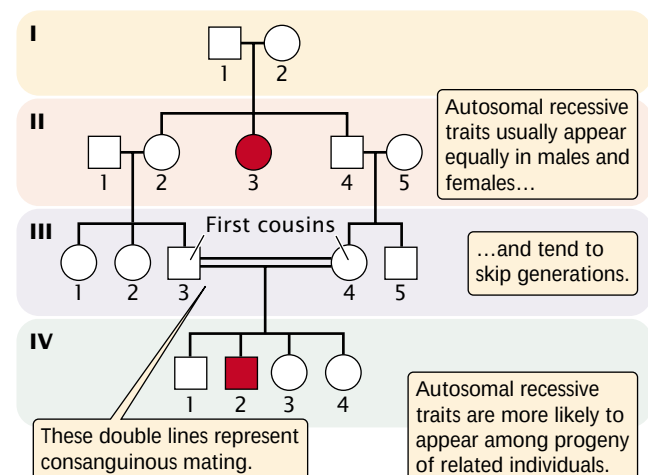
The limited number of offspring in most human families means that it is usually impossible to discern clear Mendelian ratios in a single pedigree. Pedigree analysis requires a certain amount of genetic sleuthing, based on recognizing patterns associated with different modes of inheritance. For example, autosomal dominant traits should appear with equal frequency in both sexes and should not skip generations, provided that the trait is fully penetrant (see p. 000 in Chapter 3) and not sex influenced (see p. 000 in Chapter 5).

Certain patterns may exclude the possibility of a particular mode of inheritance. For instance, a son inherits his X chromosome from his mother. If we observe that a trait is passed from father to son, we can exclude the possibility of X-linked inheritance. In the following sections, the traits discussed are assumed to be fully penetrant and rare.

Autosomal Recessive Traits

Autosomal recessive traits normally appear with equal frequency in both sexes (unless penetrance differs in males and females), and appear only when a person inherits two alleles for the trait, one from each parent. If the trait is uncommon, most parents carrying the allele are heterozygous and

unaffected; consequently, the trait appears to skip generations (FIGURE 6.4). Frequently, a recessive allele may be passed for a number of generations without the trait appearing in a pedigree. Whenever both parents are heterozygous, approximately $\frac{1}{4}$ of the offspring are expected to express the trait, but this ratio will not be obvious unless the family is large. In the rare event that both parents are affected by an autosomal recessive trait, all the offspring will be affected.



6.4 Autosomal recessive traits normally appear with equal frequency in both sexes and seem to skip generations.

When a recessive trait is rare, persons from outside the family are usually homozygous for the normal allele. Thus, when an affected person mates with someone outside the family ($aa \times AA$), usually none of the children will display the trait, although all will be carriers (i.e., heterozygous). A recessive trait is more likely to appear in a pedigree when two people within the same family mate, because there is a greater chance of both parents carrying the same recessive allele. Mating between closely related people is called **consanguinity**. In the pedigree shown in Figure 6.4, persons III-3 and III-4 are first cousins, and both are heterozygous for the recessive allele; when they mate, $\frac{1}{4}$ of their children are expected to have the recessive trait.

Concepts

Autosomal recessive traits appear with equal frequency in males and females. Affected children are commonly born to unaffected parents, and the trait tends to skip generations. Recessive traits appear more frequently among the offspring of consanguine matings.

A number of human metabolic diseases are inherited as autosomal recessive traits. One of them is Tay-Sachs disease. Children with Tay-Sachs disease appear healthy at birth but become listless and weak at about 6 months of age. Gradually, their physical and neurological conditions worsen, leading to blindness, deafness, and eventually death at 2 to 3 years of age. The disease results from the accumulation of a lipid called G_{M2} ganglioside in the brain. A normal component of brain cells, G_{M2} ganglioside is usually broken down by an enzyme called hexosaminidase A, but children with Tay-Sachs disease lack this enzyme. Excessive G_{M2} ganglioside accumulates in the brain, causing swelling and, ultimately, neurological symptoms. Heterozygotes have only

one normal copy of the hexosaminidase A allele and produce only about half the normal amount of the enzyme, but this amount is enough to ensure that G_{M2} ganglioside is broken down normally, and heterozygotes are usually healthy.

Autosomal Dominant Traits

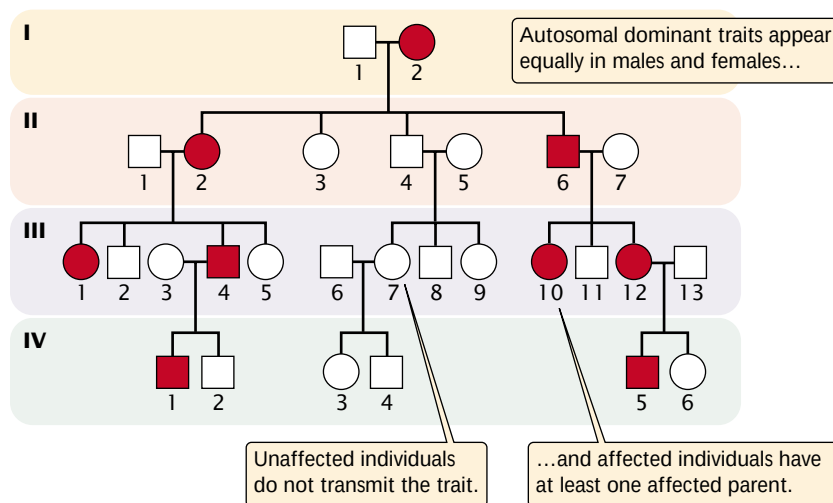
Autosomal dominant traits appear in both sexes with equal frequency, and both sexes are capable of transmitting these traits to their offspring. Every person with a dominant trait must inherit the allele from at least one parent; autosomal dominant traits therefore do not skip generations (FIGURE 6.5). Exceptions to this rule arise when people acquire the trait as a result of a new mutation or when the trait has reduced penetrance.

If an autosomal dominant allele is rare, most people displaying the trait are heterozygous. When one parent is affected and heterozygous and the other parent is unaffected, approximately $\frac{1}{2}$ of the offspring will be affected. If both parents have the trait and are heterozygous, approximately $\frac{3}{4}$ of the children will be affected. Provided the trait is fully penetrant, unaffected people do not transmit the trait to their descendants. In Figure 6.5, we see that none of the descendants of II-4 (who is unaffected) have the trait.

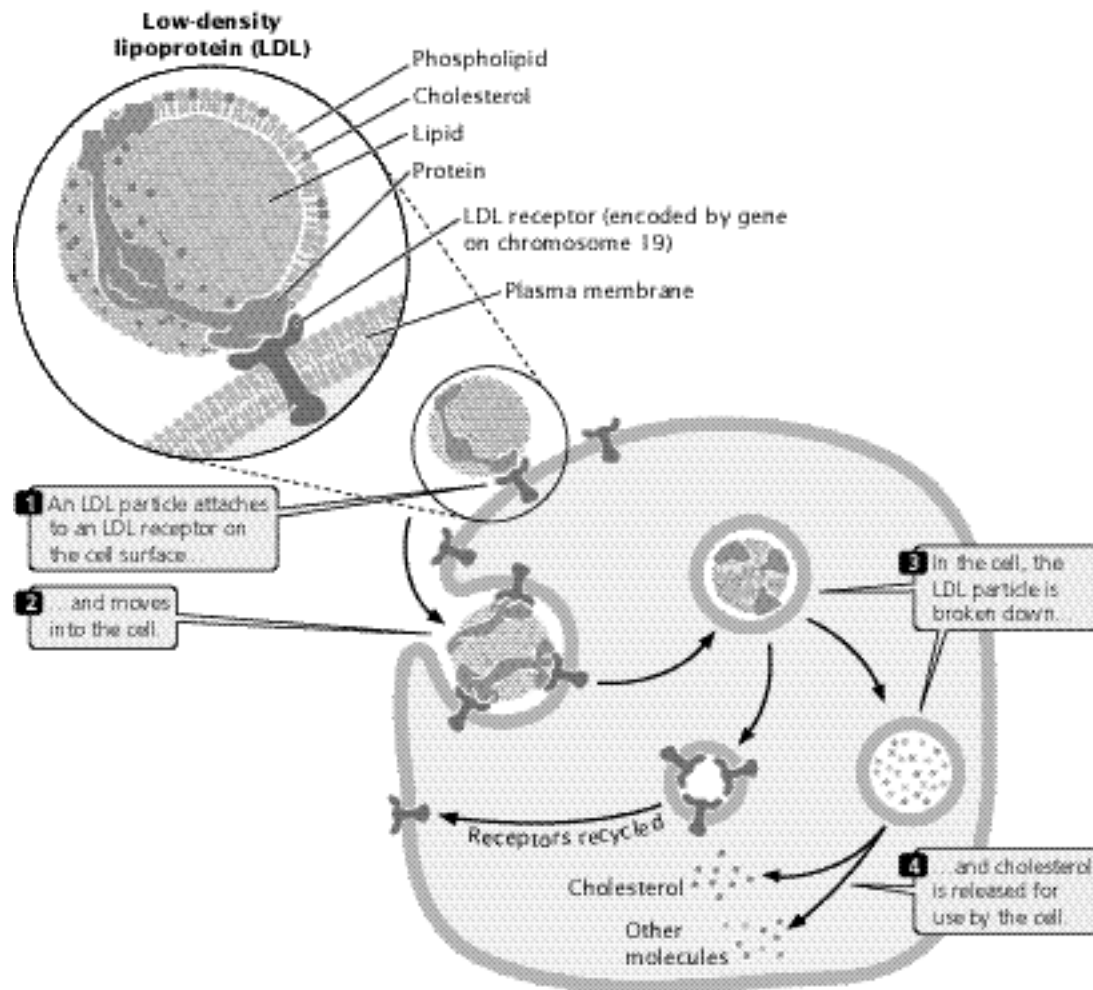
Concepts

Autosomal dominant traits appear in both sexes with equal frequency. Affected persons have an affected parent (unless they carry new mutations), and the trait does not skip generations. Unaffected persons do not transmit the trait.

One trait usually considered to be autosomal dominant is familial hypercholesterolemia, an inherited disease in which blood cholesterol is greatly elevated owing to a defect



6.5 Autosomal dominant traits normally appear with equal frequency in both sexes and do not skip generations.



6.6 Low-density lipoprotein (LDL) particles transport cholesterol.

The LDL receptor moves LDL through the cell membrane into the cytoplasm.

in cholesterol transport. Cholesterol is an essential component of cell membranes and is used in the synthesis of bile salts and several hormones. Most of our cholesterol is obtained through foods, primarily those high in saturated fats. Because cholesterol is a lipid (a nonpolar, or uncharged, compound), it is not readily soluble in the blood (a polar, or charged, solution). Cholesterol must therefore be transported throughout the body in small soluble particles called lipoproteins (FIGURE 6.6); a lipoprotein consists of a core of lipid surrounded by a shell of charged phospholipids and proteins that dissolve easily in blood. One of the principle lipoproteins in the transport of cholesterol is low-density lipoprotein (LDL). When an LDL molecule reaches a cell, it attaches to an LDL receptor, which then moves the LDL through the cell membrane into the cytoplasm, where it is broken down and its cholesterol is released for use by the cell.

Familial hypercholesterolemia is due to a defect in the gene (located on human chromosome 19) that normally

codes for the LDL receptor. The disease is usually considered an autosomal dominant disorder because heterozygotes are deficient in LDL receptors. In these people, too little cholesterol is removed from the blood, leading to elevated blood levels of cholesterol and increased risk of coronary artery disease. Persons heterozygous for familial hypercholesterolemia have blood LDL levels that are twice normal and usually have heart attacks by the age of 35. About 1 in 500 people is heterozygous for familial hypercholesterolemia and is predisposed to early coronary artery disease.

Very rarely, a person inherits two defective LDL receptor alleles. Such persons don't make *any* functional LDL receptors; their blood cholesterol levels are more than six times normal and they may suffer a heart attack as early as age 2 and almost inevitably by age 20. Because homozygotes are more severely affected than heterozygotes, familial hypercholesterolemia is said to be incompletely dominant. However, homozygotes are rarely seen (occurring with a frequency of only about 1 in 1 million people), and the

common heterozygous form of the disease appears as a simple dominant trait in most pedigrees.

X-Linked Recessive Traits

X-linked recessive traits have a distinctive pattern of inheritance (FIGURE 6.7). First, these traits appear more frequently in males, because males need inherit only a single copy of the allele to display the trait, whereas females must inherit two copies of the allele, one from each parent, to be affected. Second, because a male inherits his X chromosome from his mother, affected males are usually born to unaffected mothers who carry an allele for the trait. Because the trait is passed from unaffected female to affected male to unaffected female, it tends to skip generations (see Figure 6.7). When a woman is heterozygous, approximately $\frac{1}{2}$ of her sons will be affected and $\frac{1}{2}$ of her daughters will be unaffected carriers. For example, we know that females I-2, II-2, and III-7 in Figure 6.7 are all carriers because they transmit the trait to approximately half of their sons.

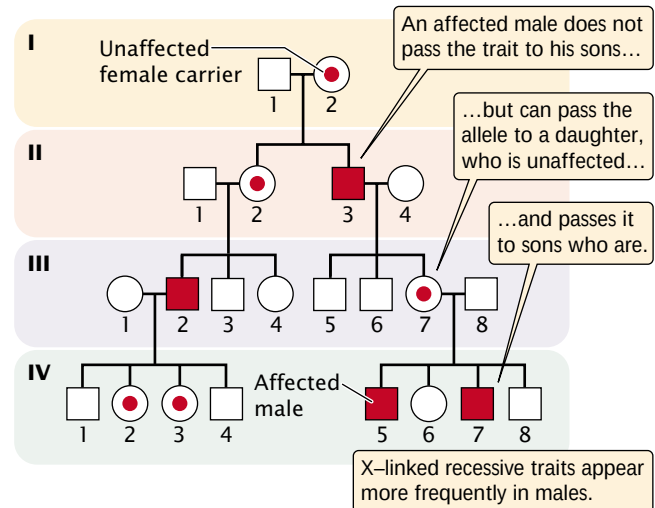
A third important characteristic of X-linked recessive traits is that they are not passed from father to son, because a son inherits his father's Y chromosome, not his X. In Figure 6.7, there is no case of a father and son who are both affected. All daughters of an affected man, however, will be carriers (if their mother is homozygous for the normal allele). When a woman displays an X-linked trait, she must be homozygous for the trait, and all of her sons will also display the trait.

Concepts

Rare X-linked recessive traits appear more often in males than in females and are not passed from father to son. Affected sons are usually born to unaffected mothers; thus X-linked recessive traits tend to skip generations.

An example of an X-linked recessive trait in humans is hemophilia A, also called classical hemophilia (FIGURE 6.8). This disease results from the absence of a protein necessary for blood to clot. The complex process of blood clotting consists of a cascade of reactions that includes more than 13 different factors. For this reason, there are several types of clotting disorders, each due to a glitch in a different step of the clotting pathway.

Hemophilia A results from abnormal or missing factor VIII, one of the proteins in the clotting cascade. The gene for factor VIII is located on the tip of the long arm of the X chromosome; so hemophilia A is an X-linked recessive disorder. People with hemophilia A bleed excessively; even small cuts and bruises can be life threatening. Spontaneous bleeding occurs in joints such as elbows, knees, and ankles, which produces pain, swelling, and erosion of



6.7 X-linked recessive traits appear more often in males and are not passed from father to son.

the bone. Fortunately, bleeding in people with hemophilia A can be now controlled by administering concentrated doses of factor VIII.

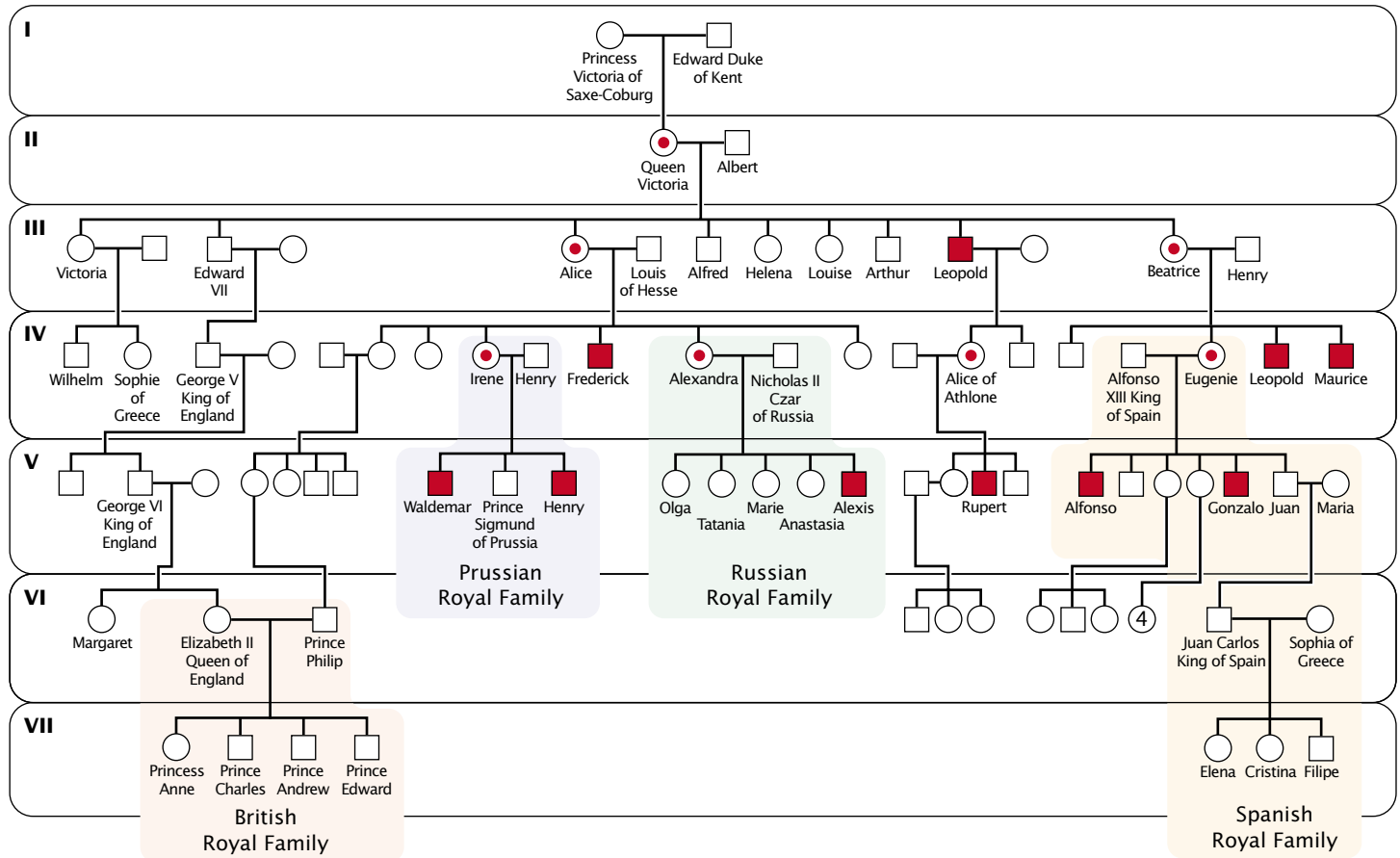
X-Linked Dominant Traits

X-linked dominant traits appear in males and females, although they often affect more females than males. As with X-linked recessive traits, a male inherits an X-linked dominant trait only from his mother—the trait is not passed from father to son. A female, on the other hand, inherits an X chromosome from both her mother and father; so females can receive an X-linked trait from either parent. Each child with an X-linked dominant trait must have an affected parent (unless the child possesses a new mutation or the trait has reduced penetrance). X-linked dominant traits do not skip generations (FIGURE 6.9); affected men pass the trait on to all their daughters and none of their sons, as is seen in the children of I-1 in Figure 6.9. In contrast, affected women (if heterozygous) pass the trait on to $\frac{1}{2}$ of their sons and $\frac{1}{2}$ of their daughters, as seen in the children of II-5 in the pedigree.

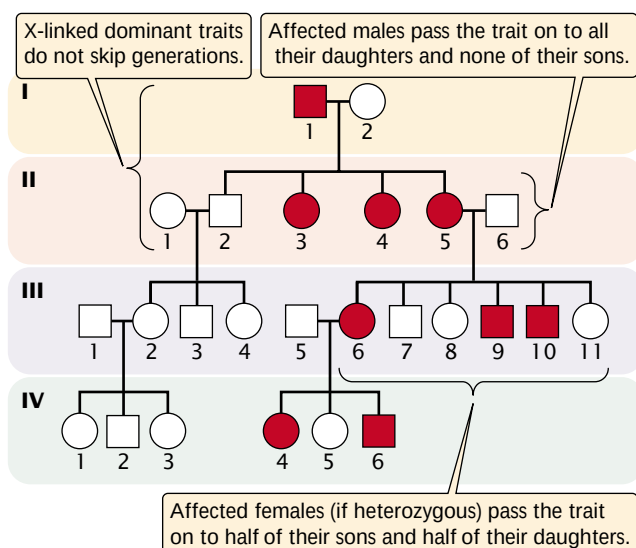
Concepts

X-linked dominant traits affect both males and females. Affected males must have affected mothers (unless they possess a new mutation), and they pass the trait on to all their daughters.

An example of an X-linked dominant trait in humans is hypophosphatemia, also called familial vitamin D-resistant rickets. People with this trait have features that superficially resemble those produced by rickets: bone deformi-



6.8 Classic hemophilia is inherited as an X-linked recessive trait. This pedigree is of hemophilia in the royal families of Europe.

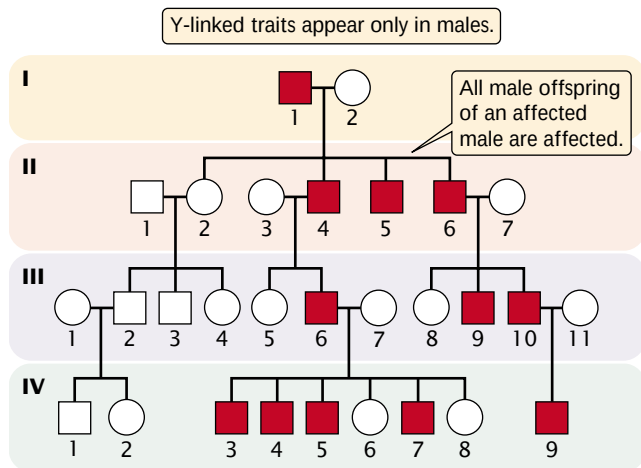


6.9 X-linked dominant traits affect both males and females. An affected male must have an affected mother.

ties, stiff spines and joints, bowed legs, and mild growth deficiencies. This disorder, however, is resistant to treatment with vitamin D, which normally cures rickets. X-linked hypophosphatemia results from the defective transport of phosphate, especially in cells of the kidneys. People with this disorder excrete large amounts of phosphate in their urine, resulting in low levels of phosphate in the blood and reduced deposition of minerals in the bone. As is common with X-linked dominant traits, males with hypophosphatemia are often more severely affected than females.

Y-Linked Traits

Y-linked traits exhibit a specific, easily recognized pattern of inheritance. Only males are affected, and the trait is passed from father to son. If a man is affected, all his male offspring should also be affected, as is the case for I-1, II-4, II-6, III-6, and III-10 of the pedigree in **FIGURE 6.10**. Y-linked traits do not skip generations. As discussed in Chapter 4, comparatively few genes reside on the human Y chromosome, and so few human traits are Y linked.



6.10 Y-linked traits appear only in males and are passed from a father to all his sons.

Concepts

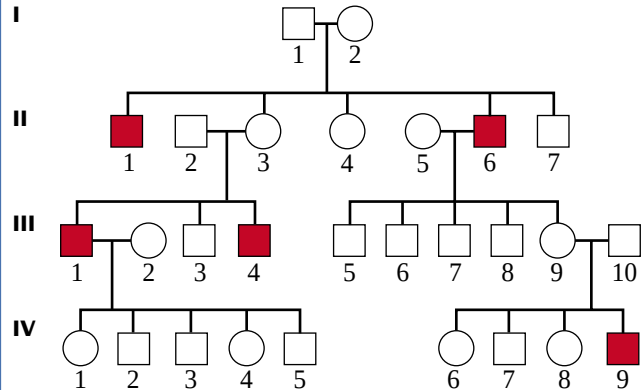
Y-linked traits appear only in males and are passed from a father to all his sons.

The major characteristics of autosomal recessive, autosomal dominant, X-linked recessive, X-linked dominant, and Y-linked traits are summarized in Table 6.1.

www.whfreeman.com/pierce The Online Mendelian Inheritance in Man, a comprehensive database of human genes and genetic disorders

Worked Problem

The following pedigree represents the inheritance of a rare disorder in an extended family. What is the most likely mode of inheritance for this disease? (Assume that the trait is fully penetrant.)



Solution

To answer this question, we should consider each mode of inheritance and determine which, if any, we can

Table 6.1 Pedigree characteristics of autosomal recessive, autosomal dominant, X-linked recessive, X-linked dominant, and Y-linked traits

Autosomal recessive trait

1. Appears in both sexes with equal frequency.
2. Trait tends to skip generations.
3. Affected offspring are usually born to unaffected parents.
4. When both parents are heterozygous, approximately 1/4 of the offspring will be affected.
5. Appears more frequently among the children of consanguine marriages.

Autosomal dominant trait

1. Appears in both sexes with equal frequency.
2. Both sexes transmit the trait to their offspring.
3. Does not skip generations.
4. Affected offspring must have an affected parent, unless they possess a new mutation.

5. When one parent is affected (heterozygous) and the other parent is unaffected, approximately 1/2 of the offspring will be affected.
6. Unaffected parents do not transmit the trait.

X-linked recessive trait

1. More males than females are affected.
2. Affected sons are usually born to unaffected mothers; thus, the trait skips generations.
3. A carrier (heterozygous) mother produces approximately 1/2 affected sons.
4. Is never passed from father to son.
5. All daughters of affected fathers are carriers.

X-linked dominant trait

1. Both males and females are affected; often more females than males are affected.
2. Does not skip generations. Affected sons must have an affected mother; affected daughters must have either an affected mother or an affected father.
3. Affected fathers will pass the trait on to all their daughters.
4. Affected mothers (if heterozygous) will pass the trait on to 1/2 of their sons and 1/2 of their daughters.

Y-linked trait

1. Only males are affected.
2. Is passed from father to all sons.
3. Does not skip generations.

eliminate. Because the trait appears only in males, autosomal dominant and autosomal recessive modes of inheritance are unlikely, because these occur equally in males and females. Additionally, autosomal dominance can be eliminated because some affected persons do not have an affected parent.

The trait is observed only among males in this pedigree, which might suggest Y-linked inheritance. However, with a Y-linked trait, affected men should pass the trait to all their sons, but here this is not the case; II-6 is an affected man who has four unaffected male offspring. We can eliminate Y-linked inheritance.

X-linked dominance can be eliminated because affected men should pass an X-linked dominant trait to all of their female offspring, and II-6 has an unaffected daughter (III-9).

X-linked recessive traits often appear more commonly in males, and affected males are usually born to unaffected female carriers; the pedigree shows this pattern of inheritance. With an X-linked trait, about half the sons of a heterozygous carrier mother should be affected. II-3 and III-9 are suspected carriers, and about $\frac{1}{2}$ of their male children (three of five) are affected. Another important characteristic of an X-linked recessive trait is that it is not passed from father-to-son. We observe no father-to-son transmission in this pedigree.

X-linked recessive is therefore the most likely mode of inheritance.

Twin Studies

Another method that geneticists use to analyze the genetics of human characteristics is twin studies. Twins come in two types: **dizygotic** (nonidentical) **twins** arise when two separate eggs are fertilized by two different sperm, producing genetically distinct zygotes; **monozygotic** (identical) **twins** result when a single egg, fertilized by a single sperm, splits early in development into two separate embryos.

Because monozygotic twins arise from a single egg and sperm (a single, “mono,” zygote), except for rare somatic mutations, they’re genetically identical, having 100% of their genes in common (FIGURE 6.11a). Dizygotic twins (FIGURE 6.11b), on the other hand, have on average only 50% of their genes in common (the same percentage that any pair of siblings has in common). Like other siblings, dizygotic twins may be of the same or different sexes. The only difference between dizygotic twins and other siblings is that dizygotic twins are the same age and shared a common uterine environment.

The frequency with which dizygotic twins are born varies among populations. Among North American Caucasians, about 7 dizygotic twin pairs are born per 1000 births but, among Japanese, the rate is only about 3 pairs per 1000 births; among Nigerians, about 40 dizygotic

(a)



(b)

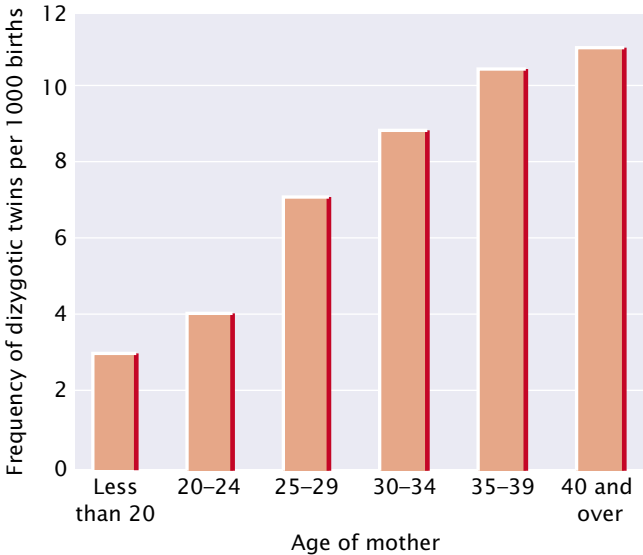


6.11 Monozygotic twins (a) are identical; dizygotic twins (b) are nonidentical. (Part a, Joe Carini/Index Stock Imagery/Picture Quest; Part b, Bruce Roberts/Photo Researchers.)

twin pairs are born per 1000 births. The rate of dizygotic twinning also varies with maternal age (FIGURE 6.12), and dizygotic twinning tends to run in families. In contrast, monozygotic twinning is relatively constant. The frequency of monozygotic twinning in most ethnic groups is about 4 twin pairs per 1000 births, and there is relatively little tendency for monozygotic twins to run in families.

Concepts

Dizygotic twins develop from two eggs fertilized by two separate sperm; they have, on average, 50% of their genes in common. Monozygotic twins develop from a single egg, fertilized by a single sperm, that splits into two embryos; they have 100% percent of their genes in common.



6.12 Older women tend to have more dizygotic twins than do younger women. Relation between the rate of dizygotic twinning and maternal age.

[Data from J. Yerushalmy and S. E. Sheeras, Human Biology 12:95–113, 1940.]

Concordance

Comparisons of dizygotic and monozygotic twins can be used to estimate the importance of genetic and environmental factors in producing differences in a characteristic. This is often done by calculating the concordance for a trait. If both members of a twin pair have a trait, the twins are said to be *concordant*; if only one member of the pair has the trait, the twins are said to be *discordant*. **Concordance** is the percentage of twin pairs that are concordant for a trait. Because identical twins have 100% of their genes in com-

mon and dizygotic twins have on average only 50% in common, genetically influenced traits should exhibit higher concordance in monozygotic twins. For instance, when one member of a monozygotic twin pair has asthma, the other twin of the pair has asthma about 48% of the time, so the monozygotic concordance for asthma is 48%. However, when a dizygotic twin has asthma, the other twin has asthma only 19% of the time (19% dizygotic concordance). The higher concordance in the monozygotic twins suggests that genes influence asthma, a finding supported by other family studies of this disease. Concordance values for several human traits and diseases are listed in Table 6.2.

The hallmark of a genetic influence on a particular characteristic is higher concordance in monozygotic twins compared with concordance in dizygotic twins. High concordance in monozygotic twins by itself does *not* signal a genetic influence. Twins normally share the same environment—they are raised in the same home, have the same friends, attend the same school—so high concordance may be due to common genes *or* to common environment. If the high concordance is due to environmental factors, then dizygotic twins, who also share the same environment, should have just as high a concordance as that of monozygotic twins. When genes influence the characteristic, however, monozygotic twin pairs should exhibit higher concordance than dizygotic twin pairs, because monozygotic twins have a greater percentage of genes in common. It is important to note that any discordance among monozygotic twins must be due to environmental factors, because monozygotic twins are genetically identical.

The use of twins in genetic research rests on the important assumption that, when there is greater concordance in monozygotic twins than in dizygotic twins, it is because monozygotic twins are more similar in their genes and not because they have experienced a more similar environment.

Table 6.2 Concordance of monozygotic and dizygotic twins for several traits			
Trait	Monozygotic Concordance (%)	Dizygotic Concordance (%)	Reference
Heart attack (males)	39	26	1
Heart attack (females)	44	14	1
Bronchial asthma	47	24	2
Cancer (all sites)	12	15	2
Epilepsy	59	19	2
Rheumatoid arthritis	32	6	3
Multiple sclerosis	28	5	4

References: (1) B. Havalld and M. Hauge, U.S. Public Health Service Publication 1103 (pp. 61–67), 1963. (2) B. Havalld and M. Hauge, *Genetics and the Epidemiology of Chronic Diseases*, U.S. Departement of Health, Education, and Welfare, 1965. (3) J. S. Lawrence, *Annals of Rheumatic Diseases* 26(1970):357–379. (4) G. C. Ebers et al, *American Journal of Human Genetics* 36(1984):495.

The degree of environmental similarity between monozygotic twins and dizygotic twins is assumed to be the same. This assumption may not always be correct, particularly for human behaviors. Because they look alike, identical twins may be treated more similarly by parents, teachers, and peers than are nonidentical twins. Evidence of this similar treatment is seen in the past tendency of parents to dress identical twins alike. In spite of this potential complication, twin studies have played a pivotal role in the study of human genetics.

Twin Studies and Obesity

To illustrate the use of twins in genetic research, let's consider a genetic study of obesity. Obesity is a serious public-health problem. About 50% of adults in affluent societies are overweight and from 15% to 25% are obese. Obesity increases the risk of a number of medical conditions, including diabetes, gallbladder disease, high blood pressure, some cancers, and heart disease. Obesity is clearly familial: when both parents are obese, 80% of their children will also become obese; when both parents are not overweight, only 15% of their children will eventually become obese. The familial nature of obesity could result from genes that influence body weight; alternatively, it could be entirely environmental, resulting from the fact that family members usually have similar diets and exercise habits.

A number of genetic studies have examined twins in an effort to untangle the genetic and environmental contributions to obesity. The largest twin study of obesity was conducted on more than 4000 pairs of twins taken from the National Academy of Sciences National Research Council twin registry. This registry is a database of almost 16,000 male twin pairs, born between 1917 and 1927, who served in the U.S. armed forces during World War II or the Korean War. Albert Stunkard and his colleagues obtained weight and height for each of the twins from medical records compiled at the time of their induction into the armed forces. Equivalent data were again collected in 1967, when the men were 40 to 50 years old. The researchers then computed how overweight each man was at induction and at middle age in 1967. Concordance values for monozygotic and dizygotic twins were then computed for several weight categories (Table 6.3).

In each weight category, monozygotic twins had significantly higher concordance than did dizygotic twins at induction and in middle age 25 years later. The researchers concluded that, among the group being studied, body weight appeared to be strongly influenced by genetic factors. Using statistics that are beyond the scope of this discussion, the researchers further concluded that genetics accounted for 77% of variation in body weight at induction and 84% at middle age in 1967. (Because a characteristic such as body weight changes in a lifetime, the effects of genes on the characteristic may vary with age.)

Table 6.3 Concordance values for body weight among monozygotic twins (MZ) and dizygotic twins (D) at induction in the armed services and at follow-up

Percent Overweight*	Concordance (%)			
	At Induction		At Follow-up in 1967	
	MZ	DZ	MZ	DZ
15	61	31	68	49
20	57	27	60	40
25	46	24	54	26
30	51	19	47	16
35	44	12	43	9
40	44	0	36	6

*Percent overweight was determined by comparing each man's actual weight with a standard recommended weight for his height.

Source: After A. J. Standard, T.T. Foch, and Z. Hrubec, A twin study of human obesity, *Journal of the American Medical Association* 256(1986):52.

This study shows that genes influence variation in body weight, yet genes *alone* do not cause obesity. In less affluent societies, obesity is rare, and no one can become overweight unless caloric intake exceeds energy expenditure. One does not inherit obesity; rather, one inherits a predisposition toward a particular body weight; geneticists say that some people are genetically more *at risk* for obesity than others.

How genes affect the risk of obesity is not yet completely understood. In 1994, scientists at Rockefeller University isolated a gene that causes an inherited form of obesity in mice (**FIGURE 6.13**). This gene encodes a protein called leptin, named after the Greek word for "thin." Leptin is produced by fat tissue and decreases appetite by affecting



6.13 Obesity in some mice is due to a defect in the gene that encodes the protein leptin. Obese mouse on the left compared with normal-sized mouse on the right. (Remi Banali/Liason.)

the hypothalamus, a part of the brain. A decrease in body fat leads to decreased leptin, which stimulates appetite; an increase in body fat leads to increased levels of leptin, which reduces appetite. Obese mice possess two mutated copies of the leptin gene and produce no functional leptin; giving leptin to these mice promotes weight loss.

The discovery of the leptin gene raised hopes that obesity in humans might be influenced by defects in the same gene and that the administration of leptin might be an effective treatment for obesity. Unfortunately, most overweight people are not deficient in leptin. Most, in fact, have elevated levels of leptin and appear to be somewhat resistant to its effects. Only a few rare cases of human obesity have been linked to genetic defects in leptin. The results of further studies have revealed that the genetic and hormonal control of body weight is quite complex; several other genes have been identified that also cause obesity in mice, and the molecular underpinnings of weight control are still being elucidated.

Concepts

Higher concordance in monozygotic twins compared with that in dizygotic twins indicates that genetic factors play a role in determining individual differences of a characteristic. Low concordance in monozygotic twins indicates that environmental factors play a significant role in the characteristic.

www.whfreeman.com/pierce

More advanced information on twin research in genetics

Adoption Studies

A third technique that geneticists use to analyze human inheritance is the study of adopted people. This approach is one of the most powerful for distinguishing the effects of genes and environment on characteristics.

For a variety of reasons, many children each year are separated from their biological parents soon after birth and adopted by adults with whom they have no genetic relationship. These adopted persons have no more genes in common with their adoptive parents than do two randomly chosen persons; however, they do share a common environment with their adoptive parents. In contrast, the adopted persons have 50% of the genes possessed by each of their biological parents but do not share the same environment with them. If adopted persons and their adoptive parents show similarities in a characteristic, these similarities can be attributed to environmental factors. If, on the other hand, adopted persons and their biological parents show similarities, these similarities are likely to be due to genetic factors.

Comparisons of adopted persons with their adoptive parents and with their biological parents can therefore help to define the roles of genetic and environmental factors in the determination of human variation.

Adoption studies assume that the environments of biological and adoptive families are independent (i.e., not more alike than would be expected by chance). This assumption may not always be correct, because adoption agencies carefully choose adoptive parents and may select a family that resembles the biological family. Offspring and their biological mother also share a common environment during prenatal development. Some of the similarity between adopted persons and their biological parents may be due to these similar environments and not due to common genetic factors.

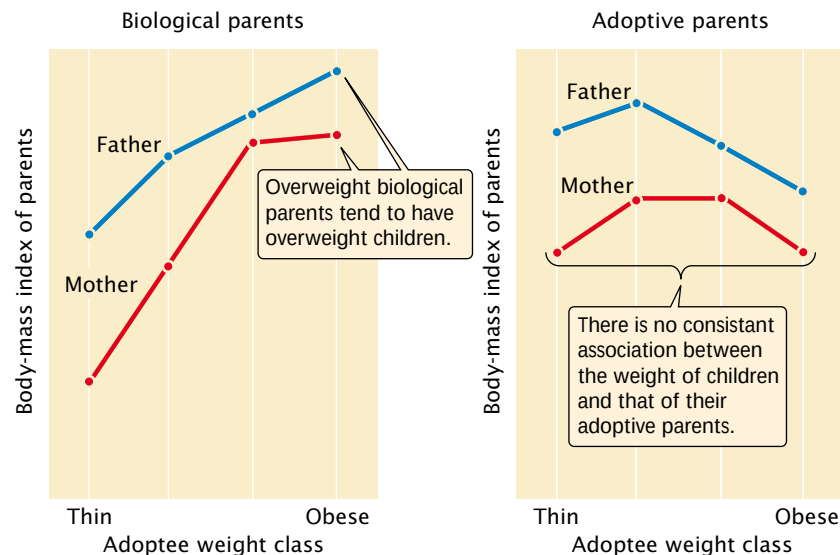
Concepts

Similarities between adopted persons and their genetically unrelated adoptive parents indicate that environmental factors affect the characteristic; similarities between adopted persons and their biological parents indicate that genetic factors influence the characteristic.

Adoption Studies and Obesity

Like twin studies, adoption studies have played an important role in demonstrating that obesity has a genetic influence. In 1986, geneticists published the results of a study of 540 people who had been adopted in Denmark between 1924 and 1947. The geneticists obtained information concerning the adult body weight and height of the adopted persons, along with the adult weight and height of their biological parents and their unrelated adoptive parents.

Geneticists used a measurement called the body-mass index to analyze the relation between the weight of the adopted persons and that of their parents. (The body-mass index, which is a measure of weight divided by height, provides a measure of weight that is independent of height.) On the basis of body-mass index, sex, and age, the adopted persons were divided into four weight classes: thin, median weight, overweight, and obese. A strong relation was found between the weight classification of the adopted persons and the body-mass index of their biological parents: obese adoptees tended to have heavier biological parents, whereas thin adoptees tended to have lighter biological parents (FIGURE 6.14). Because the only connection between the adoptees and their biological parents was the genes that they have in common, the investigators concluded that genetic factors influence adult body weight. There was no clear relation between the weight classification of adoptees and the body-mass index of their adoptive parents (see Figure 6.14), suggesting that the rearing environment has little effect on adult body weight.



6.14 Adoption studies demonstrate that obesity has a genetic influence. (Redrawn with permission of the *New England Journal of Medicine* 314:195.)

Adoption Studies and Alcoholism

Adoption studies have also been successfully used to assess the importance of genetic factors on alcoholism. Although frequently considered a moral weakness in the past, today alcoholism is more often treated as a disease or as a psychiatric condition. An estimated 10 million people in the United States are problem drinkers, and as many as 6 million are severely addicted to alcohol. Of the U.S. population, 11% are heavy drinkers and consume as much as 50% of all alcohol sold.

A large study of alcoholism was carried out on 1775 Swedish adoptees who had been separated from their mothers at an early age and raised by biologically unrelated adoptive parents. The results of this study, along with those of others, suggest that there are at least two distinct groups of alcoholics. Type I alcoholics include men and women who typically develop problems with alcohol after the age of 25 (usually in middle age). These alcoholics lose control of the ability to drink in moderation—they drink in binges—and tend to be nonaggressive during drinking bouts. Type II alcoholics consist largely of men who begin drinking before the age of 25 (often in adolescence); they actively seek out alcohol, but do not binge, and tend to be impulsive, thrill-seeking, and aggressive while drinking.

The Swedish adoption study also found that alcohol abuse among biological parents was associated with increased alcoholism in adopted persons. Type I alcoholism usually required both a genetic predisposition and exposure to a rearing environment in which alcohol was consumed. Type II alcoholism appeared to be highly hereditary; it developed primarily among males whose biological fathers also were Type II alcoholics, regardless of whether the adoptive parents

drank. A male adoptee whose biological father was a Type II alcoholic was nine times as likely to become an alcoholic as was an adoptee whose biological father was not an alcoholic.

The results of the Swedish adoption study have been corroborated by other investigations, suggesting that some people are genetically predisposed to alcoholism. However, alcoholism is a complex behavioral characteristic that is undoubtedly influenced by many factors. It would be wrong to conclude that alcoholism is strictly a genetic characteristic. Although some people may be genetically predisposed to alcohol abuse, no gene forces a person to drink, and no one becomes alcoholic without the presence of a specific environmental factor—namely, alcohol.

Genetic Counseling and Genetic Testing

Our knowledge of human genetic diseases and disorders has expanded rapidly in the past 20 years. Victor McKusick's *Mendelian Inheritance in Man* now lists more than 13,000 human genetic diseases, disorders, and traits that have a simple genetic basis. Research has provided a great deal of information about the inheritance, chromosomal location, biochemical basis, and symptoms of many of these genetic traits. This information is often useful to people who have a genetic condition.

Genetic Counseling

Genetic counseling is a new field that provides information to patients and others who are concerned about hereditary conditions. It is also an educational process that helps patients and family members deal with many aspects of a

genetic condition. Genetic counseling often includes interpreting a diagnosis of the condition; providing information about symptoms, treatment, and prognosis; helping the patient and family understand the mode of inheritance; and calculating probabilities that family members might transmit the condition to future generations. Good genetic counseling also provides information about the reproductive options that are available to those at risk for the disease. Finally, genetic counseling tries to help the patient and family cope with the psychological and physical stress that may be associated with their disorder. Clearly, all of these considerations cannot be handled by a single person; so most genetic counseling is done by a team that can include counselors, physicians, medical geneticists, and laboratory personnel. Table 6.4 lists some common reasons for seeking genetic counseling.

Genetic counseling usually begins with a diagnosis of the condition. On the bases of a physical examination, biochemical tests, chromosome analysis, family history, and other information, a physician determines the cause of the condition. An accurate diagnosis is critical, because treatment and the probability of passing on the condition may vary, depending on the diagnosis. For example, there are a number of different types of dwarfism, which may be caused by chromosome abnormalities, single-gene mutations, hormonal imbalances, or environmental factors. People who have dwarfism resulting from an autosomal dominant gene have a 50% chance of passing the condition to their children, whereas people with dwarfism caused by a rare recessive gene have a low likelihood of passing the trait to their children.

When the nature of the condition is known, a genetic counselor sits down with the patient and other family members and explains the diagnosis. A family pedigree may be constructed, and the probability of transmitting the condition to future generations can be calculated for different

family members. The counselor helps the family interpret the genetic risks and explains various reproductive options that are available, including prenatal diagnosis, artificial insemination, and in vitro fertilization. A family's decision about future pregnancies frequently depends on the magnitude of the genetic risk, the severity and effects of the condition, the importance of having children, and religious and cultural views. The genetic counselor helps the family sort through these factors and facilitates their decision making. Throughout the process, a good genetic counselor uses *nondirected* counseling, which means that he or she provides information and facilitates discussion but does not bring his or her own opinion and values into the discussion. The goal of nondirected counseling is for the family to reach its own decision on the basis of the best available information.

Genetic conditions are often perceived differently from other diseases and medical problems, because genetic conditions are intrinsic to the individual person and can be passed on to children. Such perceptions may produce feelings of guilt about past reproductive choices and intense personal dilemmas about future choices. Genetic counselors are trained to help patients and family members recognize and cope with these feelings.

Concepts



Genetic counseling is an educational process that provides patients and their families with information about a genetic condition, its medical implications, the probabilities that other family members may have the disease, and reproductive options. It also helps patients and their families cope with the psychological and physical stress associated with a genetic condition.

Table 6.3 Common reasons for seeking genetic counseling

1. A person knows of a genetic disease in the family.
2. A couple has given birth to a child with a genetic disease, birth defect, or chromosomal abnormality.
3. A couple has a child who is mentally retarded or a close relative is mentally retarded.
4. An older woman becomes pregnant or wants to become pregnant. There is disagreement about the age at which a prospective mother who has no other risk factor should seek genetic counseling; many experts suggest that any prospective mother age 35 or older should seek genetic counseling.
5. Husband and wife are closely related (e.g., first cousins).
6. A couple experiences difficulties achieving a successful pregnancy.
7. A pregnant woman is concerned about exposure to an environmental substance (drug, chemical, or virus) that causes birth defects.
8. A couple needs assistance in interpreting the results of a prenatal or other test.
9. Both parents are known carriers for a regressive genetic disease.

www.whfreeman.com/pierce

Information on genetic counseling and human genetic diseases, as well as a list of genetic counseling training programs accredited by the American Board of Genetic Counseling

Genetic Testing

Improvements in our understanding of human heredity and the identification of numerous disease-causing genes have led to the development of hundreds of tests for genetic conditions. The ultimate goal of genetic testing is to recognize the potential for a genetic condition at an early stage. In some cases, genetic testing allows early intervention that may lessen or even prevent the development of the condition. In other cases, genetic testing allows people to make informed choices about reproduction. For those who know that they are at risk for a genetic condition, genetic testing may help alleviate anxiety associated with the uncertainty of their situation. Genetic testing includes newborn screening, heterozygote screening, presymptomatic diagnosis, and prenatal testing.

Newborn screening Testing for genetic disorders in newborn infants is called **newborn screening**. Most states in the United States and many other countries require that newborn infants be tested for phenylketonuria and galactosemia. These metabolic diseases are caused by autosomal recessive alleles; if not treated at an early age, they can result in mental retardation, but early intervention—through the administration of a modified diet—prevents retardation (see p. 000 in Chapter 5). Testing is done by analyzing a drop of the infant's blood collected soon after birth. Because of widespread screening, the frequency of mental retardation due to these genetic conditions has dropped tremendously. Screening newborns for additional genetic diseases that benefit from treatment, such as sickle-cell anemia and hypothyroidism, also is common.

Heterozygote screening Testing members of a population to identify heterozygous carriers of recessive disease-causing alleles, who are healthy but have the potential to produce children with the particular disease, is termed **heterozygote screening**.

Testing for Tay-Sachs disease is a successful example of heterozygote screening. In the general population of North America, the frequency of Tay-Sachs disease is only about 1 person in 360,000. Among Ashkenazi Jews (descendants of Jewish people who settled in eastern and central Europe), the frequency is 100 times as great. A simple blood test is used to detect Ashkenazi Jews who carry the Tay-Sachs allele. If a man and woman are both heterozygotes, approximately one in four of their children is expected to have Tay-Sachs disease. A prenatal test for the Tay Sachs allele also is available. Screening programs have led to a significant decline in the number of children of Ashkenazi ancestry born with Tay-Sachs disease (now fewer than 10 children per year in the United States).

Presymptomatic testing Evaluating healthy people to determine whether they have inherited a disease-causing allele gene is known as **presymptomatic genetic testing**. For example, presymptomatic testing is available for members of families that have an autosomal dominant form of breast cancer. In this case, early identification of the disease-causing allele allows for closer surveillance and the early detection of tumors. Presymptomatic testing is also available for some genetic diseases for which no treatment is available, such as Huntington disease, an autosomal dominant disease that leads to slow physical and mental deterioration in middle age (see introduction to Chapter 5). Presymptomatic testing for untreatable conditions raises a number of social and ethical questions (Chapter 18).

Several hundred genetic diseases and disorders can now be diagnosed prenatally. The major purpose of prenatal tests is to provide families with the information that they need to make choices during pregnancies and, in some cases, to prepare for the birth of a child with a genetic condition. A number of approaches to prenatal diagnosis are described in the following sections.

Ultrasonography Some genetic conditions can be detected through direct visualization of the fetus. Such visualization is most commonly done with **ultrasonography**—usually referred to as ultrasound. In this technique, high-frequency sound is beamed into the uterus; when the sound waves encounter dense tissue, they bounce back and are transformed into a picture (FIGURE 6.15). The size of the fetus can be determined, as can genetic conditions such as neural tube defects (defects in the development of the spinal column and the skull) and skeletal abnormalities.

Amniocentesis Most prenatal testing requires fetal tissue, which can be obtained in several ways. The most widely used method is **amniocentesis**, a procedure for obtaining a



6.15 Ultrasonography can be used to detect some genetic disorders in a fetus and locate the fetus during amniocentesis and chorionic villus sampling. (SIU School of Medicine/Photo Research.)

A couple are seeking help at a clinic that offers preimplantation genetic diagnosis (PGD), which combines in vitro fertilization with molecular analysis of the DNA from a single cell of the developing embryo, and permits the selection and transfer to the uterus of embryos free of a genetic disease. Before PGD, the only alternative for those wishing to prevent the birth of a child with a serious genetic disorder was early chorionic villus sampling or amniocentesis, followed by abortion if the fetus had a disorder.

Consider a couple at risk of having a second child with severe combined immune deficiency (SCID). A child born with this condition has a seriously impaired immune system. As recently as 20 years ago, those affected died early in life, but the use of bone-marrow transplantation, which can provide the child with a supply of healthy blood stem cells, has greatly extended survival. In general, the earlier the transplantation and the closer the tissue match of the marrow donor, the better a recipient child's chances.

The couple tell the medical geneticist that they are seeking his help in identifying and transferring only embryos free of the SCID mutation so that they can begin their pregnancy knowing that it will be healthy. Some weeks later they reveal another reason for their interest in this technology: the health of their six-year-old daughter, who is affected with SCID, is on a downward course despite one partly matched bone-marrow transplant earlier in her life. Their child's best hope of survival is another bone-marrow transplant, using tissue from a compatible donor, preferably a sibling. Is it possible, they ask, to test the healthy embryos for tissue compatibility and transfer only those that match their daughter's type?

The geneticist responds that it is indeed technically possible to do so but he wonders whether helping the couple in this way is ethically appro-

priate. Is it right to conceive a child for this purpose? In addition, because tissue compatibility is not a disease, would responding to this request constitute an unwise step into a world of positive, or "enhancement," genetics, where parents' desires, not medical judgment, dictate the use of genetic knowledge?

PGD offers significant new reproductive opportunities for families or persons affected by genetic disease. However, the very power of this technology raises new ethical issues that will grow in importance as PGD and related embryo manipulation procedures become more widely available. PGD offers a technology that is medically, psychologically and, in the view of many, morally superior to the existing use of abortion for genetic selection.

The case described here is not entirely novel. Even before the advent of PGD, couples who had sought to insure the birth of a child whose HLA (human leukocyte antigen) status could be compatible with that of an existing sibling would establish a pregnancy, undergo testing, and then abort all fetuses that did not have the appropriate HLA type. Because a woman in the United States has a right to abortion for any reason through the second trimester, this option is legal. However, it is certainly not a desirable one from a medical or psychological point of view.

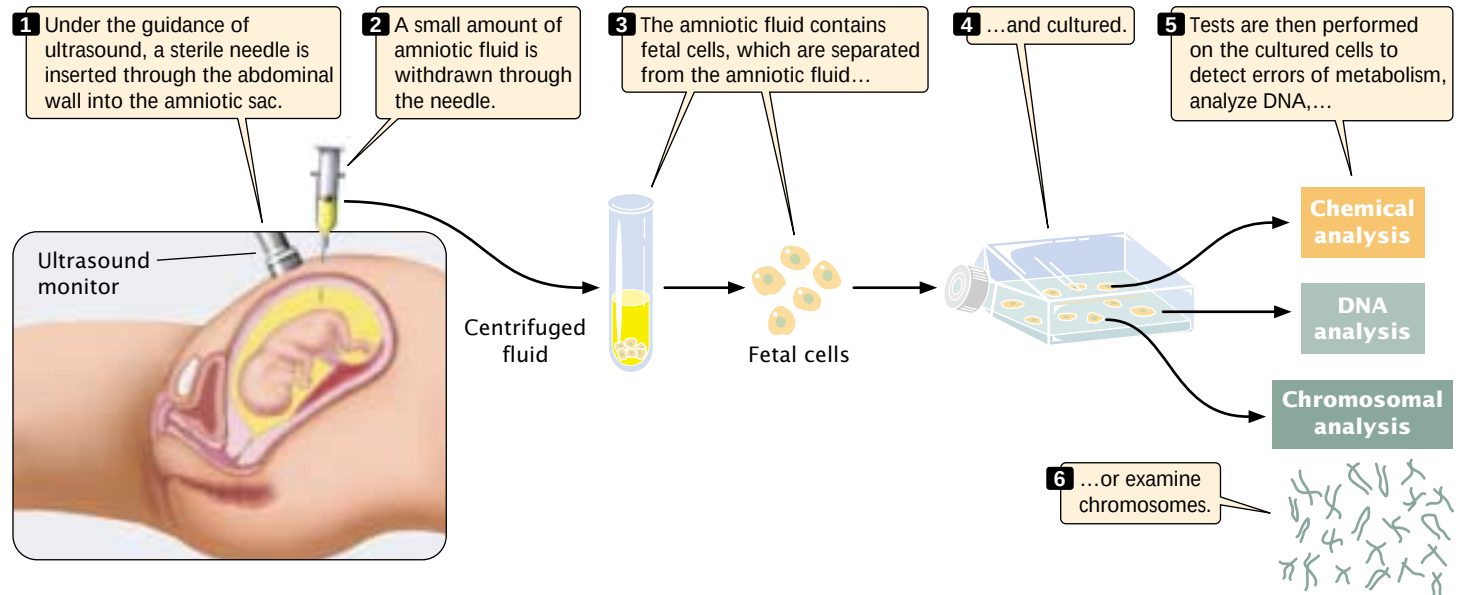
Because pregnancy is never begun, PGD avoids the emotional trauma of abortion. Although some would object to the discarding of human embryos even at this early stage, the selection of viable embryos and the discarding of others is a routine part of in vitro fertilization procedures today and raises few moral questions in the minds of most people. So, from the narrow perspective of parental decision making, the alternative described in this case is a significant medical and moral step

forward. We should not lose sight of this fact as we consider other ethically troubling aspects of the case.

This case of parental selection raises at least two distinct questions. First, even if the means of selection is relatively innocuous from a moral standpoint, is it appropriate for parents to bring a child into being at least partly for the purpose of saving the life of a sibling? A second question is whether genetic professionals should cooperate with a selection process that entails a nondisease trait.

With regard to the first question, some people believe that the parents' wishes in this situation violate the ethical principle not to use a person merely as a means to an end, as well as the modern principle of responsible parenthood, which judges each child to be of inestimable value. They also worry about future psychological harm to a child conceived in this way. Others argue that children are usually born for a specific purpose, whether it's to gratify the parents' need for a family, to cement the relationship of a couple, or whatever else. As a result, they argue that the important question is not whether the purpose for which the child was conceived is ethical but whether the parents will be able to accept the child in its own unique identity once it is born.

In response to the second question, some ethicists see any involvement in nondisease testing as a dangerous diversion of genetic testing down paths long since rejected for good reason. They see that a consensus has emerged in popular opinion and among geneticists and ethics advisory boards that nondisease characteristics should not be subject to prenatal testing. HLA testing, however well intentioned, runs counter to this consensus. Departures from these views about genetic tests could raise very difficult questions about eugenics in the future.



6.16 Amniocentesis is a procedure for obtaining fetal cells for genetic testing.

sample of amniotic fluid from a pregnant woman (FIGURE 6.16). Amniotic fluid—the substance that fills the amniotic sac and surrounds the developing fetus—contains fetal cells that can be used for genetic testing.

Amniocentesis is routinely performed as an outpatient procedure with the use of a local or no anesthetic. First, ultrasonography is used to locate the position of the fetus in the uterus. Next, a long, sterile needle is inserted through the abdominal wall into the amniotic sac (see Figure 6.16), and a small amount of amniotic fluid is withdrawn through the needle. Fetal cells are separated from the amniotic fluid and placed in a culture medium that stimulates them to grow and divide. Genetic tests are then performed on the cultured cells. Complications with amniocentesis (mostly miscarriage) are rare, arising in only about 1 in 400 procedures.

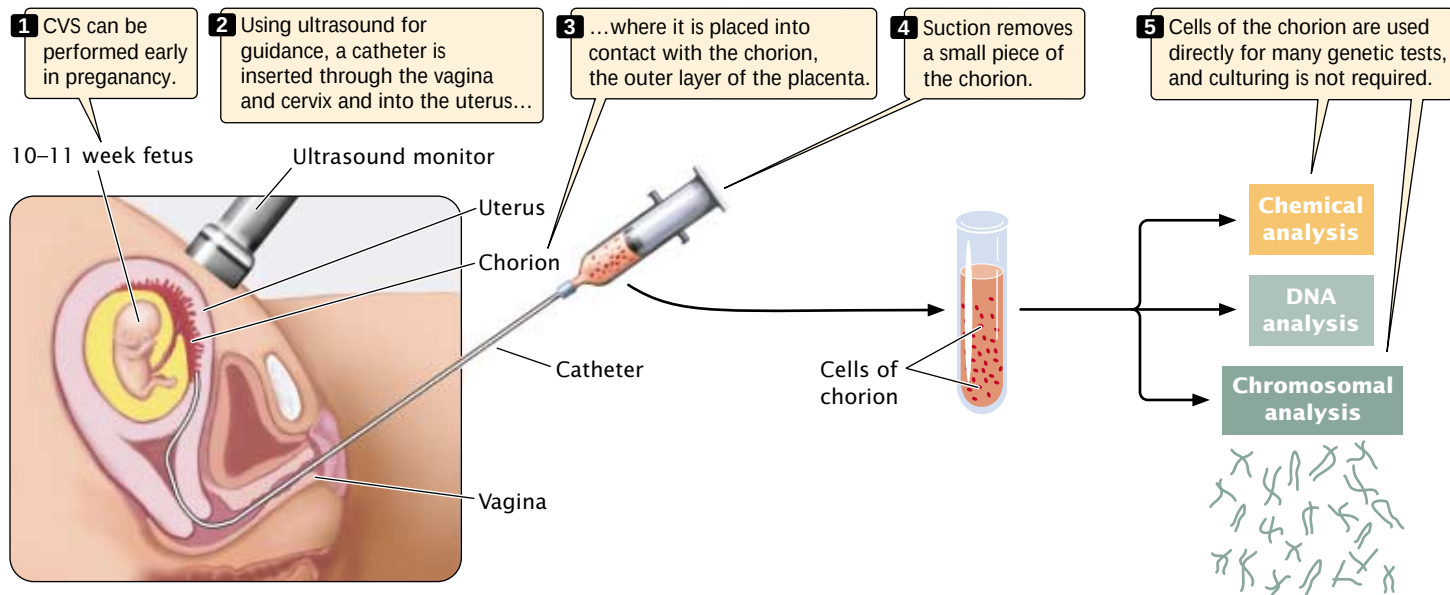
Chorionic villus sampling A major disadvantage with amniocentesis is that it is routinely performed in about the 16th week of a pregnancy, (although many obstetricians now successfully perform amniocentesis several weeks earlier). The cells obtained with amniocentesis must then be cultured before genetic tests can be performed, requiring yet more time. For these reasons, genetic information about the fetus may not be available until the 17th or 18th week of pregnancy. By this stage, abortion carries a risk of complications and may be stressful for the parents. **Chorionic villus sampling (CVS)** can be performed earlier (between the 10th and 11th weeks of pregnancy) and collects more fetal tissue, which eliminates the necessity of culturing the cells.

In CVS, a catheter—a soft plastic tube—is inserted into the vagina (FIGURE 6.17) and, with the use of ultrasound for guidance, is pushed through the cervix into the

uterus. The tip of the tube is placed into contact with the chorion, the outer layer of the placenta. Suction is then applied, and a small piece of the chorion is removed. Although the chorion is composed of fetal cells, it is a part of the placenta that is expelled from the uterus after birth; so the removal of a small sample does not endanger the fetus. The tissue that is removed contains millions of actively dividing cells that can be used directly in many genetic tests. Chorionic villus sampling has a somewhat higher risk of complication than that of amniocentesis; the results of several studies suggest that this procedure may increase the incidence of limb defects in the fetus when performed earlier than 10 weeks of gestation.

Fetal cells obtained by amniocentesis or by CVS can be used to prepare a **karyotype**, which is a picture of a complete set of metaphase chromosomes. Karyotypes can be studied for chromosome abnormalities (Chapter 9). Biochemical analyses can be conducted on fetal cells to determine the presence of particular metabolic products of genes. For genetic diseases in which the DNA sequence of the causative gene has been determined, the DNA sequence (DNA testing; Chapter 18) can be examined for defective alleles.

Maternal blood tests Some genetic conditions can be detected by performing a blood test on the mother (**maternal blood testing**). For instance, α -fetoprotein is normally produced by the fetus during development and is present in the fetal blood, the amniotic fluid, and the mother's blood during pregnancy. The level of α -fetoprotein is significantly higher than normal when the fetus has a neural-tube or one of several other disorders. Some chromosome



6.17 Chorionic villus sampling is another procedure for obtaining fetal cells for genetic testing.

abnormalities produce *lower-than-normal* levels of α -fetoprotein. Measuring the amount of α -fetoprotein in the mother's blood gives an indication of these conditions. However, because other factors affect the amount of α -fetoprotein in maternal blood, a high or low level by itself does not necessarily indicate a problem. Thus, when a blood test indicates that the amount of α -fetoprotein is abnormal, follow-up tests (additional α -fetoprotein determinations, ultrasound, amniocentesis, or all three) are usually performed.

Fetal cell sorting Prenatal tests that utilize only maternal blood are highly desirable because they are noninvasive and pose no risk to the fetus. During pregnancy, a few fetal cells are released into the mother's circulatory system, where they mix and circulate with her blood. Recent advances have made it possible to separate fetal cells from a maternal blood sample (a procedure called **fetal cell sorting**). With the use of lasers and automated cell-sorting machines, fetal cells can be detected and separated from maternal blood cells. The fetal cells obtained can be cultured for chromosome analysis or used as a source of fetal DNA for molecular testing (see p. 000 in Chapter 18).

A large number of genetic diseases can now be detected prenatally (Table 6.5), and the number is growing rapidly as new disease-causing genes are isolated. The Human Genome Project (Chapter 18) has accelerated the rate at which new genes are being isolated and new genetic tests are being developed. In spite of these advances, prenatal tests are still not available for many common genetic

diseases, and no test can guarantee that a "perfect" child will be born.

Preimplantation genetic diagnosis Prenatal genetic tests provide today's couples with increasing amounts of information about the health of their future children. New reproductive technologies also provide couples with options for using this information. One of these technologies is in vitro fertilization. In this procedure, hormones are used to induce ovulation. The ovulated eggs are surgically removed from the surface of the ovary, placed in a laboratory dish, and fertilized with sperm. The resulting embryo is then implanted into the uterus. Thousands of babies resulting from in vitro fertilization have now been born.

Genetic testing can be combined with in vitro fertilization to allow implantation of embryos that are free of a specific genetic defect. Called **preimplantation genetic diagnosis, (PGD)**, this technique allows people who carry a genetic defect to avoid producing a child with the disorder. For example, when a woman is a carrier of an X-linked recessive disease, approximately half of her sons are expected to have the disease. Through in vitro fertilization and preimplantation testing, it is possible to select an embryo without the disorder for implantation in her uterus.

The procedure begins with the production of several single-celled embryos through in vitro fertilization. The embryos are allowed to divide several times until they reach the 8 or 16-cell stage. At this point, one cell is removed from each embryo and tested for the genetic abnormality. Removing a single cell at this early stage does not harm the embryo. After determination of which embryos are free of

Table 6.5 Examples of genetic diseases and disorders that can be detected prenatally and the techniques used in their detection

Disorder	Method of Detection
Chromosome abnormalities	Examination of a karyotype from cells obtained by amniocentesis or CVS
Cleft lip and palate	Ultrasound
Cystic fibrosis	DNA analysis of cells obtained by amniocentesis or CVS
Dwarfism	Ultrasound or X-ray; some forms can be detected by DNA analysis of cells obtained by amniocentesis or CVS
Hemophilia	Fetal blood sampling* or DNA analysis of cells obtained by amniocentesis or CVS
Lesch-Nyhan syndrome (deficiency of purine metabolism leading to spasms, seizures, and compulsory self-mutilation)	Biochemical tests on cells obtained by amniocentesis or CVS
Neural-tube defects	Initial screening with maternal blood test, followed by biochemical tests on amniotic fluid obtained by amniocentesis and ultrasound
Osteogenesis imperfecta (brittle bones)	Ultrasound or X-ray
Phenylketonuria	DNA analysis of cells obtained by amniocentesis or CVS
Sickle-cell anemia	Fetal blood sampling or DNA analysis of cells obtained by amniocentesis or CVS
Tay-Sachs disease	Biochemical tests on cells obtained by amniocentesis or CVS

*A sample of fetal blood is obtained by inserting needle into the umbilical cord.

the disorder, a healthy embryo is selected and implanted in the woman's uterus.

Preimplantation genetic diagnosis requires the ability to conduct a genetic test on a single cell. Such testing is possible with the use of the polymerase chain reaction through which minute quantities of DNA can be amplified (replicated) quickly (Chapter 18). After amplification of the cell's DNA, the DNA sequence is examined. Preimplantation diagnosis is still experimental and is available at only a few research centers. Its use raises a number of ethical concerns, because it provides a means of actively selecting for or against certain genetic traits.

Concepts

Genetic testing is used to screen newborns for genetic diseases, detect persons who are heterozygous for recessive diseases, detect disease-causing alleles in those who have not yet developed symptoms of the disease, and detect defective alleles in unborn babies. Preimplantation genetic diagnosis combined with in vitro fertilization allows for selection of embryos that are free from specific genetic diseases.

www.whfreeman.com/pierce

Additional information about genetic testing

Connecting Concepts Across Chapters

This chapter builds on the basic principles of heredity that were introduced in Chapters 1 through 5, extending them to human genetic characteristics. A dominant theme of the chapter is that human inheritance is not fundamentally different from inheritance in other organisms, but the unique biological and cultural characteristics of humans require special techniques for the study of human characteristics.

Several topics introduced in this chapter are explored further in later chapters. Molecular techniques used in genetic testing and some of the ethical implications of modern genetic testing are presented in Chapter 18. Chromosome mutations and karyotypes are studied in Chapter 9. In Chapter 22, we examine additional techniques for separating genetic and environmental contributions to characteristics in humans and other organisms.

CONCEPTS SUMMARY

- There are several difficulties in applying traditional genetic techniques to the study of human traits, including the inability to conduct controlled crosses, long generation time, small family size, and the difficulty of separating genetic and environmental influences.
- A pedigree is a pictorial representation of a family history that displays the inheritance of one or more traits through several generations.
- Autosomal recessive traits typically appear with equal frequency in both sexes. If a trait is uncommon, the parents of a child with an autosomal recessive trait are usually heterozygous and unaffected; so the trait tends to skip generations. When both parents are heterozygous, approximately $\frac{1}{4}$ of their offspring will have the trait. Recessive traits are more likely to appear in families with consanguinity (mating between closely related persons).
- Autosomal dominant traits usually appear equally in both sexes and do not skip generations. When one parent is affected and heterozygous, approximately $\frac{1}{2}$ of the offspring will have the trait. When both parents are affected and heterozygous, approximately $\frac{3}{4}$ of the offspring will be affected. Unaffected people do not normally transmit an autosomal dominant trait to their offspring.
- X-linked recessive traits appear more frequently in males than in females. Affected males are usually born to females who are unaffected carriers. When a woman is a heterozygous carrier and a man is unaffected, approximately $\frac{1}{2}$ of their sons will have the trait and $\frac{1}{2}$ of their daughters will be unaffected carriers. X-linked traits are not passed from father to son.
- X-linked dominant traits appear in males and females, but more frequently in females. They do not skip generations. Affected men pass an X-linked dominant trait to all of their daughters but none of their sons. Heterozygous women pass the trait to $\frac{1}{2}$ of their sons and $\frac{1}{2}$ of their daughters.
- Y-linked traits appear only in males and are passed from father to all sons.
- Analysis of twins is an important technique for the study of human genetic characteristics. Dizygotic twins arise from two separate eggs fertilized by two separate sperm; monozygotic twins arise from a single egg, fertilized by a single sperm, that splits into two separate embryos early in development.
- Concordance is the percentage of twin pairs in which both members of the pair express a trait. Higher concordance in monozygotic than in dizygotic twins indicates a genetic influence on the trait; less than 100% concordance in monozygotic twins indicates environmental influences on the trait.
- Adoption studies are used to analyze the inheritance of human characteristics. Similarities between adopted children and their biological parents indicate the importance of genetic factors in the expression of a trait; similarities between adopted children and their genetically unrelated adoptive parents indicate the influence of environmental factors.
- Genetic counseling provides information and support to people concerned about hereditary conditions in their families.
- Genetic testing includes screening for disease-causing alleles in newborns, the detection of people heterozygous for recessive alleles, presymptomatic testing for the presence of a disease-causing allele in at-risk people, and prenatal diagnosis.
- Common techniques used for prenatal diagnosis include ultrasound, amniocentesis, chorionic villus sampling, and maternal blood sampling. Preimplantation genetic diagnosis can be used to select for embryos that are free of a genetic disease.

IMPORTANT TERMS

pedigree (p. 134)	genetic counseling (p. 145)	ultrasonography (p. 147)	maternal blood testing (p. 149)
proband (p. 135)	newborn screening (p. 147)	amniocentesis (p. 147)	fetal cell sorting (p. 150)
consanguinity (p. 136)	heterozygote screening (p. 147)	chorionic villus sampling (p. 149)	preimplantation genetic diagnosis (p. 150)
dizygotic twins (p. 141)	presymptomatic genetic testing (p. 147)	karyotype (p. 149)	
monozygotic twins (p. 141)			
concordance (p. 142)			

Worked Problems

1. Joanna has short fingers (brachydactyly). She has two older brothers who are identical twins; they both have short fingers. Joanna's two younger sisters have normal fingers. Joanna's mother has normal fingers, and her father has short fingers.

Joanna's paternal grandmother (her father's mother) has short fingers; her paternal grandfather (her father's father), who is now deceased, had normal fingers. Both of Joanna's maternal grandparents (her mother's parents) have normal fingers. Joanna

marries Tom, who has normal fingers; they adopt a son named Bill who has normal fingers. Bill's biological parents both have normal fingers. After adopting Bill, Joanna and Tom produce two children: an older daughter with short fingers and a younger son with normal fingers.

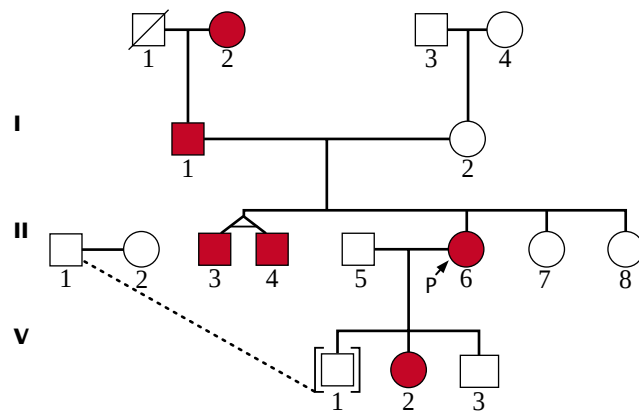
(a) Using correct symbols and labels, draw a pedigree illustrating the inheritance of short fingers in Joanna's family.

(b) What is the most likely mode of inheritance for short fingers in this family?

(c) If Joanna and Tom have another biological child, what is the probability (based on your answer to part b) that this child will have short fingers?

• Solution

(a) In the pedigree for the family, note that persons with the trait (short fingers) are indicated by filled circles (females) and filled squares (males). Joanna's identical twin brothers are connected to the line above with diagonal lines that have a horizontal line between them. The adopted child of Joanna and Tom is enclosed in brackets and is connected to the biological parents by a dashed diagonal line.



(b) The most likely mode of inheritance for short fingers in this family is autosomal dominant. The trait appears equally in males and females and does not skip generations. When one parent has the trait, it appears in approximately half of that parent's sons and daughters, although the number of children in the families is small. We can eliminate Y-linked inheritance because the trait is found in females. If short fingers were X-linked recessive, females with the trait would be expected to pass the trait to all their sons, but Joanna (III-6), who has short fingers, produced a son with normal fingers. For X-linked dominant traits, affected men should pass the trait to all their daughters; because male II-1 has short fingers and produced two daughters without short fingers (III-7 and III-8), we know that the trait cannot be X-linked dominant. It is unlikely that the trait is autosomal recessive because it does not skip generations and approximately half of the children of affected parents have the trait.

(c) If having short fingers is autosomal dominant, Tom must be homozygous (bb) because he has normal fingers. Joanna must be heterozygous (Bb) because she and Tom have produced both short- and normal-fingered offspring. In a cross between a heterozygote and homozygote, half of the progeny are expected to be heterozygous and half homozygous ($Bb \times bb \rightarrow \frac{1}{2} Bb, \frac{1}{2} bb$); so the probability that Joanna's and Tom's next biological child will have short fingers is $\frac{1}{2}$.

2. Concordance values for a series of traits were measured in monozygotic twins and dizygotic twins; the results are shown in the following table. For each trait, indicate whether the rates of concordance suggest genetic influences, environmental influences, or both. Explain your reasoning.

Characteristic	Monozygotic concordance (%)	Dizygotic concordance (%)
(a) ABO blood type	100	65
(b) Diabetes	85	36
(c) Coffee drinking	80	80
(d) Smoking	75	42
(e) Schizophrenia	53	16

• Solution

(a) The concordance of ABO blood type in the monozygotic twins is 100%. This high concordance in monozygotic twins does not, by itself, indicate a genetic basis for the trait. An important indicator of a genetic influence on the trait is lower concordance in dizygotic twins. Because concordance for ABO blood type is substantially lower in the dizygotic twins, we would be safe in concluding that genes play a role in determining differences in ABO blood types.

(b) The concordance for diabetes is substantially higher in monozygotic twins than in dizygotic twins; therefore, we can conclude that genetic factors play some role in susceptibility to diabetes. The fact that monozygotic twins show a concordance less than 100% suggests that environmental factors also play a role.

(c) Both monozygotic and dizygotic twins exhibit the same high concordance for coffee drinking; so we can conclude that there is little genetic influence on coffee drinking. The fact that monozygotic twins show a concordance less than 100% suggests that environmental factors play a role.

(d) There is lower concordance of smoking in dizygotic twins than in monozygotic twins, so genetic factors appear to influence the tendency to smoke. The fact that monozygotic twins show a concordance less than 100% suggests that environmental factors also play a role.

(e) Monozygotic twins exhibit substantially higher concordance for schizophrenia than do dizygotic twins; so we can conclude that genetic factors influence this psychiatric disorder. Because the concordance of monozygotic twins is substantially less than 100%, we can also conclude that environmental factors play a role in the disorder as well.

The New Genetics

MINING GENOMES

INTRODUCTION TO BIOINFORMATICS AND THE NATIONAL CENTER FOR BIOTECHNOLOGY INFORMATION (NCBI)

Biology and computer science merge in the field of bioinformatics, allowing biologists to ask questions and develop perspectives that

would never be possible with traditional techniques. This project will introduce you to the diverse suite of powerful interactive bioinformatics tools at the National Center for Biotechnology Information (NCBI).

COMPREHENSION QUESTIONS

- * 1. What three factors complicate the task of studying the inheritance of human characteristics?
- * 2. Describe the features that will be exhibited in a pedigree in which a trait is segregating with each of the following modes of inheritance: autosomal recessive, autosomal dominant, X-linked recessive, X-linked dominant, and Y-linked inheritance.
- * 3. What are the two types of twins and how do they arise?
4. Explain how a comparison of concordance in monozygotic and dizygotic twins can be used to determine the extent to which the expression of a trait is influenced by genes or by environmental factors.
5. How are adoption studies used to separate the effects of genes and environment in the study of human characteristics?
- * 6. What is genetic counseling?
7. Briefly define newborn screening, heterozygote screening, presymptomatic testing, and prenatal diagnosis.
- * 8. What are the differences between amniocentesis and chorionic villus sampling? What is the purpose of these two techniques?
9. What is preimplantation genetic diagnosis?

APPLICATION QUESTIONS AND PROBLEMS

- * 10. Joe is color-blind. His mother and father both have normal vision, but his mother's father (Joe's maternal grandfather) is color-blind. All Joe's other grandparents have normal color vision. Joe has three sisters—Patty, Betsy, and Lora—all with normal color vision. Joe's oldest sister, Patty, is married to a man with normal color vision; they have two children, a 9-year old color-blind boy and a 4-year-old girl with normal color vision.

(a) Using correct symbols and labels, draw a pedigree of Joe's family.

(b) What is the most likely mode of inheritance for color blindness in Joe's family?

(c) If Joe marries a woman who has no family history of color blindness, what is the probability that their first child will be a color-blind boy?

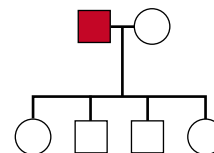
(d) If Joe marries a woman who is a carrier of the color-blind allele, what is the probability that their first child will be a color-blind boy?

(e) If Patty and her husband have another child, what is the probability that the child will be a color-blind boy?

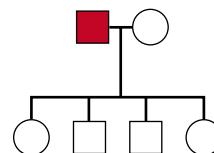
11. A man with a specific unusual genetic trait marries an unaffected woman and they have four children. Pedigrees of this family are shown in parts a through e, but the presence

or absence of the trait in the children is not indicated. For each type of inheritance, indicate how many children of each sex are expected to express the trait by filling in the appropriate circles and squares. Assume that the trait is rare and fully penetrant.

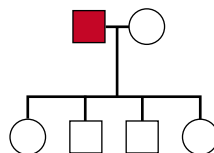
(a) Autosomal recessive trait



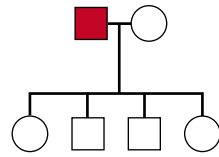
(b) Autosomal dominant trait



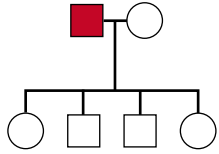
(c) X-linked recessive trait



(d) X-linked dominant trait

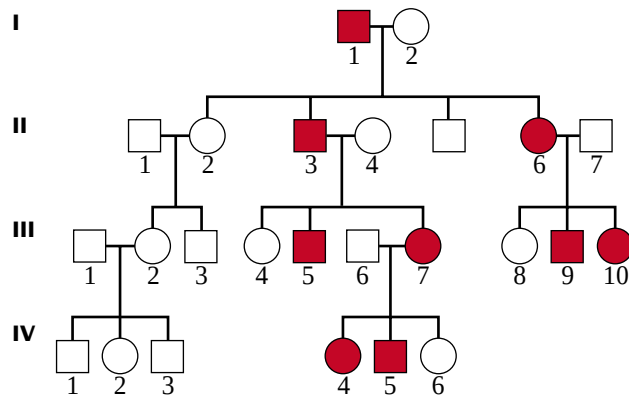


(e) Y-linked trait

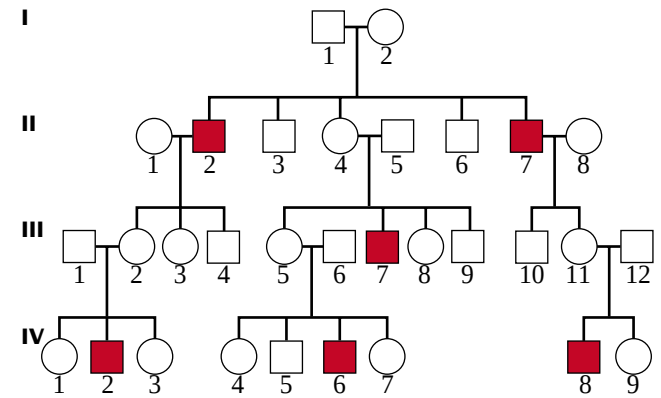


* 12. For each of the following pedigrees, give the most likely mode of inheritance, assuming that the trait is rare. Carefully explain your reasoning.

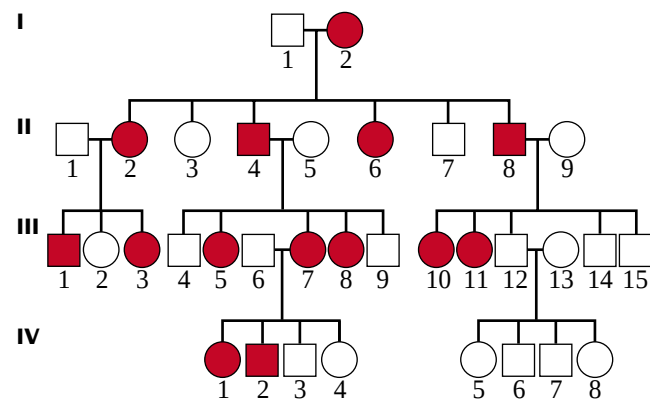
(a)



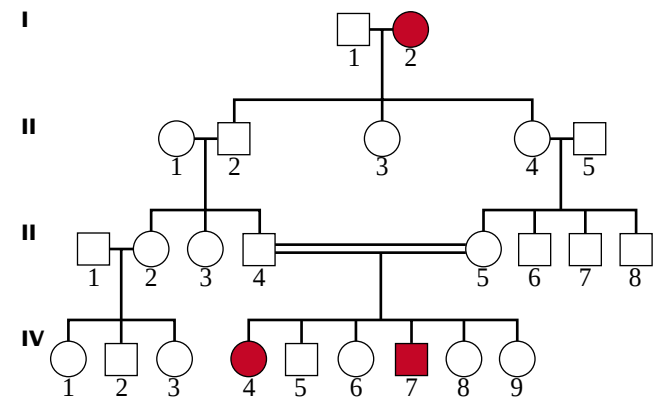
(d)



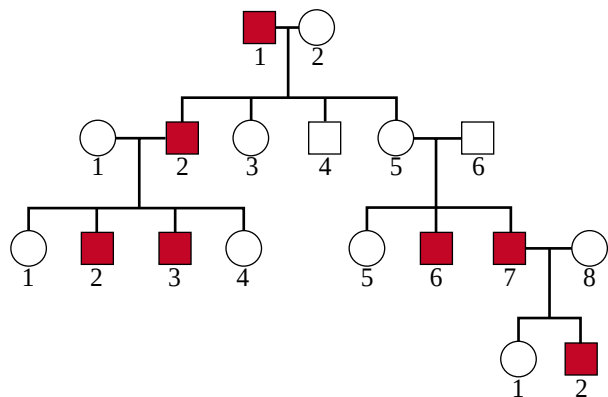
(b)



(e)



13. The trait represented in the following pedigree is expressed only in the males of the family. Is the trait Y linked? Why or why not? If you believe the trait is not Y linked, propose an alternate explanation for its inheritance.



* 14. A geneticist studies a series of characteristics in monozygotic twins and dizygotic twins, obtaining the following concordances. For each characteristic, indicate whether the rates of concordance suggest genetic influences, environmental influences, or both. Explain your reasoning.

Characteristic	Monozygotic concordance (%)	Dizygotic concordance (%)
Migraine headaches	60	30
Eye color	100	40
Measles	90	90
Clubfoot	30	10
High blood pressure	70	40
Handedness	70	70
Tuberculosis	5	5

15. In a study of schizophrenia (a mental disorder including disorganization of thought and withdrawal from reality), researchers looked at the prevalence of the disorder in the biological and adoptive parents of people who were adopted as children; they found the following results:

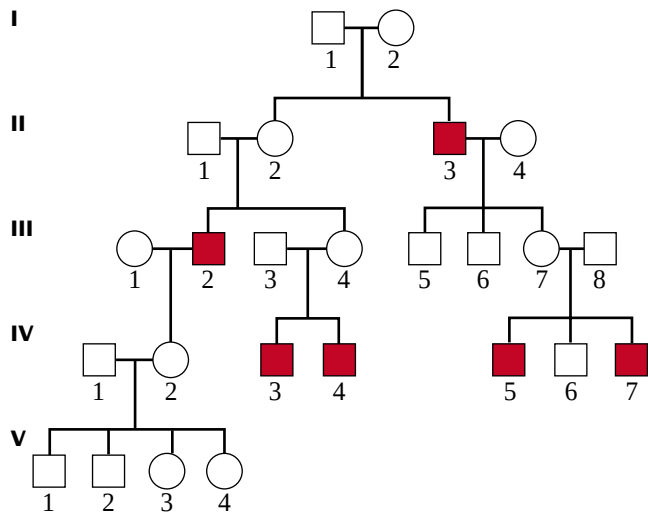
Adopted persons	Prevalence of schizophrenia (%)	
	Biological parents	Adoptive parents
With schizophrenia	12	2
Without schizophrenia	6	4

(Source: S. S. Kety, et al., The biological and adoptive families of adopted individuals who become schizophrenic: prevalence of mental illness and other characteristics, In *The Nature of Schizophrenia: New Approaches to Research and Treatment*, L. C. Wynne, R. L. Cromwell, and S. Matthysse, Eds. (New York: Wiley, 1978), pp. 25–37.)

What can you conclude from these results concerning the role of genetics in schizophrenia? Explain your reasoning.

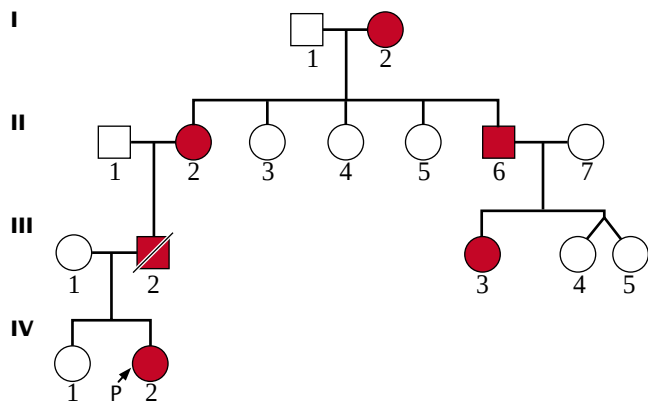
* 16. The following pedigree illustrates the inheritance of Nance-Horan syndrome, a rare genetic condition in which affected persons have cataracts and abnormally shaped teeth.

- (a) On the basis of this pedigree, what do you think is the most likely mode of inheritance for Nance-Horan syndrome?
- (b) If couple III-7 and III-8 have another child, what is the probability that the child will have Nance-Horan syndrome?
- (c) If III-2 and III-7 mated, what is the probability that one of their children would have Nance-Horan syndrome?



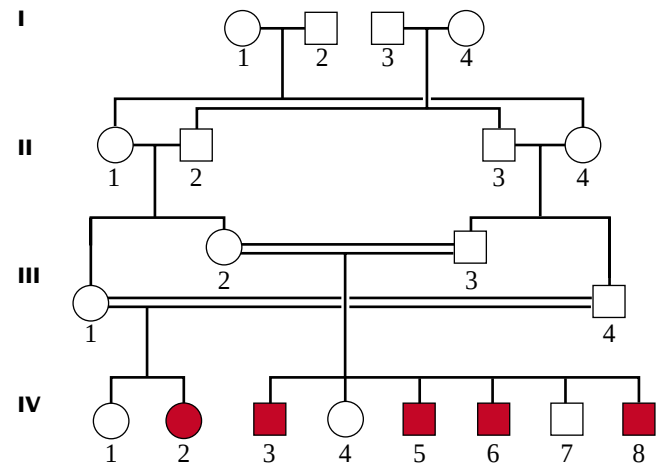
(Pedigree adapted from D. Stambolian, R. A. Lewis, K. Buetow, A. Bond, and R. Nussbaum. *American Journal of Human Genetics* 47(1990):15.)

17. The following pedigree illustrates the inheritance of ringed hair, a condition in which each hair is differentiated into light and dark zones. What mode or modes of inheritance are possible for the ringed-hair trait in this family?



(Pedigree adapted from L. M. Ashley, and R. S. Jacques, *Journal of Heredity* 41(1950):83.)

- * 18. Ectodactyly is a rare condition in which the fingers are absent and the hand is split. This condition is usually inherited as an autosomal dominant trait. Ademar Freire-Maia reported the appearance of ectodactyly in a family in Sao Paulo, Brazil, whose pedigree is shown here. Is this pedigree consistent with autosomal dominant inheritance? If not, what mode of inheritance is most likely? Explain your reasoning.
(Pedigree adapted from A. Freire-Maia, *Journal of Heredity* 62(1971):53.)



CHALLENGE QUESTIONS

19. Draw a pedigree that represent an autosomal dominant trait, sex-limited to males, and that excludes the possibility that the trait is Y linked.
20. Androgen insensitivity syndrome is a rare disorder of sexual development, in which people with an XY karyotype, genetically male, develop external female features. All persons with androgen insensitivity syndrome are infertile. In the past, some researchers proposed that androgen insensitivity syndrome is inherited as a sex-limited, autosomal dominant trait. (It is sex-limited because females cannot express the

trait.) Other investigators suggested that this disorder is inherited as a X-linked recessive trait.

Draw a pedigree that would show conclusively that androgen insensitivity syndrome is inherited as an X-linked recessive trait and that excludes the possibility that it is sex-limited, autosomal dominant. If you believe that no pedigree can conclusively differentiate between the two choices (sex-limited, X-linked recessive and sex-limited, autosomal dominant), explain why. Remember that all affected persons are infertile.

SUGGESTED READINGS

- Barsh, G. S., I. S. Farooqi, and S. O'Rahilly. 2000. Genetics of body-weight regulation. *Nature* 404:644–651.
An excellent review of the genetics of body weight in humans. This issue of *Nature* has a section on obesity, with additional review articles on obesity as a medical problem, on the molecular basis of thermogenesis, on nervous-system control of food intake, and medical strategies for treatment of obesity.
- Bennett, R. L., K. A. Steinhaus, S. B. Uhrich, C. K. O'Sullivan, R. G. Resta, D. Lochner-Doyle, D. S. Markel, V. Vincent, and J. Hamanishi. 1995. Recommendations for standardized human pedigree nomenclature. *American Journal of Human Genetics* 56:745–752.
Contains recommendations for standardized symbols used in pedigree construction.
- Brown, M. S., and J. L. Goldstein. 1984. How LDL receptors influence cholesterol and atherosclerosis. *Scientific American* 251 November: 58–66.
Excellent review of the genetics of atherosclerosis by two scientists who received the Nobel Prize for their research on atherosclerosis.

- Devor, E. J., and C. R. Cloninger. 1990. Genetics of alcoholism. *Annual Review of Genetics* 23:19–36.
A good review of how genes influence alcoholism in humans.
- Gurney, M. E., A. G. Tomasselli, and R. L. Heinrikson. 2000. Stay the executioner's hand. *Science* 288:283–284.
Reports new evidence that mutated SOD1 may be implicated in apoptosis (programmed cell death) in people with amyotrophic lateral sclerosis.
- Harper, P. S. 1998. *Practical Genetic Counseling*, 5th ed. Oxford: Butterworth Heineman.
A classic textbook on genetic counseling.
- Jorde, L. B., J. C. Carey, M. J. Bamshad, and R. L. White. 1998. *Medical Genetics*, 2d ed. St. Louis: Mosby.
A textbook on medical aspects of human genetics.
- Lewis, R. 1994. The evolution of a classical genetic tool. *Bioscience* 44:722–726.
A well-written review of the history of pedigree analysis and recent changes in symbols that have been necessitated by changing life styles and new reproductive technologies.

Mange, E. J., and A. P. Mange. 1998. *Basic Human Genetics*, 2d ed. Sunderland, MA: Sinauer.

A well-written textbook on human genetics.

MacGregor, A. J., H. Snieder, N. J. Schork, and T. D. Spector. 2000. Twins: novel uses to study complex traits and genetic diseases. *Trends in Genetics* 16:131–134.

A discussion of new methods for using twins in the study of genes.

Mahowald, M. B., M. S. Verp, and R. R. Anderson. 1998. Genetic counseling: clinical and ethical challenges. *Annual Review of Genetics* 32:547–559.

A review of genetic counseling in light of the Human Genome Project, with special consideration of the role of nondirected counseling.

McKusick, V. A. 1998. *Mendelian Inheritance in Man: A Catalog of Human Genes and Genetic Disorders*, 12th ed. Baltimore: Johns Hopkins University Press.

A comprehensive catalog of all known simple human genetic disorders and the genes responsible for them.

Pierce, B. A. 1990. *The Family Genetic Source Book*. New York: Wiley.

A book on human genetics written for the layperson; contains a catalog of more than 100 human genetic traits.

Stunkard, A. J., T. I. Sorensen, C. Hanis, T. W. Teasdale, R. Chakraborty, W. J. Schull, and F. Schulsinger. 1986. An adoption study of human obesity. *The New England Journal of Medicine* 314:193–198.

Describes the Danish adoption study of obesity.

7

Linkage, Recombination, and Eukaryotic Gene Mapping



Alfred Henry Sturtevant, an early geneticist, developed the first genetic map. (Institute Archives, California Institute of Technology.)

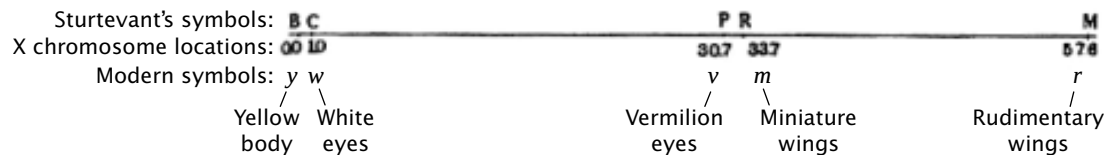
Alfred Sturtevant and the First Genetic Map

In 1909, Thomas Hunt Morgan taught the introduction to zoology class at Columbia University. Seated in the lecture hall were sophomore Alfred Henry Sturtevant and freshman Calvin Bridges. Sturtevant and Bridges were excited by Morgan's teaching style and intrigued by his interest in biological problems. They asked Morgan if they could work in his laboratory and, the following year, both young men were given desks in the "fly room," Morgan's research laboratory where the study of *Drosophila* genetics was in its infancy (see p. 000 in Chapter 4). Sturtevant, Bridges, and

Morgan's other research students virtually lived in the laboratory, raising fruit flies, designing experiments, and discussing their results.

In the course of their research, Morgan and his students observed that some pairs of genes did not segregate randomly according to Mendel's principle of independent assortment but instead tended to be inherited together. Morgan suggested that possibly the genes were located on the same chromosome and thus traveled together during meiosis. He further proposed that closely linked genes—

- Alfred Sturtevant and the First Genetic Map
- Genes That Assort Independently and Those That Don't
- Linkage and Recombination Between Two Genes
 - Notation for Crosses with Linkage
 - Complete Linkage Compared with Independent Assortment
 - Crossing Over with Linked Genes
 - Calculation of Recombination Frequency
 - Coupling and Repulsion
 - The Physical Basis of Recombination
 - Predicting the Outcome of Crosses with Linked Genes
 - Testing for Independent Assortment
 - Gene Mapping with Recombination Frequencies
 - Constructing a Genetic Map with Two-Point Testcrosses
- Linkage and Recombination Between Three Genes
 - Gene Mapping with the Three-Point Testcross
 - Gene Mapping in Humans
 - Mapping with Molecular Markers
- Physical Chromosome Mapping
 - Deletion Mapping
 - Somatic-Cell Hybridization
 - In Situ Hybridization
 - Mapping by DNA Sequencing



7.1 Sturtevant's map included five genes on the X chromosome of *Drosophila*.

The genes are yellow body (*y*), white eyes (*w*), vermilion eyes (*v*), miniature wings (*m*), and rudimentary wings (*r*). Sturtevant's original symbols for the genes are shown above the line; modern symbols are shown below with their current locations on the X chromosome.

those that are rarely shuffled by recombination—lie close together on the same chromosome, whereas loosely linked genes—those more frequently shuffled by recombination—lie farther apart.

One day in 1911, Sturtevant and Morgan were discussing independent assortment when, suddenly, Sturtevant had a flash of inspiration: variation in the strength of linkage indicated how genes were positioned along a chromosome, providing a way of mapping genes. Sturtevant went home and, neglecting his undergraduate homework, spent most of the night working out the first genetic map (FIGURE 7.1). Sturtevant's first chromosome map was remarkably accurate, and it established the basic methodology used today for mapping genes.

Alfred Sturtevant went on to become a leading geneticist. His research included gene mapping and basic mechanisms of inheritance in *Drosophila*, cytology, embryology, and evolution. Sturtevant's career was deeply influenced by his early years in the fly room, where Morgan's unique personality and the close quarters combined to stimulate intellectual excitement and the free exchange of ideas.

www.whfreeman.com/pierce

Sturtevant's life

More details about Alfred

This chapter explores the inheritance of genes located on the same chromosome. These linked genes do not strictly obey Mendel's principle of independent assortment; rather, they tend to be inherited together. This tendency requires a new approach to understanding their inheritance and predicting the types of offspring produced. A critical piece of information necessary for predicting the results of these crosses is the arrangement of the genes on the chromosomes; thus, it will be necessary to think about the relation between genes and chromosomes. A key to understanding the inheritance of linked genes is to make the conceptual connection between the genotypes in a cross and the behavior of chromosomes during meiosis.

We will begin our exploration of linkage by first comparing the inheritance of two linked genes with the inheritance of two genes that assort independently. We will then examine how crossing over breaks up linked genes. This knowledge of linkage and recombination will be used for

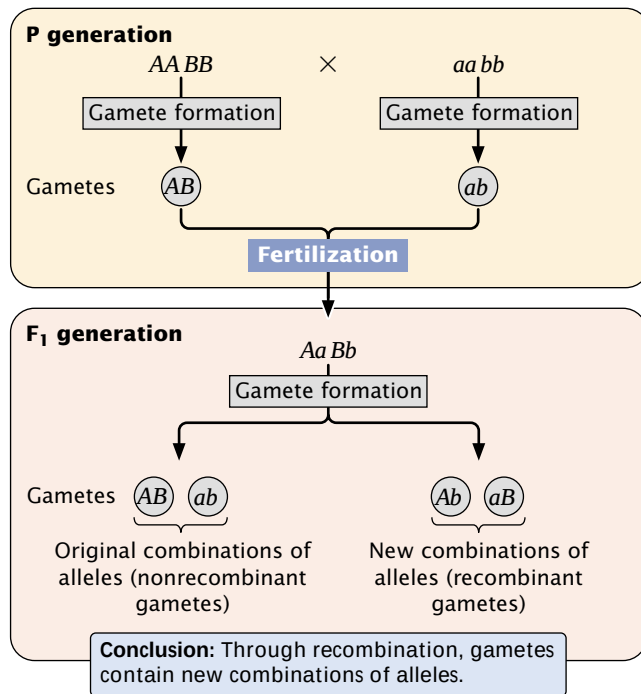
predicting the results of genetic crosses in which genes are linked and for mapping genes. The last section of the chapter focuses on physical methods of determining the chromosomal locations of genes.

Genes That Assort Independently and Those That Don't

Chapter 3 introduced Mendel's principles of segregation and independent assortment. Let's take a moment to review these two important concepts. The principle of segregation states that each diploid individual possesses two alleles that separate in meiosis, with one allele going into each gamete. The principle of independent assortment provides additional information about the process of segregation: it tells us that the two alleles separate independently of alleles at other loci.

The independent separation of alleles produces *recombination*, the sorting of alleles into new combinations. Consider a cross between individuals homozygous for two different pairs of alleles: $AABB \times aabb$. The first parent, $AABB$, produces gametes with alleles AB , and the second parent, $aabb$, produces gametes with the alleles ab , resulting in F_1 progeny with genotype $AaBb$ (FIGURE 7.2). Recombination means that, when one of the F_1 progeny reproduces, the combination of alleles in its gametes may differ from the combinations in the gametes of its parents. In other words, the F_1 may produce gametes with alleles Ab or aB in addition to gametes with AB or ab .

Mendel derived his principles of segregation and independent assortment by observing progeny of genetic crosses, but he had no idea of what biological processes produced these phenomena. In 1903, Walter Sutton proposed a biological basis for Mendel's principles, called the chromosome theory of heredity (Chapter 3). This theory holds that genes are found on chromosomes. Let's restate Mendel's two principles in terms of the chromosome theory of heredity. The principle of segregation states that each diploid individual possesses two alleles for a trait, each of which is located at the same position, or locus, on each of the two homologous chromosomes. These chromosomes segregate in meiosis, with each gamete receiving one homolog. The principle of independent assortment states that, in meiosis,

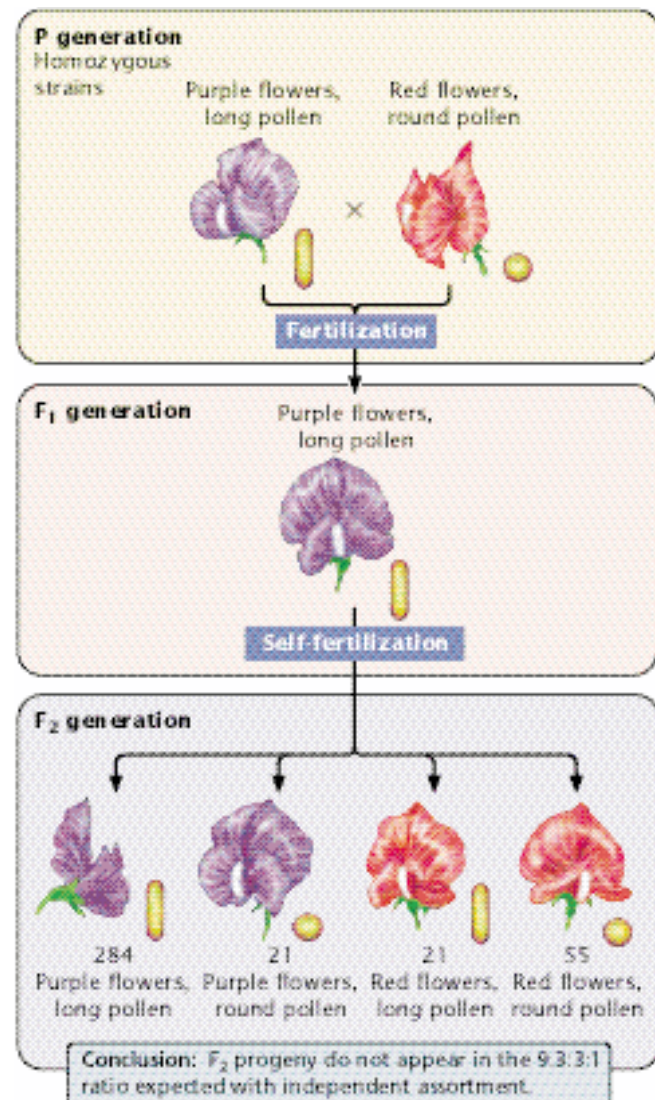


7.2 Recombination is the sorting of alleles into new combinations.

each pair of homologous chromosomes assorts independently of other homologous pairs. With this new perspective, it is easy to see that the number of chromosomes in most organisms is limited and that there are certain to be more genes than chromosomes; so some genes must be present on the same chromosome and should not assort independently.

Genes located close together on the same chromosome are called **linked genes** and belong to the same **linkage group**. As we've said, linked genes travel together during meiosis, eventually arriving at the same destination (the same gamete), and are not expected to assort independently. However, all of the characteristics examined by Mendel in peas did display independent assortment and, after the rediscovery of Mendel's work, the first genetic characteristics studied in other organisms also seemed to assort independently. How could genes be carried on a limited number of chromosomes and yet assort independently?

The apparent inconsistency between the principle of independent assortment and the chromosome theory of heredity soon disappeared, as biologists began finding genetic characteristics that did *not* assort independently. One of the first cases was reported in sweet peas by William Bateson, Edith Rebecca Saunders, and Reginald C. Punnett in 1905. They crossed a homozygous strain of peas having purple flowers and long pollen grains with a homozygous strain having red flowers and round pollen grains. All the F₁ had purple flowers and long pollen grains, indicating that

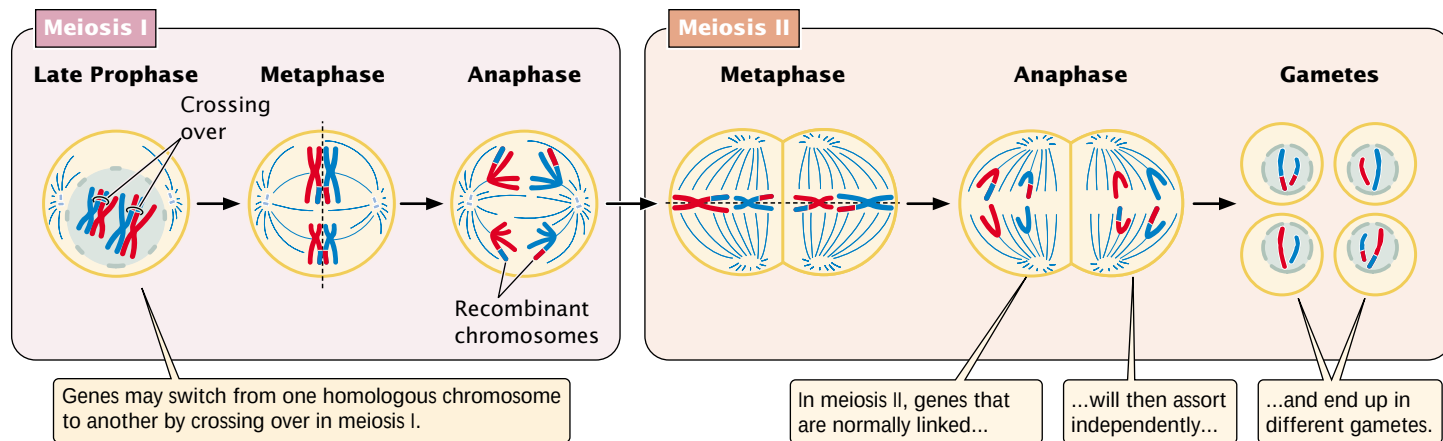


7.3 Nonindependent assortment of flower color and pollen shape in sweet peas.

purple was dominant over red and long was dominant over round. When they intercrossed the F₁, the resulting F₂ progeny did not appear in the 9:3:3:1 ratio expected with independent assortment (FIGURE 7.3). An excess of F₂ plants had purple flowers and long pollen or red flowers and round pollen (the parental phenotypes). Although Bateson, Saunders, and Punnett were unable to explain these results, we now know that the two loci that they examined lie close together on the same chromosome and therefore do not assort independently.

Linkage and Recombination Between Two Genes

Genes on the same chromosome are like passengers on a charter bus: they travel together and ultimately arrive at the



7.4 Crossing over takes place in meiosis and is responsible for recombination.

same destination. However, genes occasionally switch from one homologous chromosome to another through the process of crossing over (Chapter 2) (FIGURE 7.4). Crossing over produces recombination—it breaks up the associations of genes imposed by linkage. As will be discussed later, genes located on the same chromosome can exhibit independent assortment if they are far enough apart. In summary, linkage adds a further complication to interpretations of the results of genetic crosses. With an understanding of how linkage affects heredity, we can analyze crosses for linked genes and successfully predict the types of progeny that will be produced.

Notation for Crosses with Linkage

In analyzing crosses with linked genes, we must know not only the genotypes of the individuals crossed, but also the arrangement of the genes on the chromosomes. To keep track of this arrangement, we will introduce a new system of notation for presenting crosses with linked genes. Consider a cross between an individual homozygous for dominant alleles at two linked loci and another individual homozygous for recessive alleles at those loci. Previously, we would have written these genotypes as:

$$AABB \times aabb$$

For linked genes, however, it's necessary to write out the specific alleles as they are arranged on each of the homologous chromosomes:

$$\begin{array}{c} A & B \\ \hline A & B \end{array} \times \begin{array}{c} a & b \\ \hline a & b \end{array}$$

In this notation, each line represents one of the two homologous chromosomes. In the first parent of the cross, each homologous chromosome contains *A* and *B* alleles; in

the second parent, each homologous chromosome contains *a* and *b* alleles. Inheriting one chromosome from each parent, the F_1 progeny will have the following genotype:

$$\begin{array}{c} A & B \\ \hline a & b \end{array}$$

Here, the importance of designating the alleles on each chromosome is clear. One chromosome has the two dominant alleles *A* and *B*, whereas the homologous chromosome has the two recessive alleles *a* and *b*. The notation can be simplified by drawing only a single line, with the understanding that genes located on the same side of the line lie on the same chromosome:

$$\begin{array}{c} A & B \\ \hline a & b \end{array}$$

This notation can be simplified further by separating the alleles on each chromosome with a slash: *AB/ab*.

Remember that the two alleles at a locus are always located on different homologous chromosomes and therefore must lie on opposite sides of the line. Consequently, we would *never* write the genotypes as:

$$\begin{array}{c} A & a \\ \hline B & b \end{array}$$

because the alleles *A* and *a* can *never* be on the same chromosome.

It is also important to always keep the same order of the genes on both sides of the line; thus, we should *never* write:

$$\begin{array}{c} A & B \\ \hline b & a \end{array}$$

because this would imply that alleles A and b are allelic (at the same locus).

Complete Linkage Compared with Independent Assortment

We will first consider what happens to genes that exhibit complete linkage, meaning that they are located on the same chromosome and do not exhibit detectable crossing over. Genes rarely exhibit complete linkage but, without the complication of crossing over, the effect of linkage can be seen more clearly. We will then consider what happens when genes assort independently. Finally, we will consider the results obtained if the genes are linked but exhibit some crossing over.

A testcross reveals the effects of linkage. For example, if a heterozygous individual is test-crossed with a homozygous recessive individual ($AaBb \times aabb$), whatever alleles are present in the gametes contributed by the heterozygous parent will be expressed in the phenotype of the offspring, because the homozygous parent could not contribute dominant alleles that might mask them. Consequently, traits that appear in the progeny reveal which alleles were transmitted by the heterozygous parent.

Consider a pair of linked genes in tomato plants. One pair affects the type of leaf: an allele for mottled leaves (m) is recessive to an allele that produces normal leaves (M). Nearby on the same chromosome is another locus that determines the height of the plant: an allele for dwarf (d) is recessive to an allele for tall (D).

Testing for linkage can be done with a testcross, which requires a plant heterozygous for both traits. A geneticist might produce this heterozygous plant by crossing a variety of tomato that is homozygous for normal leaves and tall height with a variety that is homozygous for mottled leaves and dwarf height:

$$\begin{array}{c} \text{P} \quad \frac{M}{M} \frac{D}{D} \times \frac{m}{m} \frac{d}{d} \\ \downarrow \\ \text{F}_1 \quad \frac{M}{m} \frac{D}{d} \end{array}$$

The geneticist would then use these F_1 heterozygotes in a testcross, crossing them with plants homozygous for mottled leaves and dwarf height:

$$\frac{M}{m} \frac{D}{d} \times \frac{m}{m} \frac{d}{d}$$

The results of this testcross are diagrammed in [FIGURE 7.5a](#). During gamete formation, the heterozygote produces two types of gametes: some with the $\underline{M \quad D}$ chromosome

and others with the $\underline{m \quad d}$ chromosome. Because no crossing over occurs, these gametes are the only types produced by the heterozygote. Notice that these gametes contain only combinations of alleles that were present in the original parents: either the allele for normal leaves together with the allele for tall height (M and D) or the allele for mottled leaves together with the allele for dwarf height (m and d). Gametes that contain only original combinations of alleles present in the parents are **nonrecombinant gametes**, or parental gametes.

The homozygous parent in the testcross produces only one type of gamete; it contains chromosome $\underline{m \quad d}$ and pairs with one of the two gametes generated by the heterozygous parent (see Figure 7.5a). Two types of progeny result: half have normal leaves and are tall:

$$\frac{M}{m} \frac{D}{d}$$

and half have mottled leaves and are dwarf:

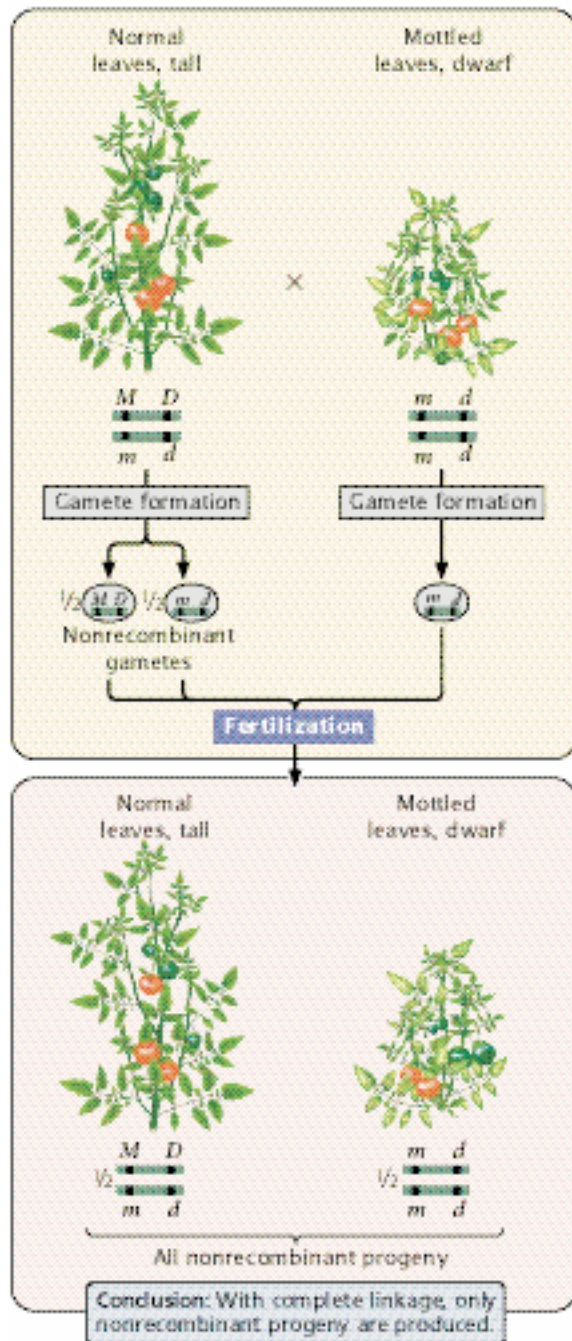
$$\frac{m}{m} \frac{d}{d}$$

These progeny display the original combinations of traits present in the P generation and are **nonrecombinant progeny**, or parental progeny. No new combinations of the two traits, such as normal leaves with dwarf or mottled leaves with tall, appear in the offspring, because the genes affecting the two characteristics are completely linked and are inherited together. New combinations of traits could arise only if the linkage between M and D or between m and d were broken.

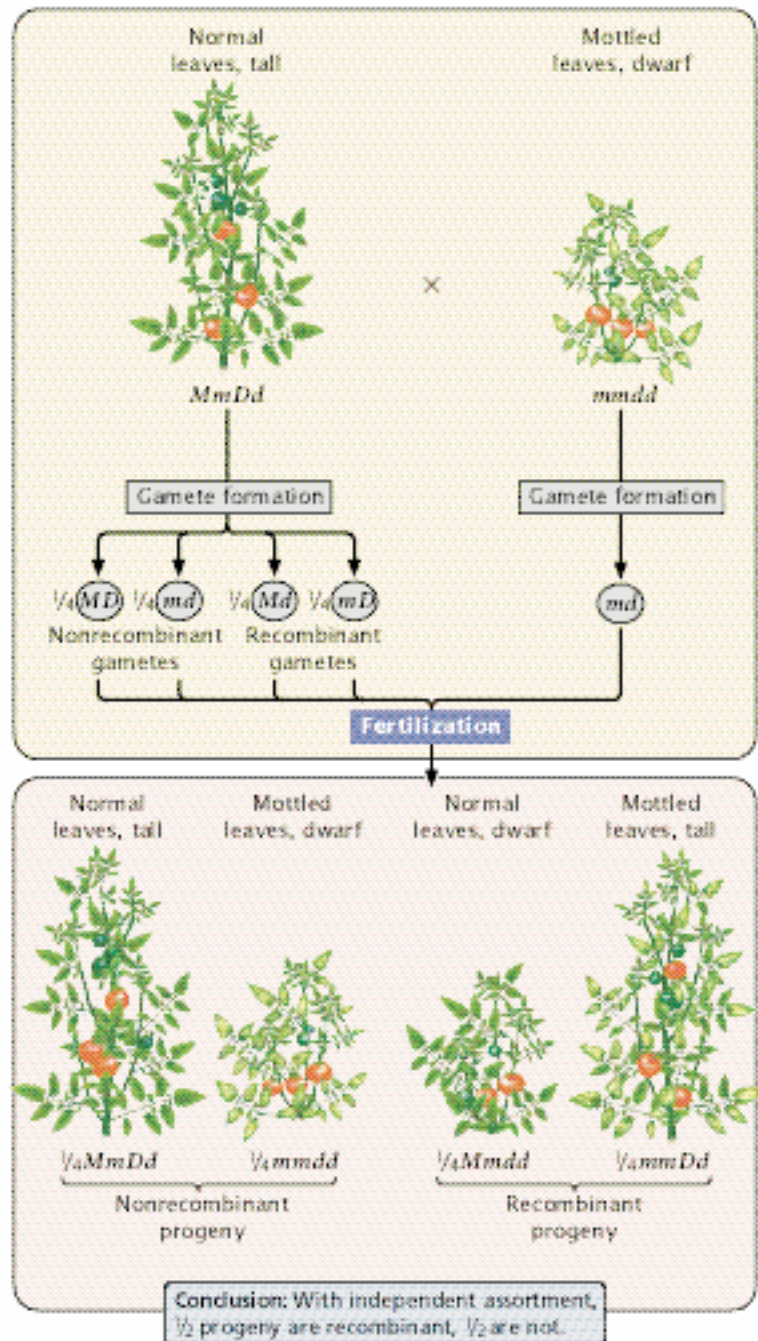
These results are distinctly different from the results that are expected when genes assort independently ([FIGURE 7.5b](#)). With independent assortment, the heterozygous plant ($MmDd$) would produce four types of gametes: two nonrecombinant gametes containing the original combinations of alleles (MD and md) and two gametes containing new combinations of alleles (Md and mD). Gametes with new combinations of alleles are called **recombinant gametes**. With independent assortment, nonrecombinant and recombinant gametes are produced in equal proportions. These four types of gametes join with the single type of gamete produced by the homozygous parent of the testcross to produce four kinds of progeny in equal proportions (see Figure 7.5b). The progeny with new combinations of traits formed from recombinant gametes are termed **recombinant progeny**.

In summary, a testcross in which one of the plants is heterozygous for two completely linked genes yields two types of progeny, each type displaying one of the original combinations of traits present in the P generation. Independent assortment, in contrast, produces two types of

(a) If genes are completely linked (no crossing over)



(b) If genes assort independently



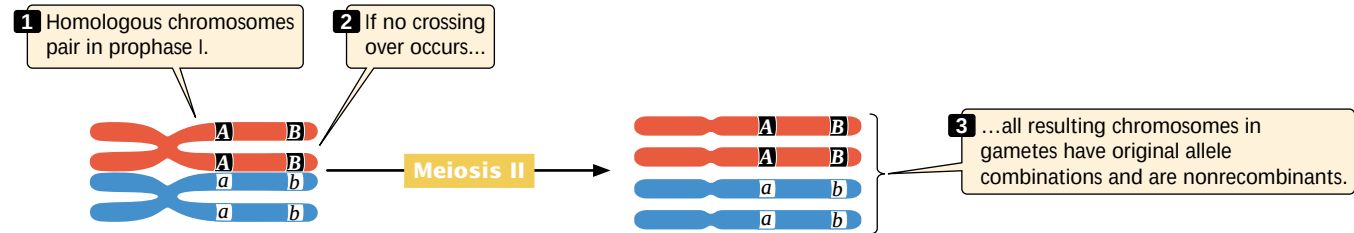
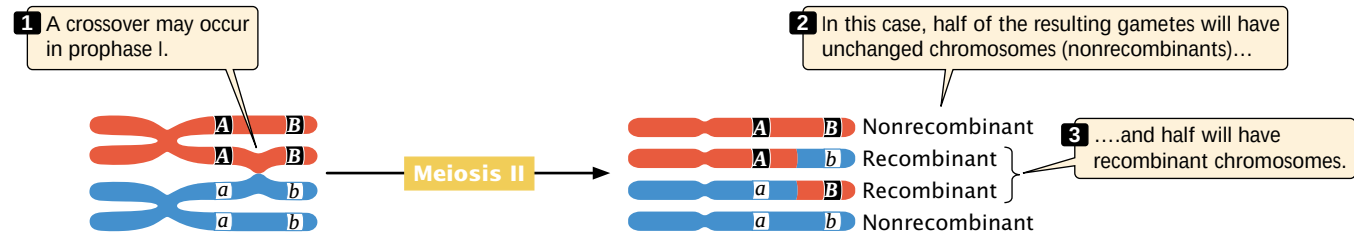
7.5 A testcross reveals the effects of linkage. Results of a testcross for two loci in tomatoes that determine leaf type and plant height.

recombinant progeny and two types of nonrecombinant progeny in equal proportions.

Crossing Over with Linked Genes

Linkage is rarely complete—usually, there is some crossing over between linked genes (incomplete linkage), producing new combinations of traits. Let's see how this occurs.

Theory The effect of crossing over on the inheritance of two linked genes is shown in **FIGURE 7.6**. Crossing over, which takes place in prophase I of meiosis, is the exchange of genetic material between nonsister chromatids (see Figures 2.15 and 2.17). After a single crossover has taken place, the two chromatids that did not participate in crossing over are unchanged; gametes that receive these chromatids are

(a) No crossing over**(b) Crossing over**

7.6 Crossing over produces half nonrecombinant gametes and half recombinant gametes.

nonrecombinants. The other two chromatids, which did participate in crossing over, now contain new combinations of alleles; gametes that receive these chromatids are recombinants. For each meiosis in which a single crossover takes place, then, two nonrecombinant gametes and two recombinant gametes will be produced. This result is the same as that produced by independent assortment (see Figure 7.5b); so, when crossing over between two loci takes place in every meiosis, it is impossible to determine whether the genes were linked and crossing over took place or whether the genes are on different chromosomes.

For closely linked genes, crossing over does not take place in every meiosis. In meioses in which there is no crossing over, only nonrecombinant gametes are produced. In meioses in which there is a single crossover, half the gametes are recombinants and half are nonrecombinants (because a single crossover only affects two of the four chromatids); so the total percentage of recombinant gametes is always half the percentage of meioses in which crossing over takes place. Even if crossing over between two genes takes place in every meiosis, only 50% of the resulting gametes will be recombinants. Thus, the frequency of recombinant gametes is always half the frequency of crossing over, and the maximum proportion of recombinant gametes is 50%.

and two nonrecombinants. The frequency of recombinant gametes is half the frequency of crossing over, and the maximum frequency of recombinant gametes is 50%.

Application Let us apply what we have learned about linkage and recombination to a cross between tomato plants that differ in the genes that code for leaf type and plant height. Assume now that these genes are linked and that some crossing over takes place between them. Suppose a geneticist carried out the testcross outlined earlier:

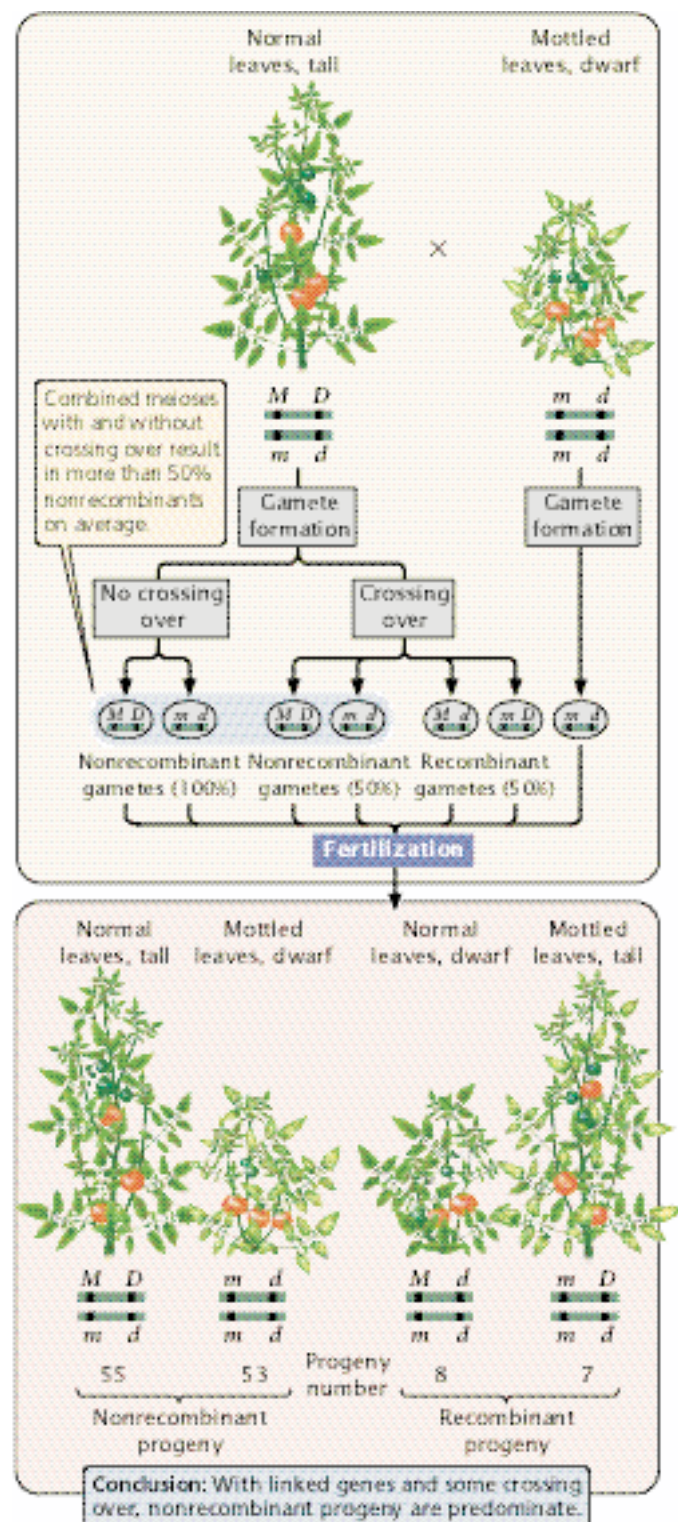
$$\frac{M}{m} \frac{D}{d} \times \frac{m}{m} \frac{d}{d}$$

When crossing over takes place between the genes for leaf type and height, two of the four gametes produced will be recombinants. When there is no crossing over, all four resulting gametes will be nonrecombinants. Thus, over all, the majority of gametes will be nonrecombinants. These gametes then unite with gametes produced by the homozygous recessive parent, which contain only the recessive alleles, resulting in mostly nonrecombinant progeny and a few recombinant progeny (**FIGURE 7.7**). In this cross, we see that 55 of the testcross progeny have normal leaves and are tall and 53 have mottled leaves and are dwarf. These plants are the nonrecombinant progeny, containing the original combinations of traits that were present in the parents. Of the 123 progeny, 15 have new combinations of traits that were not seen in the parents: 8 are normal leaved and dwarf, and 7 are mottle leaved and tall. These plants are the recombinant progeny.

Concepts

Linkage between genes causes them to be inherited together and reduces recombination; crossing over breaks up the associations of such genes. In a testcross for two linked genes, each crossover produces two recombinant gametes

The results of a cross such as the one illustrated in Figure 7.7 reveal several things. A testcross for two independently assorting genes is expected to produce a 1:1:1:1 phenotypic ratio in the progeny. The progeny of this cross clearly do not exhibit such a ratio; so we might suspect that



the genes are not assorting independently. When linked genes undergo crossing over, the result is mostly nonrecombinant progeny and fewer recombinant progeny. This result is what we observe among the progeny of the testcross illustrated in Figure 7.7; so we conclude that two genes show evidence of linkage with some crossing over.

Calculation of Recombination Frequency

The percentage of recombinant progeny produced in a cross is called the **recombination frequency**, which is calculated as follows:

$$\text{recombination frequency} = \frac{\text{number of recombinant progeny}}{\text{total number of progeny}} \times 100\%$$

In the testcross shown in Figure 7.7, 15 progeny exhibit new combinations of traits; so the recombination frequency is:

$$\frac{8 + 7}{55 + 53 + 8 + 7} \times 100\% = \frac{15}{123} \times 100\% = 12\%$$

Thus, 12% of the progeny exhibit new combinations of traits resulting from crossing over.

Coupling and Repulsion

In crosses for linked genes, the arrangement of alleles on the homologous chromosomes is critically important in determining the outcome of the cross. For example, consider the inheritance of two linked genes in the Australian blowfly, *Lucilia cuprina*. In this species, one locus determines the color of the thorax: purple thorax (p) is recessive to the normal green thorax (p^+). A second locus determines the color of the puparium: a black puparium (b) is recessive to the normal brown puparium (b^+). These loci are located close together on the second chromosome. Suppose we test-cross a fly that is heterozygous at both loci with a fly that is homozygous recessive at both. Because these genes are linked, there are two possible arrangements on the chromosomes of the heterozygous fly. The dominant alleles for green thorax (p^+) and brown puparium (b^+) might reside on the same chromosome, and the recessive alleles for purple thorax (p) and black puparium (b) might reside on the other homologous chromosome:

$$\frac{p^+ \quad b^+}{p \quad b}$$

This arrangement, in which wild-type alleles are found on one chromosome and mutant alleles are found on the other

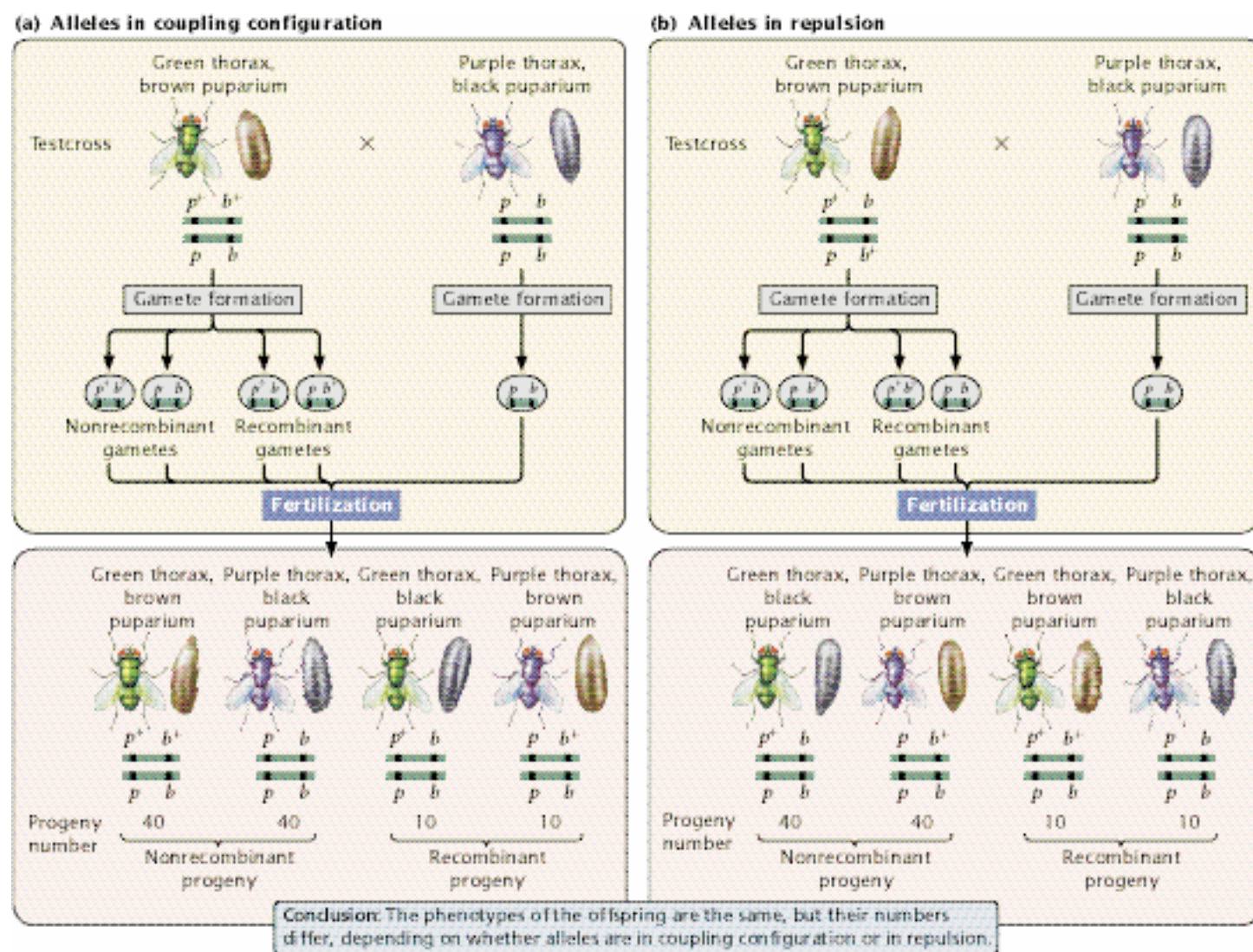
7.7 Crossing over between linked genes produces nonrecombinant and recombinant offspring. In this testcross, genes are linked and there is some crossing over. For comparison, this cross is the same as that illustrated in Figure 7.5.

chromosome, is referred to as **coupling**, or the **cis configuration**. Alternatively, one chromosome might bear the alleles for green thorax (p^+) and black puparium (b), and the other chromosome would carry the alleles for purple thorax (p) and brown puparium (b^+):

$$\begin{array}{cc} p^+ & b \\ p & b^+ \end{array}$$

This arrangement, in which each chromosome contains one wild-type and one mutant allele, is called the **repulsion** or **trans configuration**. Whether the alleles in the heterozygous parent are in coupling or repulsion determines which phenotypes will be most common among the progeny of a testcross.

When the alleles are in the coupling configuration, the most numerous progeny types are those with green thorax and brown puparium and those with purple thorax and black puparium (FIGURE 7.8a); but, when the alleles of the heterozygous parent are in repulsion, the most numerous progeny types are those with green thorax and black puparium and those with purple thorax and brown puparium (FIGURE 7.8b). Notice that the genotypes of the parents in Figure 7.8a and b are the same ($p^+p b^+b \times pp bb$) and that the dramatic difference in the phenotypic ratios of the progeny in the two crosses results entirely from the configuration—coupling or repulsion—of the chromosomes. It is essential to know the arrangement of the alleles on the chromosomes to accurately predict the outcome of crosses in which genes are linked.



7.8 The arrangement of linked genes on a chromosome (coupling or repulsion) affects the results of a testcross. Linked loci in the Australian blowfly, *Lucilia cuprina*, determine the color of the thorax and that of the puparium.

Concepts

In a cross, the arrangement of linked alleles on the chromosomes is critical for determining the outcome. When two wild-type alleles are on one homologous chromosome and two mutant alleles are on the other, they are in the coupling configuration; when each chromosome contains one wild-type allele and one mutant allele, the alleles are in repulsion.

Connecting Concepts**Relating Independent Assortment, Linkage, and Crossing Over**

We have now considered three situations concerning genes at different loci. First, the genes may be located on different chromosomes; in this case, they exhibit independent assortment and combine randomly when gametes are formed. An individual heterozygous at two loci ($AaBb$) produces four types of gametes (AB , ab , Ab , and aB) in equal proportions: two types of nonrecombinants and two types of recombinants.

Second, the genes may be completely linked—meaning that they're on the same chromosome and lie so close together that crossing over between them is rare. In this case, the genes do not recombine. An individual heterozygous for two closely linked genes in the coupling configuration:

$$\begin{array}{cc} A & B \\ \hline a & b \end{array}$$

produces only the nonrecombinant gametes containing alleles AB or ab . The alleles do not assort into new combinations such as Ab or aB .

The third situation, incomplete linkage, is intermediate between the two extremes of independent assortment and complete linkage. Here, the genes are physically linked on the same chromosome, which prevents independent assortment. However, occasional crossovers break up the linkage and allow them to recombine. With incomplete linkage, an individual heterozygous at two loci produces four types of gametes—two types of recombinants and two types of nonrecombinants—but the nonrecombinants are produced more frequently than the recombinants because crossing over does not take place in every meiosis. Linkage and crossing over are two opposing forces: linkage binds alleles at different loci together, restricting their ability to associate freely, whereas crossing over breaks the linkage and allows alleles to assort into new combinations.

Earlier in the chapter, the term recombination was defined as the sorting of alleles into new combinations. We

can now distinguish between two types of recombination that differ in the mechanism that generates these new combinations of alleles.

Interchromosomal recombination is between genes on *different* chromosomes. It arises from independent assortment—the random segregation of chromosomes in anaphase I of meiosis. **Intrachromosomal recombination** is between genes located on the *same* chromosome. It arises from crossing over—the exchange of genetic material in prophase I of meiosis. Both types of recombination produce new allele combinations in the gametes; so they cannot be distinguished by examining the types of gametes produced. Nevertheless, they can often be distinguished by the *frequencies* of types of gametes: interchromosomal recombination produces 50% nonrecombinant gametes and 50% recombinant gametes, whereas intrachromosomal recombination frequently produces less than 50% recombinant gametes. However, when the genes are very far apart on the same chromosome, intrachromosomal recombination also produces 50% recombinant gametes. The two mechanisms are then genetically indistinguishable.

Concepts

Recombination is the sorting of alleles into new combinations. Interchromosomal recombination, produced by independent assortment, is the sorting of alleles on different chromosomes into new combinations. Intrachromosomal recombination, produced by crossing over, is the sorting of alleles on the same chromosome into new combinations.

The Physical Basis of Recombination

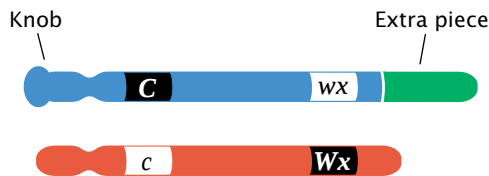
William Sutton's chromosome theory of inheritance, which stated that genes are physically located on chromosomes, was supported by Nettie Stevens and Edmund Wilson's discovery that sex was associated with a specific chromosome in insects (p. 000 in Chapter 4) and Calvin Bridges' demonstration that nondisjunction of X chromosomes was related to the inheritance of eye color in *Drosophila* (p. 000 in Chapter 4). Further evidence for the chromosome theory of heredity came in 1931, when Harriet Creighton and Barbara McClintock (FIGURE 7.9) obtained evidence that intrachromosomal recombination was the result of physical exchange between chromosomes. Creighton and McClintock discovered a strain of corn that had an abnormal chromosome 9, containing a densely staining knob at one end and a small piece of another chromosome attached to the other end. This aberrant chromosome allowed them to visually distinguish the two members of a homologous pair.

They studied the inheritance of two traits in corn determined by genes on chromosome 9: at one locus, a dom-



7.9 Barbara McClintock (left) and Harriet Creighton (right) provided evidence that genes are located on chromosomes. (Karl Maramorosch/Cold Spring Harbor Laboratory Archives.)

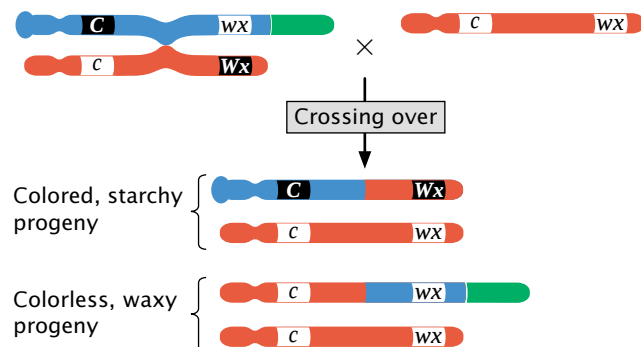
inant allele (*C*) produced colored kernels, whereas a recessive allele (*c*) produced colorless kernels; at another, linked locus, a dominant allele (*Wx*) produced starchy kernels, whereas a recessive allele (*wx*) produced waxy kernels. Creighton and McClintock obtained a plant that was heterozygous at both loci in repulsion, with the alleles for colored and waxy on the aberrant chromosome and the alleles for colorless and starchy on a normal chromosome:



They crossed this heterozygous plant with a plant that was homozygous for colorless and heterozygous for waxy:

$$\frac{C}{c} \frac{wx}{Wx} \times \frac{c}{c} \frac{Wx}{wx}$$

This cross will produce different combinations of traits in the progeny, but the only way that colorless and waxy progeny can arise is through crossing over in the doubly heterozygous parent:



Notice that, if crossing over entails physical exchange between the chromosomes, then the colorless, waxy progeny resulting from recombination should have a chromosome with an extra piece, but not a knob. Furthermore, some of the colored, starchy progeny should possess a knob but not the extra piece. This outcome is precisely what Creighton and McClintock observed, confirming the chromosomal theory of inheritance. Curt Stern provided a similar demonstration by using chromosomal markers in *Drosophila* at about the same time. We will examine the molecular basis of recombination in more detail in Chapter 12.

Predicting the Outcomes of Crosses with Linked Genes

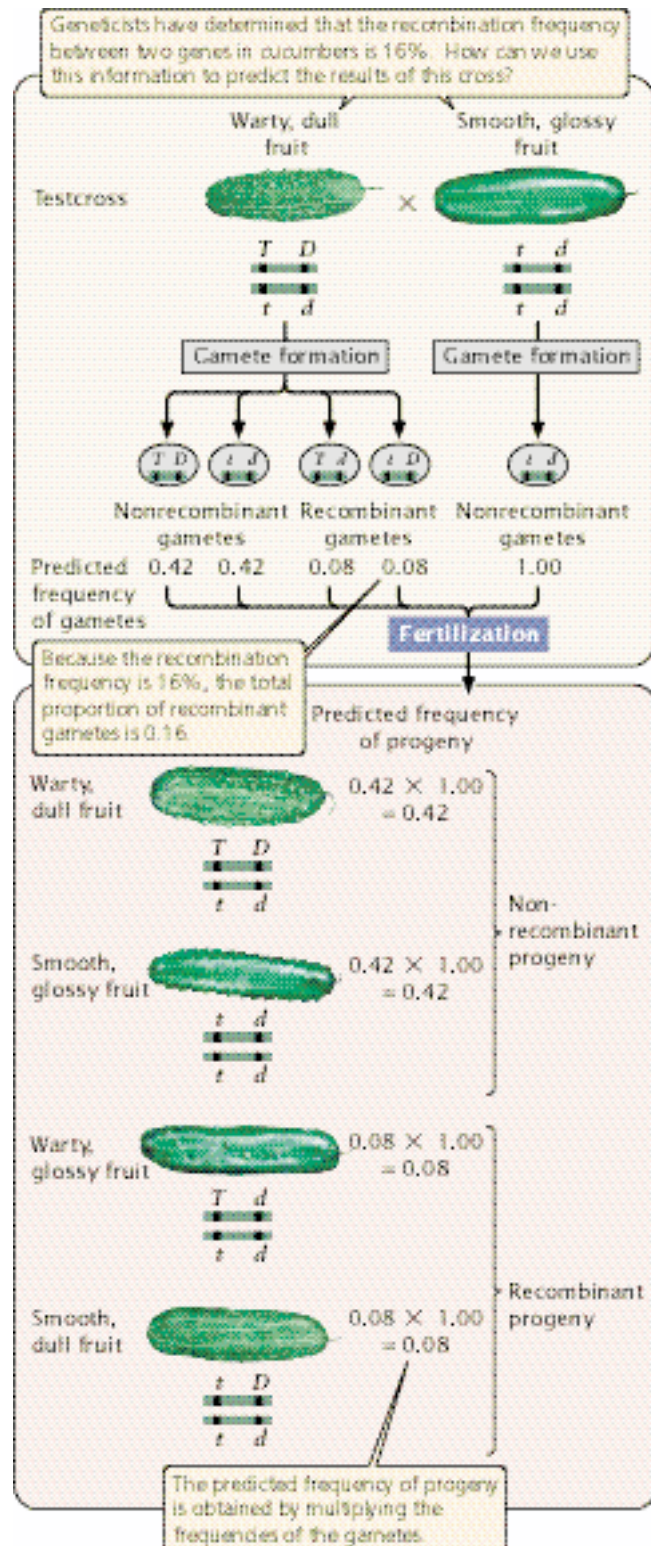
Knowing the arrangement of alleles on a chromosome allows us to predict the types of progeny that will result from a cross entailing linked genes and to determine which of these types will be the most numerous. Determining the *proportions* of the types of offspring requires an additional piece of information—the recombination frequency. The recombination frequency provides us with information about how often the alleles in the gametes appear in new combinations and allows us to predict the proportions of offspring phenotypes that will result from a specific cross entailing linked genes.

In cucumbers, smooth fruit (*t*) is recessive to warty fruit (*T*) and glossy fruit (*d*) is recessive to dull fruit (*D*). Geneticists have determined that these two genes exhibit a recombination frequency of 16%. Suppose we cross a plant homozygous for warty and dull fruit with a plant homozygous for smooth and glossy fruit and then carry out a testcross by using the F_1

$$\frac{T}{t} \frac{D}{d} \times \frac{t}{t} \frac{d}{d}$$

What types and proportions of progeny will result from this testcross?

Four types of gametes will be produced by the heterozygous parent, as shown in (FIGURE 7.10): two types of nonrecombinant gametes ($T D$ and $t d$) and two types of recombinant gametes ($T d$ and $t D$). The recombination frequency tells us that



16% of the gametes produced by the heterozygous parent will be recombinants. Because there are two types of recombinant gametes, each should arise with a frequency of $16\% / 2 = 8\%$. All the other gametes will be nonrecombinants; so they should arise with a frequency of $100\% - 16\% = 84\%$. Because there are two types of nonrecombinant gametes, each should arise with a frequency of $84\% / 2 = 42\%$. The other parent in the testcross is homozygous and therefore produces only a single type of gamete ($t d$) with a probability of 1.00.

The progeny of the cross result from the union of two gametes, producing four types of progeny (see Figure 7.10). The expected proportion of each type can be determined by using the multiplication rule, multiplying together the probability of each uniting gamete. Testcross progeny with warty and dull fruit

$$\frac{T}{t} \quad \frac{D}{d}$$

appear with a frequency of 0.42 (the probability of inheriting a gamete with chromosome $T D$ from the heterozygous parent) $\times$ 1.00 (the probability of inheriting a gamete with chromosome $t d$ from the recessive parent) = 0.42. The proportions of the other types of F_2 progeny can be calculated in a similar manner (see Figure 7.10). This method can be used for predicting the outcome of any cross with linked genes for which the recombination frequency is known.

Testing for Independent Assortment

In some crosses, the genes are obviously linked because there are clearly more nonrecombinants than recombinants. In other crosses, the difference between independent assortment and linkage is not so obvious. For example, suppose we did a testcross for two pairs of genes, such as $AaBb \times aabb$, and observed the following numbers of progeny: 54 $AaBb$, 56 $aabb$, 42 $Aabb$, and 48 $aABb$. Is this outcome a 1:1:1:1 ratio? Not exactly, but it's pretty close. Perhaps these genes are assorting independently and chance produced the slight deviations between the observed numbers and the expected 1:1:1:1 ratio. Alternatively, the genes might be linked, with considerable crossing over taking place between them, and so the number of nonrecombinants is only slightly greater than the number of recombinants. How do we distinguish between the roles of chance and of linkage in producing deviations from the results expected with independent assortment?

We encountered a similar problem in crosses in which genes were unlinked—the problem of distinguishing between deviations due to chance and those due to other

7.10 The recombination frequency allows a prediction of the proportions of offspring expected for a cross entailing linked genes.

factors. We addressed this problem (in Chapter 3) with the goodness-of-fit chi-square test, which serves to evaluate the likelihood that chance alone is responsible for deviations between observed and expected numbers. The chi-square test can also be used to test the goodness of fit between observed numbers of progeny and the numbers expected with independent assortment.

Testing for independent assortment between two linked genes requires the calculation of a series of three chi-square tests. To illustrate this analysis, we will examine the data from a cross between German cockroaches, in which yellow body (y) is recessive to brown body (y^+) and curved wings (cv) are recessive to straight wings (cv^+). A testcross ($y^+y\ cv^+cv \times yy\ cvcv$) produced the following progeny:

63	$y^+y\ cv^+cv$	brown body, straight wings
77	$yy\ cvcv$	yellow body, curved wings
28	$y^+y\ cvcv$	brown body, curved wings
32	$yy\ cv^+cv$	yellow body, straight wings
200	total progeny	

Testing ratios at each locus To determine if the genes for body color and wing shape are assorting independently, we must examine each locus separately and determine whether the observed numbers differ from the expected (we will consider why this step is necessary at the end of this section). At the first locus (for body color), the cross between heterozygote and homozygote ($y^+y \times yy$) is expected to produce $\frac{1}{2} y^+y$ brown and $\frac{1}{2} yy$ yellow progeny; so we expect 100 of each. We observe 63 + 28 = 91 brown progeny and 77 + 32 = 109 yellow progeny. Applying the chi-square test (see Chapter 3) to these observed and expected numbers, we obtain:

$$\begin{aligned}\chi^2 &= \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} \\ \chi^2 &= \frac{(91 - 100)^2}{100} + \frac{(109 - 100)^2}{100} \\ &= \frac{81}{100} + \frac{81}{100} = 0.81 + 0.81 = 1.62\end{aligned}$$

The degrees of freedom associated with the chi-square test (Chapter 3) are $n - 1$, where n equals the number of expected classes. Here, there are two expected phenotypes; so the degree of freedom is $2 - 1 = 1$. Looking up our calculated chi-square value in Table 3.4, we find that the probability associated with this chi-square value is between .30 and .20. Because the probability is above .05 (our critical probability for rejecting the hypothesis that chance produces the difference between observed and expected values), we conclude that there is no significant difference between the 1:1 ratio that we expect in the progeny of the testcross and the ratio that we observed.

We next compare the observed and expected ratios for the second locus, which determines the type of wing. At this locus, a heterozygote and homozygote also were crossed ($cv^+cv \times cvcv$) and are expected to produce $\frac{1}{2} cv^+cv$ straight-winged progeny and $\frac{1}{2} cvcv$ curved-wing progeny. We actually observe 63 + 32 = 95 straight-winged progeny and 77 + 28 = 105 curved-wing progeny; so the calculated chi-square value is:

$$\begin{aligned}\chi^2 &= \frac{(95 - 100)^2}{100} + \frac{(105 - 100)^2}{100} \\ &= \frac{25}{100} + \frac{25}{100} = 0.25 + 0.25 = 0.50\end{aligned}$$

The degree of freedom associated with this chi-square value also is $2 - 1 = 1$, and the associated probability is between .5 and .3. We again assume that there is no significant difference between what we observed and what we expected at this locus in the testcross.

Testing ratios for independent assortment We are now ready to test for the independent assortment of genes at the two loci. If the genes are assorting independently, we can use the multiplication rule to obtain the probabilities and numbers of progeny inheriting different combinations of phenotypes:

Genotypes	Expected phenotypes	Expected proportions	Expected numbers	Observed numbers
$y^+y\ cv^+cv$	brown, straight	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	50	63
$yy\ cvcv$	yellow, curved	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	50	77
$y^+y\ cvcv$	brown, curved	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	50	28
$yy\ cv^+cv$	yellow, straight	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	50	32

The observed and expected numbers of progeny can now be compared by using the chi-square test:

$$\begin{aligned}\chi^2 &= \frac{(63 - 50)^2}{50} + \frac{(77 - 50)^2}{50} + \frac{(28 - 50)^2}{50} \\ &\quad + \frac{(32 - 50)^2}{50} = 34.12\end{aligned}$$

Here, we have four expected classes of phenotypes; so the degrees of freedom equal $4 - 1 = 3$ and the associated probability is considerably less than .001. This very small probability indicates that the phenotypes are not in the proportions that we would expect if independent assortment were taking place. Our conclusion, then, is that these genes are not assorting independently and must be linked.

In summary, testing for linkage between two genes requires a series of chi-square tests: a chi-square test for the segregation of *alleles* at each individual locus, followed by a test for independent assortment between alleles at the different loci. The chi-square tests for segregation at individual loci should always be carried out before testing for independent assortment, because the probabilities expected with independent assortment are based on the probabilities expected at the separate loci. Suppose that the genes in the cockroach example were assorting independently and that some of the cockroaches with curved wings died in embryonic development; the observed proportion with curved wings was then $\frac{1}{3}$ instead of $\frac{1}{2}$. In this case, the proportion of offspring with yellow body and curved wings expected under independent assortment should be $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$ instead of $\frac{1}{4}$. Without the initial chi-square test for segregation at the curved-wing locus, we would have no way of knowing that what we expected with independent assortment was $\frac{1}{6}$ instead of $\frac{1}{4}$. If we carried out only the final test for independent assortment and assumed an expected 1:1:1:1 ratio, we would obtain a high chi-square value. We might conclude, erroneously, that the genes were linked.

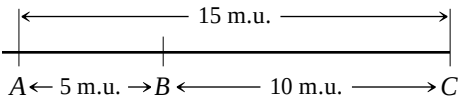
If a significant chi-square (one that has a probability less than 0.05) is obtained in either of the first two tests for segregation, then the final chi-square for independent assortment should not be carried out, because the true expected values are unknown.

Gene Mapping with Recombination Frequencies

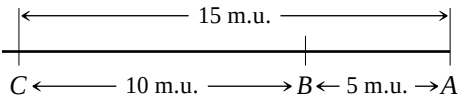
Morgan and his students developed the idea that physical distances between genes on a chromosome are related to the rates of recombination. They hypothesized that crossover events occur more or less at random up and down the chromosome and that two genes that lie far apart are more likely to undergo a crossover than are two genes that lie close together. They proposed that recombination frequencies could provide a convenient way to determine the order of genes along a chromosome and would give estimates of the relative distances between the genes. Chromosome maps calculated by using recombination frequencies are called **genetic maps**. In contrast, chromosome maps based on physical distances along the chromosome (often expressed in terms of numbers of base pairs) are called **physical maps**.

Distances on genetic maps are measured in **map units** (abbreviated m.u.); one map unit equals 1% recombination. Map units are also called **centimorgans** (cM), in honor of Thomas Hunt Morgan; one **morgan** equals 100 m.u. Genetic distances measured with recombination rates are approximately additive: if the distance from gene A to gene B is 5 m.u., the distance from gene B to gene C is 10 m.u., and the distance from gene A to gene C is 15 m.u., gene B must be located between genes A and C. On the

basis of the map distances just given, we could draw a simple genetic map for genes A, B, and C, as shown here:



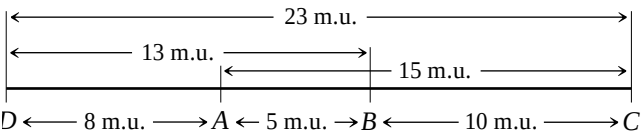
We could just as plausibly draw this map with C on the left and A on the right:



Both maps are correct and equivalent because, with information about the relative positions of only three genes, the most that we can determine is which gene lies in the middle. If we obtained distances to an additional gene, then we could position A and C relative to that gene. An additional gene D, examined through genetic crosses, might yield the following recombination frequencies:

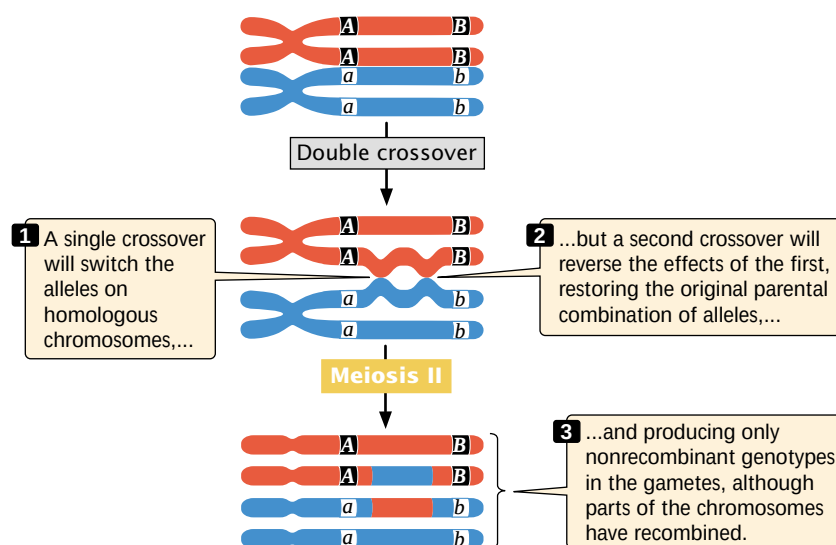
Gene pair	Recombination frequency (%)
A and D	8
B and D	13
C and D	23

Notice that C and D exhibit the greatest amount of recombination; therefore, C and D must be farthest apart, with genes A and B between them. Using the recombination frequencies and remembering that 1 m.u. = 1% recombination, we can now add D to our map:



By doing a series of crosses between pairs of genes, we can construct genetic maps showing the linkage arrangements of a number of genes.

Two points should be emphasized about constructing chromosome maps from recombination frequencies. First, recall that the recombination frequency between two genes cannot exceed 50% and that 50% is also the rate of recombination for genes located on different chromosomes. Consequently, one cannot distinguish between genes on different chromosomes and genes located far apart on the same chromosome. If genes exhibit 50% recombination, the most that can be said about them is that they belong to different groups of linked genes (different linkage groups), either on different chromosomes or far apart on the same chromosome.



A second point is that a testcross for two genes that are relatively far apart on the same chromosome tends to underestimate the true recombination frequency, because the cross does not reveal double crossovers that might take place between the two genes (FIGURE 7.11). A double crossover arises when two separate crossover events take place between the same two loci. Whereas a single crossover switches the alleles on the homologous chromosomes—producing combinations of alleles that were not present on the original parental chromosomes—a second crossover between the same two genes reverses the effects of the first, thus restoring the original parental combination of alleles (see Figure 7.11). Double crossovers produce only nonrecombinant gametes, and we cannot distinguish between the progeny produced by double crossovers and the progeny produced when there is no crossing over. However, as we shall see in the next section, it is possible to detect double crossovers if we examine a third gene that lies between the two crossovers. Because double crossovers between two genes go undetected, recombination frequencies will be underestimated whenever double crossovers take place. Double crossovers are more frequent between genes that are far apart; therefore genetic maps based on short distances are always more accurate than those based on longer distances.

Concepts

A genetic map provides the order of the genes on a chromosome and the approximate distances among the genes based on recombination frequencies. In genetic maps, 1% recombination equals 1 map unit, or 1 centimorgan. Double crossovers between two genes go undetected; so map distances between distant genes tend to underestimate genetic distances.

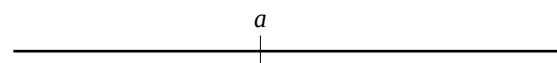
Constructing a Genetic Map with Two-Point Testcrosses

Genetic maps can be constructed by conducting a series of testcrosses between pairs of genes and examining the recombination frequencies between them. A testcross between two genes is called a **two-point testcross** or a two-point cross for short. Suppose that we carried out a series of two-point crosses for four genes, *a*, *b*, *c*, and *d*, and obtained the following recombination frequencies:

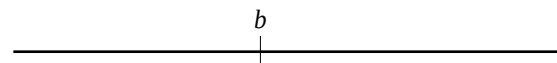
Gene loci in testcross	Recombination frequency (%)
<i>a</i> and <i>b</i>	50
<i>a</i> and <i>c</i>	50
<i>a</i> and <i>d</i>	50
<i>b</i> and <i>c</i>	20
<i>b</i> and <i>d</i>	10
<i>c</i> and <i>d</i>	28

We can begin constructing a genetic map for these genes by considering the recombination frequencies for each pair of genes. The recombination frequency between *a* and *b* is 50%, which is the recombination frequency expected with independent assortment. Genes *a* and *b* may therefore either be on different chromosomes or be very far apart on the same chromosome; so we will place them in different linkage groups with the understanding that they may or may not be on the same chromosome:

Linkage group 1

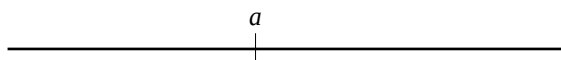


Linkage group 2

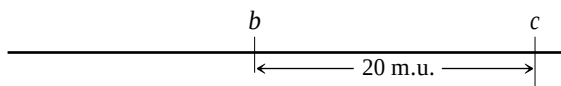


The recombination frequency between *a* and *c* is 50%, indicating that they, too, are in different linkage groups. The recombination frequency between *b* and *c* is 20%; so these genes are linked and separated by 20 map units:

Linkage group 1



Linkage group 2

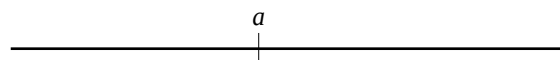


The recombination frequency between *a* and *d* is 50%, indicating that these genes belong to different linkage groups, whereas genes *b* and *d* are linked, with a recombination frequency of 10%. To decide whether gene *d* is 10 map units to the left or right of gene *b*, we must consult the *c*-to-*d* distance. If gene *d* is 10 map units to the left of gene *b*, then the distance between *d* and *c* should be 20 m.u. + 10 m.u. = 30 m.u. This distance will be only approximate because any double crossovers between the two genes will be missed and the recombination frequency will be underestimated. If, on the other hand, gene *d* lies to the right of gene *b*, then the distance between gene *d* and *c* will be much shorter, approximately 20 m.u. – 10 m.u. = 10 m.u.

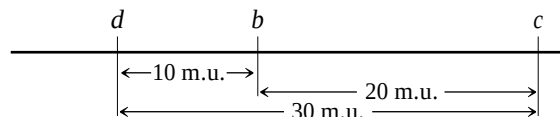
By examining the recombination frequency between *c* and *d*, we can distinguish between these two possibilities. The recombination frequency between *c* and *d* is 28%; so gene *d* must lie to the left of gene *b*. Notice that the sum of the recombination between *d* and *b* (10%) and between *b* and *c* (20%) is greater than the recombination between

d and *c* (28%). (This is what was meant by saying that recombination rates—i.e., map units—are *approximately* additive.) This discrepancy arises because double crossovers between the two outer genes go undetected, causing an underestimation of the true recombination frequency. The genetic map of these genes is now complete:

Linkage group 1



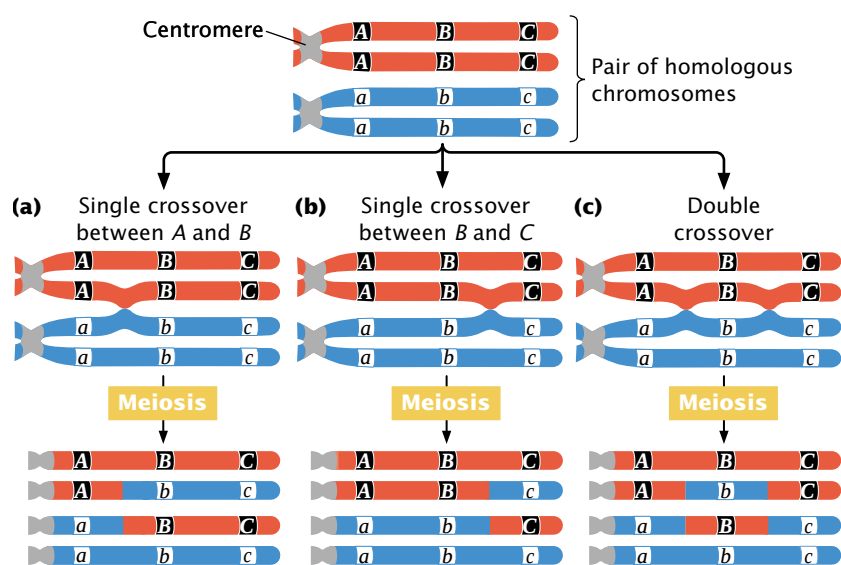
Linkage group 2



Linkage and Recombination Between Three Genes

Genetic maps can be constructed from a series of testcrosses for pairs of genes, but this approach is not particularly efficient, because numerous two-point crosses must be carried out to establish the order of the genes and because double crossovers are missed. A more efficient mapping technique is a testcross for three genes (a **three-point testcross**, or three-point cross). With a three-point cross, the order of the three genes can be established in a single set of progeny and some double crossovers can usually be detected, providing more accurate map distances.

Consider what happens when crossing over takes place among three hypothetical linked genes. **FIGURE 7.12** illustrates a pair of homologous chromosomes from an



Conclusion: Recombinant chromosomes resulting from the double crossover have only the middle gene altered.

7.12 Three types of crossovers can take place among three linked loci.

individual that is heterozygous at three loci ($AaBbCc$). Notice that the genes are in the coupling configuration; that is, all the dominant alleles are on one chromosome ($\underline{A \quad B \quad C}$) and all the recessive alleles are on the other chromosome ($\underline{a \quad b \quad c}$). Three types of crossover events can take place between these three genes: two types of single crossovers (see Figure 7.12a and b) and a double crossover (see Figure 7.12c). In each type of crossover, two of the resulting chromosomes are recombinants and two are nonrecombinants.

Notice that, in the recombinant chromosomes resulting from the double crossover, the outer two alleles are the same as in the nonrecombinants, but the middle allele is different. This result provides us with an important clue about the order of the genes. In progeny that result from a double crossover, only the middle allele should differ from the alleles present in the nonrecombinant progeny.

Gene Mapping with the Three-Point Testcross

To examine gene mapping with a three-point testcross, we will consider three recessive mutations in the fruit fly *Drosophila melanogaster*. In this species, scarlet eyes (st) are recessive to red eyes (st^+), ebony body color (e) is recessive to gray body color (e^+), and spineless (ss)—that is, the presence of small bristles—is recessive to normal bristles (ss^+). All three mutations are linked and located on the third chromosome.

We will refer to these three loci as st , e , and ss , but keep in mind that either recessive *alleles* (st , e , and ss) or the dominant *alleles* (st^+ , e^+ , and ss^+) may be present at each locus. So, when we say that there are 10 m.u. between st and ss , we mean that there are 10 m.u. between the loci at which these mutations occur; we could just as easily say that there are 10 m.u. between st^+ and ss^+ .

To map these genes, we need to determine their order on the chromosome and the genetic distances between them. First, we must set up a three-point testcross, a cross between a fly heterozygous at all three loci and a fly homozygous for recessive *alleles* at all three loci. To produce flies heterozygous for all three loci, we might cross a stock of flies that are homozygous for normal *alleles* at all three loci with flies that are homozygous for recessive *alleles* at all three loci:

$$\begin{array}{rcc}
 \text{P} & \frac{st^+ \quad e^+ \quad ss^+}{st^+ \quad e^+ \quad ss^+} & \times \quad \frac{st \quad e \quad ss}{st \quad e \quad ss} \\
 & \downarrow & \\
 \text{F}_1 & \frac{st^+ \quad e^+ \quad ss^+}{st \quad e \quad ss} &
 \end{array}$$

The order of the genes has been arbitrarily assigned because at this point we do not know which is the middle gene.

Additionally, the alleles in these heterozygotes are in coupling configuration (because all the wild-type dominant *alleles* were inherited from one parent and all the recessive mutations from the other parent), although the testcross can also be done with genes in repulsion.

In the three-point testcross, we cross the F_1 heterozygotes with flies that are homozygous for all three recessive mutations. In many organisms, it makes no difference whether the heterozygous parent in the testcross is male or female (provided that the genes are autosomal) but, in *Drosophila*, no crossing over takes place in males. Because crossing over in the heterozygous parent is essential for determining recombination frequencies, the heterozygous flies in our testcross must be female. So we mate female F_1 flies that are heterozygous for all three traits with male flies that are homozygous for all the recessive traits:

$$\frac{st^+ \quad e^+ \quad ss^+}{st \quad e \quad ss} \text{ female} \quad \times \quad \frac{st \quad e \quad ss}{st \quad e \quad ss} \text{ male}$$

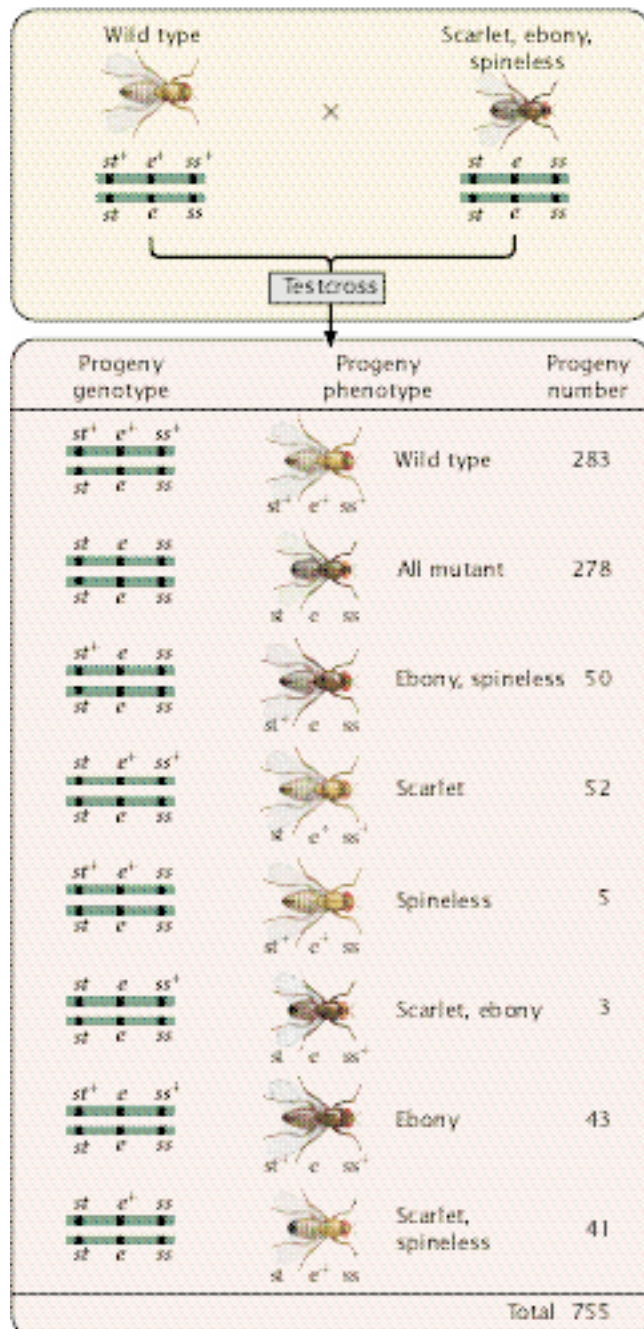
The progeny produced from this cross are listed in **FIGURE 7.13**. For each locus, two classes of progeny are produced: progeny that are heterozygous, displaying the dominant trait, and progeny that are homozygous, displaying the recessive trait. With two classes of progeny possible for each of the three loci, there will be $2^3 = 8$ classes of phenotypes possible in the progeny. In this example, all eight phenotypic classes are present but, in some three-point crosses, one or more of the phenotypes may be missing if the number of progeny is limited. Nevertheless, the absence of a particular class can provide important information about which combination of traits is least frequent and ultimately the order of the genes, as we will see.

To map the genes, we need information about where and how often crossing over has occurred. In the homozygous recessive parent, the two alleles at each locus are the same; and so crossing over will have no effect on the types of gametes produced; with or without crossing over, all gametes from this parent have a chromosome with three recessive alleles ($\underline{st \quad e \quad ss}$). In contrast, the heterozygous parent has different alleles on its two chromosomes; so crossing over can be detected. The information that we need for mapping, therefore, comes entirely from the gametes produced by the heterozygous parent. Because chromosomes contributed by the homozygous parent carry only recessive alleles, whatever alleles are present on the chromosome contributed by the heterozygous parent will be expressed in the progeny.

As a shortcut, we usually do not write out the complete genotypes of the testcross progeny, listing instead only the alleles expressed in the phenotype (as shown in Figure 7.13), which are the alleles inherited from the heterozygous parent.

Concepts

To map genes, information about the location and number of crossovers in the gametes that produced the progeny of a cross is needed. An efficient way to obtain this information is use a three-point testcross, in which an individual heterozygous at three linked loci is crossed with an individual that is homozygous recessive at the three loci.

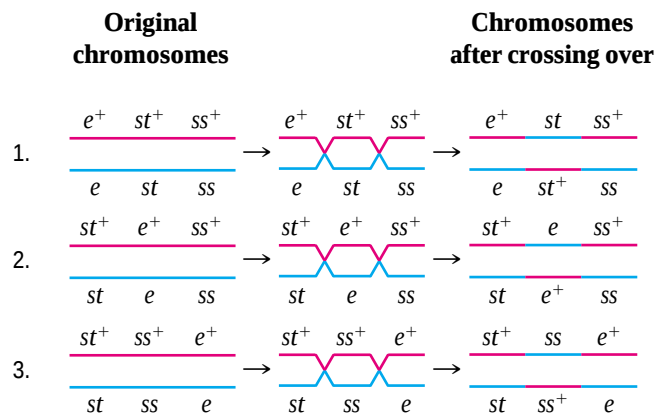


Determining the gene order The first task in mapping the genes is to determine their order on the chromosome. In Figure 7.13, we arbitrarily listed the loci in the order st , e , ss , but we had no way of knowing which of the three loci was between the other two. We can now identify the middle locus by examining the double-crossover progeny.

First, determine which progeny are the nonrecombinants—they will be the two most-numerous classes of progeny. (Even if crossing over takes place in every meiosis, the nonrecombinants will comprise at least 50% of the progeny.) Among the progeny of the testcross in Figure 7.13, the most numerous are those with all three dominant traits ($st^+ e^+ ss^+$) and those with all three recessive traits ($st e ss$).

Next, identify the double-crossover progeny. These should always be the two least-numerous phenotypes, because the probability of a double crossover is always less than the probability of a single crossover. The least-common progeny among those listed in Figure 7.13 are progeny with red eyes, gray body, and spineless bristles ($st^+ e^+ ss$) and progeny with scarlet eyes, ebony body, and normal bristles ($st e ss^+$); so they are the double-crossover progeny.

Three orders of genes are possible: the eye-color locus could be in the middle ($e st ss$), the body-color locus could be in the middle ($st e ss$), or the bristle locus could be in the middle ($st ss e$). To determine which gene is in the middle, we can draw the chromosomes of the heterozygous parent with all three possible gene orders and then see if a double crossover produces the combination of genes observed in the double-crossover progeny. The three possible gene orders and the types of progeny produced by their double crossovers are:



7.13 The results of a three-point testcross can be used to map linked genes. In this three-point testcross in *Drosophila melanogaster*, the recessive mutations scarlet eyes (st), ebony body color (e), and spineless bristles (ss) are at three linked loci. The order of the loci has been designated arbitrarily, as has the sex of the progeny flies.

The only gene order that produces chromosomes with alleles for the traits observed in the double crossovers ($st^+ e^+ ss$ and $st e ss^+$) is the third one, where the locus for bristle shape lies in the middle. Therefore, this order ($st \quad ss \quad e$) must be the correct sequence of genes on the chromosome.

With a little practice, it's possible to quickly determine which locus is in the middle without writing out all the gene orders. The phenotypes of the progeny are expressions of the alleles inherited from the heterozygous parent. Recall that, when we looked at the results of double crossovers (see Figure 7.13), only the alleles at the middle locus differed from the nonrecombinants. If we compare the nonrecombinant progeny with double-crossover progeny, they should differ only in alleles of the middle locus.

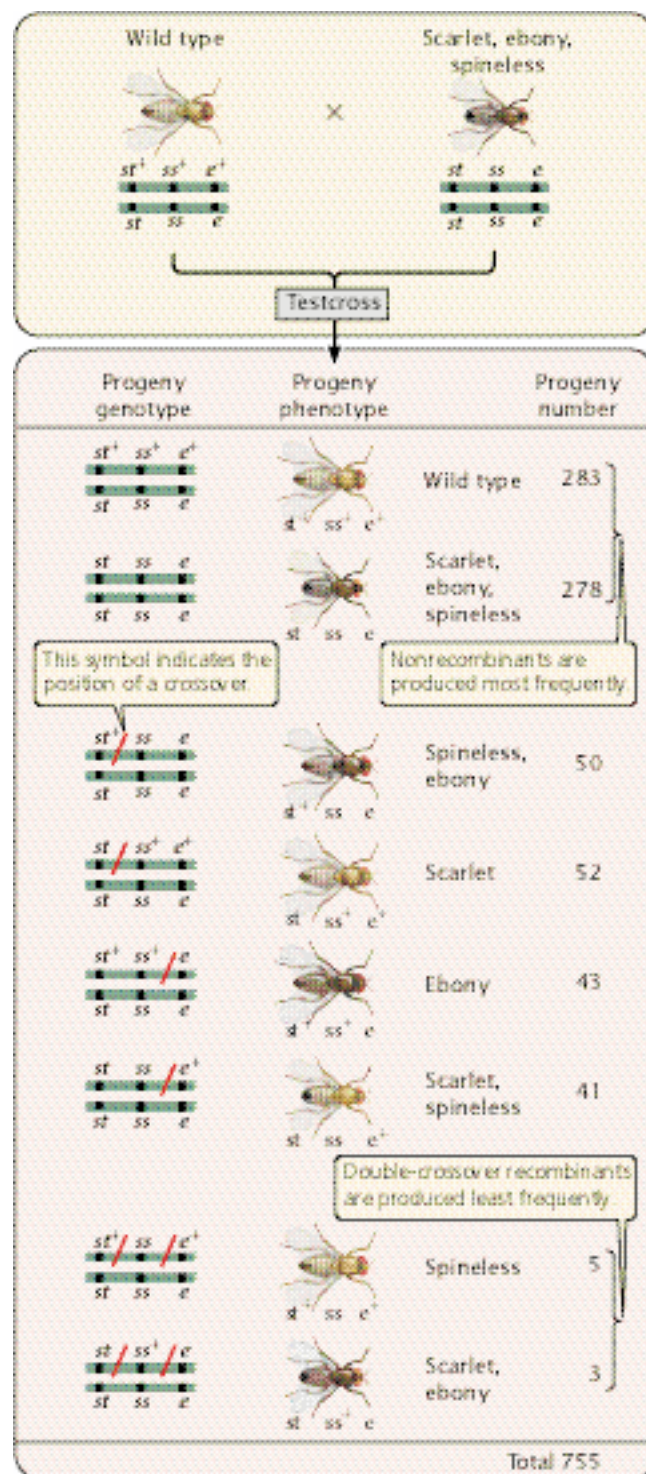
Let's compare the alleles in the double-crossover progeny $st^+ e^+ ss$ with those in the nonrecombinant progeny $st^+ e^+ ss^+$. We see that both have an allele for red eyes (st^+) and both have an allele for gray body (e^+), but the nonrecombinants have an allele for normal bristles (ss^+), whereas the double crossovers have an allele for spineless bristles (ss). Because the bristle locus is the only one that differs, it must lie in the middle. We would obtain the same results if we compared the other class of double-crossover progeny ($st e ss^+$) with other nonrecombinant progeny ($st e ss$). Again the only trait that differs is the one for bristles. Don't forget that the nonrecombinants and the double crossovers should differ only at one locus; if they differ in two loci, the wrong classes of progeny are being compared.

Concepts

To determine the middle locus in a three-point cross, compare the double-crossover progeny with the nonrecombinant progeny. The double crossovers will be the two least-common classes of phenotypes; the nonrecombinants will be the two most-common classes of phenotypes. The double-crossover progeny should have the same alleles as the nonrecombinant types at two loci and different alleles at the locus in the middle.

Determining the locations of crossovers When we know the correct order of the loci on the chromosome, we should rewrite the phenotypes of the testcross progeny in Figure 7.13 with the loci in the correct order so that we can determine where crossovers have taken place (FIGURE 7.14).

Among the eight classes of progeny, we have already identified two classes as nonrecombinants ($st^+ ss^+ e^+$ and $st ss e$) and two classes as double crossovers ($st^+ ss e^+$ and $st ss^+ e$). The other four classes include progeny that resulted from a chromosome that underwent a single crossover: two underwent single



7.14 Writing the results of a three-point testcross with the loci in the correct order allows the locations of crossovers to be determined.

These results are from the testcross illustrated in Figure 7.13, with the loci shown in the correct order. The location of a crossover is indicated with a slash (/). The sex of the progeny flies has been designated arbitrarily.

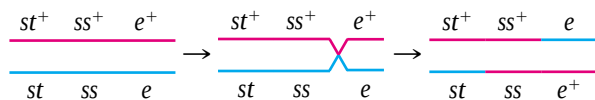
crossovers between *st* and *ss*, and two underwent single crossovers between *ss* and *e*.

To determine where the crossovers took place in these progeny, compare the alleles found in the single-crossover progeny with those found in the nonrecombinants, just as we did for the double crossovers. Some of the alleles in the single-crossover progeny are derived from one of the original (non-recombinant) chromosomes of the heterozygous parent, but at some place there is a switch (due to crossing over) and the remaining alleles are derived from the homologous non-recombinant chromosome. The position of the switch indicates where the crossover event took place. For example, consider progeny with chromosome $\underline{st^+ \quad ss \quad e}$. The first allele (st^+) came from the nonrecombinant chromosome $\underline{st^+ \quad ss^+ \quad e^+}$ and the other two alleles (*ss* and *e*) must have come from the other nonrecombinant chromosome $\underline{st \quad ss \quad e}$ through crossing over:



This same crossover also produces the $\underline{st \quad ss^+ \quad e^+}$ progeny.

This same method can be used to determine the location of crossing over in the other two types of single-crossover progeny. Crossing between *ss* and *e* produces $\underline{st^+ \quad ss^+ \quad e}$ and $\underline{st \quad ss \quad e^+}$ chromosomes:



We now know the locations of all the crossovers; their locations are marked with a slash in Figure 7.14.

Calculating the recombination frequencies Next, we can determine the map distances, which are based on the frequencies of recombination. Recombination frequency is calculated by adding up all of the recombinant progeny, dividing this number by the total number of progeny from the cross, and multiplying the number obtained by 100%. To determine the map distances accurately, we must include all crossovers (both single and double) that take place between two genes.

Recombinant progeny that possess a chromosome that underwent crossing over between the eye-color locus (*st*) and the bristle locus (*ss*) include the single crossovers ($\underline{st^+ / ss \quad e}$ and $\underline{st / ss^+ \quad e^+}$) and the two double crossovers ($\underline{st^+ / ss / e^+}$ and $\underline{st / ss^+ / e}$); see Figure 7.14. There are a total of 755 progeny; so the recombination frequency between *ss* and *st* is:

$$st-ss \text{ recombination frequency} = \frac{(50 + 52 + 5 + 3)}{755} \times 100\% = 14.6\%$$

The distance between the *st* and *ss* loci can be expressed as 14.6 m.u.

The map distance between the bristle locus (*ss*) and the body locus (*e*) is determined in the same manner. The recombinant progeny that possess a crossover between *ss* and *e* are the single crossovers $\underline{st^+ \quad ss^+ / e}$ and $\underline{st \quad ss / e^+}$, and the double crossovers $\underline{st^+ / ss / e^+}$ and $\underline{st / ss^+ / e}$. The recombination frequency is:

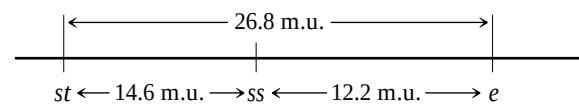
$$ss-e \text{ recombination frequency} = \frac{(43 + 41 + 5 + 3)}{755} \times 100\% = 12.2\%$$

Thus, the genetic distance between *ss* and *e* is 12.2 m.u.

Finally, calculate the genetic distance between the outer two loci, *st* and *e*. Add up all the progeny with crossovers between the two loci. These progeny include those with a single crossover between *st* and *ss*, those with a single crossover between *ss* and *e*, and the double crossovers ($\underline{st^+ / ss / e^+}$ and $\underline{st / ss^+ / e}$). Because the double crossovers have two crossovers between *st* and *e*, we must add the double crossovers *twice* to determine the recombination frequency between these two loci:

$$st-e \text{ recombination frequency} = \frac{(50 + 52 + 43 + 41 + (2 \times 5) + (2 \times 3))}{755} \times 100\% = 26.8\%$$

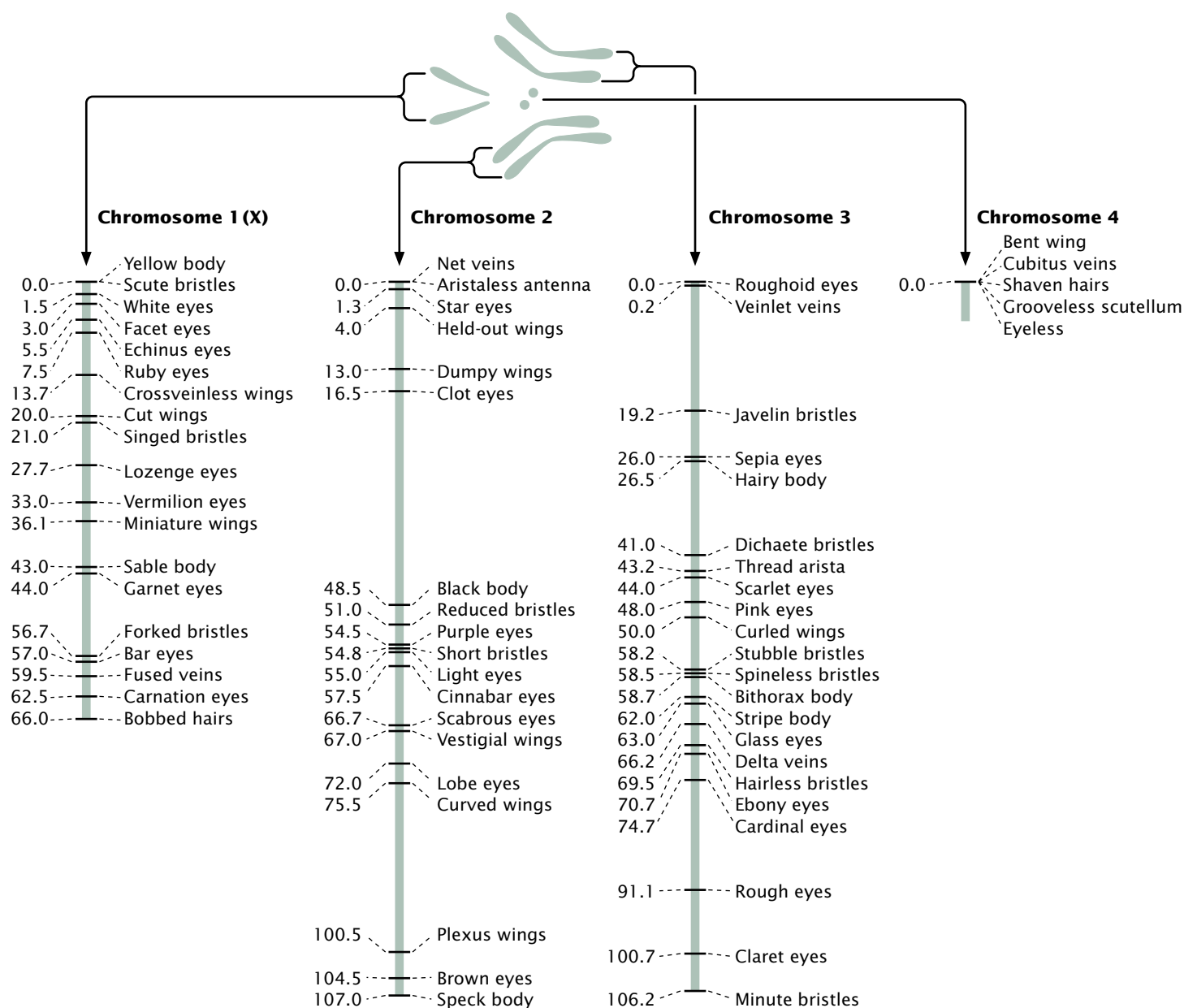
Notice that the distances between *st* and *ss* (14.6 m.u.) and between *ss* and *e* (12.2 m.u.) add up to the distance between *st* and *e* (26.8 m.u.). We can now use the map distances to draw a map of the three genes on the chromosome:



A genetic map of *D. melanogaster* is illustrated in **FIGURE 7.15**.

Interference and coefficient of coincidence Map distances give us information not only about the physical distances that separate genes, but also about the proportions of recombinant and nonrecombinant gametes that will be produced in a cross. For example, knowing that genes *st* and *ss* on the third chromosome of *D. melanogaster* are separated by a distance of 14.6 m.u. tells us that 14.6% of the gametes produced by a fly heterozygous at these two loci will be recombinants. Similarly, 12.2% of the gametes from a fly heterozygous for *ss* and *e* will be recombinants.

Theoretically, we should be able to calculate the proportion of double-recombinant gametes by using the multiplication rule of probability (Chapter 3), which states that the probability of two independent events occurring



7.15 *Drosophila melanogaster* has four linkage groups corresponding to its four pairs of chromosomes. Distances between genes within a linkage group are in map distances.

together is the multiplication of their independent probabilities. Applying this principle, we should find that the proportion (probability) of gametes with double crossovers between *st* and *e* is equal to the probability of recombination between *st* and *ss*, multiplied by the probability of recombination between *ss* and *e*, or $0.146 \times 0.122 = 0.0178$. Multiplying this probability by the total number of progeny gives us the *expected* number of double-crossover progeny from the cross: $0.0178 \times 755 = 13.4$. Only 8 double crossovers—considerably fewer than the 13 expected—were observed in the progeny of the cross (see Figure 7.13).

This phenomenon is common in eukaryotic organisms. The calculation assumes that each crossover event is independent and that the occurrence of one crossover does not influence the occurrence of another. But crossovers are frequently *not* independent events: the occurrence of one tends to inhibit additional crossovers in the same region of the chromosome, and so double crossovers are less frequent than expected.

The degree to which one crossover interferes with additional crossovers in the same region is termed the **interference**. To calculate the interference, we first determine the

coefficient of coincidence, which is the ratio of observed double crossovers to expected double crossovers:

$$\text{coefficient of coincidence} = \frac{\text{number of observed double crossovers}}{\text{number of expected double crossovers}}$$

For the loci that we mapped on the third chromosome of *D. melanogaster* (see Figure 7.14), we find that:

$$\text{coefficient of coincidence} = \frac{5 + 3}{0.146 \times 0.122 \times 755} = \frac{8}{13.4} = 0.6$$

which indicates that we are actually observing only 60% of the double crossovers that we expected on the basis of the single-crossover frequencies. The interference is calculated as:

$$\text{interference} = 1 - \text{coefficient of coincidence}$$

So the interference for our three-point cross is:

$$\text{interference} = 1 - 0.6 = 0.4$$

This value of interference tells us that 40% of the double-crossover progeny expected will not be observed because of interference. When interference is complete and no double-crossover progeny are observed, the coefficient of coincidence is 0 and the interference is 1.

Sometimes *more* double-crossover progeny appear than expected, which happens when a crossover increases the probability of another crossover occurring nearby. In this case, the coefficient of coincidence is greater than 1 and the interference will be negative.

Concepts

The coefficient of coincidence equals the number of double crossovers observed, divided by the number of double crossovers expected on the basis of the single-crossover frequencies. The interference equals $1 - \text{the coefficient of coincidence}$; it indicates the degree to which one crossover interferes with additional crossovers.

Connecting Concepts

Stepping Through the Three-Point Cross

We have now examined the three-point cross in considerable detail, seeing how the information derived from the cross can be used to map a series of three linked genes. Let's briefly review the steps required to map genes from a three-point cross.

1. Write out the phenotypes and numbers of progeny produced in the three-point cross. The progeny phenotypes will be easier to interpret if you use allelic symbols for the traits (such as $st^+ e^+ ss$).

2. Write out the genotypes of the original parents used to produce the triply heterozygous individual in the test-cross and, if known, the arrangement of the alleles on their chromosomes (coupling or repulsion).

3. Determine which phenotypic classes among the progeny are the nonrecombinants and which are the double crossovers. The nonrecombinants will be the two most-common phenotypes; the double crossovers will be the two least-common phenotypes.

4. Determine which locus lies in the middle. Compare the alleles present in the double crossovers with those present in the nonrecombinants; each class of double crossovers should be like one of the nonrecombinants for two loci and should differ for one locus. The locus that differs is the middle one.

5. Rewrite the phenotypes with genes in correct order.

6. Determine where crossovers must have taken place to give rise to the progeny phenotypes by comparing each phenotype with the phenotype of the nonrecombinant progeny.

7. Determine the recombination frequencies. Add the numbers of the progeny that possess a chromosome with a crossover between a pair of loci. Add the double crossovers to this number. Divide this sum by the total number of progeny from the cross, and multiply by 100%; the result is the recombination frequency between the loci, which is the same as the map distance.

8. Draw a map of the three loci, indicating which locus lies in the middle, and label the distances between them.

9. Determine the coefficient of coincidence and the interference. The coefficient of coincidence is the number of observed double-crossover progeny divided by the number of expected double-crossover progeny. The expected number can be obtained by multiplying the product of the two single-recombination probabilities by the total number of progeny in the cross.

Worked Problem

In *D. melanogaster*, cherub wings (*ch*), black body (*b*), and cinnabar eyes (*cn*) result from recessive alleles that are all located on chromosome 2. A homozygous wild-type fly was mated with a cherub, black, and cinnabar fly, and the resulting F_1 females were test-crossed with cherub, black, and cinnabar males. The following progeny were produced from the testcross:

<i>ch b⁺ cn</i>	105
<i>ch⁺ b⁺ cn⁺</i>	750
<i>ch⁺ b cn</i>	40
<i>ch⁺ b⁺ cn</i>	4
<i>ch b cn</i>	753
<i>ch b⁺ cn⁺</i>	41
<i>ch⁺ b cn⁺</i>	102
<i>ch b cn⁺</i>	5
total	1800

- (a) Determine the linear order of the genes on the chromosome (which gene is in the middle).
 (b) Calculate the recombinant distances between the three loci.
 (c) Determine the coefficient of coincidence and the interference for these three loci.

• **Solution**

(a) We can represent the crosses in this problem as follows:

$$\begin{array}{l}
 \text{P} \quad \frac{ch^+ \ b^+ \ cn^+}{ch^+ \ b^+ \ cn^+} \times \frac{ch \ b \ cn}{ch \ b \ cn} \\
 \quad \quad \quad \downarrow \\
 \text{F}_1 \quad \frac{ch^+ \ b^+ \ cn^+}{ch \ b \ cn} \\
 \text{Testcross} \quad \frac{ch^+ \ b^+ \ cn^+}{ch \ b \ cn} \times \frac{ch \ b \ cn}{ch \ b \ cn}
 \end{array}$$

Note that we do not know, at this point, the order of the genes; we have arbitrarily put *b* in the middle.

The next step is to determine which of the testcross progeny are nonrecombinants and which are double crossovers. The nonrecombinants should be the most-frequent phenotype; so they must be the progeny with phenotypes encoded by $\frac{ch^+ \ b^+ \ cn^+}{ch \ b \ cn}$ and $\frac{ch \ b \ cn}{ch \ b \ cn}$. These genotypes are consistent with the genotypes of the parents, which we outlined earlier. The double crossovers are the least-frequent phenotypes and are encoded by $\frac{ch^+ \ b^+ \ cn}{ch \ b \ cn}$ and $\frac{ch \ b \ cn^+}{ch \ b \ cn}$.

We can determine the gene order by comparing the alleles present in the double crossovers with those present in the nonrecombinants. The double-crossover progeny should be like one of the nonrecombinants at two loci and unlike it at one; the allele that differs should be in the middle. Compare the double-crossover progeny $\frac{ch \ b \ cn^+}{ch \ b \ cn}$ with the nonrecombinant $\frac{ch \ b \ cn}{ch \ b \ cn}$. Both have cherub wings (*ch*) and black body (*b*), but the double-crossover progeny have wild-type eyes (*cn⁺*), whereas the nonrecombinants have cinnabar eyes (*cn*). The locus that determines cinnabar eyes must be in the middle.

(b) To calculate the recombination frequencies among the genes, we first write the phenotypes of the progeny with the genes encoding them in the correct order. We have already identified the nonrecombinant and double-crossover progeny; so the other four progeny types must have resulted from single crossovers. To determine *where* single crossovers took place, we compare the alleles found in the single-crossover progeny with those in the nonrecombinants. Crossing over must have taken place where the alleles switch from those found in one nonrecombinant to those found in the other nonrecombinant. The locations of the crossovers are indicated with a slash:

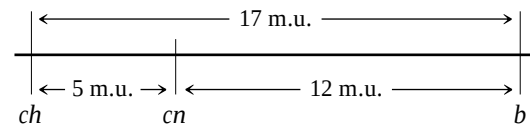
<i>ch</i>	<i>cn</i>	/	<i>b</i> ⁺	105	single crossover	
<i>ch</i> ⁺	<i>cn</i> ⁺		<i>b</i> ⁺	750	nonrecombinant	
<i>ch</i> ⁺	/	<i>cn</i>	<i>b</i>	40	single crossover	
<i>ch</i> ⁺	/	<i>cn</i>	/	<i>b</i> ⁺	4	double crossover
<i>ch</i>	<i>cn</i>		<i>b</i>	753	nonrecombinant	
<i>ch</i>	/	<i>cn</i> ⁺	<i>b</i> ⁺	41	single crossover	
<i>ch</i> ⁺	<i>cn</i> ⁺	/	<i>b</i>	102	single crossover	
<i>ch</i>	/	<i>cn</i> ⁺	/	<i>b</i>	5	double crossover
total				1800		

Next, we determine the recombination frequencies and draw a genetic map:

$$\text{ch-cn recombination frequency} = \frac{40 + 41 + 4 + 5}{1800} \times 100\% = 5\%$$

$$\text{cn-b recombination frequency} = \frac{105 + 4 + 102 + 5}{1800} \times 100\% = 12\%$$

$$\text{ch-b recombination frequency} = \frac{105 + 40 + (2 \times 4) + 41 + 102 + (2 \times 5)}{1800} \times 100\% = 17\%$$



(c) The coefficient of coincidence is the number of observed double crossovers, divided by the number of expected double crossovers. The number of expected double crossovers is obtained by multiplying the probability of a crossover between *ch* and *cn* (0.05) $\times$ the probability of a crossover between *cn* and *b* (0.12) $\times$ the total number of progeny in the cross (1800):

$$\text{coefficient of coincidence} = \frac{4 + 5}{0.05 \times 0.12 \times 1800} = 0.83$$

Finally, the interference is equal to 1 – the coefficient of coincidence:

$$\text{interference} = 1 - 0.83 = 0.17$$

Gene Mapping in Humans

Efforts in mapping the human genome are hampered by the inability to perform desired crosses and the small number of progeny in most human families. Geneticists are restricted to analyses of pedigrees, which are often incomplete and provide limited information. Nevertheless, techniques have been developed that use pedigree data to analyze linkage, and a large number of human traits have been successfully mapped with the use of these methods. Because the number of progeny from any one mating is usually small, data from several families and pedigrees are usually combined to test for independent assortment. The methods used in these types of analysis are beyond the scope of this book, but an example will illustrate how linkage can be detected from pedigree data.

One of the first documented demonstrations of linkage in humans was between the locus for nail–patella syndrome and the locus that determines the ABO blood types. Nail–patella syndrome is an autosomal dominant disorder characterized by abnormal fingernails and absent or rudimentary kneecaps. The ABO blood types are determined by an autosomal locus with multiple *alleles* (Chapter 5). Linkage between the genes encoding these traits was established in families in which both traits segregate. Part of one such family is illustrated in **FIGURE 7.16**.

Nail–patella syndrome is relatively rare; so we can assume that people having this trait are heterozygous (*Nn*); unaffected people are homozygous (*nn*). The ABO genotypes can be inferred from the phenotypes and the types of offspring produced. Person I-2 in Figure 7.16, for example,

has blood type B, which has two possible genotypes: $I^B I^B$ or $I^B i$ (see Figure 5.6). Because some of her offspring are blood type O (genotype *ii*) and must therefore have inherited an *i* allele from each parent, female I-2 must have genotype $I^B i$. Similarly, the presence of blood type O offspring in generation II indicates that male I-1, with blood type A, also must carry an *i* allele and therefore has genotype $I^A i$. The ABO and nail–patella genotypes for all persons in the pedigree are given below the squares and circles.

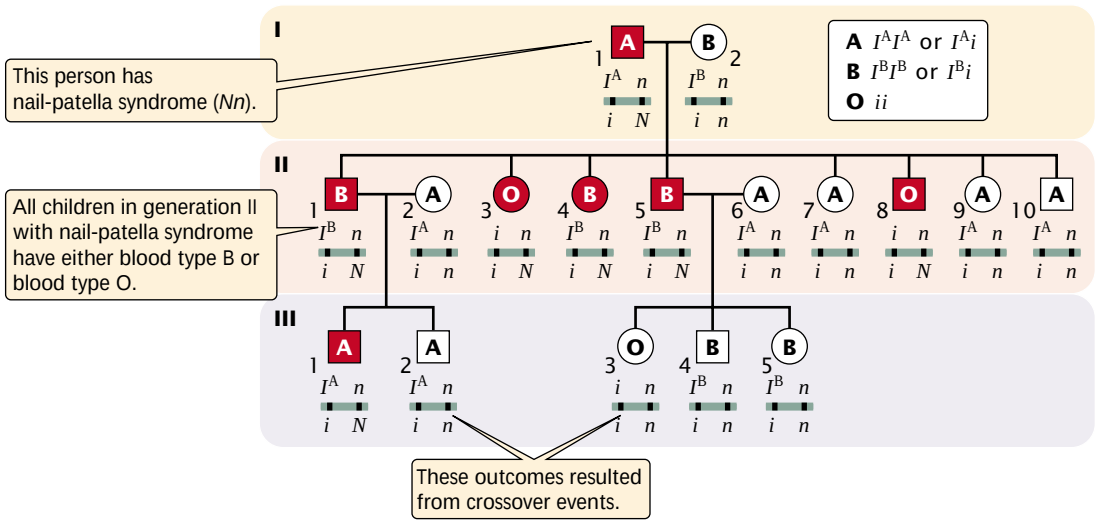
From generation II, we can see that the genes for nail–patella syndrome and the blood types do not appear to assort independently. The parents of this family are:

$$I^A i N n \times I^B i n n$$

If the genes coding for nail–patella syndrome and the ABO blood types assorted independently, we would expect that some children in generation II would have blood type A and nail–patella syndrome, inheriting both the I^A and *N* genes from their father. However, all children in generation II with nail–patella syndrome have either blood type B or blood type O; all those with blood type A have normal nails and kneecaps. This outcome indicates that the arrangements of the alleles on the chromosomes of the crossed parents are:

$$\frac{I^A}{i} \frac{n}{N} \times \frac{I^B}{i} \frac{n}{n}$$

There is no recombination among the offspring of these parents (generation II), but there are two instances of



7.16 Linkage between ABO blood types and nail–patella syndrome was established by examining families in whom both traits segregate. The pedigree shown here is for one such family. Solid circles and squares represent the presence of nail–patella syndrome; the ABO blood type is indicated in each circle or square. The genotype, inferred from phenotype, is given below each square or circle.

recombination among the persons in generation III. Individuals II-1 and II-2 have the following genotypes:

$$\frac{I^B}{i} \quad \frac{n}{N} \quad \times \quad \frac{I^A}{i} \quad \frac{n}{n}$$

Their child III-2 has blood type A and does not have nail–patella syndrome; so he must have genotype:

$$\frac{I^A}{i} \quad \frac{n}{n}$$

and must have inherited both the *i* and the *n* alleles from his father. These alleles are on different chromosomes in the father; so crossing over must have taken place. Crossing over also must have taken place to produce child III-3.

In the pedigree of Figure 7.16, there are 12 children from matings in which the genes encoding nail–patella syndrome and ABO blood types segregate; 2 of them are recombinants. On this basis, we might assume that the loci for nail–patella syndrome and ABO blood types are linked, with a recombination frequency of $2/12 = 0.167$. However, it is possible that the genes *are* assorting independently and that the small number of children just makes it seem as though the genes are linked. To determine the probability that genes are actually linked, geneticists often calculate **lod** (logarithm of odds) **scores**.

To obtain a lod score, one calculates both the probability of obtaining the observations with a specified degree of linkage and the probability of obtaining the observations with independent assortment. One then determines the ratio of these two probabilities, and the logarithm of this ratio is the lod score. Suppose that the probability of obtaining a particular set of observations with linkage and a certain recombination frequency is 0.1 and that the probability of obtaining the same observations with independent assortment is 0.0001. The ratio of these two probabilities is $0.1/0.0001 = 1000$, the logarithm of which (the lod score) is 3. Thus linkage with the specified recombination is 1000 times as likely to produce what was observed as independent assortment. A lod score of 3 or higher is usually considered convincing evidence for linkage.

Mapping with Molecular Markers

For many years, gene mapping was limited in most organisms by the availability of **genetic markers**, variable genes with easily observable phenotypes whose inheritance could be studied. Traditional genetic markers include genes that encode easily observable characteristics such as flower color, seed shape, blood types, and biochemical differences. The paucity of these types of characteristics in many organisms limited mapping efforts.

In the 1980s, new molecular techniques made it possible to examine variations in DNA itself, providing an

almost unlimited number of genetic markers that can be used for creating genetic maps and studying linkage relations. The earliest of these molecular markers consisted of restriction fragment length polymorphisms (RFLPs), variations in DNA sequence detected by cutting the DNA with restriction enzymes (see Chapter 18). Later, methods were developed for detecting variable numbers of short DNA sequences repeated in tandem, called variable number of tandem repeats (VNTRs). More recently, DNA sequencing allows the direct detection of individual variations in the DNA nucleotides, called single nucleotide polymorphisms (SNPs; see Chapter 19). All of these methods have expanded the availability of genetic markers and greatly facilitated the creation of genetic maps.

Gene mapping with molecular markers is done essentially in the same manner as mapping performed with traditional phenotypic markers: the cosegregation of two or more markers is studied and map distances are based on the rates of recombination between markers. These methods and their use in mapping are presented in more detail in Chapter 18.

Physical Chromosome Mapping

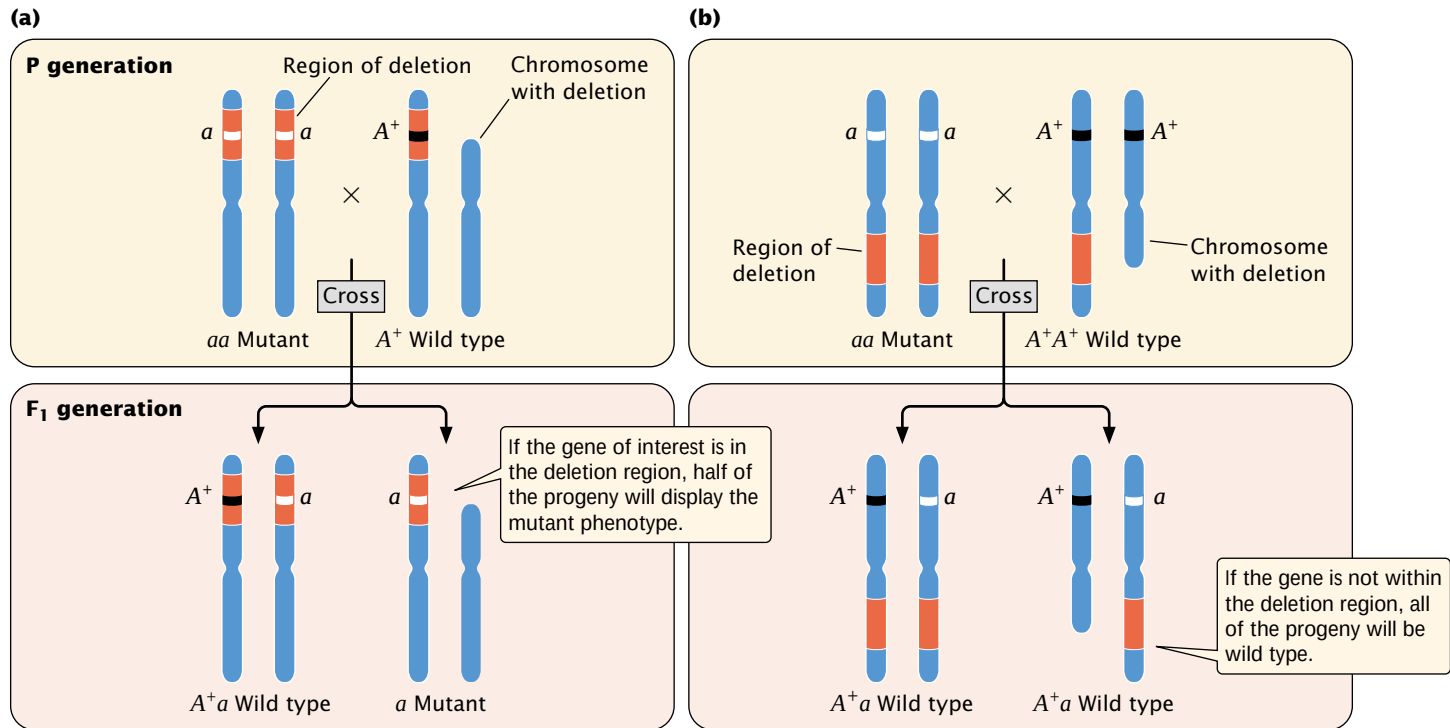
Genetic maps reveal the relative positions of genes on a chromosome on the basis of frequencies of crossing over, but they do not provide information that can allow us to place groups of linked genes on particular chromosomes. Furthermore, the units of a genetic map do not always precisely correspond to physical distances on the chromosome, because a number of factors other than physical distances between genes (such as the type and sex of the organism) can influence rates of crossing over. Because of these limitations, physical-mapping methods that do not rely on rates of crossing over have been developed.

Deletion Mapping

One method for determining the chromosomal location of a gene is **deletion mapping**. Special staining methods have been developed that make it possible to detect chromosome deletions, mutations in which a part of a chromosome is missing. Genes are assigned to regions of particular chromosomes by studying the association of a gene's phenotype or product and particular chromosome deletions.

In deletion mapping, an individual that is homozygous for a recessive mutation in the gene of interest is crossed with an individual that is heterozygous for a deletion ( **FIGURE 7.17**). If the gene of interest is in the region of the chromosome represented by the deletion (the red part of chromosome in Figure 7.17), approximately half of the progeny will display the mutant phenotype (see Figure 7.17a). If the gene is not within the deleted region, all of the progeny will be wild type (see Figure 7.17b).

Deletion mapping has been used to reveal the chromosomal locations of a number of human genes. For example,



7.17 Deletion mapping can be used to determine the chromosomal location of a gene. An individual homozygous for a recessive mutation in the gene of interest (*aa*) is crossed with an individual heterozygous for a deletion.

Duchenne muscular dystrophy is a disease that causes progressive weakening and degeneration of the muscles. From its X-linked pattern of inheritance, the mutated allele causing this disorder was known to be on the X chromosome, but its precise location was uncertain. Examination of a number of patients having Duchenne muscular dystrophy, who also possessed small deletions, allowed researchers to position the gene to a small segment of the short arm of the X chromosome.

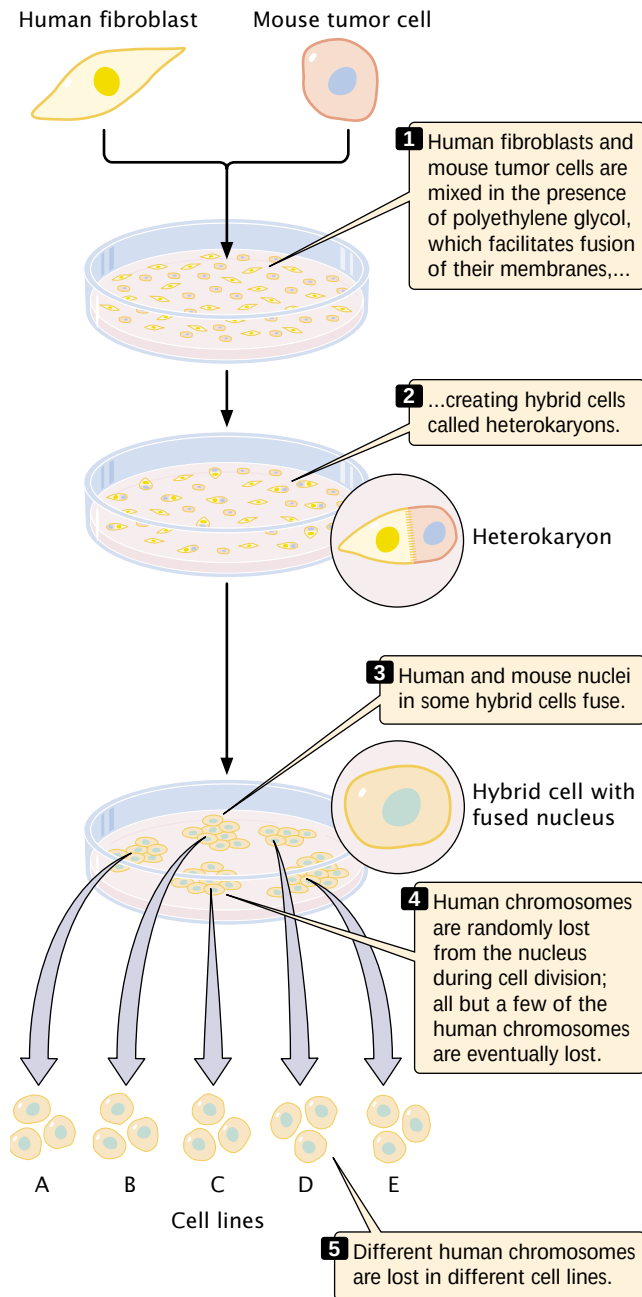
Somatic-Cell Hybridization

Another method used for positioning genes on chromosomes is **somatic cell hybridization**, which requires the fusion of different types of cells. Most mature somatic (nonsex) cells can undergo only a limited number of divisions and therefore cannot be grown continuously. However, cells that have been altered by viruses or derived from tumors that have lost the normal constraints on cell division will divide indefinitely; these types of cells can be cultured in the laboratory and are referred to as a **cell line**.

Cells from two different cell lines can be fused by treating them with polyethylene glycol or other agents that alter their plasma membranes. After fusion, the cell possesses two nuclei and is called a **heterokaryon**. The two nuclei of a heterokaryon eventually also fuse, generating a hybrid cell that contains chromosomes from both cell lines. If human

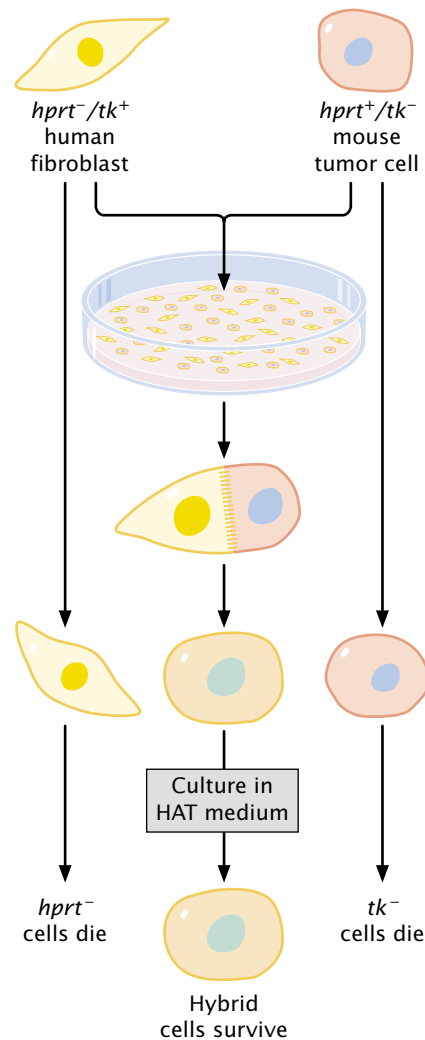
and mouse cells are mixed in the presence of polyethylene glycol, fusion results in human–mouse somatic-cell hybrids (FIGURE 7.18). The hybrid cells tend to lose chromosomes as they divide and, for reasons that are not understood, chromosomes from one of the species are lost preferentially. In human–mouse somatic-cell hybrids, the human chromosomes tend to be lost, whereas the mouse chromosomes are retained. Eventually, the chromosome number stabilizes when all but a few of the human chromosomes have been lost. Chromosome loss is random and differs among cell lines. The presence of these “extra” human chromosomes in the mouse genome makes it possible to assign human genes to specific chromosomes.

In the first step of this procedure, hybrid cells must be separated from original parental cells that have not undergone hybridization. This separation is accomplished by using a selection method that allows hybrid cells to grow while suppressing the growth of parental cells. The most commonly used method is called HAT selection (FIGURE 7.19), which stands for *hypoxanthine*, *aminopterin*, and *thymidine*, three chemicals that are used to select for hybrid cells. In the presence of HAT medium, a cell must possess two enzymes to synthesize DNA: thymidine kinase (TK) and hypoxanthine-guanine phosphoribosyl transferase (HPRT). Cells that are *tk⁻* or *hprt⁻* cannot synthesize DNA and will not grow on HAT medium. The mouse cells used in the hy-



7.18 Somatic-cell hybridization can be used to determine which chromosome contains a gene of interest.

bridization procedure are deficient in TK, but can produce HPRT (the cells are $tk^- hprt^+$); the human cells can produce TK but are deficient for HPRT (they are $tk^+ hprt^-$). On HAT medium, the mouse cells do not survive, because they are tk^- ; the human cells do not survive, because they are $hprt^-$. Hybrid cells, on the other hand, inherit the ability to make HPRT from the mouse cell and the ability to make TK from the human cell; thus, they produce both enzymes (the cells are $tk^+ hprt^+$) and will grow on HAT medium.



7.19 HAT medium can be used to separate human–mouse hybrid cells from the original hybridized cells.

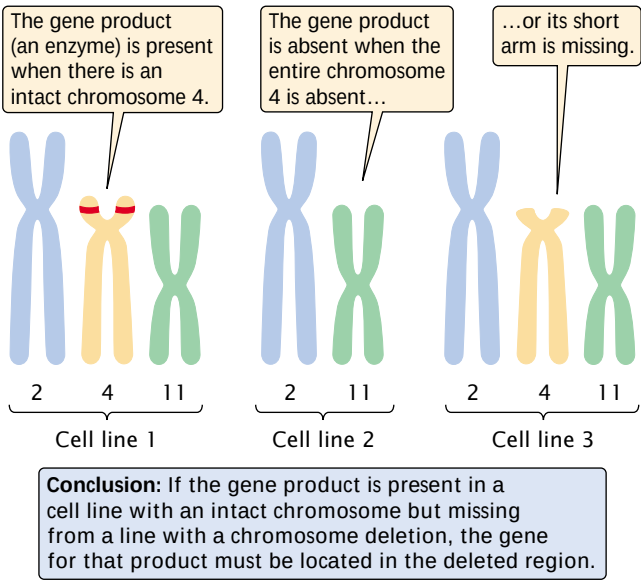
To map genes using somatic-cell hybridization requires the use of a panel of different hybrid cell lines. The cell lines of the panel differ in the human chromosomes that they have retained. For example, one cell line might possess human chromosomes 2, 4, 7, and 8, whereas another might possess chromosomes 4, 19, and 20. Each cell line in the panel is examined for evidence of a particular human gene. The human gene can be detected either by looking for the protein that it produces or by looking for the gene itself with the use of molecular probes (discussed in Chapter 18). Correlation of the presence of the gene with the presence of specific human chromosomes often allows the gene to be assigned to the correct chromosome. For example, if a gene was detected in both of the aforementioned cell lines, the gene must be on chromosome 4, because it is the only human chromosome common to both cell lines (FIGURE 7.20).

Human chromosomes present																								
Cell line	Gene product present	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	X
A	+		+		+			+	+															
B	+	+	+		+			+	+	+	+	+	+											
C	-															+		+		+			+	
D	+		+		+		+	+	+															
E	-												+								+			
F	+				+															+	+			

7.20 Somatic-cell hybridization is used to assign a gene to a particular human chromosome. A panel of six cell lines, each line containing a different subset of human chromosomes, is examined for the presence of the gene product (such as an enzyme). A plus sign means that the gene product is present; a minus sign means that the gene product is missing. Four of the cell lines (A, B, D, and F) have the gene product, indicating that the gene is present on one of the chromosomes found in these cell lines. The only chromosome common to all four of these cell lines is chromosome 4, indicating that the gene is located on this chromosome.

Two genes determined to be on the same chromosome with the use of somatic-cell hybridization are said to be **syntenic genes**. This term is used because syntenic genes may or may not exhibit linkage in the traditional genetic sense—remember that two genes can be located on the same chromosome but may be so far apart that they assort independently. Syntenic refers to genes that are physically linked, regardless of whether they exhibit genetic linkage. (Synteny is sometimes also used to refer to

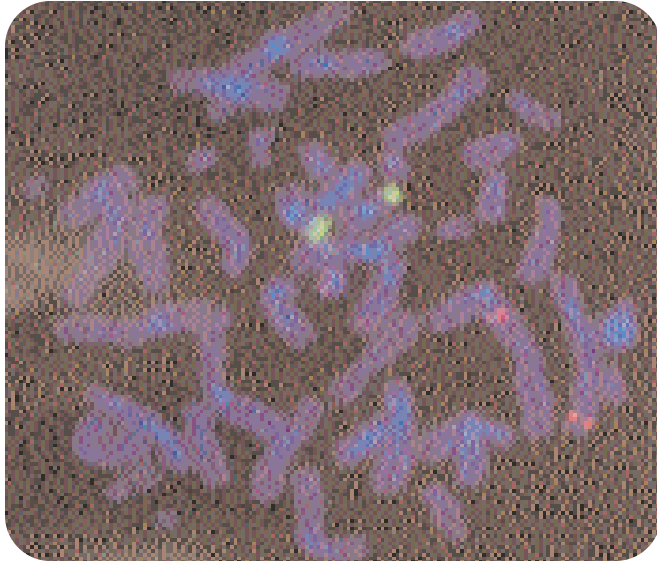
gene loci in different organisms located on a chromosome region of common evolutionary origin.) Sometimes somatic-cell hybridization can be used to position a gene on a specific part of a chromosome. Some hybrid cell lines carry a human chromosome with a chromosome mutation such as a deletion or a translocation. If the gene is present in a cell line with the intact chromosome but missing from a line with a chromosome deletion, the gene must be located in the deleted region (FIGURE 7.21) Similarly, if a gene is usually absent from a chromosome but consistently appears whenever a translocation (a piece of another chromosome that has broken off and attached itself to the chromosome in question) is present, it must be present on the translocated part of the chromosome.



7.21 Genes can be localized to a specific part of a chromosome by using somatic-cell hybridization.

In Situ Hybridization Described in more detail in Chapter 18, in situ hybridization is another method for determining the chromosomal location of a particular gene. This method requires a DNA copy of the gene or its RNA product, which is used to make a molecule (called a probe) that is complementary to the gene of interest. The probe is made radioactive or is attached to a special molecule that fluoresces under ultra-violet (UV) light and is added to chromosomes from specially treated cells that have been spread on a microscope slide. The probe binds to the complementary DNA sequence of the gene on the chromosome. The presence of radioactivity or fluorescence from the bound probe reveals the location of the gene on a particular chromosome (FIGURE 7.22a). The use of fluorescence in situ hybridiza-

(a)



(b)



7.22 In situ hybridization is another technique for determining the chromosomal location of a gene. (a) FISH technique: in this case, the bound probe reveals sequences associated with the centromere. (b) SKY technique: 24 different probes, each specific for a different human chromosome and producing a different color, identify the different human chromosomes. (Courtesy of Dr. Hesus Padilla-Nash and Dr. Thomas Ried, NIH.)

tion (FISH) has been widely used to identify the chromosomal location of human genes. In spectral karyotyping (SKY) (**FIGURE 7.22b**), a set of 24 FISH probes, each specific to a different human chromosome and attached to a molecule that fluoresces a different color, allows each chromosome in a karyotype to be identified.

www.whfreeman.com/pierce
hybridization (FISH)

More on fluorescence in situ

Mapping by DNA Sequencing

Another means of physically mapping genes is to determine the sequence of nucleotides in the DNA (DNA sequencing, Chapter 19). With this technique, physical distances between genes are measured in numbers of base pairs. Continuous sequences can be determined for only relatively small fragments of DNA; so, after sequencing, some method is still required to map the individual fragments. This mapping is often done by using the traditional gene mapping that examines rates of crossing over between molecular markers located on the fragments. It can also be accomplished by generating a set of overlapping fragments, sequencing each fragment, and then aligning the fragments by using a computer program that identifies the overlap in the sequence of adjacent fragments. With these methods, complete physical maps of entire genomes have been produced (Chapter 19).

Concepts

Physical-mapping methods determine the physical locations of genes on chromosomes and include deletion mapping, somatic-cell hybridization, in situ hybridization, and direct DNA sequencing.

Connecting Concepts Across Chapters



The principle of independent assortment states that alleles at different loci assort (separate) independently in meiosis but only if the genes are located on different chromosomes or are far apart on the same chromosome. This chapter has focused on the inheritance of genes that are physically linked on the same chromosome and do not assort independently. To predict the outcome of crosses entailing linked genes, we must consider not only the genotypes of the parents but also the physical arrangement of the alleles on the chromosomes.

An important principle learned in this chapter is that rates of recombination are related to the physical distances between genes. Crossing over is more frequent between genes that are far apart than between genes that lie close together. This fact provides the foundation for gene mapping in eukaryotic organisms: recombination frequencies are used to determine the relative order and distances between

linked genes. Gene mapping therefore requires the setting up of crosses in which recombinant progeny can be detected.

This chapter also examined several methods of physical mapping that do not rely on recombination rates but use methods to directly observe the association between genes and particular chromosomes or to position genes by determining the nucleotide sequences. Although genetic and physical distances are correlated, they are not identical, because factors other than the distances between genes can influence rates of crossing over.

Gene mapping requires a firm understanding of the behavior of chromosomes (Chapter 2) and basic principles

of heredity (Chapters 3 through 5). The discussion of gene mapping with pedigrees assumes knowledge of how families are displayed in pedigrees (Chapter 6). In Chapter 8, we will consider specialized mapping techniques used in bacteria and viruses; in Chapter 18, techniques for detecting molecular markers used in gene mapping are examined in more detail. Techniques for mapping whole genomes are discussed in Chapter 19. Chromosome mutations that play a role in deletion mapping and somatic-cell hybridization are considered in more detail in Chapter 9.

CONCEPTS SUMMARY

- Soon after Mendel's principles were rediscovered, examples of genes that did not assort independently were discovered. These genes were subsequently shown to be linked on the same chromosome.
- In a testcross for two completely linked genes (which exhibit no crossing over), only nonrecombinant progeny containing the original combinations of alleles present in the parents are produced. When two genes assort independently, recombinant progeny and nonrecombinant progeny are produced in equal proportions. When two genes are linked with some crossing over between them, more nonrecombinant progeny than recombinant progeny are produced.
- Because a single crossover between two linked genes produces two recombinant gametes and two nonrecombinant gametes, crossing over and independent assortment produce the same results.
- Recombination frequency is calculated by summing the number of recombinant progeny, dividing by the total number of progeny produced in the cross, and multiplying by 100%.
- The recombination frequency is half the frequency of crossing over, and the maximum frequency of recombinant gametes is 50%.
- When two wild-type alleles are found on one homologous chromosome and their mutant alleles are found on the other chromosome, the genes are said to be in coupling configuration. When one wild-type allele and one mutant allele are found on each homologous chromosome, the genes are said to be in repulsion. Whether genes are in coupling configuration or in repulsion determines which combination of phenotypes will be most frequent in the progeny of a testcross.
- Linkage and crossing over are two opposing forces: linkage keeps alleles at different loci together, whereas crossing over breaks up linkage and allows alleles to recombine into new associations.
- Interchromosomal recombination takes place among genes located on different chromosomes and occurs through the random segregation of chromosomes in meiosis. Intrachromosomal recombination takes place among genes located on the same chromosome and occurs through crossing over.
- Testing for independent assortment between genes requires a series of chi-square tests, in which segregation is first tested at each locus individually, followed by testing for independent assortment among genes at the different loci.
- Recombination rates can be used to determine the relative order of genes and distances between them on a chromosome. Maps based on recombination rates are called genetic maps; maps based on physical distances are called physical maps.
- One percent recombination equals one map unit, which is also a centiMorgan.
- When genes exhibit 50% recombination, they belong to different linkage groups, which may be either on different chromosomes or far apart on the same chromosome.
- Recombination rates between two genes will underestimate the true distance between them because double crossovers cannot be detected.
- Genetic maps can be constructed by examining recombination rates from a series of two-point crosses or by examining the progeny of a three-point testcross.
- Gene mapping in humans can be accomplished by examining the cosegregation of traits in pedigrees, although the inability to control crosses and the small number of progeny in many families limit mapping with this technique.
- A lod score is obtained by calculating the logarithm of the ratio of the probability of obtaining the observed progeny

with a specified degree of linkage to the probability of obtaining the observed progeny with independent assortment. A lod score of 3 or higher is usually considered evidence for linkage.

- Molecular techniques that allow the detection of variable differences in DNA sequence have greatly facilitated gene mapping.
- In deletion mapping, genes are physically associated with particular chromosomes by studying the expression of recessive mutations in heterozygotes that possess chromosome deletions.
- In somatic-cell hybridization, cells from two different cell lines (human and rodent) are fused. The resulting hybrid cells initially contain chromosomes from both species but randomly lose different human chromosomes. The hybrid cells are examined for the presence of specific genes; if a human gene is present in the hybrid cell, it must be present on one of the human chromosomes in that the cell line.
- With in situ hybridization, a radioactive or fluorescence label is added to a fragment of DNA that is complementary to a specific gene. This probe is then added to specially prepared chromosomes, where it pairs with the gene of interest. The presence of the label on a particular chromosome reveals the physical location of the gene.
- Nucleotide sequencing is another method of physically mapping genes.

IMPORTANT TERMS

linked genes (p. 161)	coupling (cis) configuration (p. 167)	map unit (m.u.) (p. 172)	deletion mapping (p. 183)
linkage group (p. 161)	repulsion (trans) configuration (p. 167)	centimorgan (p. 172)	somatic-cell hybridization (p. 184)
nonrecombinant (parental) gamete (p. 163)	interchromosomal recombination (p. 168)	morgan (p. 172)	cell line (p. 184)
nonrecombinant (parental) progeny (p. 163)	intrachromosomal recombination (p. 168)	two-point testcross (p. 173)	heterokaryon (p. 184)
recombinant gamete (p. 163)	genetic map (p. 172)	three-point testcross (p. 174)	syntenic gene (p. 186)
recombinant progeny (p. 163)	physical map (p. 172)	interference (p. 179)	
recombination frequency (p. 166)		coefficient of coincidence (p. 180)	
		lod score (p. 183)	
		genetic marker (p. 183)	

Worked Problems

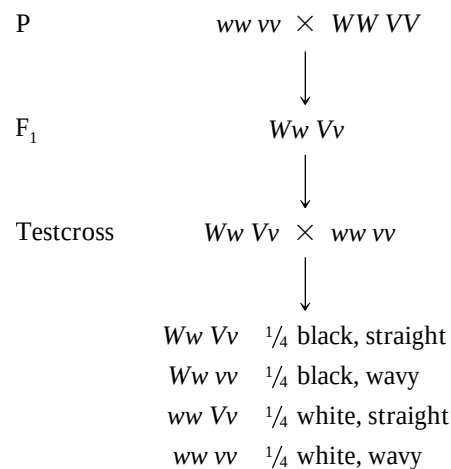
1. In guinea pigs, white coat (*w*) is recessive to black coat (*W*) and wavy hair (*v*) is recessive to straight hair (*V*). A breeder crosses a guinea pig that is homozygous for white coat and wavy hair with a guinea pig that is black with straight hair. The F_1 are then crossed with guinea pigs having white coats and wavy hair in a series of testcrosses. The following progeny are produced from these testcrosses:

black, straight	30
black, wavy	10
white, straight	12
white, wavy	31
total	83

- (a) Are the genes that determine coat color and hair type assorting independently? Carry out chi-square tests to test your hypothesis.
- (b) If the genes are not assorting independently, what is the recombination frequency between them?

• Solution

(a) Assuming independent assortment, outline the crosses conducted by the breeder:



Because a total of 83 progeny were produced in the testcrosses, we expect $\frac{1}{4} \times 83 = 20.75$ of each. The observed numbers of progeny from the testcross (30, 10, 12, 31) do not appear to fit the expected numbers (20.75, 20.75, 20.75, 20.75) well; so independent assortment may not have occurred.

To test the hypothesis, carry out a series of three chi-square tests. First, look at each locus separately and determine if the

observed numbers fit those expected from the testcross. For the locus determining coat color, crossing $Ww \times ww$ is expected to produce $\frac{1}{2}$ Ww (black) and $\frac{1}{2}$ ww (white) progeny, or 41.5 of a total of 83 progeny. Ignoring the hair type, we find that $30 + 10 = 40$ black progeny and $12 + 31 = 43$ white progeny were observed. Thus, the observed and expected values for this chi-square test are:

Phenotype	Observed	Expected
black	40	41.5
white	43	41.5

The chi-square value is:

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$
$$= \frac{(40 - 41.5)^2}{41.5} + \frac{(43 - 41.5)^2}{41.5} = 0.108$$

The degrees of freedom for the chi-square goodness-of-fit test are $n - 1$, where n equals the number of expected classes. There are two expected classes (black and white) so the degree of freedom is $2 - 1 = 1$. On the basis of the calculated chi-square value in Table 3.4, the probability associated with this chi-square is greater than .05 (the critical probability for rejecting the hypothesis that chance might be responsible for the differences between observed and expected numbers); so the black and white progeny appear to be in the 1:1 ratio expected in a testcross.

Next, compute a second chi-square value comparing the number of straight and wavy progeny with the numbers expected from the testcross. From the $Vv \times vv$, $\frac{1}{2}$ Vv (straight) and $\frac{1}{2}$ vv (wavy) progeny are expected:

Phenotype	Observed	Expected
straight	42	41.5
wavy	41	41.5

$$\chi^2 = \frac{(42 - 41.5)^2}{41.5} + \frac{(41 - 41.5)^2}{41.5} = 0.012$$

$$\text{degrees of freedom} = n - 1 = 2 - 1 = 1$$

In Table 3.4, the probability associated with this chi-square value is much greater than .05; so straight and wavy progeny are in a 1:1 ratio.

Having established that the observed numbers for each trait do not differ from the numbers expected from the testcross, we next test for independent assortment. With independent assortment, 20.75 of each phenotype are

expected; so the observed and expected numbers and the associated chi-square value are:

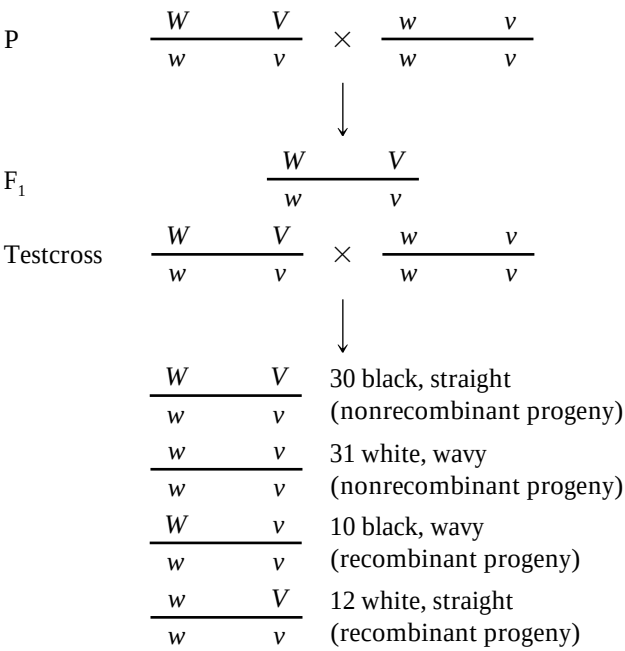
Phenotype	Observed	Expected
black, straight	30	20.75
black, wavy	10	20.75
white, straight	12	20.75
white, wavy	31	20.75

$$\chi^2 = \frac{(30 - 20.75)^2}{20.75} + \frac{(10 - 20.75)^2}{20.75} + \frac{(12 - 20.75)^2}{20.75} + \frac{(31 - 20.75)^2}{20.75}$$
$$= 118.44$$

$$\text{degrees of freedom} = n - 1 = 4 - 1 = 3$$

In Table 3.4, the associated probability is much less than .05, indicating that chance is very unlikely to be responsible for the differences between the observed numbers and the numbers expected with independent assortment. The genes for coat color and hair type have therefore not assorted independently.

(b) To determine the recombination frequencies, identify the recombinant progeny. Using the notation for linked genes, write the crosses:



The recombination frequency is:

$$\frac{\text{number of recombinant progeny}}{\text{total number progeny}} \times 100\%$$

or

$$\begin{aligned}\text{recombination frequency} &= \frac{10 + 12}{30 + 10 + 12 + 31} \times 100\% \\ &= \frac{22}{83} \times 100 = 26.5\%\end{aligned}$$

2. A series of two-point crosses entailed seven loci (*a*, *b*, *c*, *d*, *e*, *f*, and *g*), producing the following recombination frequencies. Using these recombination frequencies, map the seven loci, showing their linkage groups and the order and distances between the loci of each linkage group:

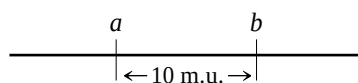
Loci	Recombination frequency (%)	Loci	Recombination frequency (%)
<i>a</i> and <i>b</i>	10	<i>c</i> and <i>d</i>	50
<i>a</i> and <i>c</i>	50	<i>c</i> and <i>e</i>	8
<i>a</i> and <i>d</i>	14	<i>c</i> and <i>f</i>	50
<i>a</i> and <i>e</i>	50	<i>c</i> and <i>g</i>	12
<i>a</i> and <i>f</i>	50	<i>d</i> and <i>e</i>	50
<i>a</i> and <i>g</i>	50	<i>d</i> and <i>f</i>	50
<i>b</i> and <i>c</i>	50	<i>d</i> and <i>g</i>	50
<i>b</i> and <i>d</i>	4	<i>e</i> and <i>f</i>	50
<i>b</i> and <i>e</i>	50	<i>e</i> and <i>g</i>	18
<i>b</i> and <i>f</i>	50	<i>f</i> and <i>g</i>	50
<i>b</i> and <i>g</i>	50		

• Solution

To work this problem, remember that 1% recombination equals 1 map unit and a recombination frequency of 50% means that genes at the two loci are assorting independently (located in different linkage groups).

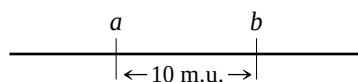
The recombination frequency between *a* and *b* is 10%; so these two loci are in the same linkage group, approximately 10 m.u. apart.

Linkage group 1

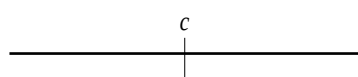


The recombination frequency between *a* and *c* is 50%; so *c* must lie in a second linkage group.

Linkage group 1

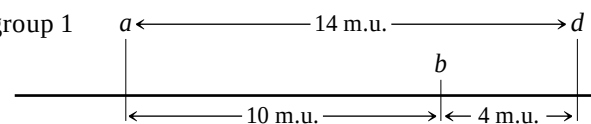


Linkage group 2

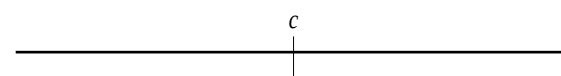


The recombination frequency between *a* and *d* is 14%; so *d* is located in linkage group 1. Is locus *d* 14 m.u. to the right or to the left of gene *a*? If *d* is 14 m.u. to the left of *a*, then the *b*-to-*d* distance should be 10 m.u. + 14 m.u. = 24 m.u. On the other hand, if *d* is to the right of *a*, then the distance between *b* and *d* should be 14 m.u. - 10 m.u. = 4 m.u. The *b*-*d* recombination frequency is 4%; so *d* is 14 m.u. to the right of *a*. The updated map is:

Linkage group 1

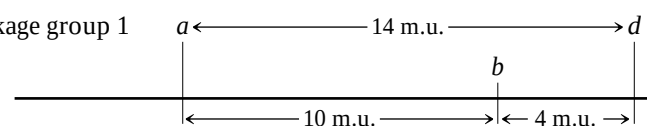


Linkage group 2

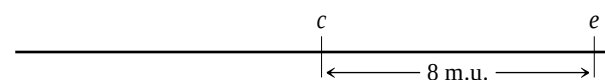


The recombination frequencies between each of loci *a*, *b*, and *d*, and locus *e* are all 50%; so *e* is not in linkage group 1 with *a*, *b*, and *d*. The recombination frequency between *e* and *c* is 8 m.u.; so *e* is in linkage group 2:

Linkage group 1

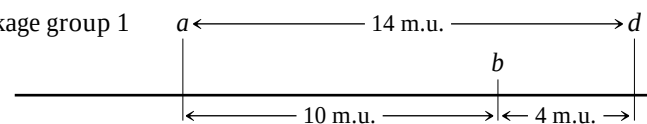


Linkage group 2

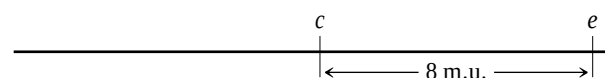


There is 50% recombination between *f* and all the other genes; so *f* must belong to a third linkage group:

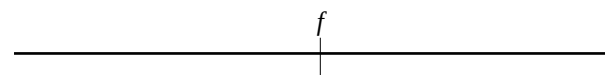
Linkage group 1



Linkage group 2

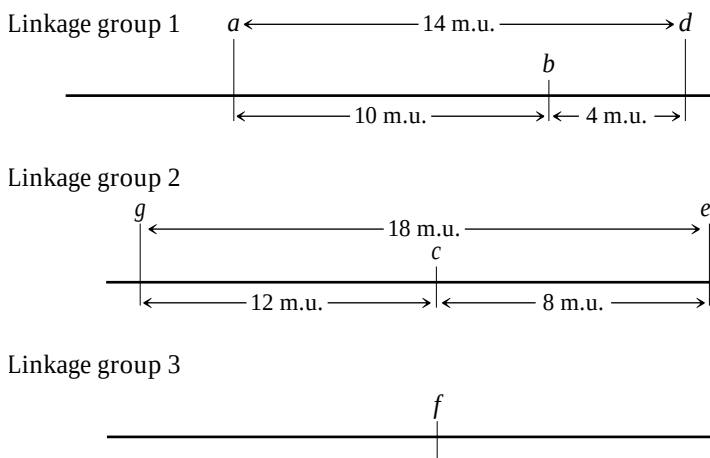


Linkage group 3



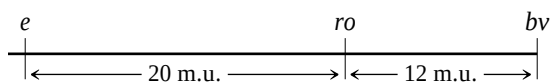
Finally, position locus *g* with respect to the other genes. The recombination frequencies between *g* and loci *a*, *b*, and *d* are all 50%; so *g* is not in linkage group 1. The recombination

frequency between *g* and *c* is 12 m.u.; so *g* is a part of linkage group 2. To determine whether *g* is 12 map units to the right or left of *c*, consult the *g*–*e* recombination frequency. Because this recombination frequency is 18%, *g* must lie to the left of *c*:



Note that the *g*–*e* distance (18 m.u.) is shorter than the sum of the *g*–*c* (12 m.u.) and *c*–*e* distances (8 m.u.), because of undetectable double crossovers between *g* and *e*.

3. Ebony body color (*e*), rough eyes (*ro*), and brevis bristles (*bv*) are three recessive mutations that occur in fruit flies. The loci for these mutations have been mapped and are separated by the following map distances:

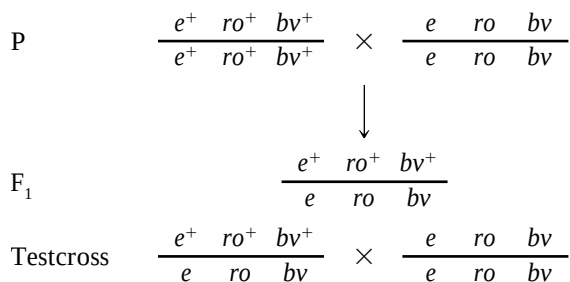


The interference between these genes is 0.4.

A fly with ebony body, rough eyes, and brevis bristles is crossed with a fly that is homozygous for the wild-type traits. The resulting F_1 females are test-crossed with males that have ebony body, rough eyes, and brevis bristles; 1800 progeny are produced. Give the phenotypes and expected numbers of phenotypes in the progeny of the testcross.

• Solution

The crosses are:



In this case, we know that *ro* is the middle locus because the genes have been mapped. Eight classes of progeny will be produced from this cross:

$e^+ \ ro^+ \ bv^+$	nonrecombinant
$e \ ro \ bv$	nonrecombinant
$e^+ \ / \ ro \ bv$	single crossover between <i>e</i> and <i>ro</i>
$e \ / \ ro^+ \ bv^+$	single crossover between <i>e</i> and <i>ro</i>
$e^+ \ ro^+ \ / \ bv$	single crossover between <i>ro</i> and <i>bv</i>
$e \ ro \ / \ bv^+$	single crossover between <i>ro</i> and <i>bv</i>
$e^+ \ / \ ro \ / \ bv^+$	double crossover
$e \ / \ ro^+ \ / \ bv$	double crossover

To determine the numbers of each type, use the map distances, starting with the double crossovers. The expected number of double crossovers is equal to the product of the single-crossover probabilities:

$$\begin{aligned} \text{expected number of double crossovers} &= 0.20 \times 0.12 \times 1800 \\ &= 43.2 \end{aligned}$$

However, some interference occurs; so the observed number of double crossovers will be less than the expected. The interference is $1 - \text{coefficient of coincidence}$; so the coefficient of coincidence is:

$$\text{coefficient of coincidence} = 1 - \text{interference}$$

The interference is given as 0.4; so the coefficient of coincidence equals $1 - 0.4 = 0.6$. Recall that the coefficient of coincidence is:

$$\text{coefficient of coincidence} = \frac{\text{number of observed double crossovers}}{\text{number of expected double crossovers}}$$

Rearranging this equation, we obtain:

$$\text{number of observed double crossovers} = \text{coefficient of coincidence} \times \text{number of expected double crossovers}$$

$$\text{number of observed double crossovers} = 0.6 \times 43.2 = 26$$

A total of 26 double crossovers should be observed. Because there are two classes of double crossovers ($e^+ \ / \ ro \ / \ bv^+$ and $e \ / \ ro^+ \ / \ bv$), we should observe 13 of each.

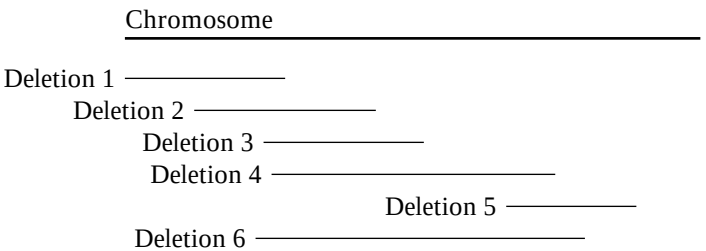
Next, we determine the number of single-crossover progeny. The genetic map indicates that there are 20 m.u. between *e* and *ro*; so 360 progeny (20% of 1800) are expected to have resulted from recombination between these two loci. Some of them will be single-crossover progeny and some will be double-crossover progeny. We have already determined that the number of double-crossover progeny is 26; so the number of progeny resulting from a single crossover between *e* and *ro* is $360 - 26 = 334$, which will

be divided equally between the two single-crossover phenotypes ($\underline{e / ro^+ \quad bv^+}$ and $\underline{e^+ / ro \quad bv}$).

There are 12 map units between *ro* and *bv*; so the number of progeny resulting from recombination between these two genes is $0.12 \times 1800 = 216$. Again, some of these recombinants will be single-crossover progeny and some will be double-crossover progeny. To determine the number of progeny resulting from a single crossover, subtract the double crossovers: $216 - 26 = 190$. These single-crossover progeny will be divided between the two single-crossover phenotypes ($\underline{e^+ ro^+ / bv}$ and $\underline{e ro / bv^+}$); so there will be $190/2 = 95$ of each. The remaining progeny will be nonrecombinants, and they can be obtained by subtraction: $1800 - 26 - 334 - 190 = 1250$; there are two nonrecombinants ($\underline{e^+ ro^+ \quad bv^+}$ and $\underline{e ro \quad bv}$); so there will be $1250/2 = 625$ of each. The numbers of the various phenotypes are listed here:

$e^+ \quad ro^+ \quad bv^+$	625	nonrecombinant
$e \quad ro \quad bv$	625	nonrecombinant
$e^+ / ro \quad bv$	167	single crossover between <i>e</i> and <i>ro</i>
$e / ro^+ \quad bv^+$	167	single crossover between <i>e</i> and <i>ro</i>
$e^+ \quad ro^+ / bv$	95	single crossover between <i>ro</i> and <i>bv</i>
$e \quad ro / bv^+$	95	single crossover between <i>ro</i> and <i>bv</i>
$e^+ / ro / bv^+$	13	double crossover
$e / ro^+ / bv$	13	double crossover
total	1800	

4. The locations of six deletions have been mapped to the *Drosophila* chromosome as shown in the following diagram. Recessive mutations *a, b, c, d, e, f,* and *g* are known to be located in the same regions as the deletions, but the order of the mutations on the chromosome is not known. When flies homozygous for the recessive mutations are crossed with flies homozygous for the deletions, the following results are obtained, where the letter “m” represents a mutant phenotype and a plus sign (+) represents the wild type. On the basis of these data, determine the relative order of the seven mutant genes on the chromosome:

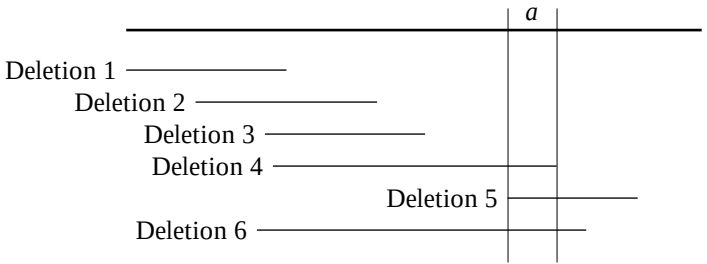


Deletion	Mutations						
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
1	+	m	m	m	+	+	+
2	+	+	m	m	+	+	+
3	+	+	+	m	m	+	+
4	m	+	+	m	m	+	+
5	m	+	+	+	+	m	m
6	m	+	+	m	m	m	+

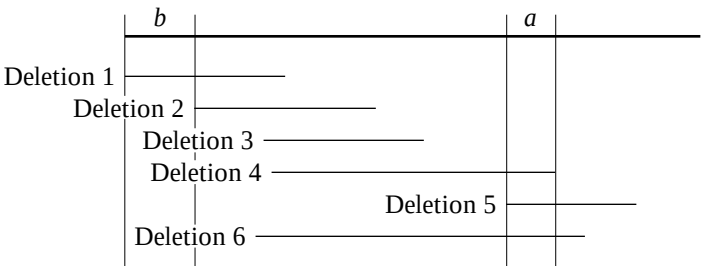
• Solution

The offspring of the cross will be heterozygous, possessing one chromosome with the deletion and wild-type alleles and one chromosome without the deletion and recessive mutant alleles. For loci within the deleted region, only the recessive mutations will be present in the offspring, which will exhibit the mutant phenotype. The presence of a mutant trait in the offspring therefore indicates that the locus for that trait is within the region covered by the deletion. We can map the genes by examining the expression of the recessive mutations in the flies with different deletions.

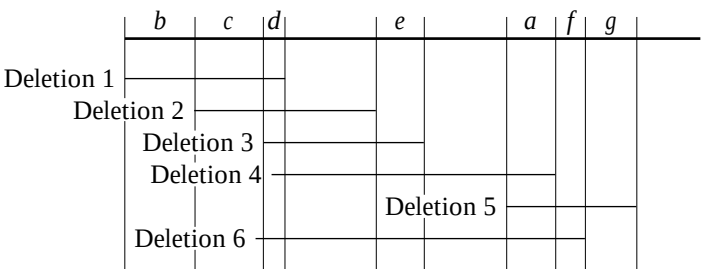
Mutation *a* is expressed in flies with deletions 4, 5, and 6 but not in flies with other deletions; so *a* must be in the area that is unique to deletions 4, 5, and 6:



Mutation *b* is expressed only when deletion 1 is present; so it must be located in the region of the chromosome covered by deletion 1 and none of the other deletions:



Using this procedure, we can map the remaining mutations. For each mutation, we look for the area of overlap among deletions that express the mutations and exclude any areas of overlap that are covered by other deletions that do not express the mutation:



5. A panel of cell lines was created from mouse–human somatic-cell fusions. Each line was examined for the presence of human chromosomes and for the production of human haptoglobin (a protein). The following results were obtained:

Cell line	Human haptoglobin	Human chromosomes						
		1	2	3	14	15	16	21
A	–	+	–	+	–	+	–	–
B	+	+	–	+	–	–	+	–
C	+	+	–	–	–	+	+	–
D	–	+	+	–	–	+	–	–

On the basis of these results, which human chromosome carries the gene for haptoglobin?

• Solution

Examine those cell lines that are positive for human haptoglobin and see what chromosomes they have in common. Lines B and C produce human haptoglobin; the chromosomes that they have in common are 1 and 16. Next, examine all lines that possess chromosomes 1 and 16 and determine whether they produce haptoglobin. Chromosome 1 is found in cell lines A, B, C, and D. If the gene for human haptoglobin were found on chromosome 1, human haptoglobin would be present in all of these cell lines. However, lines A and D do not produce human haptoglobin; so the gene cannot be on chromosome 1. Chromosome 16 is found only in cell lines B and C, and only these lines produce human haptoglobin; so the gene for human haptoglobin lies on chromosome 16.

COMPREHENSION QUESTIONS

- * 1. What does the term recombination mean? What are two causes of recombination?
- * 2. In a testcross for two genes, what types of gametes are produced with **(a)** complete linkage, **(b)** independent assortment, and **(c)** incomplete linkage?
3. What effect does crossing over have on linkage?
4. Why is the frequency of recombinant gametes always half the frequency of crossing over?
- * 5. What is the difference between genes in coupling configuration and genes in repulsion? What effect does the arrangement of linked genes (whether they are in coupling configuration or in repulsion) have on the results of a cross?
6. How does one test to see if two genes are linked?
7. What is the difference between a genetic map and a physical map?
- * 8. Why do calculated recombination frequencies between pairs of loci that are located relatively far apart underestimate the true genetic distances between loci?
9. Explain how one can determine which of three linked loci is the middle locus from the progeny of a three-point testcross.
- *10. What does the interference tell us about the effect of one crossover on another?
11. List some of the methods for physically mapping genes and explain how they are used to position genes on chromosomes.
12. What is a lod score and how is it calculated?

APPLICATION QUESTIONS AND PROBLEMS

- *13. In the snail *Cepaea nemoralis*, an autosomal allele causing a banded shell (B^B) is recessive to the allele for unbanded shell (B^O). Genes at a different locus determine the background color of the shell; here, yellow (C^Y) is recessive to brown (C^{Bw}). A banded, yellow snail is crossed with a homozygous brown, unbanded snail. The F_1 are then crossed with banded, yellow snails (a testcross).
 - (a)** What will be the results of the testcross if the loci that control banding and color are linked with no crossing over?
 - (b)** What will be the results of the testcross if the loci assort independently?
 - (c)** What will be the results of the testcross if the loci are linked and 20 map units apart?
- *14. In silkmoths (*Bombyx mori*) red eyes (re) and white-banded wing (wb) are encoded by two mutant alleles that are recessive to those that produce wild-type traits (re^+ and wb^+); these two genes are on the same chromosome. A moth homozygous for red eyes and white-banded wings is crossed with a moth homozygous for the wild-type traits. The F_1 have normal eyes and normal wings. The F_1 are crossed with moths that have red eyes and white-banded wings in a testcross. The progeny of this testcross are:

wild-type eyes, wild-type wings	418
red eyes, wild-type wings	19
wild-type eyes, white-banded wings	16
red eyes, white-banded wings	426

(a) What phenotypic proportions would be expected if the genes for red eyes and white-banded wings were located on different chromosomes?

(b) What is the genetic distance between the genes for red eyes and white-banded wings?

- *15. A geneticist discovers a new mutation in *Drosophila melanogaster* that causes the flies to shake and quiver. She calls this mutation spastic (*sps*) and determines that spastic is due to an autosomal recessive gene. She wants to determine if the spastic gene is linked to the recessive gene for vestigial wings (*vg*). She crosses a fly homozygous for spastic and vestigial traits with a fly homozygous for the wild-type traits and then uses the resulting F_1 females in a testcross. She obtains the following flies from this testcross.

<i>vg</i> ⁺	<i>sps</i> ⁺	230
<i>vg</i>	<i>sps</i>	224
<i>vg</i>	<i>sps</i> ⁺	97
<i>vg</i> ⁺	<i>sps</i>	99
total		650

Are the genes that cause vestigial wings and the spastic mutation linked? Do a series of chi-square tests to determine if the genes have assorted independently.

16. In cucumbers, heart-shaped leaves (*hl*) are recessive to normal leaves (*Hl*) and having many fruit spines (*ns*) is recessive to having few fruit spines (*Nl*). The genes for leaf shape and number of spines are located on the same chromosome; mapping experiments indicate that they are 32.6 map units apart. A cucumber plant having heart-shaped leaves and many spines is crossed with a plant that is homozygous for normal leaves and few spines. The F_1 are crossed with plants that have heart-shaped leaves and many spines. What phenotypes and proportions are expected in the progeny of this cross?
- *17. In tomatoes, tall (*D*) is dominant over dwarf (*d*) and smooth fruit (*P*) is dominant over pubescent fruit (*p*), which is covered with fine hairs. A farmer has two tall and smooth tomato plants, which we will call plant A and plant B. The farmer crosses plants A and B with the same dwarf and pubescent plant and obtains the following numbers of progeny:

	Progeny of	
	Plant A	Plant B
<i>Dd Pp</i>	122	2
<i>Dd pp</i>	6	82
<i>dd Pp</i>	4	82
<i>dd pp</i>	124	4

(a) What are the genotypes of plant A and plant B?

(b) Are the loci that determine height of the plant and pubescence linked? If so, what is the map distance between them?

(c) Explain why different proportions of progeny are produced when plant A and plant B are crossed with the same dwarf pubescent plant.

18. A cross between individuals with genotypes $a^+ a b^+ b \times aa bb$ produces the following progeny:

$a^+ a b^+ b$	83
$a^+ a bb$	21
$aa b^+ b$	19
$aa bb$	77

(a) Does the evidence indicate that the *a* and *b* loci are linked?

(b) What is the map distance between *a* and *b*?

(c) Are the alleles in the parent with genotype $a^+ a b^+ b$ in coupling configuration or repulsion? How do you know?

19. In tomatoes, dwarf (*d*) is recessive to tall (*D*) and opaque (light green) leaves (*op*) are recessive to green leaves (*Op*). The loci that determine the height and leaf color are linked and separated by a distance of 7 m.u. For each of the following crosses, determine the phenotypes and proportions of progeny produced.

$$\begin{aligned}
 \text{(a)} & \frac{D}{d} \frac{Op}{op} \times \frac{d}{d} \frac{op}{op} \\
 \text{(b)} & \frac{D}{d} \frac{op}{Op} \times \frac{d}{d} \frac{op}{op} \\
 \text{(c)} & \frac{D}{d} \frac{Op}{op} \times \frac{D}{d} \frac{Op}{op} \\
 \text{(d)} & \frac{D}{d} \frac{op}{Op} \times \frac{D}{d} \frac{op}{Op}
 \end{aligned}$$

- *20. In *Drosophila melanogaster*, ebony body (*e*) and rough eyes (*ro*) are encoded by autosomal recessive genes found on chromosome 3; they are separated by 20 map units. The gene that encodes forked bristles (*f*) is X-linked recessive and assort independently of *e* and *ro*. Give the phenotypes of progeny and their expected proportions when each of the following genotypes is test-crossed.

$$\begin{aligned}
 \text{(a)} & \frac{e^+}{e} \frac{ro^+}{ro} \frac{f^+}{f} \\
 \text{(b)} & \frac{e^+}{e} \frac{ro}{ro^+} \frac{f}{f}
 \end{aligned}$$

- *21. A series of two-point crosses were carried out among seven loci (*a*, *b*, *c*, *d*, *e*, *f*, and *g*), producing the following recombination frequencies. Map the seven loci, showing

their linkage groups, the order of the loci in each linkage group, and the distances between the loci of each group:

Loci	Recombination frequency (%)	Loci	Recombination frequency (%)
<i>a</i> and <i>b</i>	50	<i>c</i> and <i>d</i>	50
<i>a</i> and <i>c</i>	50	<i>c</i> and <i>e</i>	26
<i>a</i> and <i>d</i>	12	<i>c</i> and <i>f</i>	50
<i>a</i> and <i>e</i>	50	<i>c</i> and <i>g</i>	50
<i>a</i> and <i>f</i>	50	<i>d</i> and <i>e</i>	50
<i>a</i> and <i>g</i>	4	<i>d</i> and <i>f</i>	50
<i>b</i> and <i>c</i>	10	<i>d</i> and <i>g</i>	8
<i>b</i> and <i>d</i>	50	<i>e</i> and <i>f</i>	50
<i>b</i> and <i>e</i>	18	<i>e</i> and <i>g</i>	50
<i>b</i> and <i>f</i>	50	<i>f</i> and <i>g</i>	50
<i>b</i> and <i>g</i>	50		

*22. Waxy endosperm (*wx*), shrunken endosperm (*sh*), and yellow seedling (*v*) are encoded by three recessive genes in corn that are linked on chromosome 5. A corn plant homozygous for all three recessive alleles is crossed with a plant homozygous for all the dominant alleles. The resulting *F*₁ are then crossed with a plant homozygous for the recessive alleles in a three-point testcross. The progeny of the testcross are:

<i>wx sh V</i>	87
<i>Wx Sh v</i>	94
<i>Wx Sh V</i>	3,479
<i>wx sh v</i>	3,478
<i>Wx sh V</i>	1,515
<i>wx Sh v</i>	1,531
<i>wx Sh V</i>	292
<i>Wx sh v</i>	280
total	10,756

- (a) Determine order of these genes on the chromosome.

(b) Calculate the map distances between the genes.

(c) Determine the coefficient of coincidence and the interference among these genes.
23. Fine spines (*s*), smooth fruit (*tu*), and uniform fruit color (*u*) are three recessive traits in cucumbers whose genes are linked on the same chromosome. A cucumber plant heterozygous for all three traits is used in a testcross, and the following progeny are produced from this testcross:

<i>S U Tu</i>	2
<i>s u Tu</i>	70
<i>S u Tu</i>	21
<i>s u tu</i>	4
<i>S U tu</i>	82
<i>s U tu</i>	21
<i>s U Tu</i>	13
<i>S u tu</i>	17
total	230

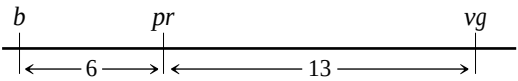
- (a) Determine the order of these genes on the chromosome.

(b) Calculate the map distances between the genes.

(c) Determine the coefficient of coincidence and the interference among these genes.

(d) List the genes found on each chromosome in the parents used in the testcross.

*24. In *Drosophila melanogaster*, black body (*b*) is recessive to gray body (*b*⁺), purple eyes (*pr*) are recessive to red eyes (*pr*⁺), and vestigial wings (*vg*) are recessive to normal wings (*vg*⁺). The loci coding for these traits are linked, with the following map distances:



The interference among these genes is 0.5. A fly with black body, purple eyes, and vestigial wings is crossed with a fly homozygous for gray body, red eyes, and normal wings. The female progeny are then crossed with males that have black body, purple eyes, and vestigial wings. If 1000 progeny are produced from this testcross, what will be the phenotypes and proportions of the progeny?

*25. The locations of six deletions have been mapped to the *Drosophila* chromosome shown here. Recessive mutations *a*, *b*, *c*, *d*, *e*, and *f* are known to be located in the same region as the deletions, but the order of the mutations on the chromosome is not known. When flies homozygous for the recessive mutations are crossed with flies homozygous for the deletions, the following results are obtained, where “m” represents a mutant phenotype and a plus sign (+) represents the wild type. On the basis of these data, determine the relative order of the seven mutant genes on the chromosome:

Chromosome						
Deletion 1	_____					
Deletion 2	_____					
Deletion 3	_____					
Deletion 4	_____					
Deletion 5	_____					
Deletion 6	_____					
Deletion	Mutations					
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
1	m	+	m	+	+	m
2	m	+	+	+	+	+
3	+	m	m	m	m	+
4	+	+	m	m	m	+
5	+	+	+	m	m	+
6	+	m	+	m	+	+

26. A panel of cell lines was created from mouse–human somatic-cell fusions. Each line was examined for the presence of human chromosomes and for the production of an enzyme. The following results were obtained:

Cell line	Enzyme	Human chromosomes													
		1	2	3	4	5	6	7	8	9	10	17	22		
A	–	+	–	–	–	+	–	–	–	–	–	+	–		
B	+	+	+	–	–	–	–	–	+	–	–	+	+		
C	–	+	–	–	–	+	–	–	–	–	–	–	+		
D	–	–	–	–	+	–	–	–	–	–	–	–	–		
E	+	+	–	–	–	–	–	–	+	–	+	+	–		

On the basis of these results, which chromosome has the gene that codes for the enzyme?

*27. A panel of cell lines was created from mouse–human somatic-cell fusions. Each line was examined for the presence of human chromosomes and for the production of three enzymes. The following results were obtained.

Cell line	Enzyme			Human chromosomes									
	1	2	3	4	8	9	12	15	16	17	22	X	
A	+	–	+	–	–	+	–	+	+	–	–	+	
B	+	–	–	–	–	+	–	–	+	+	–	–	
C	–	+	+	+	–	–	–	–	–	+	–	+	
D	–	+	+	+	+	–	–	–	+	–	–	+	

On the basis of these results, give the chromosome location of enzyme 1, enzyme 2, and enzyme 3.

CHALLENGE QUESTION

28. In calculating map distances, we did not concern ourselves with whether double crossovers were two stranded, three stranded, or four stranded; yet, these different types of double crossovers produce different types of gametes. Can you explain why we do not need to determine how many

strands take part in double crossovers in diploid organisms? (Hint: Draw out the types of gametes produced by the different types of double crossovers and see how they contribute to the determination of map distances.)

SUGGESTED READINGS

- Creighton, H. B., and B. McClintock. 1931. A correlation of cytological and genetical crossing over in *Zea mays*. *Proceedings of the National Academy of Science U. S. A.* 17:492–497.
Paper reporting Creighton and McClintock's finding that crossing over is associated with exchange of chromosome segments.
- Crow, J. 1988. A diamond anniversary: the first genetic map. *Genetics* 118:1–3.
A brief review of the history of Sturtevant's first genetic map.
- Ruddle, F. H., and R. S. Kucherlapati. 1974. Hybrid cells and human genes. *Scientific American* 231(1):36–44.
A readable review of somatic-cell hybridization.

- Stern, C. 1936. Somatic crossing over and segregation in *Drosophila melanogaster*. *Genetics* 21:625–631.
Stern's finding, similar to Creighton and McClintock's, of a correlation between crossing over and physical exchange of chromosome segments.
- Sturtevant, A. H. 1913. The linear arrangement of six sex-linked factors in *Drosophila*, as shown by their mode of association. *Journal of Experimental Zoology* 14:43–59.
Sturtevant's report of the first genetic map.

8

Bacterial and Viral Genetic Systems



A map of London in 1856. (Mylne, Robert W., *Map of the Contours of London and Its Environs*, Published by Edward Stanford, Charing Cross, London. Engraved and Printed from Stone by Waterlow and Sons, 1856.)

- Pump Handles and Cholera Genes
- Bacterial Genetics
 - Techniques for the Study of Bacteria
 - The Bacterial Genome
 - Plasmids
 - Gene Transfer in Bacteria
 - Conjugation
 - Natural Gene Transfer and Antibiotic Resistance
 - Transformation in Bacteria
 - Bacterial Genome Sequences
- Viral Genetics
 - Techniques for the Study of Bacteriophages
 - Gene Mapping in Phages
 - Transduction: Using Phages to Map Bacterial Genes
 - Fine-Structure Analysis of Bacteriophage Genes
 - Overlapping Genes
 - RNA Viruses
 - Prions: Pathogens Without Genes

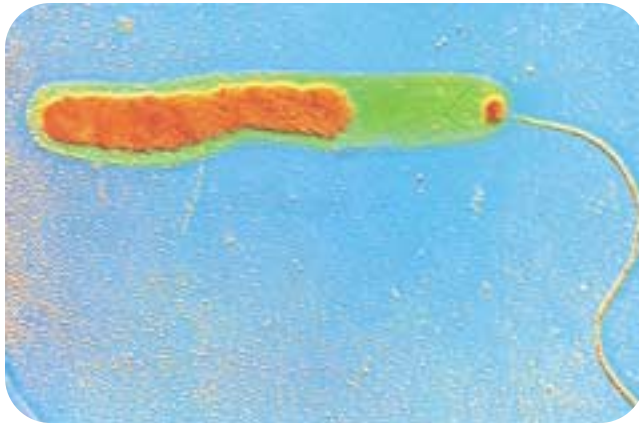
Pump Handles and Cholera Genes

On the night of August 31 in 1854, a terrible epidemic of cholera broke out in the Soho neighborhood of London. Hundreds of residents were stricken with severe diarrhea and vomiting and, in the next three days, 127 people living on or near Broad Street died. By September 10, the number of fatalities had climbed to more than 500. It was the worst outbreak of cholera ever seen in England. Residents of Soho fled their homes in terror, leaving businesses closed, homes locked, and streets deserted.

The Soho epidemic was witnessed firsthand by Dr. John Snow, a physician who lived on Sackville Street and saw the devastating effects and rapid spread of the disease. He had conducted research on cholera and suspected that it was spread through the water supply, but most medical authorities dismissed his suspicion.

Snow conducted a thorough survey of the Soho neighborhood, identifying all those who were sick with cholera. He plotted the locations of the cases on a map and observed that they clustered around one particular water pump located on Broad Street. Cholera cases did not cluster around other nearby water pumps. Snow contacted the parish officials and convinced them to remove the handle to the Broad Street pump and, with this simple action, the spread of cholera stopped dramatically. Snow later conducted additional studies of cholera outbreaks in London and established that the disease was spread in water contaminated with sewage.

Cholera has existed in Asia for at least 1000 years but, at the time of the Soho epidemic, it was a relatively new disease in England. Today, cholera is recognized as a severe infection of the intestine caused by *Vibrio cholerae* (FIGURE 8.1). This bacterium produces a potent endotoxin that induces



8.1 *Vibrio cholerae* is the bacterium that causes cholera. (CMRI/Science Photo Library/Photo Researchers.)

copious diarrhea and vomiting. If untreated, the condition can lead to serious dehydration and death. Although the number of cholera deaths has dropped dramatically since the advent of oral rehydration treatment and antibiotics, the disease continues to be a serious public-health problem, particularly in areas that lack modern water-supply systems. A recent epidemic in South Africa infected more than 25,000 people in a 6-month period.

Many of cholera's secrets have now been revealed through the sequencing of *V. cholerae*'s genome. Most bacteria have a single circular chromosome, but *V. cholerae* has two. One of the most significant findings to emerge from the sequencing of the *V. cholerae* genome is that many of the bacterium's genes for pathogenesis were acquired from other bacteria. *Vibrio cholerae* apparently has a long history

of exchanging genetic material with other bacteria and viruses, and it seems that much of this acquired DNA is responsible for its virulence. The gene that encodes the cholera toxin, for example, is found imbedded in a viral genome that infected the bacterium and became a permanent part of its genome long ago. *Vibrio cholerae* illustrates the importance of gene exchange between bacteria and viruses, a major theme of this chapter.

Since the 1940s, the genetic systems of bacteria and viruses have contributed to the discovery of many important concepts in genetics. The study of molecular genetics initially focused almost entirely on their genes; today, bacteria and viruses are still essential tools for probing the nature of genes in more-complex organisms, in part because they possess a number of characteristics that make them suitable for genetic studies (Table 8.1).

The genetic systems of bacteria and viruses are also studied because these organisms play important roles in human society. They have been harnessed to produce a number of economically important substances, and they are of immense medical significance, causing many human diseases. In this chapter, we focus on several unique aspects of bacterial and viral genetic systems. Important processes of gene transfer and recombination, like those that contributed to the pathogenesis of the cholera bacterium, will be described, and we will see how these processes can be used to map bacterial and viral genes.

www.whfreeman.com/pierce

Information about John Snow and his contributions to the study of cholera

Bacterial Genetics

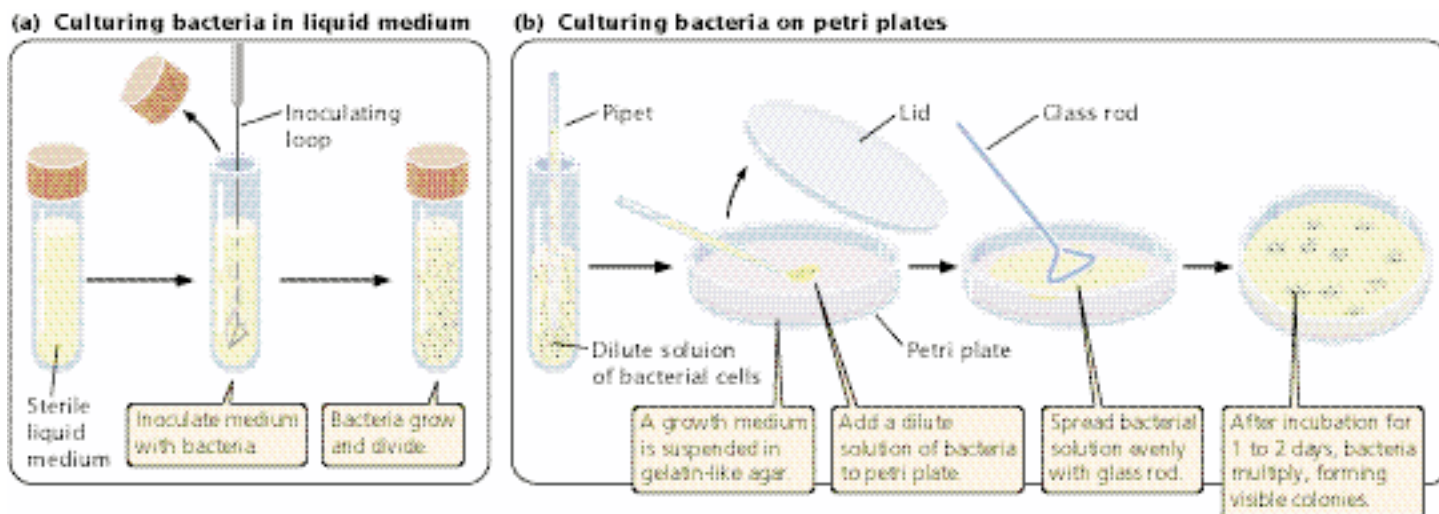
Heredity in bacteria is fundamentally similar to heredity in more-complex organisms, but the bacterial haploid genome and their small size (which makes observation of their phenotypes difficult) require different approaches and methods. First, we will consider how bacteria are studied and then examine several processes that transfer genes from one bacterium to another.

Techniques for the Study of Bacteria

Microbiologists have defined the nutritional needs of a number of bacteria and developed culture media for growing them in the laboratory. Culture media typically contain a carbon source, essential elements such as nitrogen and phosphorus, certain vitamins, and other required ions and nutrients. Wild-type (prototrophic) bacteria can use these simple ingredients to synthesize all the compounds that they need for growth and reproduction. A medium that contains only the nutrients required by prototrophic bacteria is termed **minimal medium**. Mutant strains called auxotrophs lack one or more enzymes necessary for metabolizing nutrients or synthesizing essential molecules and will grow only on medium supplemented with one or more

Table 8.1 Advantages of using bacteria and viruses for genetic studies

1. Reproduction is rapid.
2. Many progeny are produced.
3. Haploid genome allows all mutations to be expressed directly.
4. Asexual reproduction simplifies the isolation of genetically pure strains.
5. Growth in the laboratory is easy and requires little space.
6. Genomes are small.
7. Techniques are available for isolating and manipulating their genes.
8. They have medical importance.
9. They can be genetically engineered to produce substances of commercial value.



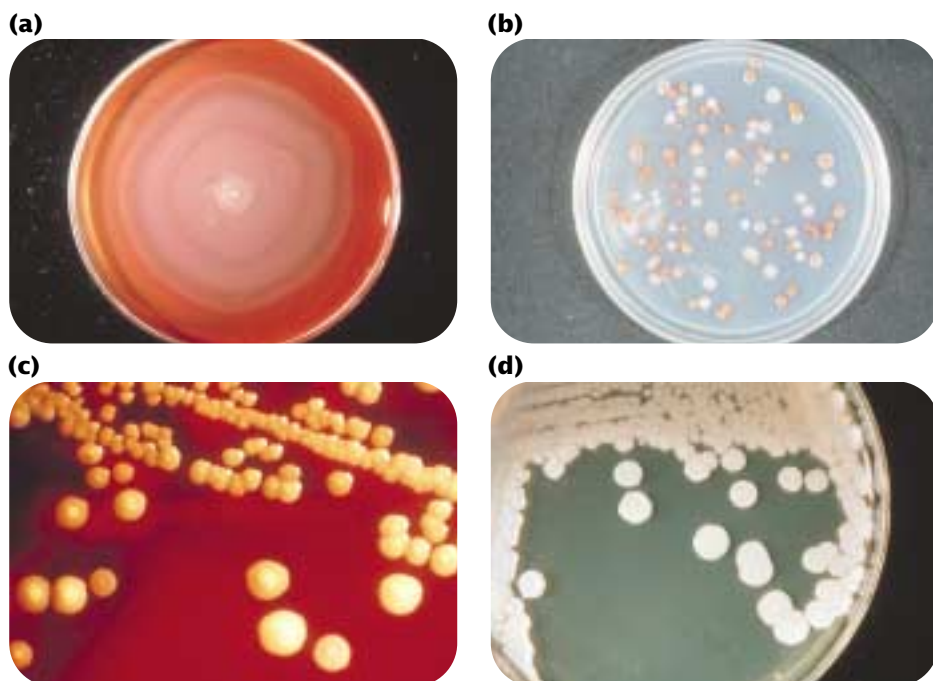
8.2 Bacteria can be grown in liquid medium.

nutrients. For example, auxotrophic strains that are unable to synthesize the amino acid leucine will not grow on minimal medium but *will* grow on medium to which leucine has been added. **Complete medium** contains all the substances required by bacteria for growth and reproduction.

Cultures of bacteria are often grown in test tubes that contain sterile liquid medium (FIGURE 8.2a). A few bacteria are added to the tube, and they grow and divide until all the nutrients are used up or—more commonly—until the concentration of their waste products becomes toxic. Bacteria are also grown in petri plates (FIGURE 8.2b). Growth medium suspended in agar is poured into the bottom half of the petri plate, providing a solid, gel-like base

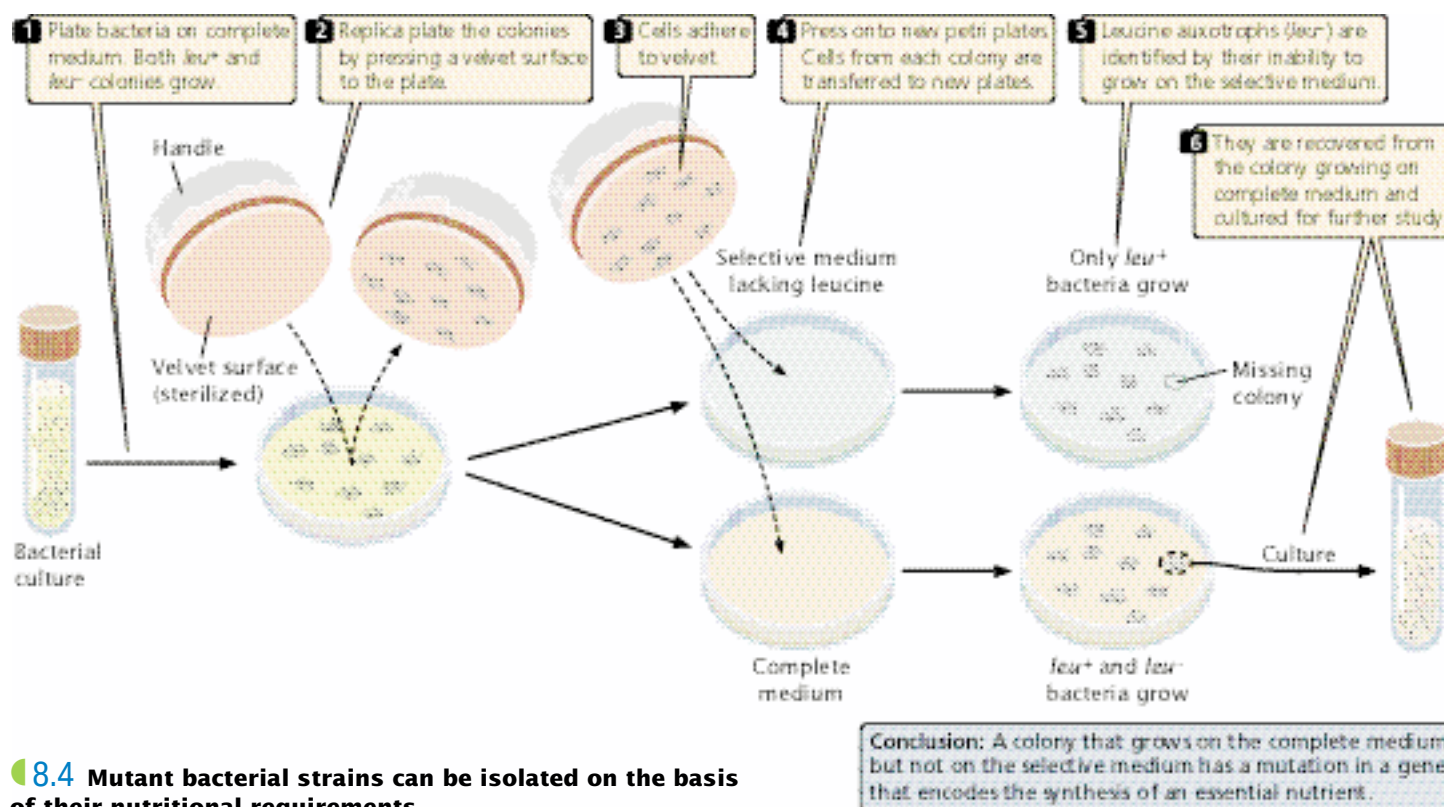
for bacterial growth. The chief advantage of this method is that it allows one to isolate and count bacteria, which individually are too small to see without a microscope. In a process called plating, a dilute solution of bacteria is spread over the surface of an agar-filled petri plate. As each bacterium grows and divides, it gives rise to a visible clump of genetically identical cells (a **colony**). Genetically pure strains of the bacteria can be isolated by collecting bacteria from a single colony and transferring them to a new test tube or petri plate.

Because individual bacteria are too small to be seen directly, it is often easier to study phenotypes that affect the appearance of the colony (FIGURE 8.3) or can be de-



8.3 Bacteria can be grown on solid media and show a variety of phenotypes.

(a) Smooth, circular raised surface. (b) Granular, circular raised surface. (c) Elevated folds on a flat colony with irregular edges. (d) Irregular elevations on a raised colony with an undulating edge. (Parts a and d, Biophoto Associates/Photo Researchers; part b, Dr. E. Bottone/Peter Arnold; part c, Larry Jensen/Visuals Unlimited.)



8.4 Mutant bacterial strains can be isolated on the basis of their nutritional requirements.

ected by simple chemical tests. Nutritional requirements of the bacteria are used to detect some commonly studied phenotypes. Suppose we want to detect auxotrophic mutants that cannot synthesize leucine (leu^- mutants). We first spread the bacteria on a petri plate containing complete medium that includes leucine; so both prototrophs that have the leu^+ allele and auxotrophs that have leu^- alleles will grow on it (FIGURE 8.4). Next, using a technique called replica plating, we transfer a few cells from each of the colonies on the original plate to two new replica plates, one containing complete medium and the other containing *selective medium* that lacks leucine. The leu^+ bacteria will grow on both media, but the leu^- mutants will grow only on the complete medium, because they cannot synthesize the leucine that is absent from the selective medium. Any colony that grows on complete medium but not on this selective medium consists of leu^- bacteria. The auxotrophic mutants growing on complete medium can then be cultured for further study.

The Bacterial Genome

Bacteria are unicellular organisms that lack a nuclear membrane (FIGURE 8.5). Most bacterial genomes consist

8.5 Most bacterial cells possess a single, circular chromosome not bounded by a nuclear membrane. (K.G. Murti/Visuals Unlimited.)



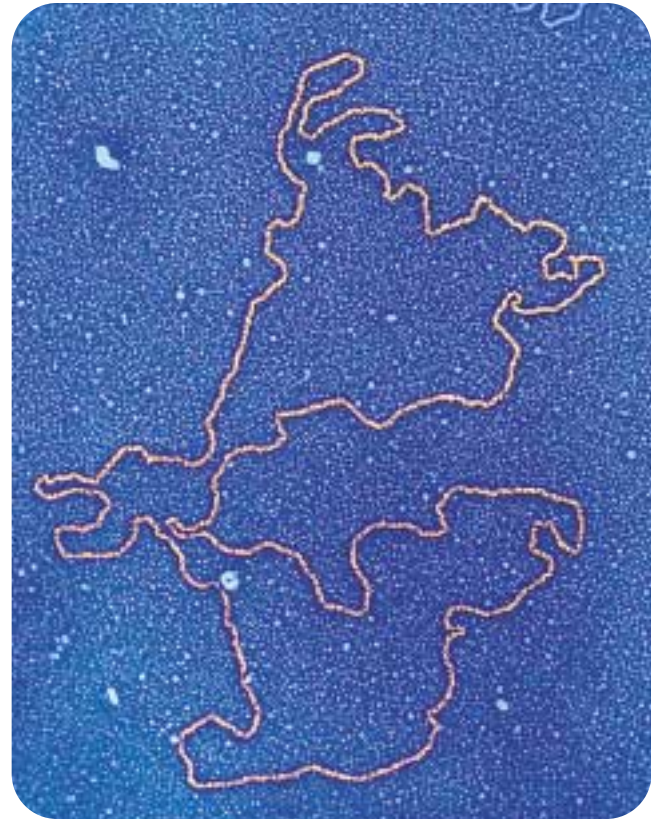
of a circular chromosome that contains a single DNA molecule several million base pairs in length. For example, the genome of *E. coli* has approximately 4.6 million base pairs of DNA (such as *V. cholerae*). However, some bacteria contain multiple chromosomes, and a few even have linear chromosomes. Bacterial chromosomes are usually organized efficiently, with little DNA between genes.

www.whfreeman.com/pierce General information on bacteria, bacterial structure, and major groups of bacteria, with some great pictures

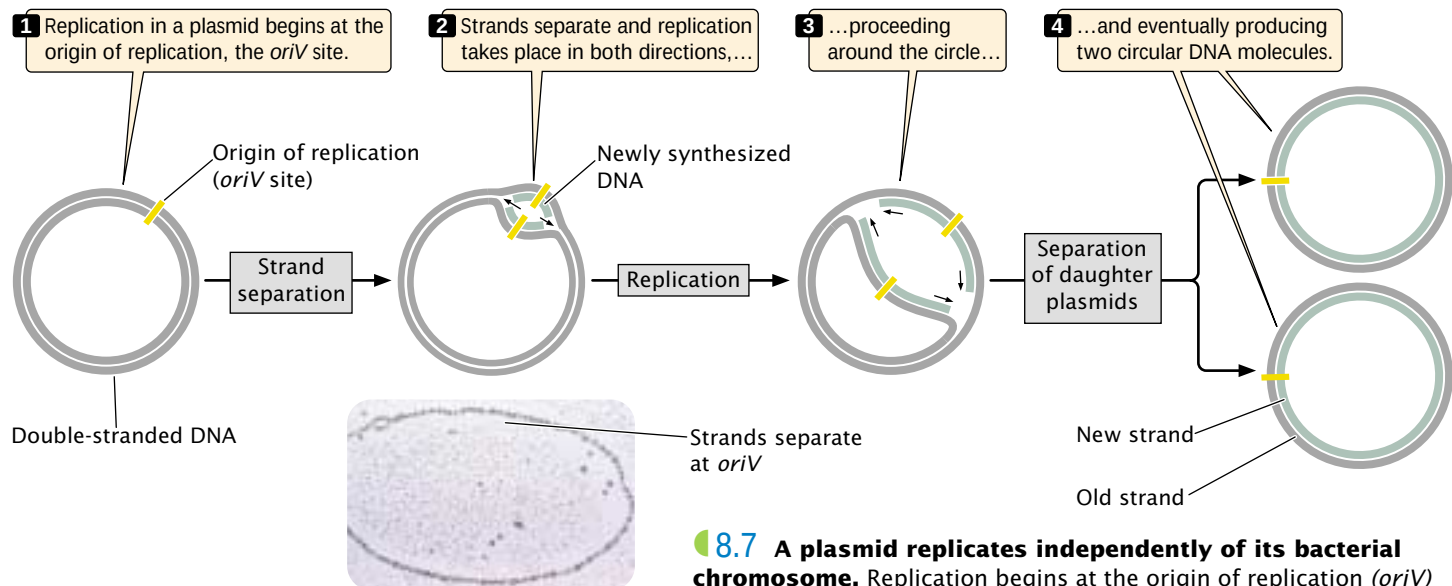
Plasmids

In addition to having a chromosome, many bacteria possess **plasmids**, small, circular DNA molecules (FIGURE 8.6). Some plasmids are present in many copies per cell, whereas others are present in only one or two copies. In general, plasmids carry genes that are not essential to bacterial function but that may play an important role in the life cycle and growth of their bacterial hosts. Some plasmids promote mating between bacteria; others contain genes that kill other bacteria. Of great importance, plasmids are used extensively in genetic engineering (Chapter 18) and some of them play a role in the spread of antibiotic resistance among bacteria.

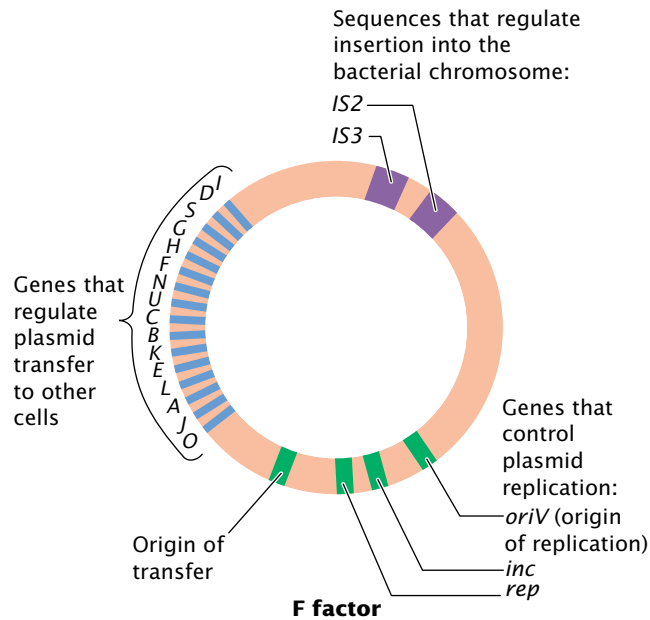
Most plasmids are circular and several thousand base pairs in length, although plasmids consisting of several hundred thousand base pairs also have been found. Possessing its own origin of replication, a plasmid replicates independently of the bacterial chromosome. Replication proceeds from the origin in one or two directions until the entire plasmid is copied. In FIGURE 8.7, the origin of replication is *oriV*. A few plasmids have multiple replication origins.



8.6 *Neisseria gonorrhoeae* (a bacterium that causes gonorrhea), like many other bacteria, contains plasmids in addition to its chromosome. The connected plasmids (indicated by the arrow) have just replicated. (A. B. Dowsett/Science Photo Library/Photo Researchers.)



8.7 A plasmid replicates independently of its bacterial chromosome. Replication begins at the origin of replication (*oriV*) and continues around the circle. In this diagram, replication is taking place in both directions; in some plasmids, replication is in one direction only. (Photo from Photo Researchers.)



8.8 The F factor, a circular episome of *E. coli*, contains a number of genes that regulate transfer into the bacterial cell and insertion into the bacterial chromosome. It contains a number of genes that regulate its transfer to other cells and that control replication. Replication is initiated at *oriV*. Insertion sequences (Chapter 11) *IS3* and *IS2* control insertion into the bacterial chromosome and excision from it.

Episomes are plasmids that are capable of either freely replicating or integrating into the bacterial chromosomes. The **F** (fertility) **factor** of *E. coli* (FIGURE 8.8) is an episome that controls mating and gene exchange between *E. coli* cells, as will be discussed in the next section.

Concepts

The typical bacterial genome consists of a single circular chromosome that contains several million base pairs. Some bacterial genes may be present on plasmids, which are small, circular DNA molecules that replicate independently of the bacterial chromosome.

Gene Transfer in Bacteria

For many years, bacteria were thought to reproduce only by simple binary fission, in which one cell splits into two identical cells without any exchange or recombination of genetic material. In 1946, Joshua Lederberg and Edward Tatum demonstrated that bacteria can transfer and recombine genetic information. This finding paved the way for the use of bacteria as model genetic organisms. Bacteria exchange genetic material by three different mechanisms,

all entailing some type of DNA transfer and recombination between the transferred DNA and the bacterial chromosome.

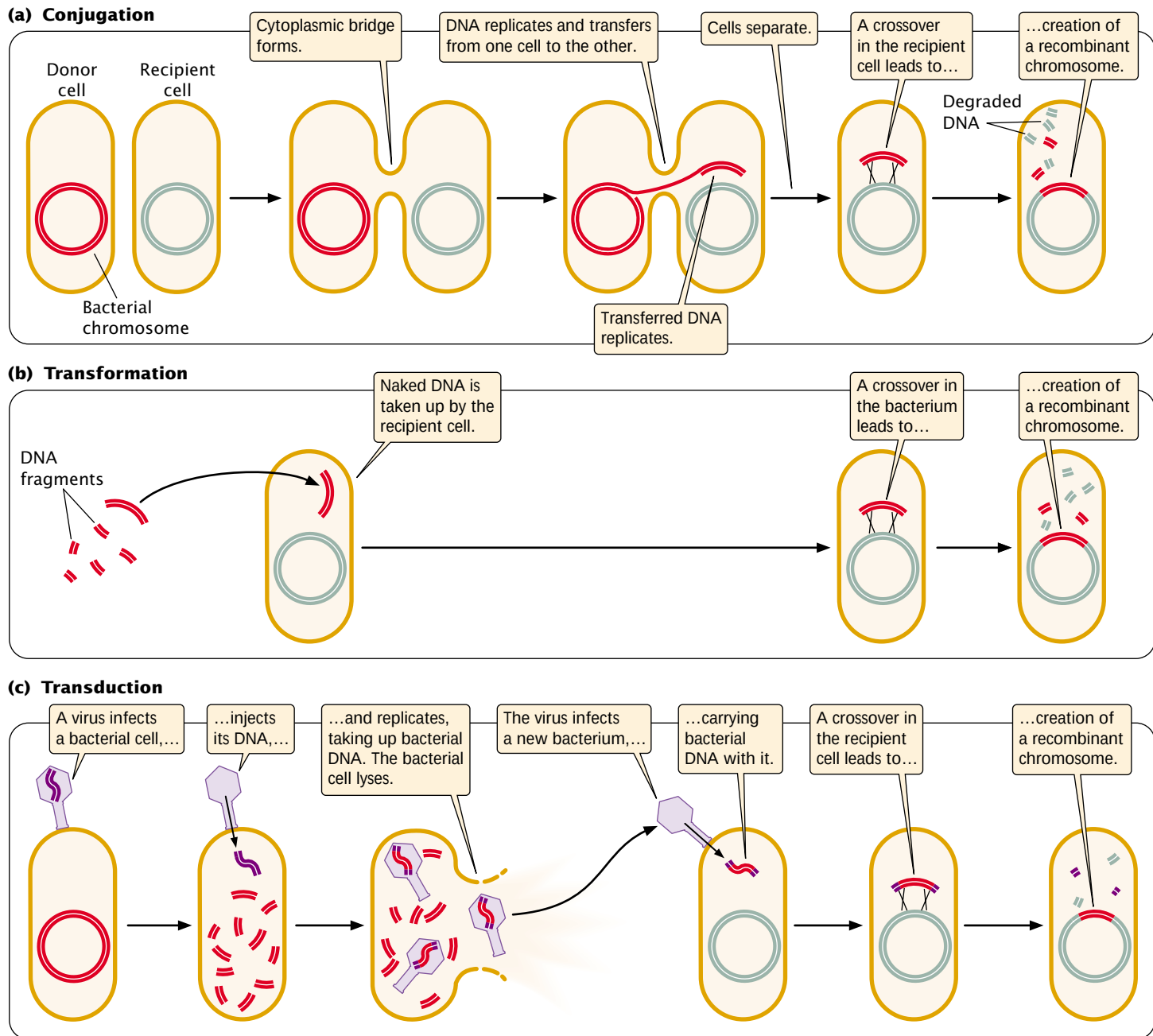
1. **Conjugation** (FIGURE 8.9a) is the direct transfer of genetic material from one bacterium to another. In conjugation, two bacteria lie close together and a connection forms between them. A plasmid or a part of the bacterial chromosome passes from one cell (the donor) to the other (the recipient). Subsequent to conjugation, crossing over takes place between homologous sequences in the transferred DNA and the chromosome of the recipient cell. In conjugation, DNA is transferred only from donor to recipient, with no reciprocal exchange of genetic material.
2. In **transformation** (FIGURE 8.9b), DNA in the medium surrounding a bacterium is taken up. After transformation, recombination may take place between the introduced genes and those of the bacterial chromosome.
3. In **transduction** (FIGURE 8.9c), bacterial viruses (bacteriophages) carry DNA from one bacterium to another. Once inside the bacterium, the newly introduced DNA may undergo recombination with the bacterial chromosome.

Not all bacterial species exhibit all three types of genetic transfer. Conjugation is more frequent for some bacteria than for others. Transformation takes place to a limited extent in many bacteria, but laboratory techniques have been developed that increase the rate of DNA uptake. Most bacteriophages have a limited host range; so transduction is normally between bacteria of the same or closely related species only.

These processes of genetic exchange in bacteria differ from the sexual reproduction of diploid eukaryotes in two important ways. First, DNA exchange and reproduction are not coupled in bacteria. Second, donated genetic material that is not recombined into the host DNA is usually degraded and so the recipient cell remains haploid. Each type of genetic transfer can be used to map genes, as will be discussed in the following sections.

Concepts

DNA may be transferred between bacterial cells through conjugation, transformation, or transduction. Each type of genetic transfer consists of a one-way movement of genetic information to the recipient cell, sometimes followed by recombination. These processes are not connected to cellular reproduction in bacteria.

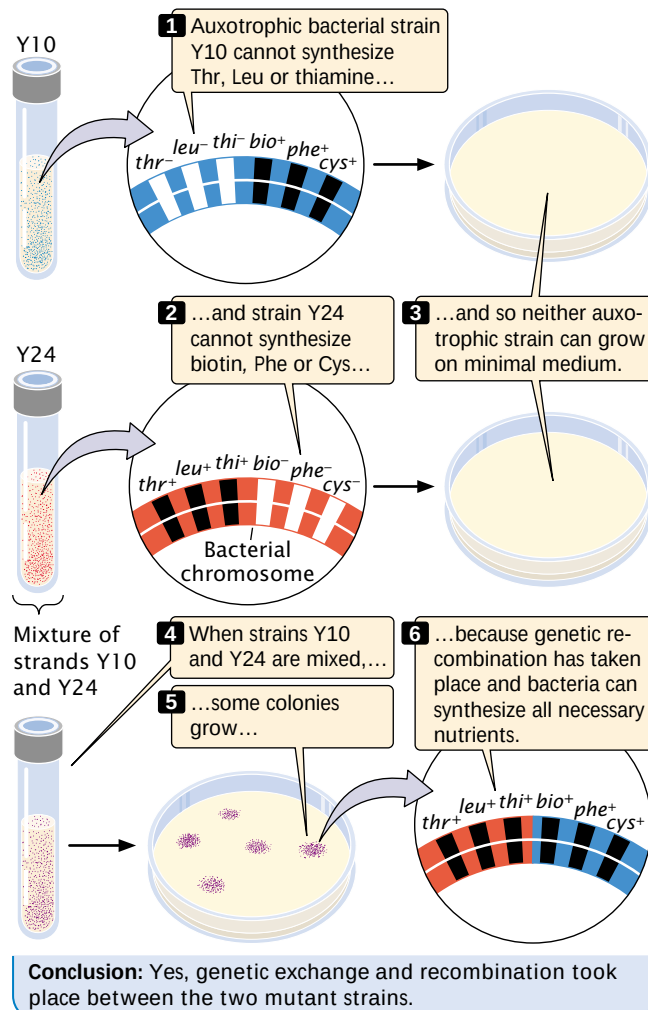


8.9 Conjugation, transformation, and transduction are three processes of gene transfer in bacteria. All three processes require transferred DNA to undergo recombination with the bacterial chromosome for the transferred DNA to be stably inherited.

Conjugation

In the course of their research, Lederberg and Tatum studied strains of *E. coli* possessing auxotrophic mutations. The Y10 strain required the amino acids threonine (and genotypically was *thr*⁻) and leucine (*leu*⁻) and the vitamin thiamine (*thi*⁻) for growth but did not require the vitamin biotin (*bio*⁺) or the amino acids phenylalanine (*phe*⁺) and cysteine (*cys*⁺); the genotype of this strain can be written

as: *thr*⁻ *leu*⁻ *thi*⁻ *bio*⁺ *phe*⁺ *cys*⁺. The Y24 strain required biotin, phenylalanine, and cysteine in its medium, but it did not require threonine, leucine, or thiamine; its genotype was: *thr*⁺ *leu*⁺ *thi*⁺ *bio*⁻ *phe*⁻ *cys*⁻. In one experiment, Lederberg and Tatum mixed Y10 and Y24 bacteria together and plated them on minimal medium (FIGURE 8.10). Each strain was also plated separately on minimal medium.

Experiment**Question:** Do bacteria exchange genetic information?

8.10 Lederberg and Tatum's experiment demonstrated that bacteria undergo genetic exchange.

Alone, neither Y10 nor Y24 grew on minimal medium. Strain Y10 was unable to grow, because it required threonine, leucine, and thiamine, which were absent in the minimal medium; strain Y24 was unable to grow, because it required biotin, phenylalanine, and cysteine, which also were absent from the minimal medium. When Lederberg and Tatum mixed the two strains, however, a few colonies did grow on the minimal medium. These prototrophic bacteria must have had genotype $thr^+ leu^+ thi^+ bio^+ phe^+ cys^+$. Where had they come from?

If mutations were responsible for the prototrophic colonies, then some colonies should also have grown on the plates containing Y10 or Y24 alone, but no bacteria grew on these plates. Multiple simultaneous mutations ($thr^- \rightarrow thr^+$,

$leu^- \rightarrow leu^+$, and $thi^- \rightarrow thi^+$ in strain Y10 or $bio^- \rightarrow bio^+$, $phe^- \rightarrow phe^+$, and $cys^- \rightarrow cys^+$ in strain Y24) would have been required for either strain to become prototrophic by mutation, which was very improbable. Lederberg and Tatum concluded that some type of genetic transfer and recombination had taken place:

Auxotrophic strain

Y10 $thr^- leu^- thi^- bio^+ phe^+ cys^+$ Y24 $thr^+ leu^+ thi^+ bio^- phe^- cys^-$

$$\begin{array}{c} thr^- leu^- thi^- bio^+ phe^+ cys^+ \\ thr^+ leu^+ thi^+ bio^- phe^- cys^- \end{array}$$

$$\begin{array}{c} thr^- leu^- thi^- bio^+ phe^+ cys^+ \\ thr^+ leu^+ thi^+ bio^- phe^- cys^- \end{array}$$

Prototrophic strain

 $thr^+ leu^+ thi^+ bio^+ phe^+ cys^+$

What they did not know was *how* it had taken place.

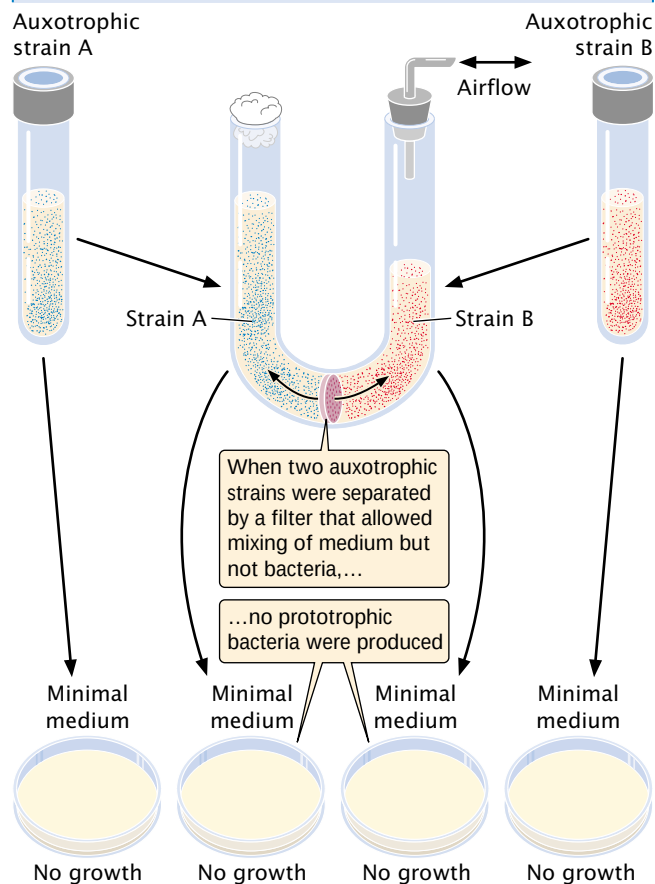
To study this problem, Bernard Davis constructed a U-shaped tube (FIGURE 8.11) that was divided into two compartments by a filter having fine pores. This filter allowed liquid medium to pass from one side of the tube to the other, but the pores of the filter were too small to allow passage of bacteria. Two auxotrophic strains of bacteria were placed on opposite sides of the filter, and suction was applied alternately to the ends of the U-tube, causing the medium to flow back and forth between the two compartments. Despite hours of incubation in the U-tube, bacteria plated out on minimal medium did not grow; there had been no genetic exchange between the strains. The exchange of bacterial genes clearly required direct contact between the bacterial cells. This type of genetic exchange entailing cell-to-cell contact in bacteria is called conjugation.

F⁺ and F⁻ cells In most bacteria, conjugation depends on a fertility (F) factor that is present in the donor cell and absent in the recipient cell. Cells that contain F are referred to as F⁺, and cells lacking F are F⁻.

The F factor contains an origin of replication and a number of genes required for conjugation (see Figure 8.8). For example, some of these genes encode sex **pili** (singular, pilus), slender extensions of the cell membrane. A cell containing F produces the sex pili, which makes contact with a receptor on an F⁻ cell (FIGURE 8.12) and pulls the two cells together. DNA is then transferred from the F⁺ cell to the F⁻ cell. Conjugation can take place only between a cell that possesses F and a cell that lacks F.

Experiment

Question: How did the genetic exchange seen in Lederberg and Tatum's experiment take place?



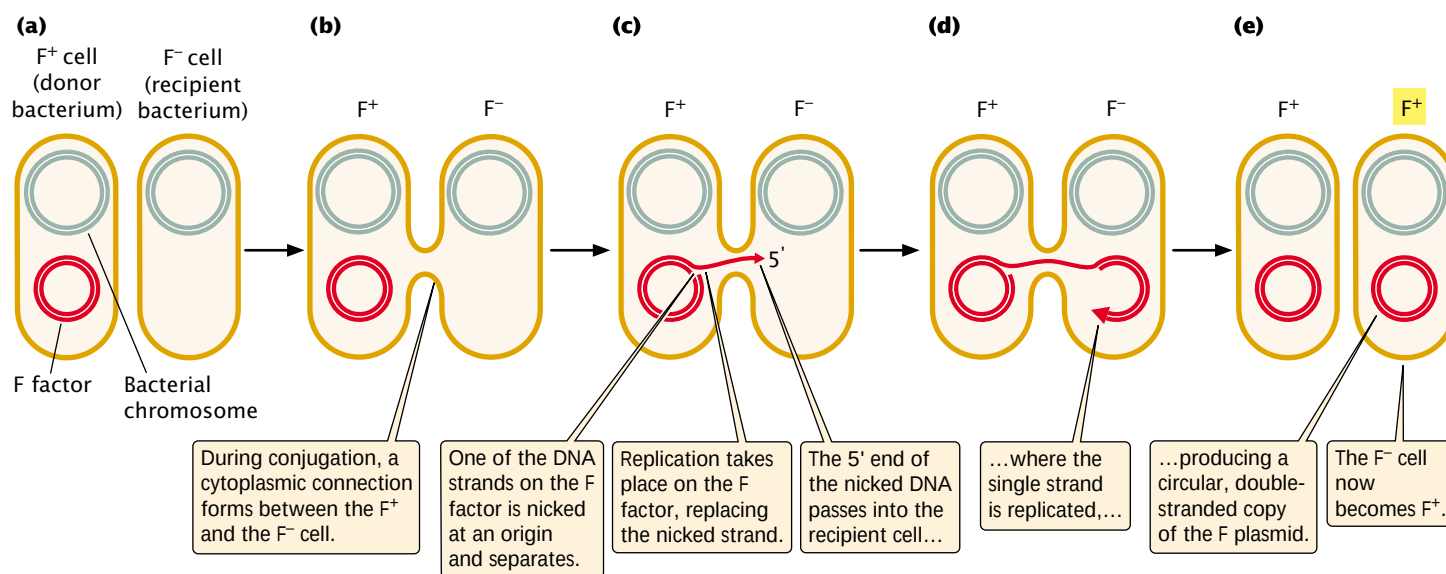
Conclusion: Genetic exchange requires direct contact between bacterial cells.

8.11 Davis's U-tube experiment.

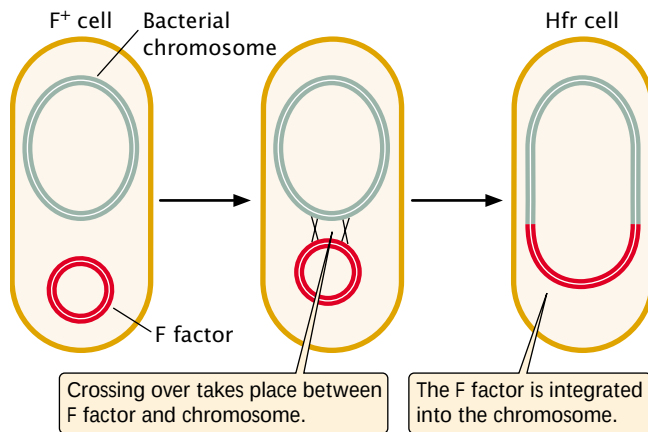
In most cases, the only genes transferred during conjugation between an F^+ and F^- cell are those on the F factor (FIGURE 8.13a and b). Transfer is initiated when one of the DNA strands on the F factor is nicked at an origin (*oriT*). One end of the nicked DNA separates from the circle and passes into the recipient cell (FIGURE 8.13c). Replication takes place on the nicked strand, proceeding around the circular plasmid and replacing the transferred strand (FIGURE 8.13d). Because the plasmid in the F^+ cell is always nicked at the *oriT* site, this site always enters the recipient cell first, followed by the rest of the plasmid. Thus, the transfer of genetic material has a defined direction. Once inside the recipient cell, the single strand



8.12 A sex pilus connects F^+ and F^- cells during bacterial conjugation. (Dr. Dennis Kunkel/Phototake.)



8.13 The F factor is transferred during conjugation between an F^+ and F^- cell.



8.14 The F factor is integrated into the bacterial chromosome in an Hfr cell.

is replicated, producing a circular, double-stranded copy of the F plasmid (FIGURE 8.13e). If the entire F factor is transferred to the recipient F^- cell, that cell becomes an F^+ cell.

Hfr cells Conjugation transfers genetic material in the F plasmid from F^+ to F^- cells but does not account for the transfer of chromosomal genes observed by Lederberg and Tatum. In Hfr (high-frequency) strains, the F factor is integrated into the bacterial chromosome (FIGURE 8.14). Hfr cells behave as F^+ cells, forming sex pili and undergoing conjugation with F^- cells.

In conjugation between Hfr and F^- cells (FIGURE 8.15a), the integrated F factor is nicked, and the end of the nicked strand moves into the F^- cell (FIGURE 8.15b), just as it does in conjugation between F^+ and F^- cells. In the Hfr

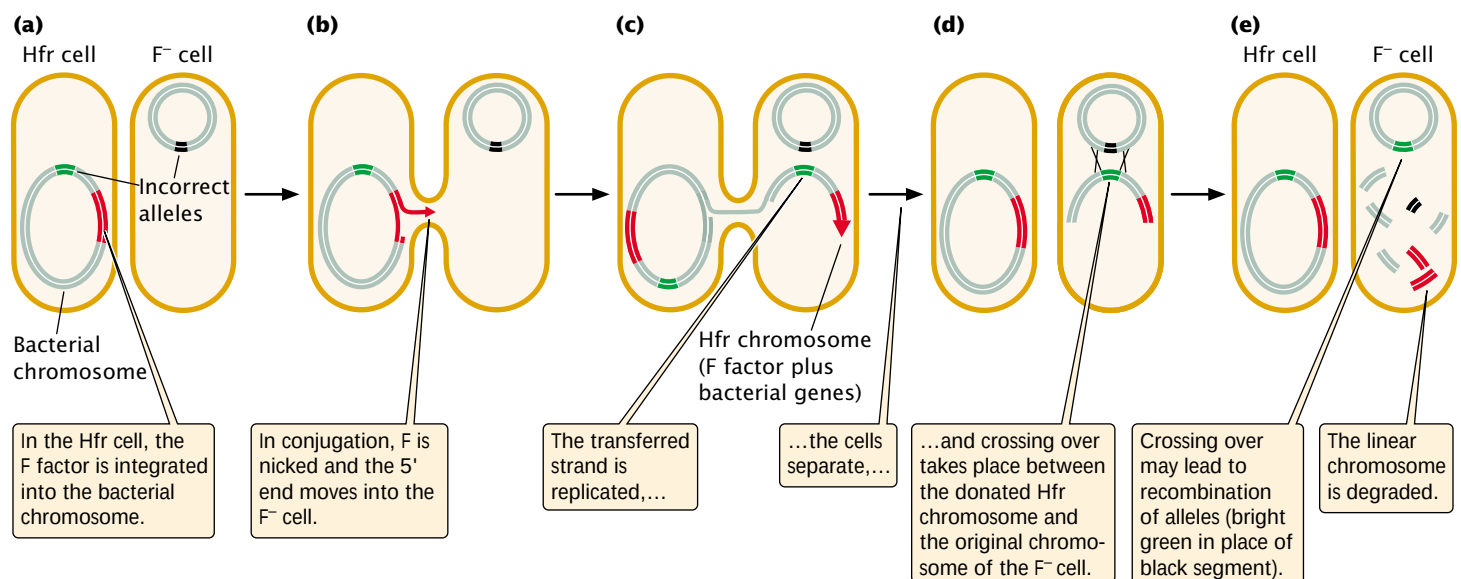
cells, the F factor is linked to the bacterial chromosome, so the chromosome follows it into the recipient cell. How much of the bacterial chromosome is transferred depends on the length of time that the two cells remain in conjugation.

Once inside the recipient cell, the donor DNA strand is replicated (FIGURE 8.15c), and crossing over between it and the original chromosome of the F^- cell (FIGURE 8.15d) may take place. This gene transfer between Hfr and F^- cells is how the recombinant prototrophic cells observed by Lederberg and Tatum were produced. When crossing over has taken place in the recipient cell, the donated chromosome is degraded, and the recombinant recipient chromosome remains (FIGURE 8.15e) to be replicated and passed to later generations by binary fission.

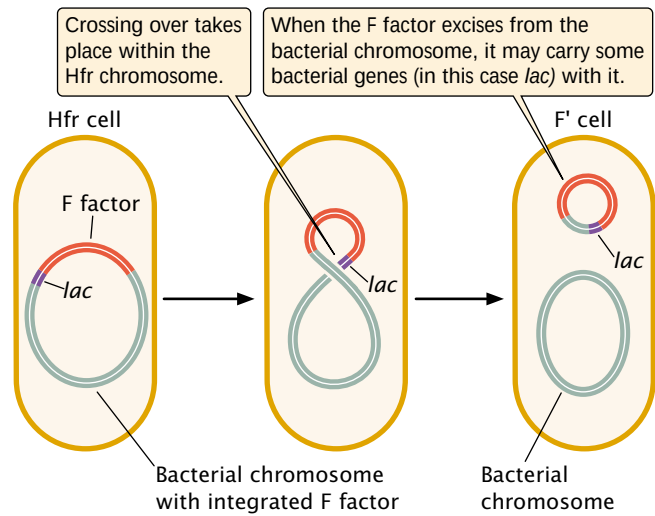
In a mating of $Hfr \times F^-$, the F^- cell almost never becomes F^+ or Hfr, because the F factor is nicked in the middle during the initiation of strand transfer, placing part of F at the beginning and part at the end of the strand to be transferred. To become F^+ or Hfr, the recipient cell must receive the entire F factor, requiring that the entire bacterial chromosome is transferred. This event happens rarely, because most conjugating cells break apart before the entire chromosome has been transferred.

The F plasmid in F^+ cells integrates into the bacterial chromosome, causing an F^+ cell to become Hfr, at a frequency of only about 1/10,000. This low frequency accounts for the low rate of recombination observed by Lederberg and Tatum in their F^+ cells. The F factor is excised from the bacterial chromosome at a similarly low rate, causing a few Hfr cells to become F^+ .

F^- cells When an F factor does excise from the bacterial chromosome, a small amount of the bacterial chromosome



8.15 Bacterial genes may be transferred from an Hfr cell to an F^- cell in conjugation.



8.16 An Hfr cell may be converted into an F' cell when the F factor excises from the bacterial chromosome and carries bacterial genes with it.

may be removed with it, and these chromosomal genes will then be carried with the F plasmid (FIGURE 8.16). Cells containing an F plasmid with some bacterial genes are called F prime (F'). For example, if an F factor integrates into a chromosome adjacent to the chromosome's *lac* operon, the F factor may pick up *lac* genes when it excises, becoming F'*lac*. F' cells can conjugate with F⁻ cells, given that they possess the F plasmid with all the genetic information necessary for conjugation and gene transfer. Characteristics of different mating types of *E. coli* (cells with different types of F) are summarized in Table 8.2.

During conjugation between an F'*lac* cell and an F⁻ cell, the F plasmid is transferred to the F⁻ cell, which means that any genes on the F plasmid, including those from the bacterial chromosome, may be transferred to F⁻ recipient cells. This process is called *sexduction*. It produces partial diploids, or *merozygotes*, which are cells with two copies of some genes,

Table 8.2 Characteristics of *E. coli* cells with different types of F factor

Type	F Factor Characteristics	Role in Conjugation
F ⁺	Present as separate circular DNA	Donor
F ⁻	Absent	Recipient
Hfr	Present, integrated into bacterial chromosome	High-frequency donor
F'	Present as separate circular DNA, carrying some bacterial genes	Donor

Table 8.3 Results of conjugation between cells with different F factors

Conjugating Cells	Cell Types Present After Conjugation
F ⁺ × F ⁻	Two F ⁺ cells (F ⁻ cell becomes F ⁺)
Hfr × F ⁻	One F ⁺ cell and one F ⁻ (no change)*
F' × F ⁻	Two F' cells (F ⁻ cell becomes F')

*Rarely, the F⁻ cell becomes F⁺ in an Hfr × F⁻ conjugation if the entire chromosome is transferred during conjugation.

one on the bacterial chromosome and one on the newly introduced F plasmid. The outcomes of conjugation between different mating types of *E. coli* are summarized in Table 8.3.

Concepts

Conjugation in *E. coli* is controlled by an episome called the F factor. Cells containing F (F⁺ cells) are donors during gene transfer; cells without F (F⁻ cells) are recipients. Hfr cells possess F integrated into the bacterial chromosome; they donate DNA to F⁻ cells at a high frequency. F' cells contain a copy of F with some bacterial genes.

Mapping bacterial genes with interrupted conjugation

The transfer of DNA that takes place during conjugation between Hfr and F⁻ cells allows bacterial genes to be mapped. During conjugation, the chromosome of the Hfr cell is transferred to the F⁻ cell. Transfer of the entire *E. coli* chromosome requires about 100 minutes; if conjugation is interrupted before 100 minutes have elapsed, only part of the chromosome will pass into the F⁻ cell and have an opportunity to recombine with the recipient chromosome. Chromosome transfer always begins within the integrated F factor and proceeds in a continuous direction; so genes are transferred according to their sequence on the chromosome. The time required for individual genes to be transferred indicates their relative positions on the chromosome. In most genetic maps, distances are expressed as percent recombination; but, in bacterial maps constructed with interrupted conjugation, the basic unit of distance is a minute.

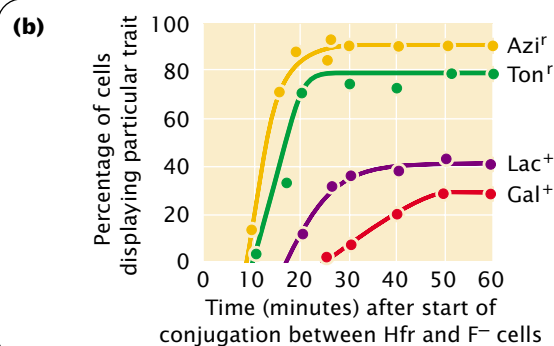
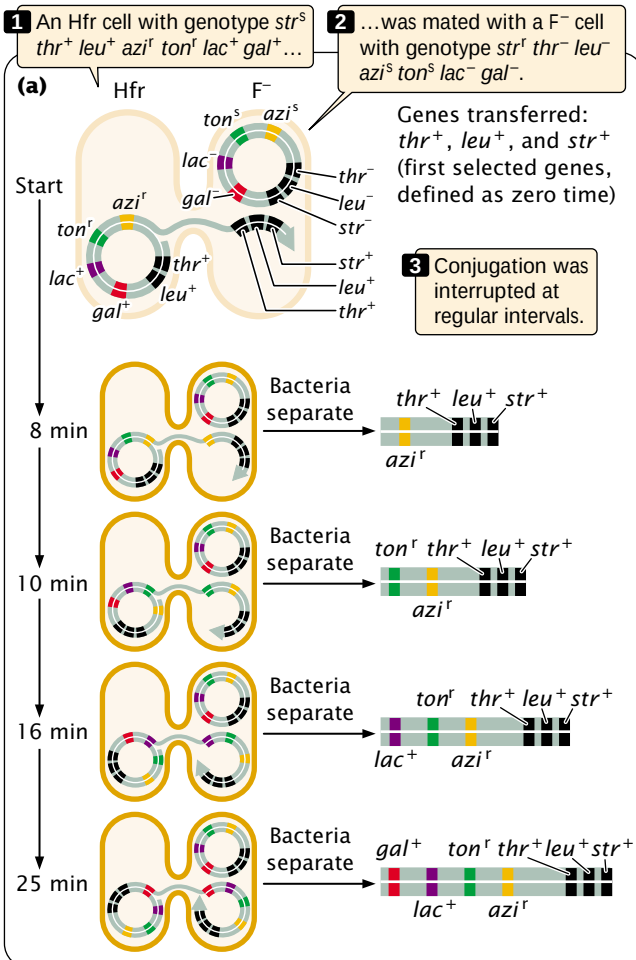
Worked Problem

To illustrate the method of mapping genes with interrupted conjugation, let's look at a cross analyzed by François Jacob and Elie Wollman, who first developed this method of gene mapping (FIGURE 8.17a). They used donor Hfr cells that were sensitive to the antibiotic streptomycin (genotype *str*^s); resistant to sodium azide (*azi*^r) and infection by bacteriophage T1 (*ton*^r); prototrophic for threonine (*thr*⁺) and

leucine (*leu*⁺); and able to break down lactose (*lac*⁺) and galactose (*gal*⁺). They used F⁻ recipient cells that were resistant to streptomycin (*str*^r); sensitive to sodium azide (*azi*^s)

Experiment

Question: How can interrupted conjugation be used to map bacterial genes?



Conclusion: The transfer times indicate the order and relative distances between genes and can be used to construct a genetic map.

8.17 Jacob and Wollman used interrupted conjugation to map bacterial genes.

and to infection by bacteriophage T1 (*ton*^s); auxotrophic for threonine (*thr*⁻) and leucine (*leu*⁻); and unable to break-down lactose (*lac*⁻) and galactose (*gal*⁻). Thus, the genotypes of the donor and recipient cells were:

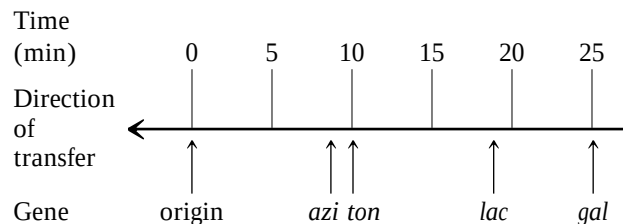
Donor Hfr cells: Hfr *str*^s *thr*⁺ *leu*⁺ *azi*^r *ton*^r *lac*⁺ *gal*⁺
 Recipient F⁻ cells: F⁻ *str*^r *thr*⁻ *leu*⁻ *azi*^s *ton*^s *lac*⁻ *gal*⁻

The two strains were mixed in nutrient medium and allowed to conjugate. After a few minutes, the medium was diluted to prevent any new pairings. At regular intervals, a sample of cells was removed and agitated vigorously in a kitchen blender to halt all conjugation and DNA transfer. The cells were plated on a selective medium that contained streptomycin and lacked leucine and threonine. The original donor cells were streptomycin sensitive (*str*^s) and would not grow on this medium. The F⁻ recipient cells were auxotrophic for leucine and threonine and also failed to grow on this medium. Only cells that underwent conjugation and received at least the *leu*⁺ and *thr*⁺ genes from the Hfr donors could grow on the selective medium. All *str*^r *leu*⁺ *thr*⁺ cells were then tested for the presence of other genes that might have been transferred from the donor Hfr strain.

All of the cells that grow on the selective medium are *str*^r *leu*⁺ *thr*⁺; so we know that these genes were transferred. The percentage of *str*^r *leu*⁺ *thr*⁺ exconjugates receiving specific alleles (*azi*^r, *ton*^r, *lac*⁺, and *gal*⁺) from the Hfr strain are plotted against the duration of conjugation (FIGURE 8.17b). What is the order and distances among the genes?

• Solution

The first donor gene to appear in all of these exconjugates (at about 9 minutes) was *azi*^r. Gene *ton*^r appeared next (after about 10 minutes), followed by *lac*⁺ (at about 18 minutes) and by *gal*⁺ (after 25 minutes). These transfer times indicate the order and relative distances among the genes (FIGURE 8.17b).



Notice that the maximum frequency of exconjugates decreased for the more distant genes. For example, about 90% of the exconjugates received the *azi*^r allele, but only about 30% received the *gal*⁺ allele. The lower percentage for *gal*⁺ is due to the fact that some conjugating cells spontaneously broke apart before they were disrupted by the blender. The probability of spontaneous disruption increases with time; so fewer cells had an opportunity to receive genes that were transferred later.

Directional transfer and mapping Different Hfr strains have the F factor integrated into the bacterial chromosome at different sites and in different orientations. Gene transfer always begins within F, and the orientation and position of F determine the direction and starting point of gene transfer. In **FIGURE 8.18a**, strain Hfr1 has F integrated between *leu* and *azi*; the orientation of F at this site dictates that gene transfer will proceed in a counterclockwise direction around the circular chromosome. Genes from this strain will be transferred in the order of:

← *leu*–*thr*–*thi*–*his*–*gal*–*lac*–*pro*–*azi*

Strain Hfr5 has F integrated between the *thi* and *his* genes (**FIGURE 8.18b**) and in the opposite orientation. Here gene transfer will proceed in a clockwise direction:

← *thi*–*thr*–*leu*–*azi*–*pro*–*lac*–*gal*–*his*

Although the starting point and direction of transfer may differ between two strains, the relative distance in time between any two pairs of genes is constant.

Notice that the order of gene transfer is not the same for different Hfr strains (**FIGURE 8.19a**). For example, *azi* is transferred just after *leu* in strain HfrH, but long after *leu* in strain Hfr1. Aligning the sequences (**FIGURE 8.19b**) shows that the two genes on either side of *azi* are always the same: *leu* and *pro*. That they are the same makes sense when one recognizes that the bacterial chromosome is circular and the starting point of transfer varies from strain to strain. These data provided the first evidence that the bacterial chromosome is circular (**FIGURE 8.19c**).

Concepts

Conjugation can be used to map bacterial genes by mixing Hfr and F[–] cells that differ in genotype and interrupting conjugation at regular intervals. The amount of time required for individual genes to be transferred from the Hfr to the F[–] cells indicates the relative positions of the genes on the bacterial chromosome.

Natural Gene Transfer and Antibiotic Resistance

Many pathogenic bacteria have developed resistance to antibiotics, particularly in environments where antibiotics are routinely used, such as hospitals and fish farms. (Massive amounts of antibiotics are often used in aquaculture to prevent infection in the fish and enhance their growth.) The continual presence of antibiotics in these environments selects for resistant bacteria, which reduces the effectiveness of antibiotic treatment for medically important infections.

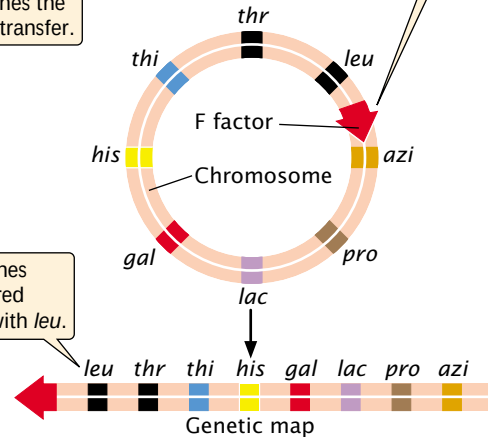
Antibiotic resistance in bacteria frequently results from the action of genes located on *R plasmids*, small circular

(a) Hfr1

1 Transfer always begins within F, and the orientation of F determines the direction of transfer.

2 In Hfr1, F is integrated between the *leu* and *azi* genes;...

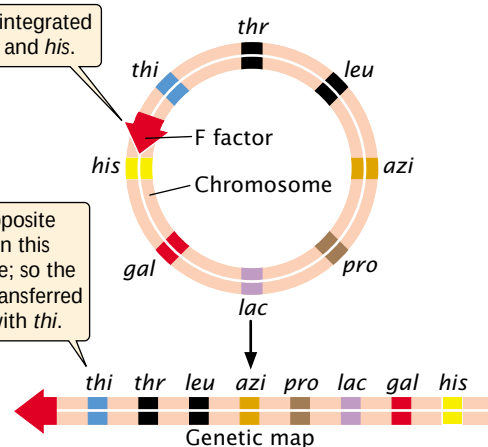
3 ...so the genes are transferred beginning with *leu*.



(b) Hfr5

4 In Hfr5, F is integrated between *thi* and *his*.

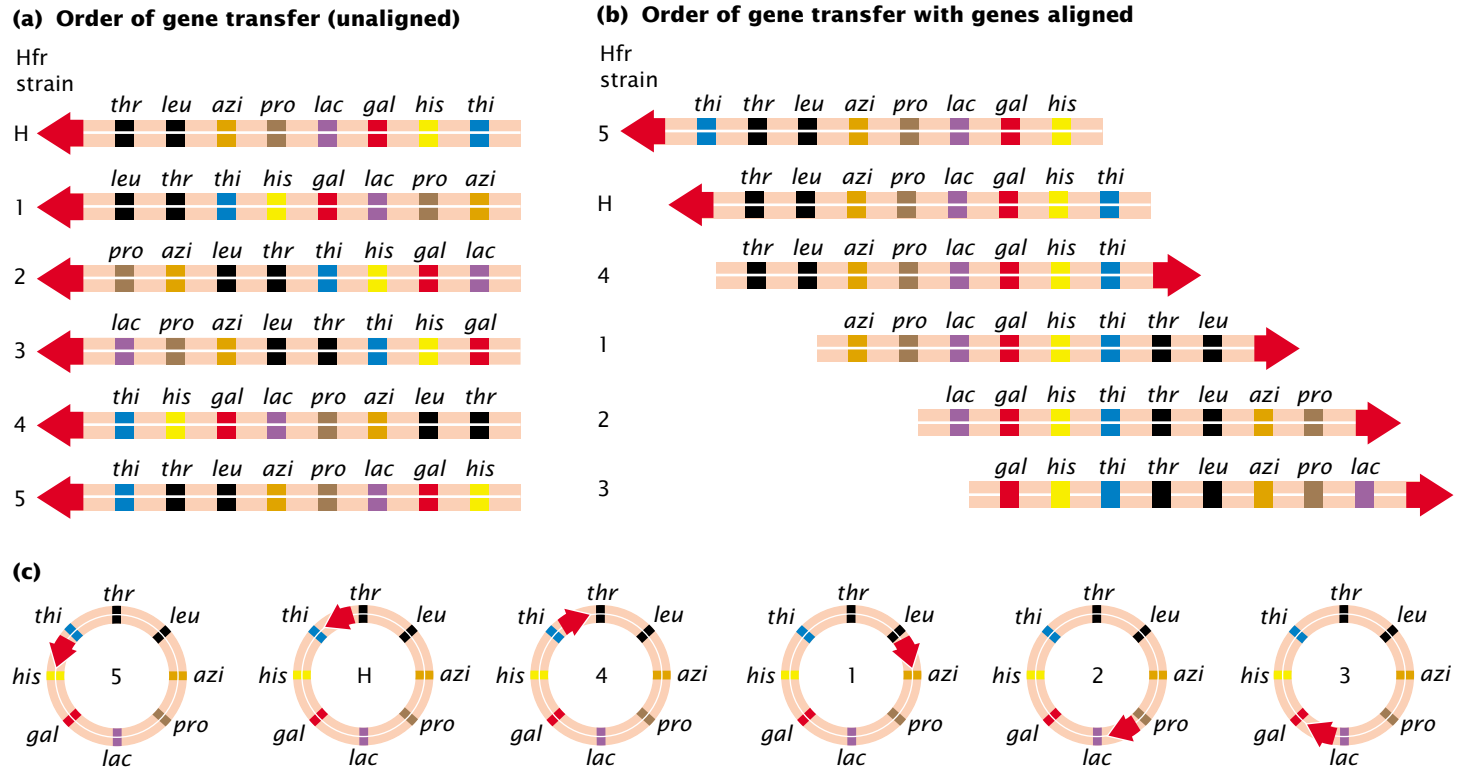
5 F has the opposite orientation in this chromosome; so the genes are transferred beginning with *thi*.



8.18 The orientation of the F factor in an Hfr strain determines the direction of gene transfer. Arrowheads indicate the origin and direction of transfer.

plasmids that can be transferred by conjugation. R plasmids have evolved in the past 50 years (since the beginning of widespread use of antibiotics), and some convey resistance to several antibiotics simultaneously. Irony but plausible sources of some of the resistance genes found in R plasmids are the microbes that produce antibiotics in the first place.

The results of recent studies demonstrate that R plasmids and their resistance genes are transferred among bacteria in a variety of natural environments. In one study, plasmids carrying genes for resistance to multiple antibiotics were transferred from a cow udder infected with *E. coli* to a human strain of *E. coli* on a hand towel: a farmer wiping his hands after milking an infected cow might unwittingly transfer antibiotic resistance from bovine- to human-inhabiting microbes. Conjugation taking place in minced meat on a cutting board allowed R plasmids to be passed from porcine



Conclusion: The order of the genes on the chromosome is the same, but the position and orientation of the F factor differs among the strains.

8.19 The order of gene transfer in a series of different Hfr strains indicates that the *E. coli* chromosome is circular.

(pig) to human *E. coli*. The transfer of R plasmids also occurs in sewage, soil, lake water, and marine sediments.

Perhaps most significantly, the transfer of R plasmids is not restricted to bacteria of the same or even related species. R plasmids with multiple antibiotic resistances have been transferred in marine waters from *E. coli* and other human-inhabiting bacteria (in sewage) to the fish bacterium *Aeromonas salmonicida* and then back to *E. coli* through raw salmon chopped on a cutting board. These results indicate that R plasmids can spread easily through the environment, passing among related and unrelated bacteria in a variety of common situations. That they can do so underscores both the importance of limiting antibiotic use to treating medically important infections and the importance of hygiene in everyday life.

Transformation in Bacteria

A second way that DNA can be transferred between bacteria is through transformation (see Figure 8.9b). Transformation played an important role in the initial identification of DNA as the genetic material, which will be discussed in Chapter 10.

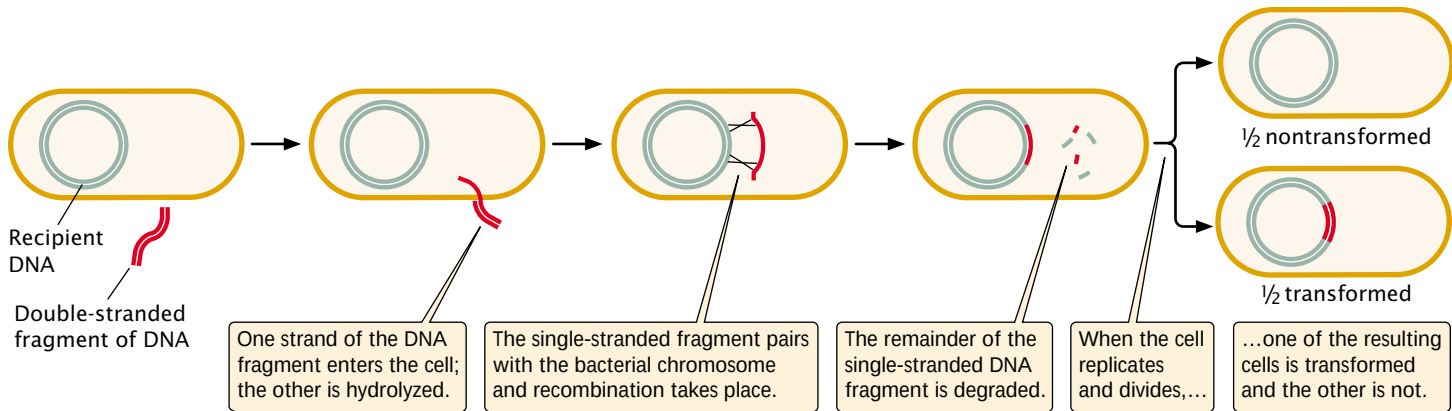
Transformation requires both the uptake of DNA from the surrounding medium and its incorporation into the bacterial chromosome or a plasmid. It may occur naturally

when dead bacteria break up and release DNA fragments into the environment. In soil and marine environments, this means may be an important route of genetic exchange for some bacteria.

Cells that take up DNA are said to be **competent**. Some species of bacteria take up DNA more easily than do others; competence is influenced by growth stage, the concentration of available DNA, and the composition of the medium. The uptake of DNA fragments into a competent bacterial cell appears to be a random process. The DNA need not even be bacterial: virtually any type of DNA (bacterial or otherwise) can be transferred to competent cells under the appropriate conditions.

As a DNA fragment enters the cell in the course of transformation (see Figure 8.20), one of the strands is hydrolyzed, whereas the other strand associates with proteins as it moves across the membrane. Once inside the cell, this single strand may pair with a homologous region and become integrated into the bacterial chromosome. This integration requires two crossover events, after which the remaining single-stranded DNA is degraded by bacterial enzymes.

Bacterial geneticists have developed techniques to increase the frequency of transformation in the laboratory



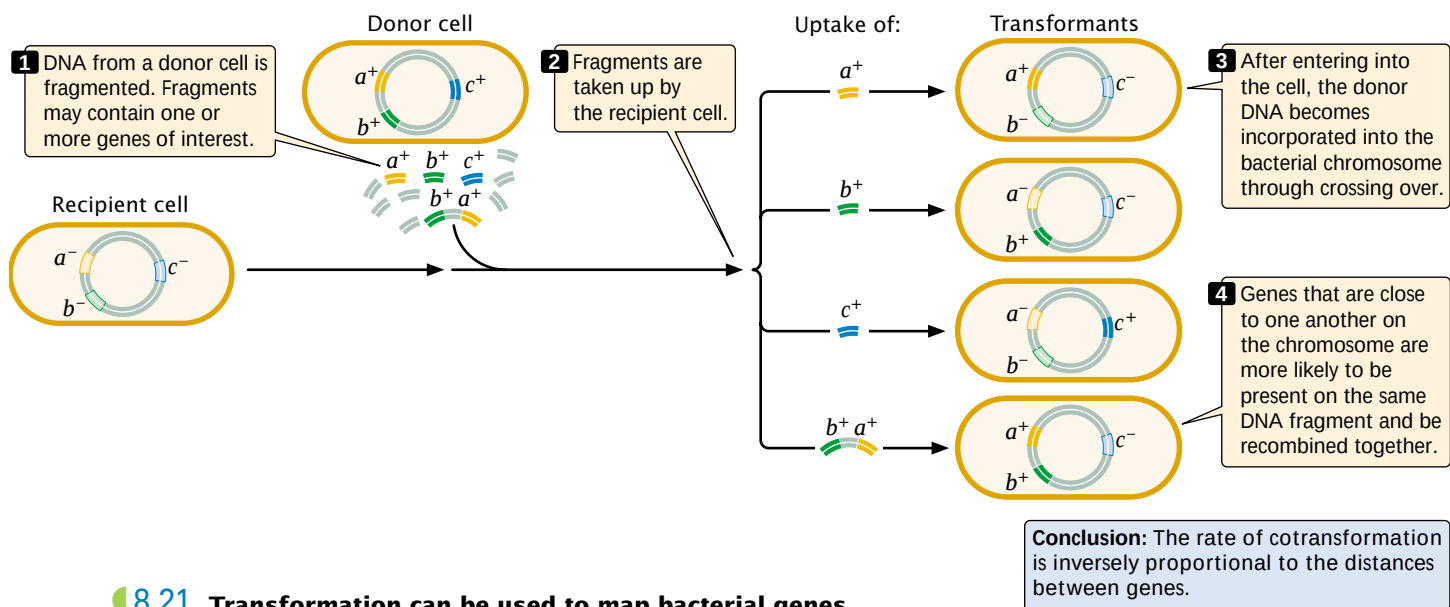
8.20 Genes can be transferred between bacteria through transformation.

in order to introduce particular DNA fragments into cells. They have developed strains of bacteria that are more competent than wild-type cells. Treatment with calcium chloride, heat shock, or an electrical field makes bacterial membranes more porous and permeable to DNA, and the efficiency of transformation can also be increased by using high concentrations of DNA. These techniques make it possible to transform bacteria such as *E. coli*, which are not naturally competent.

Transformation, like conjugation, is used to map bacterial genes, especially in those species that do not undergo conjugation or transduction (see Figure 8.9a and c). Transformation mapping requires two strains of bacteria that differ in several genetic traits; for example, the recipient strain might be $a^- b^- c^-$ (auxotrophic for three nutrients), with the donor cell being prototrophic with alleles $a^+ b^+ c^+$.

DNA from the donor strain is isolated and purified. The recipient strain is treated to increase competency, and DNA from the donor strain is added to the medium. Fragments of the donor DNA enter the recipient cells and undergo recombination with homologous DNA sequences on the bacterial chromosome. Cells that receive genetic material through transformation are called **transformants**.

Genes can be mapped by observing the rate at which two or more genes are transferred together (**cotransformed**) in transformation. When the DNA is fragmented during isolation, genes that are physically close on the chromosome are more likely to be present on the same DNA fragment and transferred together, as shown for genes a^+ and b^+ in **FIGURE 8.21**. Genes that are far apart are unlikely to be present on the same DNA fragment and rarely will be transferred together. Once inside the cell, DNA becomes



8.21 Transformation can be used to map bacterial genes.

incorporated into the bacterial chromosome through recombination. If two genes are close together on the same fragment, any two crossovers are likely to occur on either side of the two genes, allowing both to become part of the recipient chromosome. If the two genes are far apart, there may be one crossover between them, allowing one gene but not the other to recombine with the bacterial chromosome. Thus, two genes are more likely to be transferred together when they are close together on the chromosome, and genes located far apart are rarely cotransformed. Therefore, the frequency of cotransformation can be used to map bacterial genes. If genes *a* and *b* are frequently cotransformed, and genes *b* and *c* are frequently cotransformed, but genes *a* and *c* are rarely cotransformed, then gene *b* must be between *a* and *c*—the gene order is *a b c*.

Concepts

Genes can be mapped in bacteria by taking advantage of transformation, the ability of cells to take up DNA from the environment and incorporate it into their chromosomes through crossing over. The relative rate at which pairs of genes are cotransformed indicates the distance between them: the higher the rate of cotransformation, the closer the genes are on the bacterial chromosome.

Bacterial Genome Sequences

Genetic maps serve as the foundation for more detailed information provided by DNA sequencing, such as gene content and organization (see Chapter 19 for a discussion of gene sequencing).

Geneticists have now determined the complete nucleotide sequence of a number of bacterial genomes. The genome of *E. coli*, one of the most widely studied of all bacteria, is a single circular DNA molecule approximately 1 mm in length. It consists of 4,638,858 nucleotides and an estimated 4300 genes, more than half of which have no known function. These “orphan genes” may play important roles in adapting to unusual environments, coordinating metabolic pathways, organizing the chromosome, or communicating with other bacterial cells.

A number of other bacterial genomes have been completely sequenced (see Table 19.2), and many additional microbial sequencing projects are underway. A substantial proportion of genes in all bacteria have no known function. Certain genes, particularly those with related functions, tend to reside next to one another, but these clusters are in very different locations in different species, suggesting that bacterial genomes are constantly being reshuffled. Comparisons of the gene sequences of pathogenic and nonpathogenic bacteria are helping to identify genes implicated in disease and

may suggest new targets for antibiotics and other antimicrobial agents.

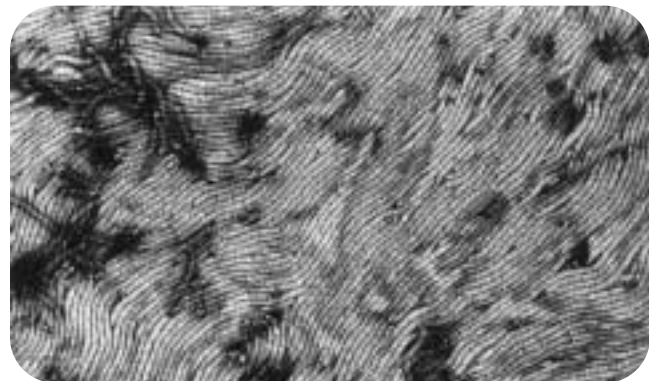
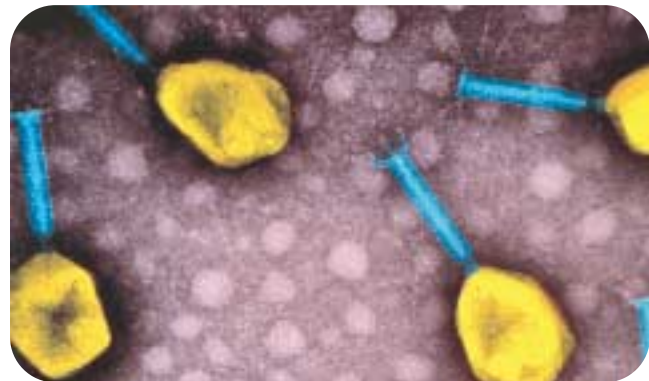
www.whfreeman.com/pierce

For a current list of completed and partial microbial genome projects and a list of microbial genome projects funded by the U.S. Department of Energy

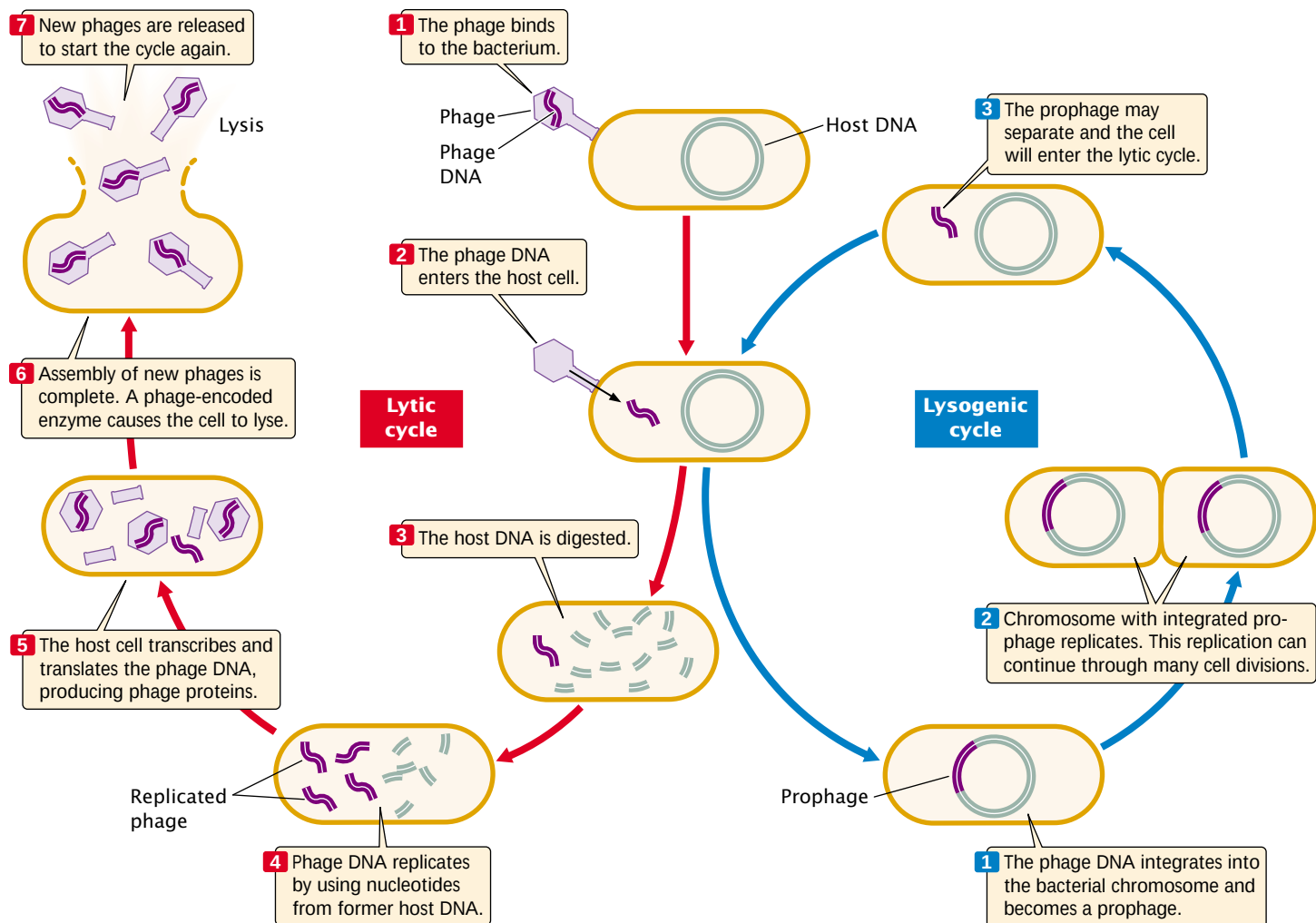
Viral Genetics

All organisms—plants, animals, fungi, and bacteria—are infected by viruses. A **virus** is a simple replicating structure made up of nucleic acid surrounded by a protein coat (see Figure 2.3). Viruses come in a great variety of shapes and sizes (FIGURE 8.22). Some have DNA as their genetic material, whereas others have RNA; the nucleic acid may be double stranded or single stranded, linear or circular. Not surprisingly, viruses reproduce in a number of different ways.

Bacteriophages (phages) have played a central role in genetic research since the late 1940s. They are ideal for many types of genetic research because they have small and easily manageable genomes, reproduce rapidly, and produce large numbers of progeny. Bacteriophages have two alternative life cycles: the lytic and the lysogenic cycles. In the lytic cycle, a phage attaches to a receptor on the bacterial cell wall and



8.22 Viruses come in a great variety of shapes and sizes. (Top, Dr. Dennis Kunkel/Phototake; bottom, R.W. Horne/Photo Researchers.)



8.23 Bacteriophages have two alternating life cycles—lytic and lysogenic.

injects its DNA into the cell (FIGURE 8.23). Once inside the cell, the phage DNA is replicated, transcribed, and translated, producing more phage DNA and phage proteins. New phage particles are assembled from these components. The phages then produce an enzyme that breaks open the cell, releasing the new phages. **Virulent phages** reproduce strictly through the lytic cycle and always kill their host cells.

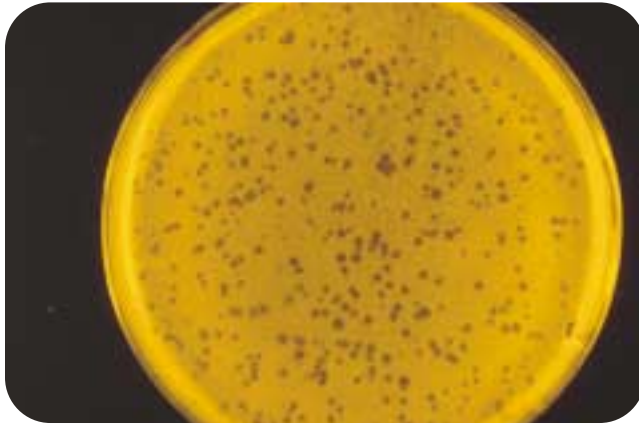
Temperate phage can utilize either the lytic or the lysogenic cycle. The lysogenic cycle begins like the lytic cycle (see Figure 8.23) but, inside the cell, the phage DNA integrates into the bacterial chromosome, where it remains as an inactive **prophage**. The prophage is replicated along with the bacterial DNA and is passed on when the bacterium divides. Certain stimuli cause the prophage to dissociate from the bacterial chromosome and enter into the lytic cycle, producing new phage particles and lysing the cell.

www.whfreeman.com/pierce

For additional information on viruses

Techniques for the Study of Bacteriophages

Viruses reproduce only within host cells; so bacteriophages must be cultured in bacterial cells. To do so, phages and bacteria are mixed together and plated on solid medium in a petri plate. A high concentration of bacteria is used so that the colonies grow into one another and produce a continuous layer of bacteria, or “lawn,” on the agar. An individual phage infects a single bacterial cell and goes through its lytic cycle. Many new phages are released from the lysed cell and infect additional cells; the cycle is then repeated. The bacteria grow on solid medium; so the diffusion of the phages is restricted and only nearby cells are infected. After several rounds of phage reproduction, a clear patch of lysed cells (a **plaque**) appears on the plate (FIGURE 8.24). Each plaque represents a single phage that multiplied and lysed many cells. Plating a known volume of a dilute solution of phages on a bacterial lawn and counting the number of plaques that appear can be used to determine the original concentration of phage in the solution.



8.24 Plaques are clear patches of lysed cells on a lawn of bacteria. (E.C.S. Chan/Visuals Unlimited.)

Concepts

Viral genomes may be DNA or RNA, circular or linear, and double or single stranded.
Bacteriophages are used in many types of genetic research.

Gene Mapping in Phages

Mapping genes in bacteriophage requires the application of the same principles as those applied to mapping genes in eukaryotic organisms (Chapter 7). Crosses are made between viruses that differ in two or more genes, and recombinant progeny phage are identified and counted. The proportion of recombinant progeny is then used to estimate the distances between the genes and their linear order on the chromosome.

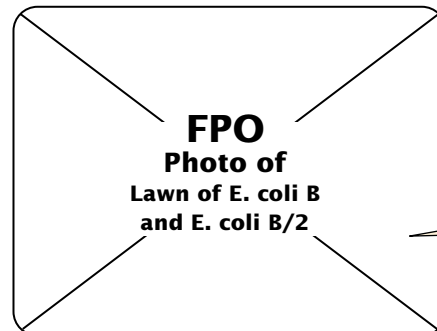
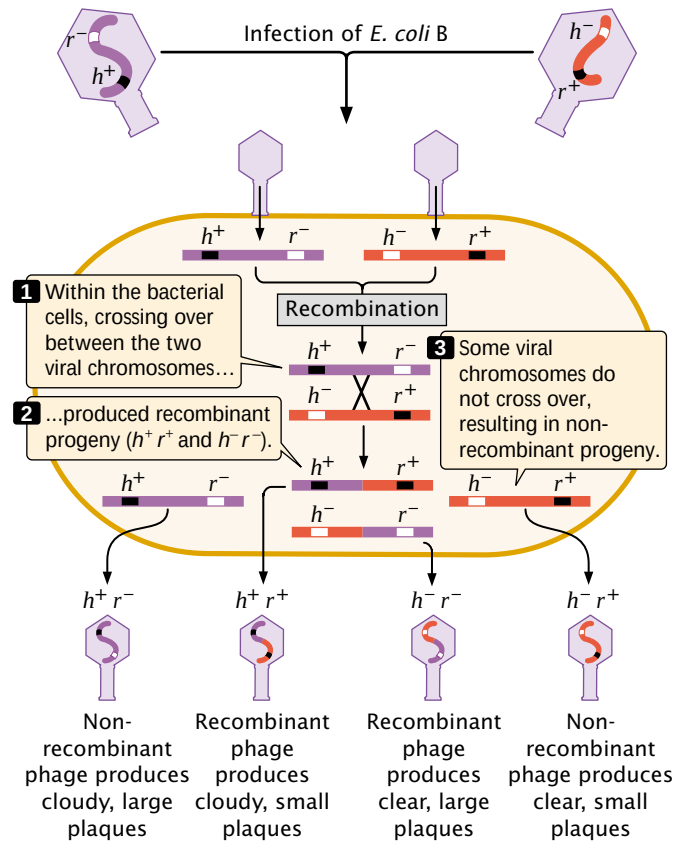
In 1949, Alfred Hershey and Raquel Rotman examined rates of recombination between genes in two strains of the T2 bacteriophage that differed in plaque appearance and host range (the bacterial strains that the phages could infect). One strain was able to infect and lyse type B *E. coli* cells but not B/2 cells (normal host range, h^+) and produced an abnormal plaque that was large with distinct borders (r^-). The second strain was able to infect and lyse both B and B/2 cells (mutant host range, h^-) and produced normal plaques that were small with fuzzy borders (r^+).

Hershey and Rotman crossed the $h^+ r^-$ and $h^- r^+$ strains of T2 by infecting type B *E. coli* cells with a mixture of the two strains. They used a high concentration of phages so that most cells could be simultaneously infected by both strains (FIGURE 8.25). Homologous recombination occasionally took place between the chromosomes of the different strains, producing $h^+ r^+$ and $h^- r^-$ chromosomes, which

8.25 Hershey and Rotman developed a technique for mapping viral genes. (Photo from G.S. Stent, Molecular Biology of Bacterial Viruses. Copyright © 1963 by W.H. Freeman and Company.)

Experiment

Question: How can we determine the position of a gene on a phage chromosome?



4 Progeny phages were then plated on a mixture of *E. coli* B and *E. coli* B/2 cells,...

Results of a cross for the h and r genes in phage T2 ($h^+ r^- \times h^- r^+$)

Genotype	Plaques	Designation
$h^- r^+$	42	Parental progeny 76%
$h^+ r^-$	34	
$h^+ r^+$	12	Recombinant 24%
$h^- r^-$	12	

5 ...which allowed all four genotypes of progeny to be identified.

6 The percentage of recombinant progeny allowed the h^- and r^- mutants to be mapped.

$$RF = \frac{\text{recombinant plaques}}{\text{total plaques}} = \frac{(h^+ r^+) + (h^- r^-)}{\text{total plaques}}$$

Conclusion: The recombination frequency indicates that the distance between h and r genes is 24%.

Table 8.4 Progeny phage produced from $h^-r^+ \times h^+r^-$

Phenotype	Genotype
Clear and small	h^-r^+
Cloudy and large	h^+r^-
Cloudy and small	h^+r^+
Clear and large	h^-r^-

were then packaged into new phage particles. When the cells lysed, the recombinant phages were released, along with the nonrecombinant h^+r^- phages and h^-r^+ phages.

Hershey and Rotman diluted and plated the progeny phages on a bacterial lawn that consisted of a mixture of B and B/2 cells. Phages carrying the h^+ allele (which conferred the ability to infect only B cells) produced a cloudy plaque because the B/2 cells did not lyse. Phages carrying the h^- allele produced a clear plaque because all the cells within the plaque were lysed. The r^+ phages produced small plaques, whereas the r^- phages produced large plaques. The genotypes of these progeny phages could therefore be determined by the appearance of the plaque (see Figure 8.25 and Table 8.4).

In this type of phage cross, the recombination frequency (RF) between the two genes can be calculated by using the following formula:

$$RF = \frac{\text{recombinant plaques}}{\text{total plaques}}$$

In Hershey and Rotman's cross, the recombinant plaques were h^+r^+ and h^-r^- ; so the recombination frequency was

$$RF = \frac{(h^+r^+) + (h^-r^-)}{\text{total plaques}}$$

Recombination frequencies can be used to determine the distances and orders of genes on the phage chromosome, just as recombination frequencies are used to map genes in eukaryotes.

Concepts

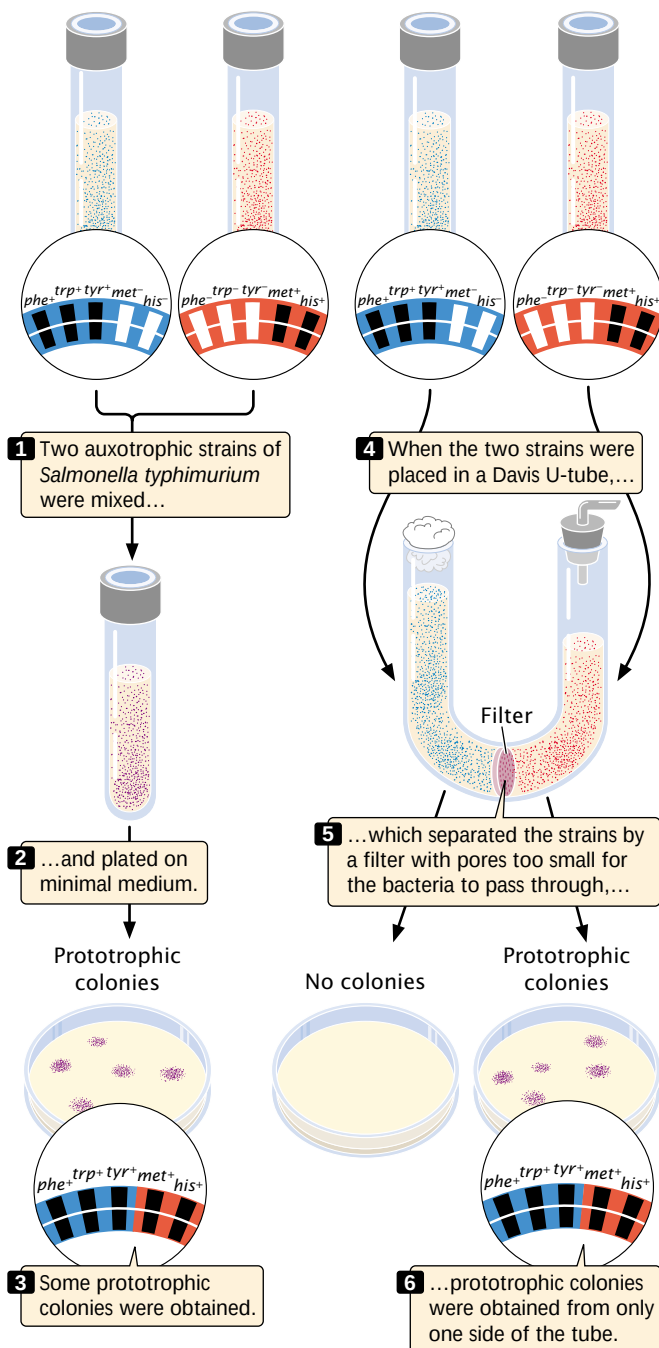
To map phage genes, bacterial cells are infected with viruses that differ in two or more genes. Recombinant plaques are counted, and rates of recombination are used to determine the linear order of the genes on the chromosome and the distances between them.

Transduction: Using Phages to Map Bacterial Genes

In the discussion of bacterial genetics, we identified three mechanisms of gene transfer: conjugation, transformation, and transduction (see Figure 8.9). Let's take a closer look at transduction, in which genes are transferred between bacte-

Experiment

Question: Does genetic exchange between bacteria always require cell-to-cell contact?



Conclusion: Genetic exchange did not take place via conjugation. A phage was later shown to be the agent of transfer.

8.26 The Lederberg and Zinder experiment.

ria by viruses. In **generalized transduction**, any gene may be transferred. In **specialized transduction**, only a few genes are transferred.

Generalized transduction Joshua Lederberg and Norton Zinder discovered generalized transduction in 1952. They were trying to produce recombination in the bacterium *Salmonella typhimurium* by conjugation. They mixed a strain of *S. typhimurium* that was *phe*⁺ *trp*⁺ *tyr*⁺ *met*⁻ *his*⁻ with a strain that was *phe*⁻ *trp*⁻ *tyr*⁻ *met*⁺ *his*⁺ (FIGURE 8.26) and plated them on minimal medium. A few prototrophic recombinants (*phe*⁺ *trp*⁺ *tyr*⁺ *met*⁺ *his*⁺) appeared, suggesting that conjugation had taken place. However, when they tested the two strains in a U-shaped tube similar to the one used by Davis, some *phe*⁺ *trp*⁺ *tyr*⁺ *met*⁺ *his*⁺ prototrophs were obtained on one side of the tube (compare Figure 8.26 with Figure 8.11). This apparatus separated the two strains by a filter with pores too small for the passage of bacteria; so how were genes being transferred between bacteria in the absence of conjugation? The results of subsequent studies revealed that the agent of transfer was a bacteriophage.

In the lytic cycle of phage reproduction, the bacterial chromosome is broken into random fragments (FIGURE 8.27). For some types of bacteriophage, a piece of the bacterial chromosome occasionally gets packaged into a phage coat instead of phage DNA; these phage particles are called **transducing phages**. The transducing phage infects a new cell, releasing the bacterial DNA, and the introduced genes may then become integrated into the bacterial chromosome by a double crossover. Bacterial genes can, by this process, be moved from one bacterial strain to another, producing recombinant bacteria called **transductants**.

Not all phages are capable of transduction, a rare event that requires (1) that the phage degrade the bacterial chromosome; (2) that the process of packaging DNA into the phage protein not be specific for phage DNA; and (3) that the bacterial genes transferred by the virus recombine with the chromosome in the recipient cell.

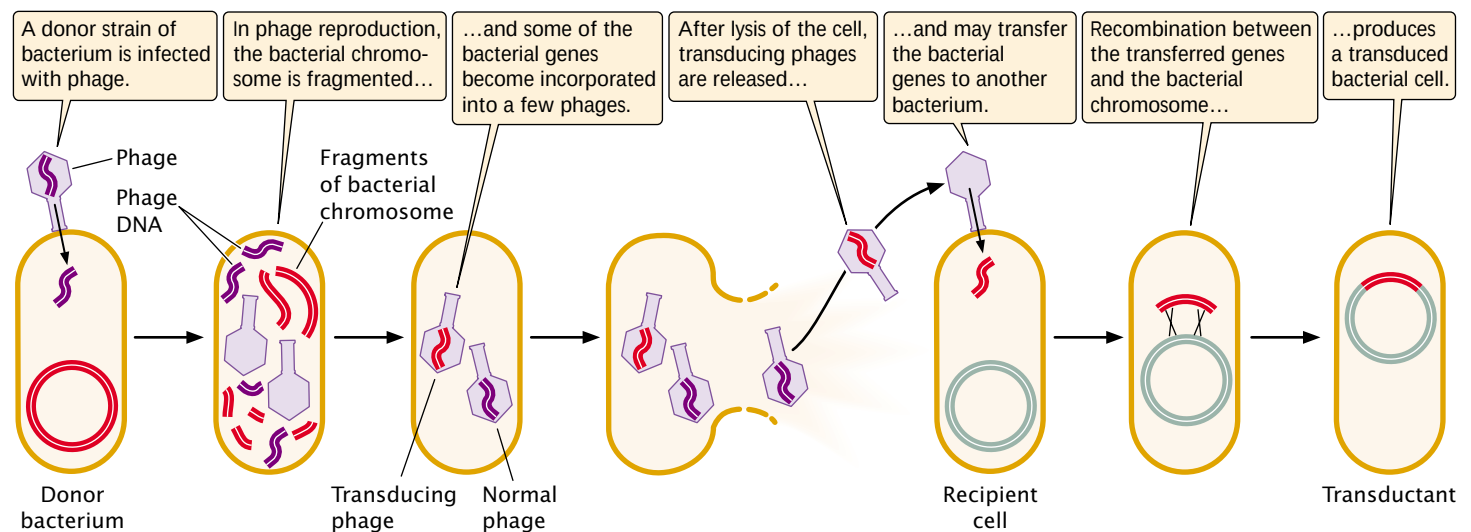
Because of the limited size of a phage particle, only about 1% of the bacterial chromosome can be transduced. Only genes located close together on the bacterial chromosome will be transferred together (**cotransduced**). The overall rate of transduction ranges from only about 1 in 100,000 to 1 in 1,000,000. Because the chance of a cell being transduced by two separate phages is exceedingly small, any cotransduced genes are usually located close together on the bacterial chromosome. Thus, rates of cotransduction, like rates of cotransformation, give an indication of the physical distances between genes on a bacterial chromosome.

To map genes by using transduction, two bacterial strains with different alleles at several loci are used. The donor strain is infected with phages (FIGURE 8.28), which reproduce within the cell. When the phages have lysed the donor cells, a suspension of the progeny phage is mixed with a recipient strain of bacteria, which are then plated on several different kinds of media to determine the phenotypes of the transducing progeny phages.

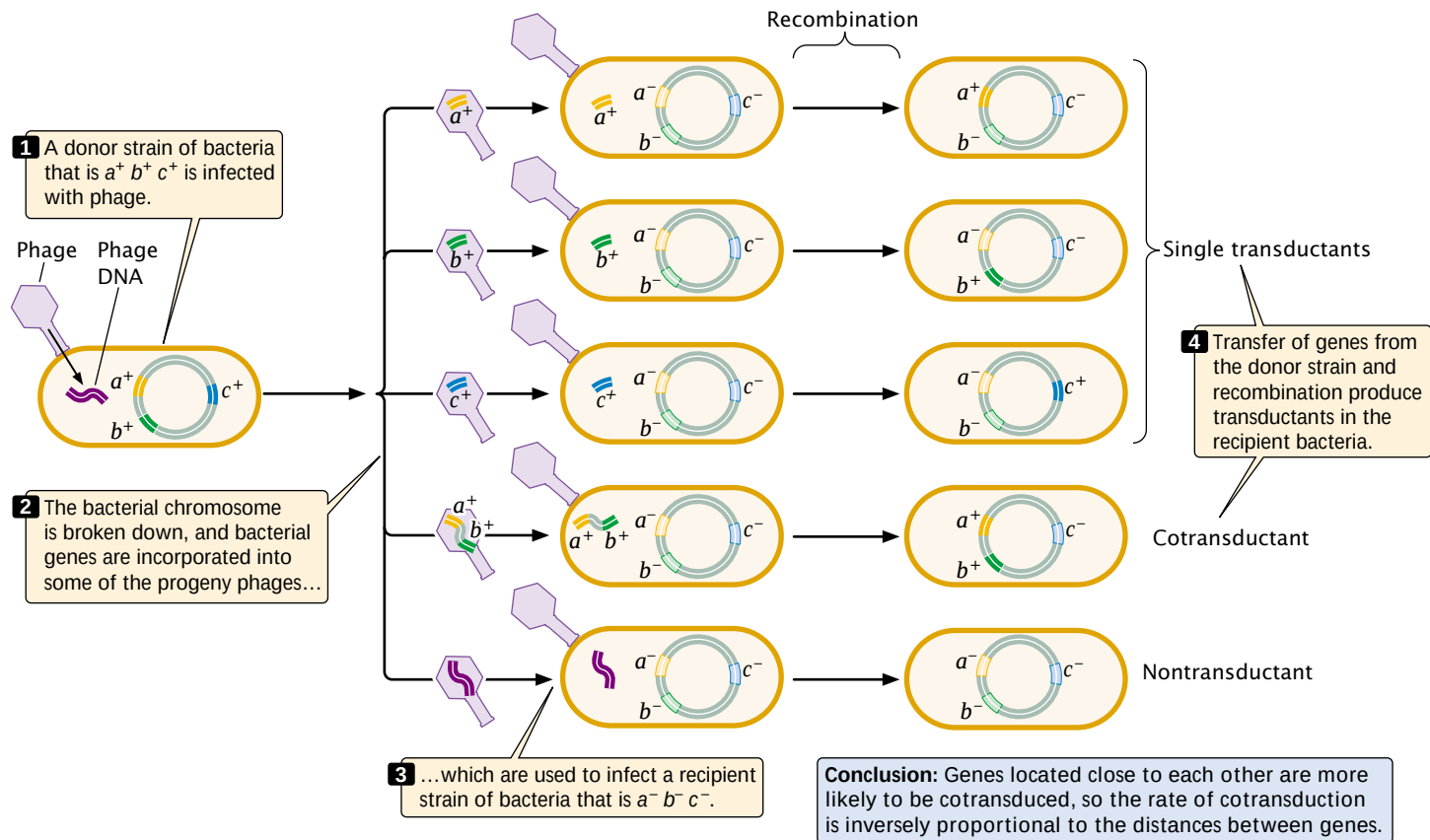
Concepts

In transduction, bacterial genes become packaged into a viral coat, are transferred to another bacterium by the virus, and become incorporated into the bacterial chromosome by crossing over. Bacterial genes can be mapped with the use of generalized transduction.

Specialized transduction Like generalized transduction, specialized transduction requires gene transfer from one bacterium to another through phages, but here only genes near particular sites on the bacterial chromosome are transferred. This process requires lysogenic bacteriophages. The



8.27 Genes can be transferred from one bacterium to another through generalized transduction.



8.28 Generalized transduction can be used to map genes.

prophage may imperfectly excise from the bacterial chromosome, carrying with it a small part of the bacterial DNA adjacent to the site of prophage integration. A phage carrying this DNA will then inject it into another bacterial cell in the next round of infection. This process resembles the situation in F' cells, where the F plasmid carries genes from one bacterium into another (see Figure 8.16).

One of the best-studied examples of specialized transduction is in bacteriophage lambda (λ), which integrates into the *E. coli* chromosome at the **attachment** (*att*) site. The phage DNA contains a site similar to the *att* site; a single crossover integrates the phage DNA into the bacterial chromosome (FIGURE 8.29a). The λ prophage is excised through a similar crossover that reverses the process (FIGURE 8.29b and c).

An error in excision may cause genes on either side of the bacterial *att* site to be excised along with some of the phage DNA (FIGURE 8.29d and e). In *E. coli*, these genes are usually the *gal* (galactose fermentation) and *bio* (biotin biosynthesis) genes. When a transducing phage carrying the *gal* gene infects another bacterium, the gene may integrate into the bacterial chromosome along with the prophage (FIGURE 8.29f), giving the bacterial chromosome two copies of the *gal* gene (FIGURE 8.29g). These transductants are

unstable, because the prophage DNA may excise from the chromosome, carrying the introduced gene with it. Stable transductants are produced when the *gal* gene in the phage is exchanged for the *gal* gene in the chromosome through a double crossover (FIGURE 8.29h).

Concepts

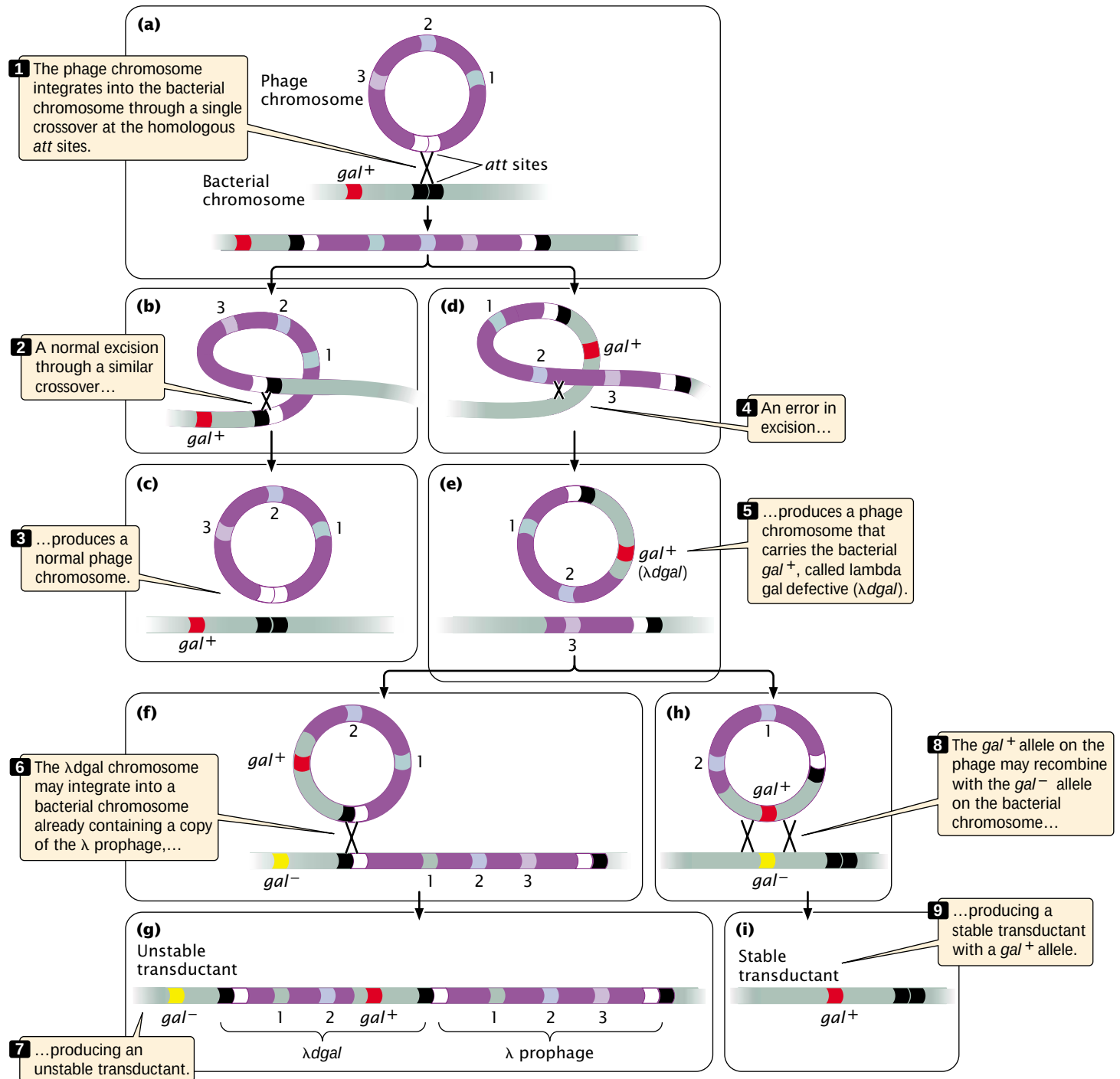
Specialized transduction transfers only those bacterial genes located near the site of prophage insertion.

Connecting Concepts

Three Methods for Mapping Bacterial Genes

Three methods of mapping bacterial genes have now been outlined: (1) interrupted conjugation; (2) transformation; and (3) transduction. These methods have important similarities and differences.

Mapping with interrupted conjugation is based on the time required for genes to be transferred from one



8.29 Bacteria can exchange genes through specialized transduction.

Segments 1, 2, and 3 represent genes on the phage chromosome.

bacterium to another by means of cell-to-cell contact. The key to this technique is that the bacterial chromosome itself is transferred, and the order of genes and the time required for their transfer provide information about the positions of the genes on the chromosome. In contrast with other mapping methods, the distance between genes is measured

not in recombination frequencies but units of time required for genes to be transferred. Here, the basic unit of conjugation mapping is a minute.

In gene mapping with transformation, DNA from the donor strain is isolated, broken up, and mixed with the recipient strain. Some fragments pass into the recipient cells,

where the transformed DNA may recombine with the bacterial chromosome. The unit of transfer here is a random fragment of the chromosome. Loci that are close together on the donor chromosome tend to be on the same DNA fragment; so the rates of cotransformation provide information about the relative positions of genes on the chromosome.

Transduction mapping also relies on the transfer of genes between bacteria that differ in two or more traits, but here the vehicle of gene transfer is a bacteriophage. In a number of respects, transduction mapping is similar to transformation mapping. Small fragments of DNA are carried by the phage from donor to recipient bacteria, and the rates of cotransduction, like the rates of cotransformation, provide information about the relative distances between the genes.

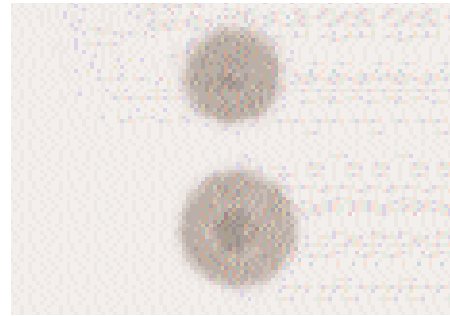
All of the methods use a common strategy for mapping bacterial genes. The movement of genes from donor to recipient is detected by using strains that differ in two or more traits, and the transfer of one gene relative to the transfer of others is examined. Additionally, all three methods rely on recombination between the transferred DNA and the bacterial chromosome. In mapping with interrupted conjugation, the relative order and timing of gene transfer provide the information necessary to map the genes; in transformation and transduction, the rate of cotransfer provides this information.

In conclusion, the same basic strategies are used for mapping with interrupted conjugation, transformation, and transduction. The methods differ principally in their mechanisms of transfer: in conjugation mapping, DNA is transferred through contact between bacteria; in transformation, DNA is transferred as small naked fragments; and, in transduction, DNA is transferred by bacteriophages.

Fine-Structure Analysis of Bacteriophage Genes

In the 1950s and 1960s, Seymour Benzer conducted a series of experiments to examine the structure of a gene. Because no molecular techniques were available at the time for directly examining nucleotide sequences, Benzer was forced to infer gene structure from analyses of mutations and their effects. The results of his studies showed that different mutational sites *within* a single gene could be mapped (**intragenic mapping**) by using techniques similar to those just described. Different sites within a single gene are very close together; so recombination between them takes place at a very low frequency. Because large numbers of progeny are required to detect these recombination events, Benzer used the bacteriophage T4, which reproduces rapidly and produces large numbers of progeny.

Benzer's mapping techniques Wild-type T4 phages normally produce small plaques with rough edges when

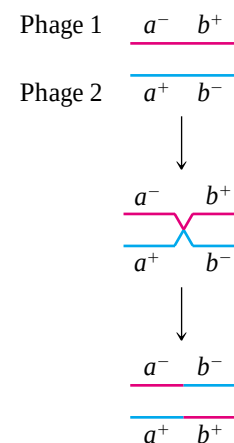


8.30 T4 phage *rII* mutants produce distinct plaques when grown on *E. coli* B cells. (Dr. D. P. Snustad, College of Biological Sciences, University of Minnesota.)

grown on a lawn of *E. coli* bacteria. Certain mutants, called *r* for rapid lysis, produce larger plaques with sharply defined edges. Benzer isolated phages with a number of different *r* mutations, concentrating on one particular subgroup called *rII* mutants.

Wild-type T4 phages produce typical plaques (FIGURE 8.30) on *E. coli* strains B and K. In contrast, the *rII* mutants produce *r* plaques on strain B and do not form plaques at all on strain K. Benzer recognized the *r* mutants by their distinctive plaques when grown on *E. coli* B. He then collected lysate from these plaques and used it to infect *E. coli* K. Phages that did not produce plaques on *E. coli* K were defined as the *rII* type.

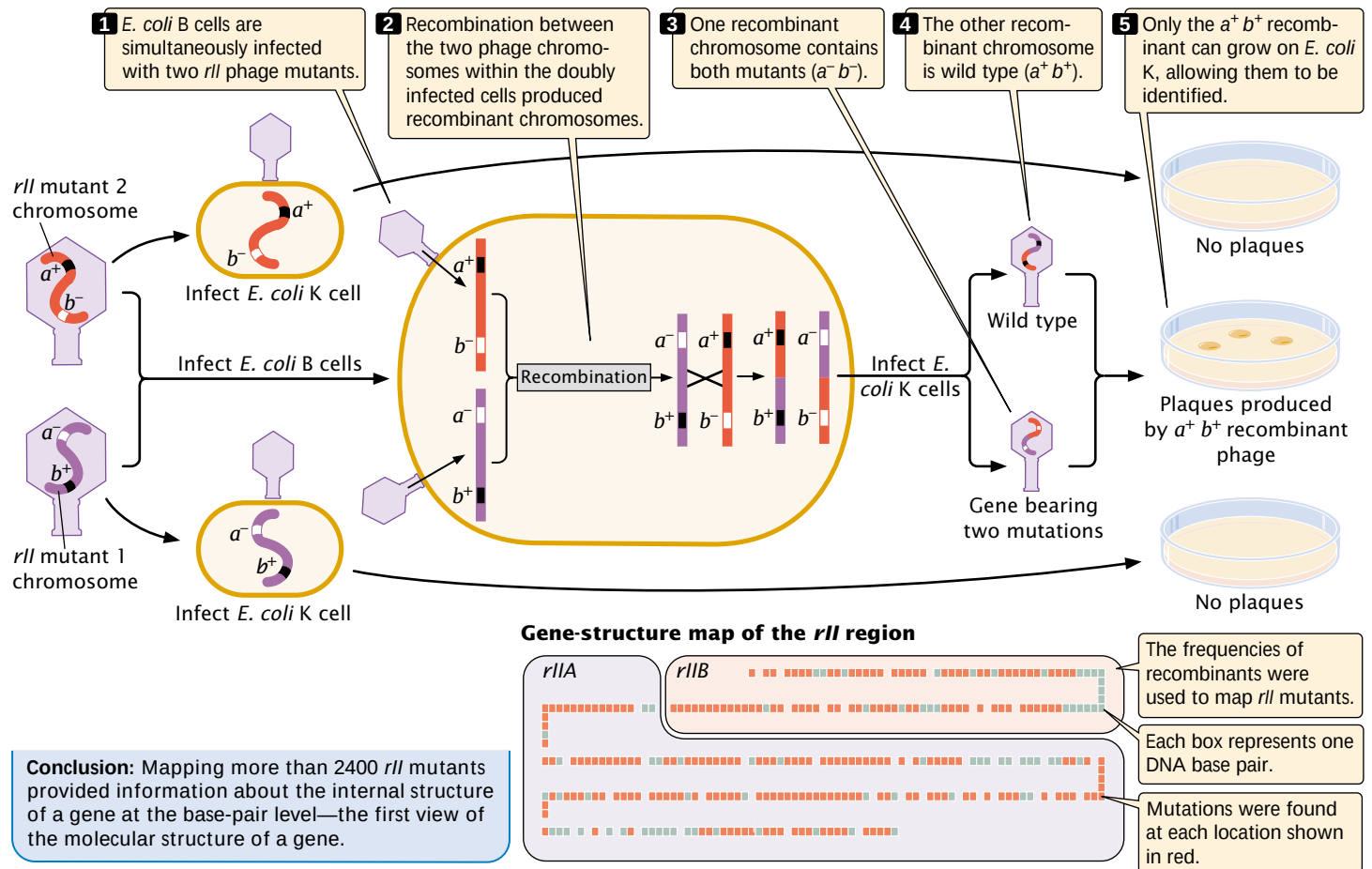
Benzer collected thousands of *rII* mutations. He simultaneously infected bacterial cells with two different mutants and looked for recombinant progeny (FIGURE 8.31). Consider two *rII* mutations, a^- and b^- , and their wild-type alleles, a^+ and b^+ . Benzer infected *E. coli* B cells with two different strains of phages, one $a^- b^+$ and the other $a^+ b^-$ (Figure 8.31, step 1). While reproducing within the B cells, a few phages of the two strains recombined (Figure 8.31, step 2). A single crossover produces two recombinant chromosomes; one with genotype $a^+ b^+$ and the other with genotype $a^- b^-$:



The resulting recombinant chromosomes, along with the nonrecombinant (parental) chromosomes, were incorporated

Experiment

Question: How can *rII* phage mutants be mapped and what can they reveal about the structure of the gene?



8.31 Benzer developed a procedure for mapping *rII* mutants. Two different *rII* mutants ($a^- b^-$ and $a^+ b^-$) are isolated on *E. coli* B cells. Neither will grow on *E. coli* K cells. Only the $a^+ b^+$ recombinant can grow on *E. coli* K, allowing these recombinants to be identified. *rIIA* and *rIIB* refer to different parts of the gene.

into progeny phages (Figure 8.31, steps 3 and 4), which were then used to infect *E. coli* K cells. The resulting plaques were examined to determine the genotype of the infecting phage.

The *rII* mutants would not grow on *E. coli* K, but wild-type phages could; so progeny phages with the recombinant genotype $a^+ b^+$ produced plaques on *E. coli* K (Figure 8.31, step 5). Each recombination event produces an equal number of double mutants ($a^- b^-$) and wild-type chromosomes ($a^+ b^+$); so the number of recombinant progeny should be twice the number of wild-type plaques that appeared on *E. coli* K. The recombination frequency between the two *rII* mutants would be:

$$\text{recombination frequency} = \frac{2 \times \text{number of plaques on } E. coli K}{\text{total number of plaques on } E. coli B}$$

Benzer was able to detect a single recombinant among billions of progeny phages, allowing very low rates of recombination to be detected. Recombination frequencies are proportional to physical distances along the chromosome (p. 000 in Chapter 7), revealing the positions of the different mutations within the *rII* region of the phage chromosome. In this way, Benzer eventually mapped more than 2400 *rII* mutations, many corresponding to single base pairs in the viral DNA. His work provided the first molecular view of a gene.

Concepts

In a series of experiments with the bacteriophage T4, Seymour Benzer showed that recombination could occur within a single gene and created the first molecular map of a gene.

Complementation experiments At the time Benzer was conducting his experiments, the relationship between genes and DNA structure was unknown. A gene had been defined as a functional unit of heredity that coded for a phenotype. To test whether different *rII* mutations belonged to different functional genes, Benzer used the complementation (cis-trans) test (see pp. xxx in Chapter 5).

Individuals heterozygous for two mutations may have the mutations in trans,

$$\frac{a^+}{a^-} \quad \frac{b^-}{b^+}$$

meaning that they are located on different chromosomes, or in cis, meaning that they are located on the same chromosome

$$\frac{a^-}{b^+} \quad \frac{a^-}{b^+}$$

(see p. xxx in Chapter 7). Suppose that the a^- and b^- mutations occur at different loci, which code for different proteins. In the trans heterozygote

$$\frac{a^+}{a^-} \quad \frac{b^-}{b^+}$$

one chromosome has a functional allele at the *a* locus ($a^+ b^-$) and the other chromosome has a functional allele at the *b* locus ($a^- b^+$); since a^- and b^- are recessive mutations, both A and B proteins will be produced. The two mutations complement each other, so the presence of wild type trait in the trans heterozygote indicates that these mutations belong to different complementation groups—they come from different loci.

Suppose the two mutations occur within a single locus that codes for one protein. In the trans heterozygote, one chromosome fails to produce a functional protein because it has a defect at the *b* site ($a^+ b^-$) and the other chromosome fails to produce a functional protein because it has a defect at the *a* site ($a^- b^+$). No functional protein is produced by either chromosome, and the trans heterozygote has a mutant phenotype—the mutations are unable to complement each other.

The heterozygous individual used in complementation testing must have the mutations in the trans configuration. When the mutations are in the cis configuration:

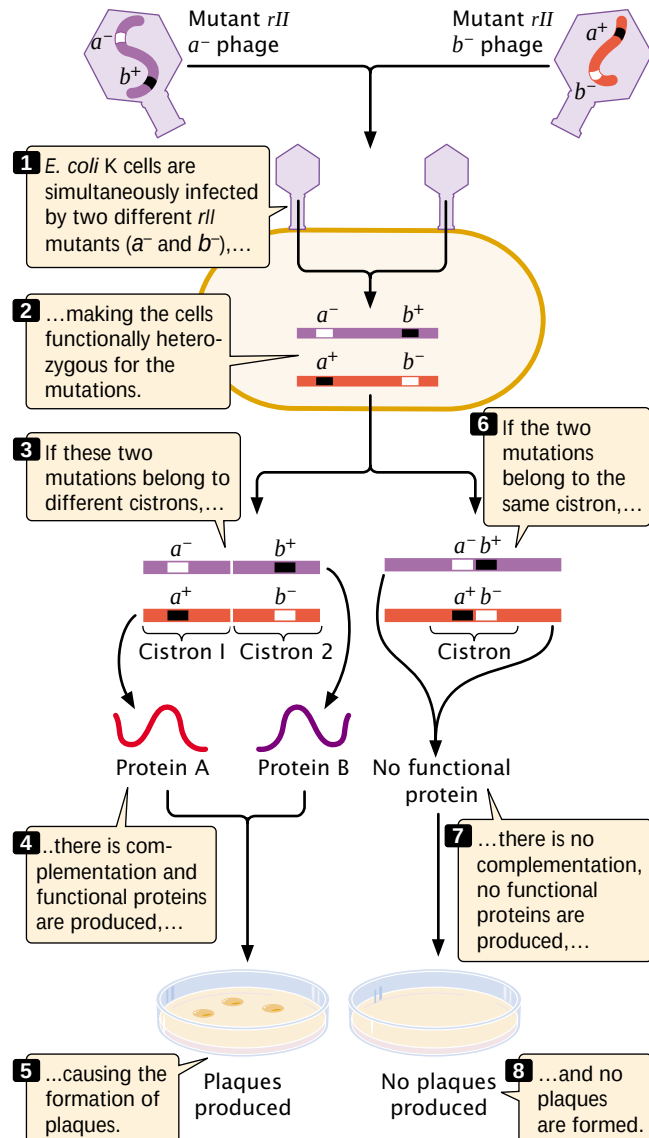
$$\frac{a^-}{a^+} \quad \frac{b^-}{b^+}$$

heterozygotes will have a wild-type phenotype regardless of whether the two mutations occur at the same locus or at different loci, because one chromosome ($a^+ b^+$) is mutation free.

To carry out the complementation test in bacteriophage, Benzer infected cells of *E. coli* K with large numbers of two mutant strains of phage (FIGURE 8.32, step 1). We will refer to the two mutations as *rIIa* ($a^- b^+$) and *rIIb* ($a^+ b^-$). Cells infected with both mutants:

Experiment

Question: How do we determine whether two different *rII* mutants occur at the same locus?

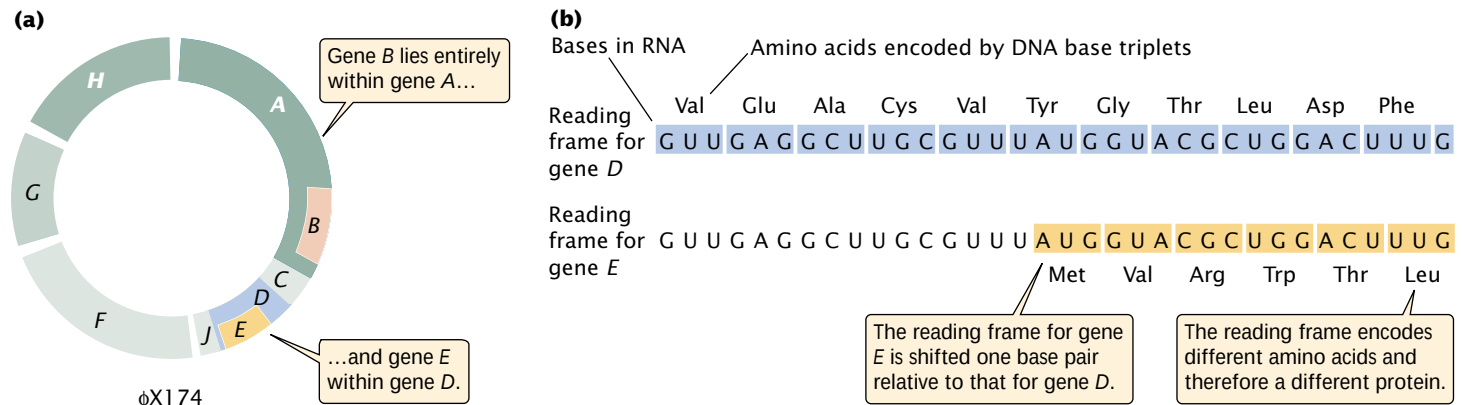


Conclusion: The complementation test indicates whether two mutations occur at the same locus or at different loci.

8.32 Complementation tests are used to determine whether different mutations are at the same functional gene.

$$\frac{a^-}{a^+} \quad \frac{b^+}{b^-}$$

were effectively heterozygous for the phage genes, with the mutations in the trans configuration (FIGURE 8.32, step 2). In the complementation testing, the phenotypes of progeny phages were examined on the K strain, rather than the B strain as illustrated in Figure 8.31.



8.33 The genome of bacteriophage ϕ X174 contains overlapping genes. The genome contains nine genes (A through J).

If the *rIIa* and *rIIb* mutations occur at different loci that code for different proteins then, in bacterial cells infected by both mutants, the wild-type sequences on the chromosome opposite each mutation will overcome the effects of the recessive mutations; the phages will produce normal plaques on *E. coli* K cells (FIGURE 8.32, steps 3, 4, and 5). (Benzer coined the term *cistron* to designate a functional gene defined by the complementation test.) If, on the other hand, the mutations occur at the same locus, no functional protein is produced by either chromosome, and no plaques develop in the *E. coli* K cells (FIGURE 8.32 steps 6, 7, and 8). Thus, the absence of plaques indicates that the two mutations occur at the same locus.

In the complementation test, the *cis* heterozygote is used as a control. Benzer simultaneously infected bacteria with wild-type phage ($a^+ b^+$) and with phage carrying both mutations ($a^- b^-$). This test also produced cells that were heterozygous and *cis* for the phage genes:

$$\frac{a^+}{a^-} \quad \frac{b^+}{b^-}$$

Regardless of whether the *rIIa* and *rIIb* mutations are in the same functional unit, these cells contain a copy of the wild-type phage chromosome ($a^+ b^+$) and will produce normal plaques in *E. coli* K.

Benzer carried out complementation testing on many pairs of *rII* mutants. He found that the *rII* region consists of two loci, designated *rIIA* and *rIIB*. Mutations belonging to the *rIIA* and *rIIB* groups complemented each other, but mutations in the *rIIA* group did not complement others in *rIIA*; nor did mutations in the *rIIB* group complement others in *rIIB*.

Concepts

Benzer used the complementation test to distinguish between functional genes (loci).

At the time of Benzer's research, many geneticists believed that genes were indivisible and that recombination could not take place within them. Benzer demonstrated that intragenic recombination did indeed take place (although at a very low rate) and gave geneticists their first glimpse at the structure of an individual gene.

Overlapping Genes

The first viral genome to be completely sequenced, that of bacteriophage ϕ X174, revealed surprising information: the nucleotide sequences of several genes overlapped. This genome encodes nine proteins (FIGURE 8.33). Two of the genes are nested within other genes; in both cases, the same DNA sequence codes for two different proteins by using different reading frames (see p. 000 in Chapter 13). In five of the ϕ X174 genes, the initiation codon of one gene overlaps the termination codon of another.

The results of subsequent studies revealed that overlapping genes are found in a number of viruses and bacteria. Viral genome size is strictly limited by the capacity of the viral protein coat; so there is strong selective pressure for economic use of the DNA.

Concepts

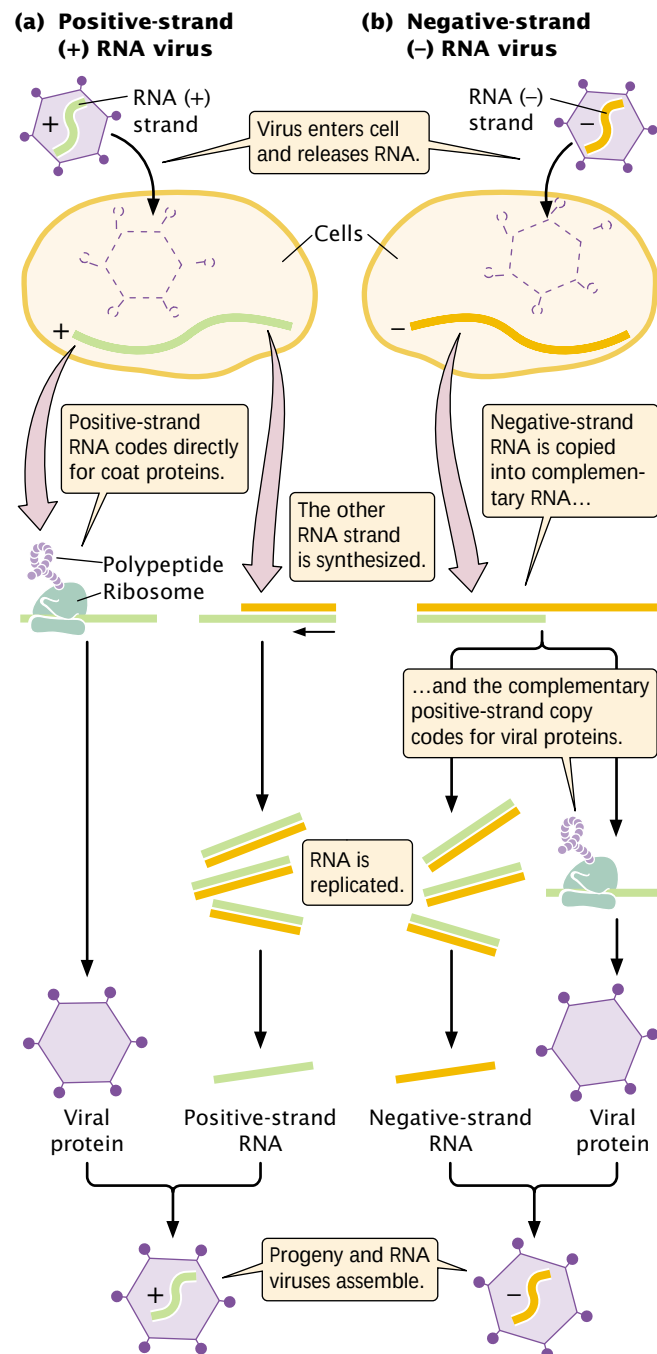
Some viruses contain overlapping genes, in which the same base sequence specifies more than one protein.

RNA Viruses

Viral genomes may be encoded in either DNA or RNA. Some medically important human viruses have RNA as their genetic material, including those that cause influenza, common colds, polio, and AIDS. Almost all viruses that infect plants have RNA genomes. The medical and economic importance of RNA viruses has encouraged their study.

RNA viruses, like bacteriophages, reproduce by infecting cells and making copies of themselves. Most use RNA-dependent RNA polymerases encoded by their own genes.

In **positive-strand RNA viruses**, the genomic RNA molecule carried inside the viral particle codes directly for viral proteins (● **FIGURE 8.34a**). In **negative-strand RNA viruses**, the virus first makes a complementary copy of its RNA genome, which is then translated into viral proteins (● **FIGURE 8.34b**).



8.34 The process of reproduction differs in positive-strand RNA viruses and negative-strand RNA viruses.

RNA viruses capable of integrating into the genome of their hosts, much as temperate phages insert themselves into bacterial chromosomes, are called **retroviruses** (● **FIGURE 8.35a**). Because the retroviral genome is RNA, whereas that of the host is DNA, a retrovirus must produce **reverse transcriptase**, an enzyme that synthesizes complementary DNA (cDNA) from either an RNA or a DNA template. A retrovirus uses reverse transcriptase to make a double-stranded DNA copy from its single-stranded RNA genome. The DNA copy then integrates into the host chromosome to form a **provirus**, which is replicated by host enzymes when the host chromosome is duplicated (● **FIGURE 8.35b**).

When conditions are appropriate, the provirus undergoes transcription to produce numerous copies of the original RNA genome. This RNA codes for viral proteins and serves as genomic RNA for new viral particles. As these viruses escape the cell, they collect patches of the cell membrane to use as their envelopes.

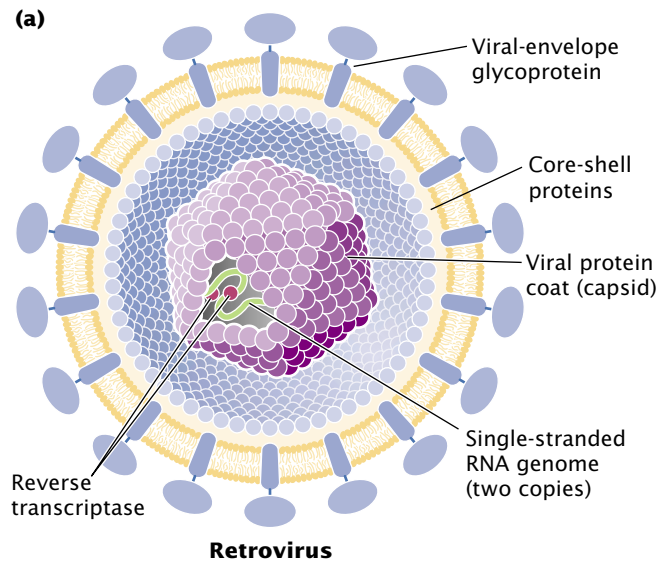
All known retroviral genomes have in common three genes: *gag*, *pol*, and *env* (● **FIGURE 8.36**), each encoding a precursor protein that is cleaved into two or more functional proteins. The *gag* gene encodes the three or four proteins that make up the viral capsid. The *pol* gene codes for reverse transcriptase and an enzyme, called **integrase**, that inserts the viral DNA into the host chromosome. The *env* gene codes for the glycoproteins, which appear on the viral envelope that surrounds the viral capsid.

Some retroviruses contain **oncogenes** (Chapter 20) that may stimulate cell division and cause the formation of tumors. The first retrovirus to be isolated, the Rous sarcoma virus, was originally recognized by its ability to produce connective-tissue tumors (sarcomas) in chickens.

The human immunodeficiency virus (HIV) is a retrovirus that causes acquired immune deficiency syndrome. AIDS was first recognized in 1982, when a number of homosexual males in the United States began to exhibit symptoms of a new immune-system-deficiency disease. In that year, Robert Gallo proposed that AIDS was caused by a retrovirus. Between 1983 and 1984, as the AIDS epidemic became widespread, the HIV retrovirus was isolated from AIDS patients.

HIV is thought to have appeared first in Africa in the 1950s or 1960s. It is closely related to several retroviruses found in monkeys and may have evolved when a monkey retrovirus mutated and infected humans. HIV is transmitted by sexual contact between humans and through any type of blood-to-blood contact, such as that caused by the sharing of dirty needles by drug addicts. Until screening tests could identify HIV-infected blood, transfusions and clotting factors used by hemophiliacs also were sources of infection.

HIV principally attacks a class of blood cells called helper T lymphocytes (● **FIGURE 8.37**). HIV enters a helper T cell, undergoes reverse transcription, and integrates into the chromosome. The virus reproduces rapidly, destroying the T cell as new virus particles escape from the cell. Because helper T cells are central to immune function and



8.35 A retrovirus uses reverse transcription to incorporate its RNA into the host DNA. (a) Structure of a typical retrovirus. Two copies of the single-stranded RNA genome and reverse transcriptase enzyme are shown enclosed within a protein capsid. The capsid is surrounded by a viral envelope that is studded with viral glycoproteins. (b) The retrovirus life cycle.

are destroyed in the infection, AIDS patients have a diminished immune response—most AIDS patients die of secondary infections that develop because they have lost the ability to fight off pathogens.

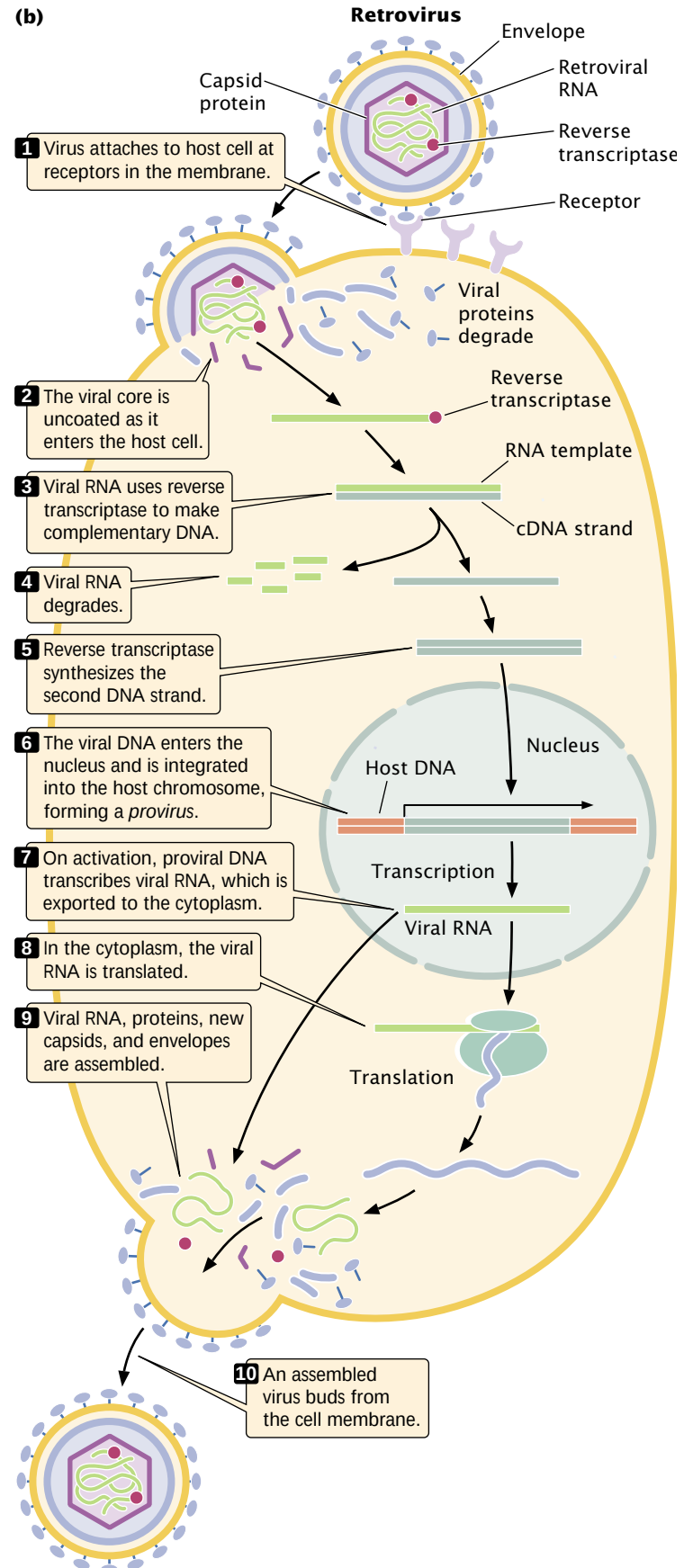
The HIV genome is 9749 nucleotides long and carries *gag*, *pol*, *env*, and six other genes that regulate the life cycle of the virus. HIV's reverse transcriptase is very error prone, giving the virus a high mutation rate and allowing it to evolve rapidly, even within a single host. This rapid evolution makes the development of an effective vaccine against HIV particularly difficult.

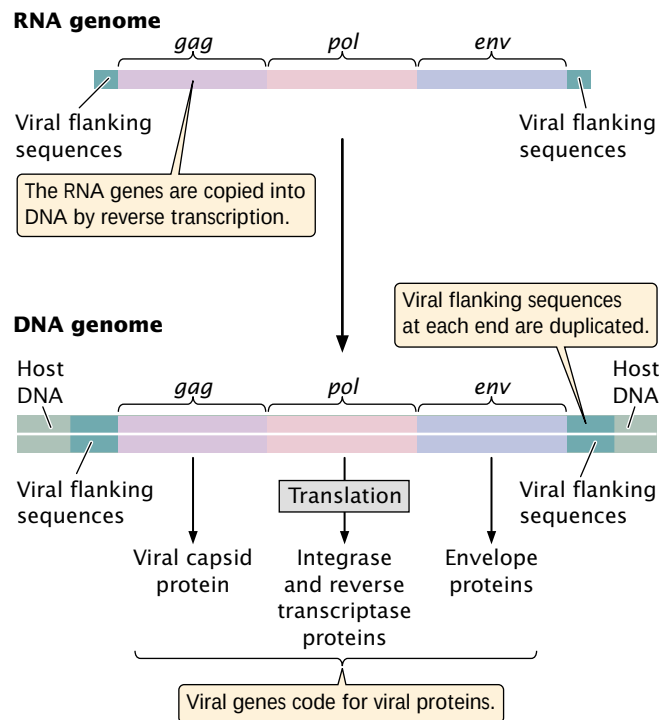
Concepts

Retrovirus is an RNA virus that integrates into its host chromosome by making a DNA copy of its RNA genome through the process of reverse transcription. Human immunodeficiency virus, the causative agent of AIDS, is a retrovirus.

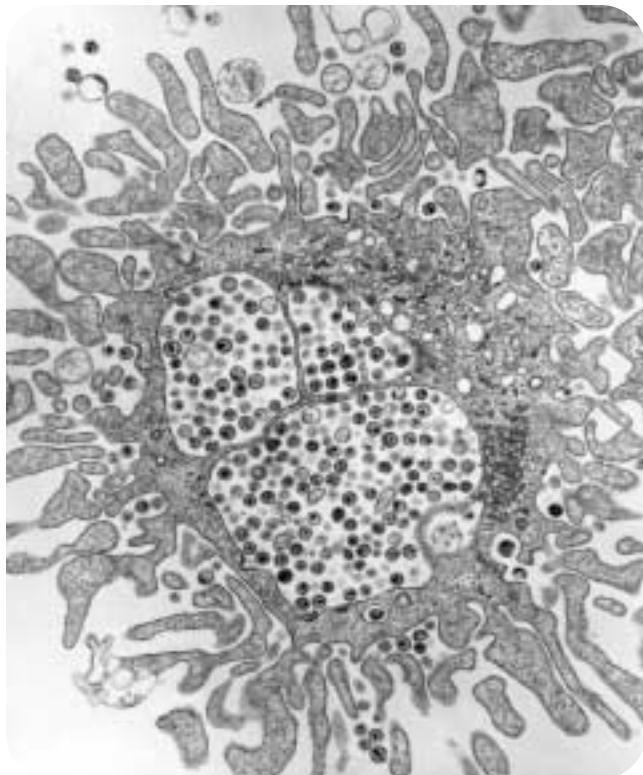
Prions: Pathogens Without Genes

In 1997, Stanley B. Prusiner was awarded the Nobel Prize in Physiology or Medicine for his discovery and characterization of **prions**, a novel class of pathogens that cause several rare neurodegenerative diseases and that appear to replicate without any genes. Initially, Prusiner's proposal that prions were composed entirely of protein and lacked any trace of nucleic acid was met with skepticism. One of the foundations of modern biology is that all living things possess





8.36 The typical genome of a retrovirus contains *gag*, *pol*, and *env* genes.



8.37 HIV principally attacks T lymphocytes. Electron micrograph showing a T cell infected with HIV. (Courtesy of Dr. Hans Gelderblom.)

hereditary information in the form of DNA or RNA, and so how are prions able to reproduce without nucleic acid?

Prions were first recognized as unusual infectious agents that cause scrapie, a disease of sheep that destroys the brain. In 1982, Prusiner purified the scrapie pathogen and reported that it consisted entirely of protein. Prusiner and his colleagues eventually showed that the prion protein (PrP) is derived from a normal protein that is encoded by a gene found throughout eukaryotes, including yeast. Normal PrP (PrP^C) is folded into a helical shape, but the protein can also fold into a flattened β sheet that causes scrapie (PrP^{Sc}) (FIGURE 8.38). When PrP^{Sc} is present, it interacts with and causes PrP^C to fold into the disease-causing form of the protein; infection with PrP^{Sc} converts an individual's normal PrP protein into abnormal PrP that forms prions. Accumulation of the PrP^{Sc} in the brain appears to be responsible for the neurological degeneration associated with diseases caused by prions. This explanation for prion diseases, called the "protein only" hypothesis, is not universally accepted; some scientists still believe that these diseases are caused by an as-of-yet unisolated virus.

Prions cause scrapie, bovine spongiform encephalopathy (BSE, or "mad cow" disease), and kuru, an exotic disorder spread among New Guinea aborigines by ritualistic cannibalism. They also play a role in some inherited human neurodegenerative disorders, including Creutzfeldt-Jakob disease and Gerstmann-Scheinker disease. In these inherited diseases, the PrP gene is mutated and produces a type of PrP that is more susceptible to folding into PrP^{Sc}. Nearly all those who carry such a mutated gene eventually produce prions and get the disease.

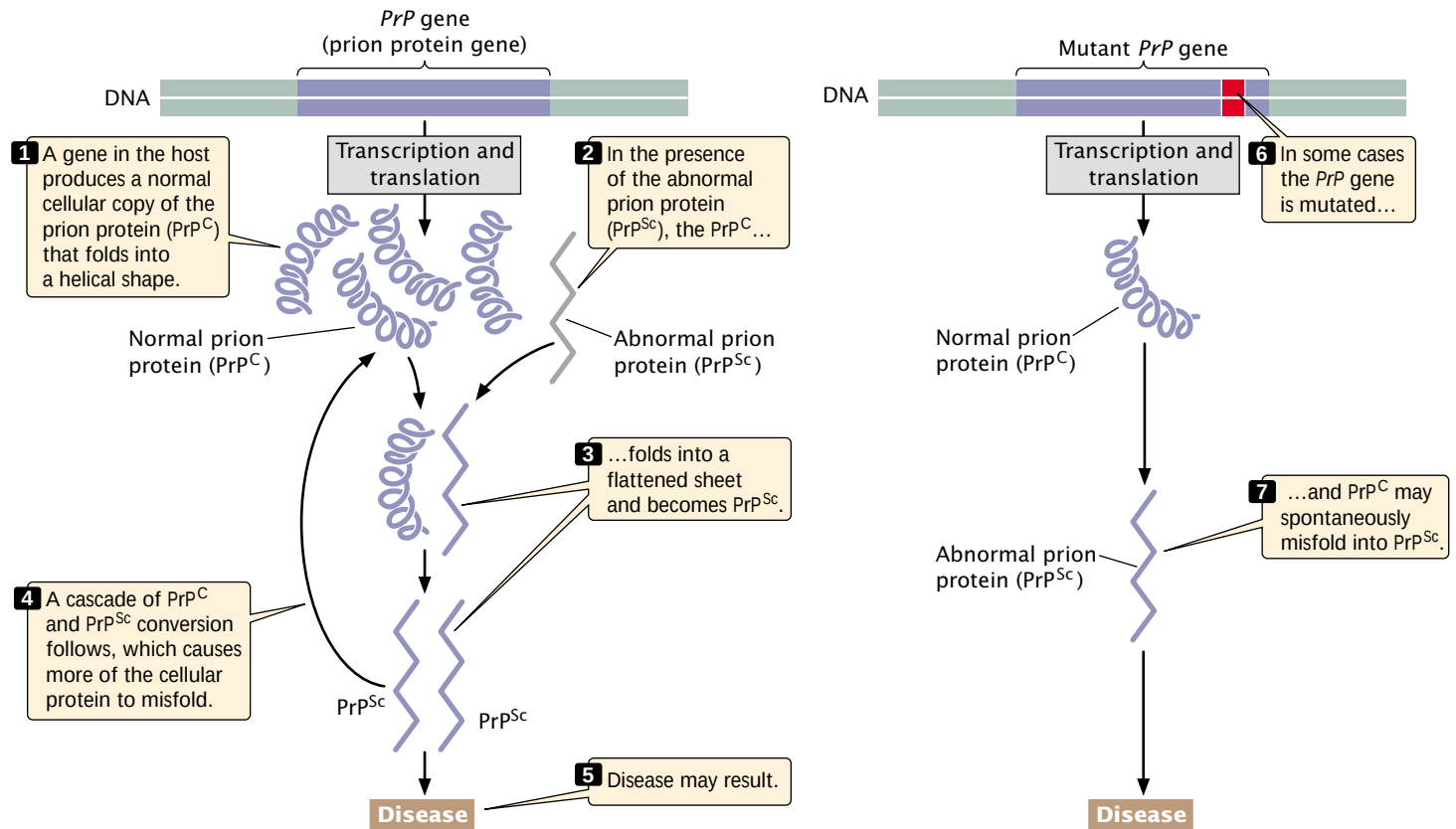
Some cases of human prion diseases have been traced to injections of growth hormone, which until recently was obtained from the brains of human cadavers infected with prions. In England, an epidemic of mad cow disease erupted in the late 1980s, the origin of which was traced to cattle feed containing the remains of sheep infected with scrapie.

www.whfreeman.com/pierce For more information on prions, prion-caused diseases, and Stanley Prusiner's account of his hunt for the secret of prions.

Connecting Concepts Across Chapters

Bacteria and viruses have been used extensively in the study of genetics: their rapid reproduction, large numbers of progeny, small haploid genomes, and medical importance make them ideal organisms for many types of genetic investigations.

This chapter examined some of the techniques used to study and map bacterial and viral genomes. Some of these methods are an extension of the principles of recombination and gene mapping explored in Chapter 7.



8.38 The protein-only hypothesis describes a method for the replication of prions.

Bacterial reproduction was discussed in Chapter 2, and a number of the principles and techniques covered in this chapter are linked to topics in future chapters. Bacterial chromosomes will be considered in more detail in Chapter 11, and bacterial replication, transcription, translation, and gene regulation will be the topics of Chapters 12 through 16. Bacteria are central to recombinant DNA technology, the topic of Chapter 18, where they are often used in mass producing specific DNA fragments. Many of the tools of recombinant DNA technology, including plasmids, restriction enzymes, DNA polymerases, and many other enzymes, have been isolated and engineered from natural components of bacterial

cells. Engineered viruses are common vehicles for delivering genes to host cells.

Some transposable genetic elements (discussed in Chapter 11) are closely related to viruses, and considerable evidence suggests that viruses evolved from such elements. Because their mutations are easily isolated, bacteria also play an important role in the study of gene mutations, a topic examined in Chapter 17. Chapter 20 deals with mitochondrial and chloroplast DNA, which in many respects are more similar to bacterial DNA than to the nuclear DNA of the cells in which these organelles are found. Finally, viruses cause some cancers, and the role of viral genes in cancer development is studied in Chapter 21.

CONCEPTS SUMMARY

- Bacteria and viruses are well suited to genetic studies: they are small, have a small haploid genome, undergo rapid reproduction, and produce large numbers of progeny through asexual reproduction. When spread on a petri plate, individual bacteria grow into colonies of identical cells that can be easily seen.
- The bacterial genome normally consists of a single, circular molecule of double-stranded DNA.
- Plasmids are small pieces of bacterial DNA that can replicate independently of the large chromosome. Episomes are plasmids that can exist either in a freely replicating state or can integrate into the bacterial chromosome.

- DNA may be transferred between bacteria by means of conjugation, transformation, and transduction.
- Conjugation is the union and the transfer of genetic material between two bacterial cells and is controlled by a fertility factor called F, which is an episome. F^+ cells are donors, and F^- cells are recipients during conjugation. An Hfr cell has F incorporated into the bacterial chromosome. An F' cell has an F factor that has excised from the bacterial genome and carries some bacterial genes.
- The rate at which individual genes are transferred from Hfr to F^- cells during conjugation provides information about the order and distance between the genes on the bacterial chromosome.
- In transformation, bacteria take up DNA from their environment. Frequencies of cotransformation provide information about the physical distances between chromosomal genes.
- Viruses are replicating structures with DNA or RNA genomes that may be double stranded or single stranded, linear or circular. Bacteriophages are viruses that infect bacteria. An individual phage can be identified when it enters a bacterial cell, multiplies, and eventually produces a patch of lysed bacterial cells (a plaque) on an agar plate.
- Phage genes can be mapped by infecting bacterial cells with two different strains of phage. The numbers of recombinant plaques produced by the progeny phages are used to estimate recombination rates between phage genes.
- In generalized transduction, bacterial genes become incorporated into phage coats and are transferred to other bacteria during phage infection. Rates of cotransduction can be used to determine the order and distance between genes on the bacterial chromosome.
- In specialized transduction, DNA near the site of phage integration on the bacterial chromosome is transferred from one bacterium to another.
- Benzer mapped a large number of mutations that occurred within the *rII* region of phage T4 and showed that intragenic recombination takes place. The results of his complementation studies demonstrated that the *rII* region consists of two functional units (cistrons).
- A number of viruses have RNA genomes. In positive-strand viruses, the RNA genome codes directly for viral proteins; in negative-strand viruses, a complementary copy of the genome is translated to form viral proteins. Retroviruses encode a reverse transcriptase enzyme used to make a DNA copy of the viral genome, which then integrates into the host genome as a provirus.
- Prions are infectious agents consisting only of protein; they are thought to cause disease by altering the shape of proteins encoded by the host genome.

IMPORTANT TERMS

minimal medium (p. 199)	competent cell (p. 211)	specialized transduction (p. 216)	negative-strand RNA virus (p. 223)
complete medium (p. 200)	transformant (p. 212)	transducing phage (p. 217)	retrovirus (p. 224)
colony (p. 200)	cotransformation (p. 212)	transductants (p. 217)	reverse transcriptase (p. 224)
plasmid (p. 202)	virus (p. 213)	cotransduction (p. 217)	provirus (p. 224)
episome (p. 203)	virulent phage (p. 214)	attachment site (p. 218)	integrase (p. 224)
F factor (p. 203)	temperate phage (p. 214)	intragenic mapping (p. 220)	oncogene (p. 224)
conjugation (p. 203)	prophage (p. 214)	positive-strand RNA virus (p. 223)	prion (p. 225)
transformation (p. 203)	plaque (p. 214)		
transduction (p. 203)	generalized transduction (p. 216)		
pili (p. 205)			

Worked Problems

1. DNA from a strain of bacteria with genotype $a^+ b^+ c^+ d^+ e^+$ was isolated and used to transform a strain of bacteria that was $a^- b^- c^- d^- e^-$. The transformed cells were tested for the presence of donated genes. The following genes were cotransformed:

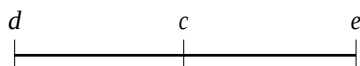
- a^+ and d^+
- b^+ and e^+
- c^+ and d^+
- c^+ and e^+

What is the order of genes a , b , c , d , and e on the bacterial chromosome?

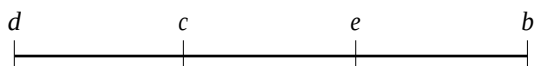
• Solution

The rate at which genes are cotransformed is inversely proportional to the distance between them: genes that are close together are frequently cotransformed, whereas genes that are far apart are rarely cotransformed. In this transformation experiment, gene c^+ is cotransformed with both genes e^+ and d^+ , but genes e^+ and d^+

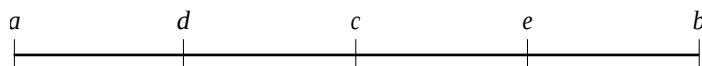
are not cotransformed; therefore the *c* locus must be between the *d* and *e* loci:



Gene *e*⁺ is also cotransformed with gene *b*⁺; so the *e* and *b* loci must be located close together. Locus *b* could be on either side of locus *e*. To determine whether locus *b* is on the same side of *e* as locus *c*, we look to see whether genes *b*⁺ and *c*⁺ are cotransformed. They are not; so locus *b* must be on the opposite side of *e* from *c*:



Gene *a*⁺ is cotransformed with gene *d*⁺; so they must be located close together. If locus *a* were located on the same side of *d* as locus *c*, then genes *a*⁺ and *c*⁺ would be cotransformed. Because these genes display no cotransformation, locus *a* must be on the opposite side of locus *d*:



2. Consider three genes in *E. coli*: *thr*⁺ (the ability to synthesize threonine), *ara*⁺ (the ability to metabolize arabinose), and *leu*⁺ (the ability to synthesize leucine). All three of these genes are close together on the *E. coli* chromosome. Phages are grown in a *thr*⁺ *ara*⁺ *leu*⁺ strain of bacteria (the donor strain). The phage lysate is collected and used to infect a strain of bacteria that is *thr*[−] *ara*[−] *leu*[−]. The recipient bacteria are then tested on medium lacking leucine. Bacteria that grow and form colonies on this medium (*leu*⁺ transductants) are then replica plated onto medium lacking threonine and medium lacking arabinose to see which are *thr*⁺ and which are *ara*⁺.

Another group of recipient bacteria are tested on medium lacking threonine. Bacteria that grew and formed colonies on this medium (*thr*⁺ transductants) were then replica plated onto medium lacking leucine and medium lacking arabinose to see

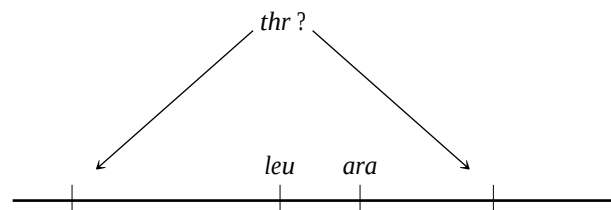
which are *ara*⁺ and which are *leu*⁺. Results from these experiments are as follows:

Selected marker	Cells with cotransduced genes (3%)
<i>leu</i> ⁺	3 <i>thr</i> ⁺ 76 <i>ara</i> ⁺
<i>thr</i> ⁺	3 <i>leu</i> ⁺ 0 <i>ara</i> ⁺

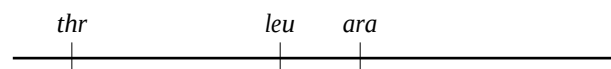
How are the loci arranged on the chromosome?

• Solution

Notice that, when we select for *leu*⁺ (the top half of the table), most of the selected cells also are *ara*⁺. This finding indicates that the *leu* and *ara* genes are located close together, because they are usually cotransduced. In contrast, *thr*⁺ is only rarely cotransduced with *leu*⁺, indicating that *leu* and *thr* are much farther apart. On the basis of these observations, we know that *leu* and *ara* are closer together than are *leu* and *thr*, but we don't yet know the order of three genes—whether *thr* is on the same side of *ara* as *leu* or on the opposite side, as shown here:



We can determine the position of *thr* with respect to the other two genes by looking at the cotransduction frequencies when *thr*⁺ is selected (the bottom half of the table). Notice that, although the cotransduction frequency for *thr* and *leu* also is 3%, no *thr*⁺ *ara*⁺ cotransductants are observed. This finding indicates that *thr* is closer to *leu* than to *ara*, and therefore *thr* must be to the left of *leu*, as shown here:



COMPREHENSION QUESTIONS

- * 1. List some of the characteristics that make bacteria and viruses ideal organisms for many types of genetic studies.
2. Explain how auxotrophic bacteria are isolated.
3. Briefly explain the differences between F⁺, F[−], Hfr, and F' cells.
- * 4. What types of matings are possible between F⁺, F[−], Hfr, and F' cells? What outcomes do these matings produce? What is the role of F factor in conjugation?
- * 5. Explain how interrupted conjugation, transformation, and transduction can be used to map bacterial genes. How are these methods similar and how are they different?
6. What types of genomes do viruses have?
7. Briefly describe the differences between the lytic cycle of virulent phages and the lysogenic cycle of temperate phages.
8. Briefly explain how genes in phages are mapped.

- * 9. How does specialized transduction differ from generalized transduction?
- *10. Briefly explain the method used by Benzer to determine whether two different mutations occurred at the same locus.
11. What is the difference between a positive-strand RNA virus and a negative-strand RNA virus?
- *12. Explain how a retrovirus, which has an RNA genome, is able to integrate its genetic material into that of a host having a DNA genome.
13. Briefly describe the genetic structure of a typical retrovirus.

APPLICATION QUESTIONS AND PROBLEMS

- *14. John Smith is a pig farmer. For the past 5 years, Smith has been adding vitamins and low doses of antibiotics to his pig food; he says that these supplements enhance the growth of the pigs. Within the past year, however, several of his pigs died from infections of common bacteria, which failed to respond to large doses of antibiotics. Can you offer an explanation for the increased rate of mortality due to infection in Smith's pigs? What advice might you offer Smith to prevent this problem in the future?
15. Rarely, conjugation of Hfr and F^- cells produces two Hfr cells. Explain how this occurs.
- *16. A strain of Hfr cells that is sensitive to the antibiotic streptomycin (str^s) has the genotype $gal^+ his^+ bio^+ pur^+ gly^+$. These cells were mixed with an F^- strain that is resistant to streptomycin (str^r) and has genotype $gal^- his^- bio^- pur^- gly^-$. The cells were allowed to undergo conjugation. At regular intervals, a sample of cells was removed and conjugation was interrupted by placing the sample in a blender. The cells were then plated on medium that contains streptomycin. The cells that grew on this medium were then tested for the presence of genes transferred from the Hfr strain. Genes from the donor Hfr strain first appeared in the recipient F^- strain at the times listed here. On the basis of these data, give the order of the genes on the bacterial chromosome and indicate the minimum distances between them:

gly^+	3 minutes
his^+	14 minutes
bio^+	35 minutes
gal^+	36 minutes
pur^+	38 minutes

- *17. A series of Hfr strains that have genotype $m^+ n^+ o^+ p^+ q^+ r^+$ are mixed with an F^- strain that has genotype $m^- n^- o^- p^- q^- r^-$. Conjugation is interrupted at regular intervals and the order of appearance of genes from the Hfr strain is determined in the recipient cells. The order of gene transfer for each Hfr strain is:

Hfr5	$m^+ q^+ p^+ n^+ r^+ o^+$
Hfr4	$n^+ r^+ o^+ m^+ q^+ p^+$
Hfr1	$o^+ m^+ q^+ p^+ n^+ r^+$
Hfr9	$q^+ m^+ o^+ r^+ n^+ p^+$

What is the order of genes on the circular bacterial chromosome? For each Hfr strain, give the location of the F factor in the chromosome and its polarity.

- *18. Crosses of three different Hfr strains with separate samples of an F^- strain are carried out, and the following mapping data are provided from studies of interrupted conjugation:

Appearance of Genes in F^- cells						
Hfr1:	Genes	b^+	d^+	c^+	f^+	g^+
	Time	3	5	16	27	59
Hfr2:	Genes	e^+	f^+	c^+	d^+	b^+
	Time	6	24	35	46	48
Hfr3:	Genes	d^+	c^+	f^+	e^+	g^+
	Time	4	15	26	44	58

Construct a genetic map for these genes, indicating their order on the bacterial chromosome and the distances between them.

19. DNA from a strain of *Bacillus subtilis* with the genotype $trp^+ tyr^+$ is used to transform a recipient strain with the genotype $trp^- tyr^-$. The following numbers of transformed cells were recovered:

Genotype	Number of transformed cells
$trp^+ tyr^-$	154
$trp^- tyr^+$	312
$trp^+ tyr^+$	354

What do these results suggest about the linkage of the *trp* and *tyr* genes?

20. DNA from a strain of *Bacillus subtilis* with genotype $a^+ b^+ c^+ d^+ e^+$ is used to transform a strain with genotype $a^- b^- c^- d^- e^-$. Pairs of genes are checked for cotransformation and the following results are obtained:

Pair of genes	Cotransformation
a^+ and b^+	no
a^+ and c^+	no
a^+ and d^+	yes
a^+ and e^+	yes
b^+ and c^+	yes
b^+ and d^+	no
b^+ and e^+	yes
c^+ and d^+	no
c^+ and e^+	yes
d^+ and e^+	no

On the basis of these results, what is the order of the genes on the bacterial chromosome?

21. DNA from a bacterial strain that is *his⁺ leu⁺ lac⁺* is used to transform a strain that is *his⁻ leu⁻ lac⁻*. The following percentages of cells were transformed:

Donor Strain	Recipient strain	Genotype of transformed cells	Percentage
<i>his⁺ leu⁺ lac⁺</i>	<i>his⁻ leu⁻ lac⁻</i>	<i>his⁺ leu⁺ lac⁺</i>	0.02
		<i>his⁺ leu⁺ lac⁻</i>	0.00
		<i>his⁺ leu⁻ lac⁺</i>	2.00
		<i>his⁺ leu⁻ lac⁻</i>	4.00
		<i>his⁻ leu⁺ lac⁺</i>	0.10
		<i>his⁻ leu⁻ lac⁺</i>	3.00
		<i>his⁻ leu⁺ lac⁻</i>	1.50

- (a) What conclusions can you make about that order of these three genes on the chromosome?
(b) Which two genes are closest?

22. Two mutations that affect plaque morphology in phages (*a⁻* and *b⁻*) have been isolated. Phages carrying both mutations (*a⁻ b⁻*) are mixed with wild-type phages (*a⁺ b⁺*) and added to a culture of bacterial cells. Subsequent to infection and lysis, samples of the phage lysate are collected and cultured on bacterial cells. The following numbers of plaques are observed:

Plaque phenotype	Number
<i>a⁺ b⁺</i>	2043
<i>a⁺ b⁻</i>	320
<i>a⁻ b⁺</i>	357
<i>a⁻ b⁻</i>	2134

What is the frequency of recombination between the *a* and *b* genes?

- *23. A geneticist isolates two mutations in bacteriophage. One mutation causes the clear plaques (*c*) and the other produces minute plaques (*m*). Previous mapping experiments have established that the genes responsible for these two mutations are 8 map units apart. The geneticist mixes phages with genotype *c⁺ m⁺* and genotype *c⁻ m⁻* and uses the mixture to infect bacterial cells. She collects the progeny phages and cultures a sample of them on plated bacteria. A total of 1000 plaques are observed. What numbers of the different types of plaques (*c⁺ m⁺*, *c⁻ m⁻*, *c⁺ m⁻*, *c⁻ m⁺*) should she expect to see?
24. The geneticist carries out the same experiment described in Problem 23, but this time she mixes phages with genotypes *c⁺ m⁻* and *c⁻ m⁺*. What results are expected with this cross?
- *25. A geneticist isolates two *r* mutants (*r₁₃* and *r₂*) that cause rapid lysis. He carries out the following crosses and counts the number of plaques listed here:

Genotype of parental phage	Progeny	Number of plaques
$h^+ r_{13}^- \times h^- r_{13}^+$	<i>h⁺ r₁₃⁺</i>	1
	<i>h⁻ r₁₃⁺</i>	104
	<i>h⁺ r₁₃⁻</i>	110
	<i>h⁻ r₁₃⁻</i>	2
	total	216
$h^+ r_2^- \times h^- r_2^+$	<i>h⁺ r₂⁺</i>	6
	<i>h⁻ r₂⁺</i>	86
	<i>h⁺ r₂⁻</i>	81
	<i>h⁻ r₂⁻</i>	7
	total	180

- (a) Calculate the recombination frequencies between *r₂* and *h* and between *r₁₃* and *h*.
(b) Draw all possible linkage maps for these three genes.
- *26. *E. coli* cells are simultaneously infected with two strains of phage λ. One strain has a mutant host range, is temperature sensitive, and produces clear plaques (genotype = *h st c*); another strain carries the wild-type alleles (genotype = *h⁺ st⁺ c⁺*). Progeny phage are collected from the lysed cells and are plated on bacteria. The genotypes of the progeny phage are:

Progeny phage genotype	Number of plaques
<i>h⁺ c⁺ t⁺</i>	321
<i>h c t</i>	338
<i>h⁺ c t</i>	26
<i>h c⁺ t⁺</i>	30
<i>h⁺ c t⁺</i>	106
<i>h c⁺ t</i>	110
<i>h⁺ c⁺ t</i>	5
<i>h c t⁺</i>	6

- (a) Determine the order of the three genes on the phage chromosome.
(b) Determine the map distances between the genes.
(c) Determine the coefficient of coincidence and the interference (see p. 000 in Chapter 7).
27. A donor strain of bacteria with genes *a⁺ b⁺ c⁺* is infected with phages to map the donor chromosome with generalized transduction. The phage lysate from the bacterial cells is collected and used to infect a second strain of bacteria that are *a⁻ b⁻ c⁻*. Bacteria with the *a⁺* gene are selected and the percentage of cells with cotransduced *b⁺* and *c⁺* genes are recorded.

Donor	Recipient	Selected gene	Cells with cotransduced gene (%)
<i>a⁺ b⁺ c⁺</i>	<i>a⁻ b⁻ c⁻</i>	<i>a⁺</i>	25 <i>b⁺</i>
		<i>a⁺</i>	3 <i>c⁺</i>

Is the *b* or *c* gene closer to *a*? Explain your reasoning.

28. A donor strain of bacteria with genotype *leu*⁺ *gal*[−] *pro*⁺ is infected with phages. The phage lysate from the bacterial cells is collected and used to infect a second strain of bacteria that are *leu*[−] *gal*⁺ *pro*[−]. The second strain is selected for *leu*⁺, and the following cotransduction data are obtained:

Donor	Recipient	Selected gene	Cells with cotransduced gene (%)
<i>leu</i> ⁺ <i>gal</i> [−] <i>pro</i> ⁺	<i>leu</i> [−] <i>gal</i> ⁺ <i>pro</i> [−]	<i>leu</i> ⁺	47 <i>pro</i> ⁺
		<i>leu</i> ⁺	26 <i>gal</i> [−]

Which genes are closest, *leu* and *gal* or *leu* and *pro*?

29. A geneticist isolates two new mutations from the *rII* region of bacteriophage T4, called *rII*_x and *rII*_y. *E. coli* B cells are simultaneously infected with phages carrying the *rII*_x mutation and with phages carrying the *rII*_y mutation. After the cells have lysed, samples of the phage lysate are collected. One sample is grown on *E. coli* K cells and a second sample on *E. coli* B cells. There are 8322 plaques on *E. coli* B and 3 plaques on *E. coli* K. What is the recombination frequency between these two mutations?

CHALLENGE QUESTIONS

31. As a summer project, a microbiology student independently isolates two mutations in *E. coli* that are auxotrophic for glycine (*gly*[−]). The student wants to know whether these two mutants occur at the same cistron. Outline a procedure that the student could use to determine whether these two *gly*[−] mutations occur within the same cistron.
32. A group of genetics students mix two auxotrophic strains of bacteria: one is *leu*⁺ *trp*⁺ *his*[−] *met*[−] and the other is *leu*[−] *trp*[−] *his*⁺ *met*⁺. After mixing the two strains, they plate the

bacteria on minimal medium and observe a few prototrophic colonies (*leu*⁺ *trp*⁺ *his*⁺ *met*⁺). They assume that some gene transfer has taken place between the two strains. How can they determine whether the transfer of genes is due to conjugation, transduction, or transformation?

SUGGESTED READINGS

Aguzzi, A., and C. Weissman. 1997. Prion research: the next frontiers. *Nature* 389:795–798.
A review of research into the nature of prions.

Benzer, S. 1962. The fine structure of the gene. *Scientific American* 206(1):70–84.
A good summary of Benzer’s methodology for intragenic mapping, written by Benzer.

Birge, E. A. 2000. *Bacterial and Bacteriophage Genetics*, 4th ed. New York: Springer-Verlag.
An excellent textbook on the genetics of bacteria and bacteriophage.

Cole, L. A. 1996. The specter of biological weapons. *Scientific American* 275(6):60–65.
Reviews germ warfare and what can be done to discourage it.

30. A geneticist is working with a new bacteriophage called phage Y3 that infects *E. coli*. He has isolated eight mutant phages that fail to produce plaques when grown on *E. coli* strain K. To determine whether these mutations occur at the same functional gene, he simultaneously infects *E. coli* K cells with paired combinations of the mutants and looks to see whether plaques are formed. He obtains the following results. (A plus sign means that plaques were formed on *E. coli* K; a minus sign means that no plaques were formed on *E. coli* K.)

Mutant	1	2	3	4	5	6	7	8
1								
2	+							
3	+	+						
4	+	−	+					
5	−	+	+	+				
6	−	+	+	+	−			
7	+	−	+	−	+	+		
8	−	+	+	+	−	−	+	

- (a) To how many functional genes (cistrons) do these mutations belong?
- (b) Which mutations belong to the same functional gene?

Dale, J. 1998. *Molecular Genetics of Bacteria*, 3rd ed. New York: Wiley.
A concise summary of basic and molecular genetics of bacteria and bacteriophage.

Davies, J. 1994. Inactivation of antibiotics and the dissemination of resistance genes. *Science* 264:275–282.
Reviews the crisis of antibiotic resistance in bacteria, with particular emphasis on the physiology and genetics of resistance.

Doolittle, R. F. 1998. Microbial genomes opened up. *Nature* 392:339–342.
Discussion of sequence data on bacterial genomes and what this information provides.

Fraser, C. M., J. A. Eisen, and S. L. Salzberg. 2000. Microbial genome sequencing. *Nature* 406:799–803.
A short review of DNA sequencing of bacterial genomes.

- Heidelberg, J. F., et al. 2000. DNA sequence of both chromosomes of the cholera pathogen *Vibrio cholera*. *Nature* 406:477–483.
Report of the sequencing and analysis of the genome of the bacterium that causes cholera.
- Hershey, A. D., and R. Rotman. 1942. Genetic recombination between host-range and plaque-type mutants of bacteriophage in single bacterial cells. *Genetics* 34:44–71.
Original report of Hershey and Rotman's mapping experiments with phage.
- Ippen-Ihler, K. A., and E. G. Minkley, Jr. 1986. The conjugation system of F, the fertility factor of *Escherichia coli*. *Annual Review of Genetics* 20:593–624.
A detailed review of the F factor.
- Kruse, H., and H. Sørum. 1994. Transfer of multiple drug resistance plasmids between bacteria of diverse origins in natural microenvironments. *Applied and Environmental Microbiology* 60:4015–4021.
Reports experiments demonstrating the transfer of R plasmids between diverse bacteria under natural conditions.
- Lederberg, J., and E. L. Tatum. 1946. Gene recombination in *Escherichia coli*. *Nature* 158:558.
One of the original descriptions of Lederberg and Tatum's discovery of gene transfer in bacteria. A slightly different set of experiments showing the same result were published in 1946 in *Cold Spring Harbor Symposium on Quantitative Biology* 11:113–114.
- Meselson, M., G. Guillemin, M. Hugh-Jones, A. Langmuir, I. Popova, A. Shelokov, and O. Yampolskaya. 1994. The Sverdlovsk anthrax outbreak of 1979. *Science* 266:1202–1208.
Report of the epidemiological studies concerning the anthrax outbreak in Sverdlovsk.
- Miller, R. V. 1998. Bacterial gene swapping in nature. *Scientific American* 278(1):66–71.
Discusses the importance of gene transfer by conjugation, transformation, and transduction in nature.
- Novick, R. P. 1980. Plasmids. *Scientific American* 243(6):103–124.
A good summary of plasmids and their importance in drug resistance.
- Pace, N. R. 1997. A molecular view of microbial diversity and the biosphere. *Science* 276:734–740.
Good review of the diversity and classification of bacteria based on DNA sequence data.
- Scientific American*. 1998. Volume 279, issue 1.
This issue contains a special report with a number of articles on HIV and AIDS.
- Walsh, C. 2000. Molecular mechanisms that confer antibacterial drug resistance. *Nature* 406:775–781.
A very good review of how antibiotic resistance develops and how antibiotics can be developed that are less likely to be resisted by bacteria.
- Wollman, E. L., F. Jacob, and W. Hayes. 1962. Conjugation and genetic recombination in *Escherichia coli* K-12. *Cold Spring Harbor Symposium on Quantitative Biology* 21:141–162.
Original work on the use of interrupted conjugation to map genes in *E. coli*.

9

Chromosome Variation



Chapter 9 Opening Photo legend not supplied. Space for 2 lines of legend x 26 pica width to come to set this position.
(Charles Palek/Animals Animals.)

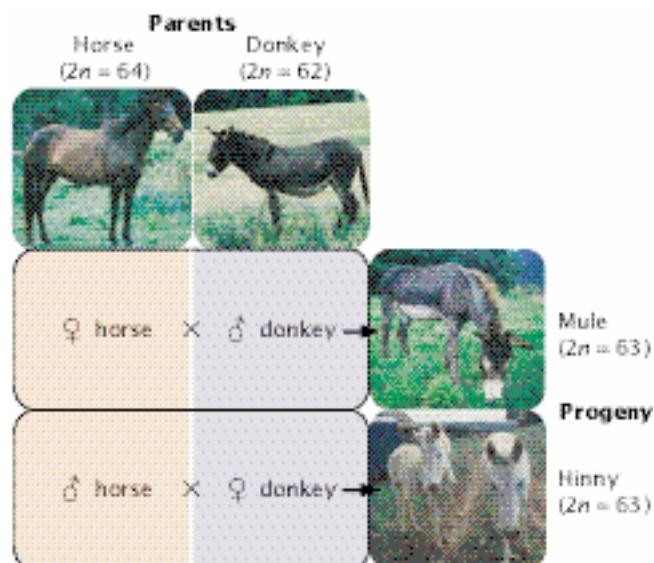
Once in a Blue Moon

One of the best-known facts of genetics is that a cross between a horse and a donkey produces a mule. Actually, it's a cross between a *female* horse and a *male* donkey that produces the mule; the reciprocal cross, between a male horse and a female donkey, produces a hinny, which has smaller ears and a bushy tail, like a horse (FIGURE 9.1). Both mules and hinnies are sterile because horses and donkeys are different species with different numbers of chromosomes: a horse has 64 chromosomes, whereas a donkey has only 62. There are also considerable differences in the sizes and shapes of the chromosomes that horses and donkeys have in common. A mule inherits 32 chromosomes from its horse mother and 31 chromosomes from its donkey father, giving the mule a chromosome number of 63. The maternal and paternal chromosomes of a mule are not homologous, and so they do not pair and separate properly in meiosis; consequently, a mule's gametes are abnormal and the animal is sterile.

In spite of the conventional wisdom that mules are sterile, reports of female mules with foals have surfaced over the years, although many of them can be attributed to mistaken identification. In several instances, a chromosome check of the alleged fertile mule has demonstrated that she is actually a donkey. In other instances, analyses of genetic markers in both mule and foal demonstrated that the foal was not the offspring of the mule; female mules are capable of lactation and sometimes they adopt the foal of a nearby horse or donkey.

In the summer of 1985, a female mule named Krause, who was pastured with a male donkey, was observed with a newborn foal. There were no other female horses or donkeys in the pasture; so it seemed unlikely that the mule had adopted the foal. Blood samples were collected from Krause, her horse and donkey parents, and her male foal, which was appropriately named Blue Moon. A team of geneticists led by Oliver Ryder of the San Diego Zoo examined their chromosomal makeup and analyzed 17 genetic markers from the blood samples.

- Once in a Blue Moon
- Chromosome Variation
 - Chromosome Morphology
 - Types of Chromosome Mutations
- Chromosome Rearrangements
 - Duplications
 - Deletions
 - Inversions
 - Translocations
- Aneuploidy
 - Types of Aneuploidy
 - Effects of Aneuploidy
 - Aneuploidy in Humans
 - Uniparental Disomy
 - Mosaicism
- Polyploidy
 - Autopolyploidy
 - Allopolyploidy
 - The Significance of Polyploidy
- Chromosome Mutations and Cancer



9.1 A cross between a female horse and a male donkey produces a mule; a cross between a male horse and a female donkey produces a hinny.

(Clockwise from top left, Bonnie Rauch/Photo Researchers; R.J. Erwin/Photo Researchers; Bruce Gaylord/Visuals Unlimited; Bill Kamin/Visuals Unlimited).

Krause's karyotype revealed that she was indeed a mule, with 63 chromosomes and blood type genes that were a mixture of those found in donkeys and horses. Blue Moon also had 63 chromosomes and, like his mother, he possessed both donkey and horse genes (FIGURE 9.2). Remarkably, he seemed to have inherited the entire set of horse chromosomes that were present in his mother. A mule's horse and donkey chromosomes would be expected to segregate randomly when the mule produces its own gametes; so Blue Moon *should* have inherited a mixture of horse and donkey chromosomes from his mother. The genetic markers that Ryder and his colleagues studied suggested that random segregation had not occurred. Krause and Blue Moon were therefore not only mother and son, but also sister and brother because they have the same father and they inherited the same maternal genes. The mechanism that allowed Krause to pass only horse chromosomes to

her son is not known; possibly all Krause's donkey chromosomes passed into the polar body during the first division of meiosis (see Figure 2.22), leaving the oocyte with only horse chromosomes.

Krause later gave birth to another male foal named White Lightning. Like his brother, White Lightning possessed mule chromosomes and appeared to have inherited only horse chromosomes from his mother. Additional reports of fertile female mules support the idea that their offspring inherit only horse chromosomes from their mother. When the father of a mule's offspring is a horse, the offspring is horselike in appearance, because it apparently inherits horse chromosomes from both of its parents. When the father of a mule's offspring is a donkey, however, the offspring is mulelike in appearance, because it inherits horse chromosomes from its mule mother and donkey chromosomes from its father.

Most species have a characteristic number of chromosomes, each with a distinct size and structure, and all the tissues of an organism (except for gametes) generally have the same set of chromosomes. Nevertheless, variations in chromosome number and structure do periodically arise. Individual chromosomes may lose or gain parts; the sequence of genes within a chromosome may become altered; whole chromosomes can even be lost or gained. These variations in the number and structure of chromosomes are termed **chromosome mutations**, and they frequently play an important role in evolution.

We begin this chapter by briefly reviewing some basic concepts of chromosome structure, which we learned in Chapter 2. We then consider the different types of chromosome mutations, their definitions, their features, and their phenotypic effects. Finally, we examine the role of chromosome mutations in cancer.

www.whfreeman.com/pierce

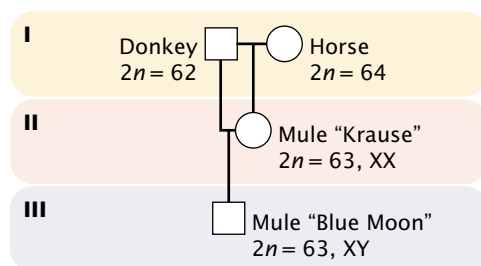
For more information on mules

Chromosome Variation

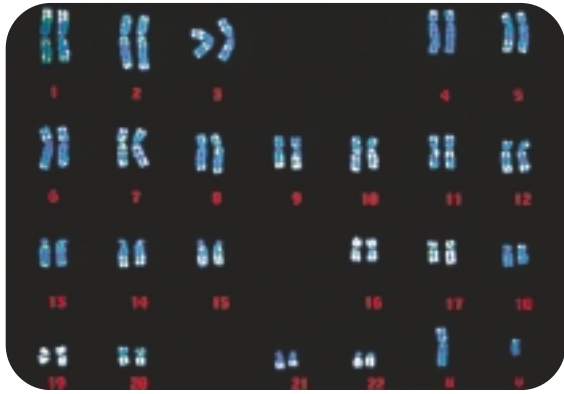
Before we consider the different types of chromosome mutations, their effects, and how they arise, we will review the basics of chromosome structure.

Chromosome Morphology

Each functional chromosome has a centromere, where spindle fibers attach, and two telomeres that stabilize the chromosome (see Figure 2.7). Chromosomes are classified into four basic types: **metacentric**, in which the centromere is located approximately in the middle, and so the chromosome has two arms of equal length; **submetacentric**, in which the centromere is displaced toward one end, creating a long arm and a short arm; **acrocentric**, in which the centromere is near one end, producing a long arm and a knob, or satellite, at the other; and **telocentric**, in which the centromere is at or very near the end of the chromosome (see



9.2 Blue Moon resulted from a cross between a fertile mule and a donkey. The probable pedigree of Blue Moon, the foal of a fertile mule, is shown.



9.3 A human karyotype consists of 46 chromosomes. A karyotype for a male is shown here; a karyotype for a female would have two X chromosomes. (ISM/Phototake).

Figure 2.8). On human chromosomes, the short arm is designated by the letter p and the long arm by the letter q.

The complete set of chromosomes that an organism possesses is called its *karyotype* and is usually presented as a picture of metaphase chromosomes lined up in descending order of their size (◀ **FIGURE 9.3**). Karyotypes are prepared from actively dividing cells, such as white blood cells, bone marrow cells, or cells from meristematic tissues of plants. After treatment with a chemical (such as colchicine) that prevents them from entering anaphase, the cells are chemically preserved, spread on a microscope slide, stained, and photographed. The photograph is then enlarged, and the individual chromosomes are cut out and arranged in a karyotype. For human chromosomes, karyotypes are often

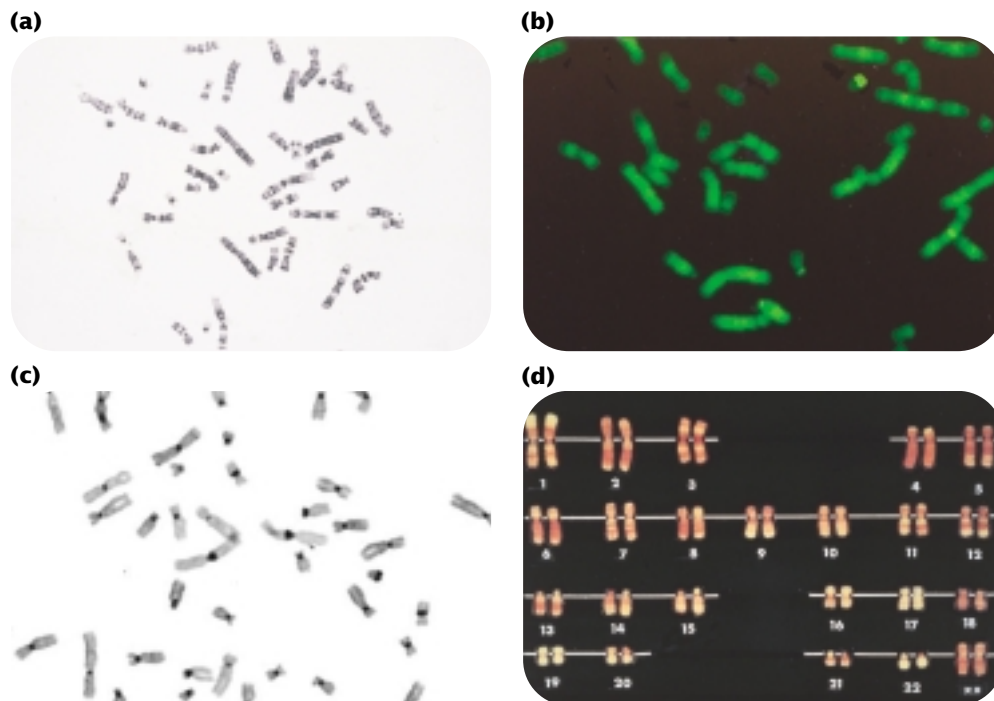
routinely prepared by automated machines, which scan a slide with a video camera attached to a microscope, looking for chromosome spreads. When a spread has been located, the camera takes a picture of the chromosomes, the image is digitized, and the chromosomes are sorted and arranged electronically by a computer.

Preparation and staining techniques have been developed to help distinguish among chromosomes of similar size and shape. For instance, chromosomes may be treated with enzymes that partly digest them; staining with a special dye called Giemsa reveals G bands, which distinguish areas of DNA that are rich in adenine–thymine base pairs (◀ **FIGURE 9.4a**). Q bands (◀ **FIGURE 9.4b**) are revealed by staining chromosomes with quinacrine mustard and viewing the chromosomes under UV light. Other techniques reveal C bands (◀ **FIGURE 9.4c**), which are regions of DNA occupied by centromeric heterochromatin, and R bands (◀ **FIGURE 9.4d**), which are rich in guanine–cytosine base pairs.

www.whfreeman.com/pierce Pictures of karyotypes, including specific chromosome abnormalities, the analysis of human karyotypes, and links to a number of Web sites on chromosomes

Types of Chromosome Mutations

Chromosome mutations can be grouped into three basic categories: chromosome rearrangements, aneuploids, and polyploids. **Chromosome rearrangements** alter the structure of chromosomes; for example, a piece of a chromosome might be duplicated, deleted, or inverted. In **aneuploidy**, the *number* of chromosomes is altered: one or more individual chromosomes are added or deleted. In **polyploidy**, one or more



9.4 Chromosome banding is revealed by special staining techniques.

(Part a, Leonard Lessin/Peter Arnold; parts b and c, Dr. Dorothy Warburton, HICC, Columbia University; part d, Dr. Ram Verma/Phototake).